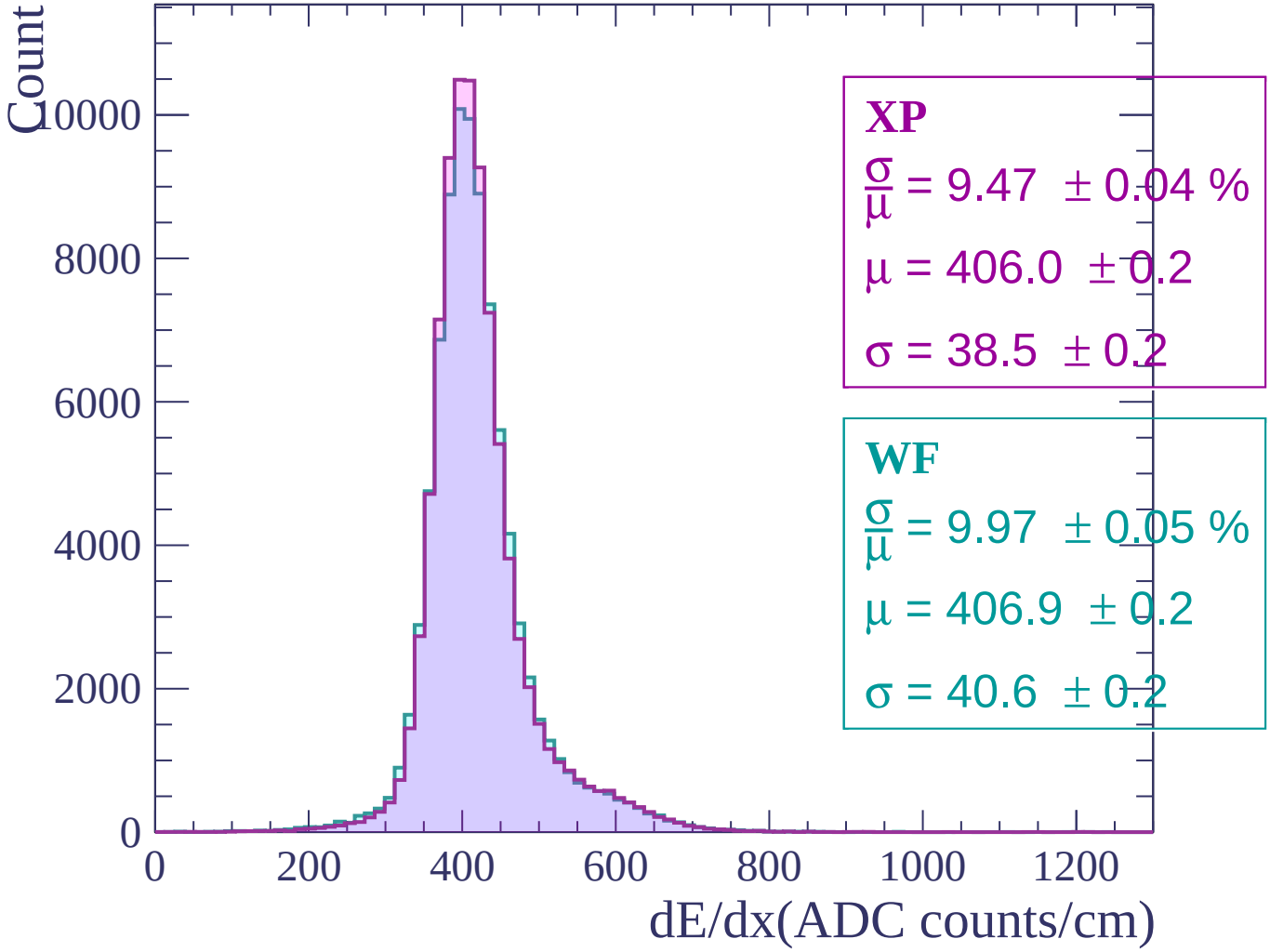
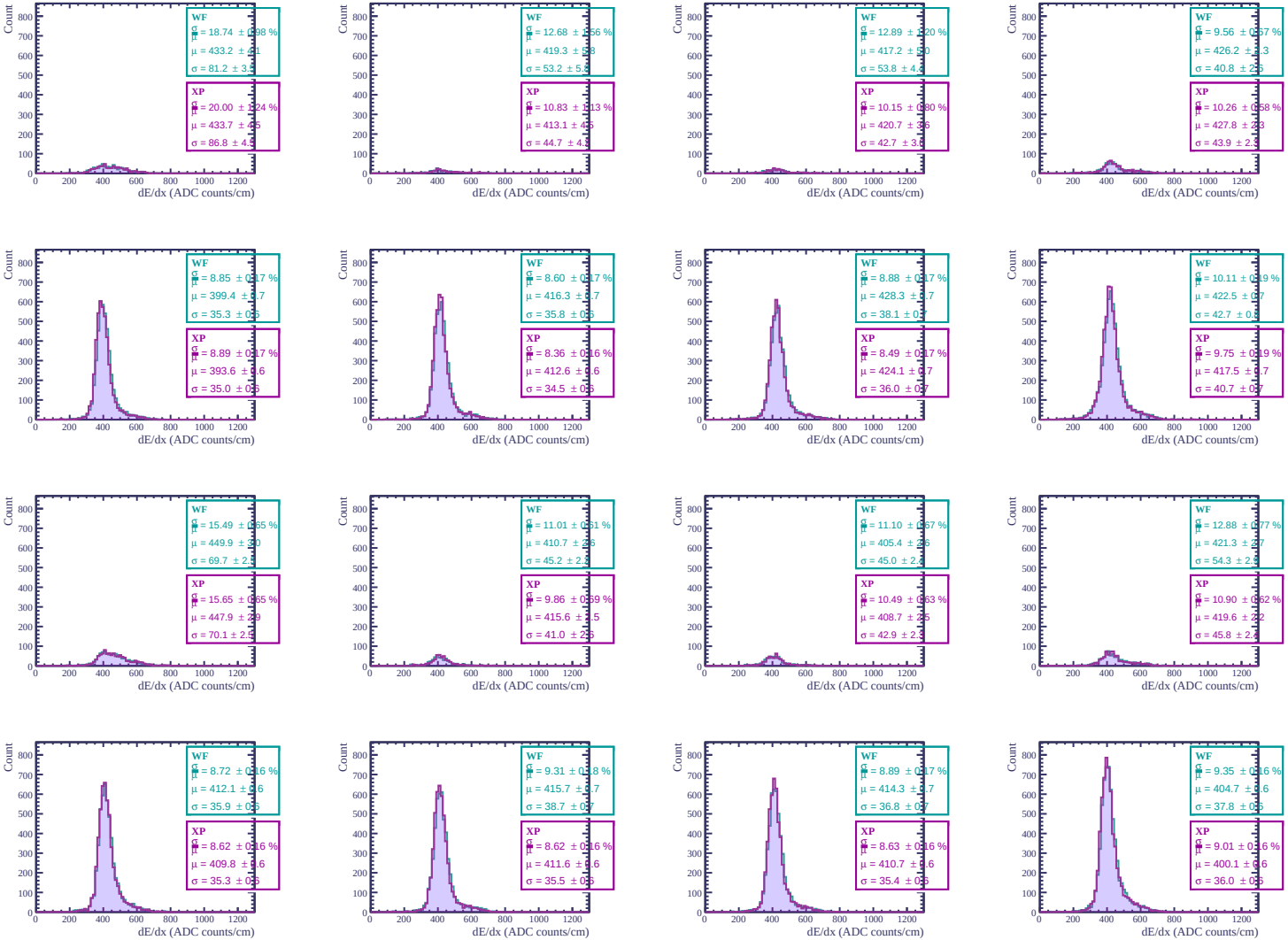
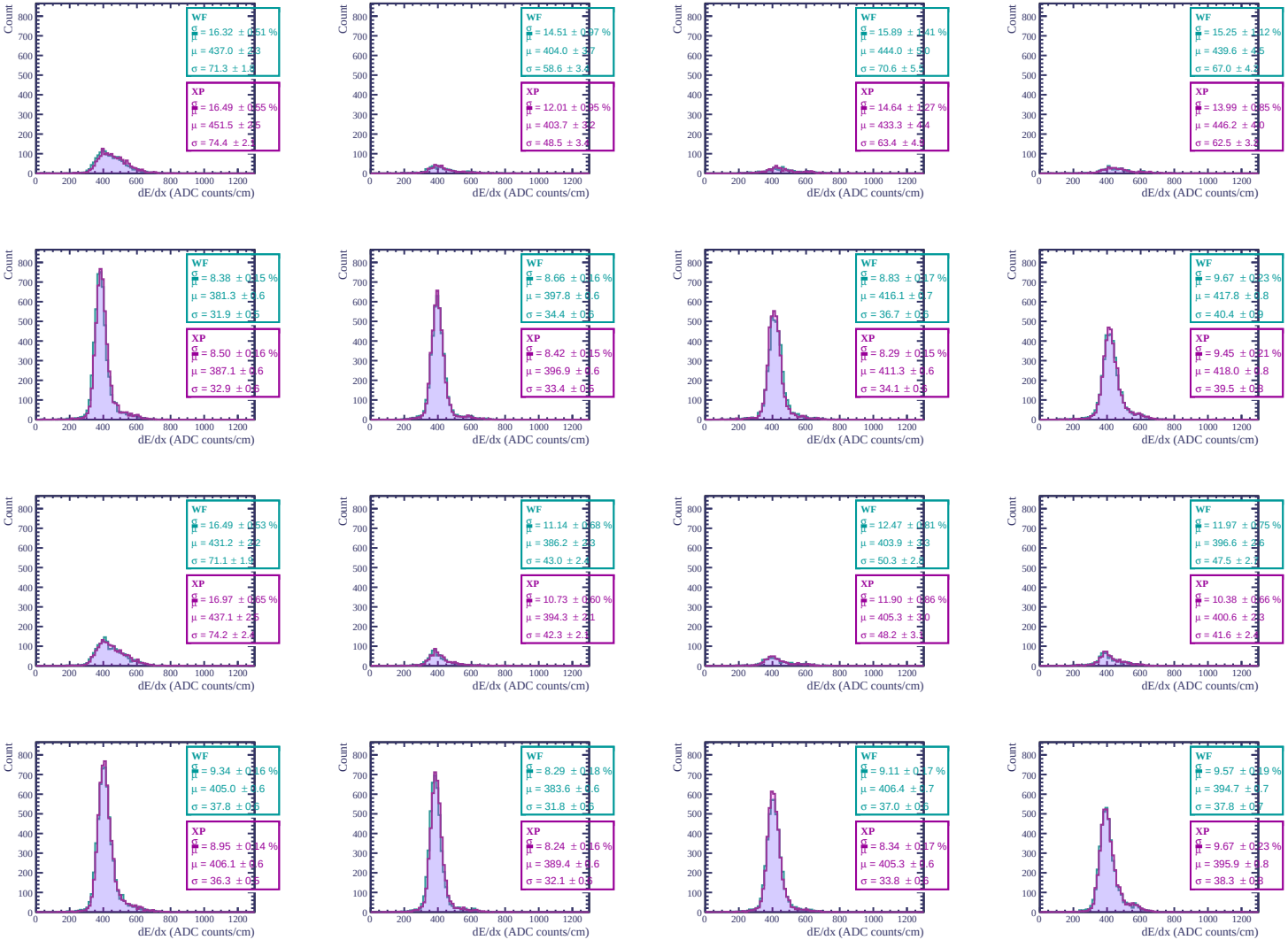
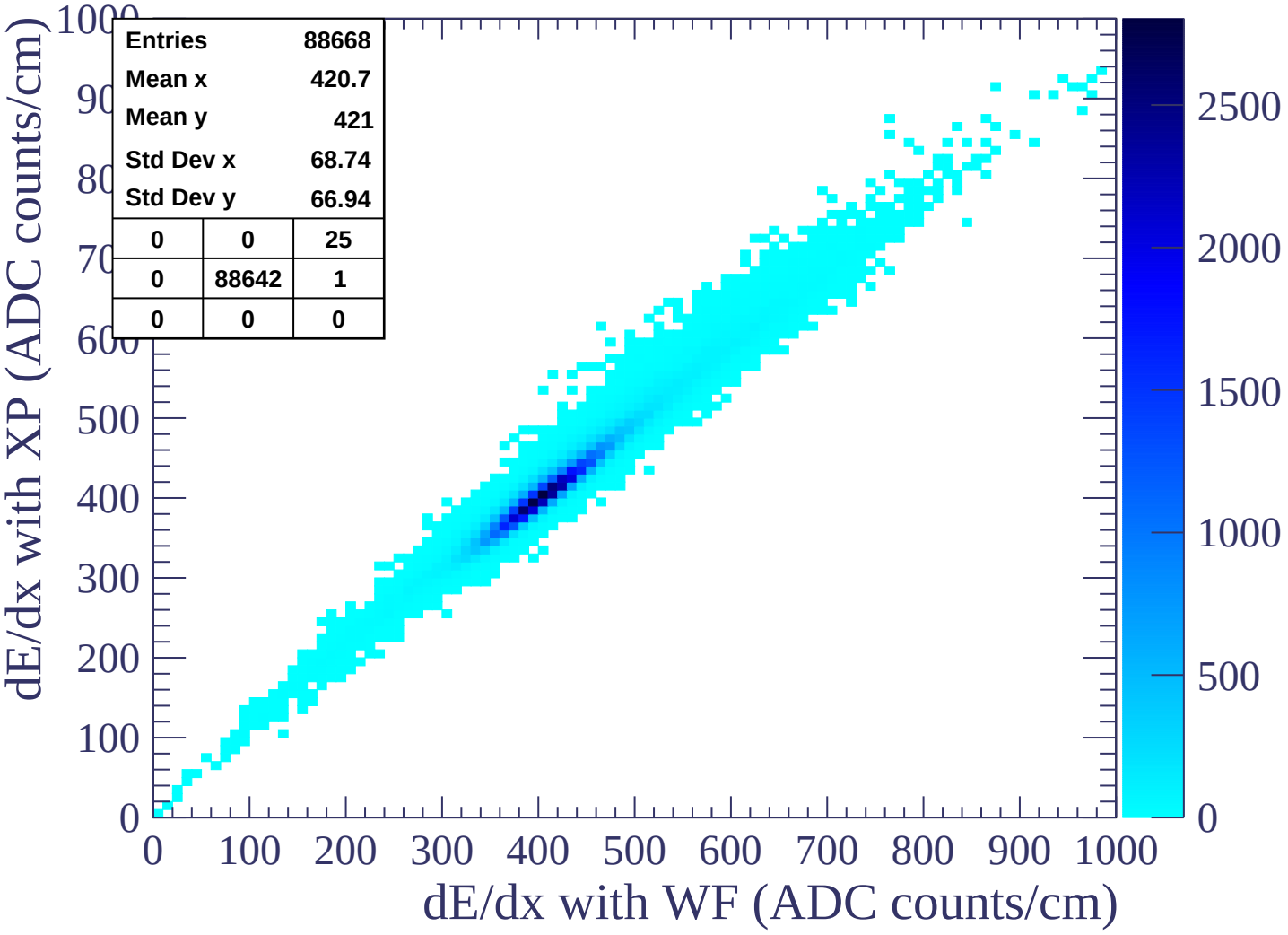
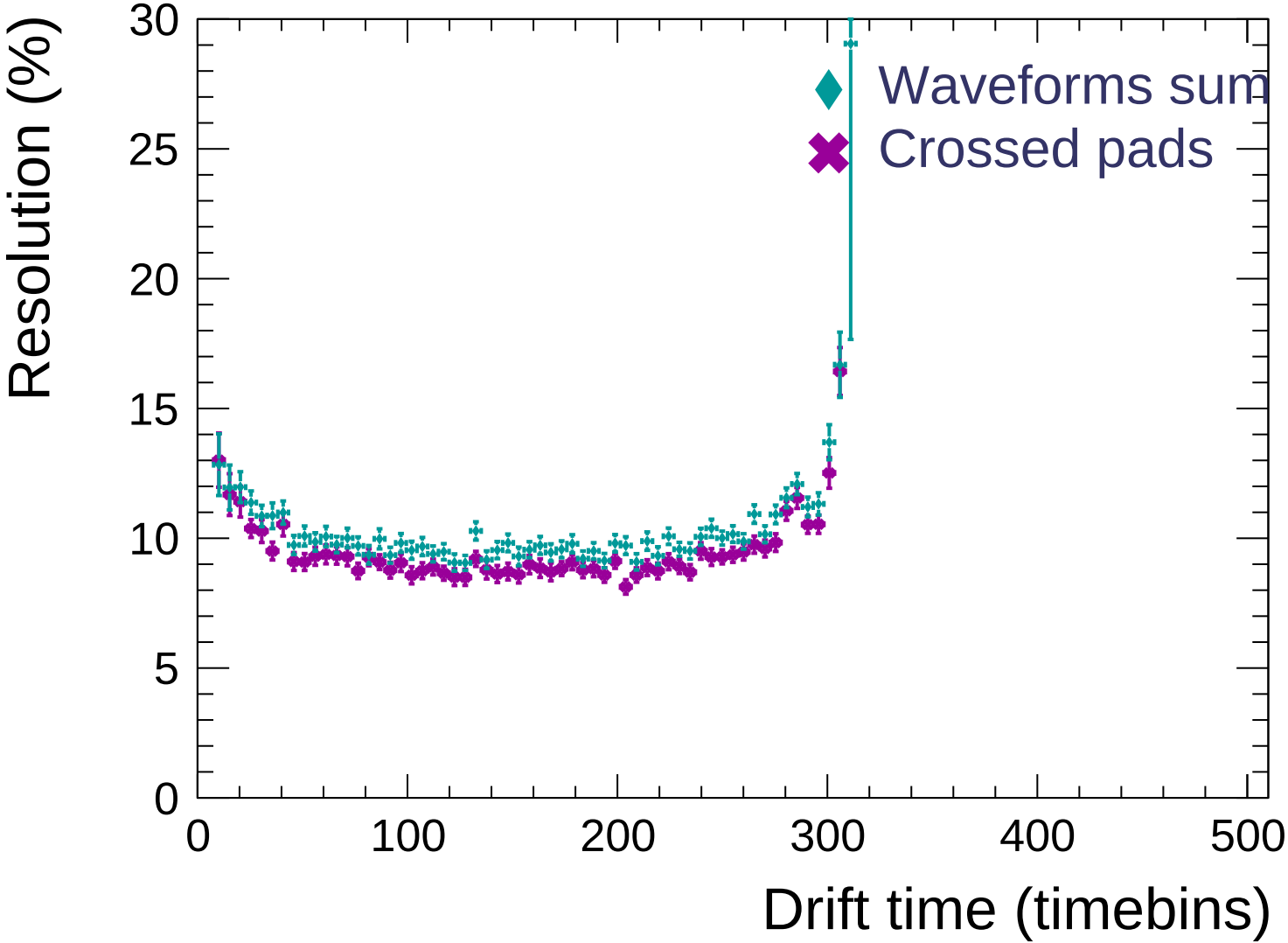
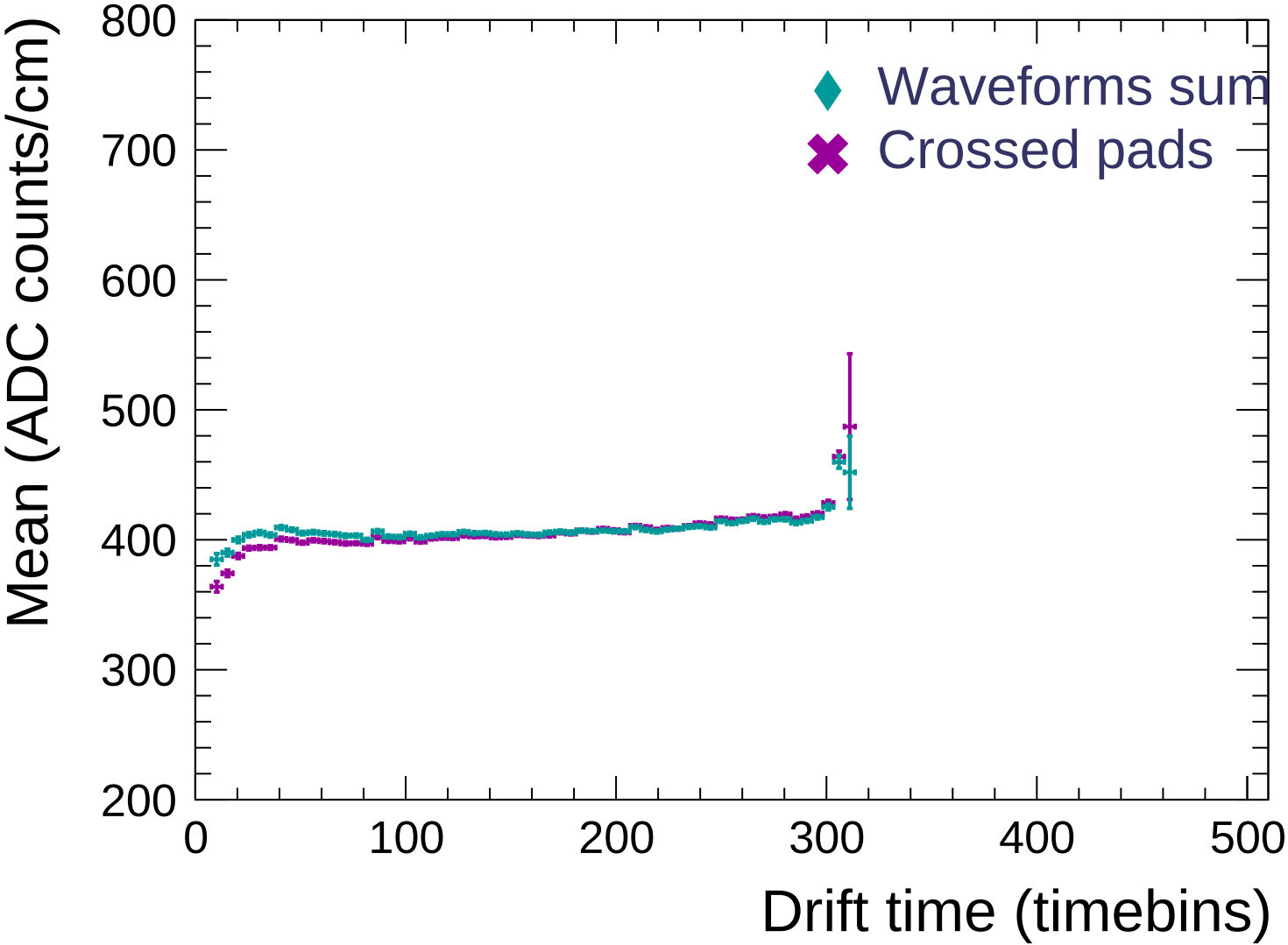

Count
10000
8000
6000
4000
2000
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP
σ/μ = 9.47 ± 0.04 %
μ = 406.0 ± 0.2
σ = 38.5 ± 0.2
WF
σ/μ = 9.97 ± 0.05 %
μ = 406.9 ± 0.2
σ = 40.6 ± 0.2

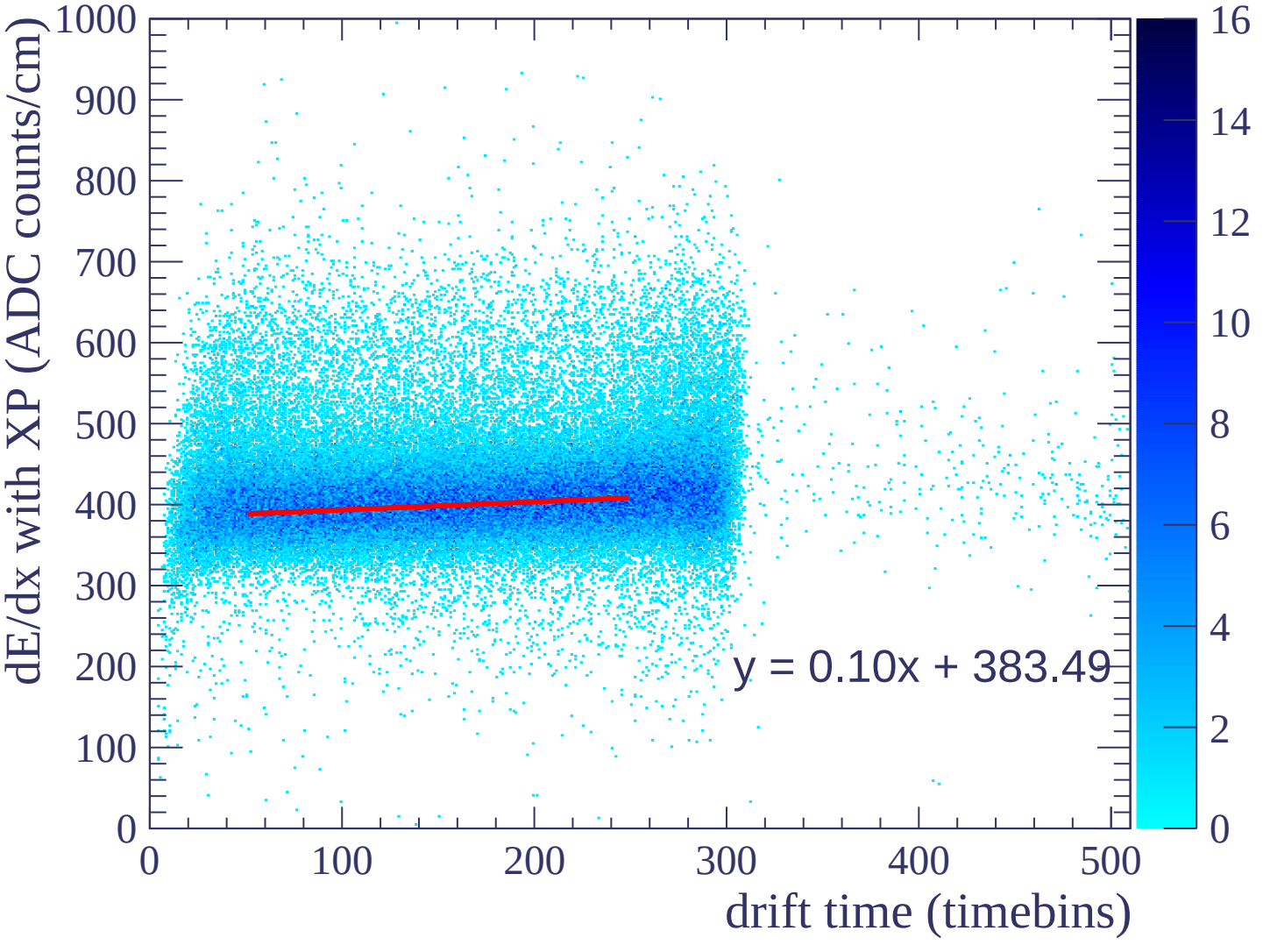

Count
dE/dx (ADC counts/cm)
WF
σ/μ = 18.74 ± 0.98 %
μ = 433.2 ± 1.1
σ = 81.2 ± 3.6
XP
σ/μ = 20.00 ± 0.24 %
μ = 433.7 ± 1.5
σ = 86.8 ± 4.1
WF
σ/μ = 12.68 ± 0.56 %
μ = 419.3 ± 1.3
σ = 53.2 ± 5.3
XP
σ/μ = 10.83 ± 0.13 %
μ = 413.1 ± 1.3
σ = 44.7 ± 4.3
WF
σ/μ = 12.89 ± 0.20 %
μ = 417.2 ± 1.0
σ = 53.8 ± 4.2
XP
σ/μ = 10.15 ± 0.30 %
μ = 420.7 ± 1.6
σ = 42.7 ± 3.6
WF
σ/μ = 9.56 ± 0.37 %
μ = 426.2 ± 1.3
σ = 40.8 ± 2.6
XP
σ/μ = 10.26 ± 0.58 %
μ = 427.8 ± 2.3
σ = 43.9 ± 2.9
WF
σ/μ = 8.85 ± 0.17 %
μ = 399.4 ± 1.7
σ = 35.3 ± 0.6
XP
σ/μ = 8.89 ± 0.17 %
μ = 393.6 ± 1.6
σ = 35.0 ± 0.6
WF
σ/μ = 8.60 ± 0.17 %
μ = 416.3 ± 1.7
σ = 35.8 ± 0.6
XP
σ/μ = 8.36 ± 0.16 %
μ = 412.6 ± 1.6
σ = 34.5 ± 0.6
WF
σ/μ = 8.88 ± 0.17 %
μ = 428.3 ± 1.7
σ = 38.1 ± 0.7
XP
σ/μ = 8.49 ± 0.17 %
μ = 424.1 ± 1.7
σ = 36.0 ± 0.7
WF
σ/μ = 10.11 ± 0.19 %
μ = 422.5 ± 1.7
σ = 42.7 ± 0.8
XP
σ/μ = 9.75 ± 0.19 %
μ = 417.5 ± 1.7
σ = 40.7 ± 0.7
WF
σ/μ = 15.49 ± 0.65 %
μ = 449.9 ± 3.0
σ = 69.7 ± 2.5
XP
σ/μ = 15.65 ± 0.65 %
μ = 447.9 ± 2.9
σ = 70.1 ± 2.5
WF
σ/μ = 11.01 ± 0.51 %
μ = 410.7 ± 2.6
σ = 45.2 ± 2.3
XP
σ/μ = 9.86 ± 0.39 %
μ = 415.6 ± 2.5
σ = 41.0 ± 2.3
WF
σ/μ = 11.10 ± 0.67 %
μ = 405.4 ± 2.6
σ = 45.0 ± 2.3
XP
σ/μ = 10.49 ± 0.63 %
μ = 408.7 ± 2.5
σ = 42.9 ± 2.3
WF
σ/μ = 12.88 ± 0.77 %
μ = 421.3 ± 2.7
σ = 54.3 ± 2.9
XP
σ/μ = 10.90 ± 0.62 %
μ = 419.6 ± 2.2
σ = 45.8 ± 2.4
WF
σ/μ = 8.72 ± 0.16 %
μ = 412.1 ± 1.6
σ = 35.9 ± 0.6
XP
σ/μ = 8.62 ± 0.16 %
μ = 409.8 ± 1.6
σ = 35.3 ± 0.6
WF
σ/μ = 9.31 ± 0.18 %
μ = 415.7 ± 1.7
σ = 38.7 ± 0.7
XP
σ/μ = 8.62 ± 0.16 %
μ = 411.6 ± 1.6
σ = 35.5 ± 0.6
WF
σ/μ = 8.89 ± 0.17 %
μ = 414.3 ± 1.7
σ = 36.8 ± 0.7
XP
σ/μ = 8.63 ± 0.16 %
μ = 410.7 ± 1.6
σ = 35.4 ± 0.6
WF
σ/μ = 9.35 ± 0.16 %
μ = 404.7 ± 1.6
σ = 37.8 ± 0.6
XP
σ/μ = 9.01 ± 0.16 %
μ = 400.1 ± 1.6
σ = 36.0 ± 0.6

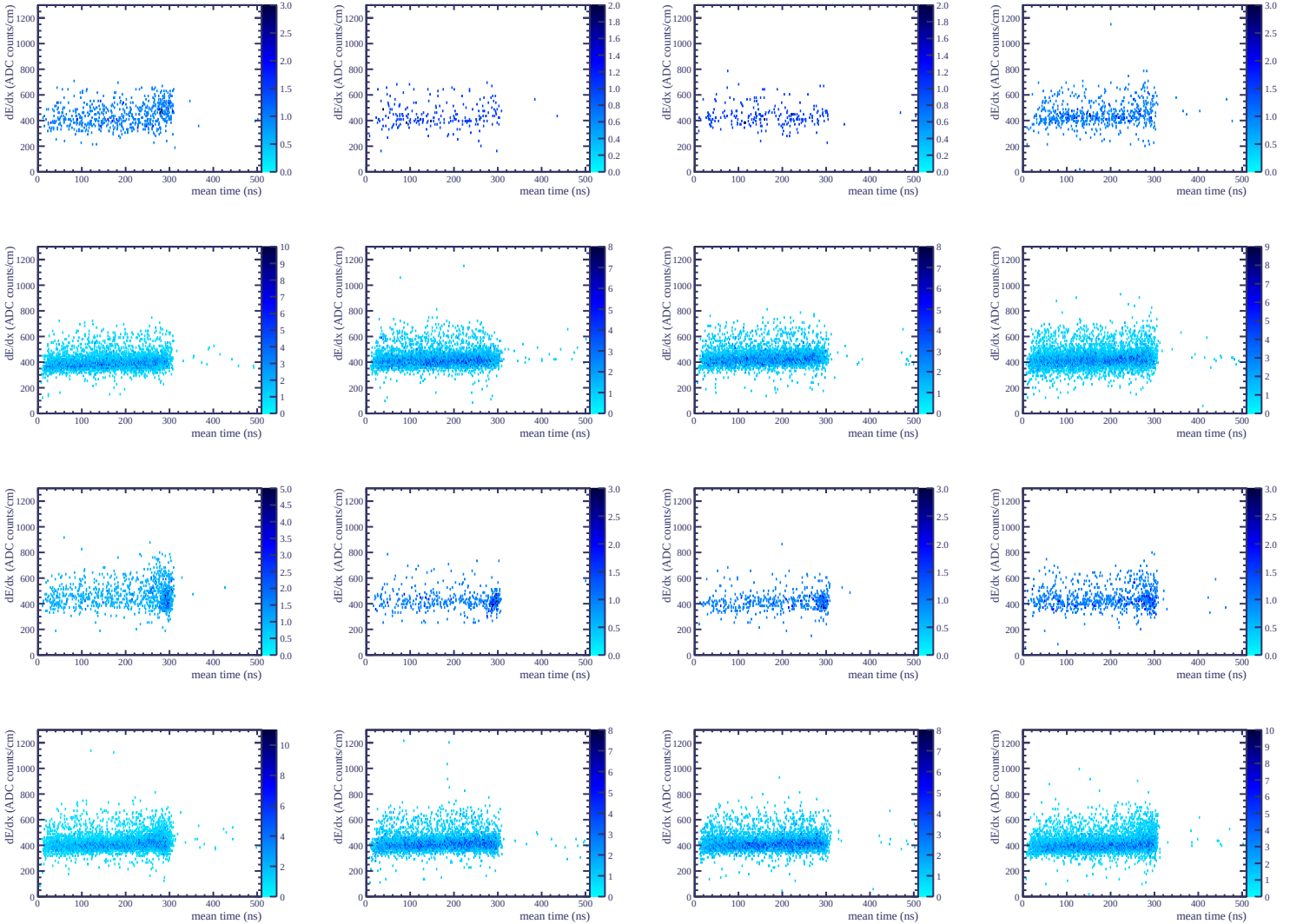

Count
dE/dx (ADC counts/cm)
WF
σ/μ = 16.32 ± 0.51 %
μ = 437.0 ± 0.3
σ = 71.3 ± 1.0
XP
σ/μ = 16.49 ± 0.55 %
μ = 451.5 ± 0.5
σ = 74.4 ± 2.1
WF
σ/μ = 14.51 ± 0.37 %
μ = 404.0 ± 0.7
σ = 58.6 ± 3.4
XP
σ/μ = 12.01 ± 0.35 %
μ = 403.7 ± 0.2
σ = 48.5 ± 3.4
WF
σ/μ = 15.89 ± 0.41 %
μ = 444.0 ± 0.0
σ = 70.6 ± 5.1
XP
σ/μ = 14.64 ± 0.27 %
μ = 433.3 ± 0.4
σ = 63.4 ± 4.4
WF
σ/μ = 15.25 ± 0.12 %
μ = 439.6 ± 0.5
σ = 67.0 ± 4.5
XP
σ/μ = 13.99 ± 0.35 %
μ = 446.2 ± 0.0
σ = 62.5 ± 3.4
WF
σ/μ = 8.38 ± 0.15 %
μ = 381.3 ± 0.6
σ = 31.9 ± 0.5
XP
σ/μ = 8.50 ± 0.16 %
μ = 387.1 ± 0.6
σ = 32.9 ± 0.5
WF
σ/μ = 8.66 ± 0.16 %
μ = 397.8 ± 0.6
σ = 34.4 ± 0.5
XP
σ/μ = 8.42 ± 0.15 %
μ = 396.9 ± 0.6
σ = 33.4 ± 0.5
WF
σ/μ = 8.83 ± 0.17 %
μ = 416.1 ± 0.7
σ = 36.7 ± 0.5
XP
σ/μ = 8.29 ± 0.15 %
μ = 411.3 ± 0.6
σ = 34.1 ± 0.5
WF
σ/μ = 9.67 ± 0.23 %
μ = 417.8 ± 0.8
σ = 40.4 ± 0.9
XP
σ/μ = 9.45 ± 0.21 %
μ = 418.0 ± 0.8
σ = 39.5 ± 0.8
WF
σ/μ = 16.49 ± 0.53 %
μ = 431.2 ± 0.2
σ = 71.1 ± 1.9
XP
σ/μ = 16.97 ± 0.55 %
μ = 437.1 ± 0.3
σ = 74.2 ± 2.4
WF
σ/μ = 11.14 ± 0.58 %
μ = 386.2 ± 0.3
σ = 43.0 ± 2.2
XP
σ/μ = 10.73 ± 0.50 %
μ = 394.3 ± 0.1
σ = 42.3 ± 2.5
WF
σ/μ = 12.47 ± 0.81 %
μ = 403.9 ± 0.3
σ = 50.3 ± 2.8
XP
σ/μ = 11.90 ± 0.86 %
μ = 405.3 ± 0.0
σ = 48.2 ± 3.5
WF
σ/μ = 11.97 ± 0.75 %
μ = 396.6 ± 0.6
σ = 47.5 ± 2.3
XP
σ/μ = 10.38 ± 0.56 %
μ = 400.6 ± 0.3
σ = 41.6 ± 2.4
WF
σ/μ = 9.34 ± 0.16 %
μ = 405.0 ± 0.6
σ = 37.8 ± 0.5
XP
σ/μ = 8.95 ± 0.14 %
μ = 406.1 ± 0.6
σ = 36.3 ± 0.5
WF
σ/μ = 8.29 ± 0.18 %
μ = 383.6 ± 0.6
σ = 31.8 ± 0.5
XP
σ/μ = 8.24 ± 0.16 %
μ = 389.4 ± 0.6
σ = 32.1 ± 0.5
WF
σ/μ = 9.11 ± 0.17 %
μ = 406.4 ± 0.7
σ = 37.0 ± 0.5
XP
σ/μ = 8.34 ± 0.17 %
μ = 405.3 ± 0.6
σ = 33.8 ± 0.5
WF
σ/μ = 9.57 ± 0.19 %
μ = 394.7 ± 0.7
σ = 37.8 ± 0.7
XP
σ/μ = 9.67 ± 0.23 %
μ = 395.9 ± 0.8
σ = 38.3 ± 0.8

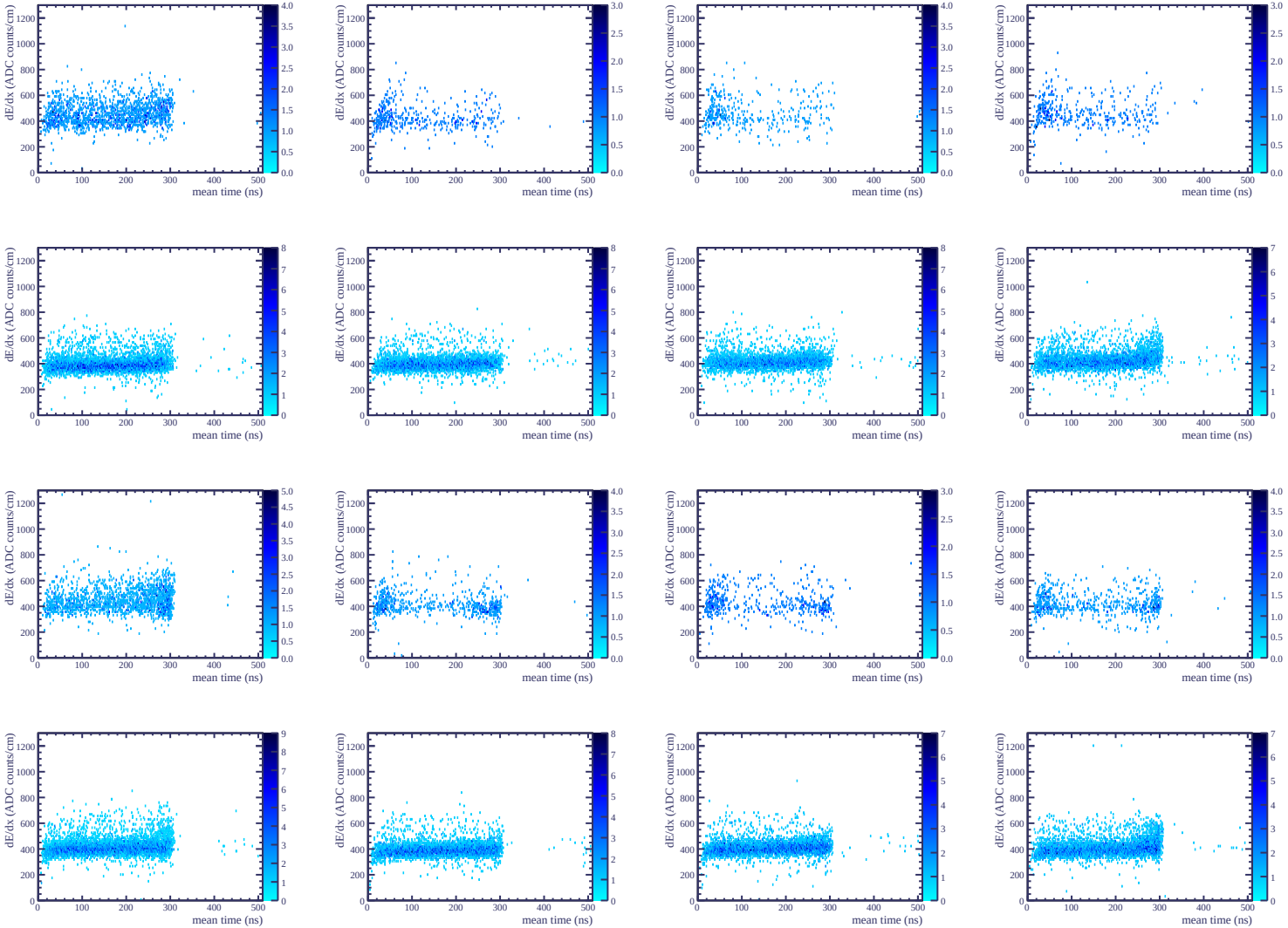

Entries 88668
Mean x 420.7
Mean y 421
Std Dev x 68.74
Std Dev y 66.94
0 0 25
0 88642 1
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
2500
2000
1500
1000
500

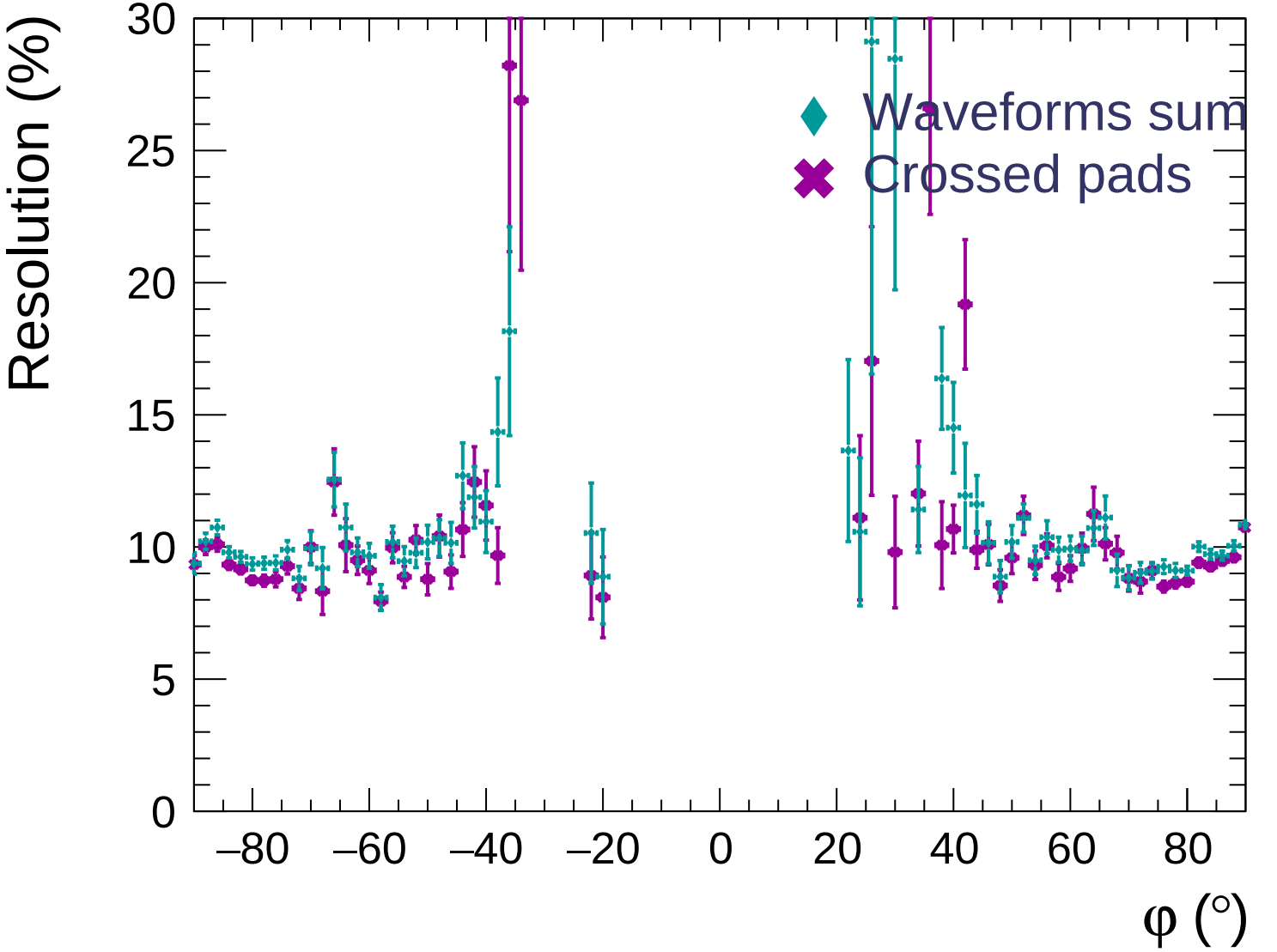

Resolution (%)
30
25
20
15
10
5
0
0
100
200
300
400
500
Drift time (timebins)
Waveforms sum
Crossed pads

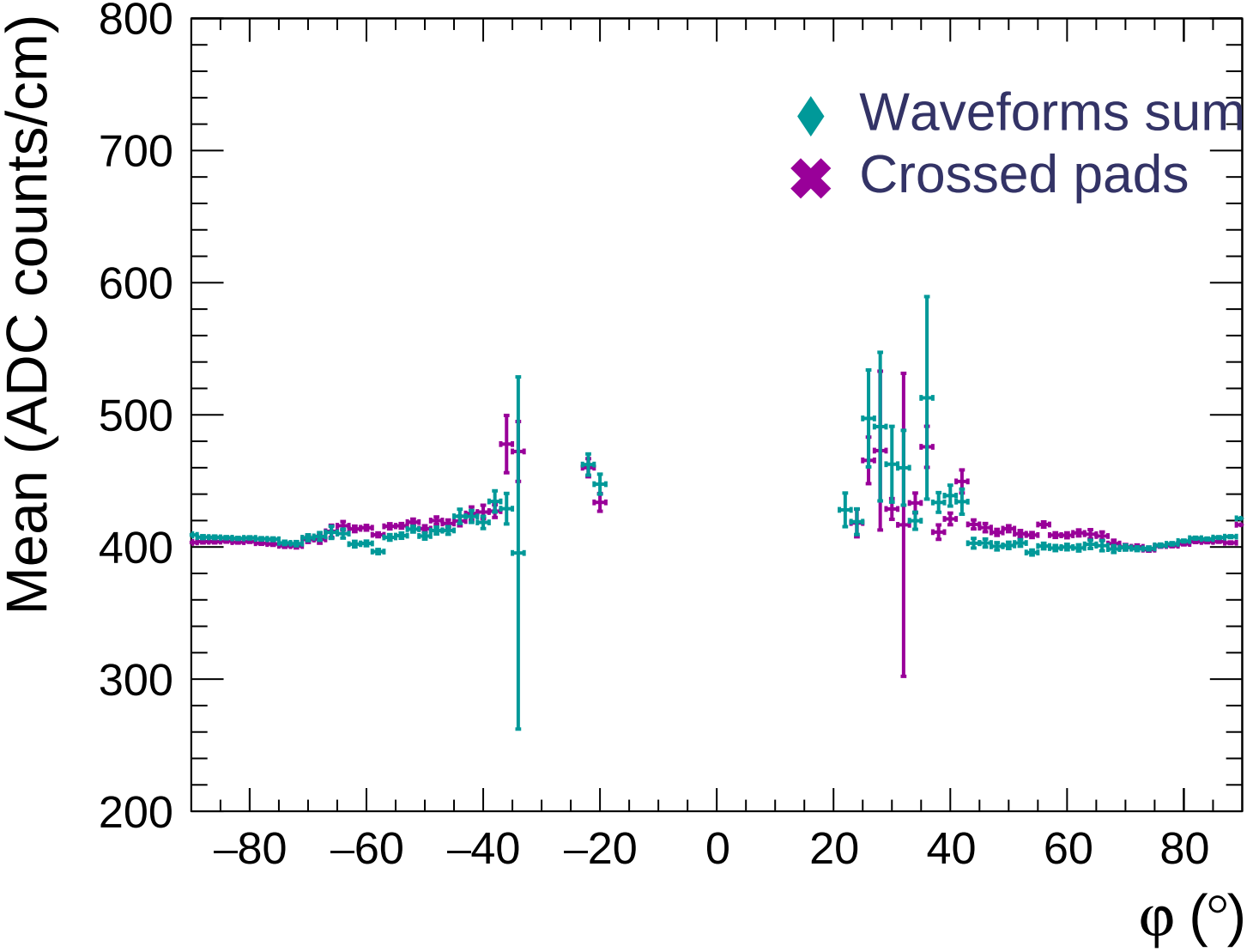

Waveforms sum
Crossed pads
Mean (ADC counts/cm)
Drift time (timebins)
800
700
600
500
400
300
200
0
100
200
300
400
500

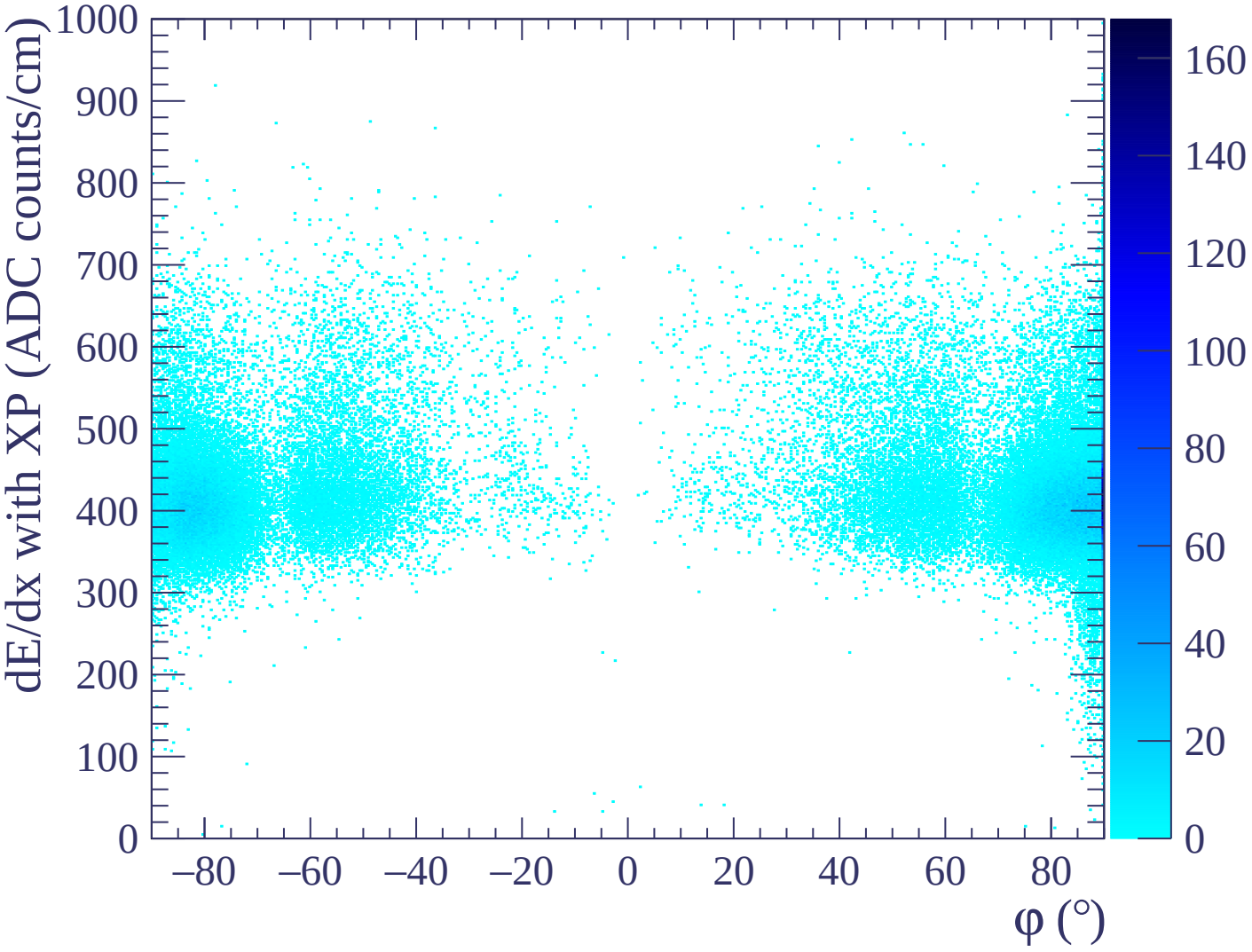

dE/dx with XP (ADC counts/cm)
drift time (timebins)
y = 0.10x + 383.49
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
2
4
6
8
10
12
14
16

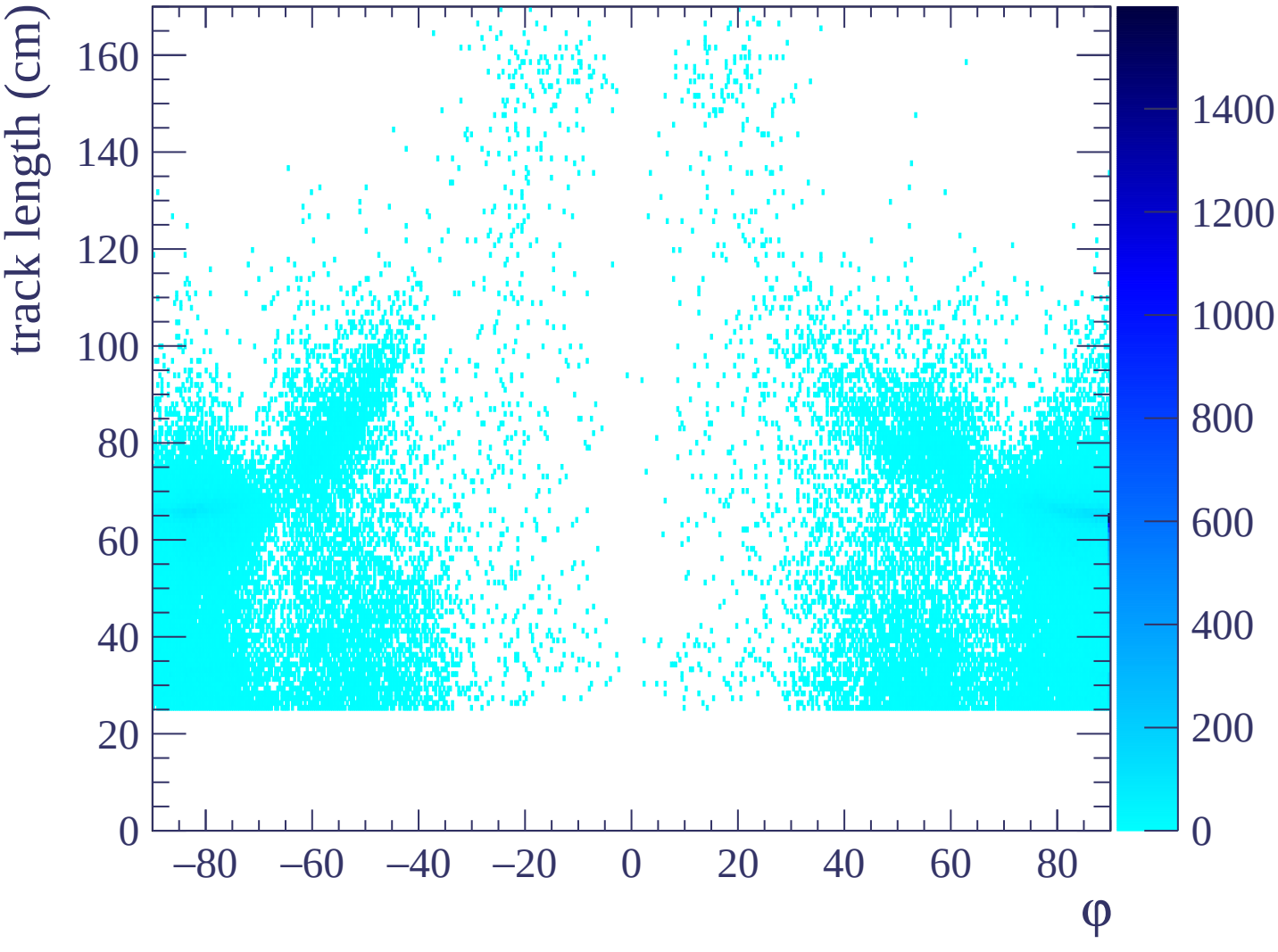

dE/dx (ADC counts/cm)
mean time (ns)

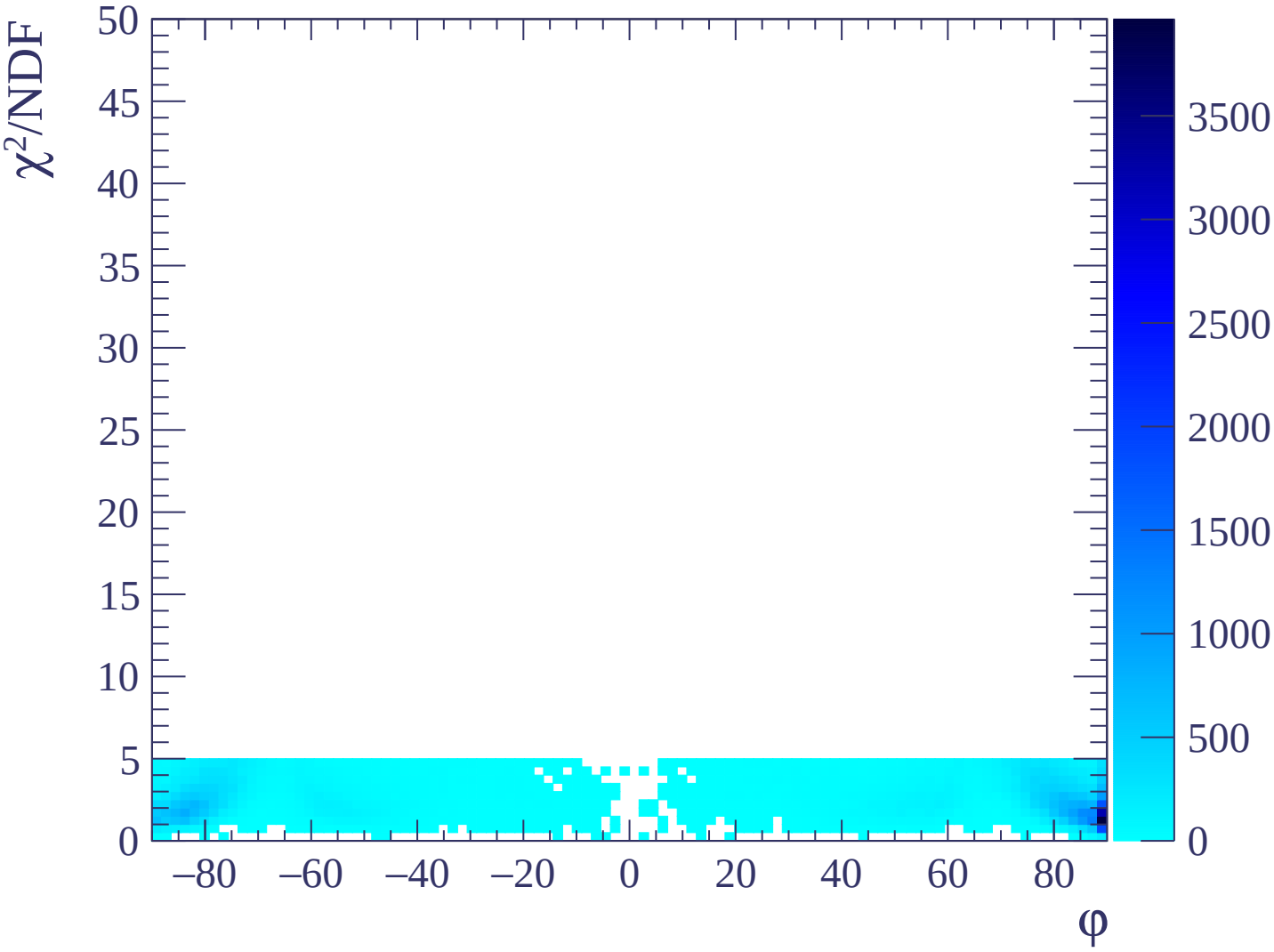

dE/dx (ADC counts/cm)
mean time (ns)

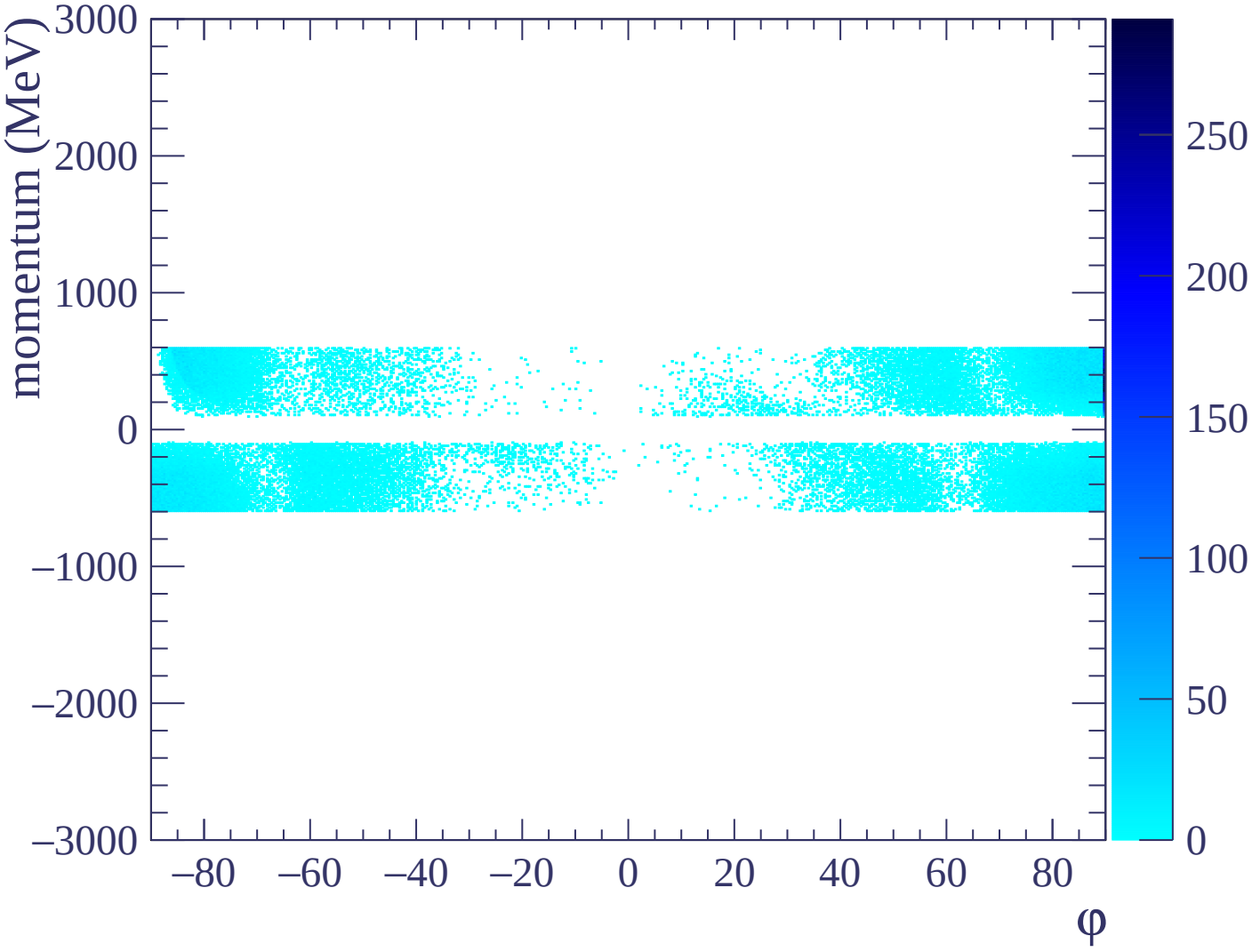

Resolution (%)
30
25
20
15
10
5
0
–80
–60
–40
–20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads

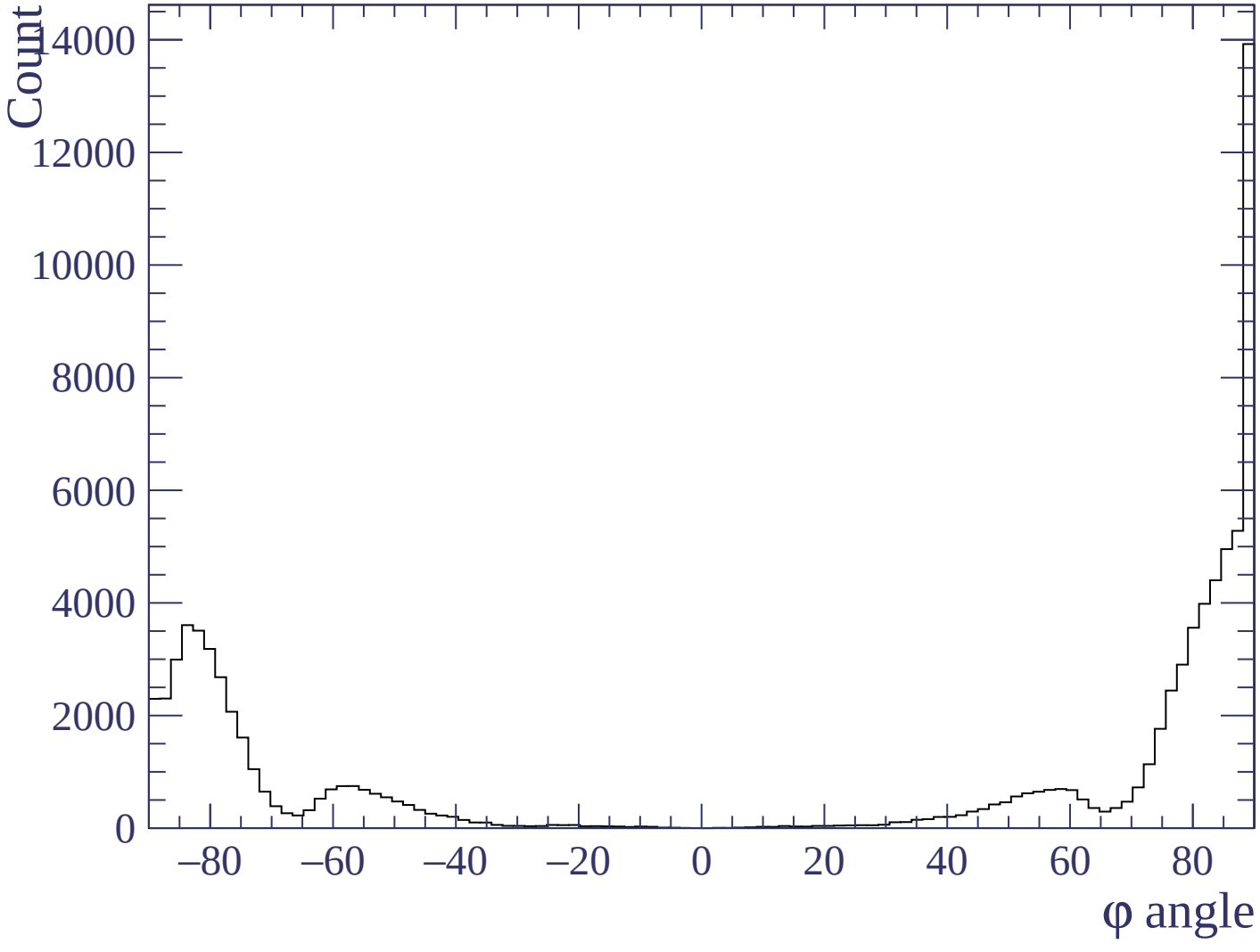

Mean (ADC counts/cm)
800
700
600
500
400
300
200
–80
–60
–40
–20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads

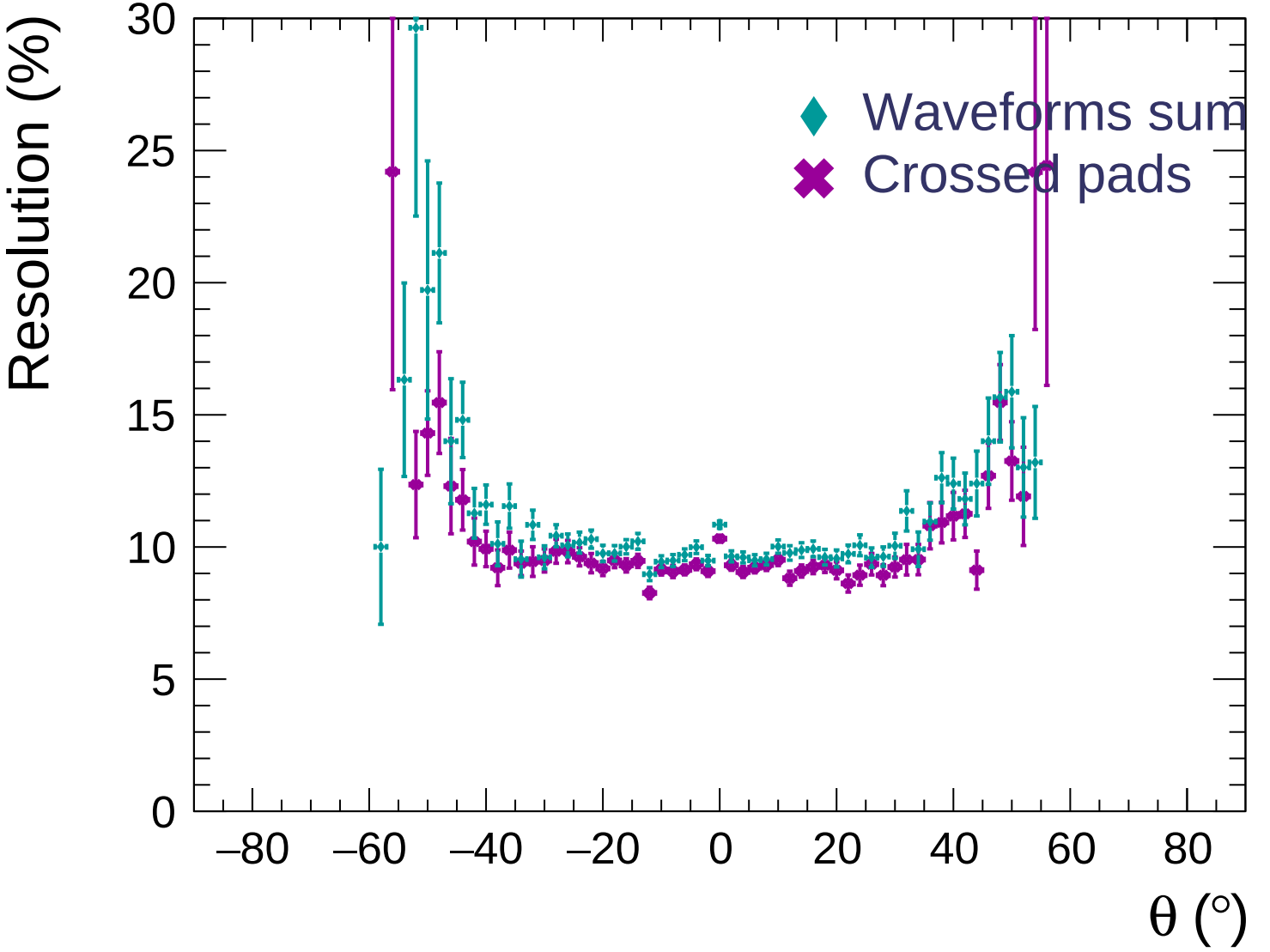

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
160
140
120
100
80
60
40
20
0

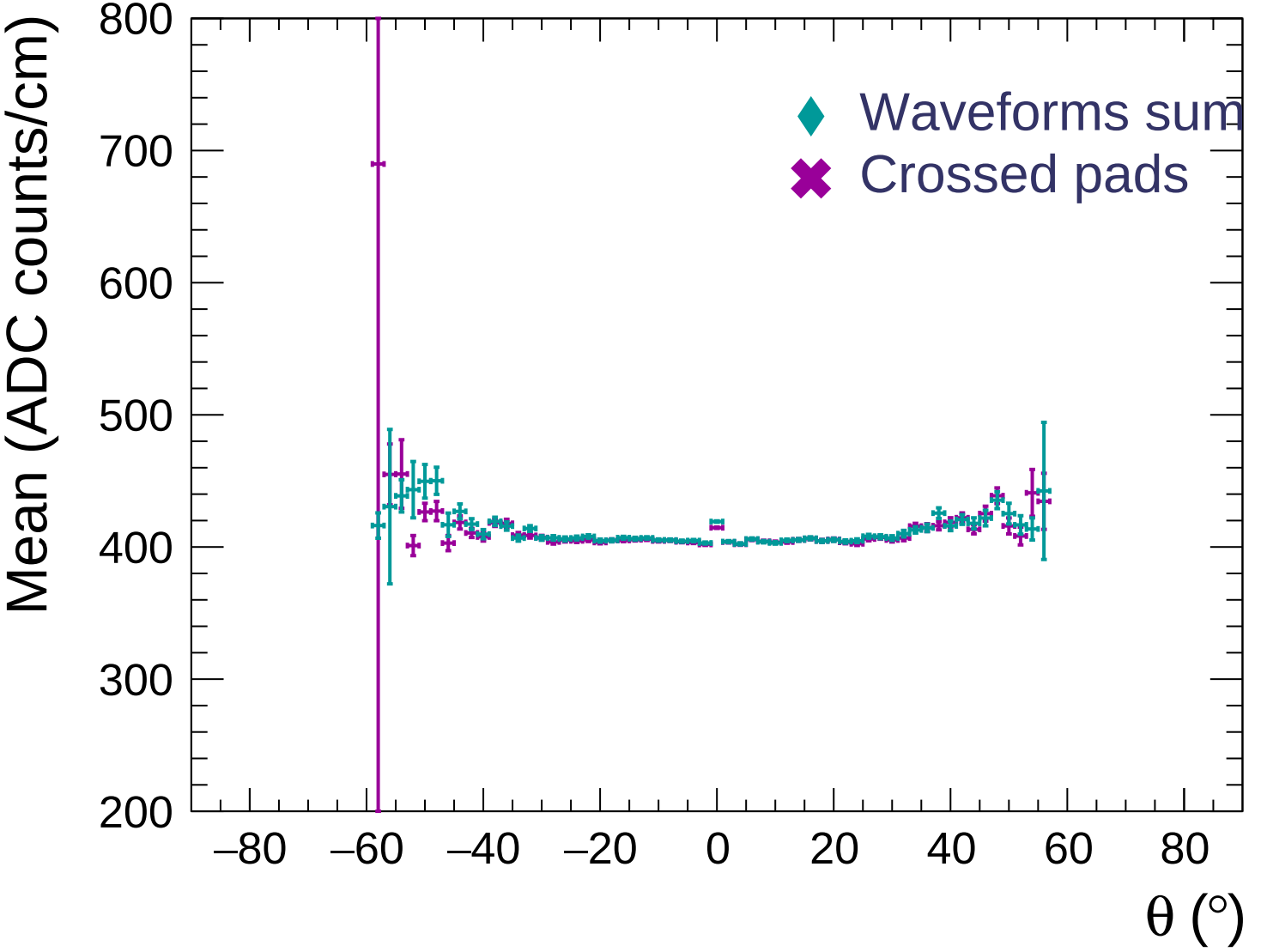

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
1400
1200
1000
800
600
400
200
0

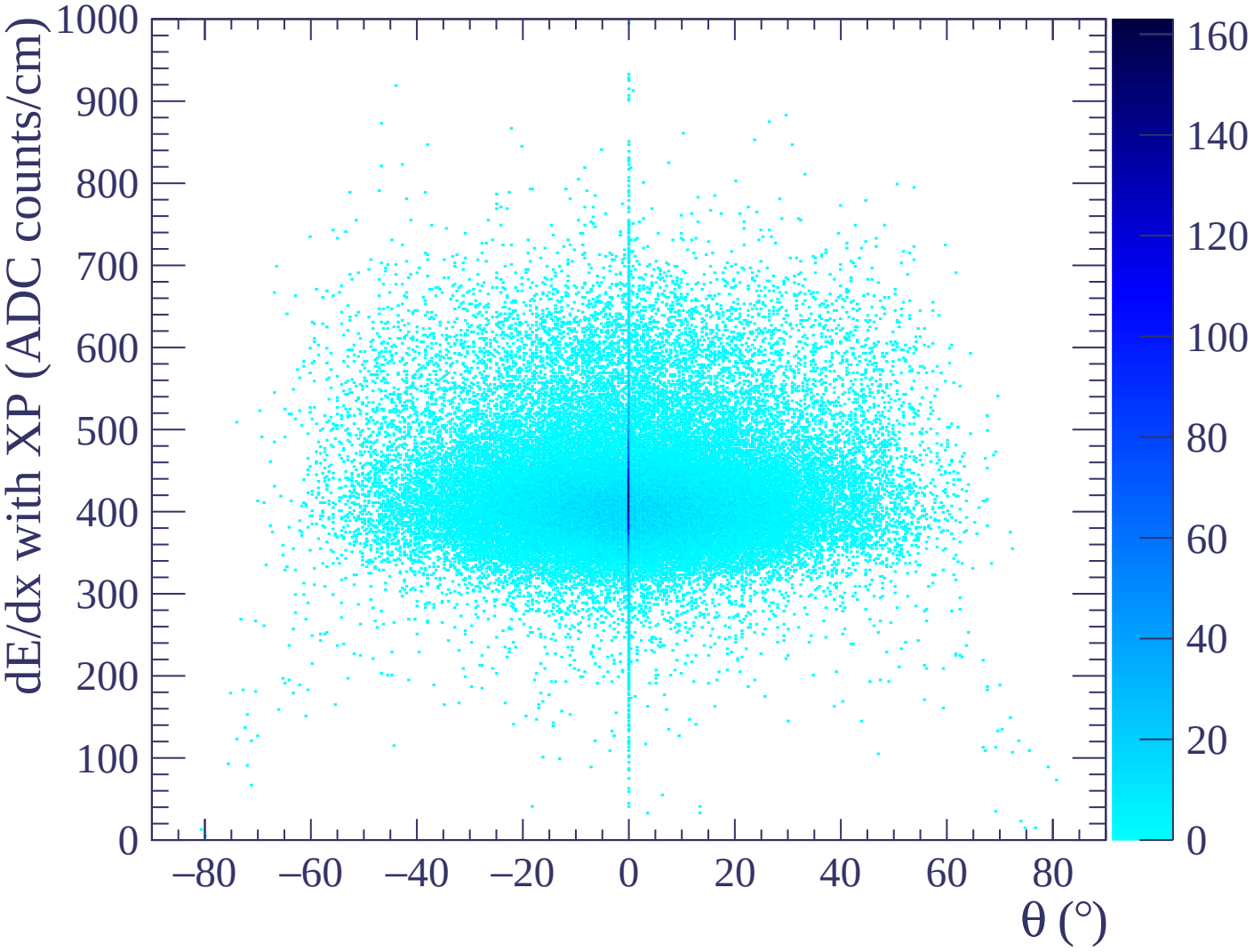

χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
3500
3000
2500
2000
1500
1000
500
0

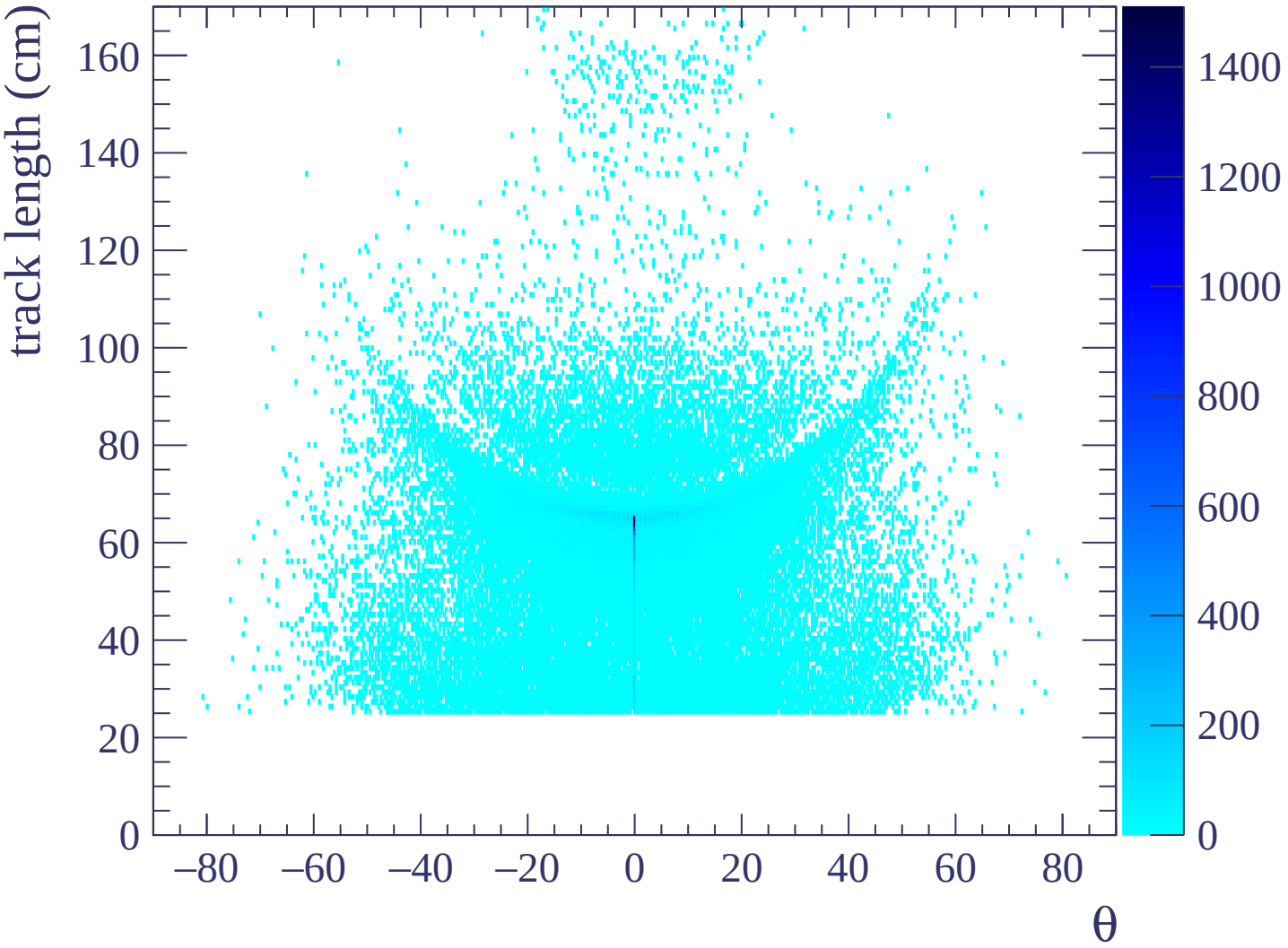

momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−80
−60
−40
−20
0
20
40
60
80
φ
250
200
150
100
50
0

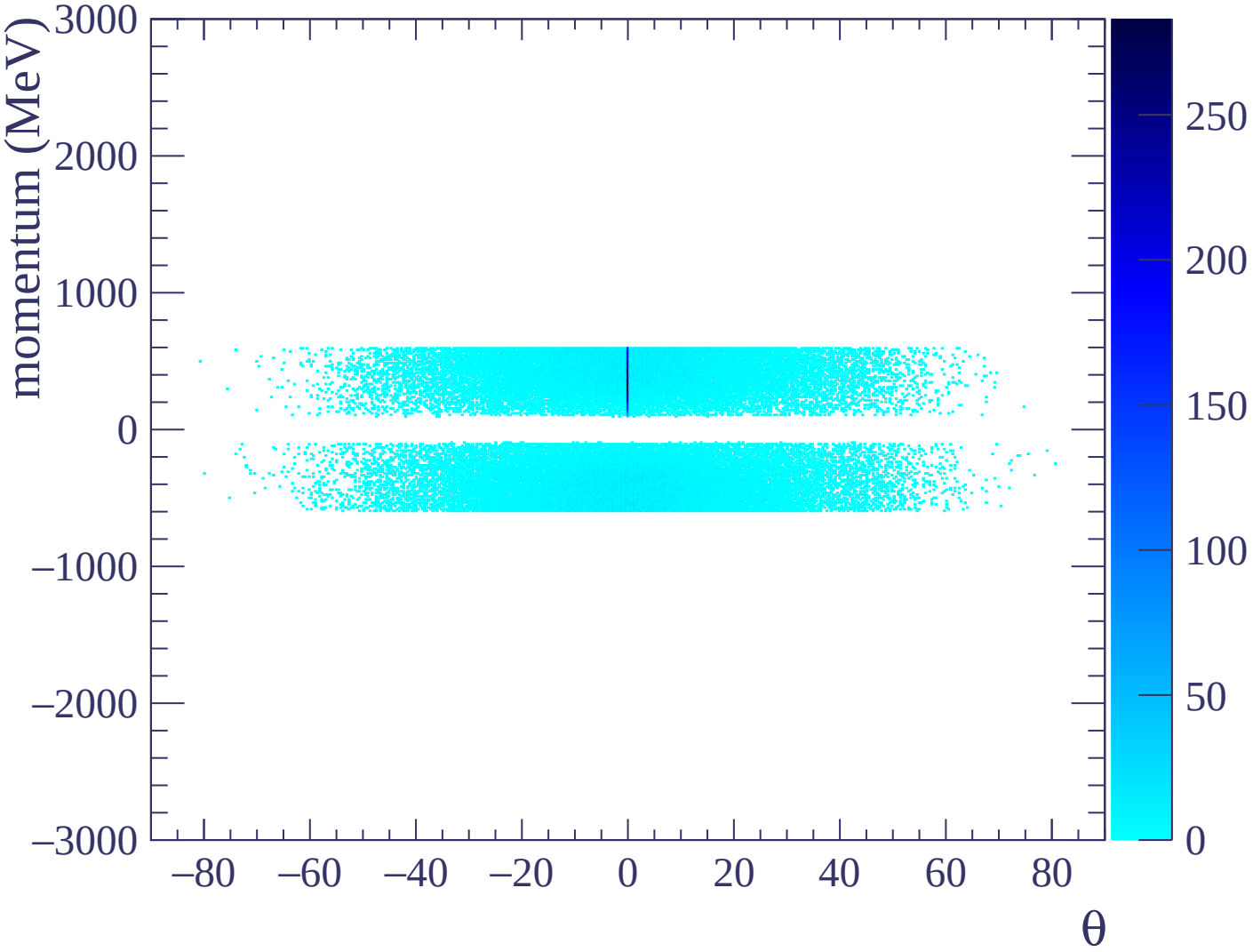

Count
14000
12000
10000
8000
6000
4000
2000
0
–80
–60
–40
–20
0
20
40
60
80
φ angle

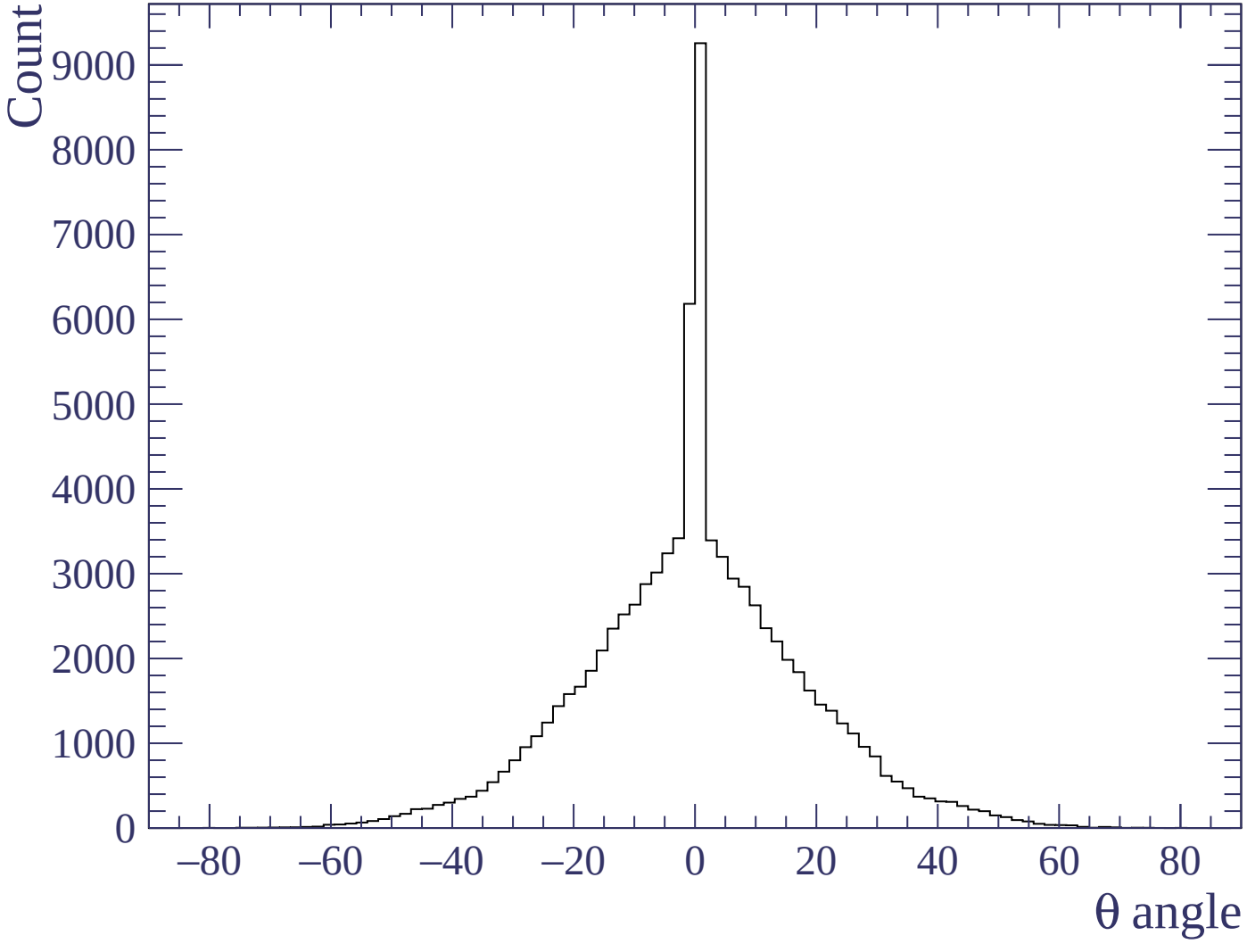

Resolution (%)
30
25
20
15
10
5
0
–80
–60
–40
–20
0
20
40
60
80
θ (°)
Waveforms sum
Crossed pads

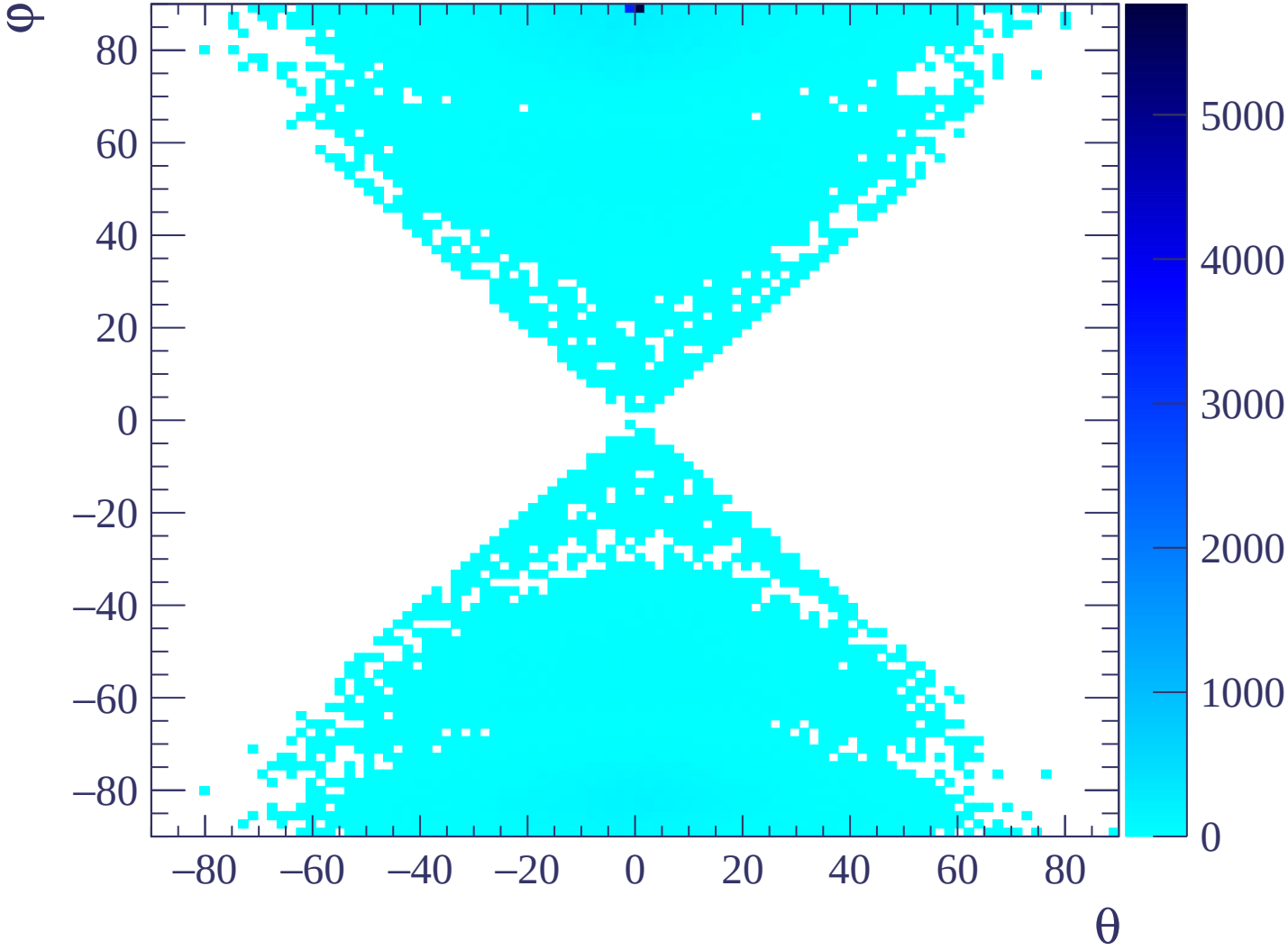

Waveforms sum
Crossed pads
Mean (ADC counts/cm)
800
700
600
500
400
300
200
–80
–60
–40
–20
0
20
40
60
80
θ (°)

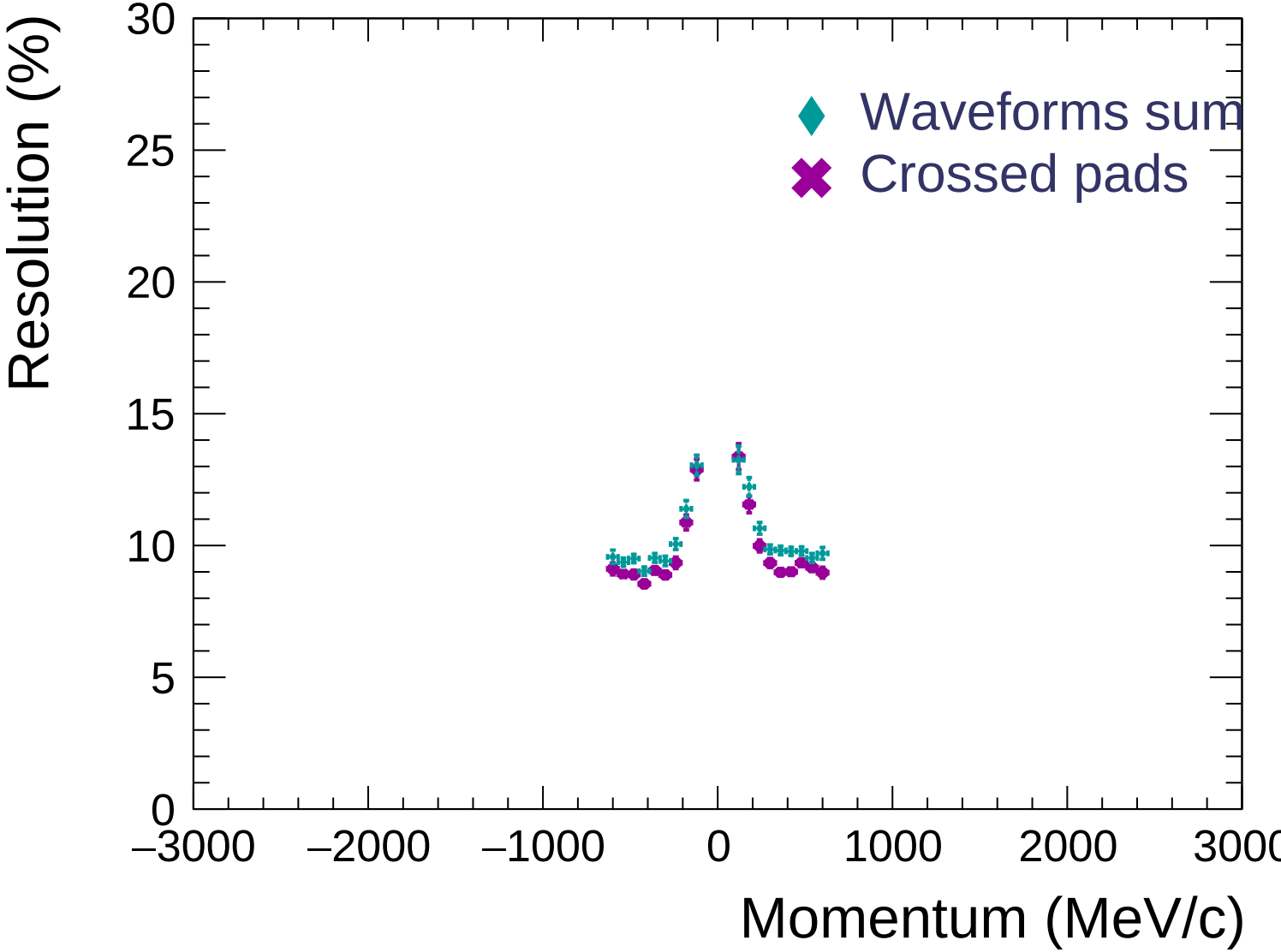

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
160
140
120
100
80
60
40
20
0

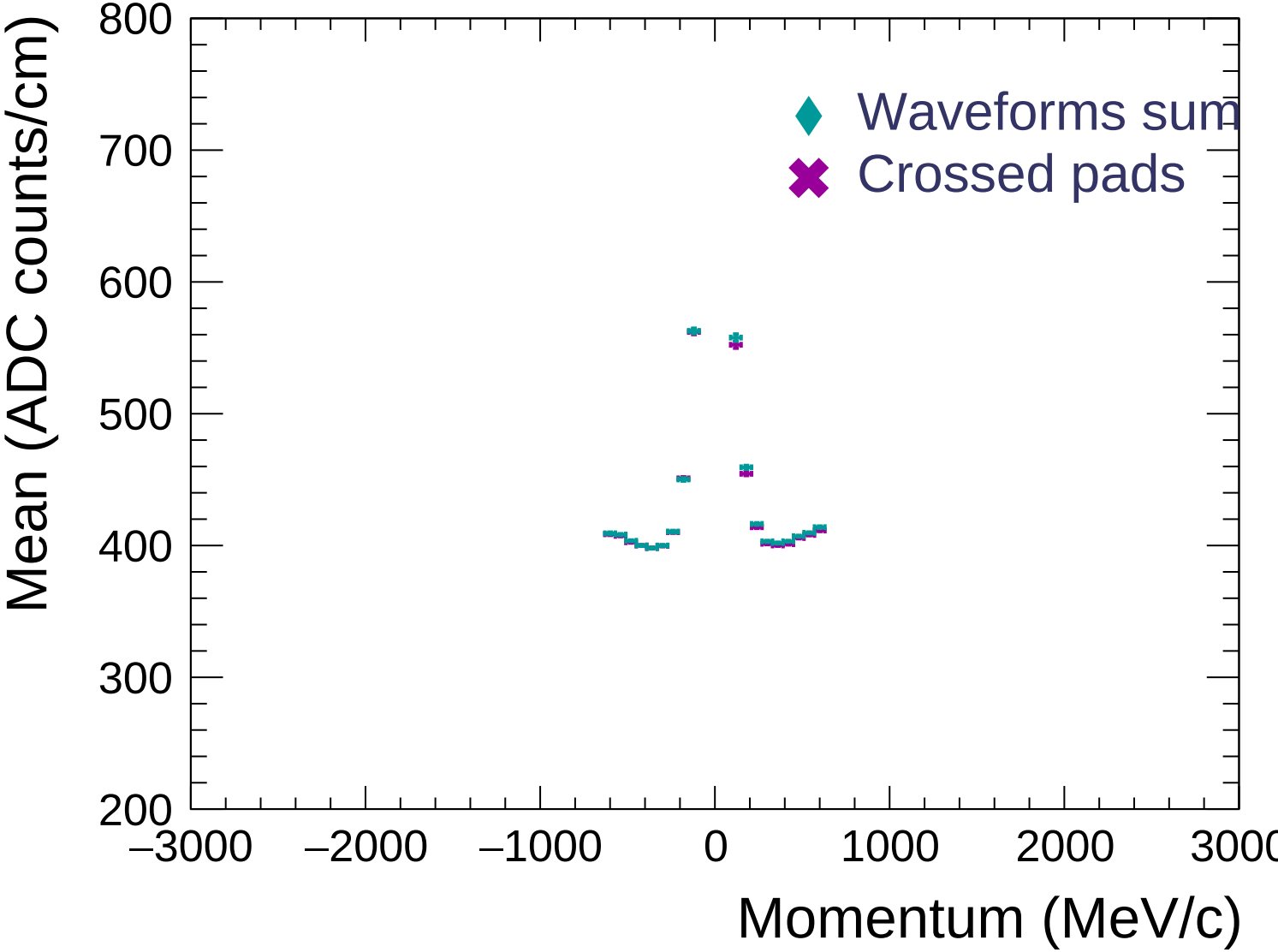

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
1400
1200
1000
800
600
400
200
0

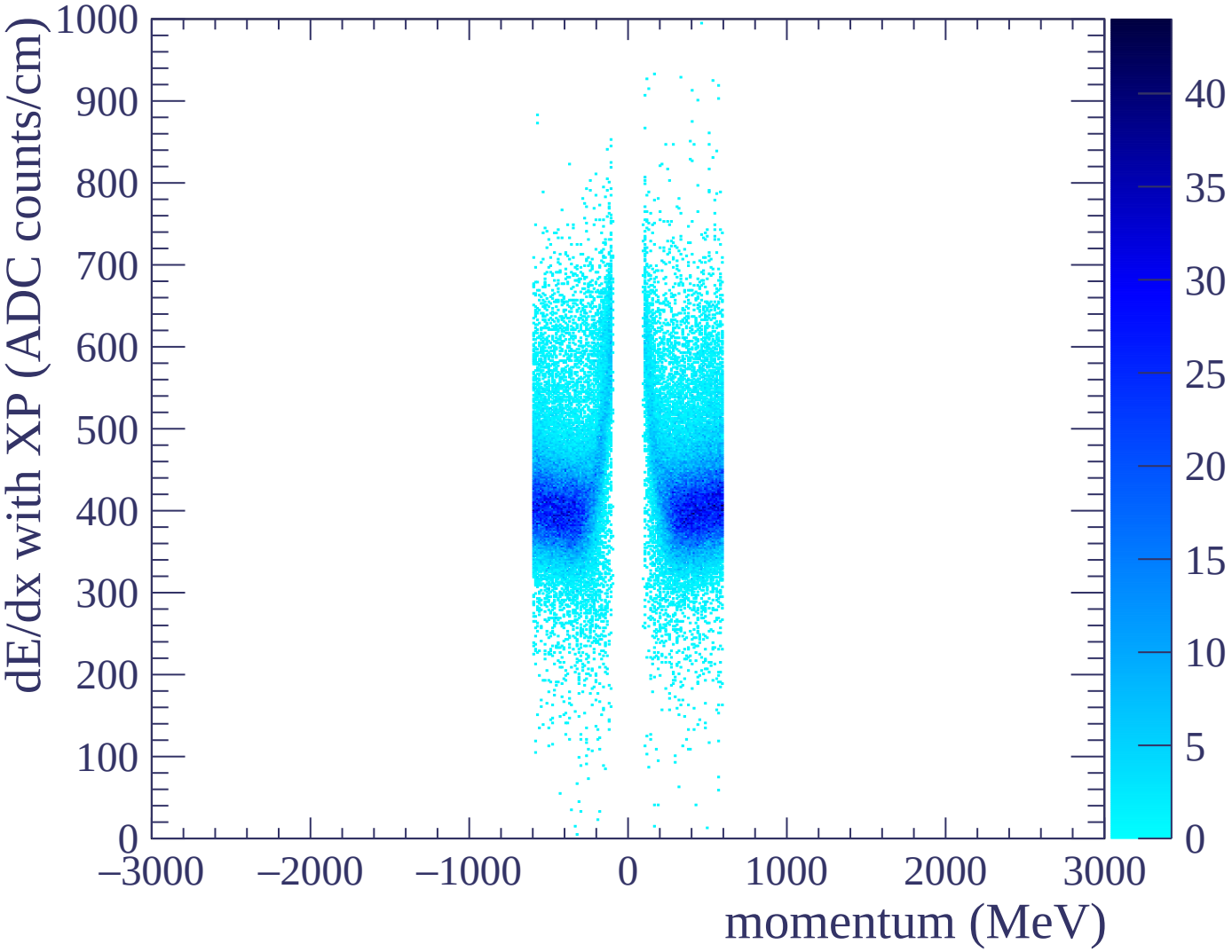

momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−80
−60
−40
−20
0
20
40
60
80
θ
250
200
150
100
50
0

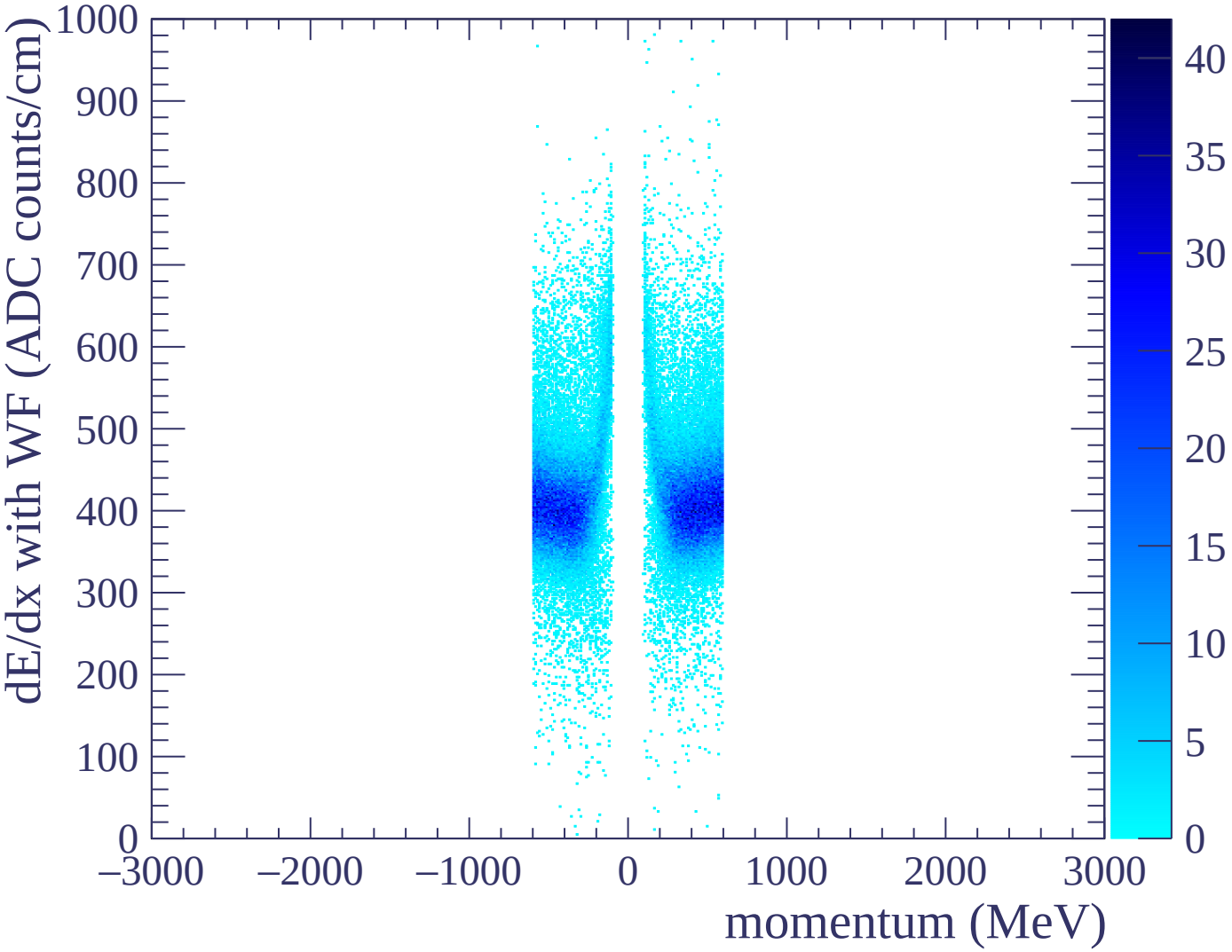

Count
9000
8000
7000
6000
5000
4000
3000
2000
1000
0
–80
–60
–40
–20
0
20
40
60
80
θ angle

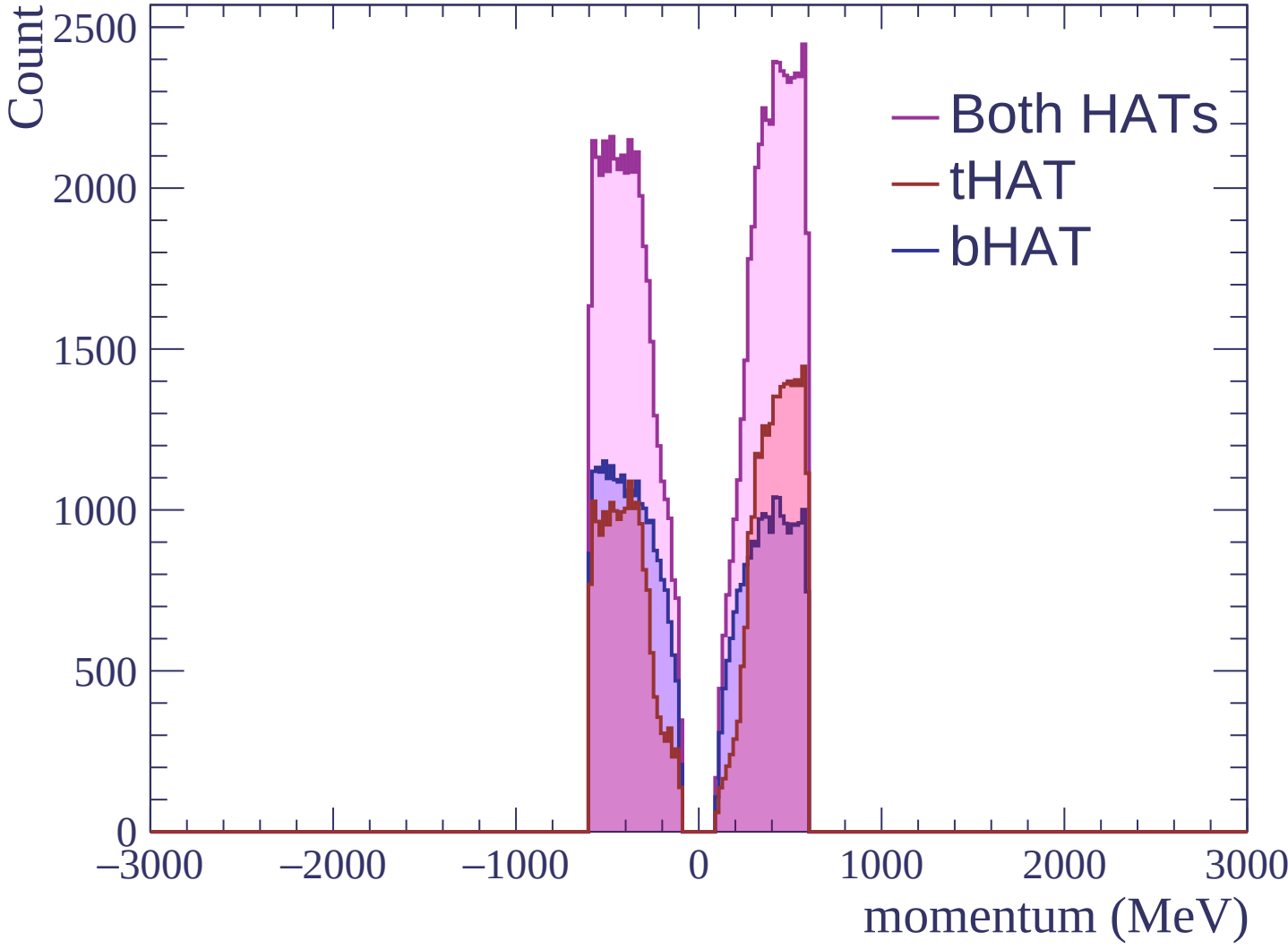

φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
5000
4000
3000
2000
1000
0

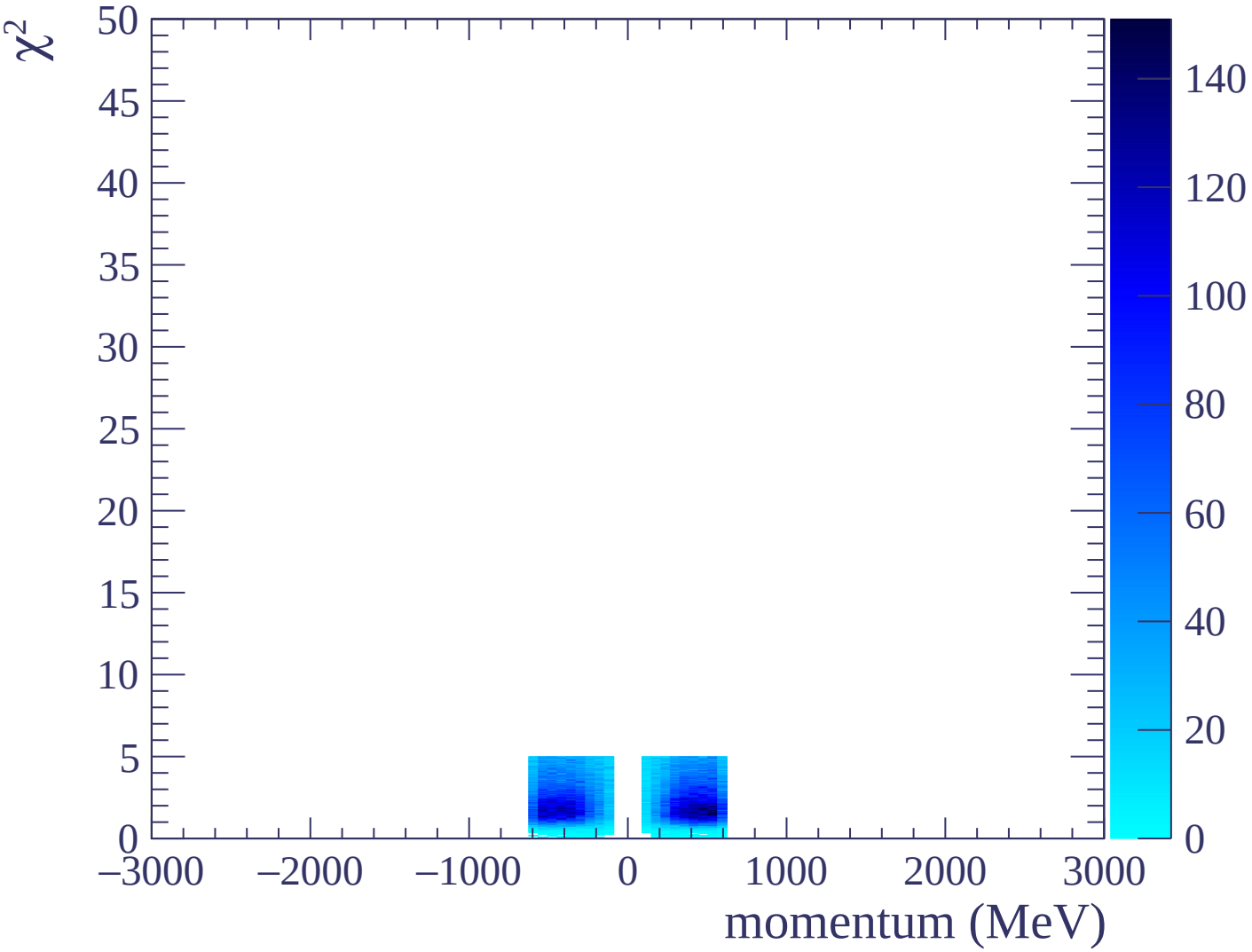

Resolution (%)
30
25
20
15
10
5
0
–3000
–2000
–1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads

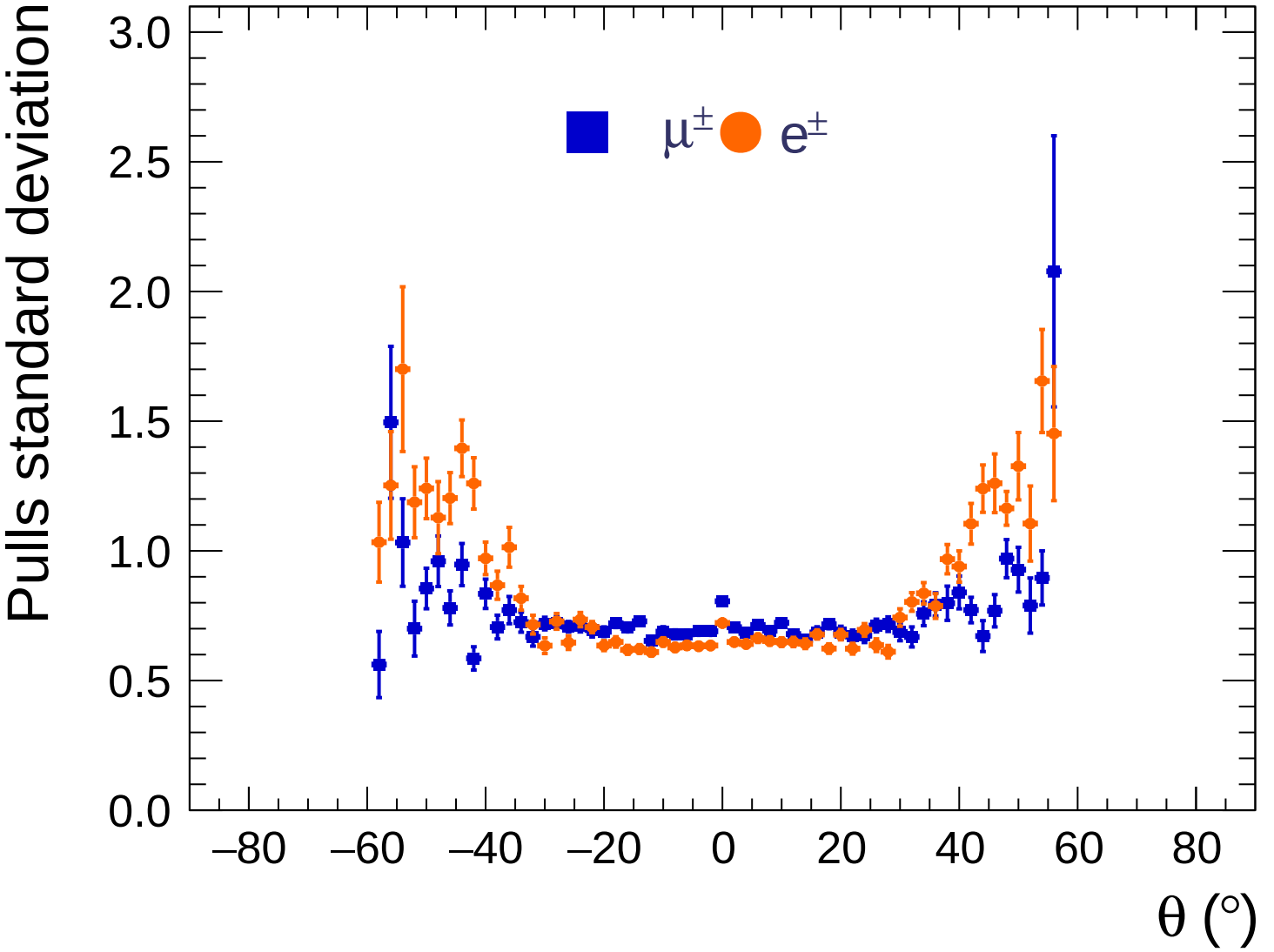

Waveforms sum
Crossed pads
Mean (ADC counts/cm)
800
700
600
500
400
300
200
–3000
–2000
–1000
0
1000
2000
3000
Momentum (MeV/c)

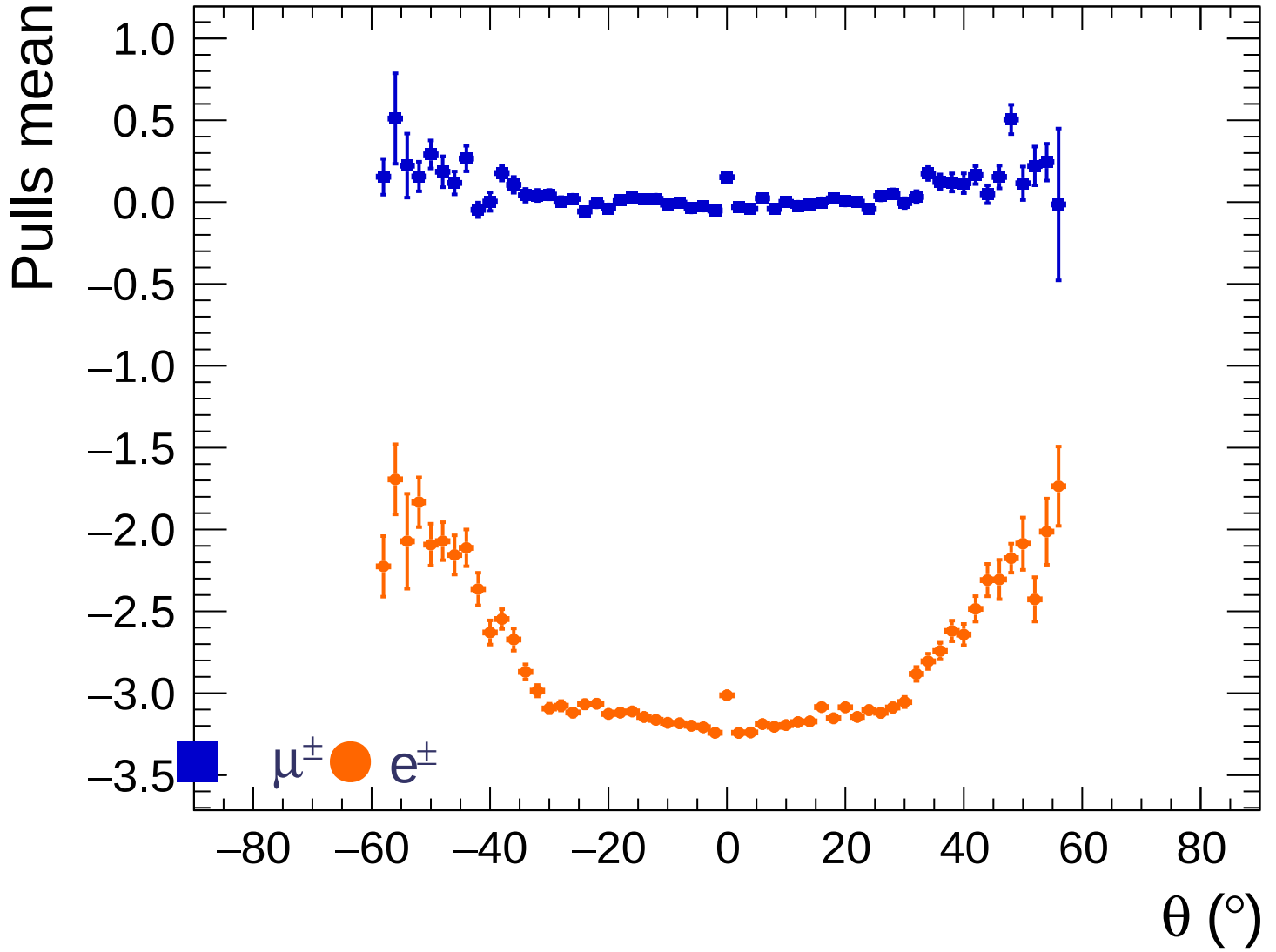

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
40
35
30
25
20
15
10
5
0

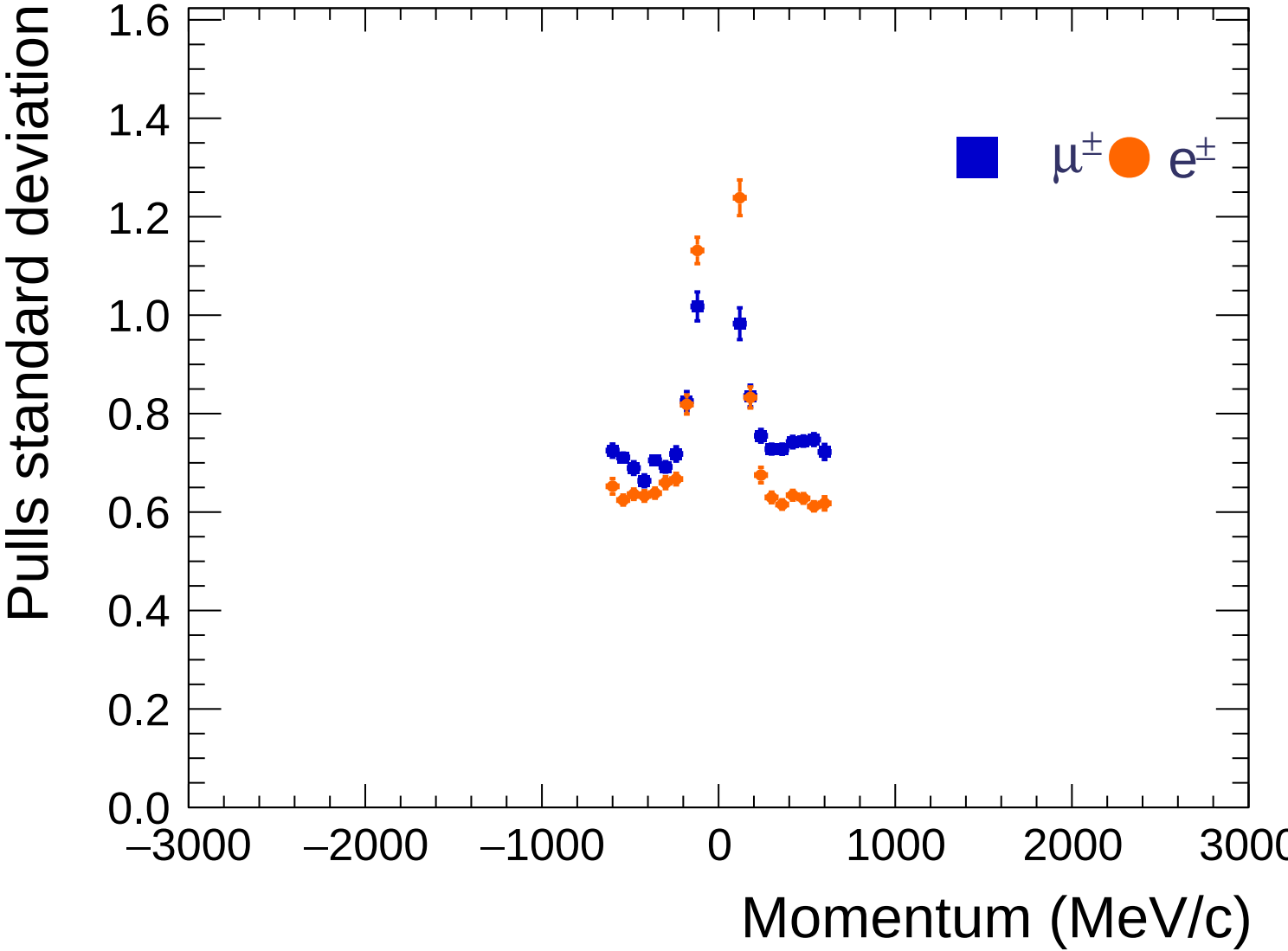

dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
40
35
30
25
20
15
10
5
0

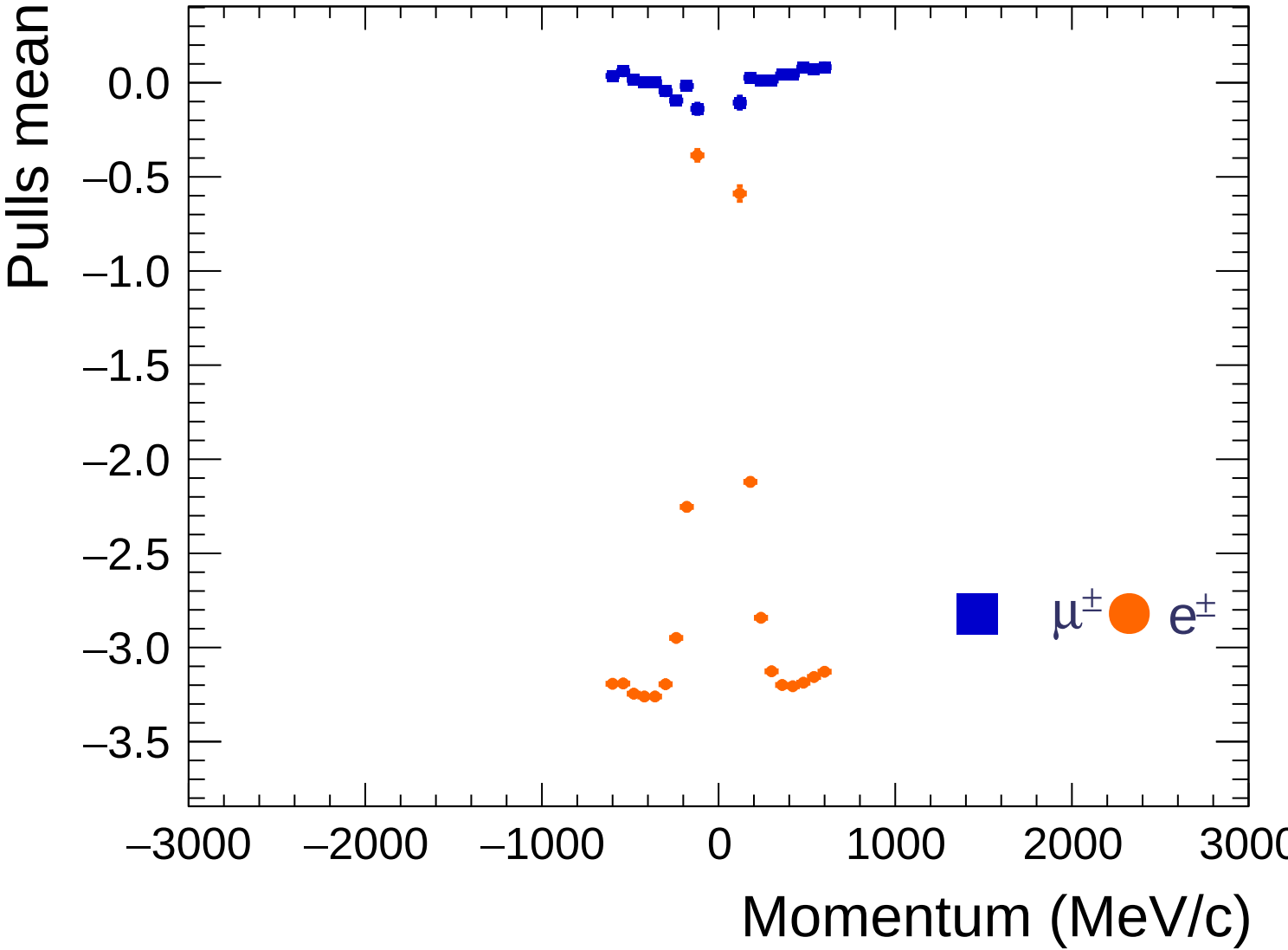

Count
2500
2000
1500
1000
500
0
–3000
–2000
–1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT

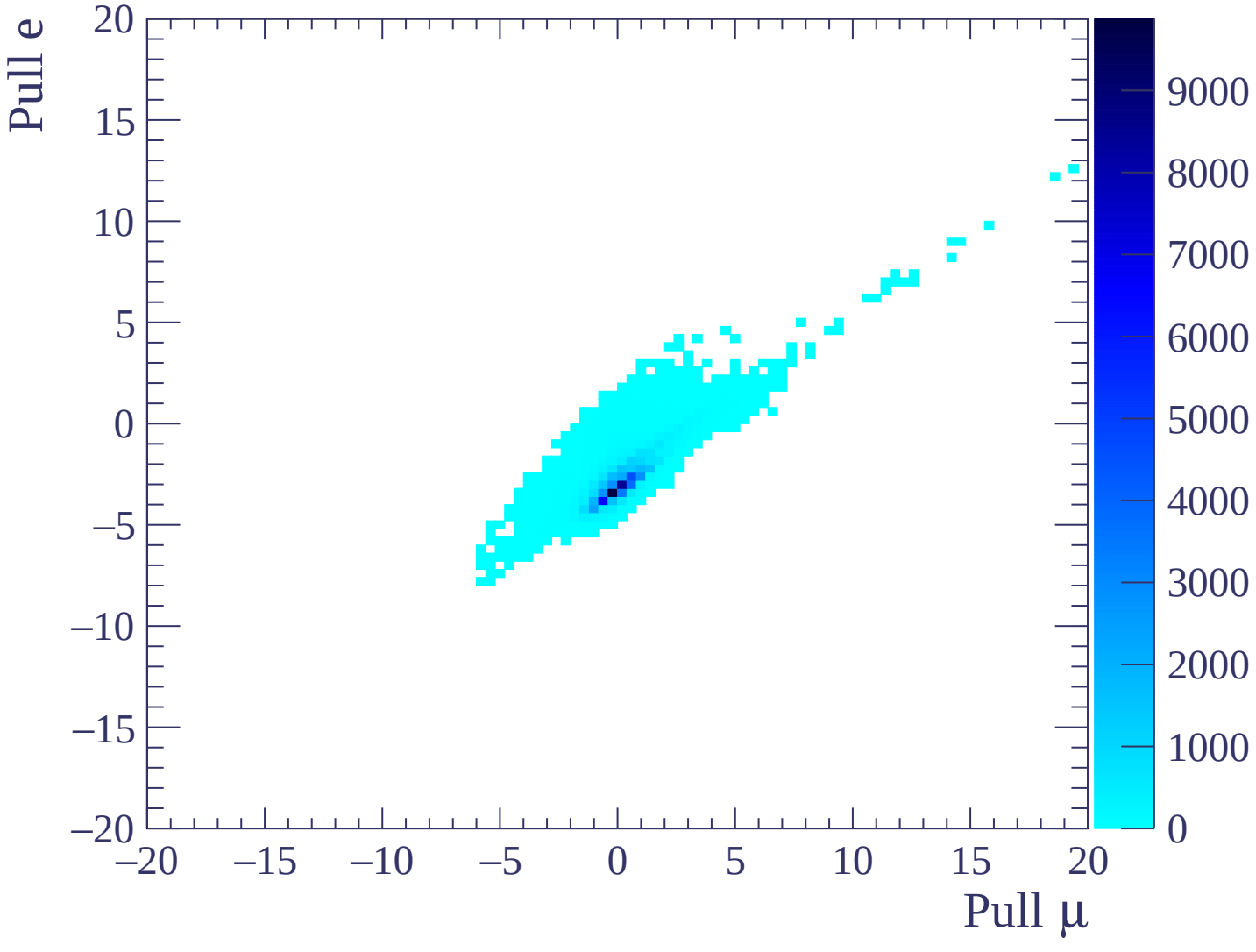

χ^2
50
45
40
35
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
140
120
100
80
60
40
20
0

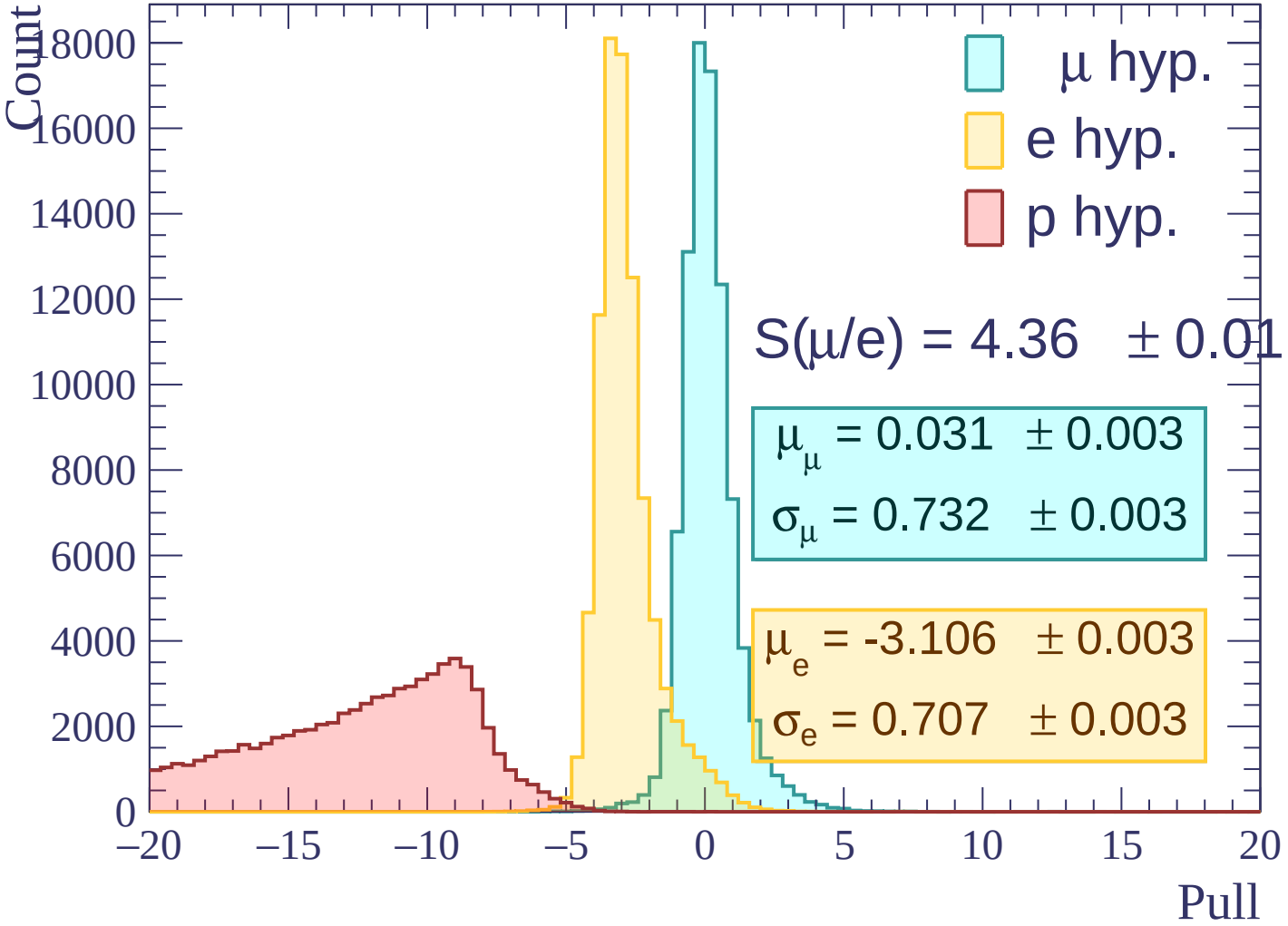

Pulls standard deviation
3.0
2.5
2.0
1.5
1.0
0.5
0.0
–80
–60
–40
–20
0
20
40
60
80
θ (°)
μ±
e±

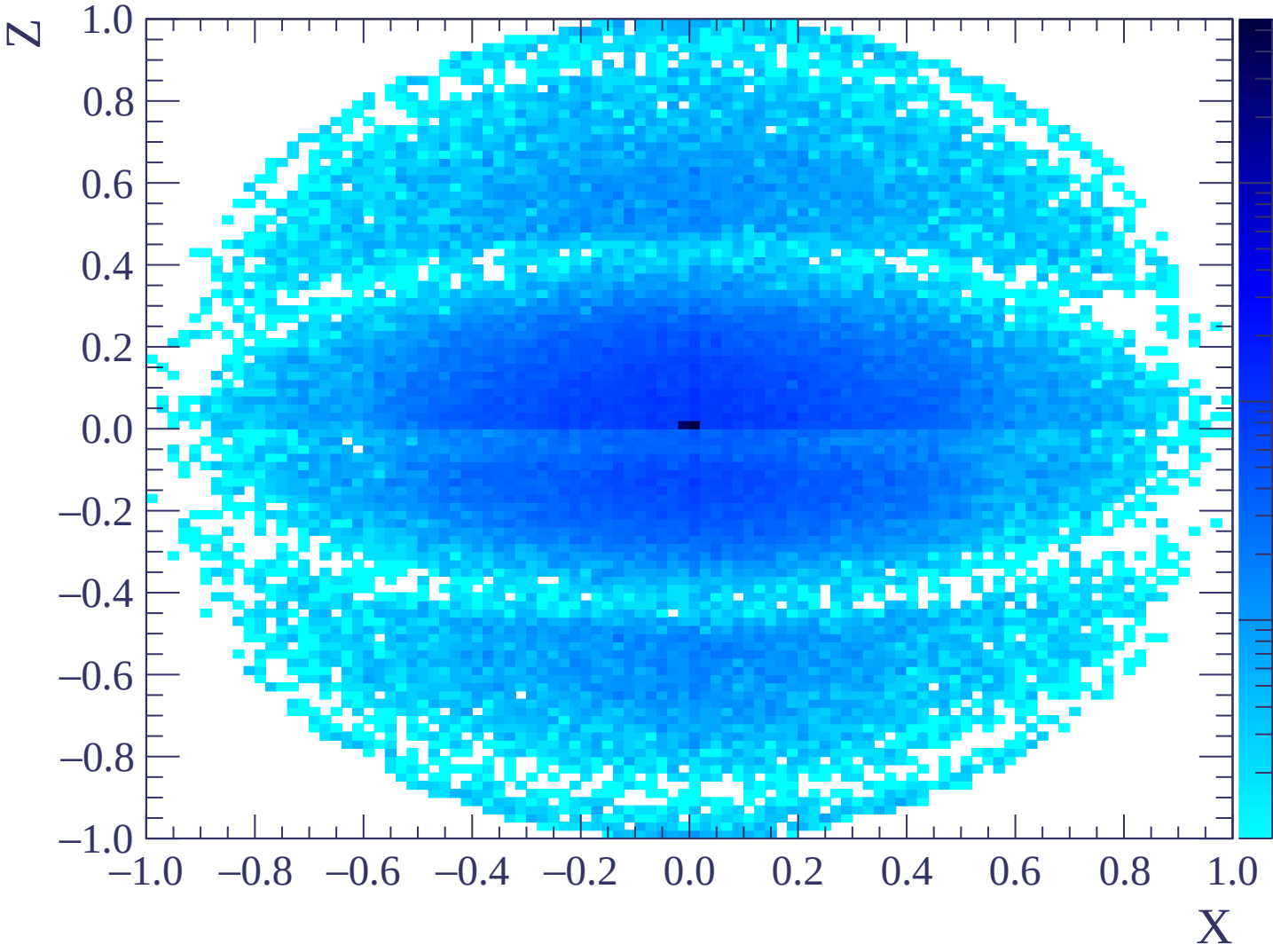

Pulls mean
1.0
0.5
0.0
–0.5
–1.0
–1.5
–2.0
–2.5
–3.0
–3.5
μ±
e±
–80
–60
–40
–20
0
20
40
60
80
θ (°)

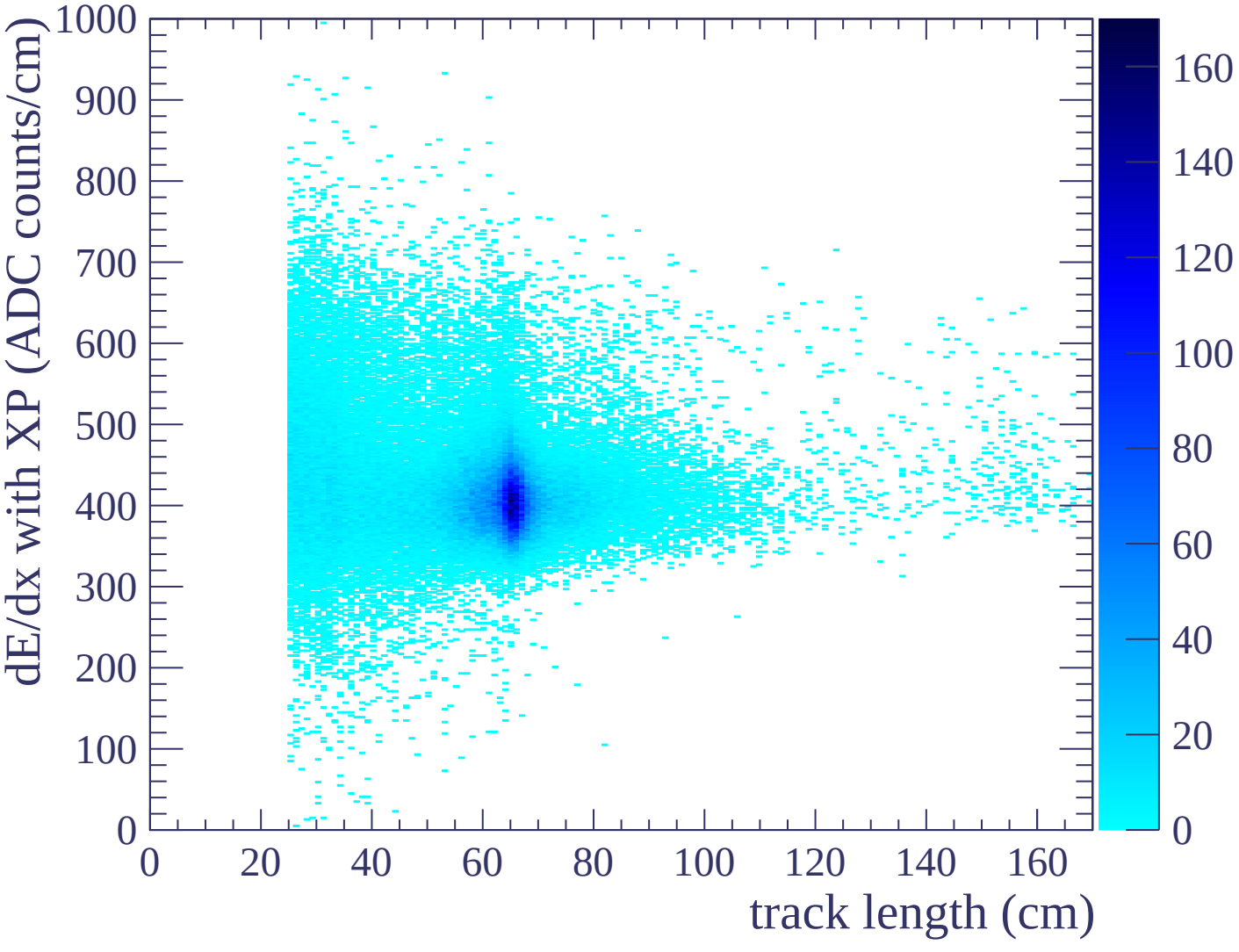

Pulls standard deviation
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
–3000
–2000
–1000
0
1000
2000
3000
Momentum (MeV/c)
μ±
e±

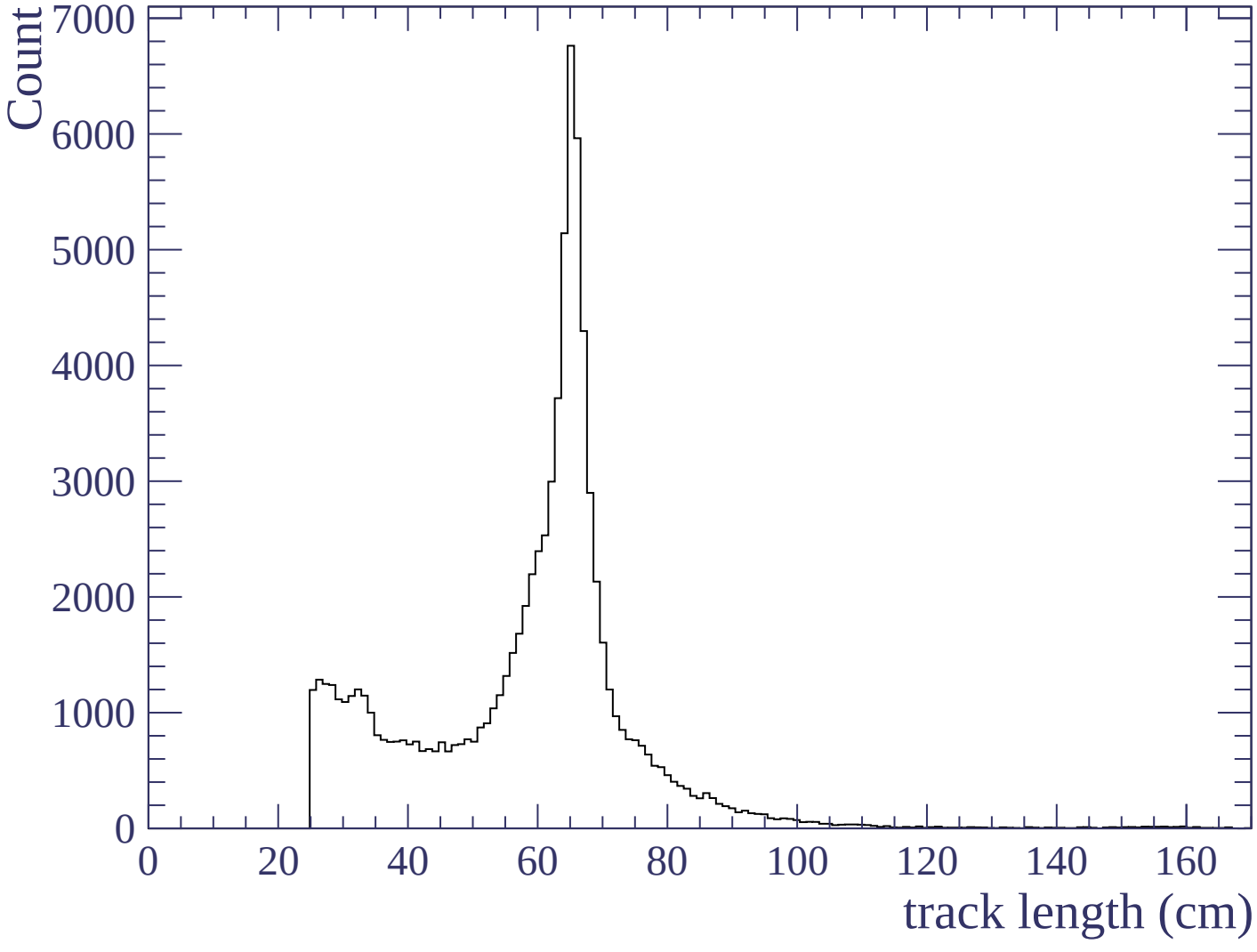

Pulls mean
0.0
–0.5
–1.0
–1.5
–2.0
–2.5
–3.0
–3.5
–3000
–2000
–1000
0
1000
2000
3000
Momentum (MeV/c)
μ±
e±

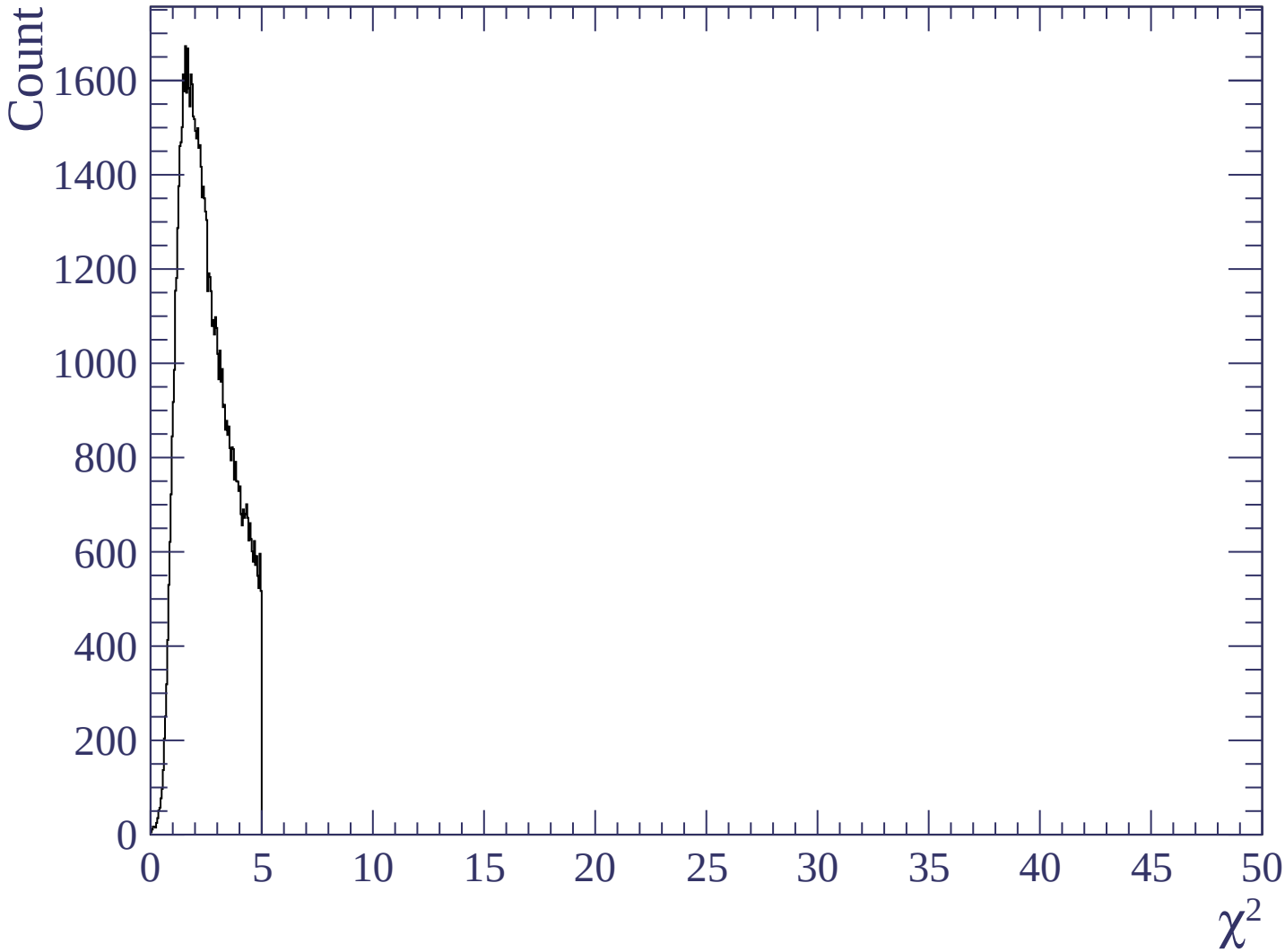

Pull e
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
Pull μ
9000
8000
7000
6000
5000
4000
3000
2000
1000
0

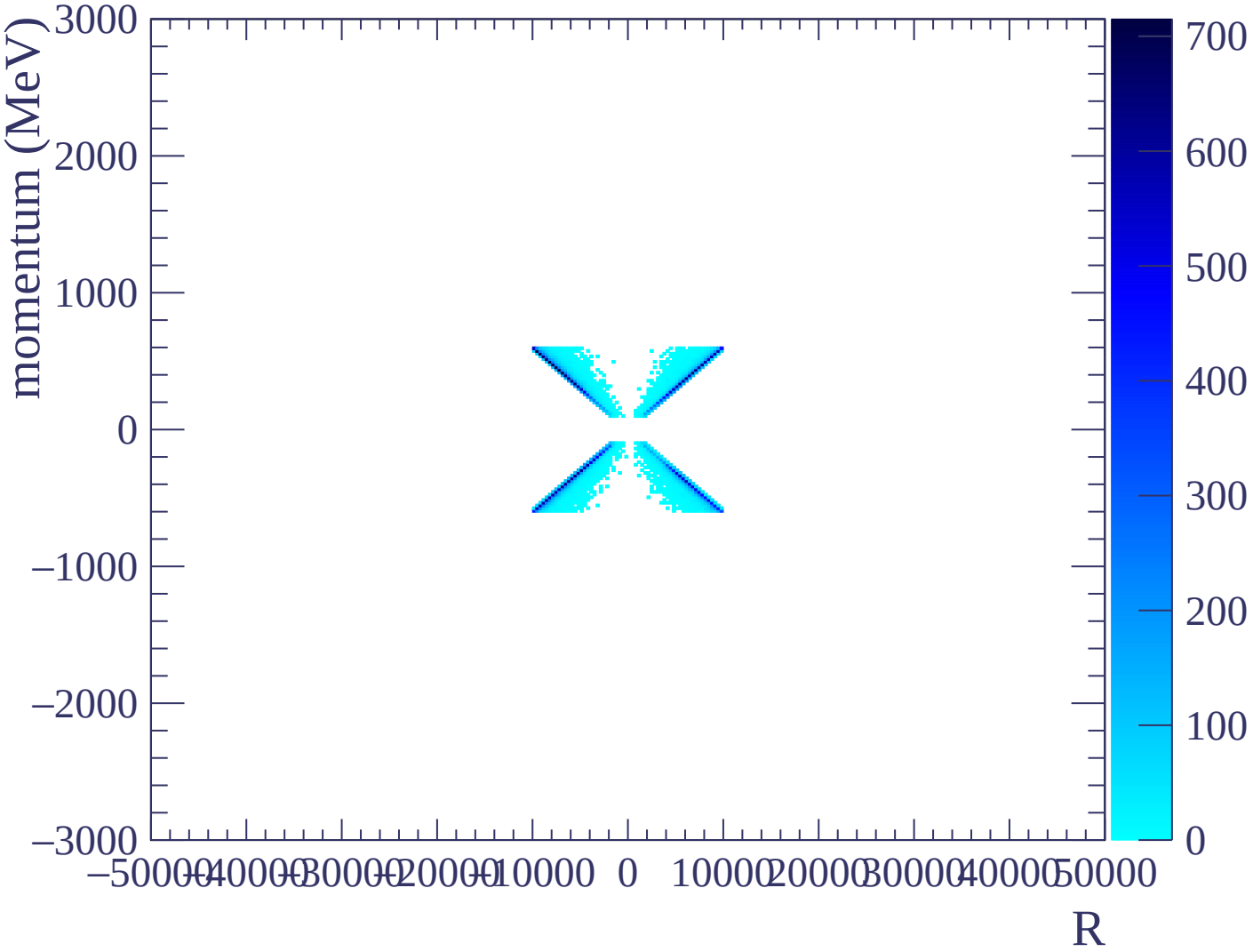

Count
18000
16000
14000
12000
10000
8000
6000
4000
2000
0
–20
–15
–10
–5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 4.36 ± 0.01
μμ = 0.031 ± 0.003
σμ = 0.732 ± 0.003
μe = -3.106 ± 0.003
σe = 0.707 ± 0.003

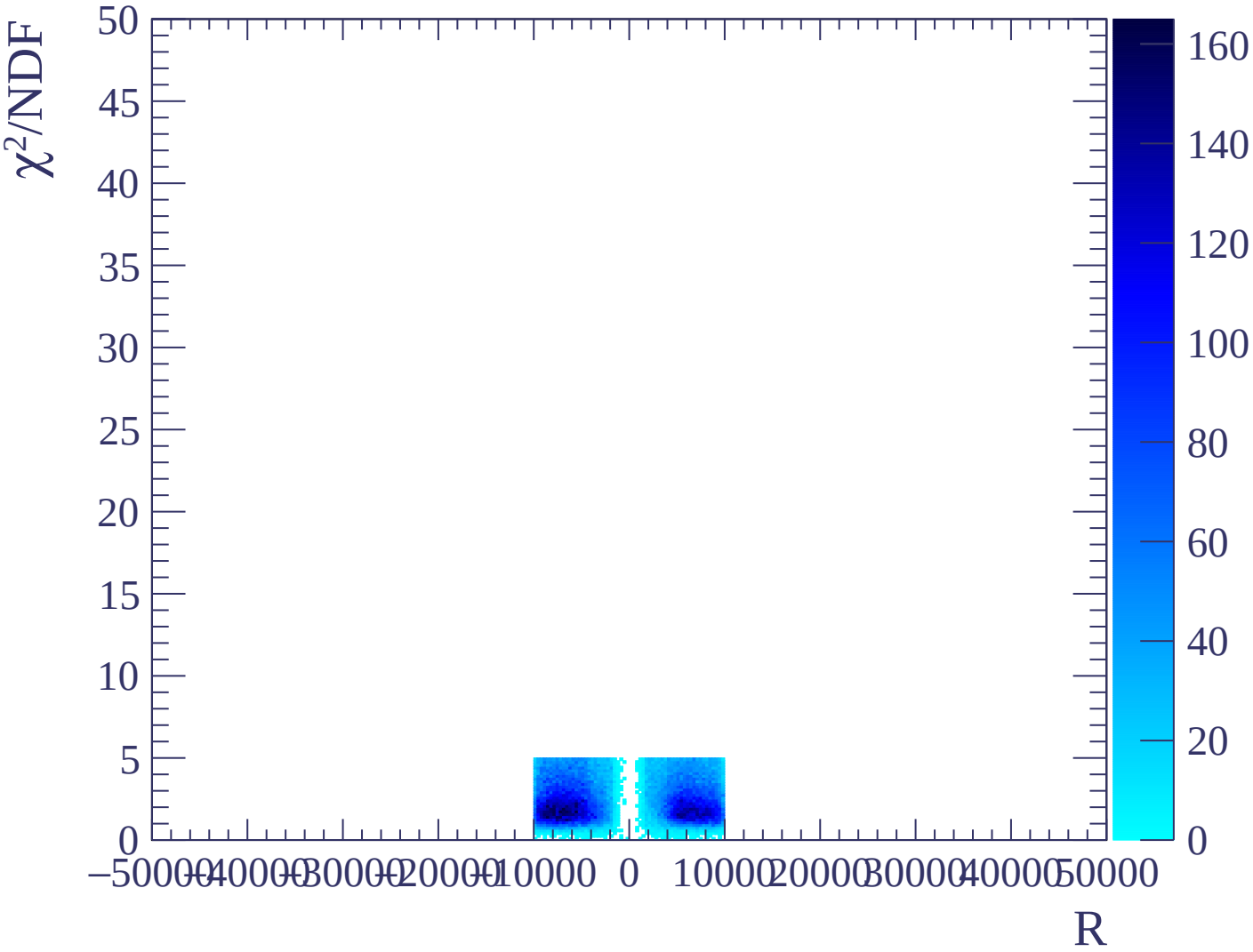

Z
1.0
0.8
0.6
0.4
0.2
0.0
–0.2
–0.4
–0.6
–0.8
–1.0
–1.0
–0.8
–0.6
–0.4
–0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

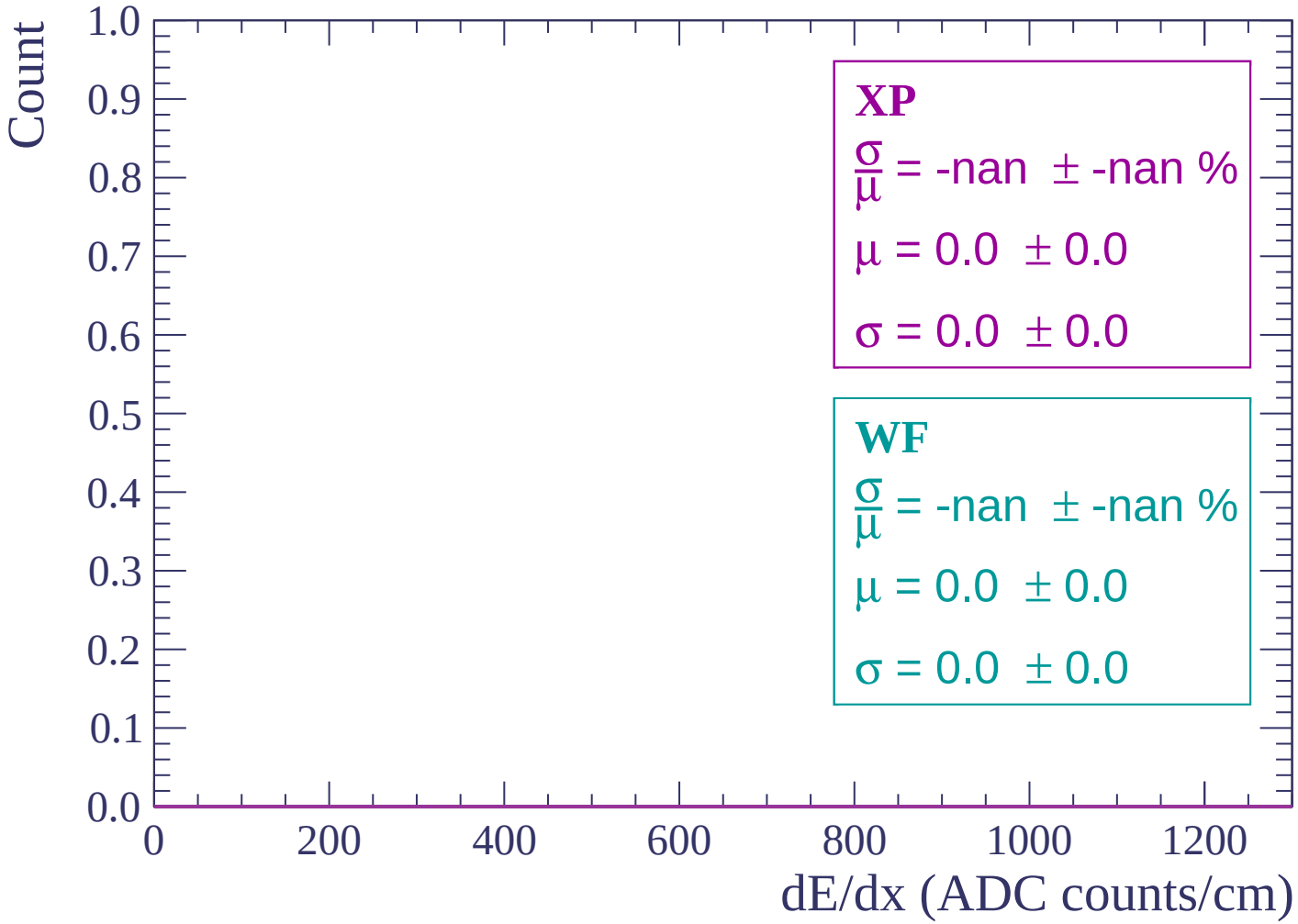

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
20
40
60
80
100
120
140
160
track length (cm)
160
140
120
100
80
60
40
20
0

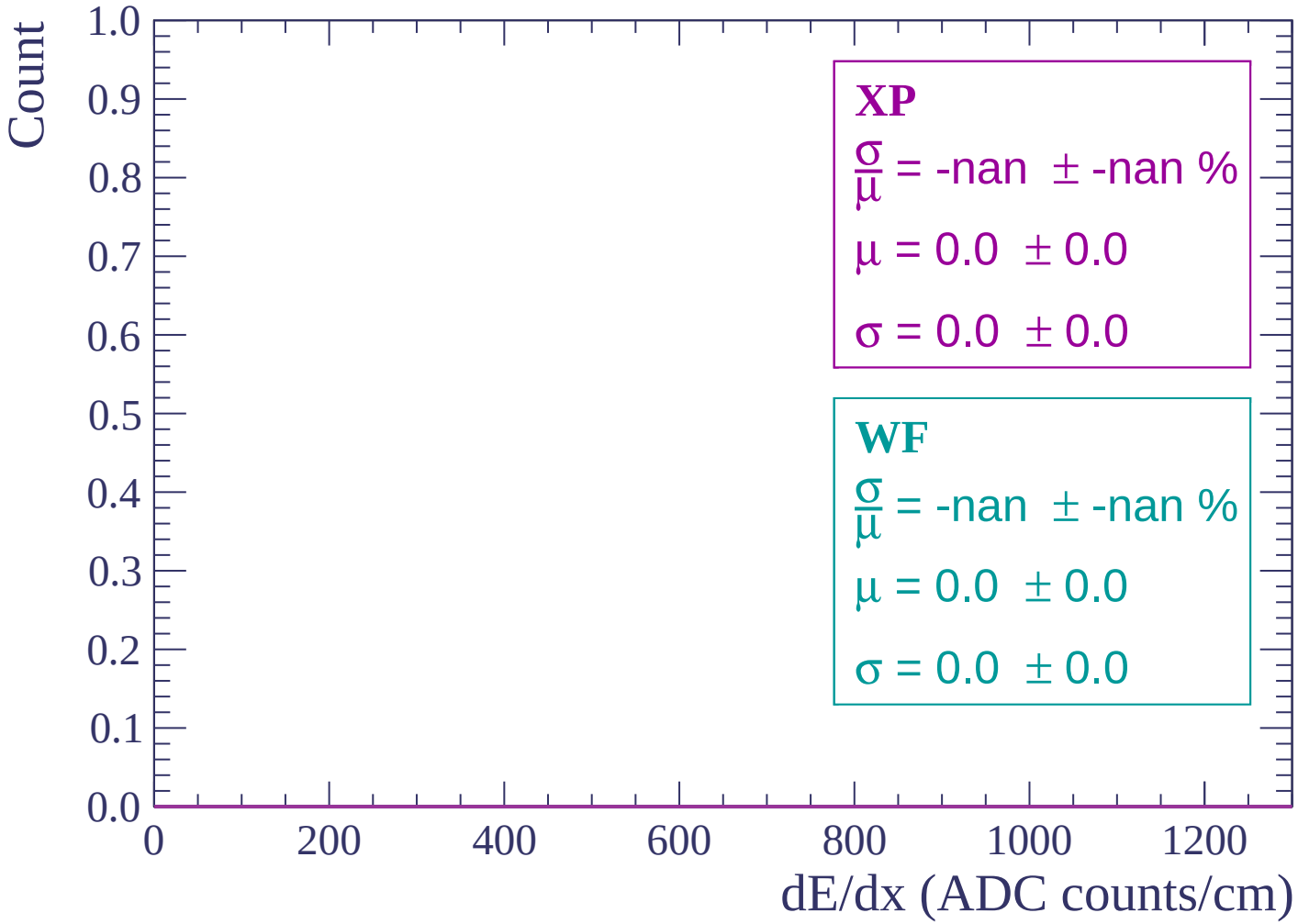

Count
7000
6000
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
140
160
track length (cm)

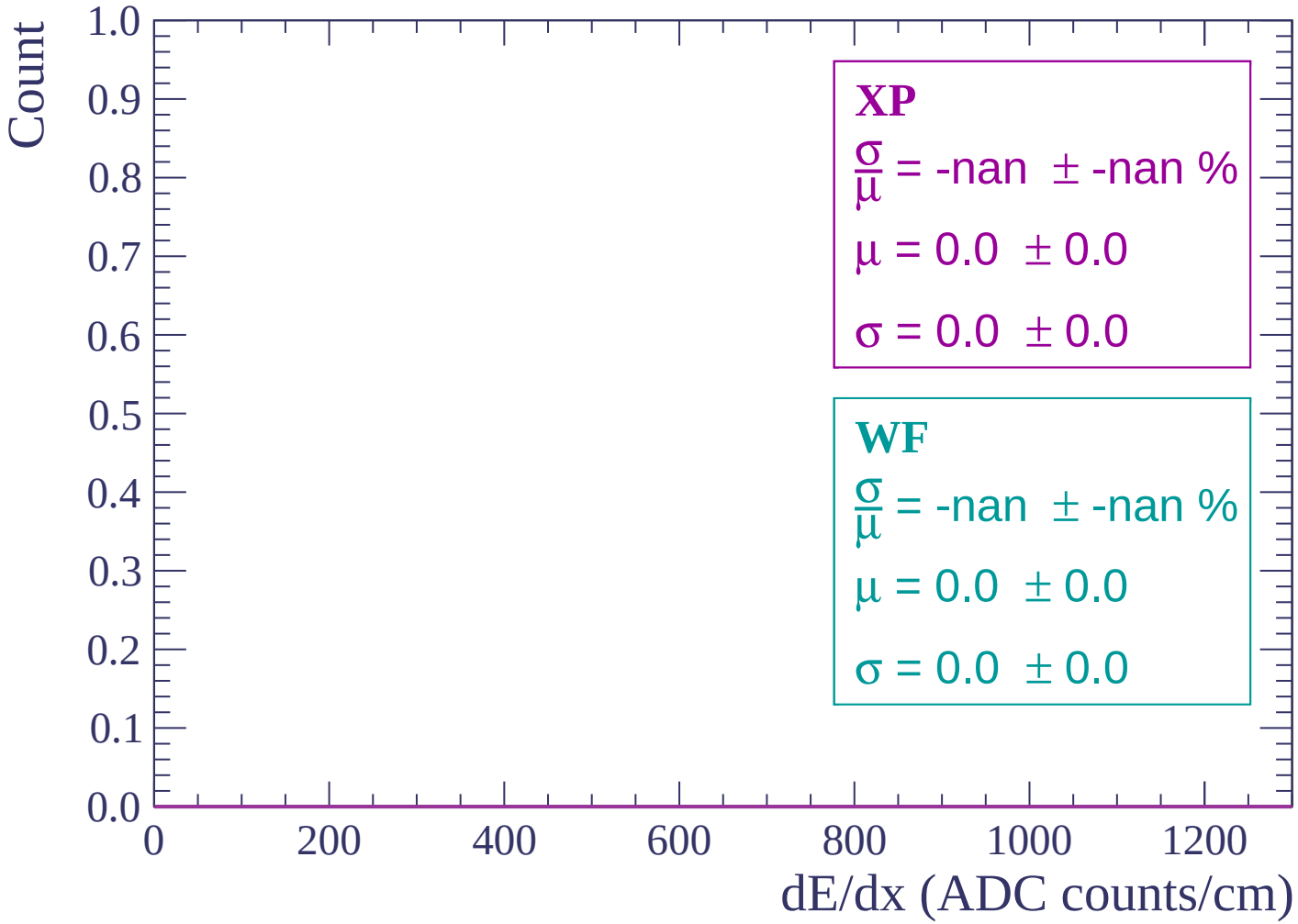

Count
1600
1400
1200
1000
800
600
400
200
0
0
5
10
15
20
25
30
35
40
45
50
χ^2

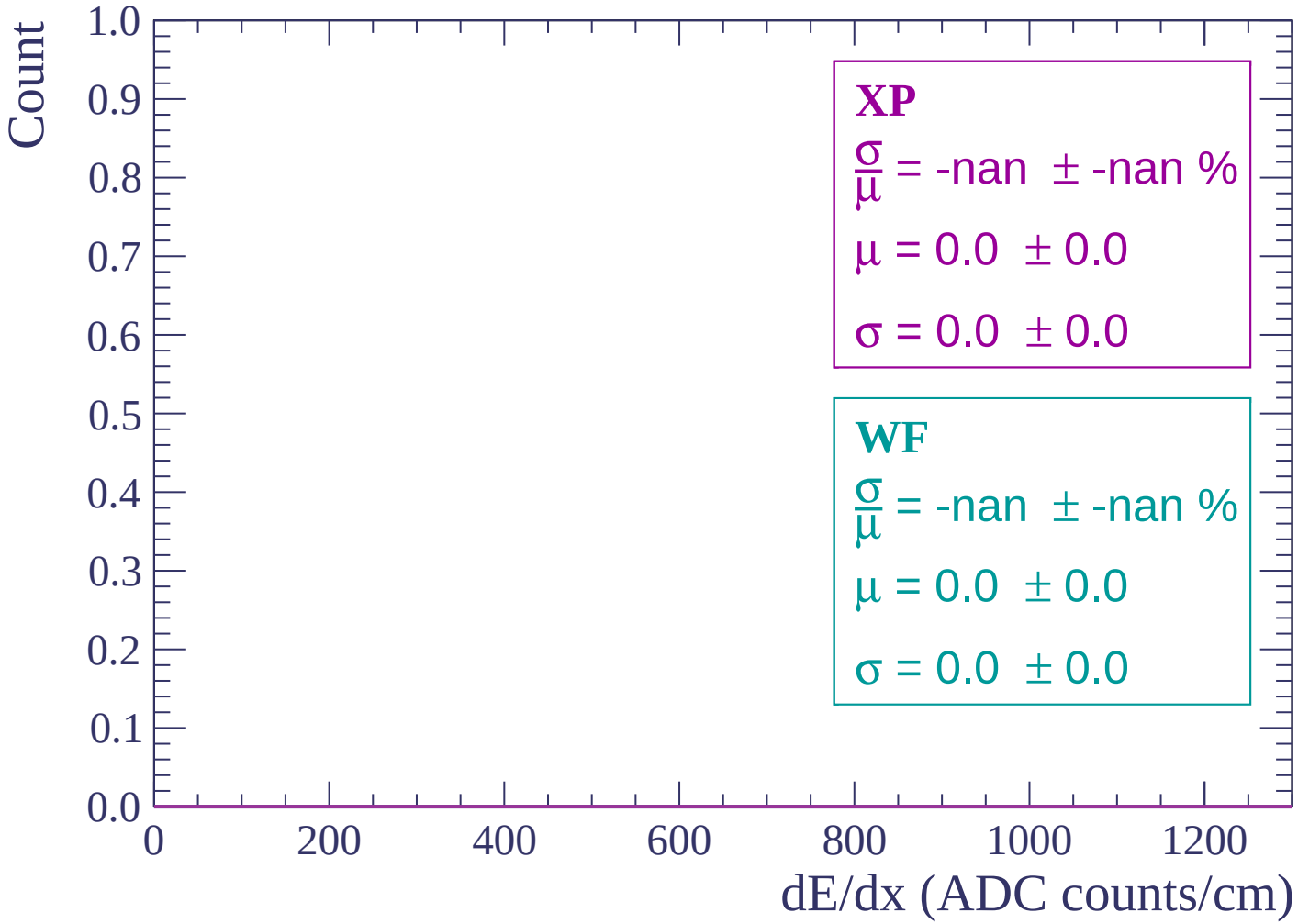

momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−50000
−40000
−30000
−20000
−10000
0
10000
20000
30000
40000
50000
R
700
600
500
400
300
200
100
0

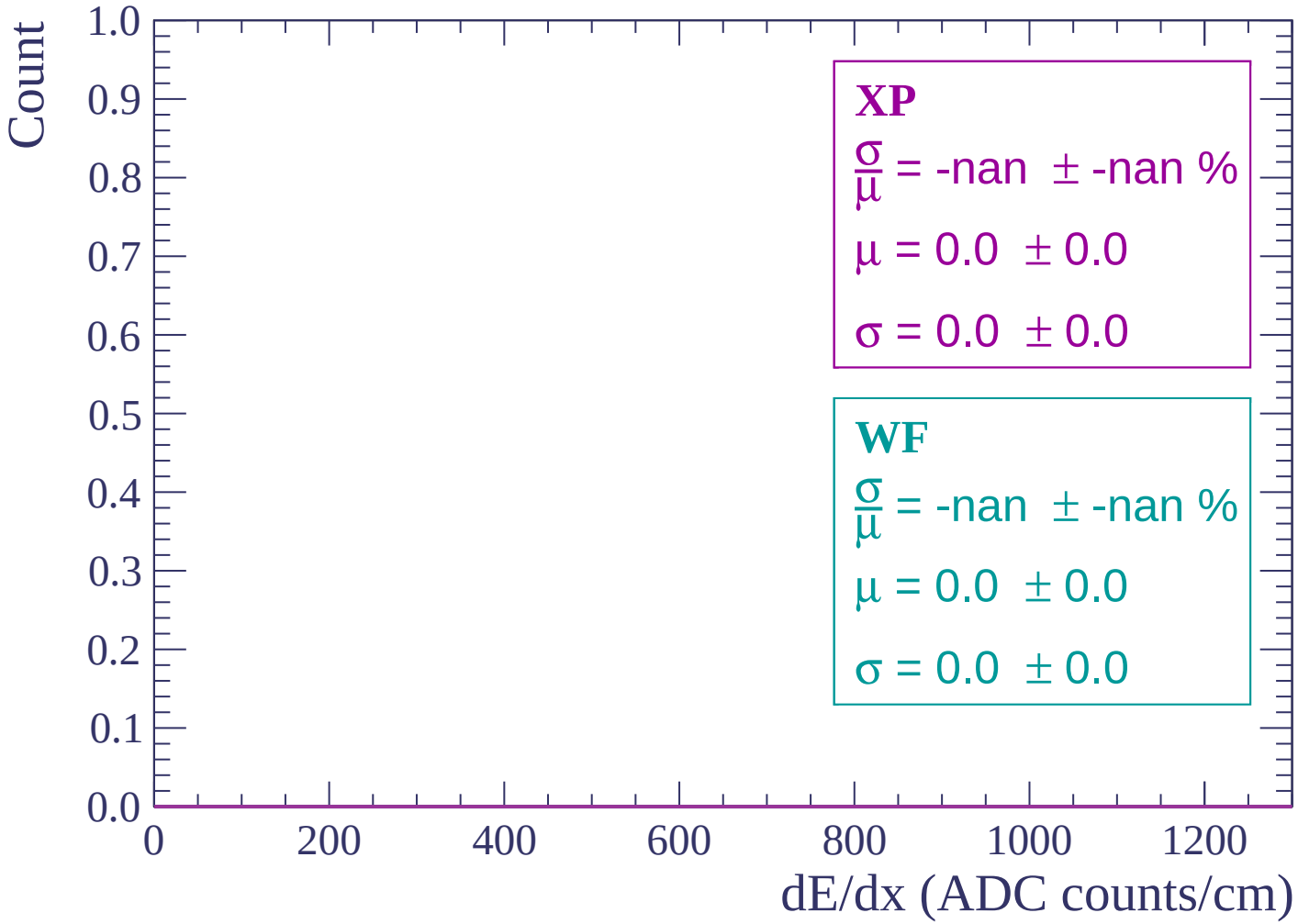

χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50000
−40000
−30000
−20000
−10000
0
10000
20000
30000
40000
50000
R
160
140
120
100
80
60
40
20
0

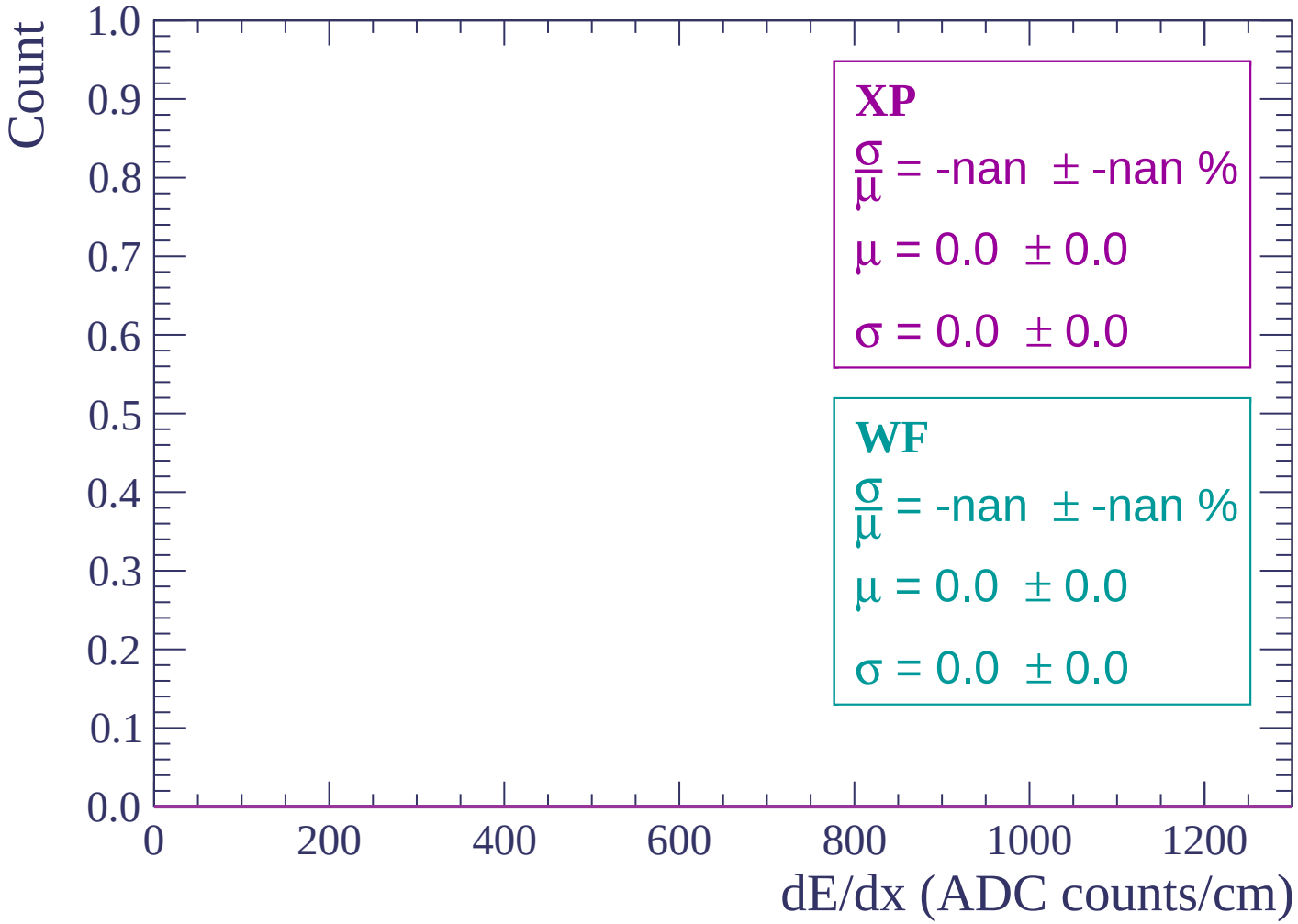

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

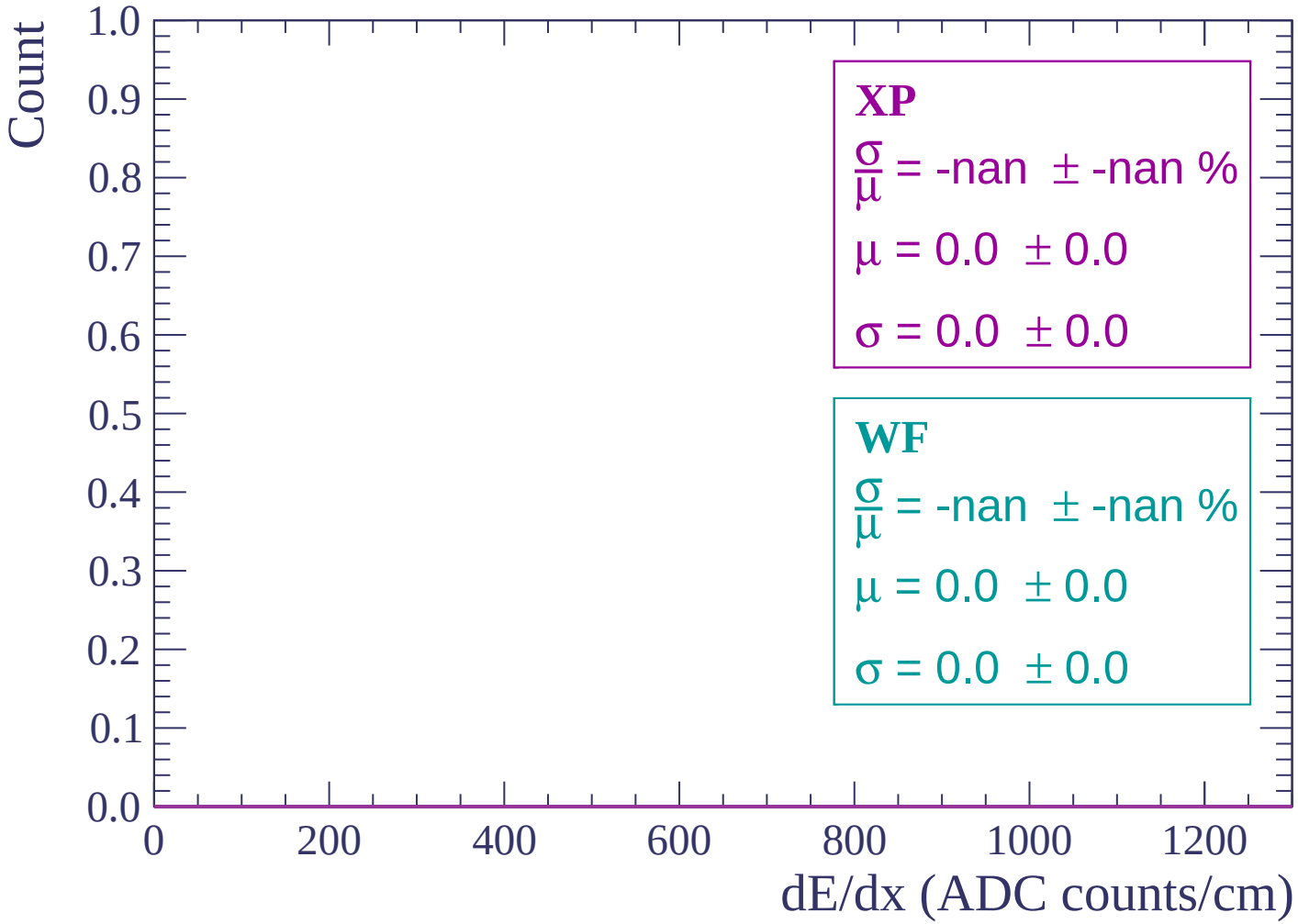

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

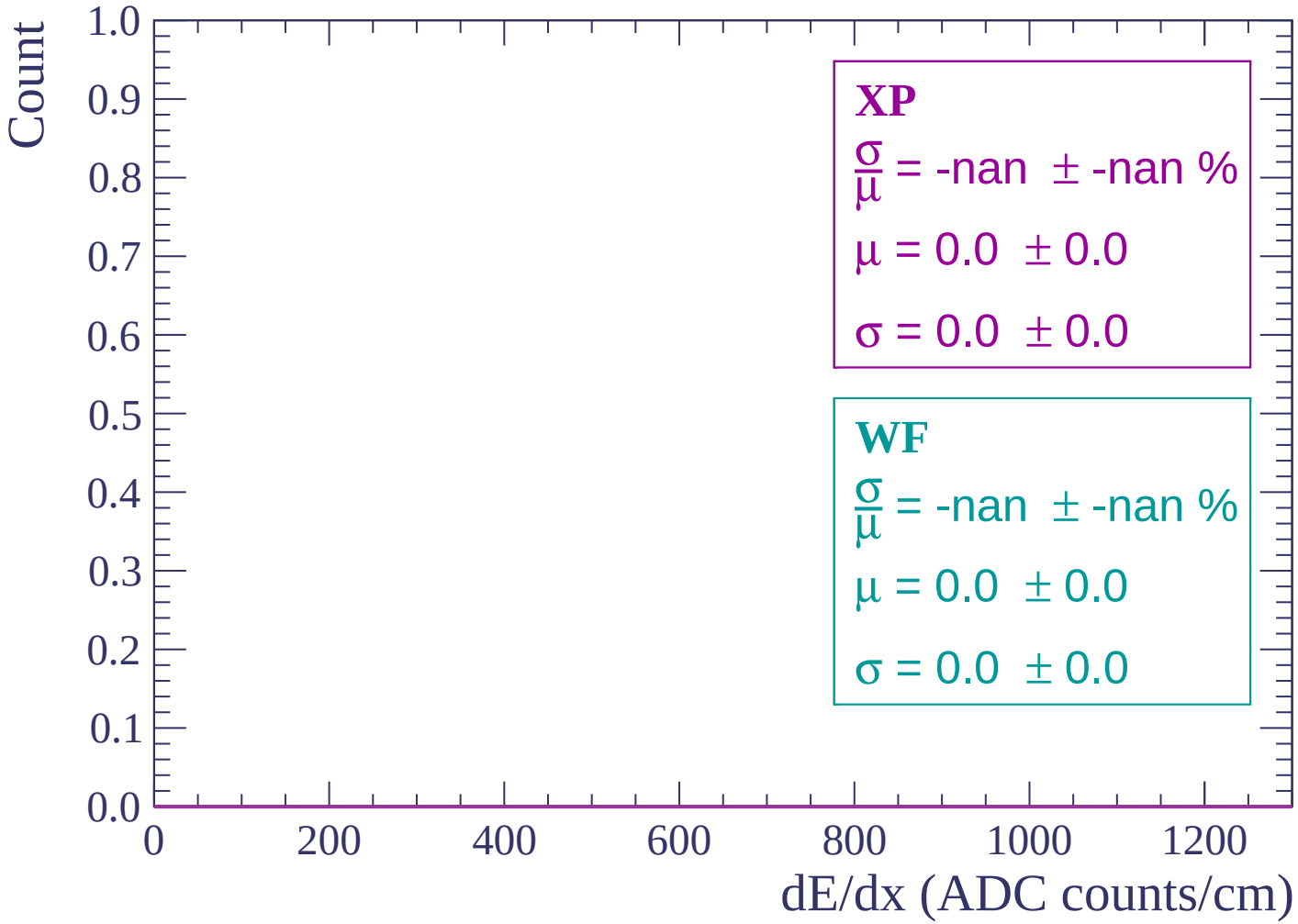

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

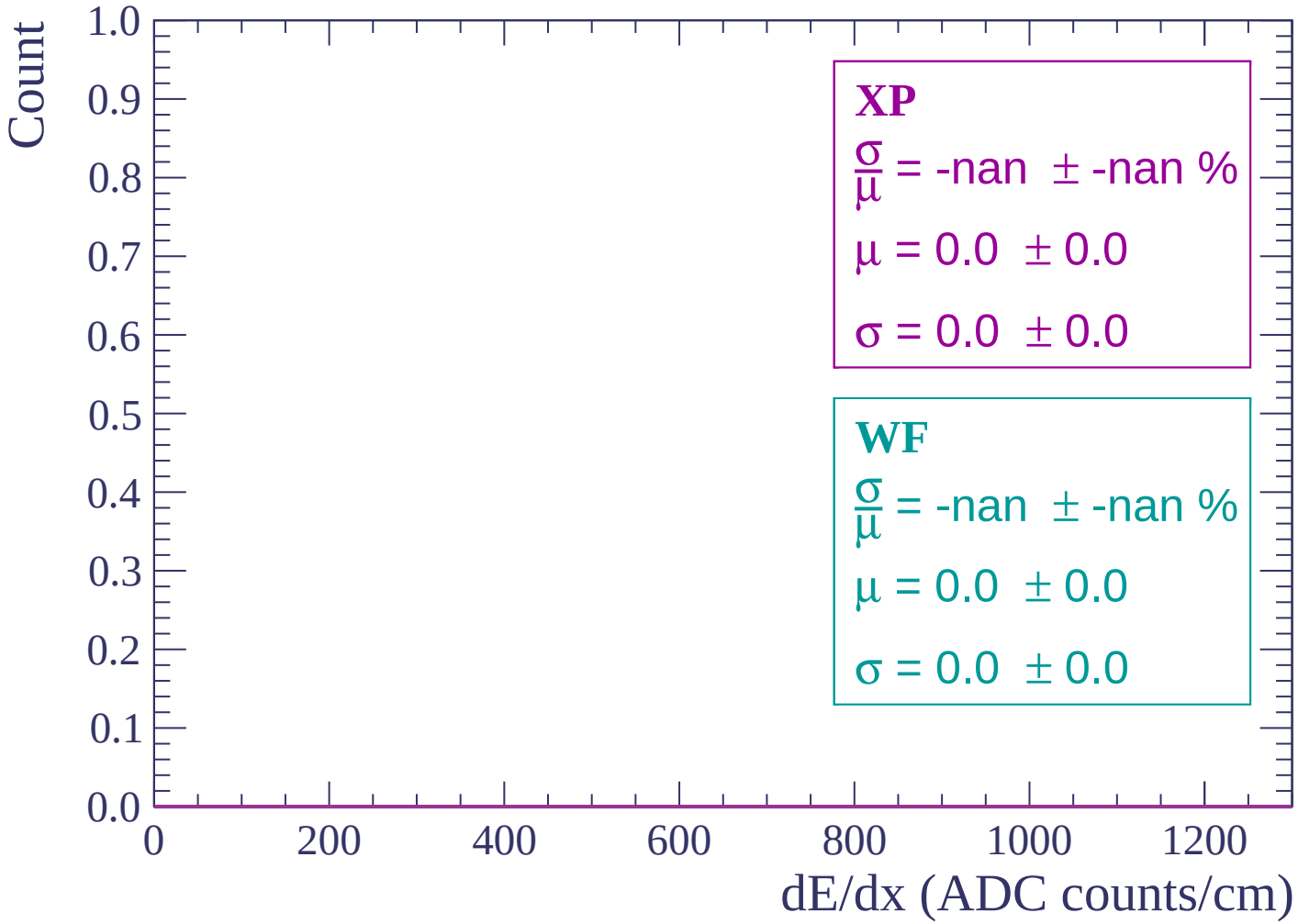

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

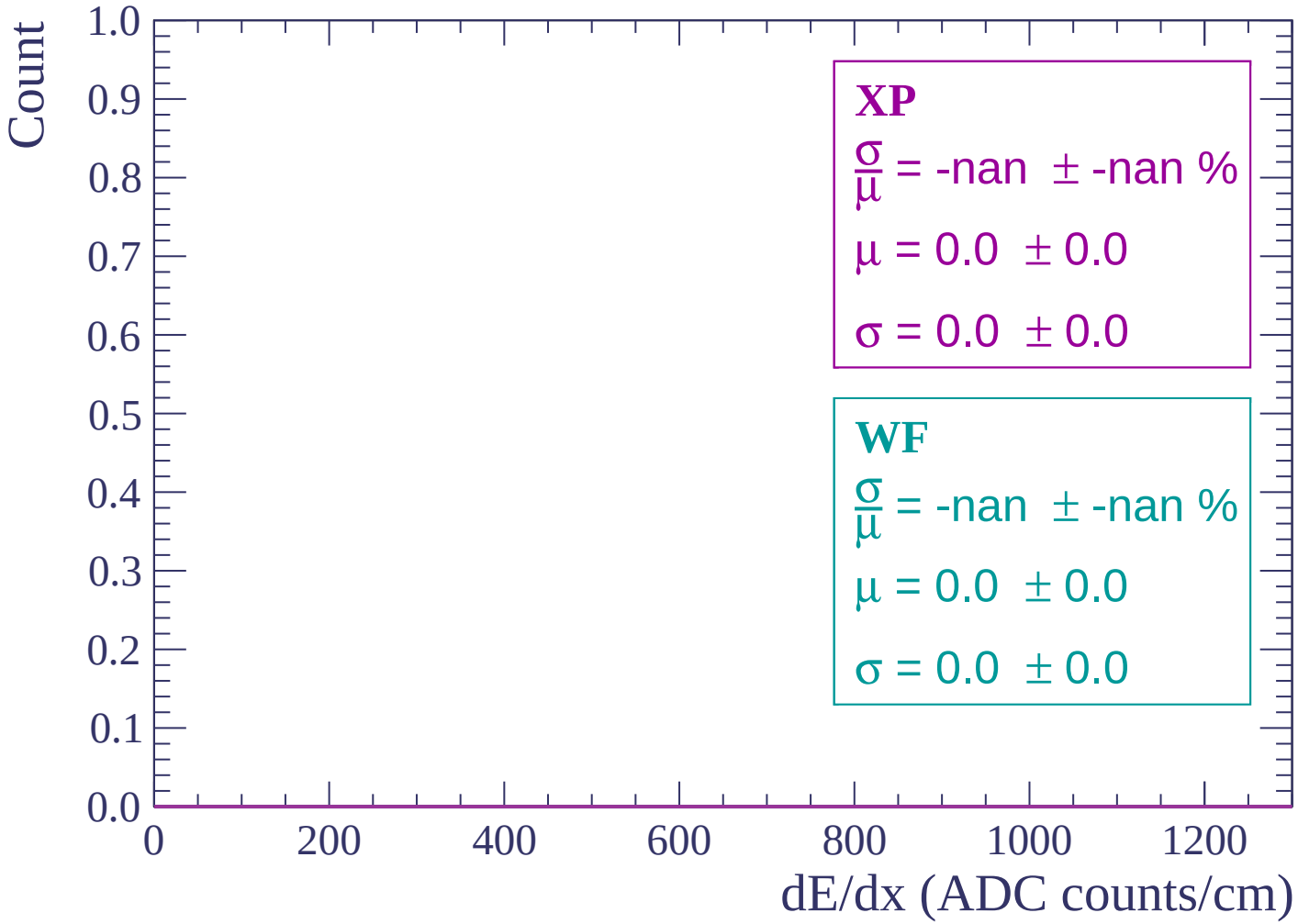

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

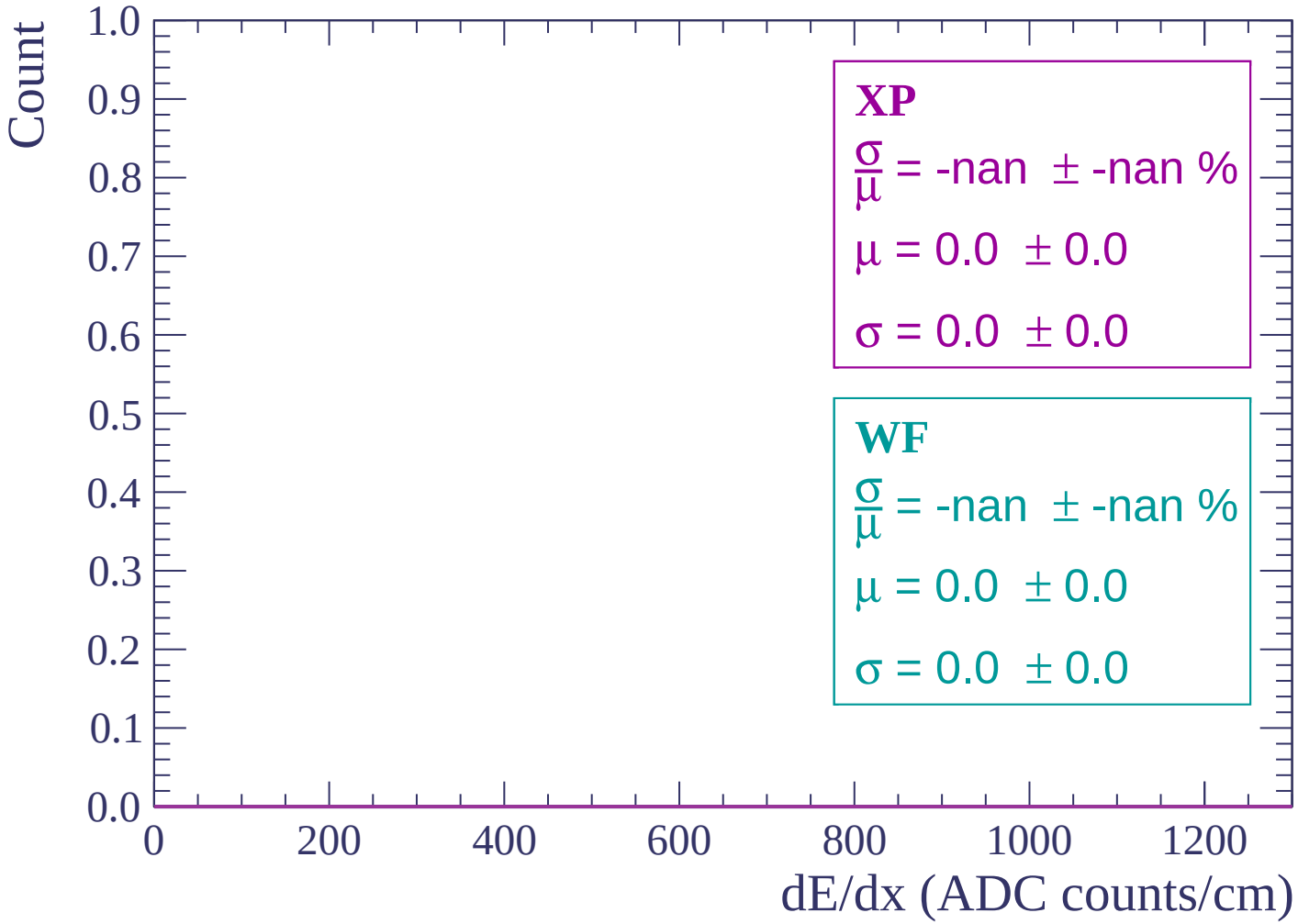

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

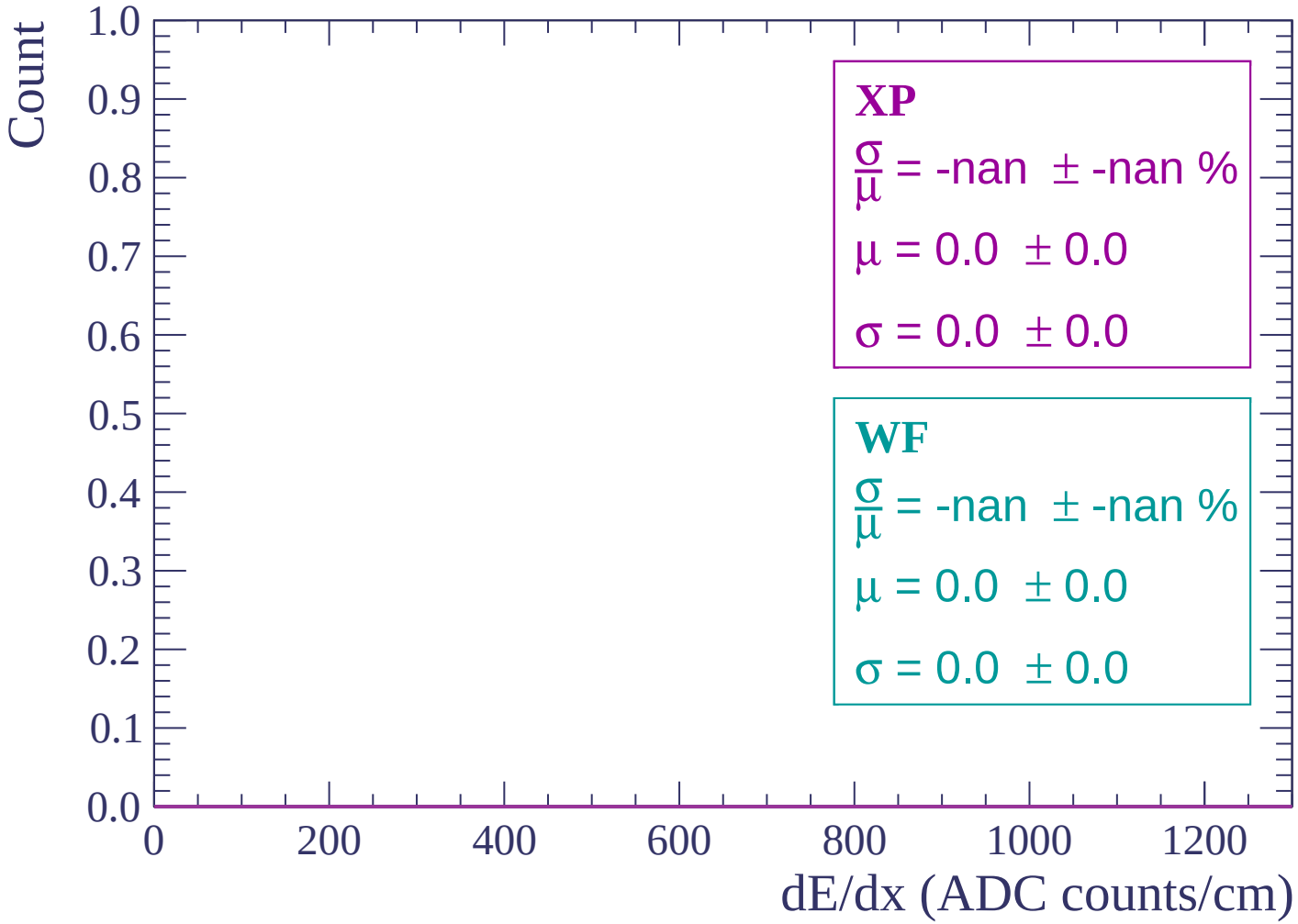

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

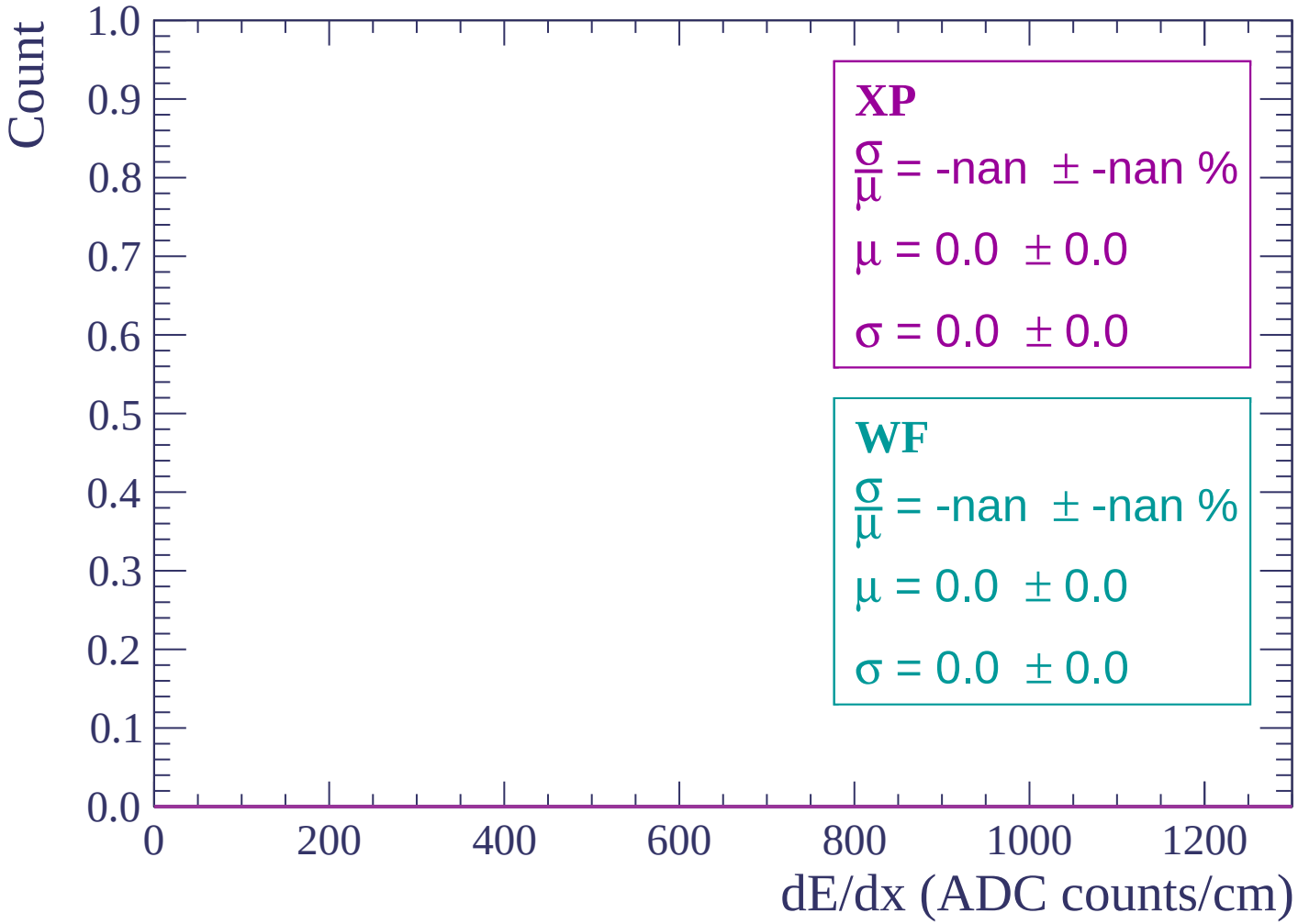

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

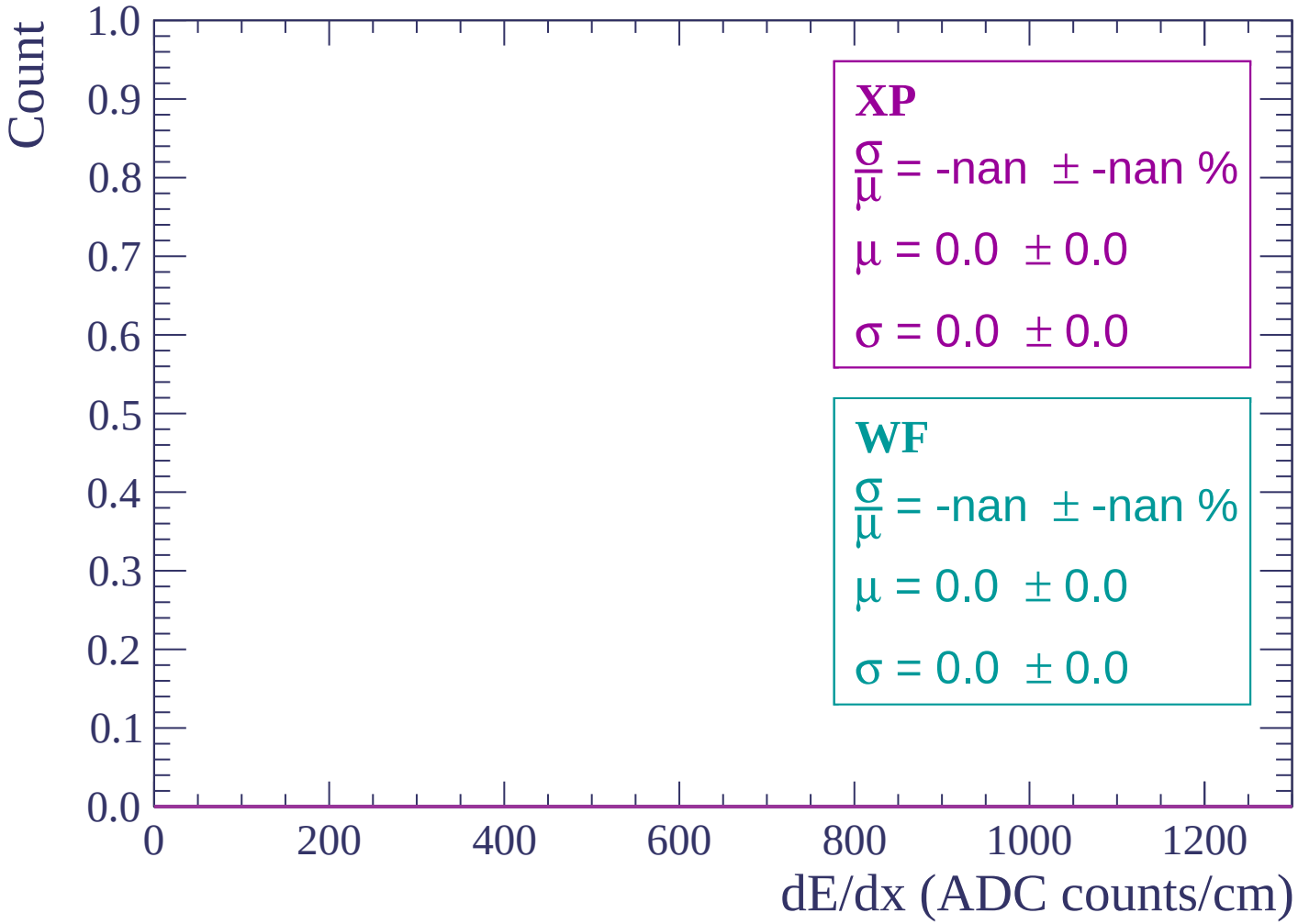

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

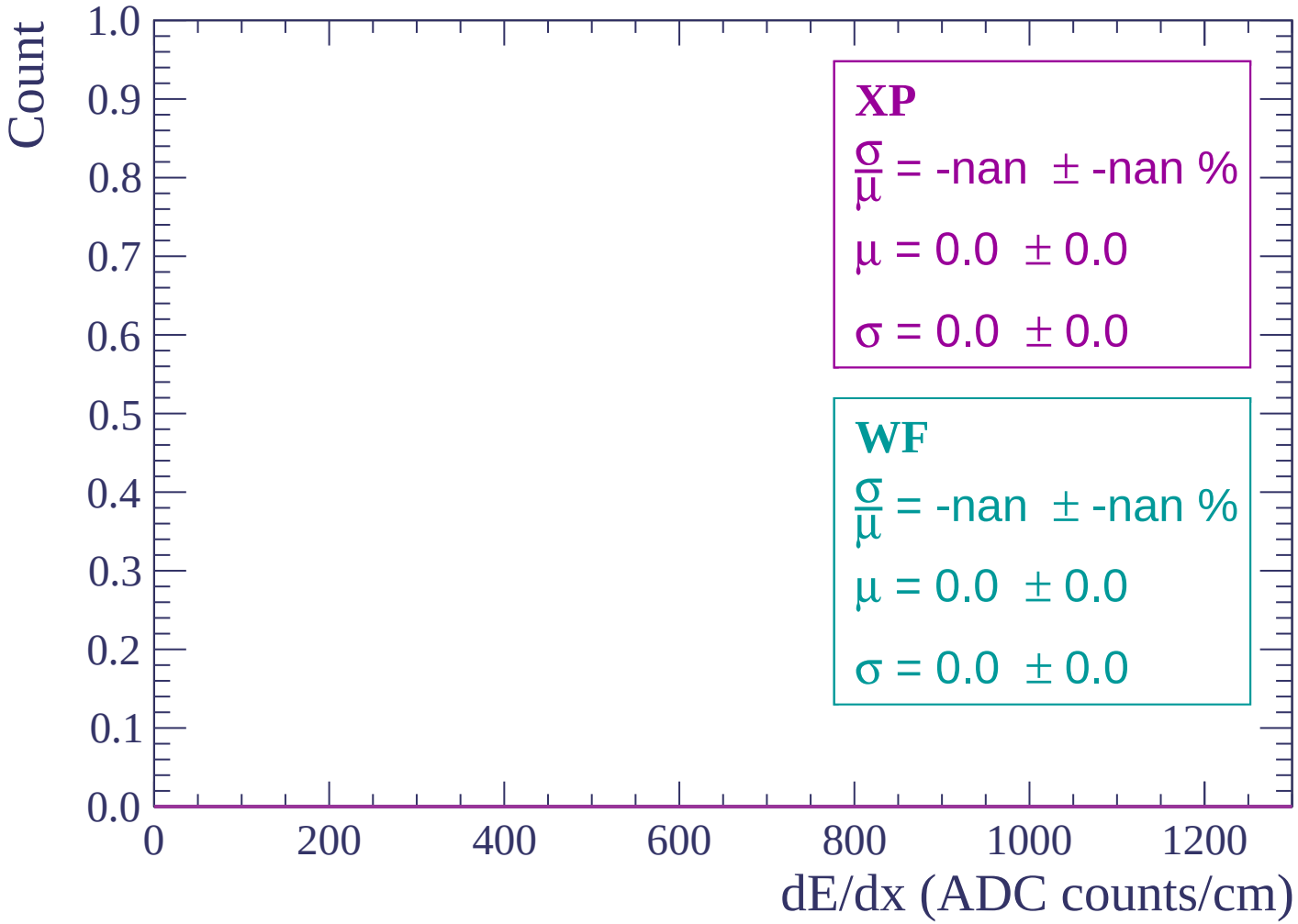

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

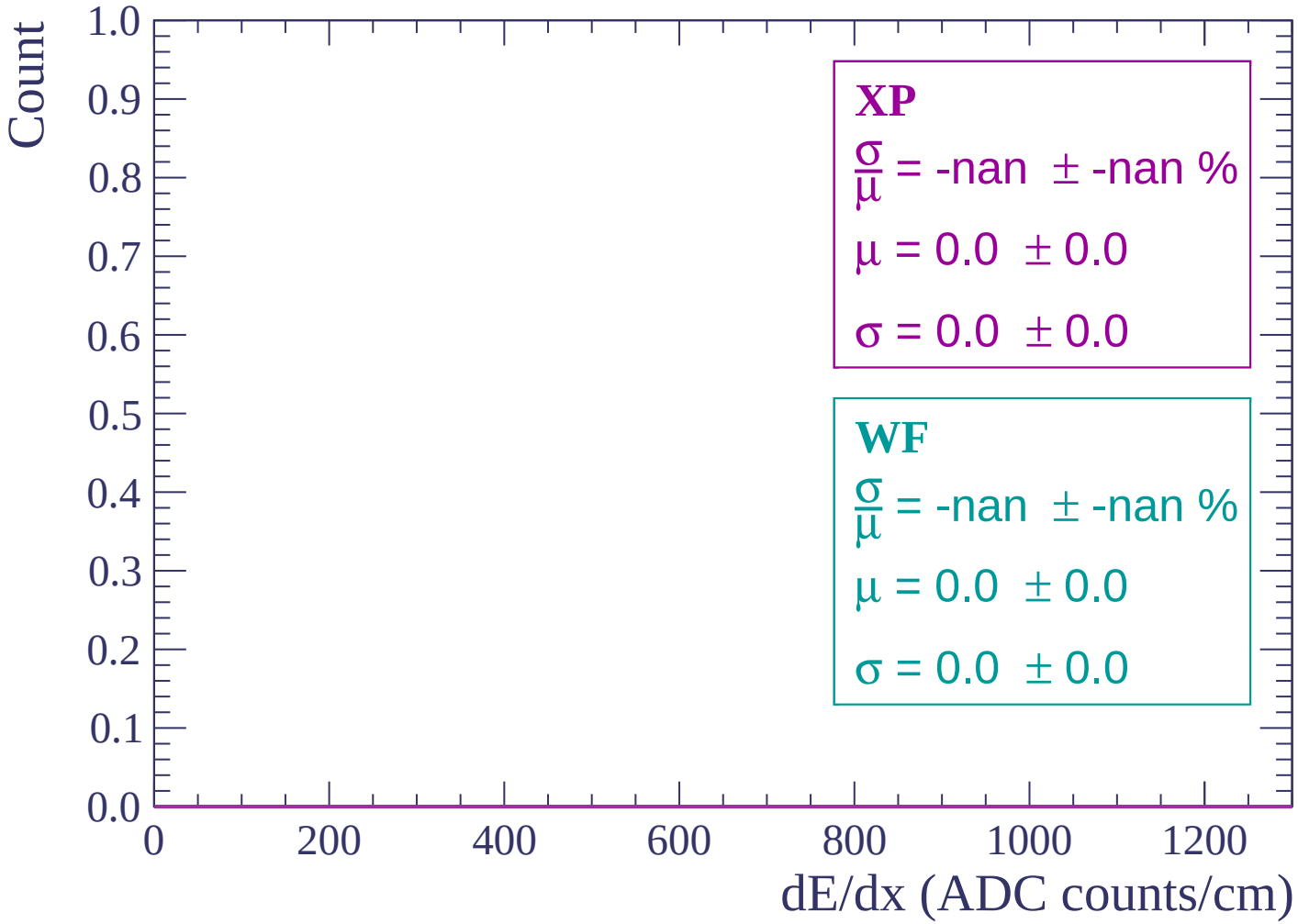

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0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dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
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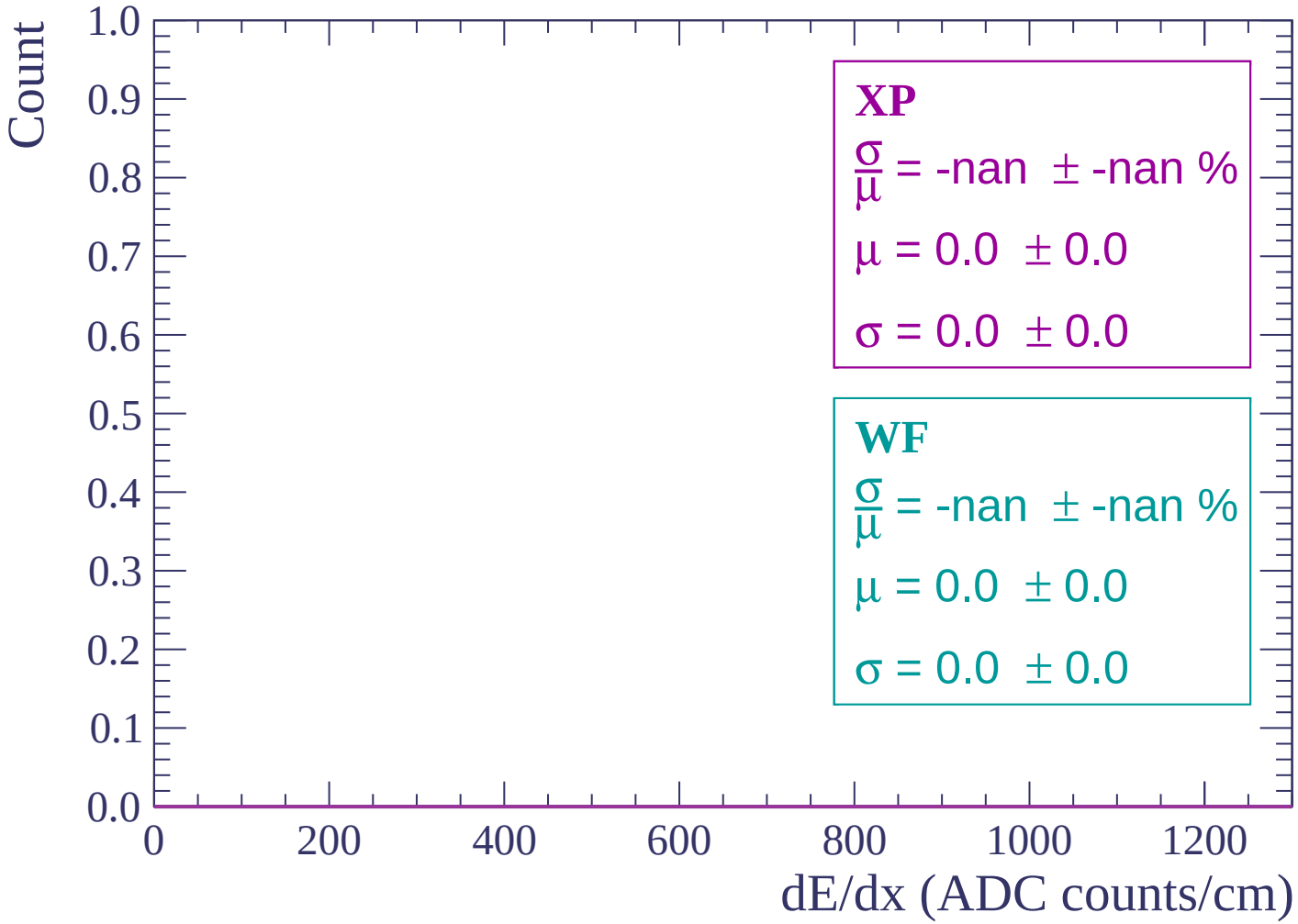
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0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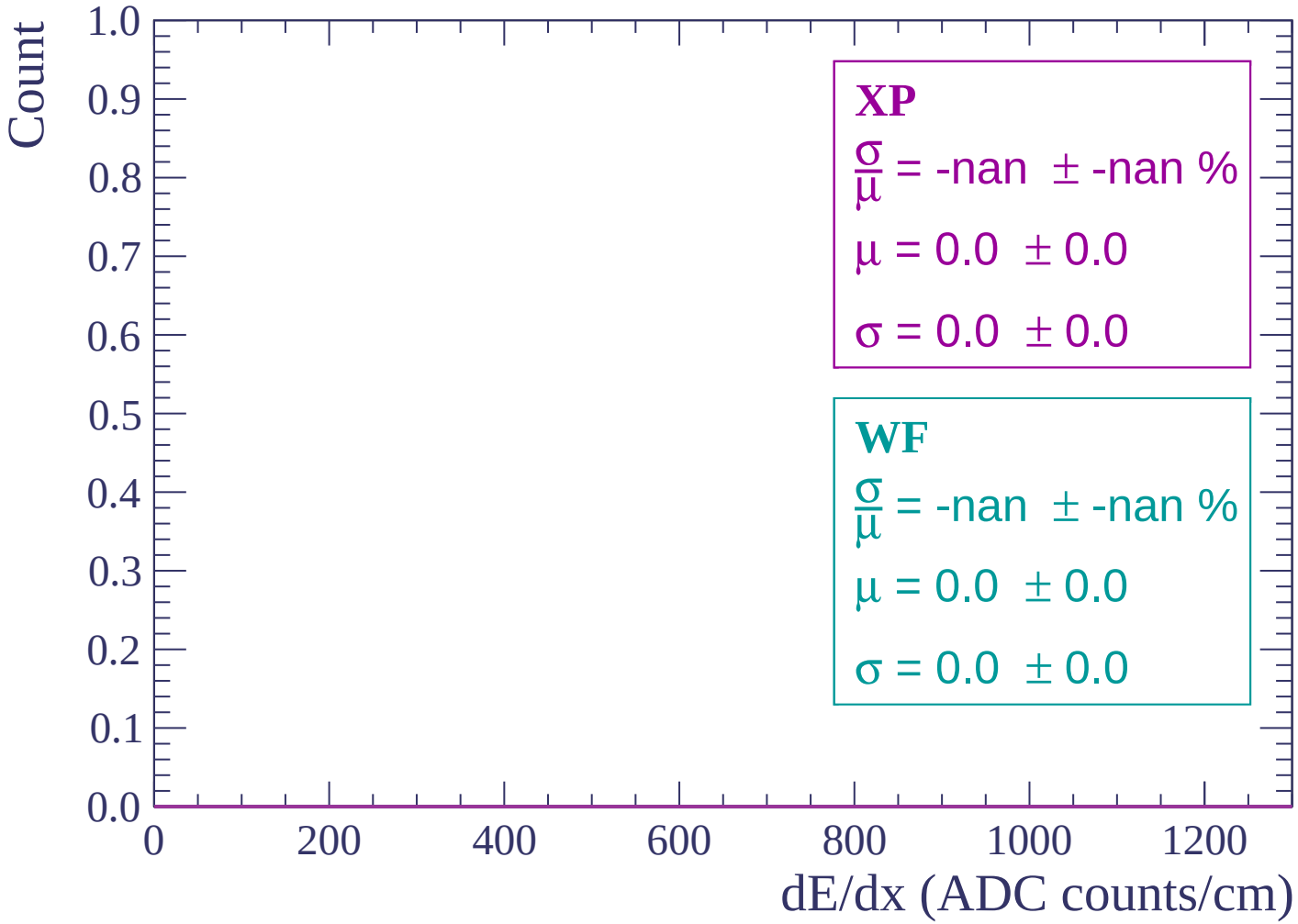
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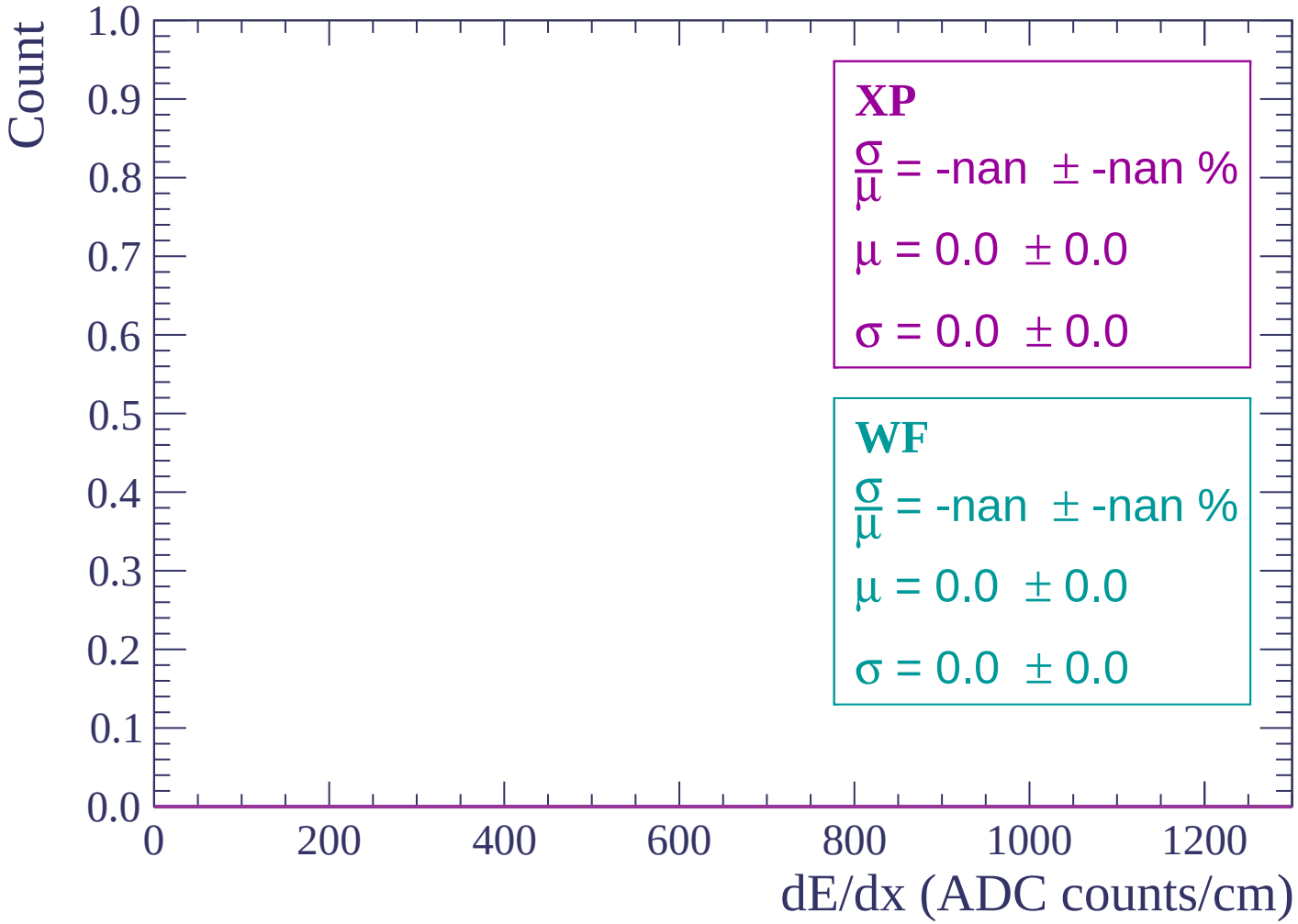
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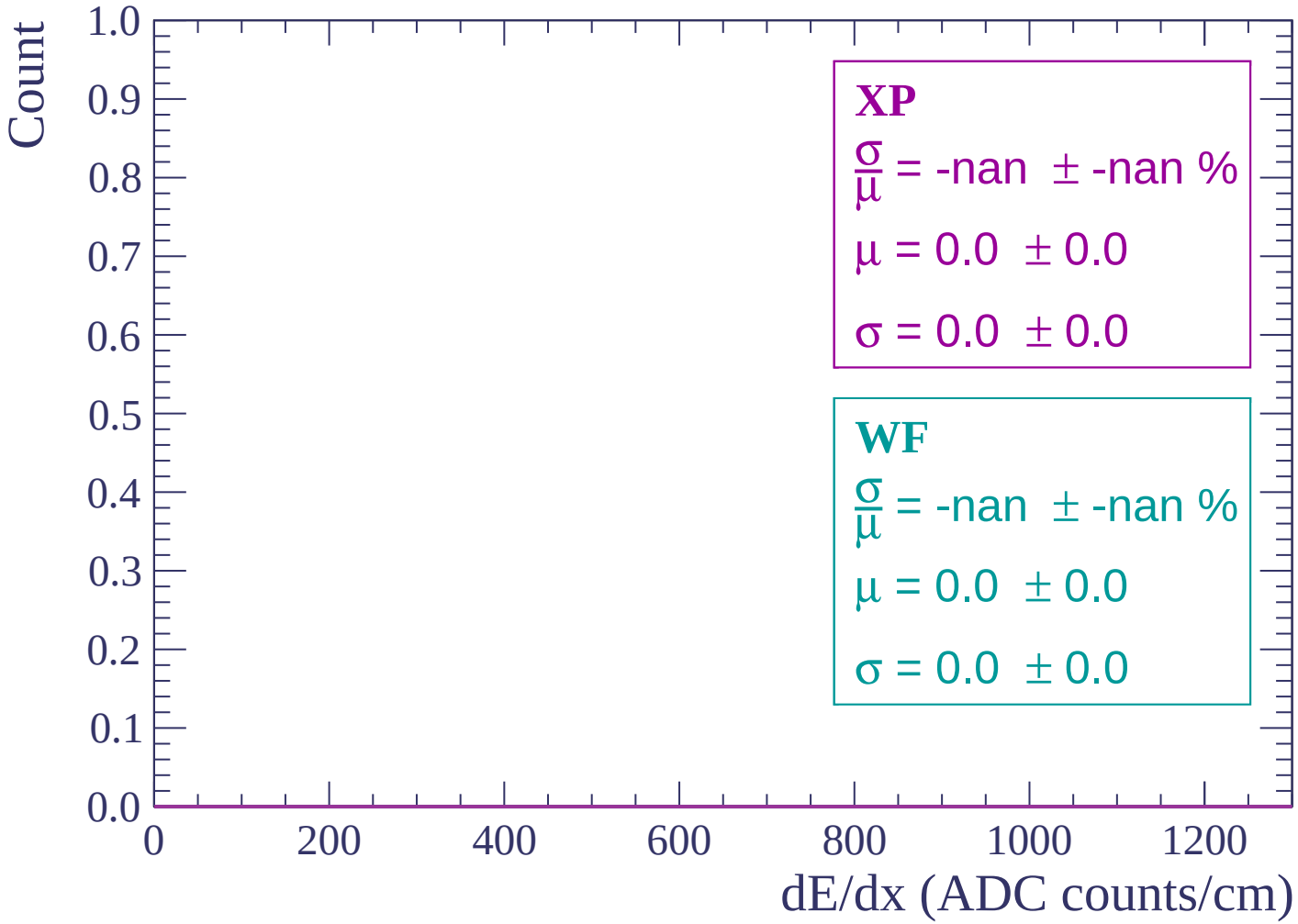
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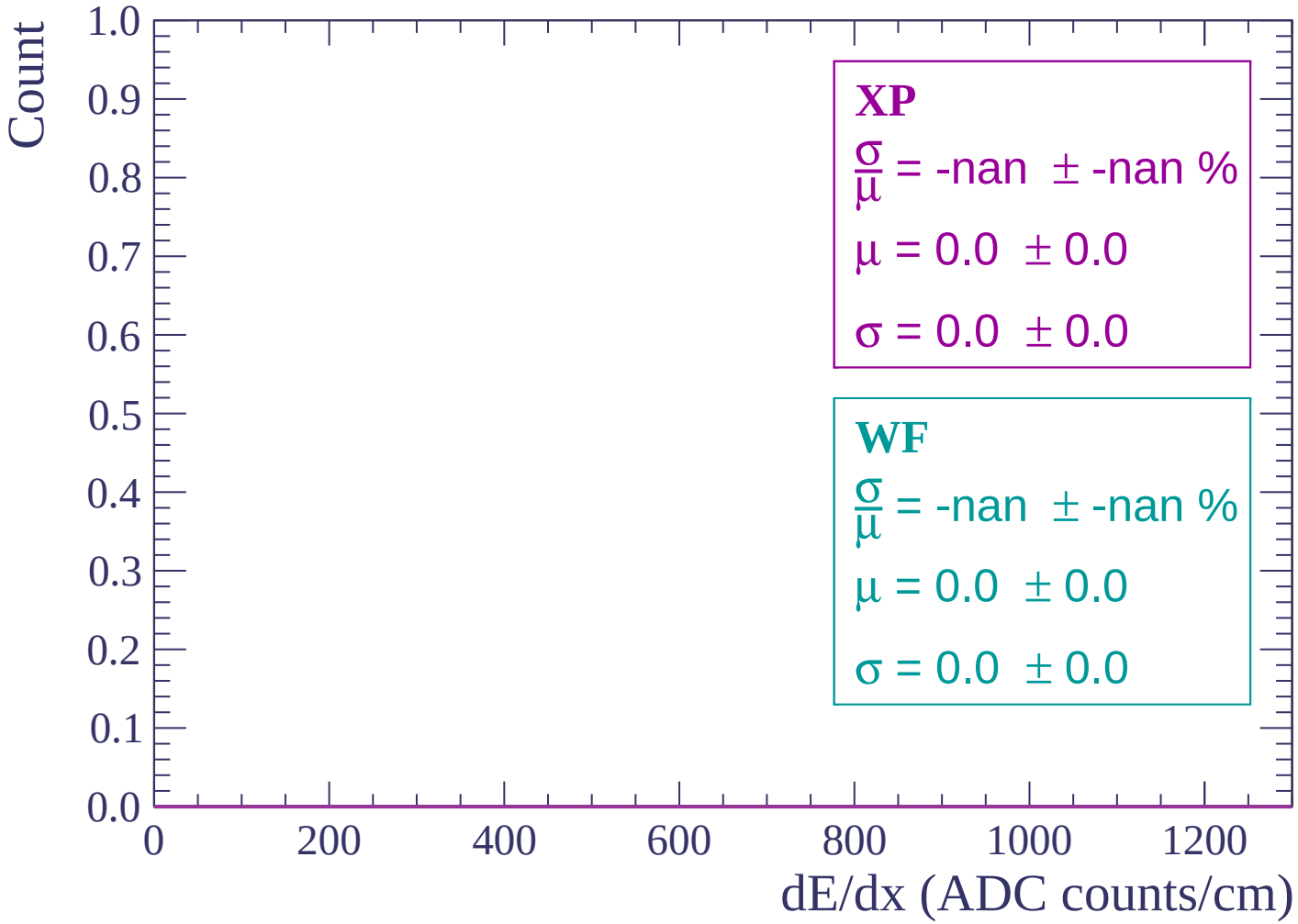
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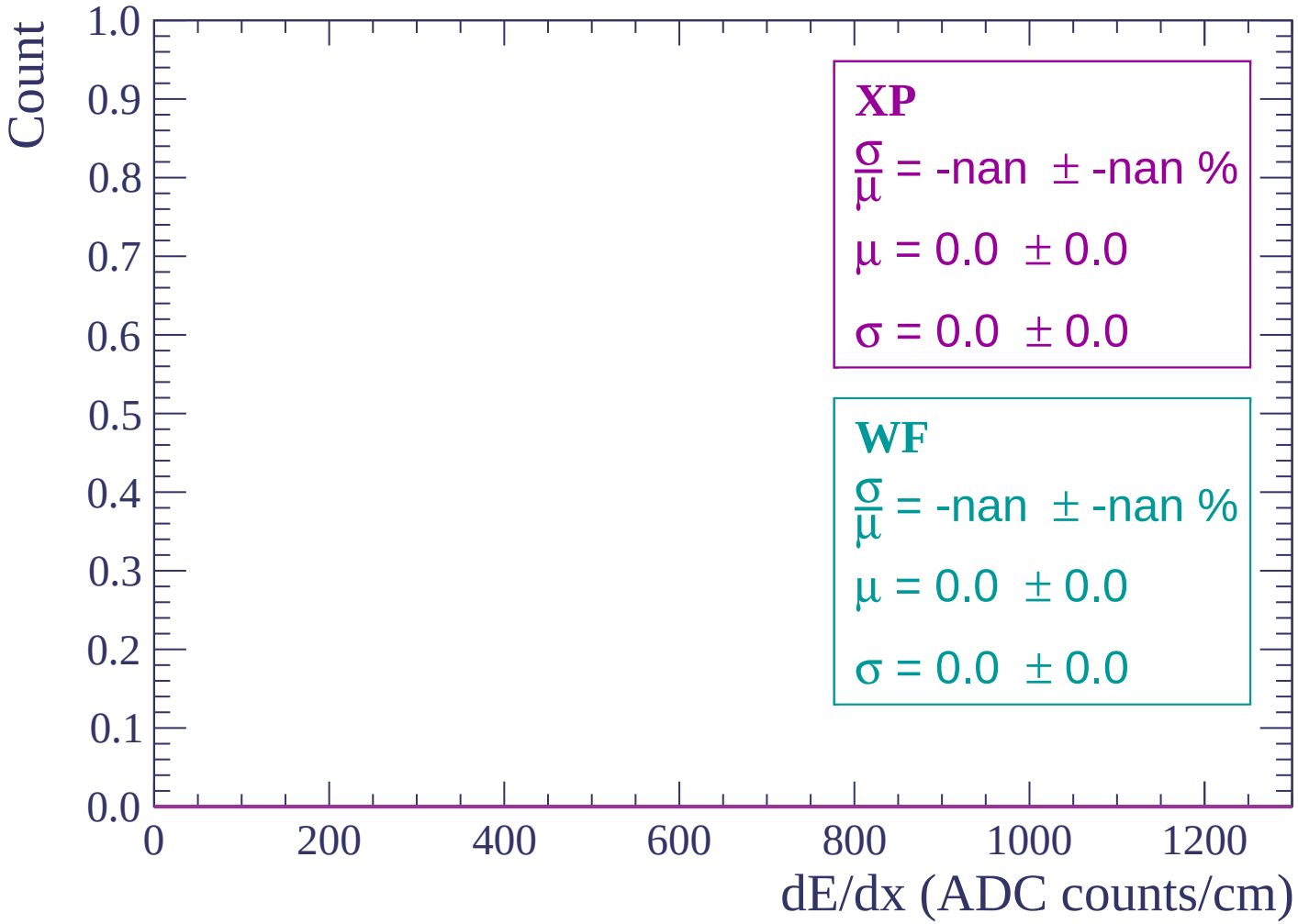
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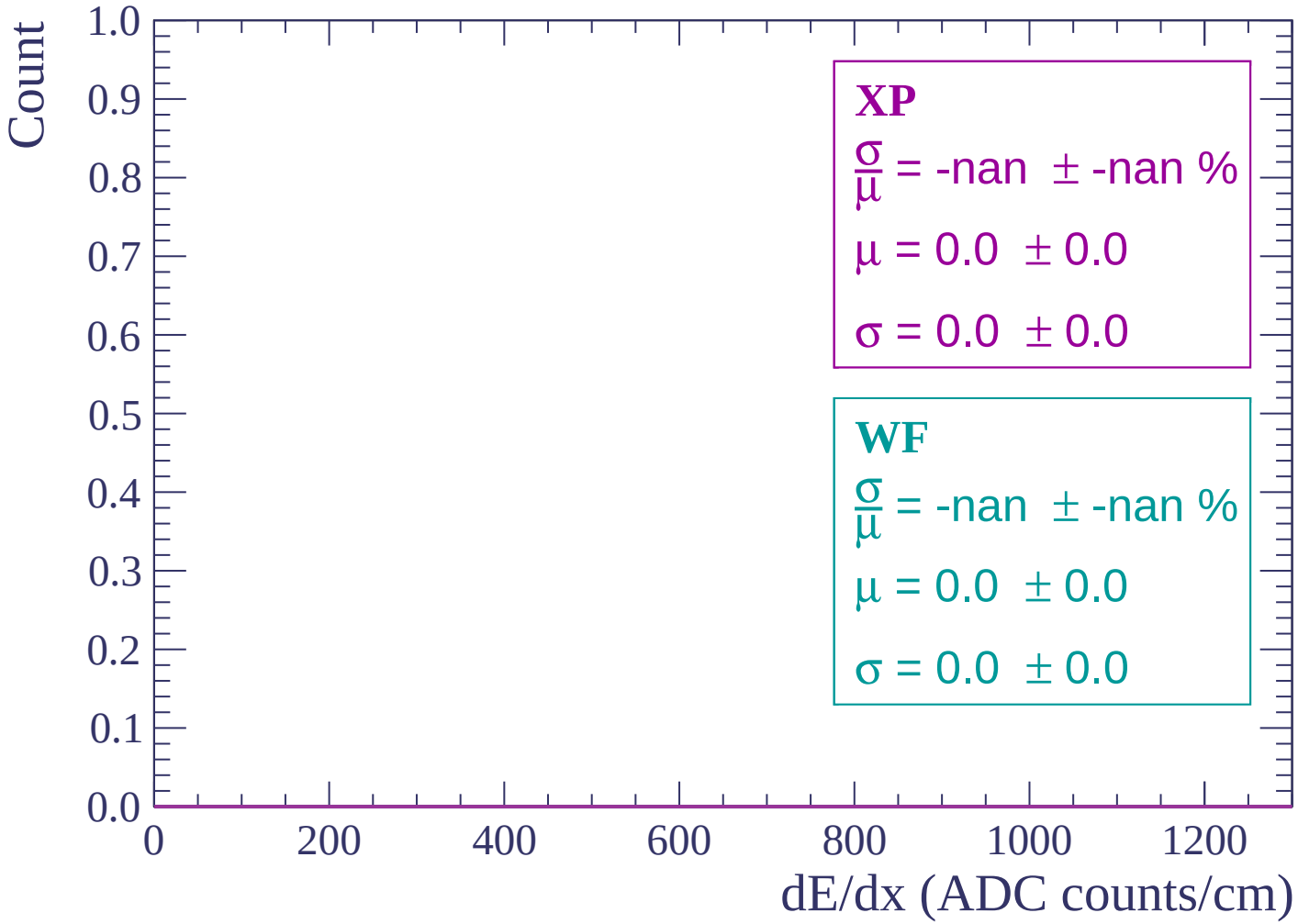
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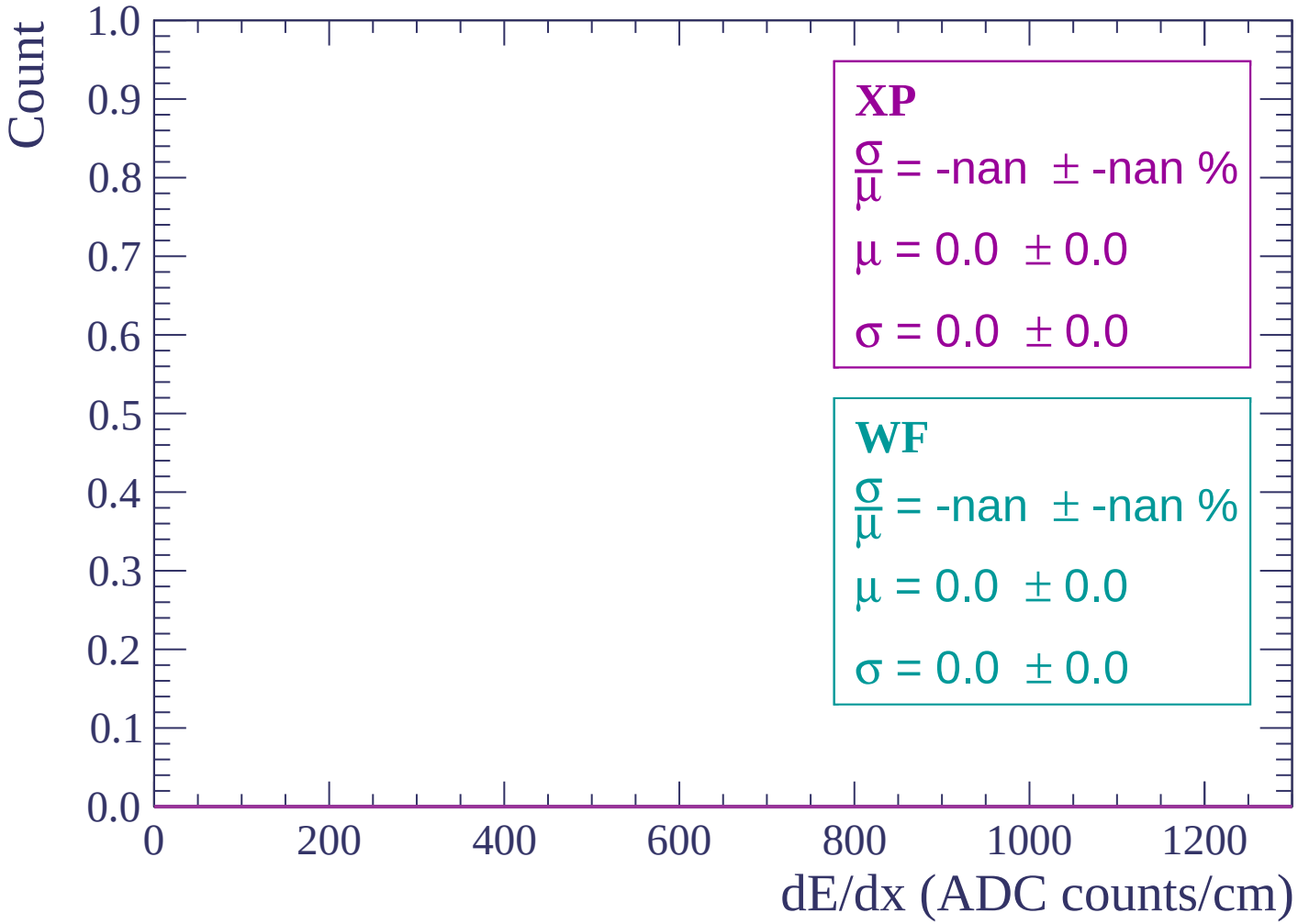
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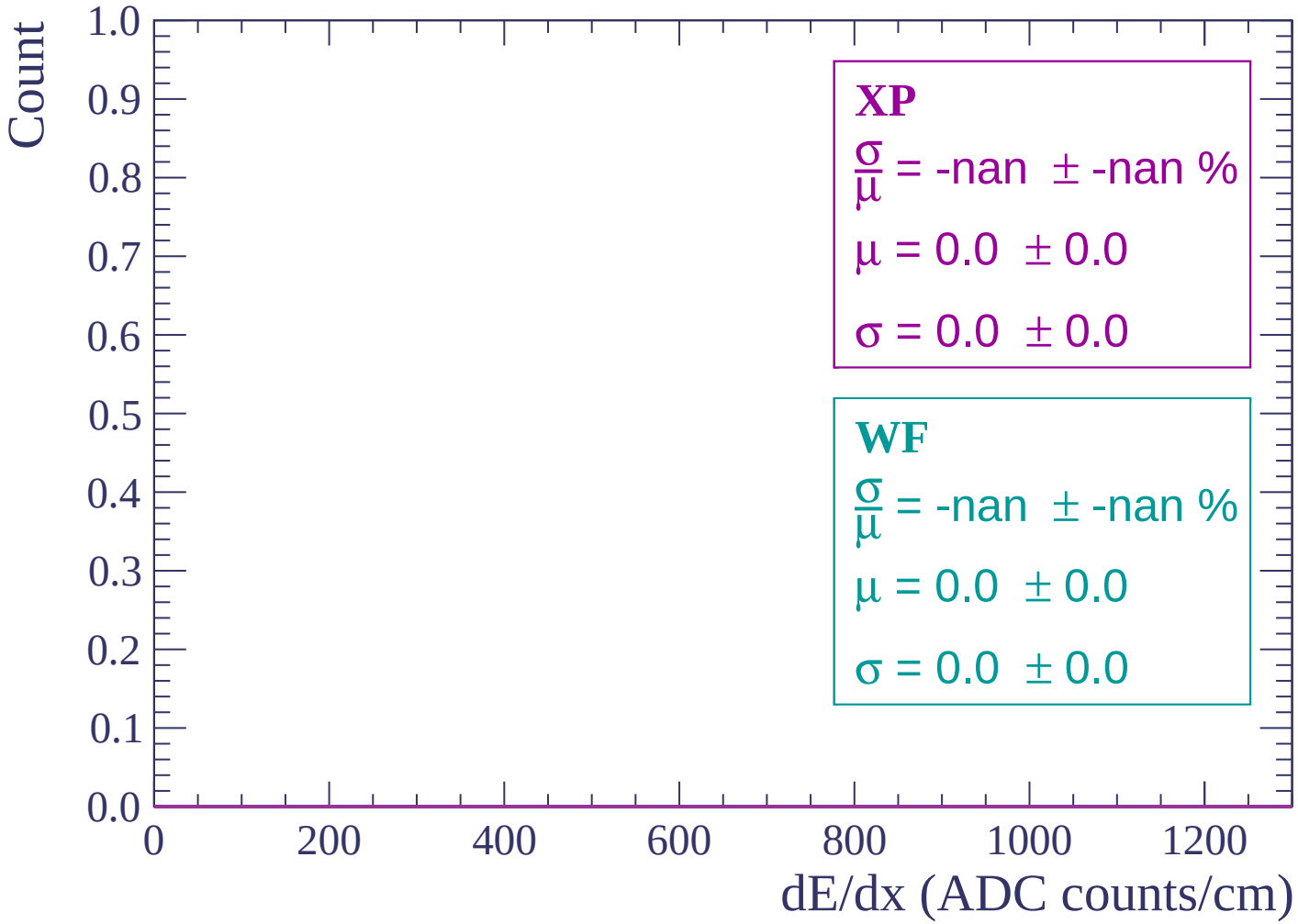
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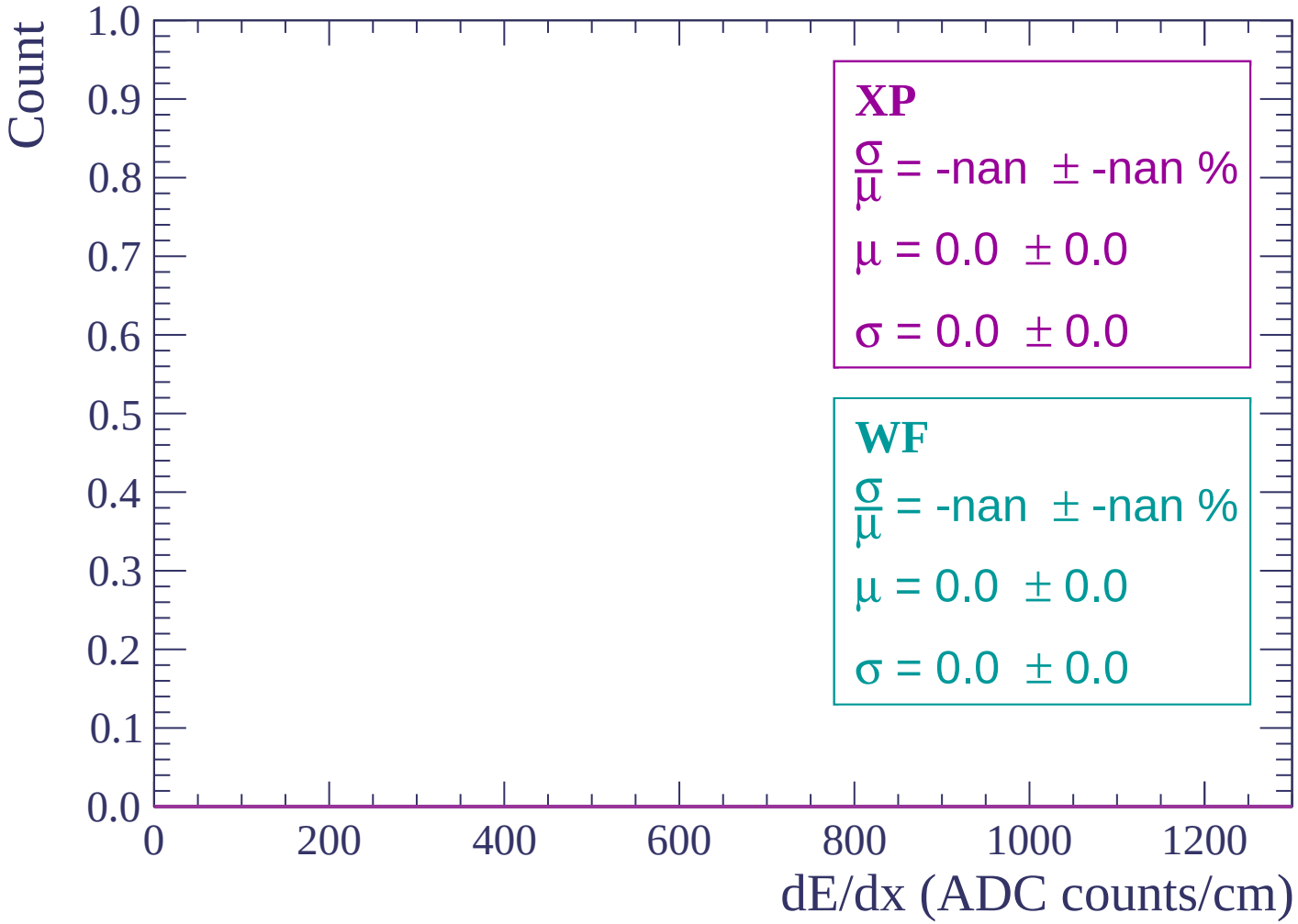
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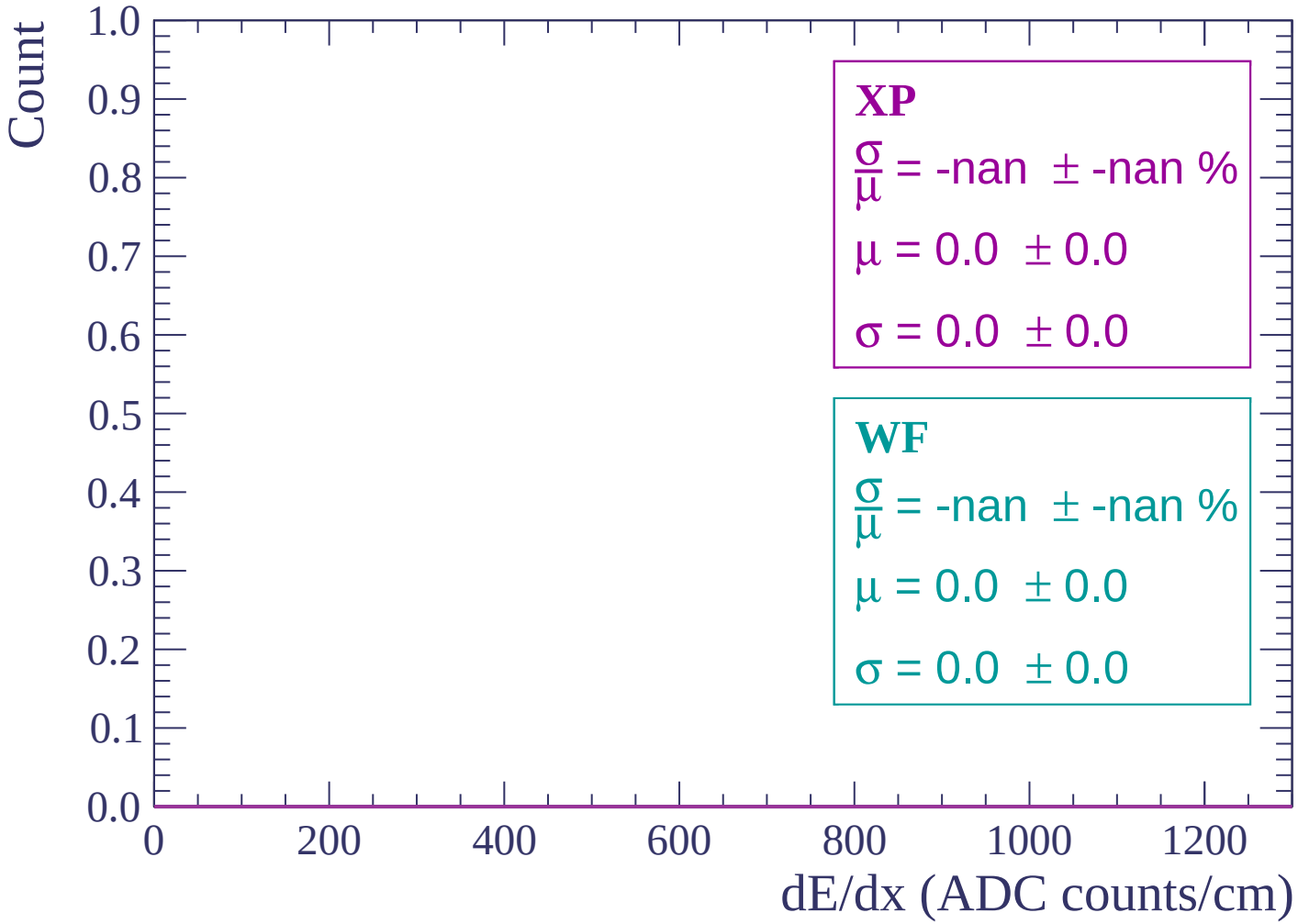
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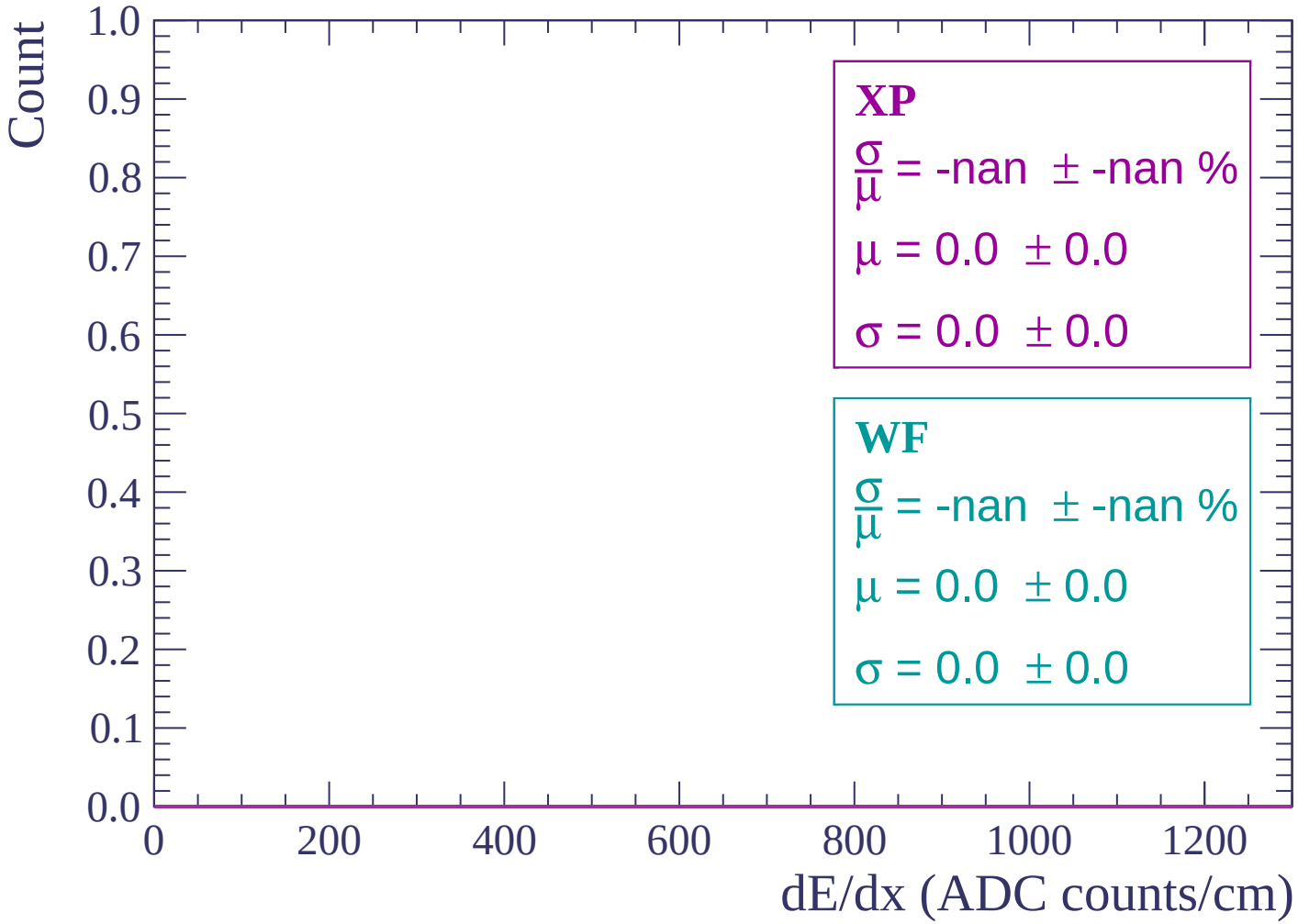
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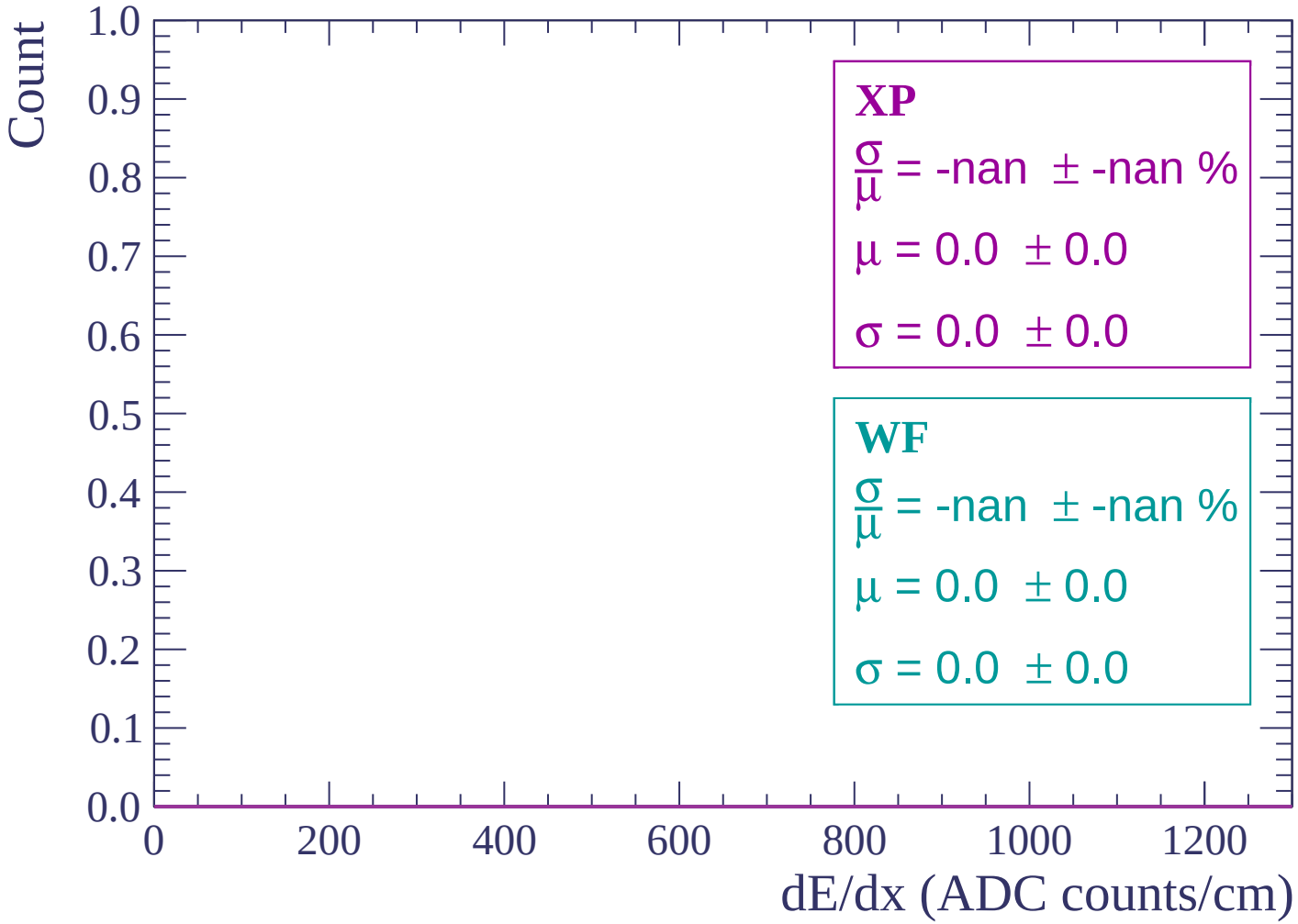
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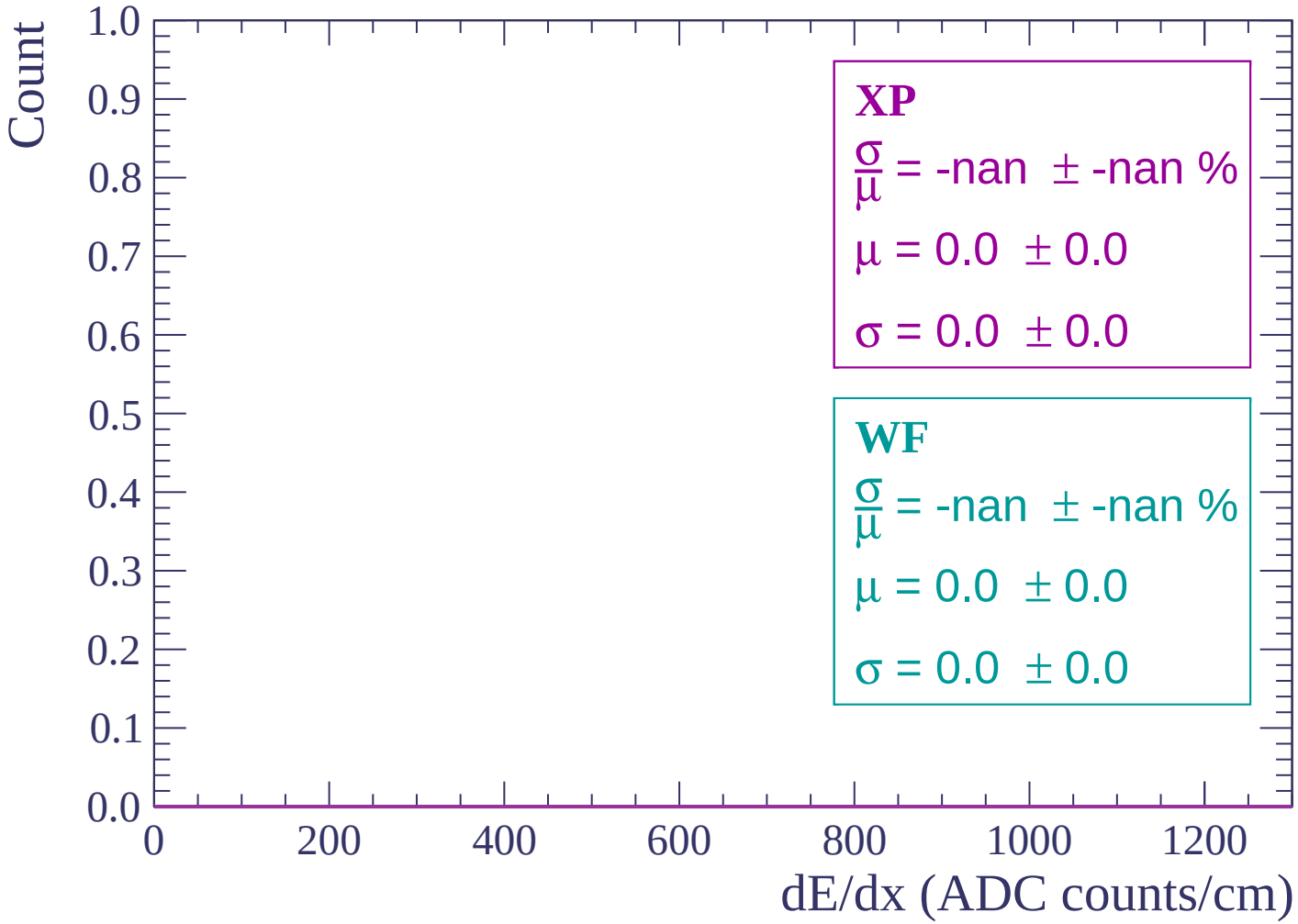
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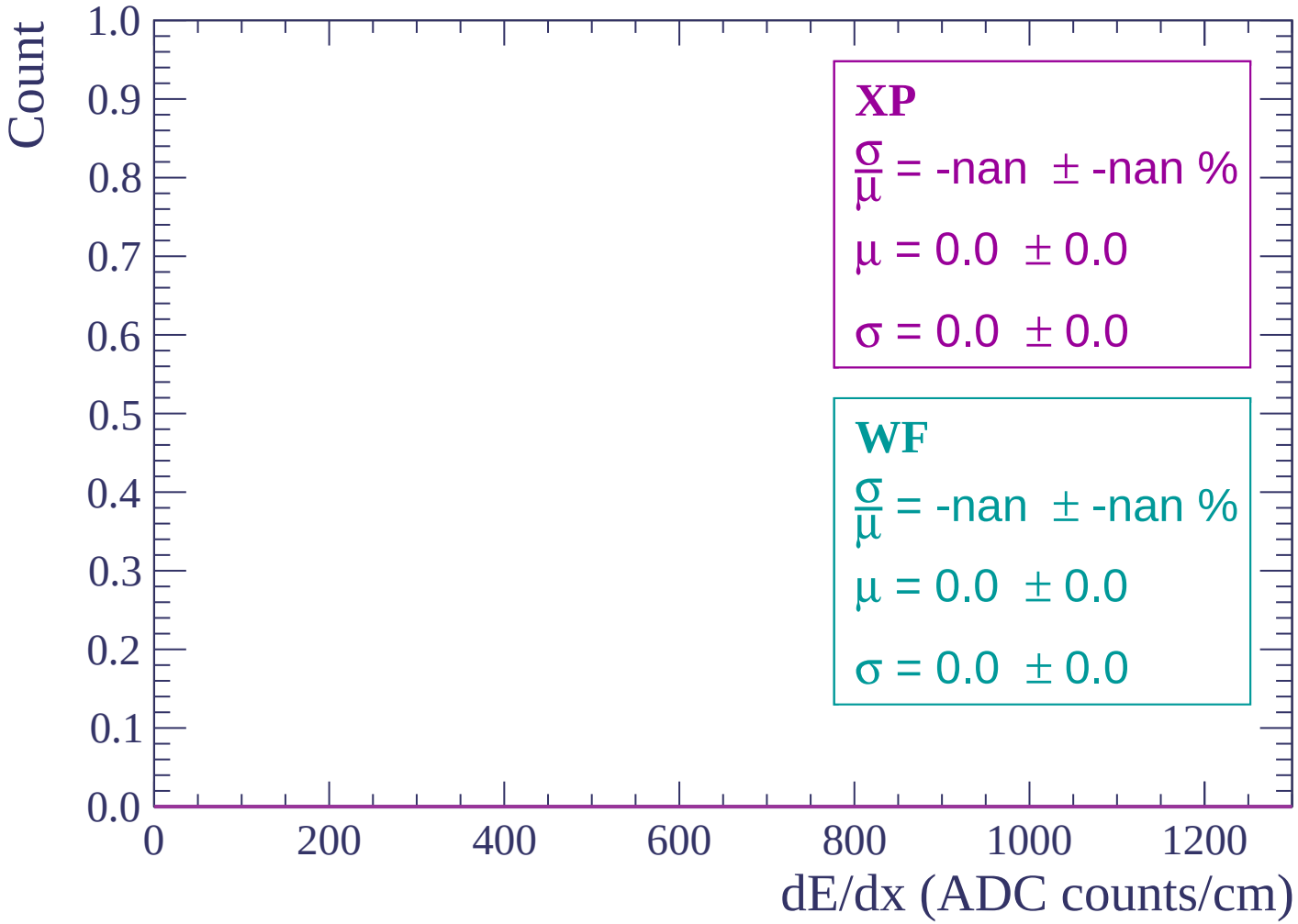
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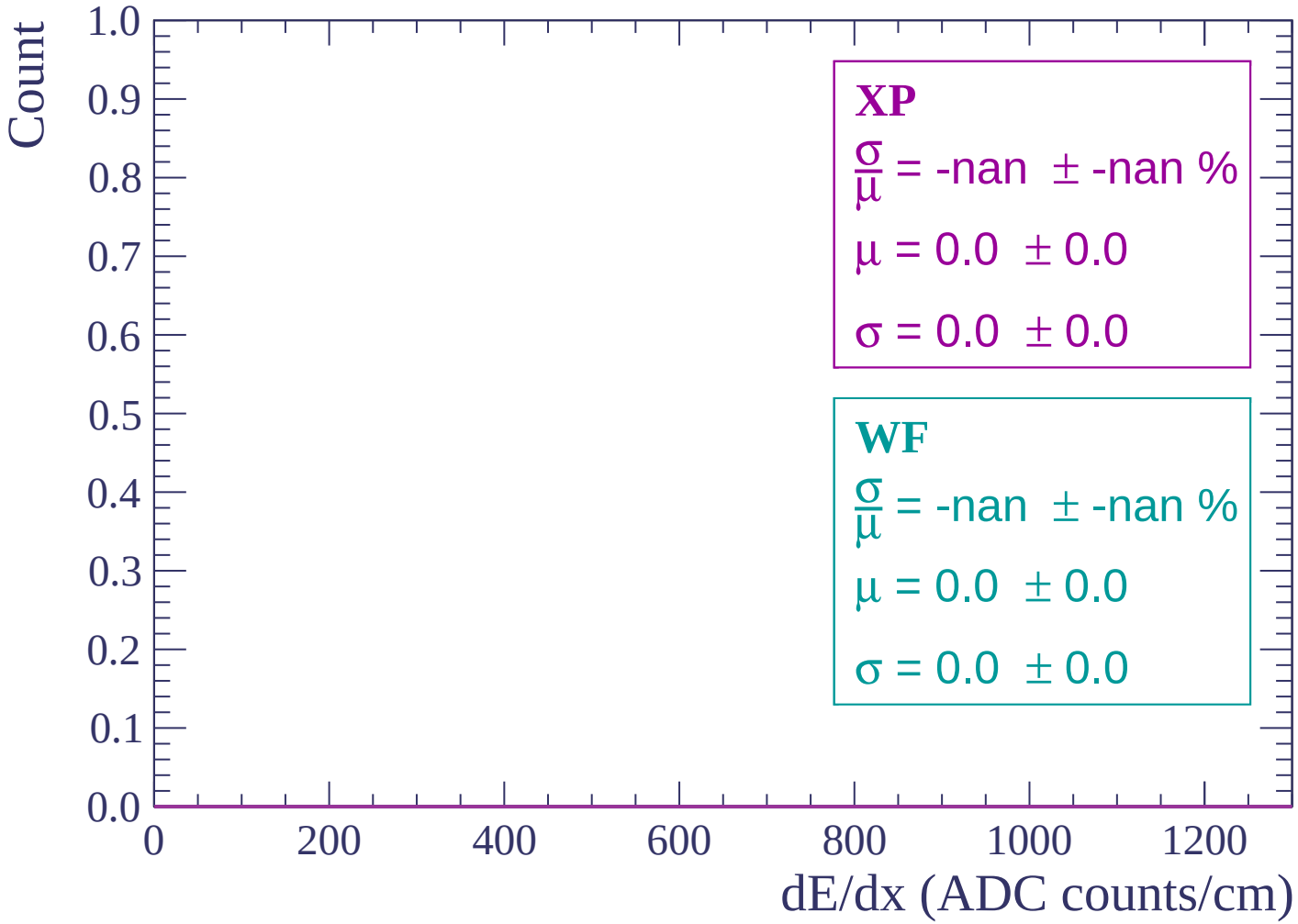
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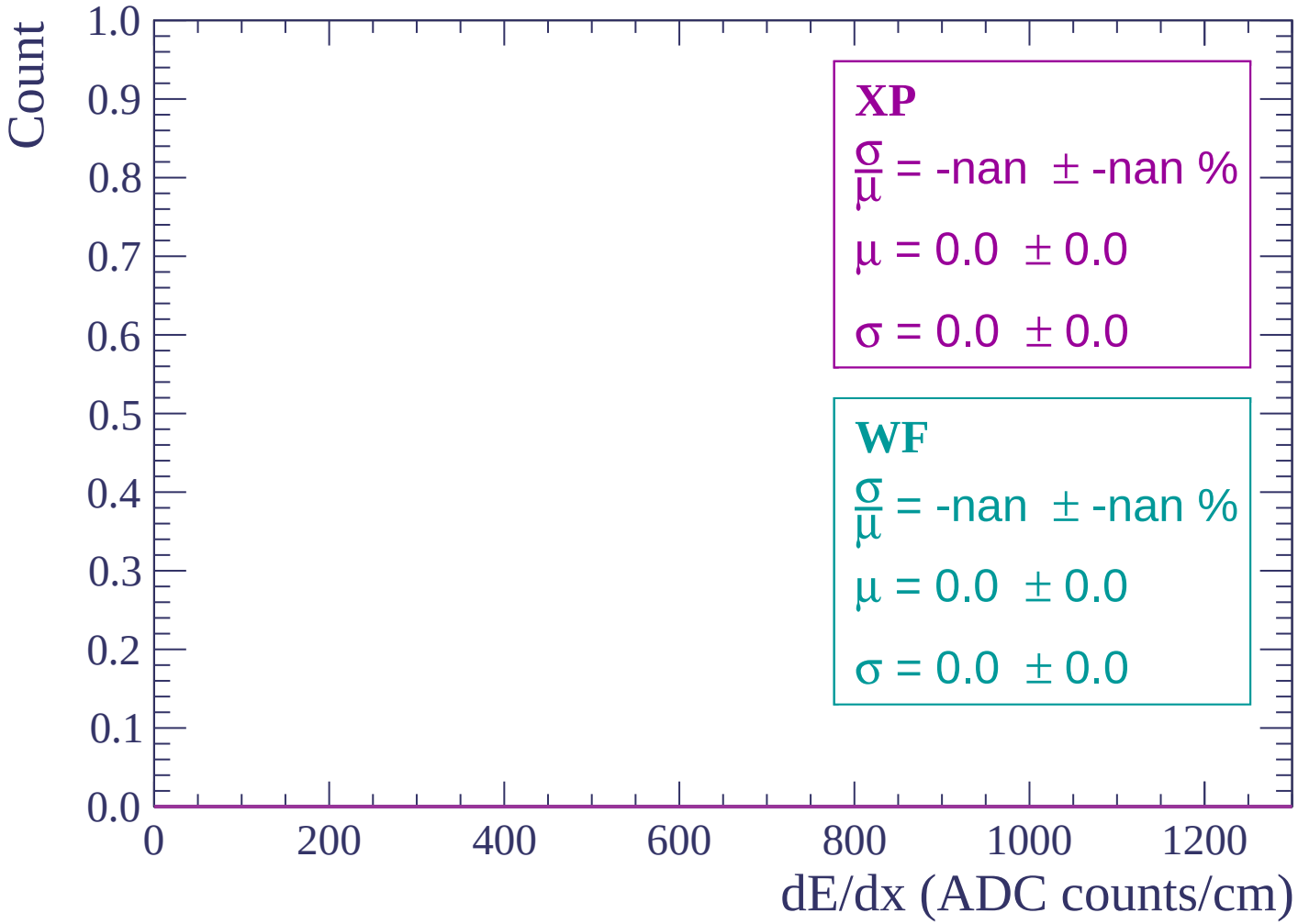
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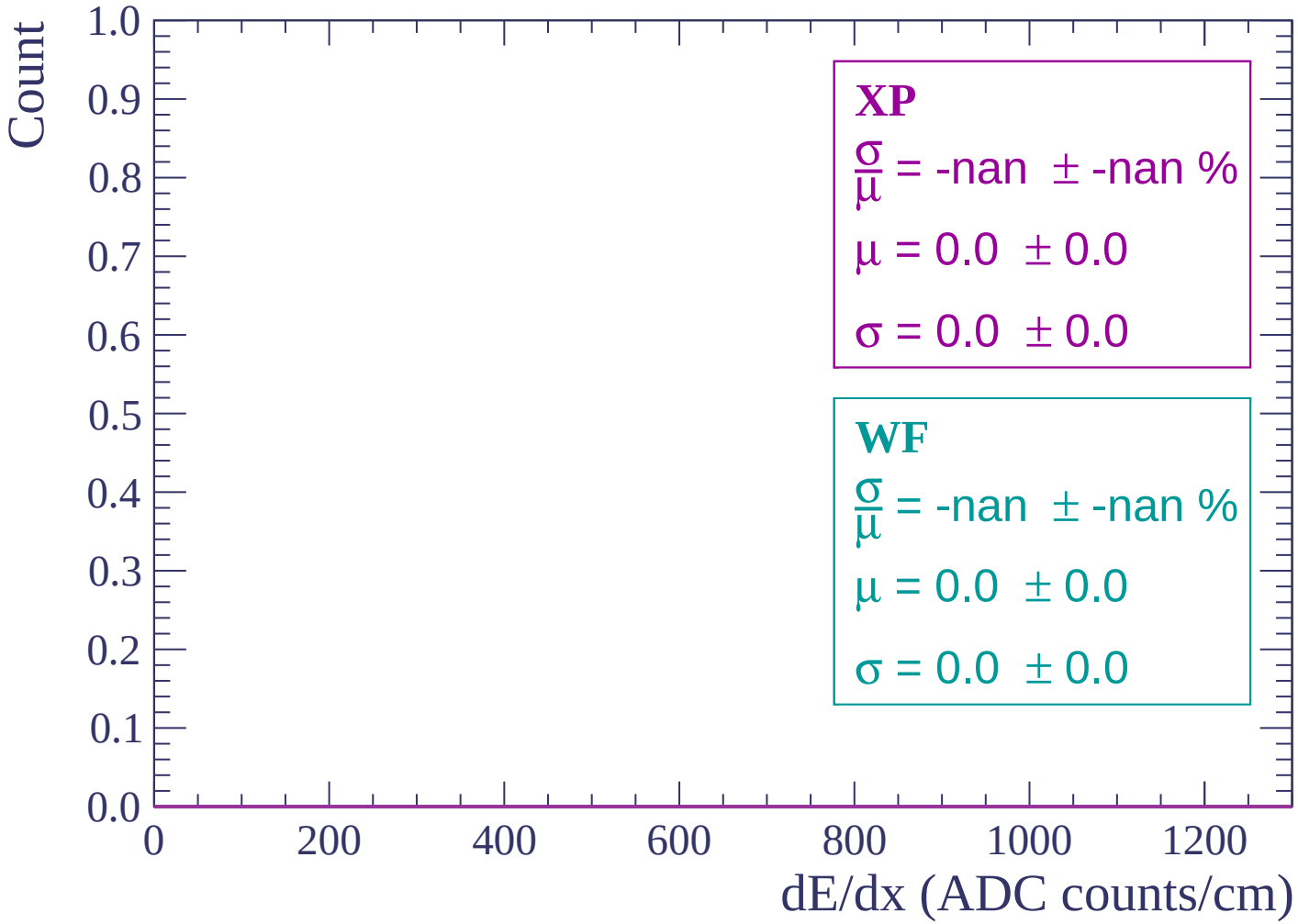
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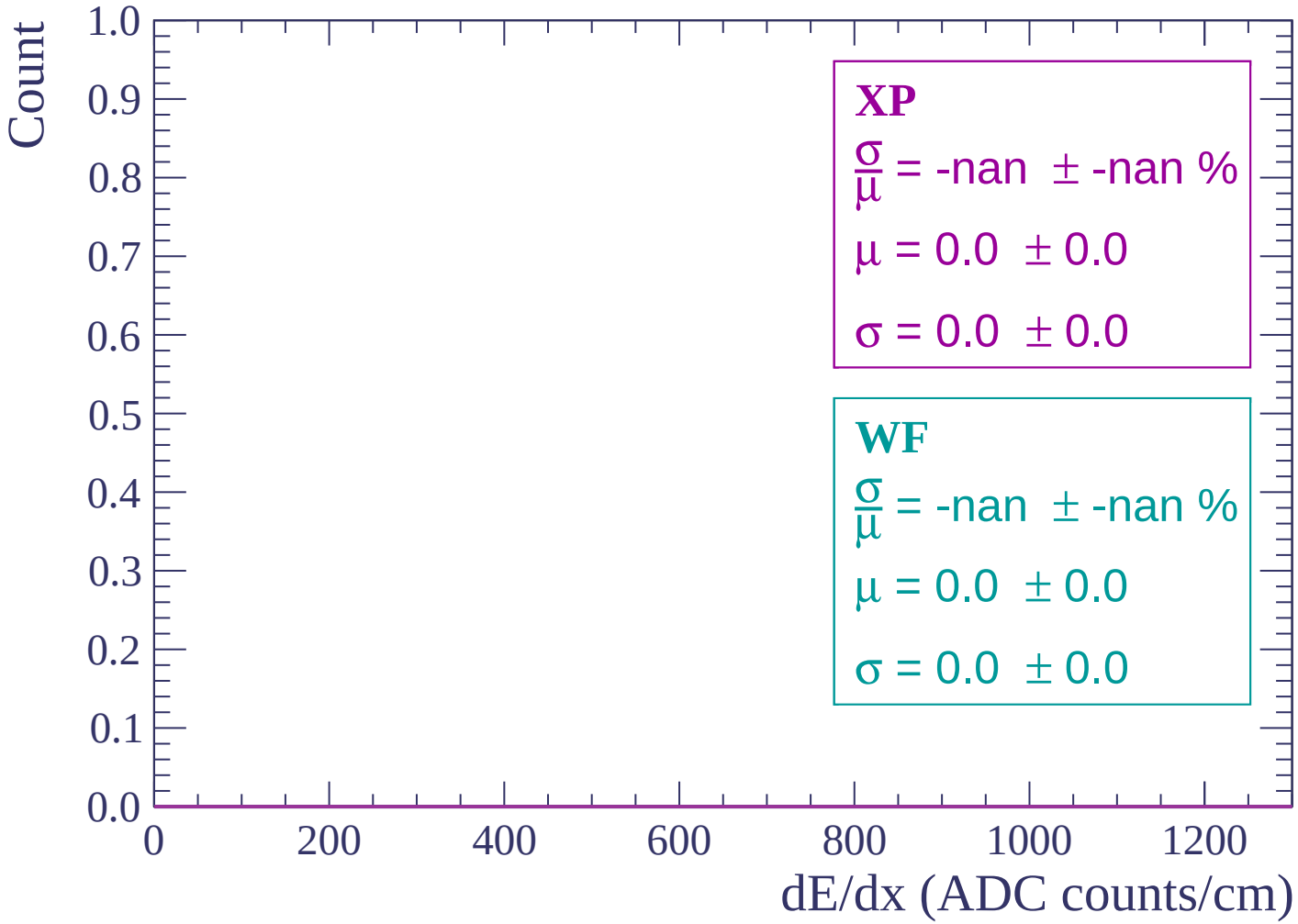
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dE/dx (ADC counts/cm)
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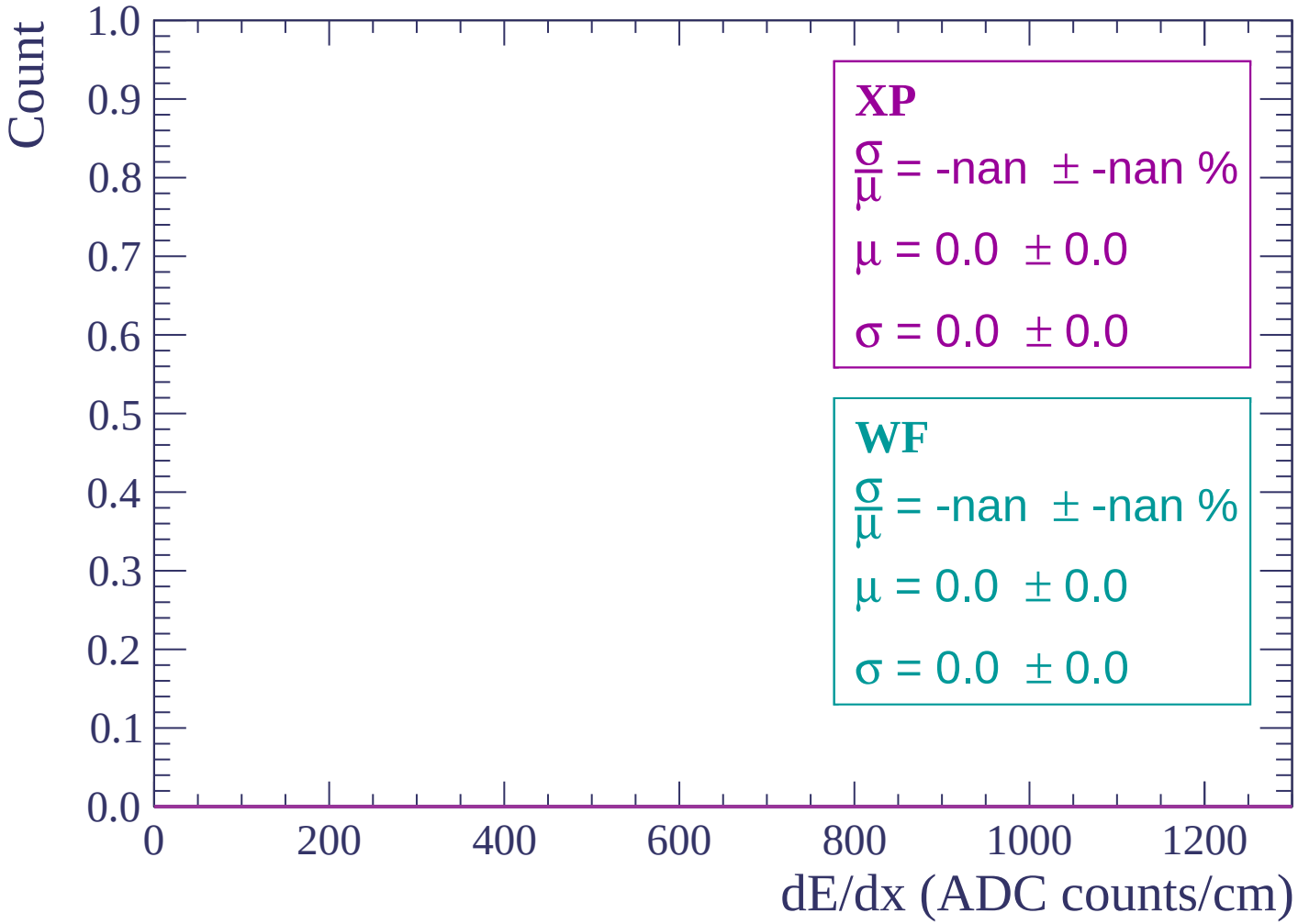
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dE/dx (ADC counts/cm)
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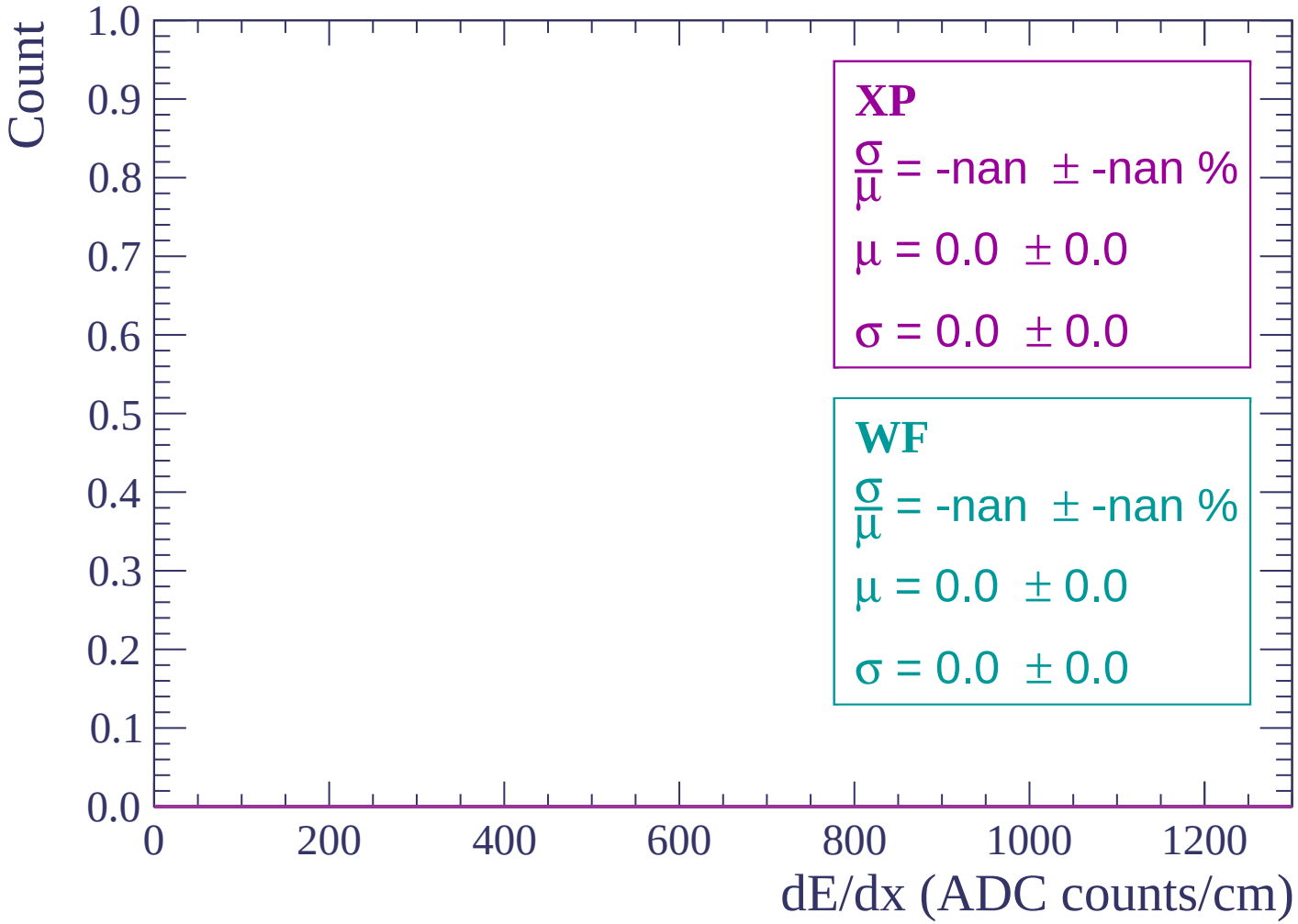
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dE/dx (ADC counts/cm)
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σ/μ = -nan ± -nan %
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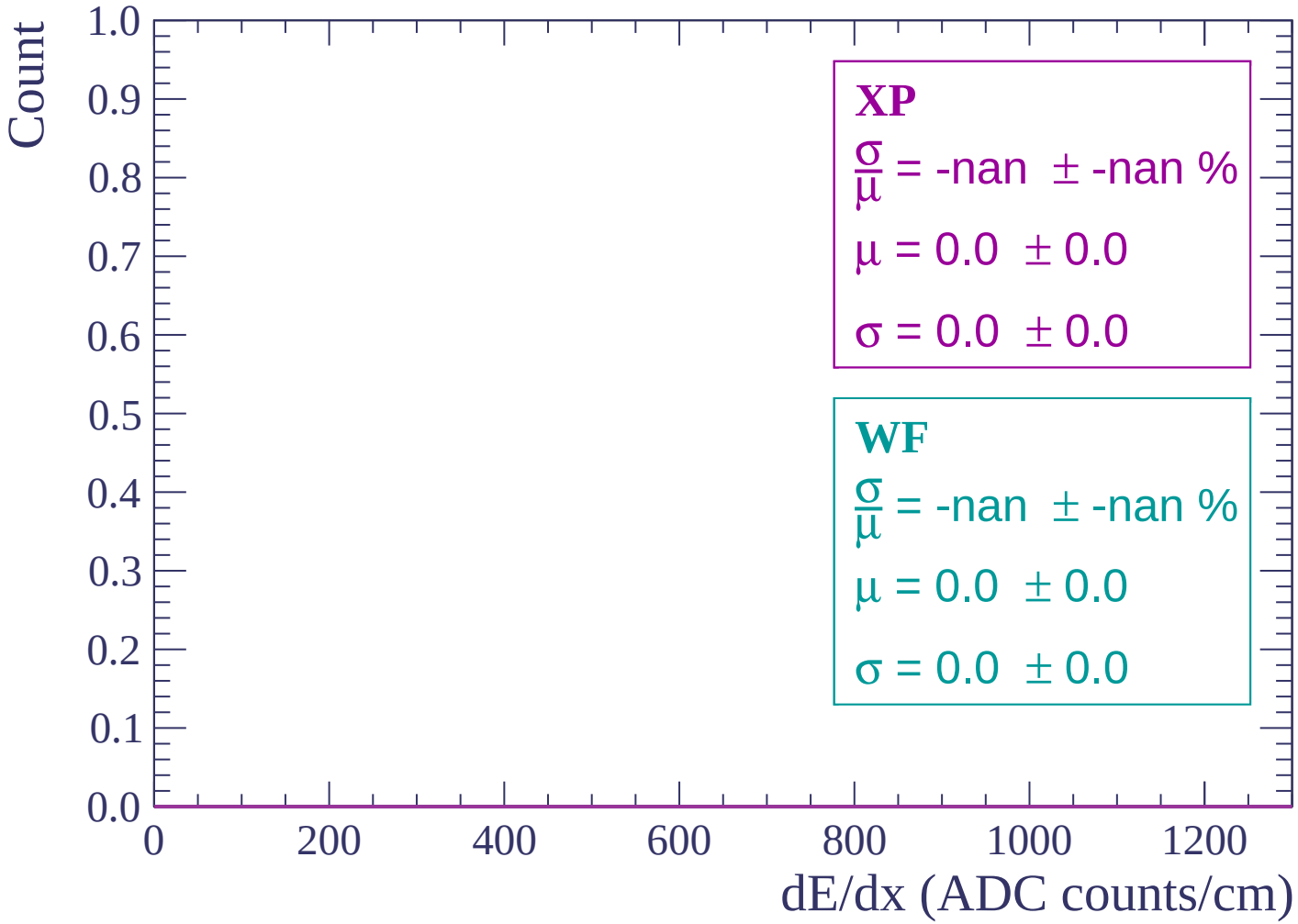
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1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
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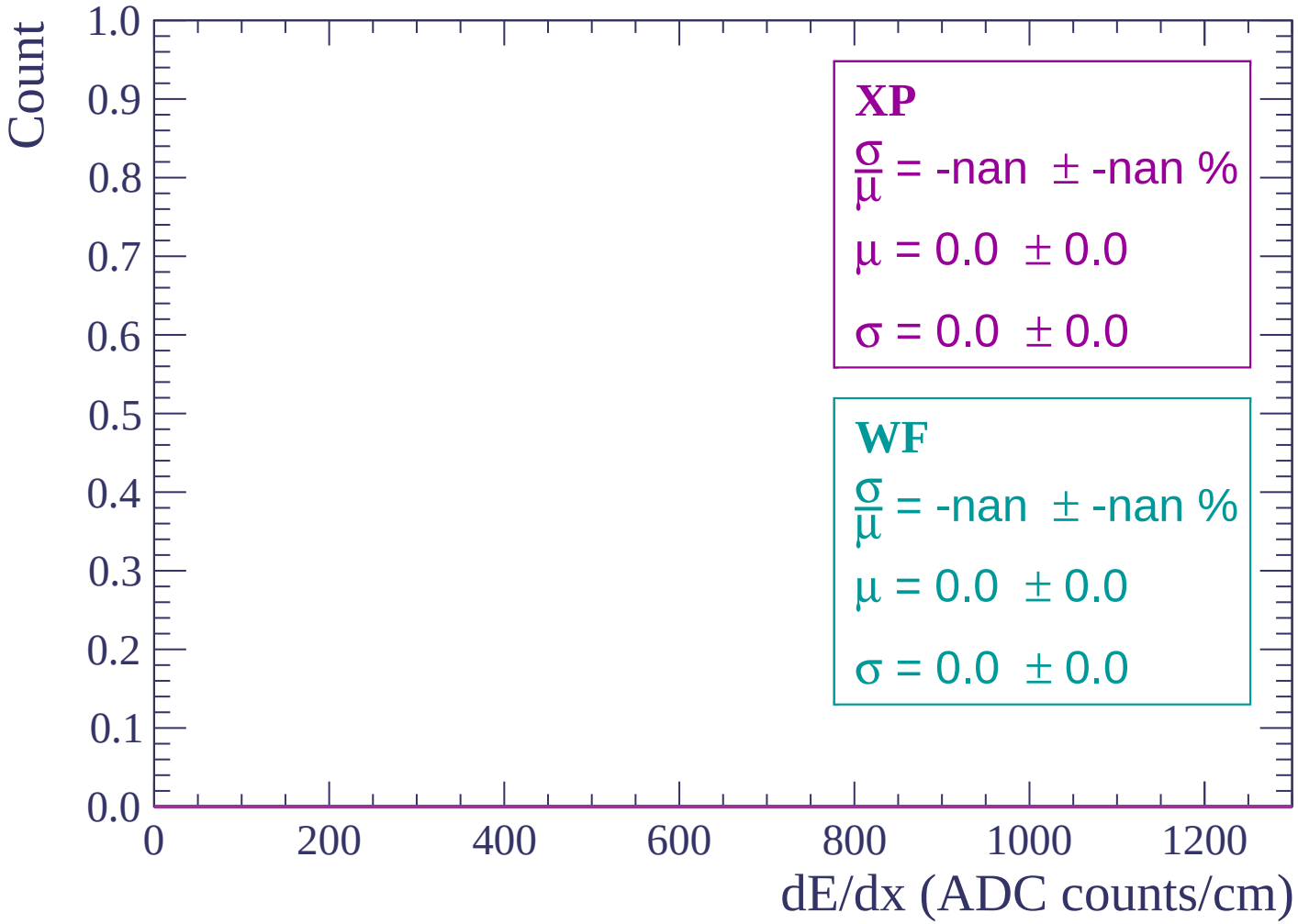
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200
400
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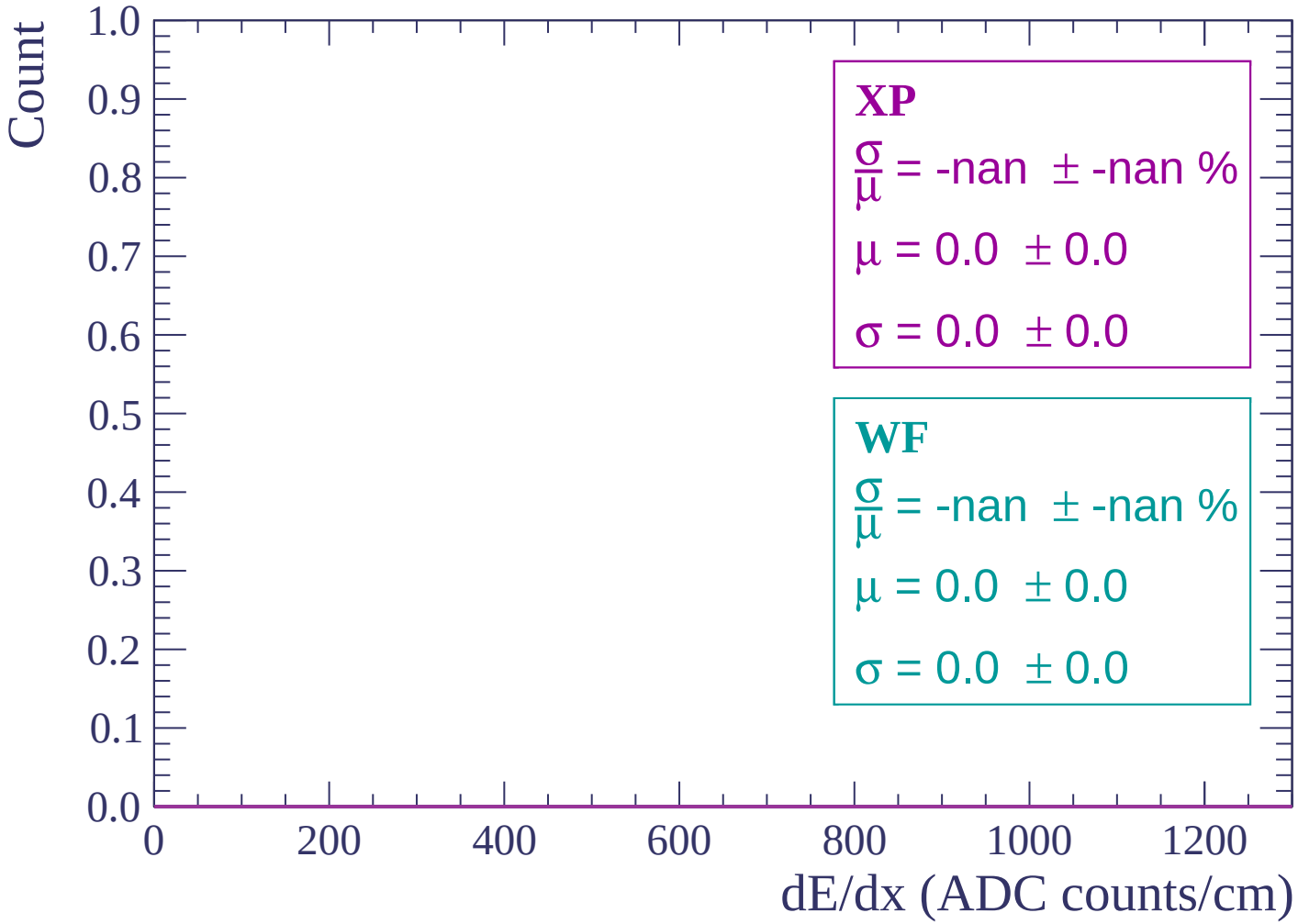
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0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200
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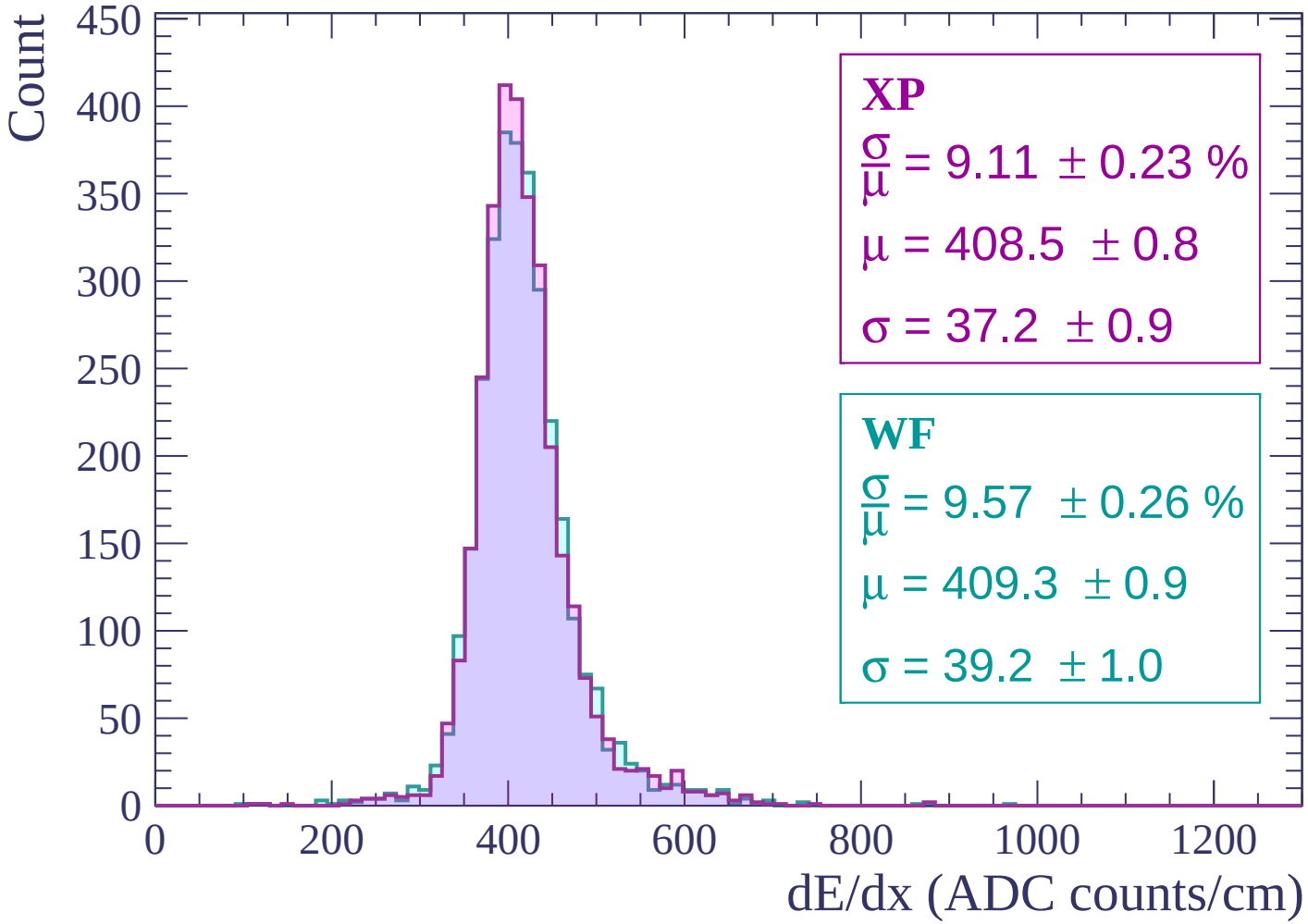
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
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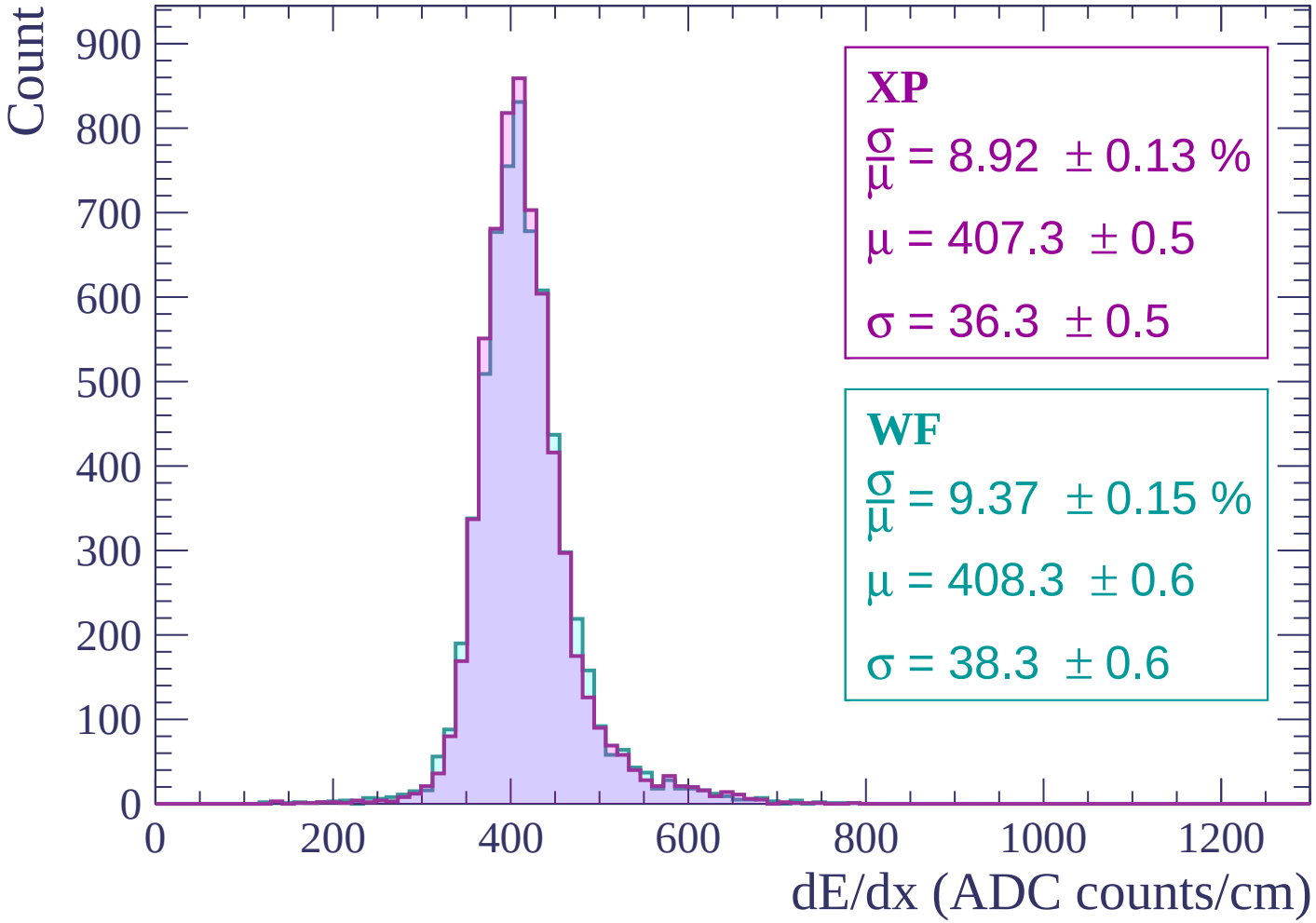
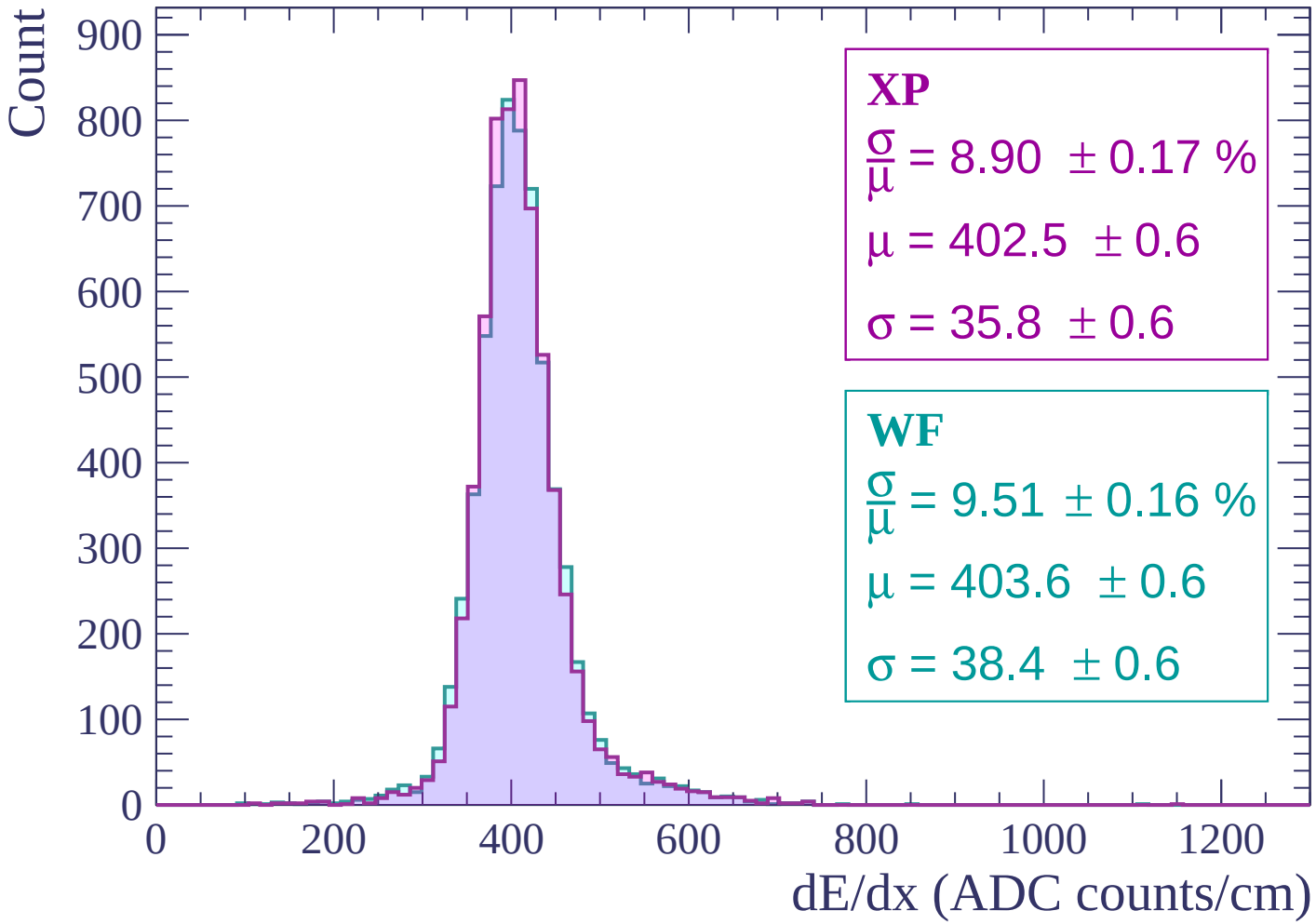
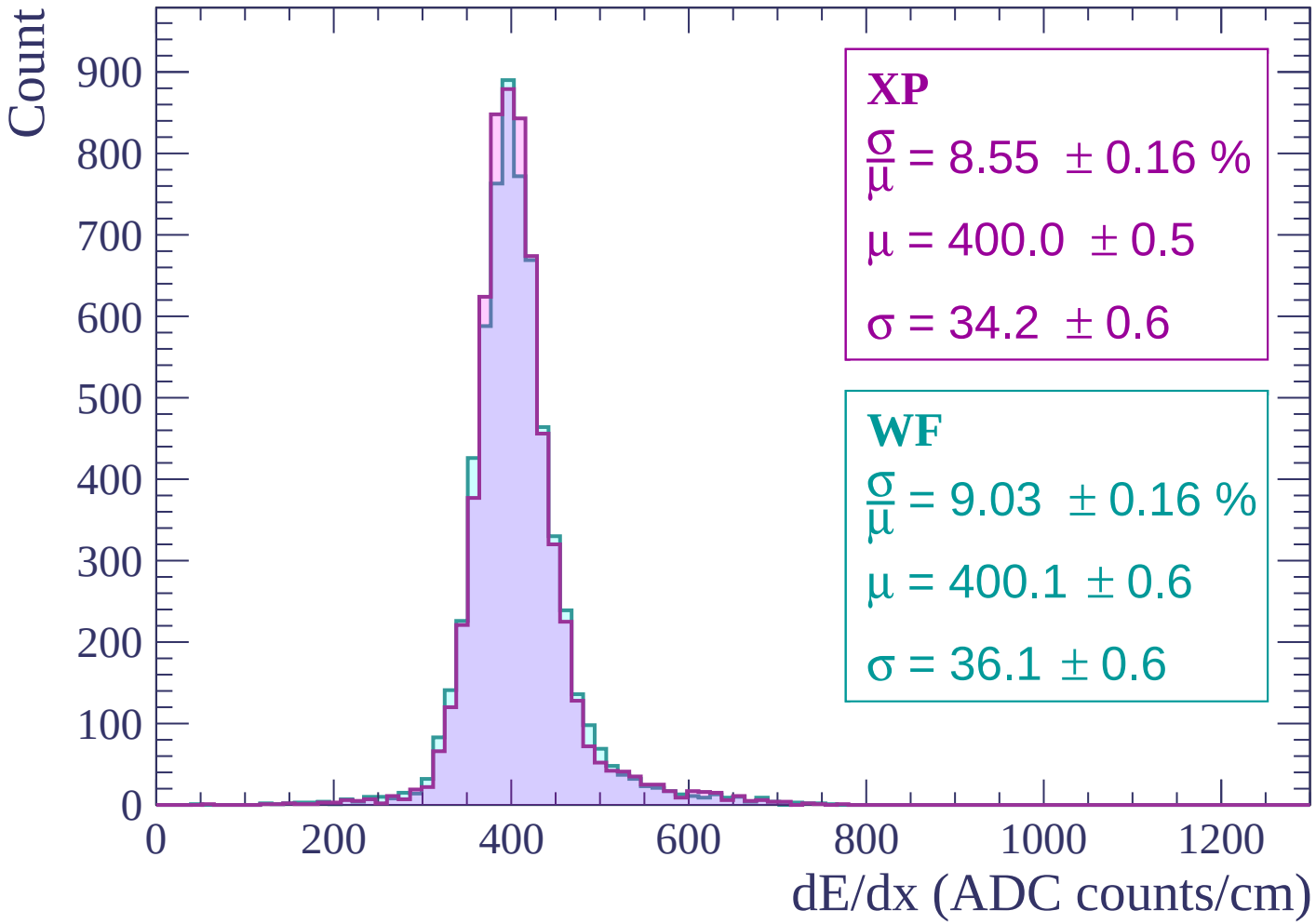
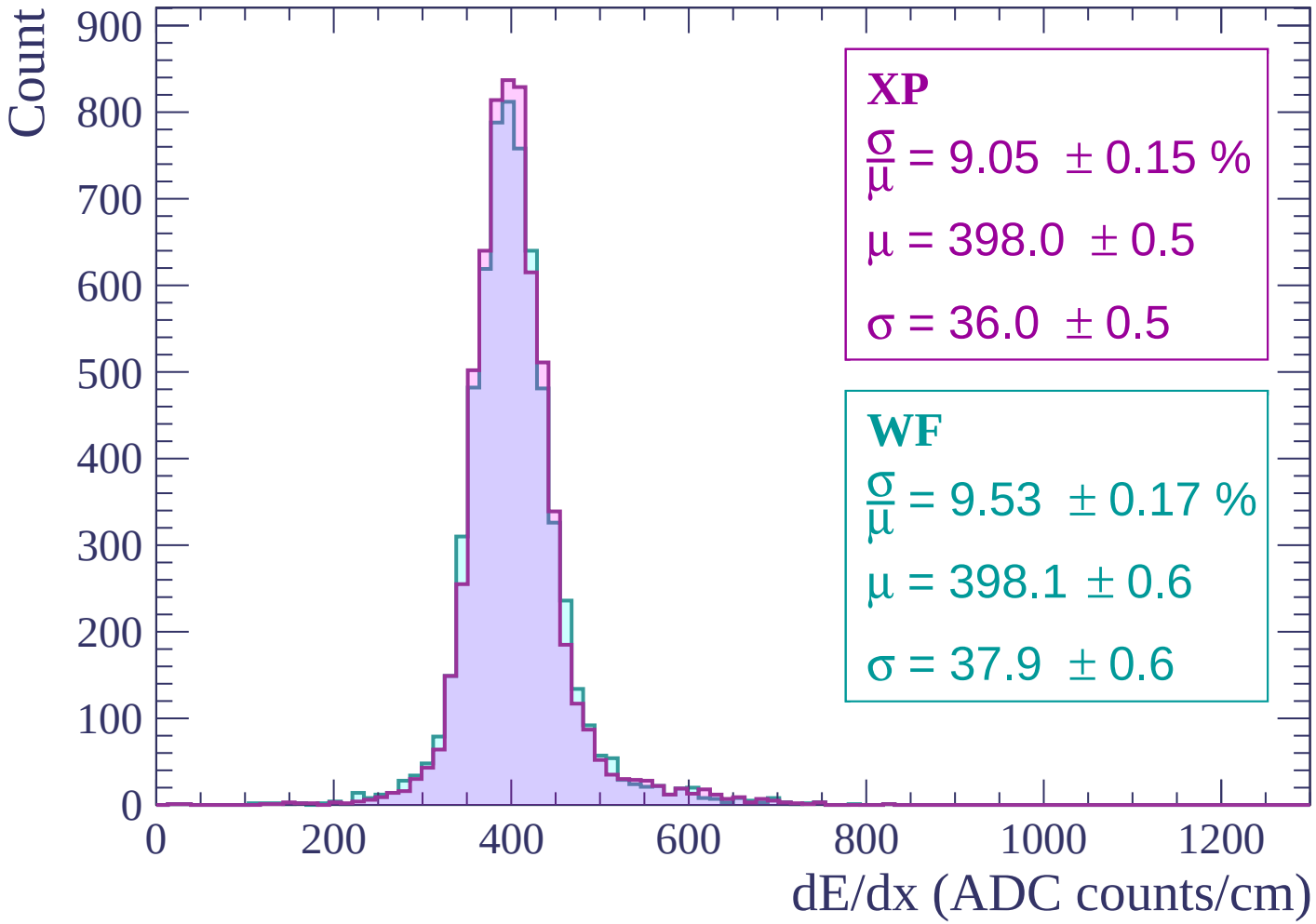
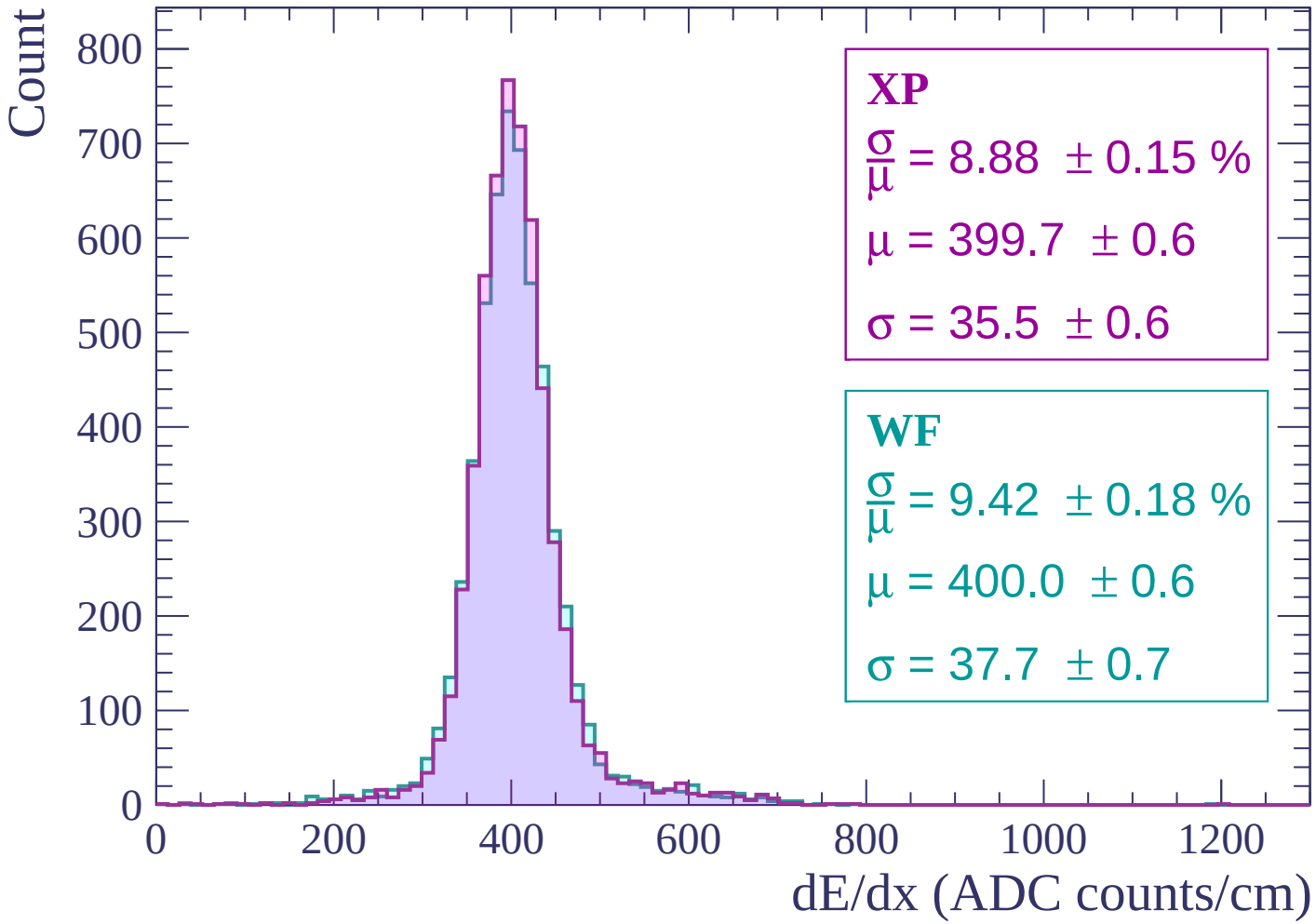
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.11 ± 0.23 %
μ = 408.5 ± 0.8
σ = 37.2 ± 0.9
WF
σ/μ = 9.57 ± 0.26 %
μ = 409.3 ± 0.9
σ = 39.2 ± 1.0

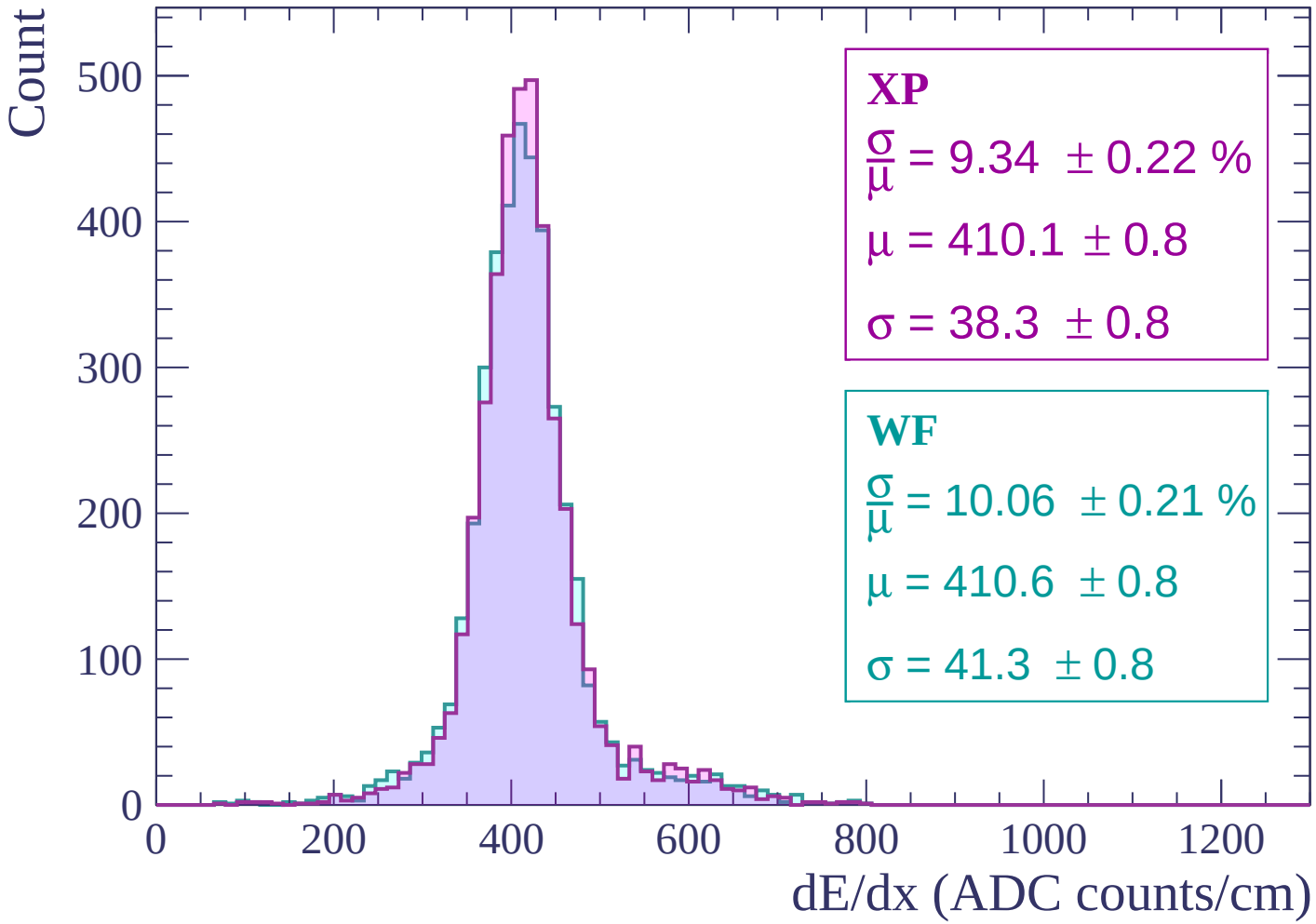

Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.92 ± 0.13 %
μ = 407.3 ± 0.5
σ = 36.3 ± 0.5
WF
σ/μ = 9.37 ± 0.15 %
μ = 408.3 ± 0.6
σ = 38.3 ± 0.6

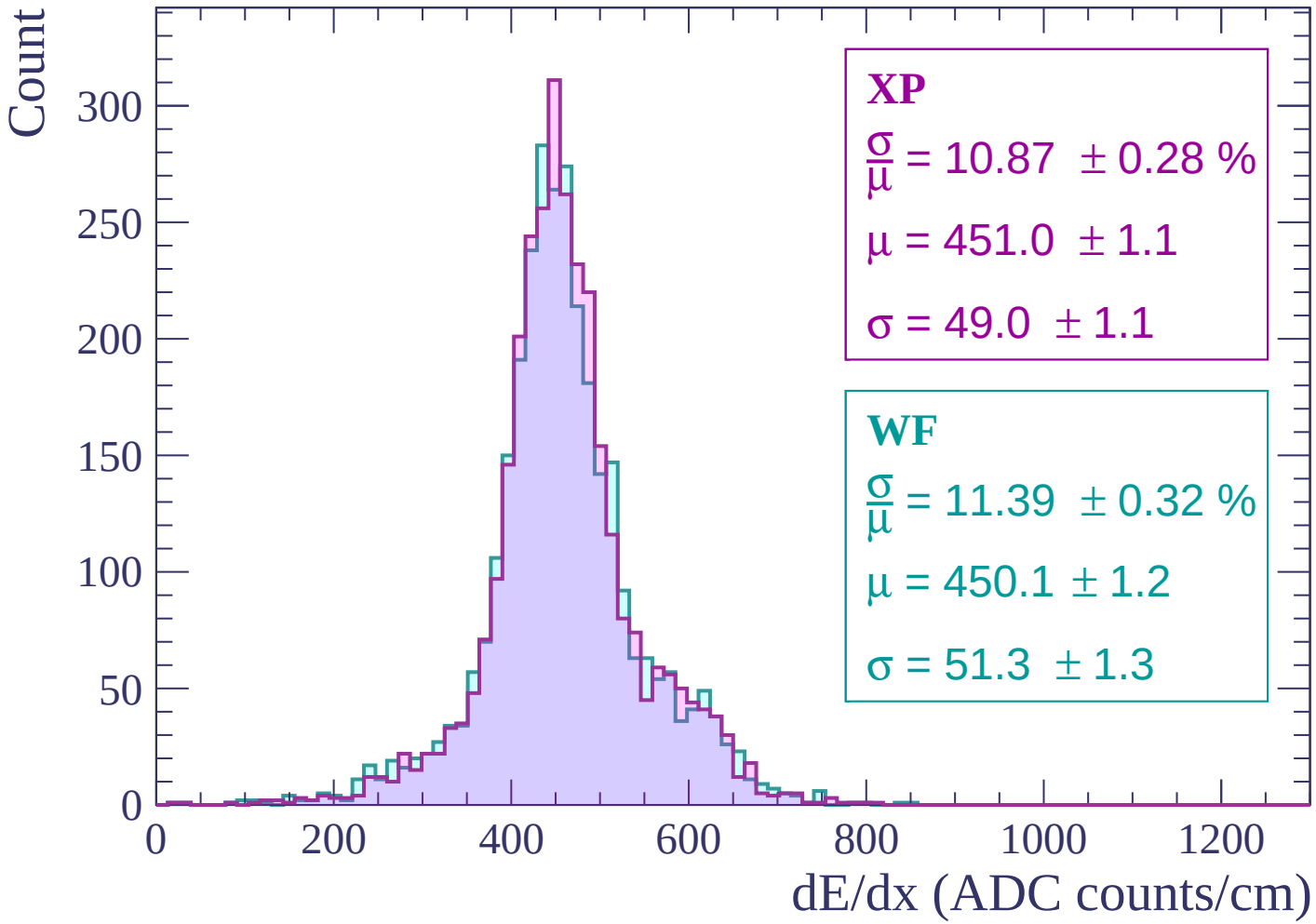

Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.90 ± 0.17 %
μ = 402.5 ± 0.6
σ = 35.8 ± 0.6
WF
σ/μ = 9.51 ± 0.16 %
μ = 403.6 ± 0.6
σ = 38.4 ± 0.6

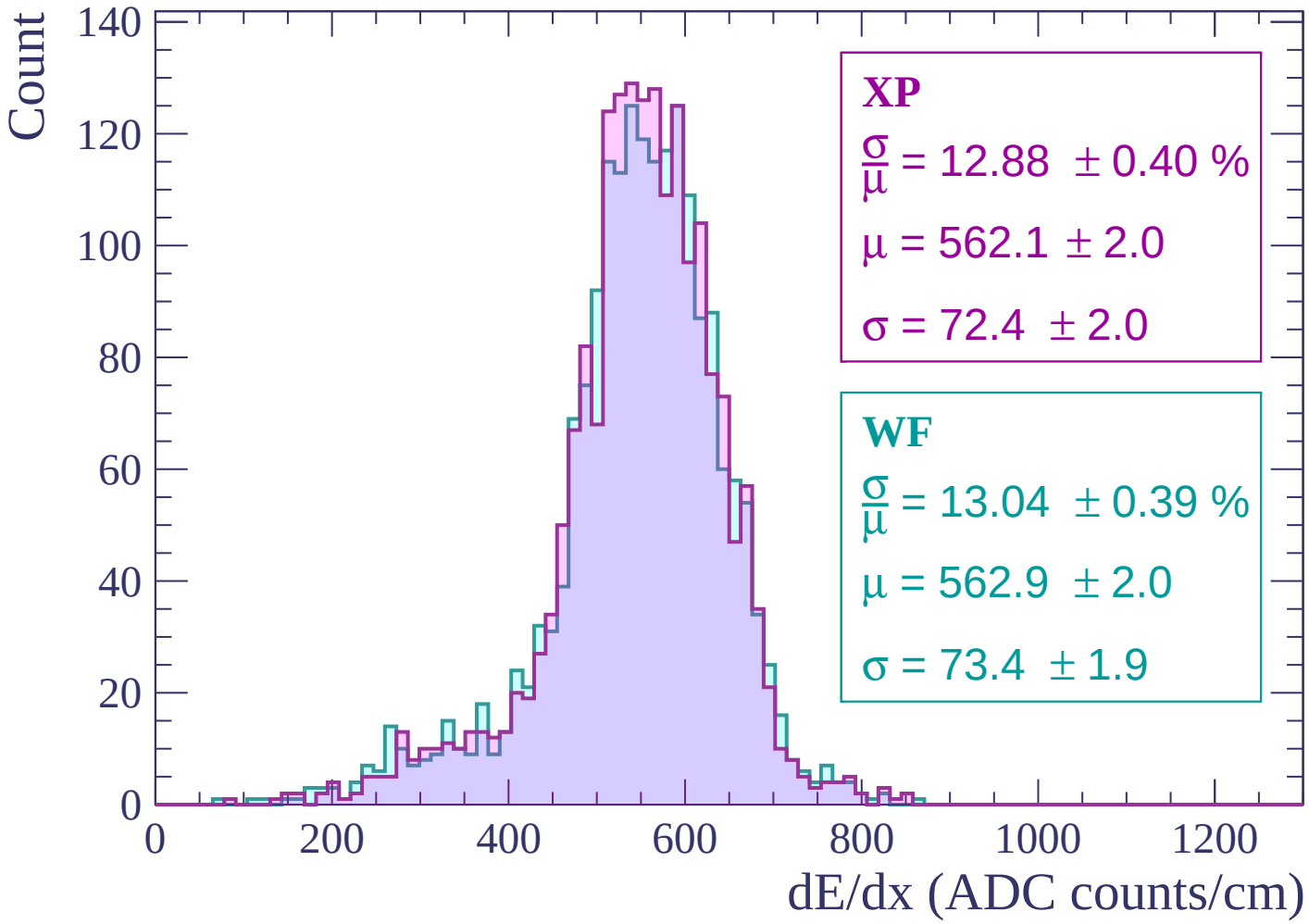

Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.55 ± 0.16 %
μ = 400.0 ± 0.5
σ = 34.2 ± 0.6
WF
σ/μ = 9.03 ± 0.16 %
μ = 400.1 ± 0.6
σ = 36.1 ± 0.6

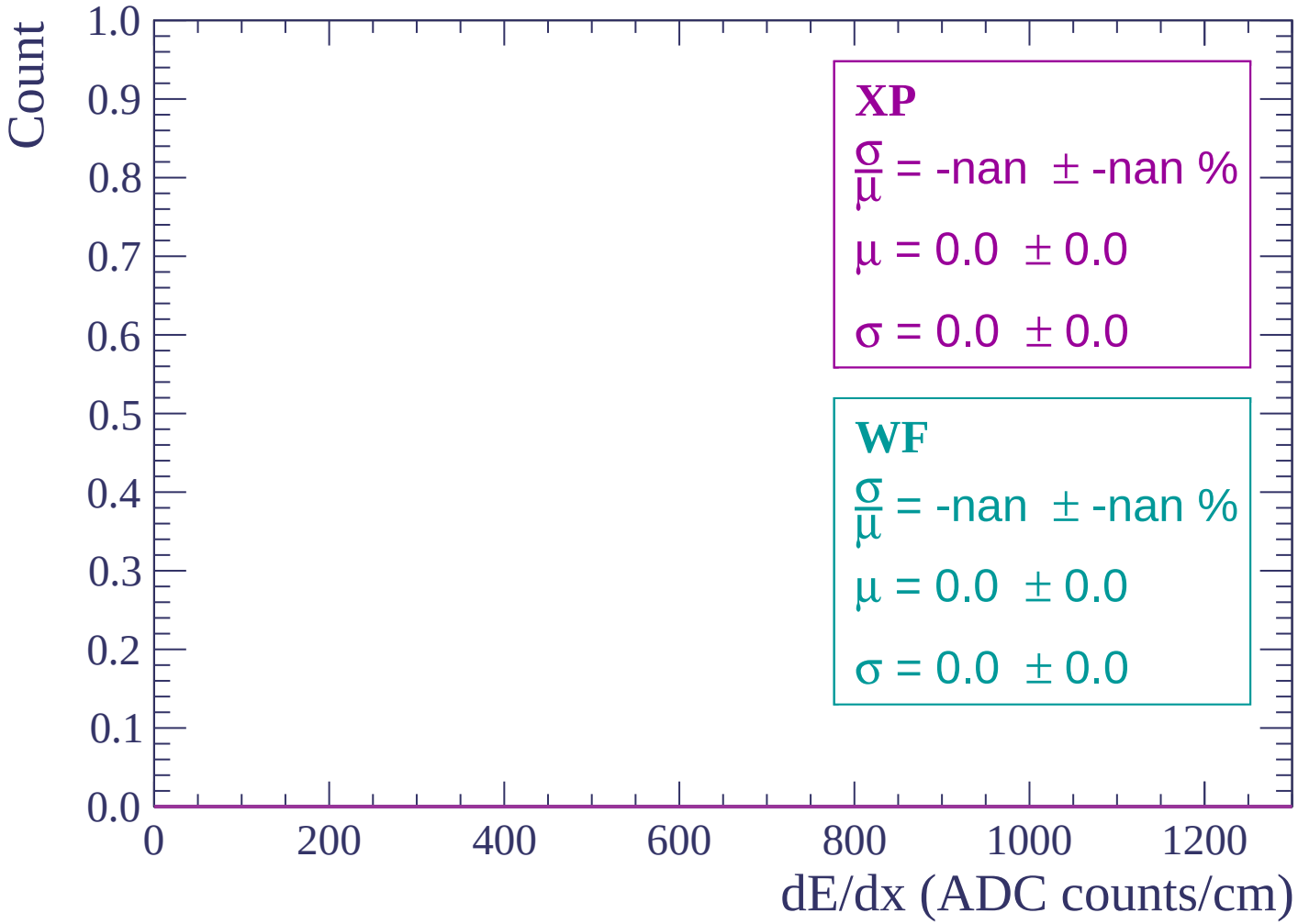

Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.05 ± 0.15 %
μ = 398.0 ± 0.5
σ = 36.0 ± 0.5
WF
σ/μ = 9.53 ± 0.17 %
μ = 398.1 ± 0.6
σ = 37.9 ± 0.6

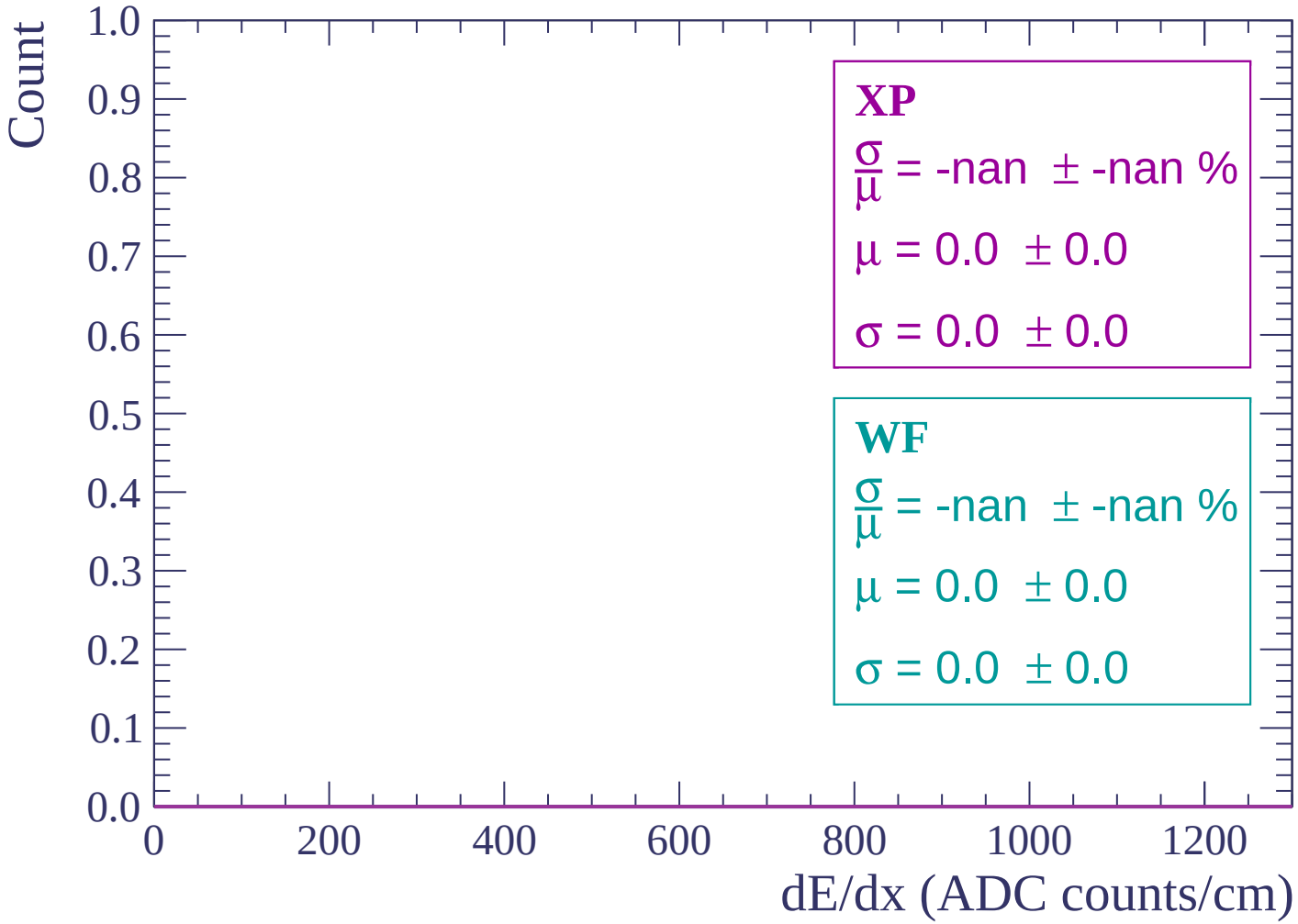

Count
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.88 ± 0.15 %
μ = 399.7 ± 0.6
σ = 35.5 ± 0.6
WF
σ/μ = 9.42 ± 0.18 %
μ = 400.0 ± 0.6
σ = 37.7 ± 0.7

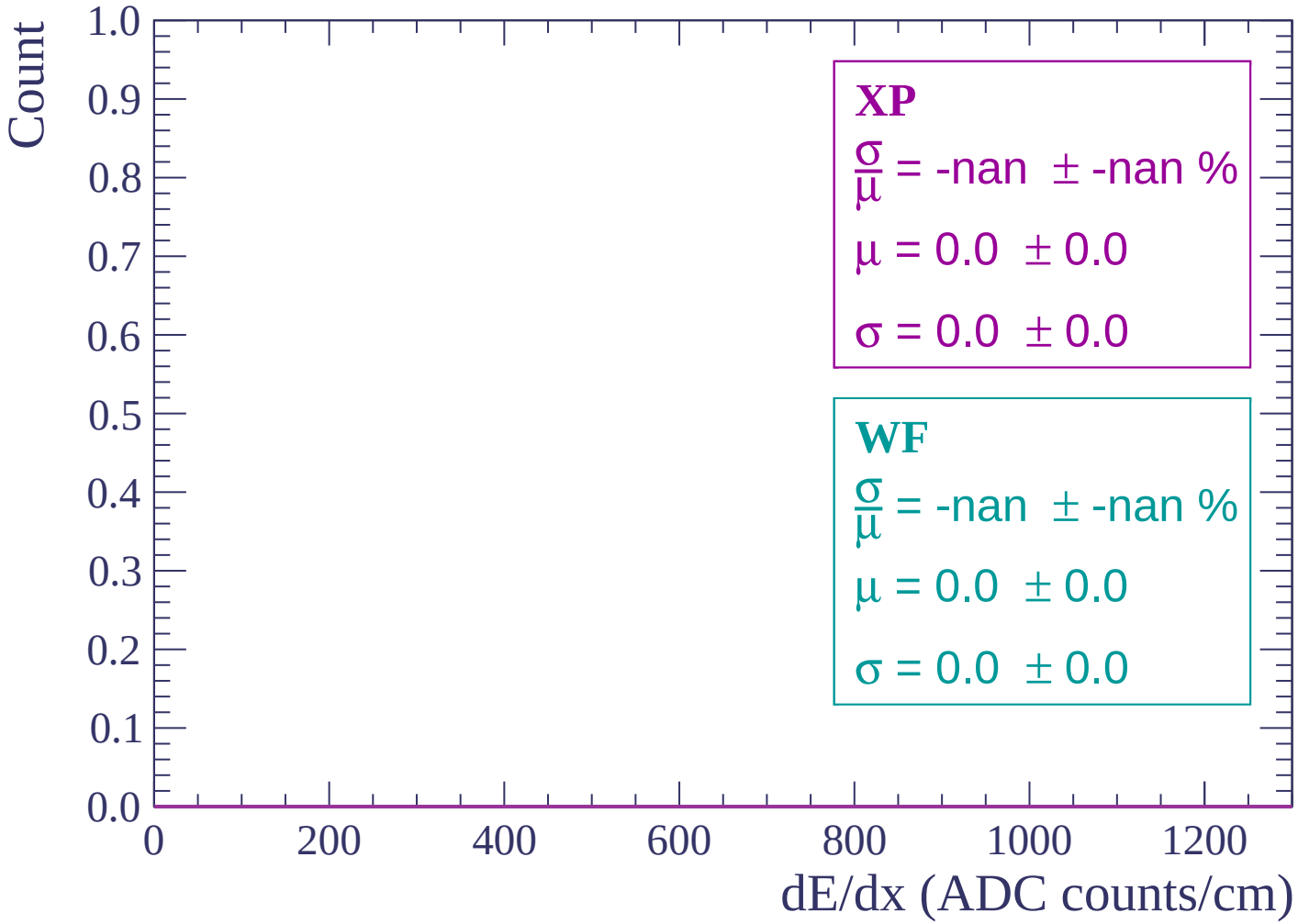

Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.34 ± 0.22 %
μ = 410.1 ± 0.8
σ = 38.3 ± 0.8
WF
σ/μ = 10.06 ± 0.21 %
μ = 410.6 ± 0.8
σ = 41.3 ± 0.8

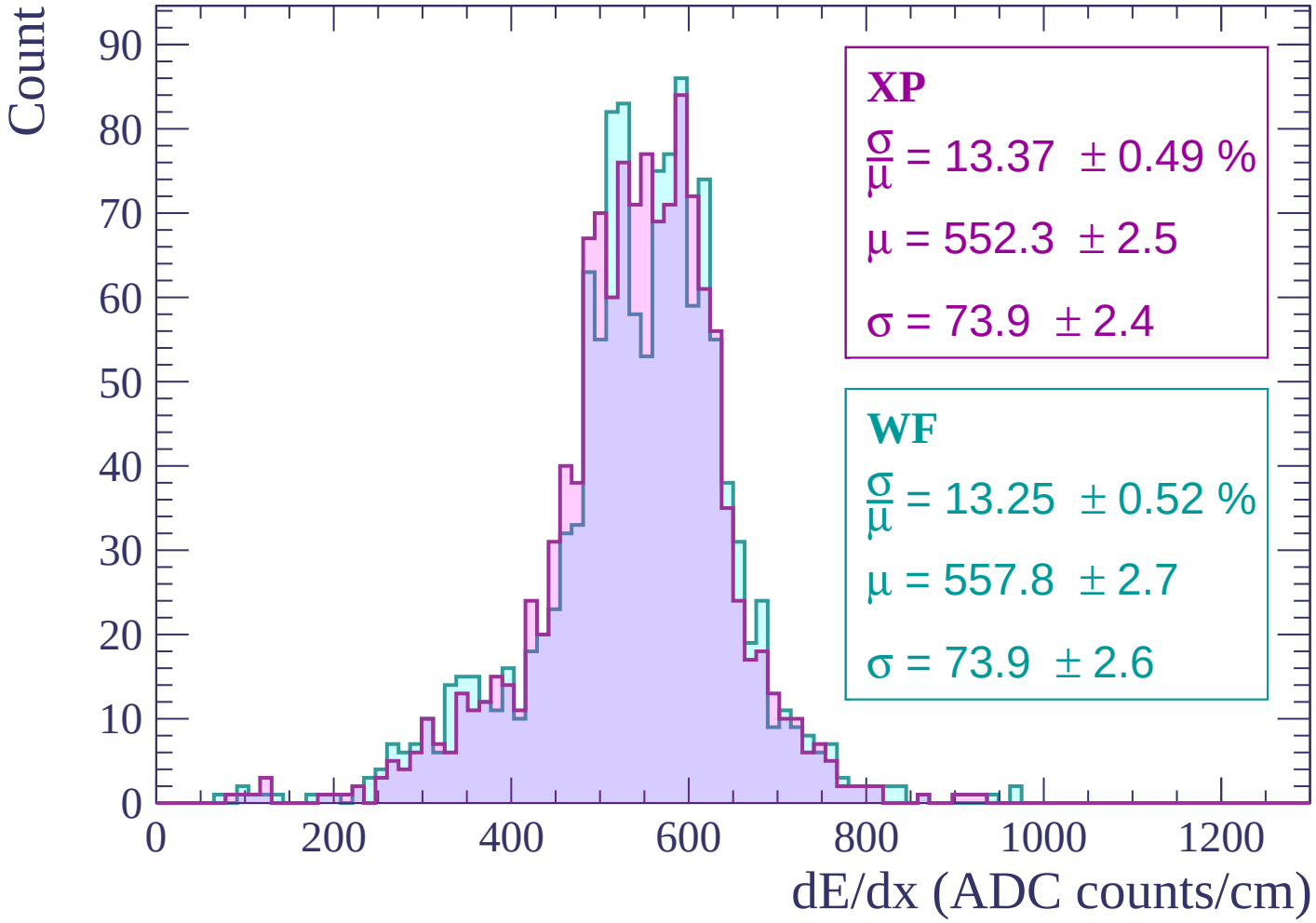

Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.87 ± 0.28 %
μ = 451.0 ± 1.1
σ = 49.0 ± 1.1
WF
σ/μ = 11.39 ± 0.32 %
μ = 450.1 ± 1.2
σ = 51.3 ± 1.3

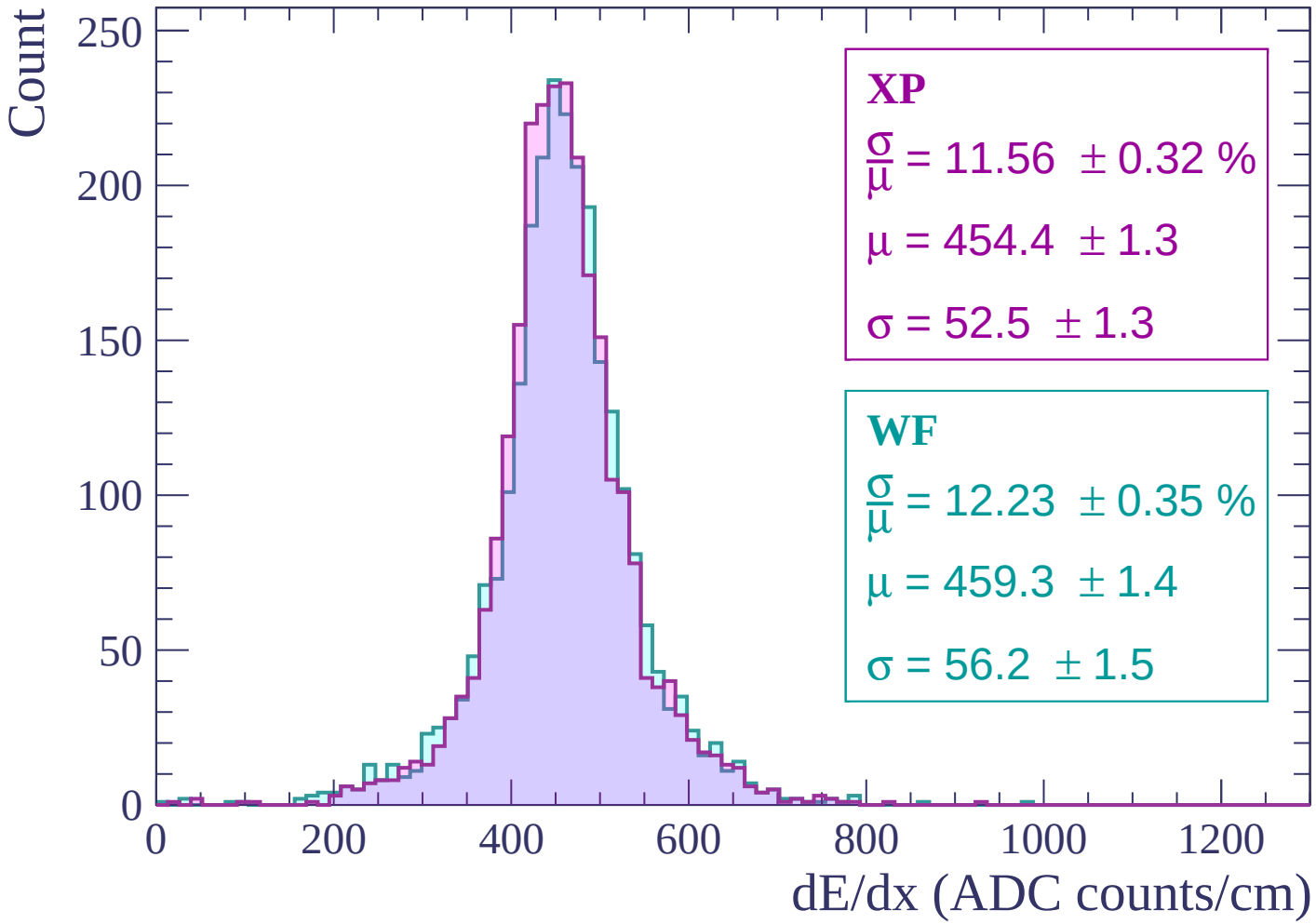

Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.88 ± 0.40 %
μ = 562.1 ± 2.0
σ = 72.4 ± 2.0
WF
σ/μ = 13.04 ± 0.39 %
μ = 562.9 ± 2.0
σ = 73.4 ± 1.9

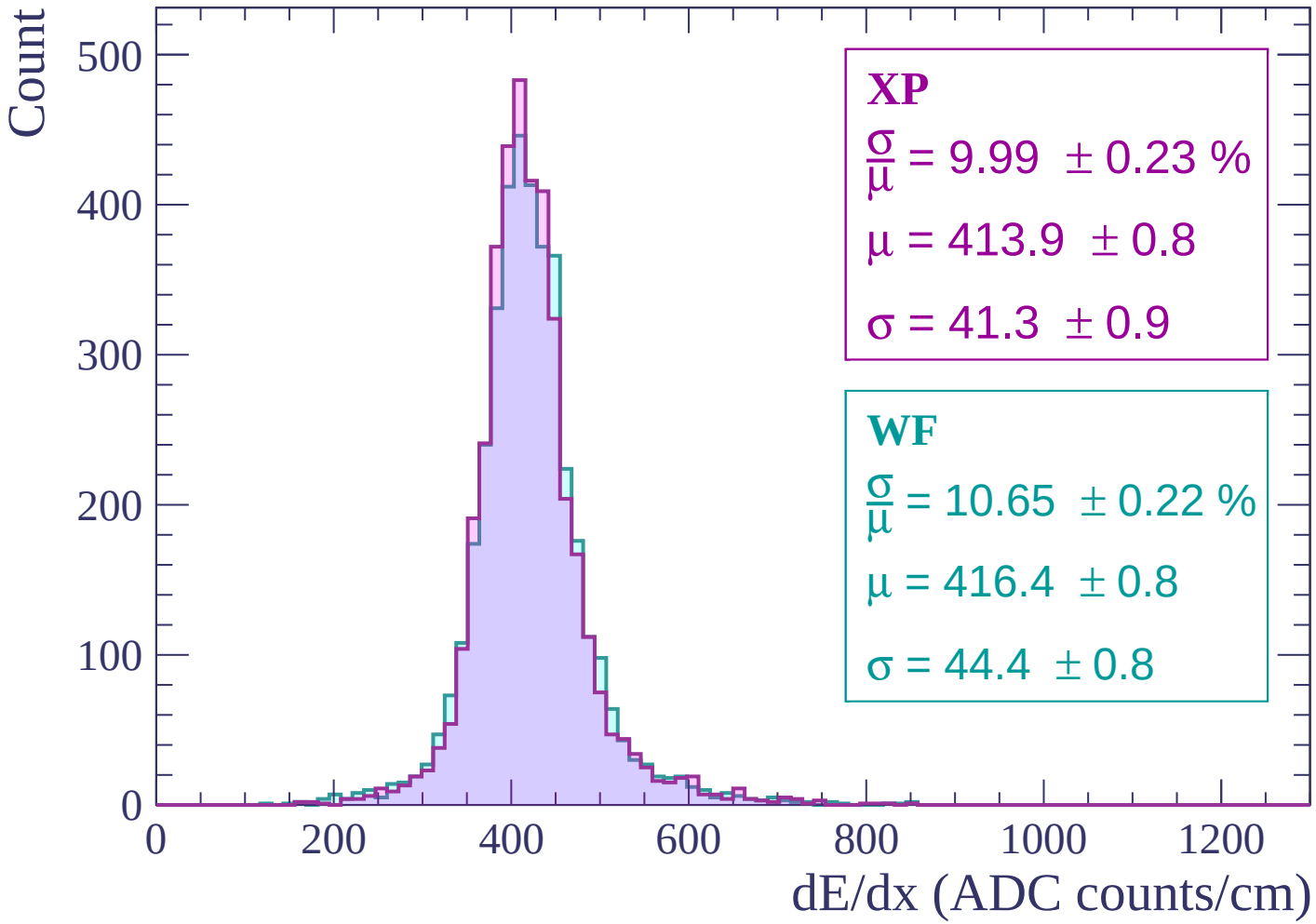

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

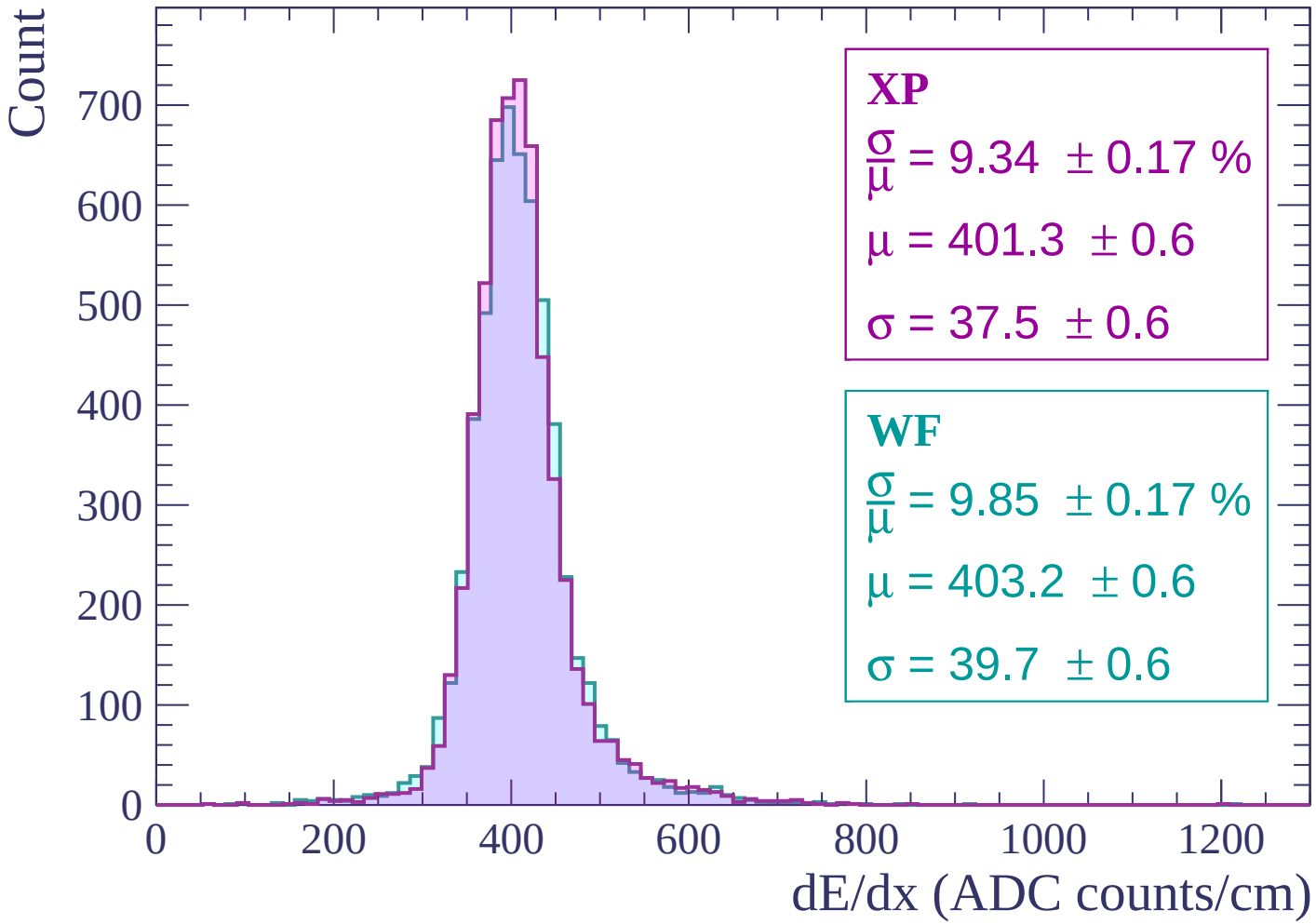

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

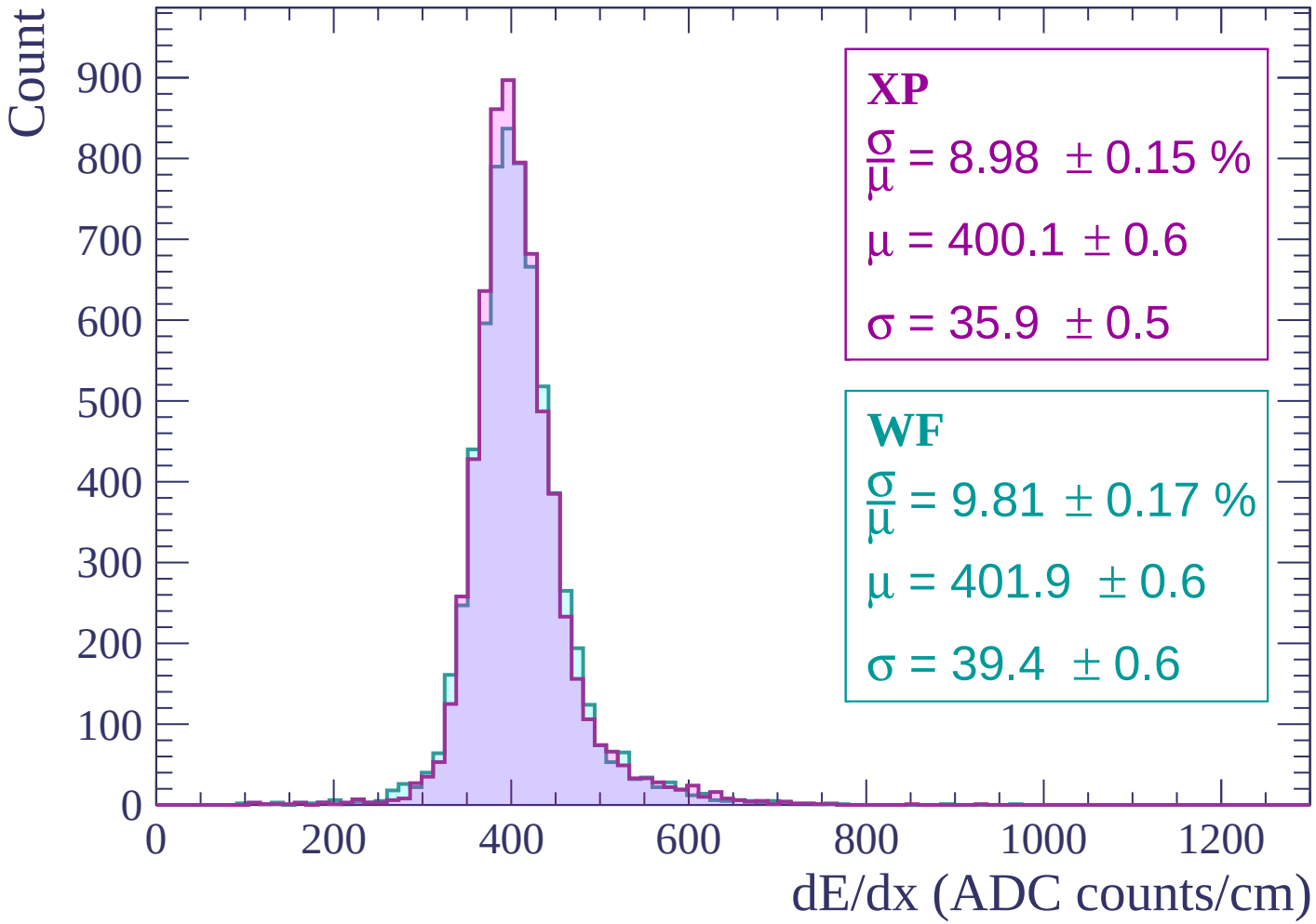

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

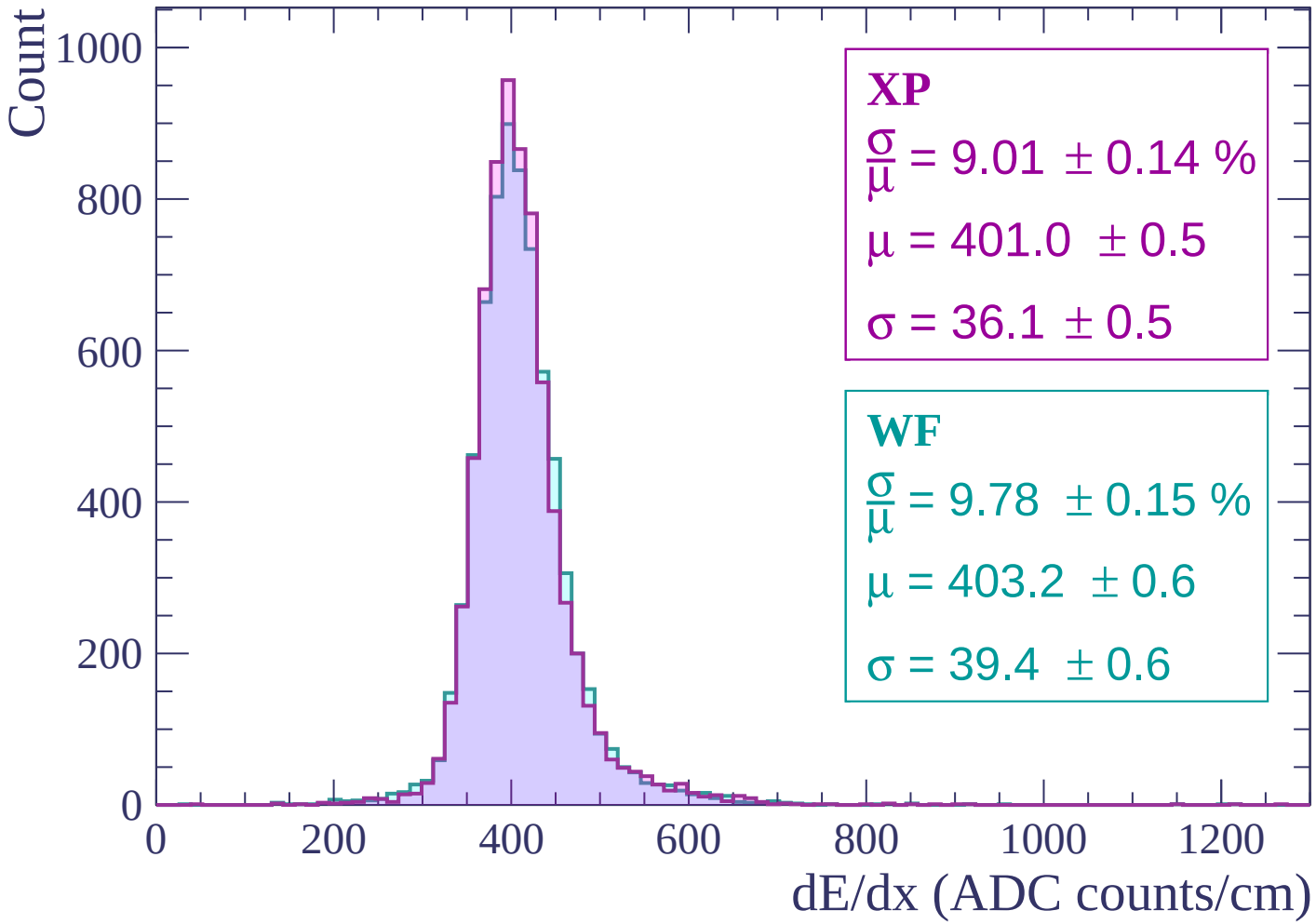

Count
90
80
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.37 ± 0.49 %
μ = 552.3 ± 2.5
σ = 73.9 ± 2.4
WF
σ/μ = 13.25 ± 0.52 %
μ = 557.8 ± 2.7
σ = 73.9 ± 2.6

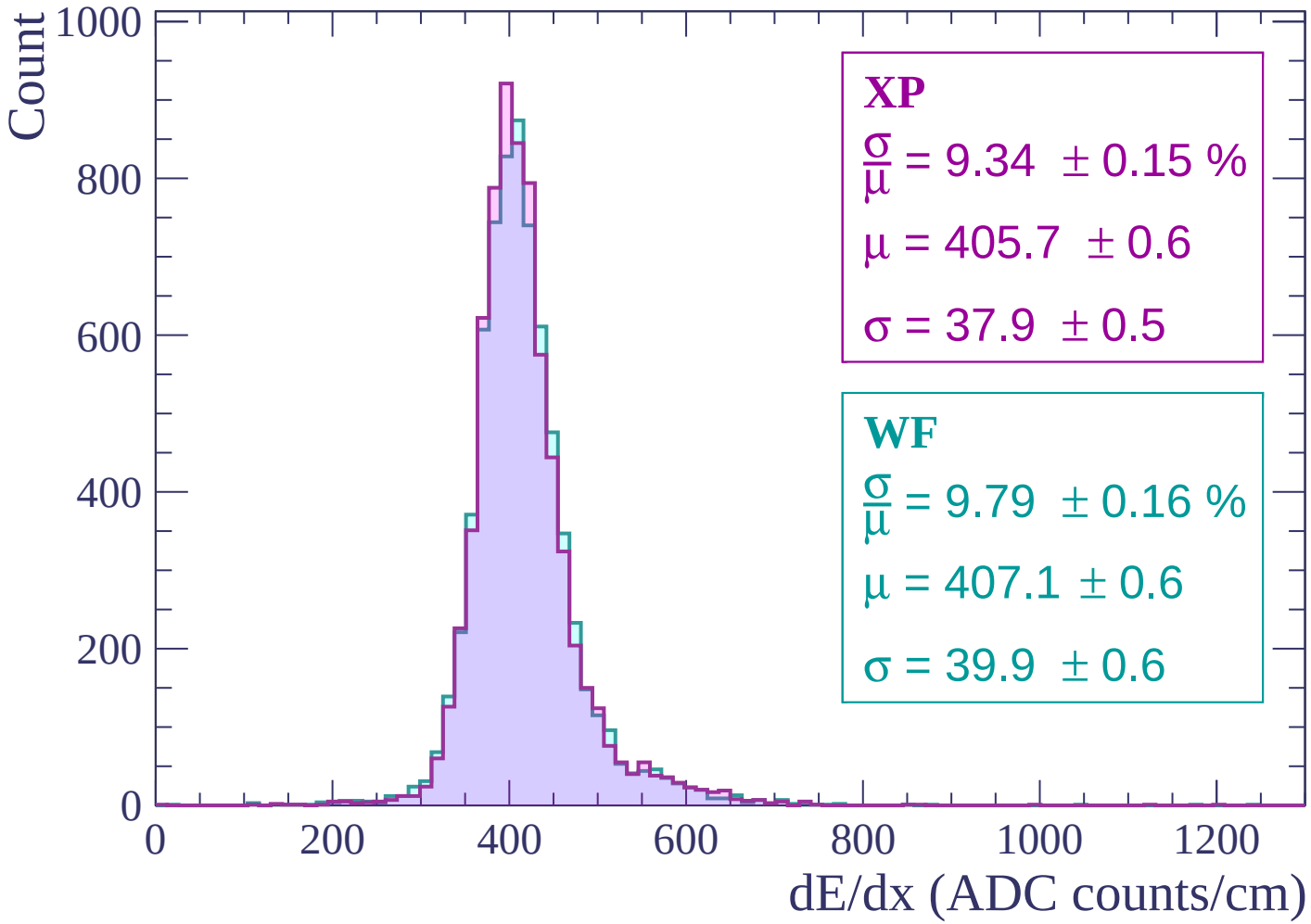

Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.56 ± 0.32 %
μ = 454.4 ± 1.3
σ = 52.5 ± 1.3
WF
σ/μ = 12.23 ± 0.35 %
μ = 459.3 ± 1.4
σ = 56.2 ± 1.5

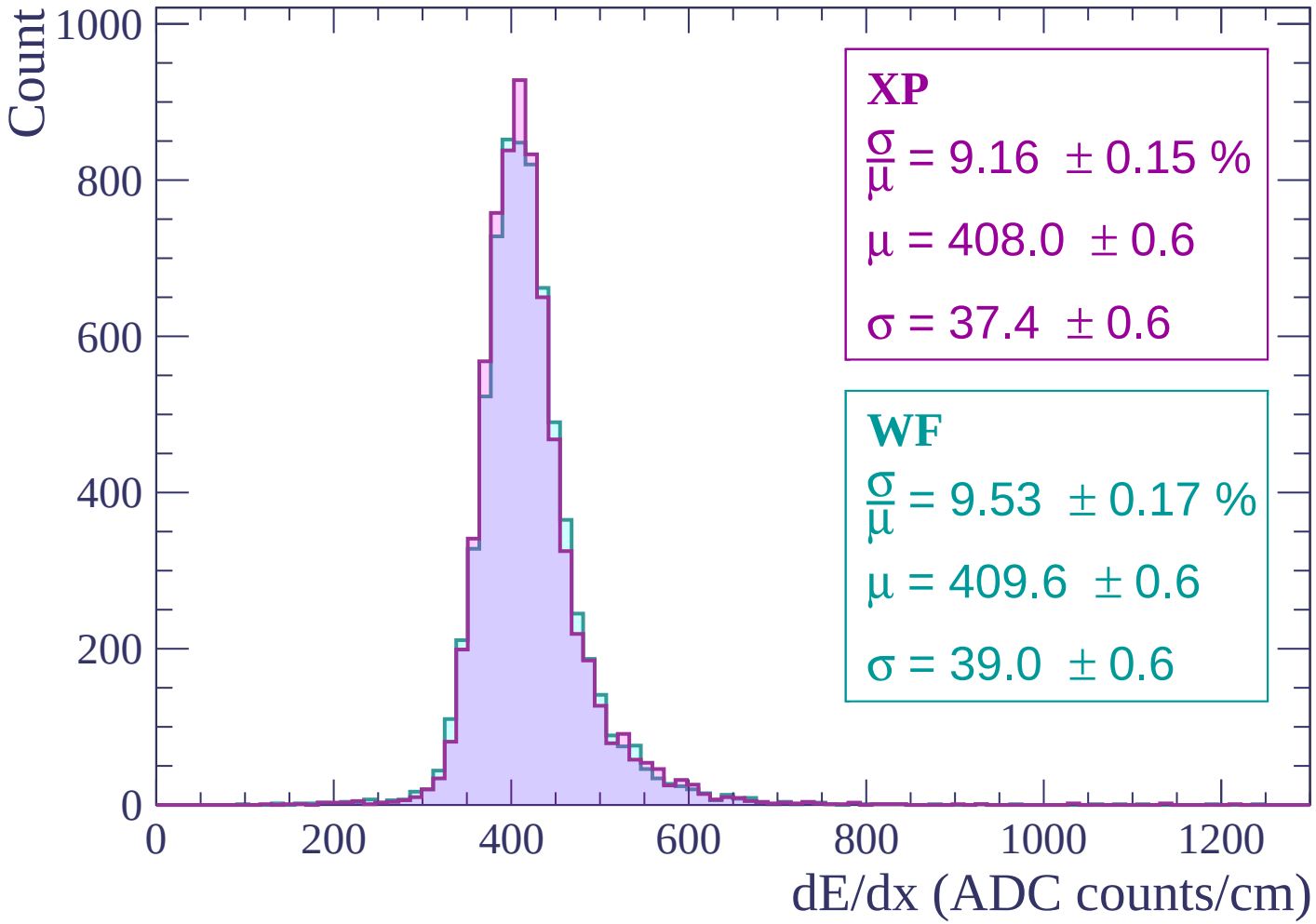

Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.99 ± 0.23 %
μ = 413.9 ± 0.8
σ = 41.3 ± 0.9
WF
σ/μ = 10.65 ± 0.22 %
μ = 416.4 ± 0.8
σ = 44.4 ± 0.8

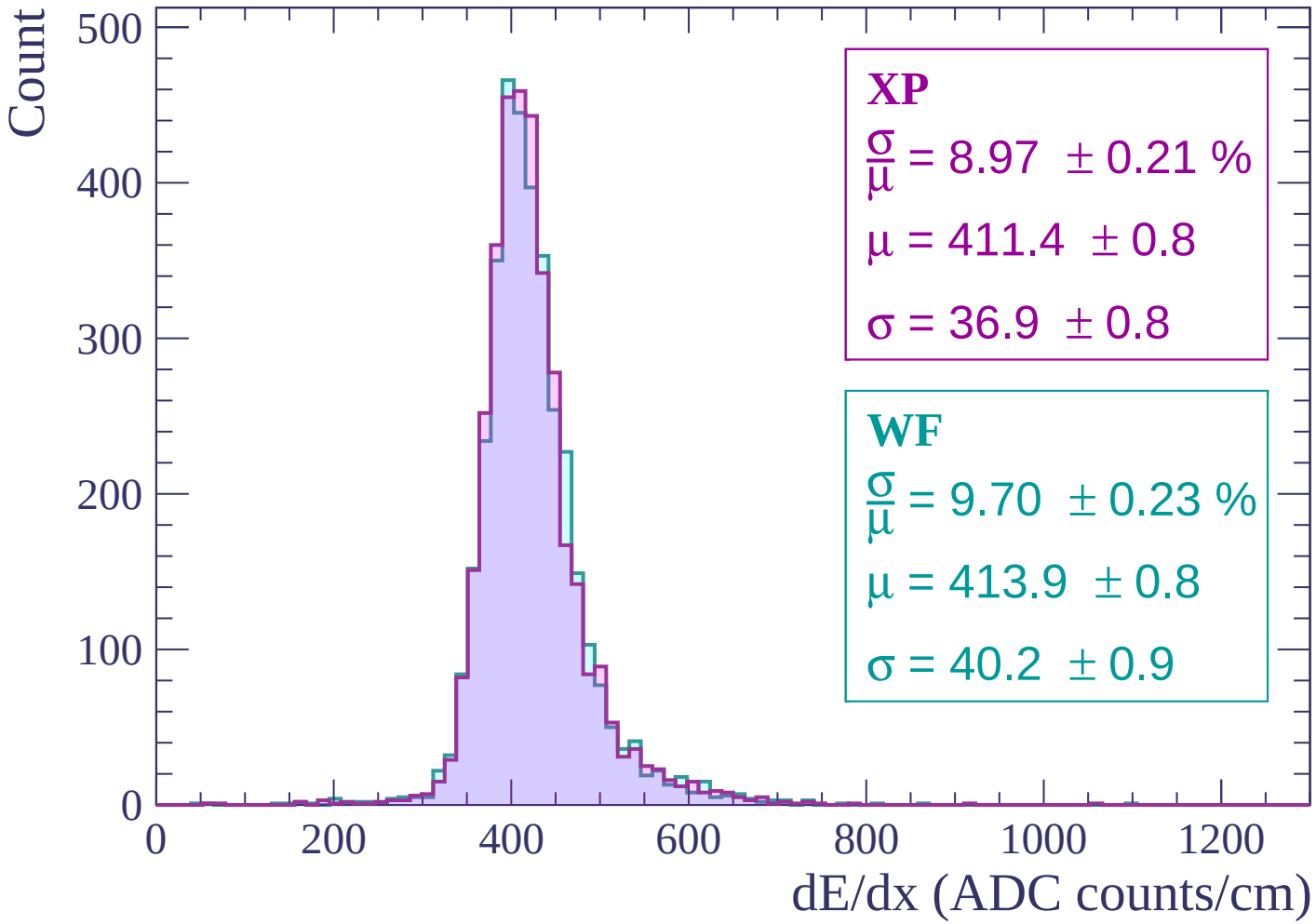

Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.34 ± 0.17 %
μ = 401.3 ± 0.6
σ = 37.5 ± 0.6
WF
σ/μ = 9.85 ± 0.17 %
μ = 403.2 ± 0.6
σ = 39.7 ± 0.6

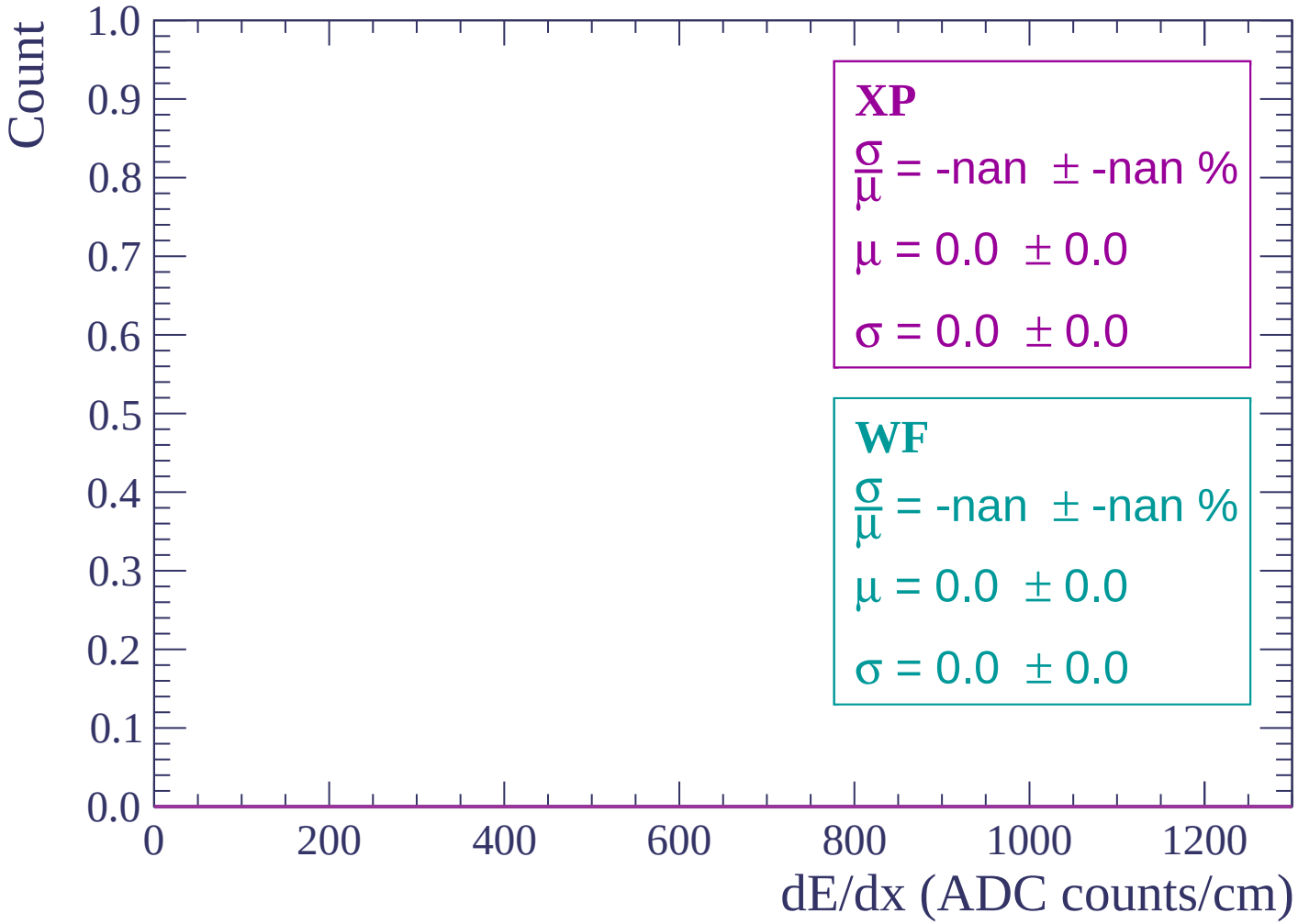

Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.98 ± 0.15 %
μ = 400.1 ± 0.6
σ = 35.9 ± 0.5
WF
σ/μ = 9.81 ± 0.17 %
μ = 401.9 ± 0.6
σ = 39.4 ± 0.6

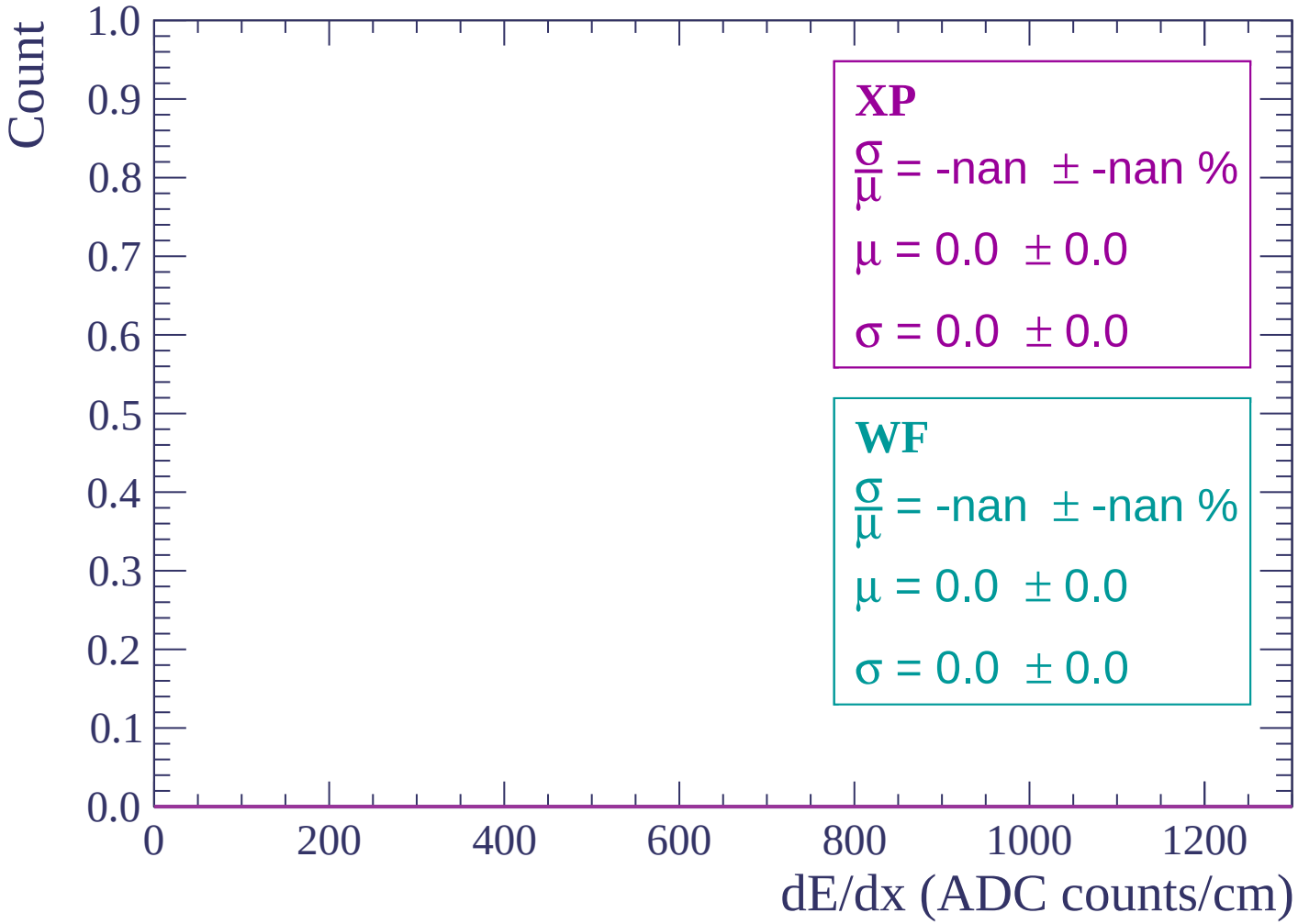

Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.01 ± 0.14 %
μ = 401.0 ± 0.5
σ = 36.1 ± 0.5
WF
σ/μ = 9.78 ± 0.15 %
μ = 403.2 ± 0.6
σ = 39.4 ± 0.6

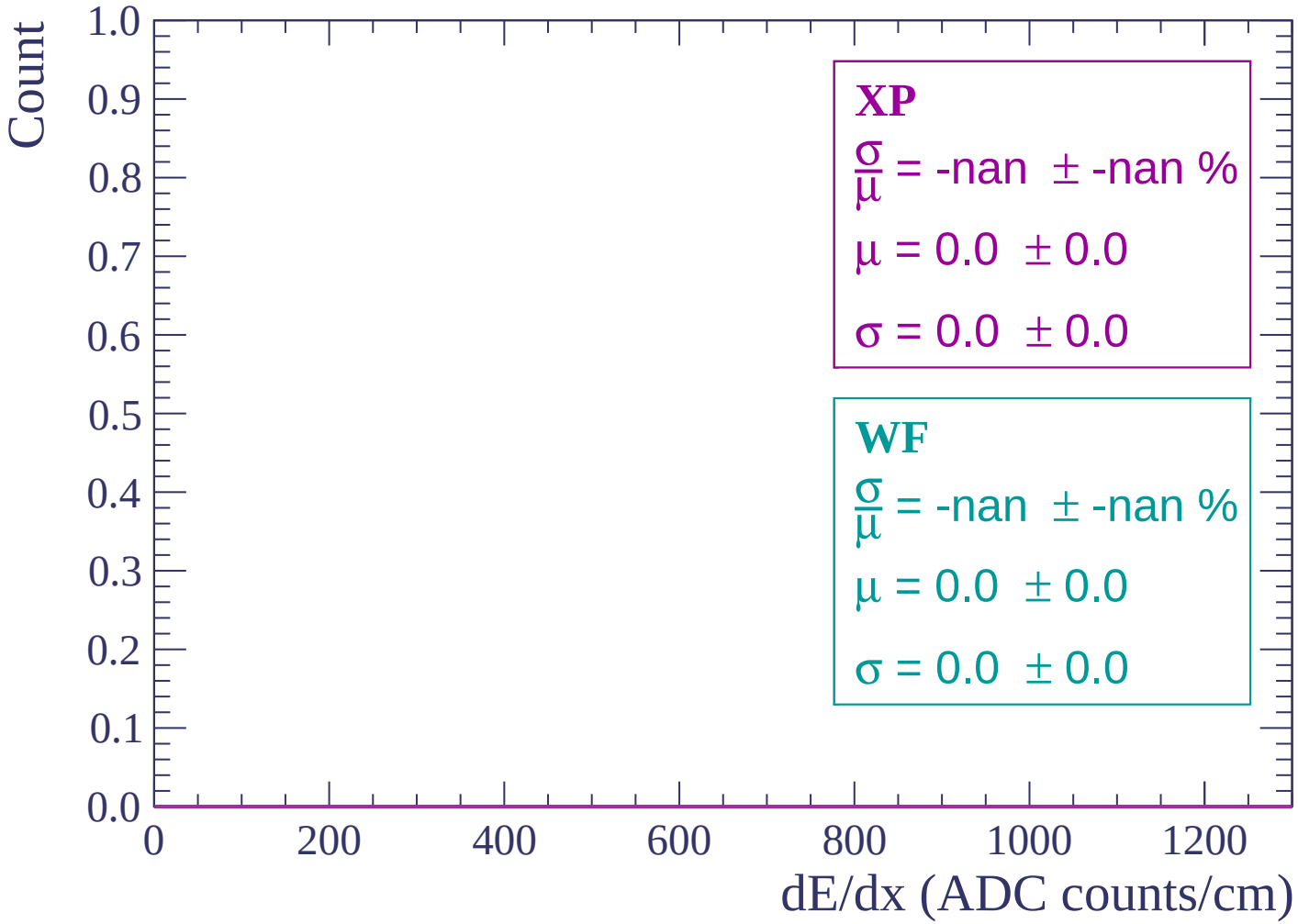

Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.34 ± 0.15 %
μ = 405.7 ± 0.6
σ = 37.9 ± 0.5
WF
σ/μ = 9.79 ± 0.16 %
μ = 407.1 ± 0.6
σ = 39.9 ± 0.6

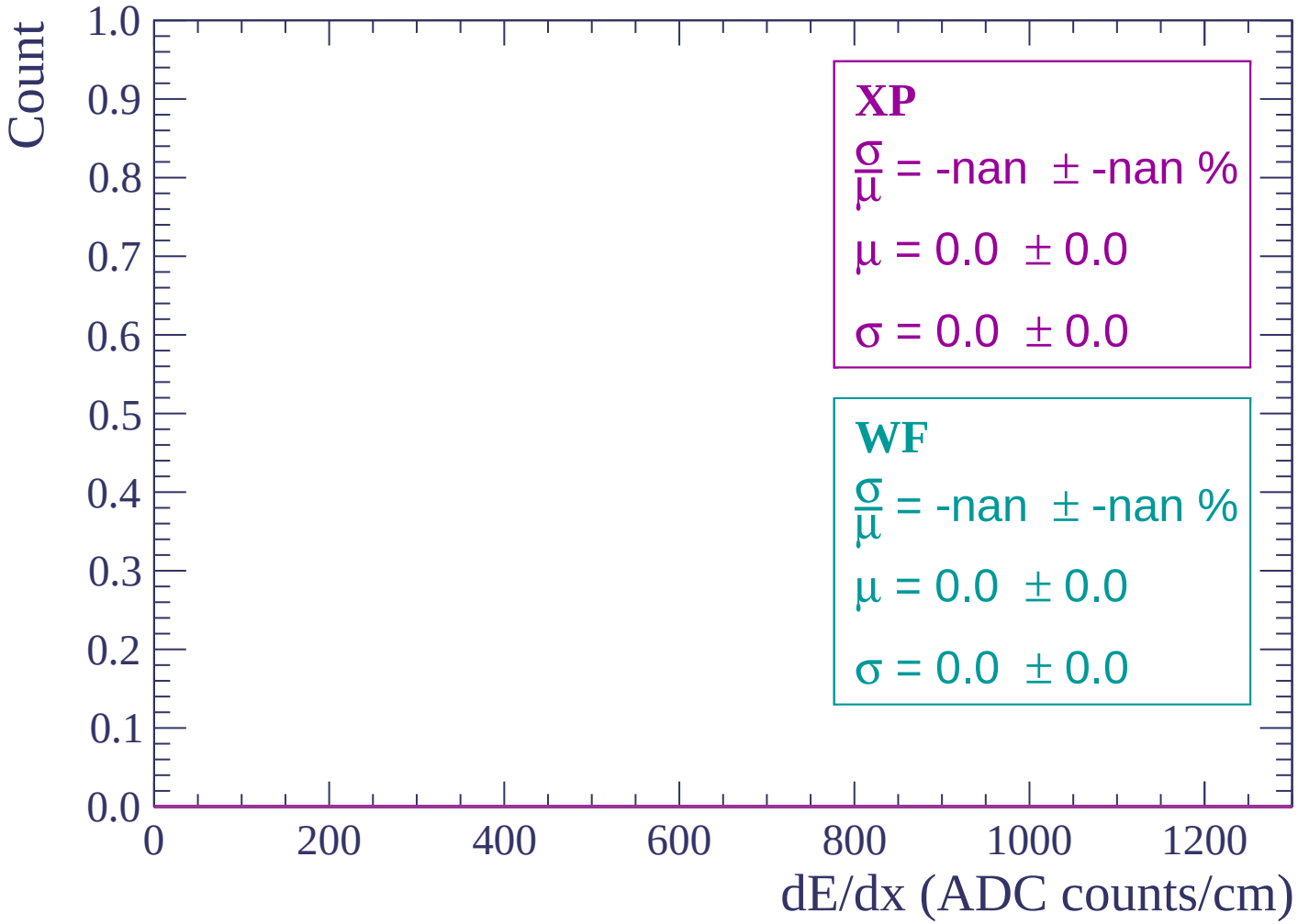

Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.16 ± 0.15 %
μ = 408.0 ± 0.6
σ = 37.4 ± 0.6
WF
σ/μ = 9.53 ± 0.17 %
μ = 409.6 ± 0.6
σ = 39.0 ± 0.6

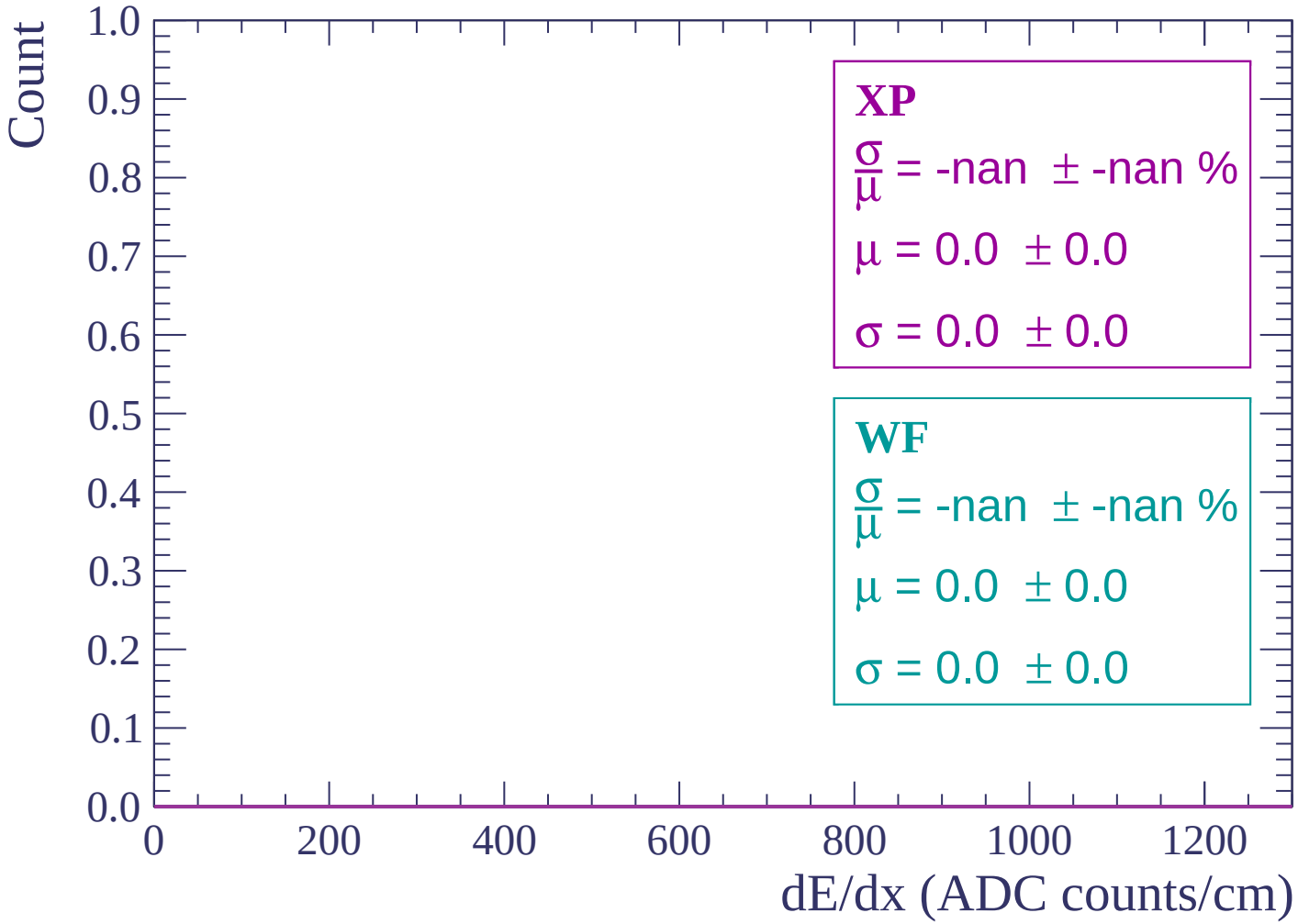

Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.97 ± 0.21 %
μ = 411.4 ± 0.8
σ = 36.9 ± 0.8
WF
σ/μ = 9.70 ± 0.23 %
μ = 413.9 ± 0.8
σ = 40.2 ± 0.9

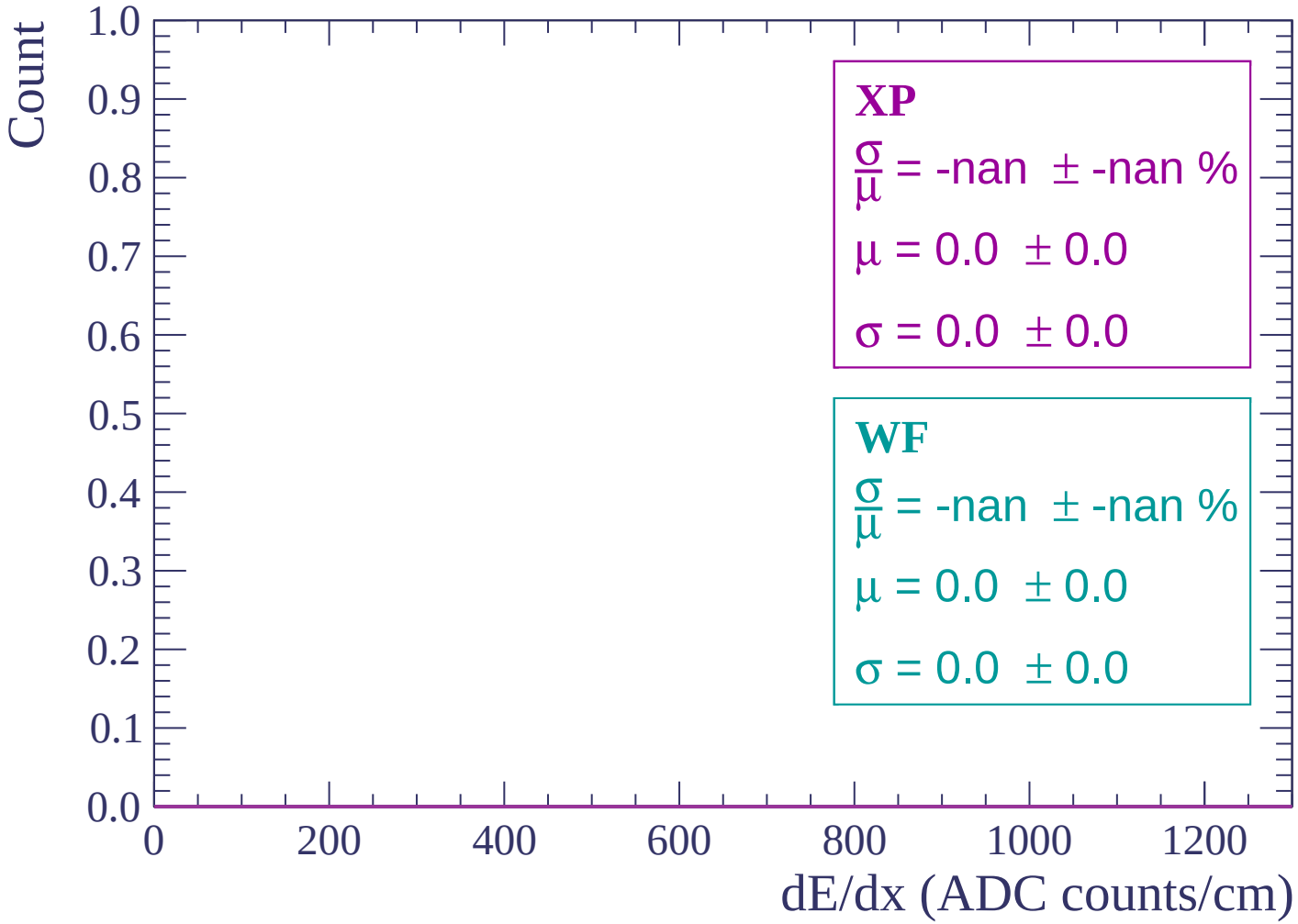

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

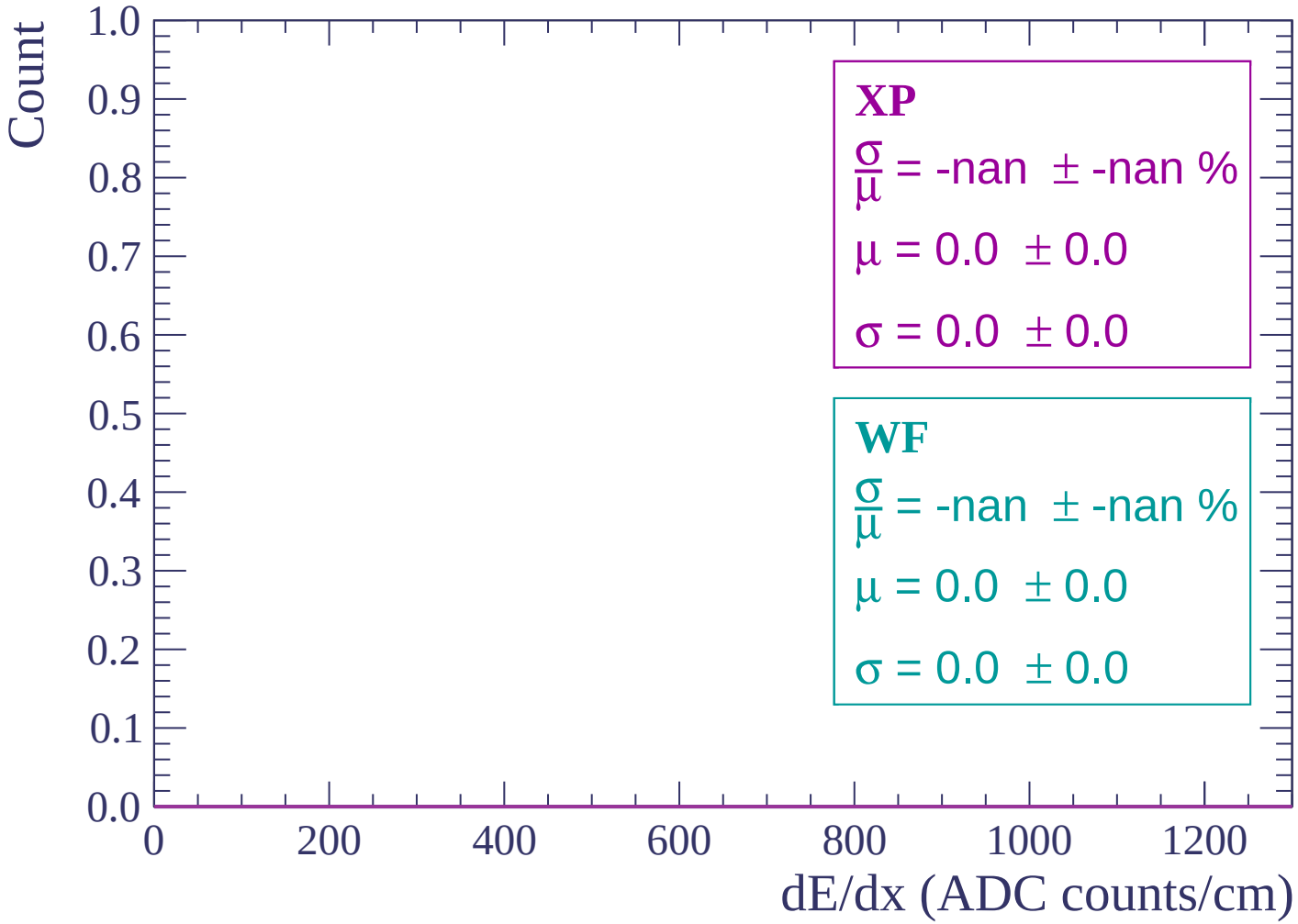

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

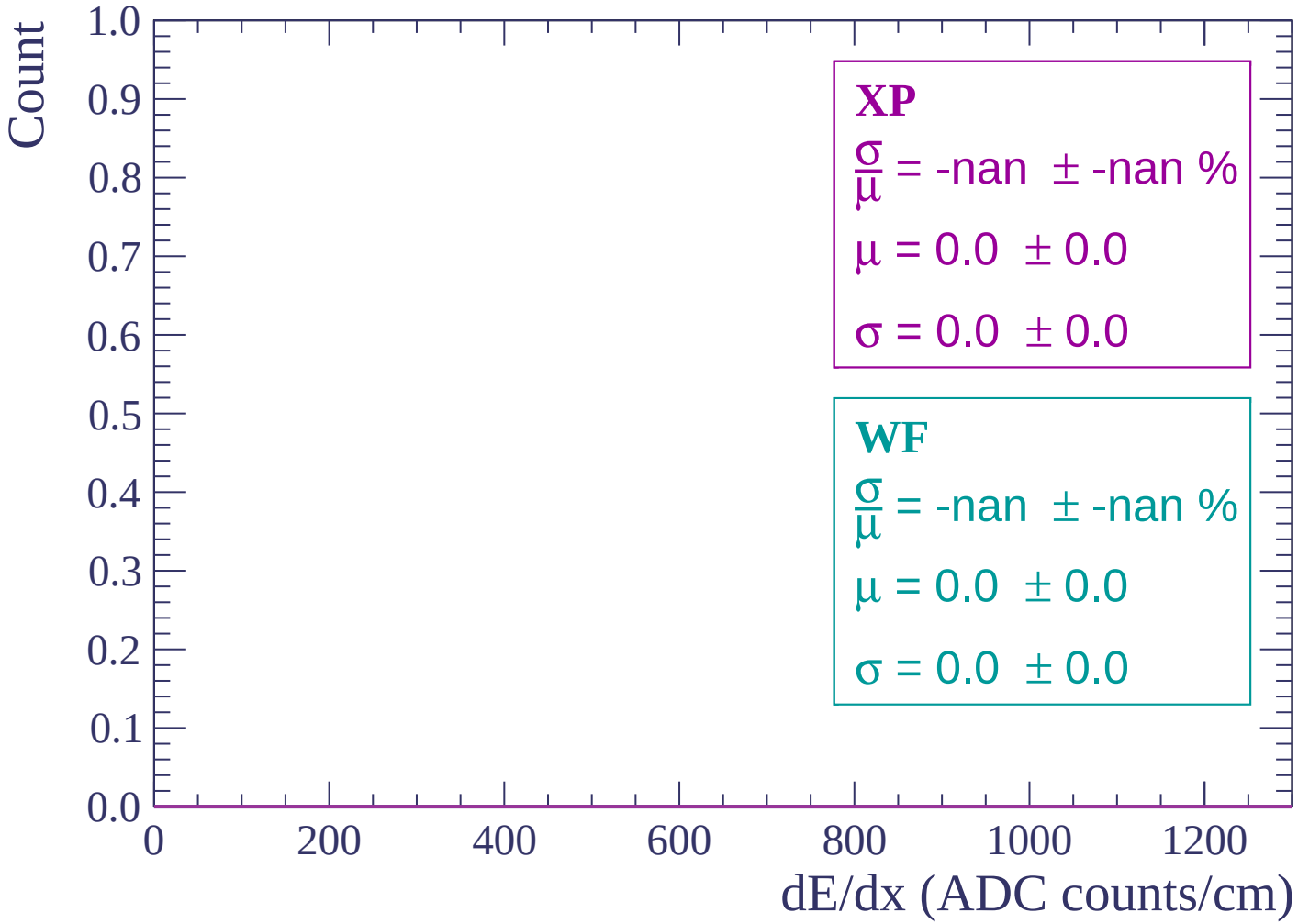

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

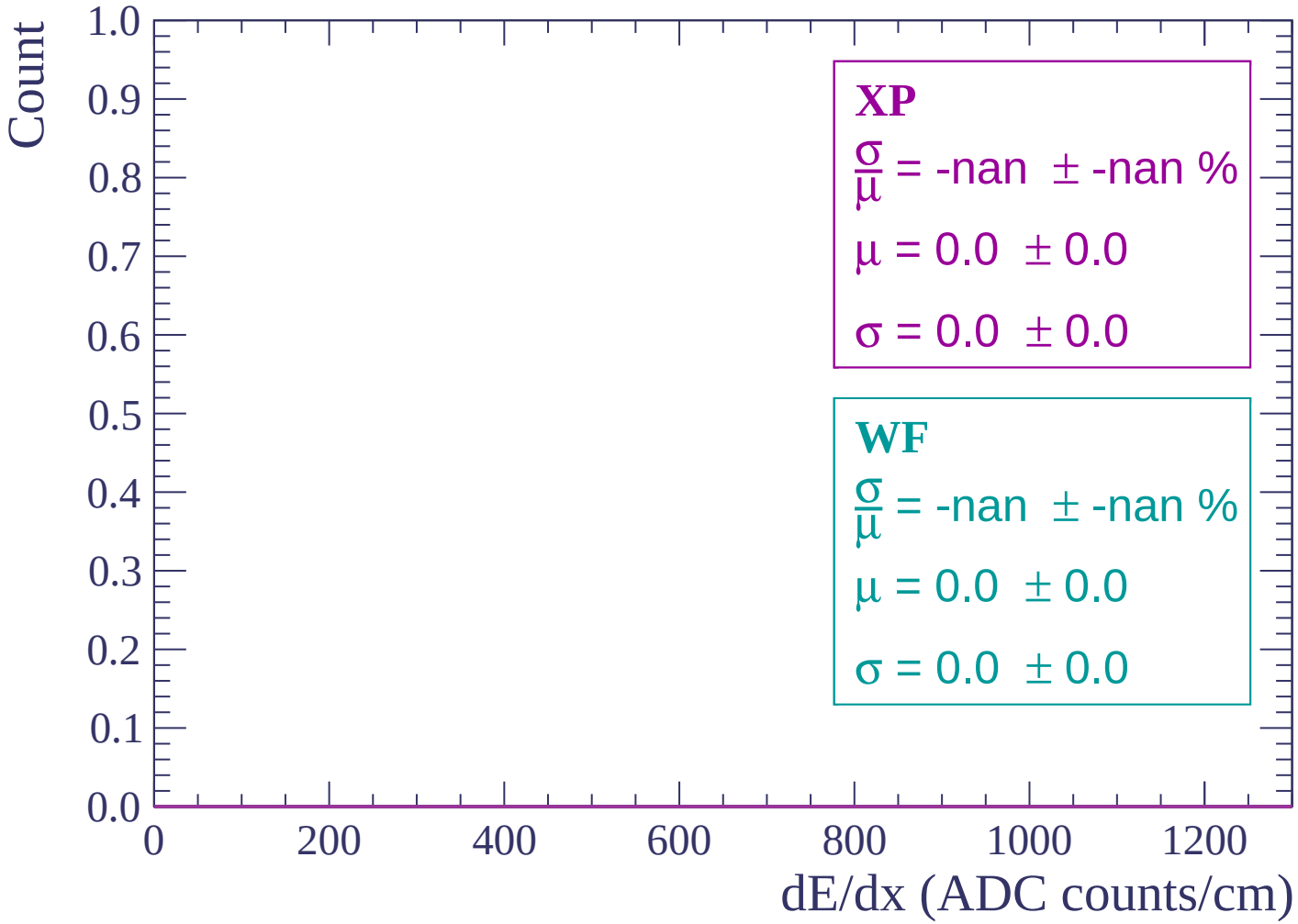

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0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dE/dx (ADC counts/cm)
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σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
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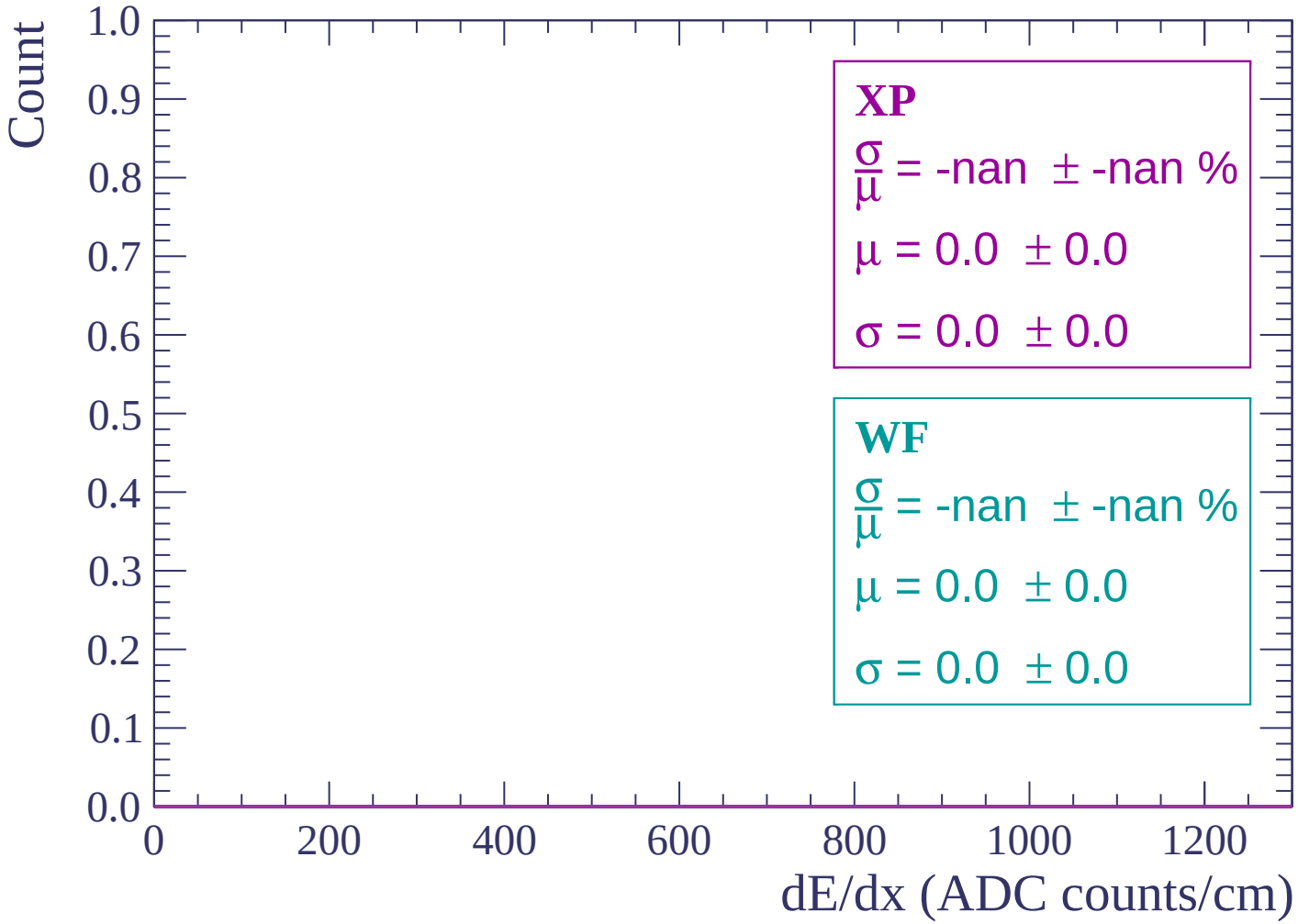
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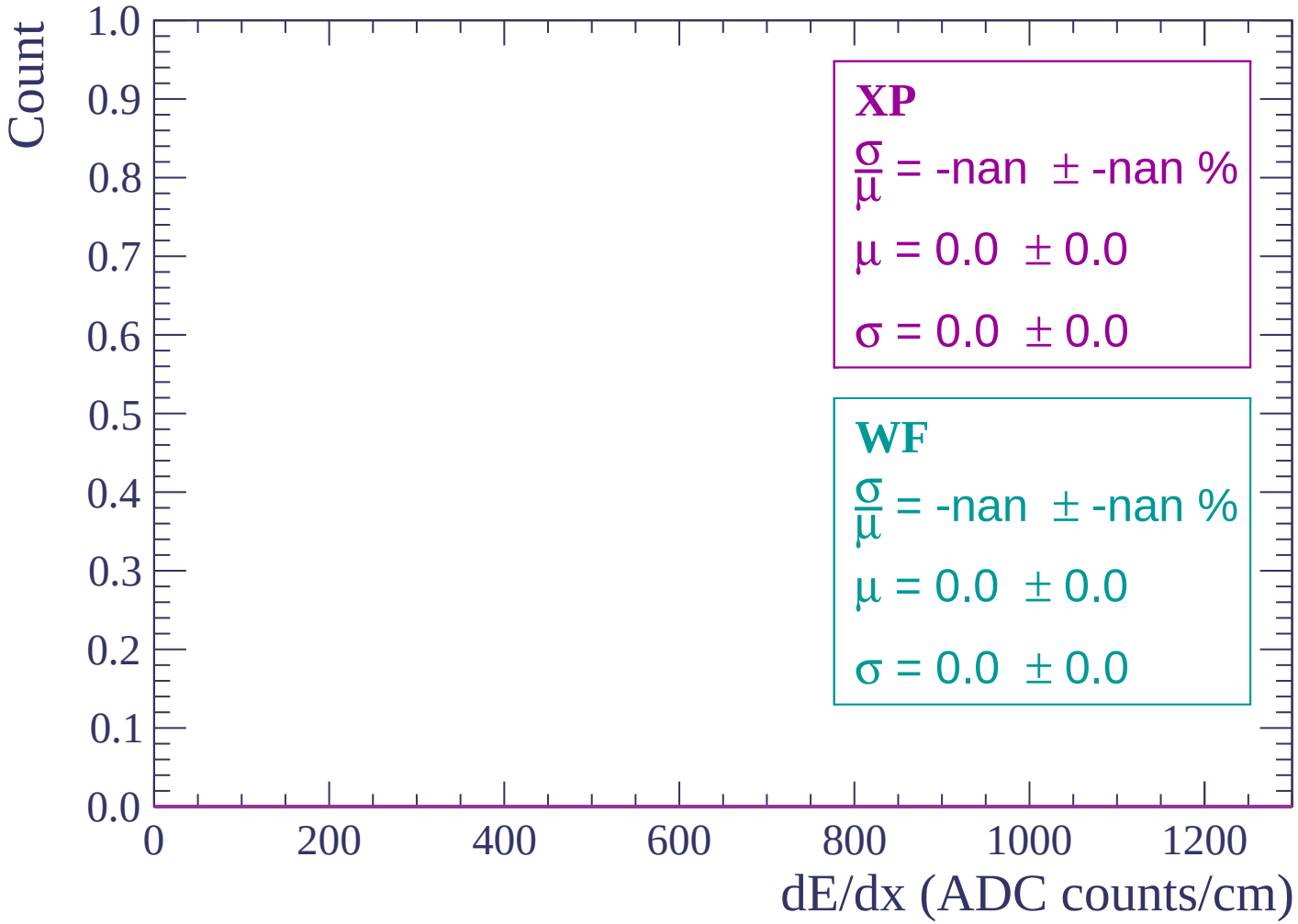
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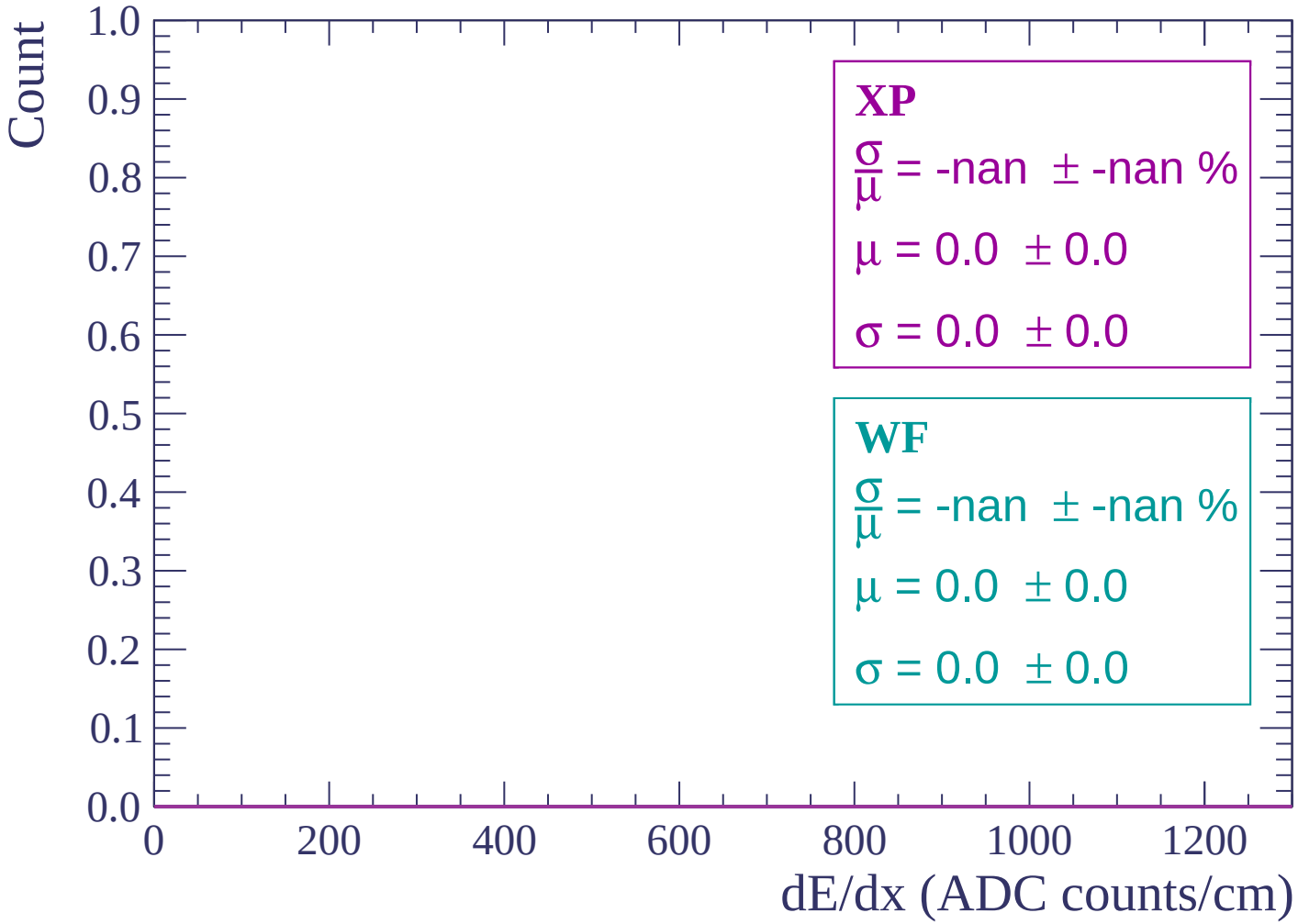
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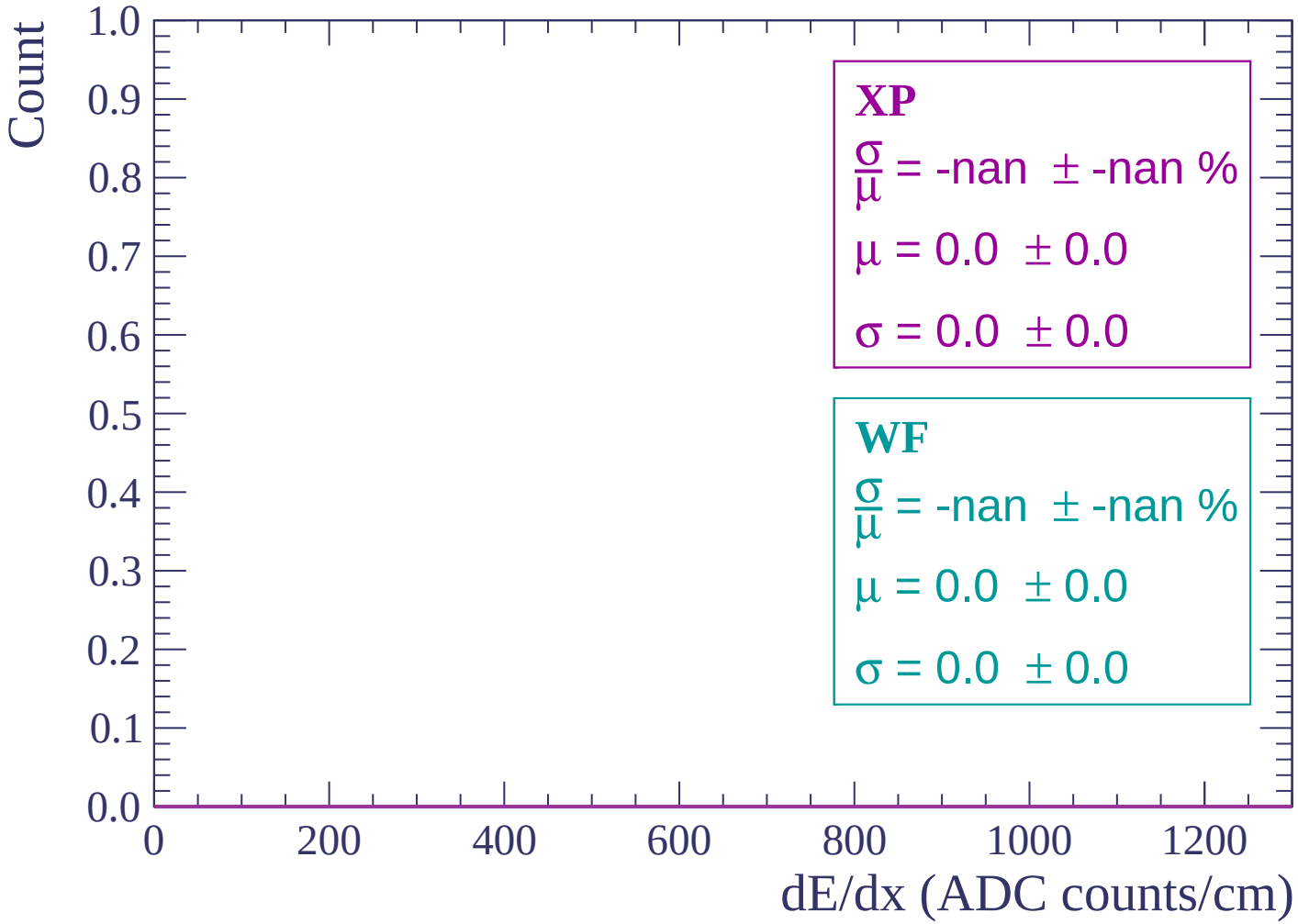
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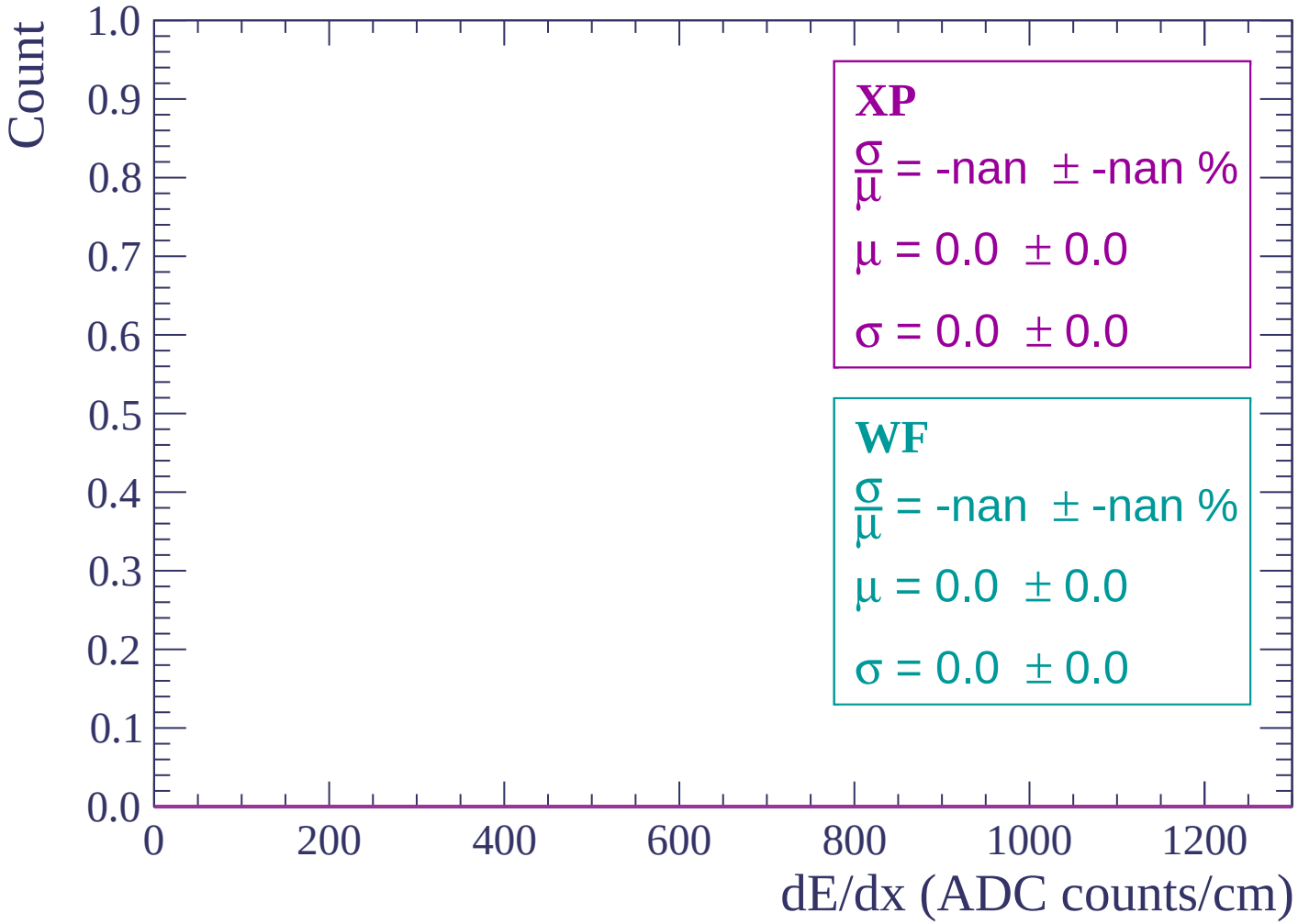
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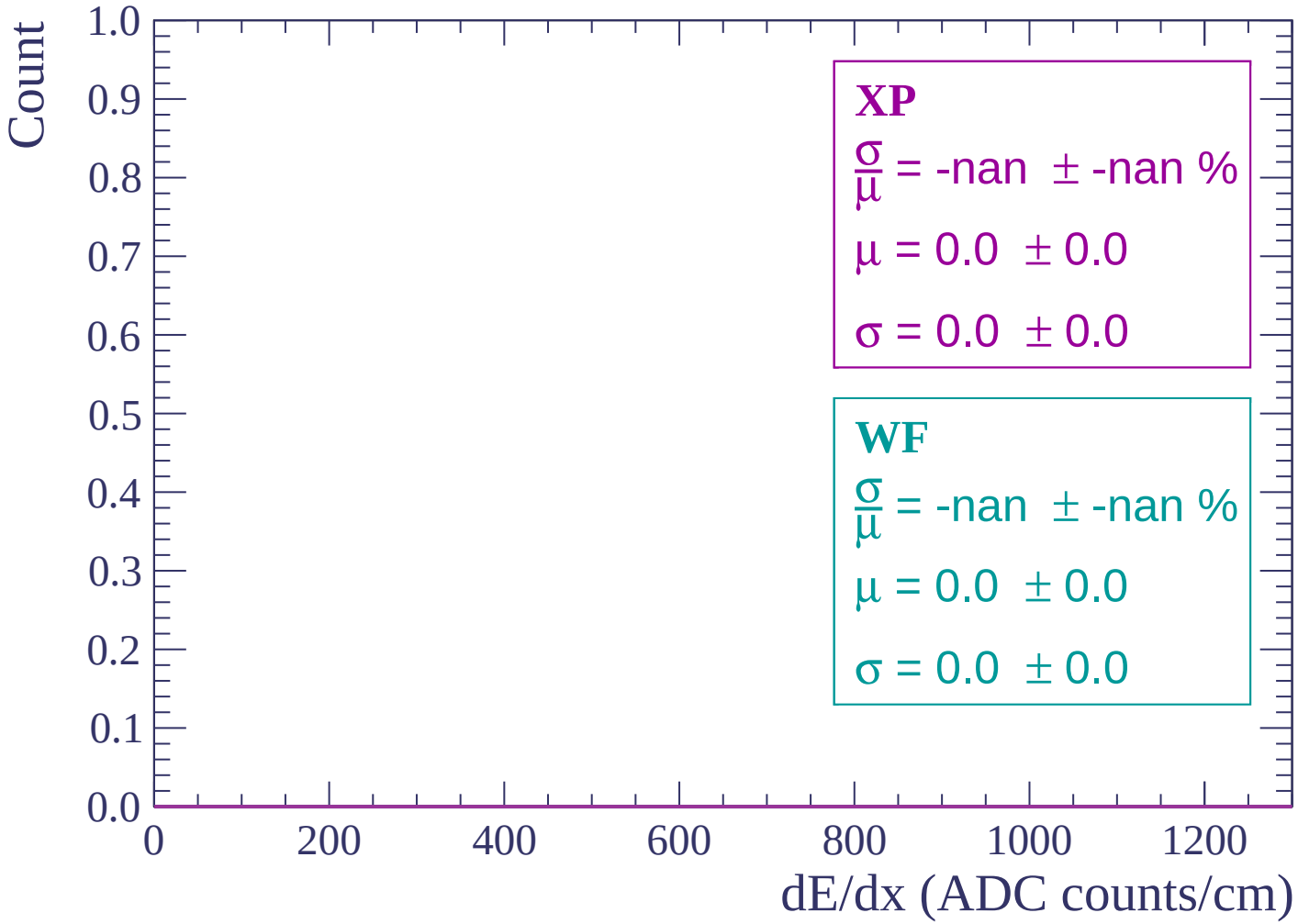
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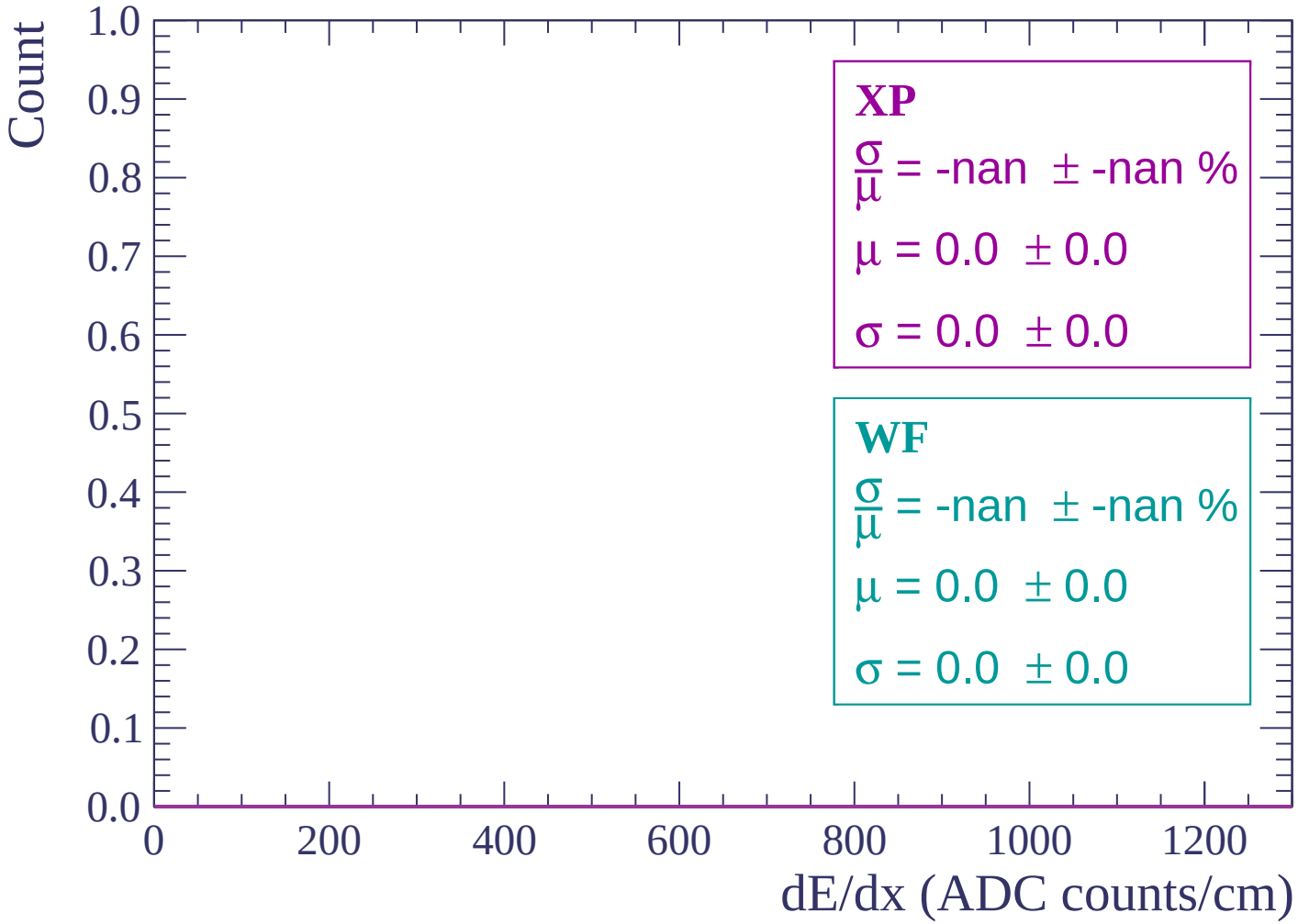
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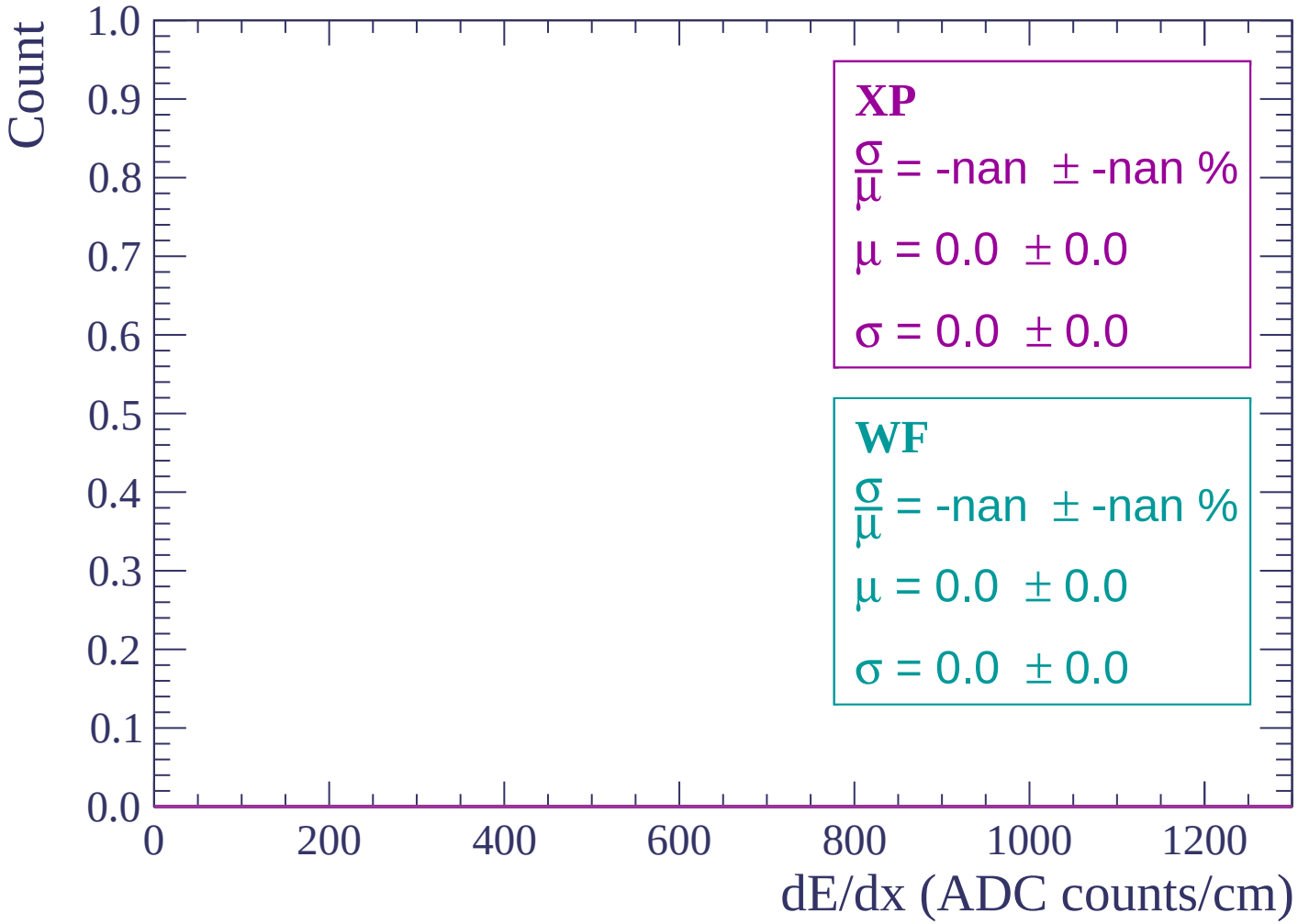
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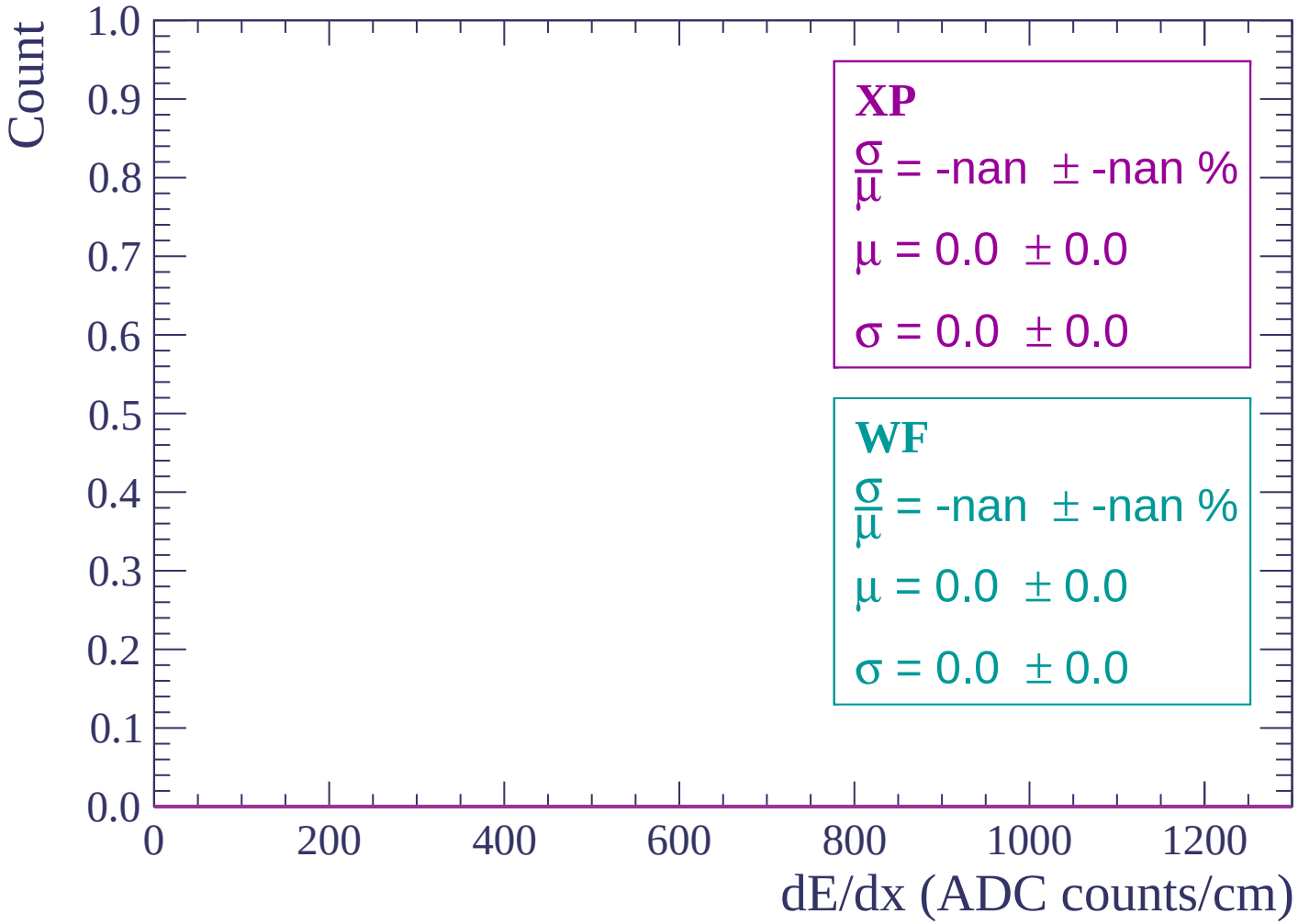
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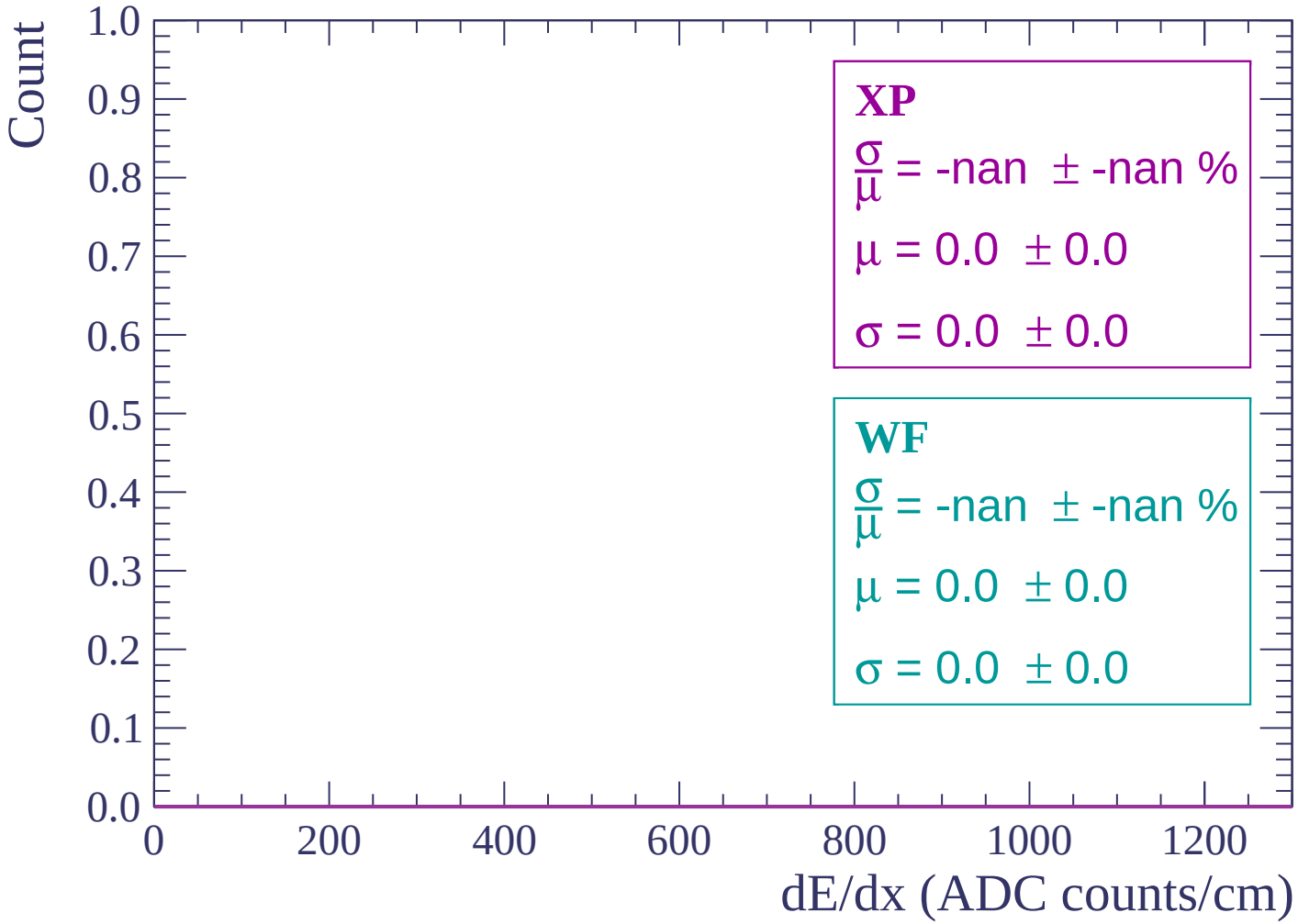
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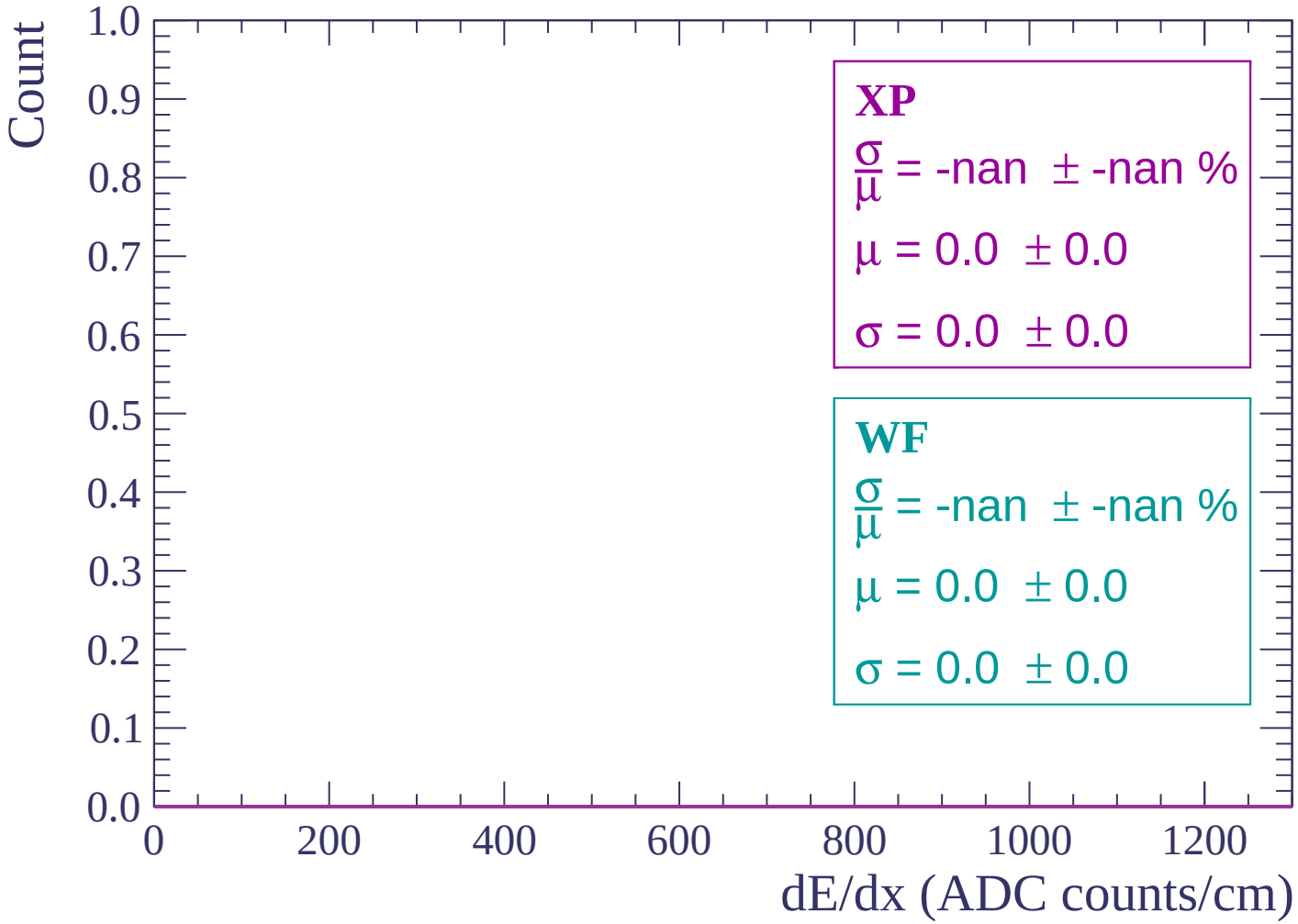
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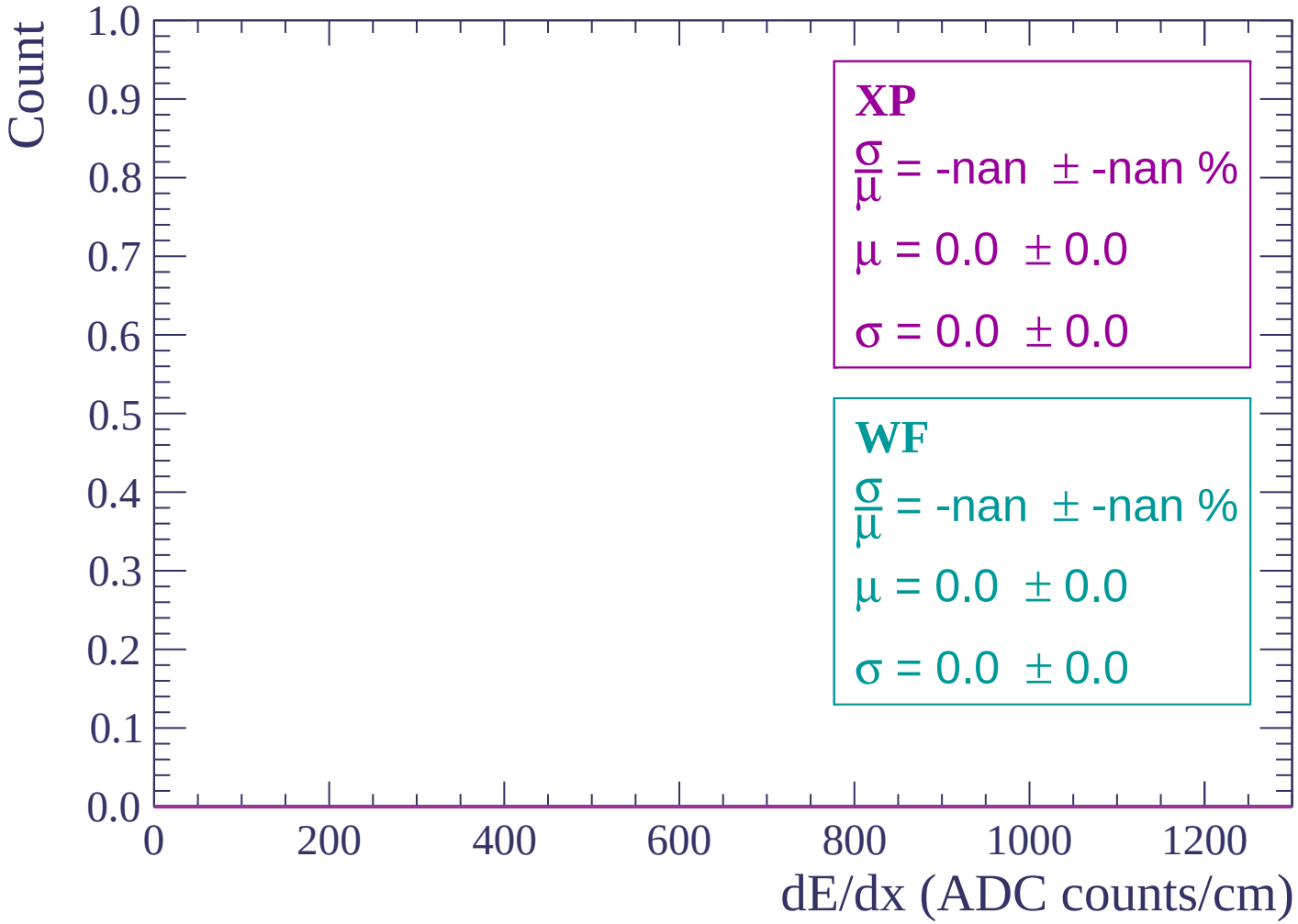
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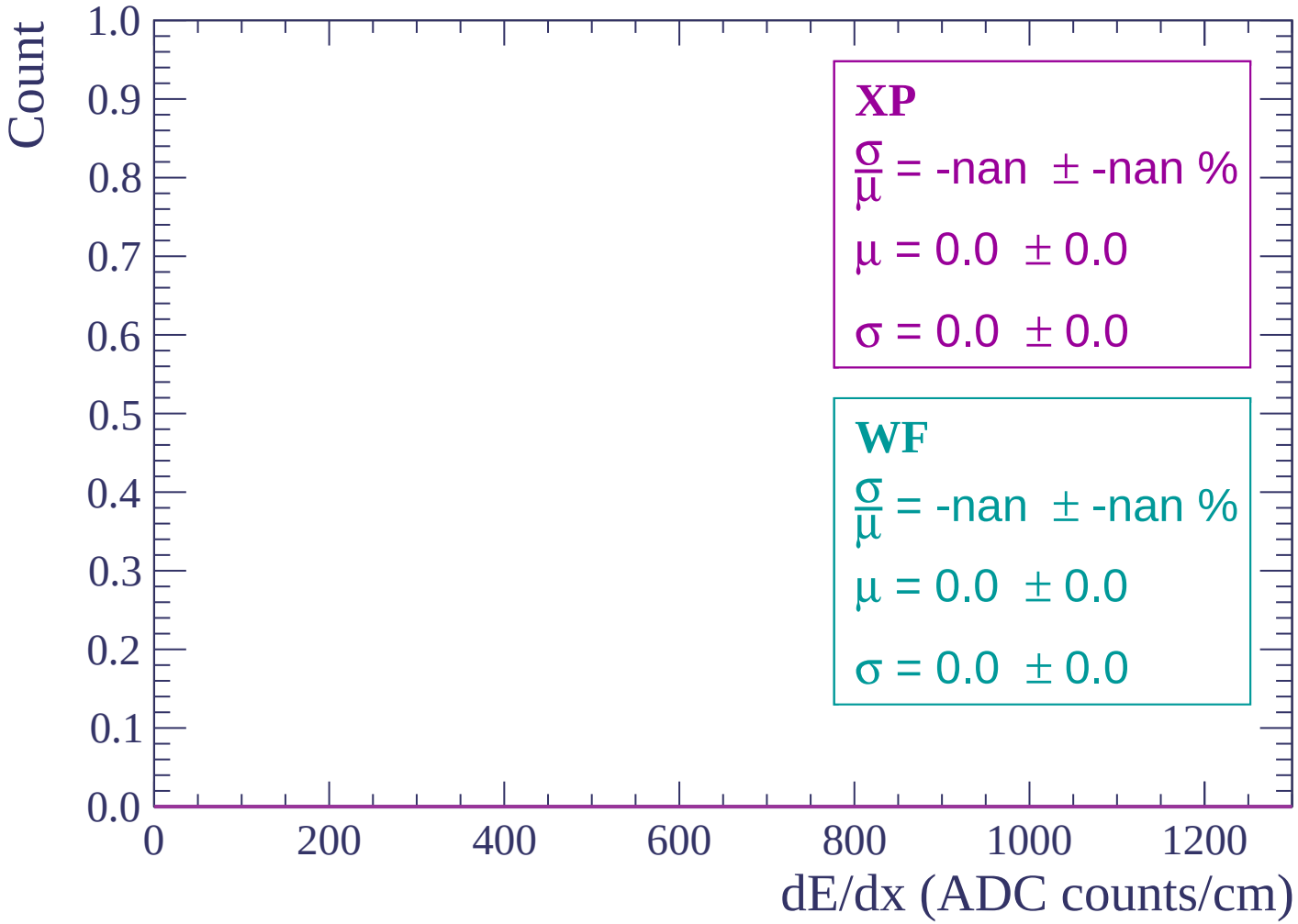
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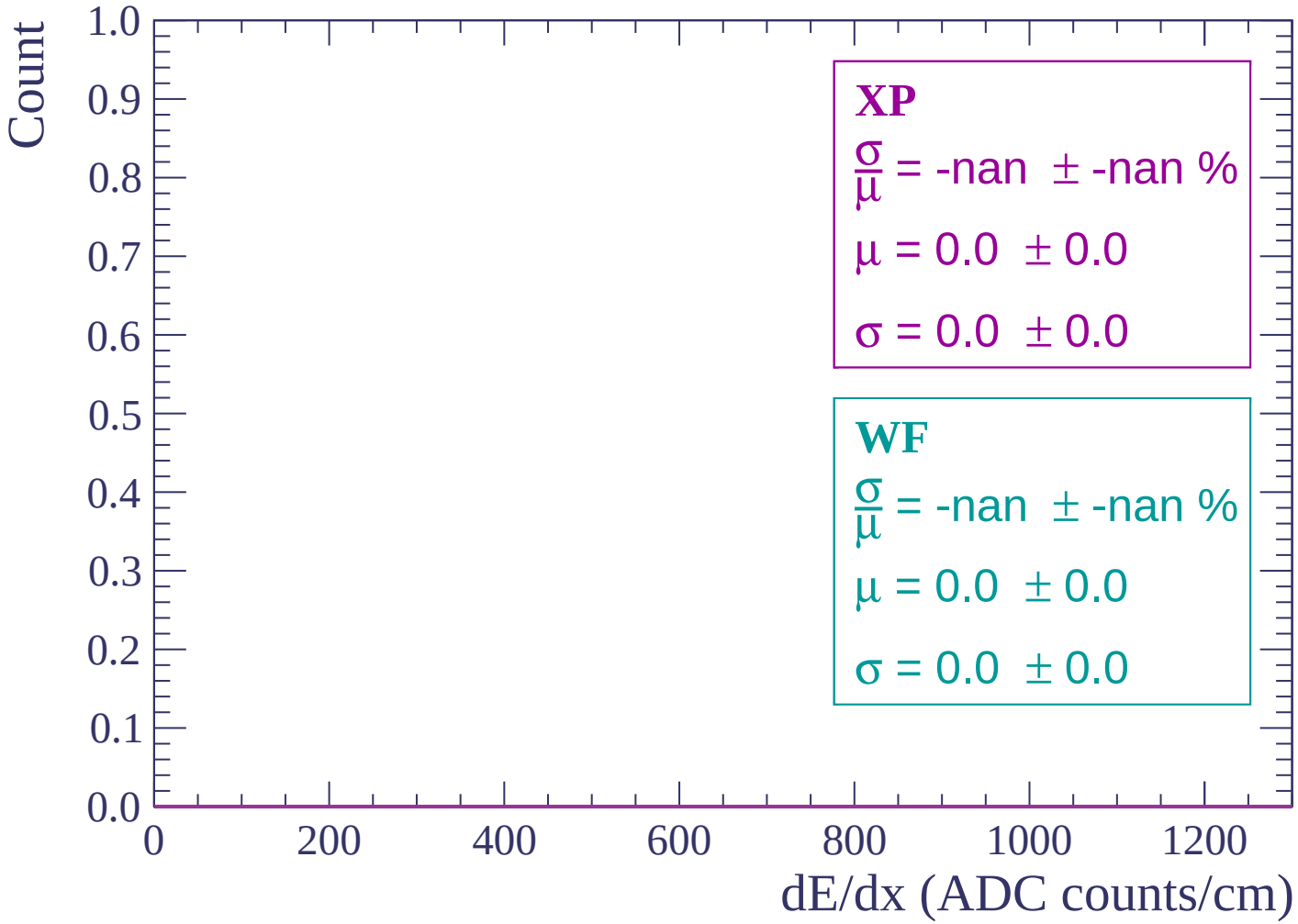
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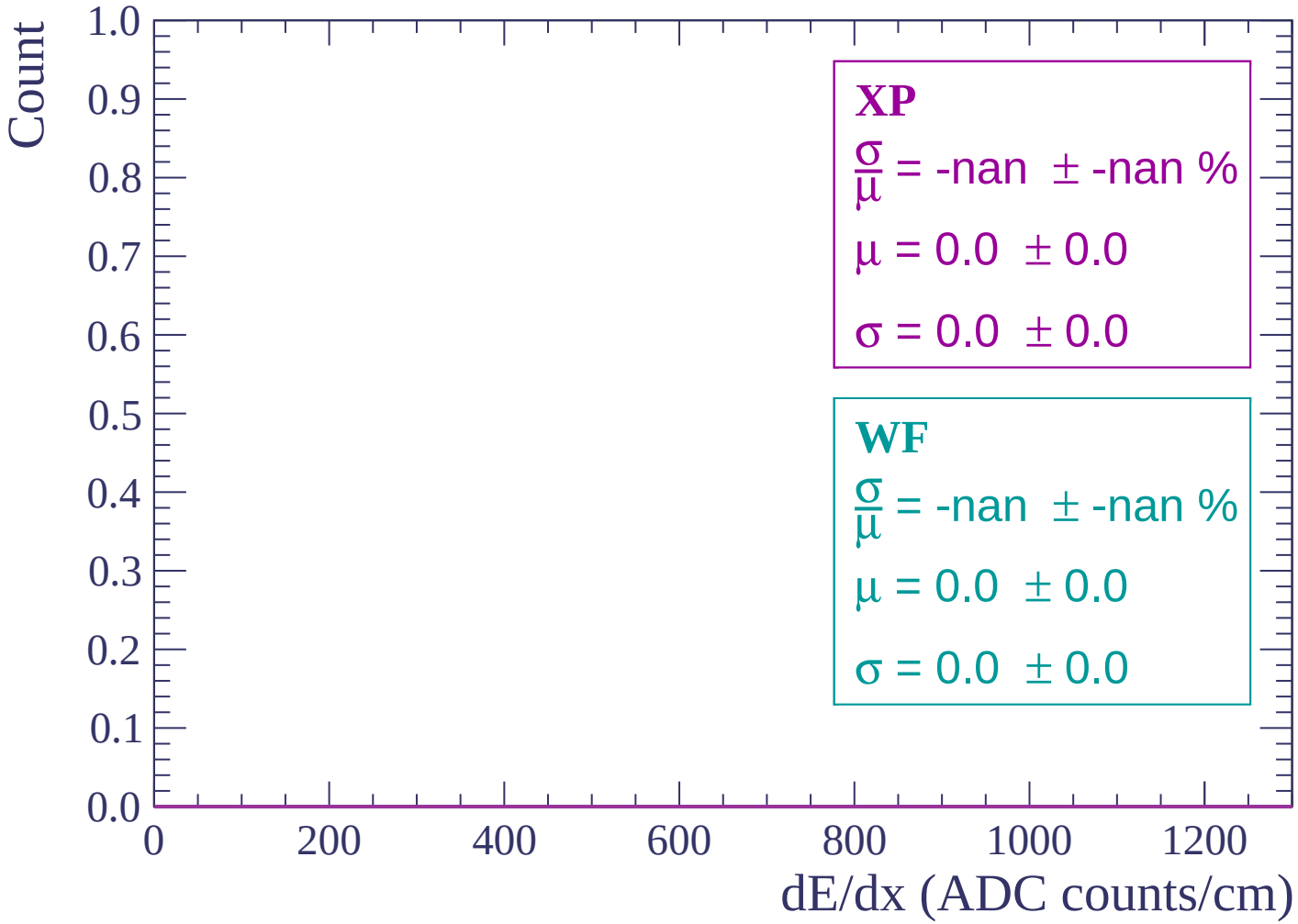
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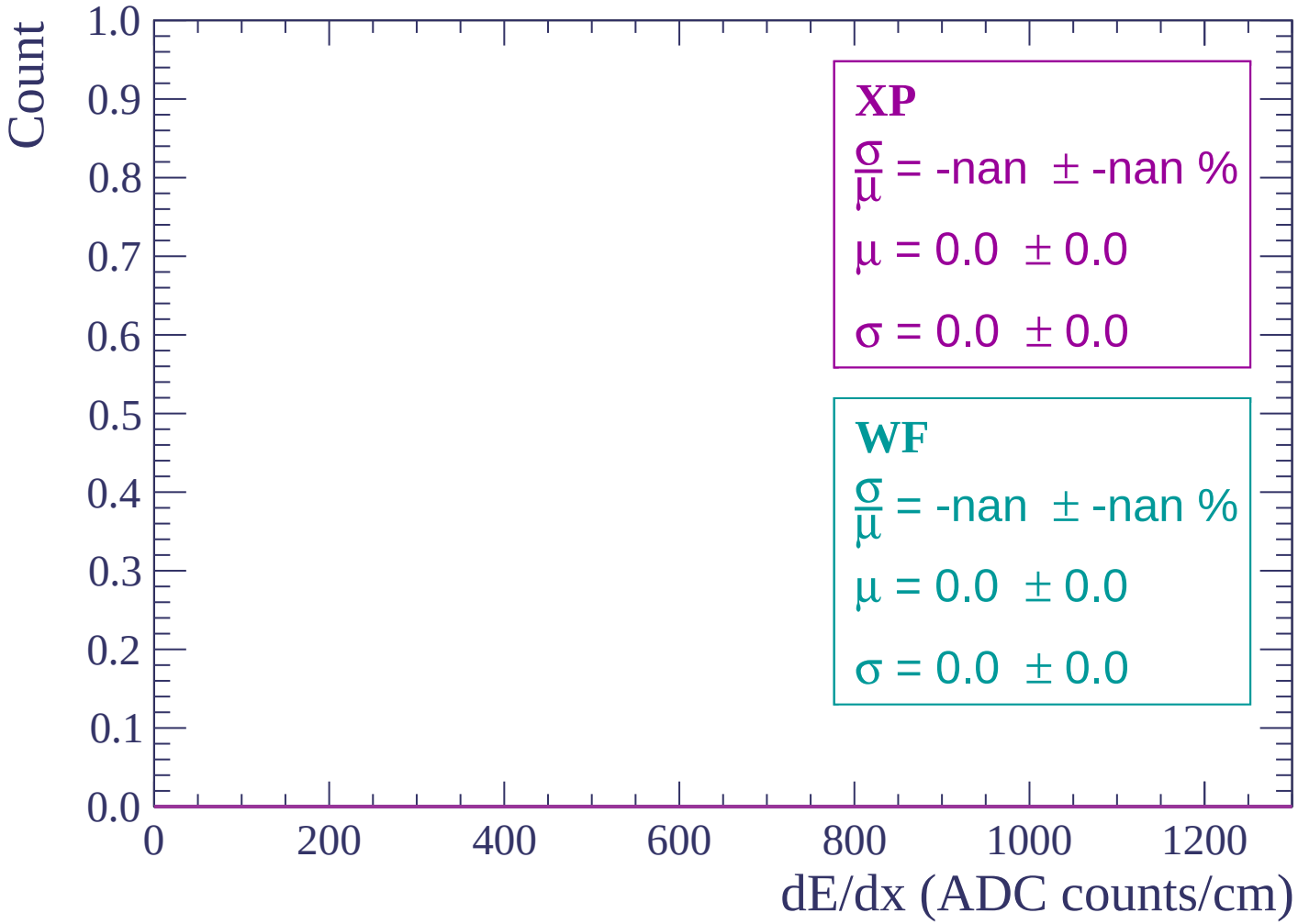
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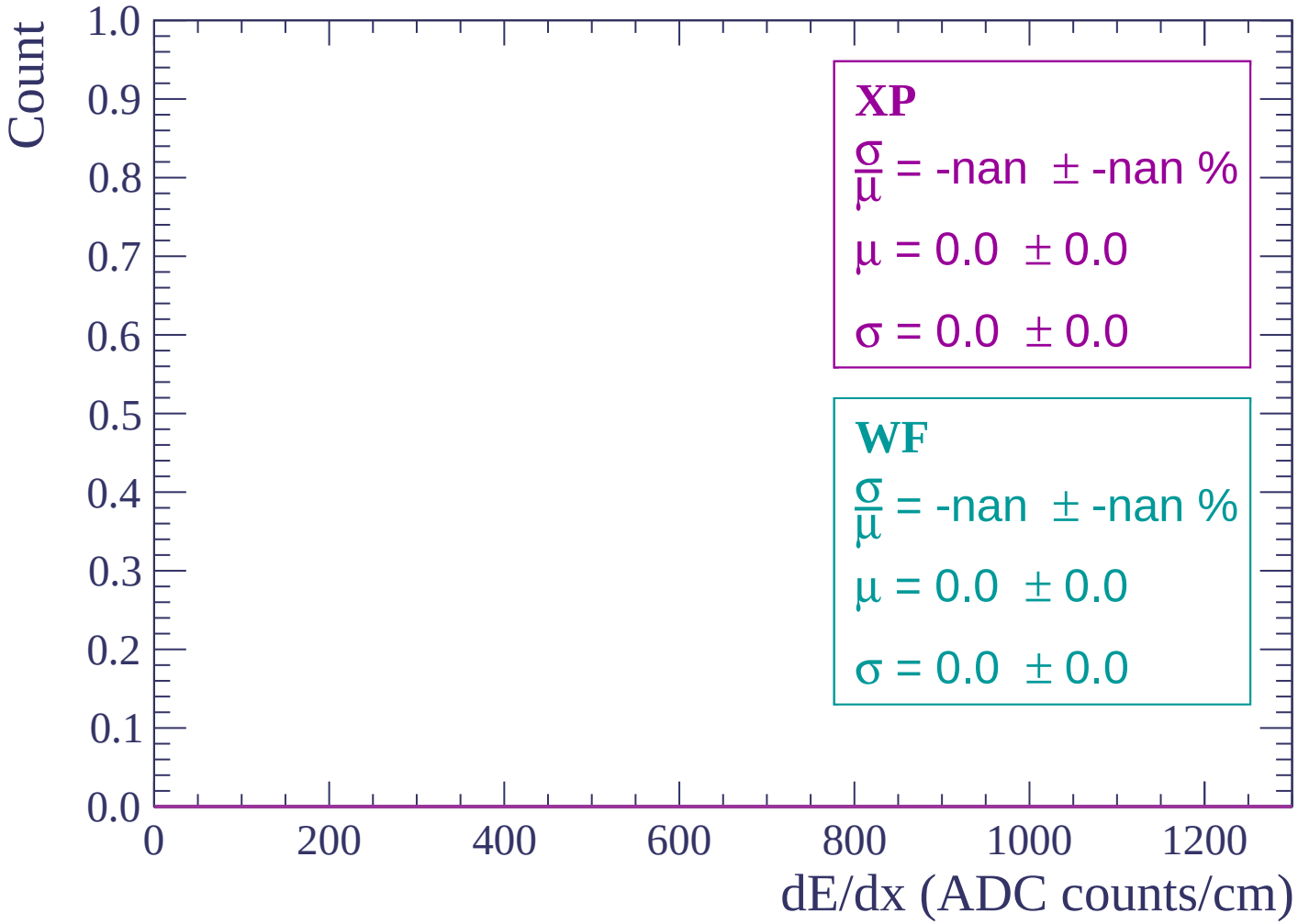
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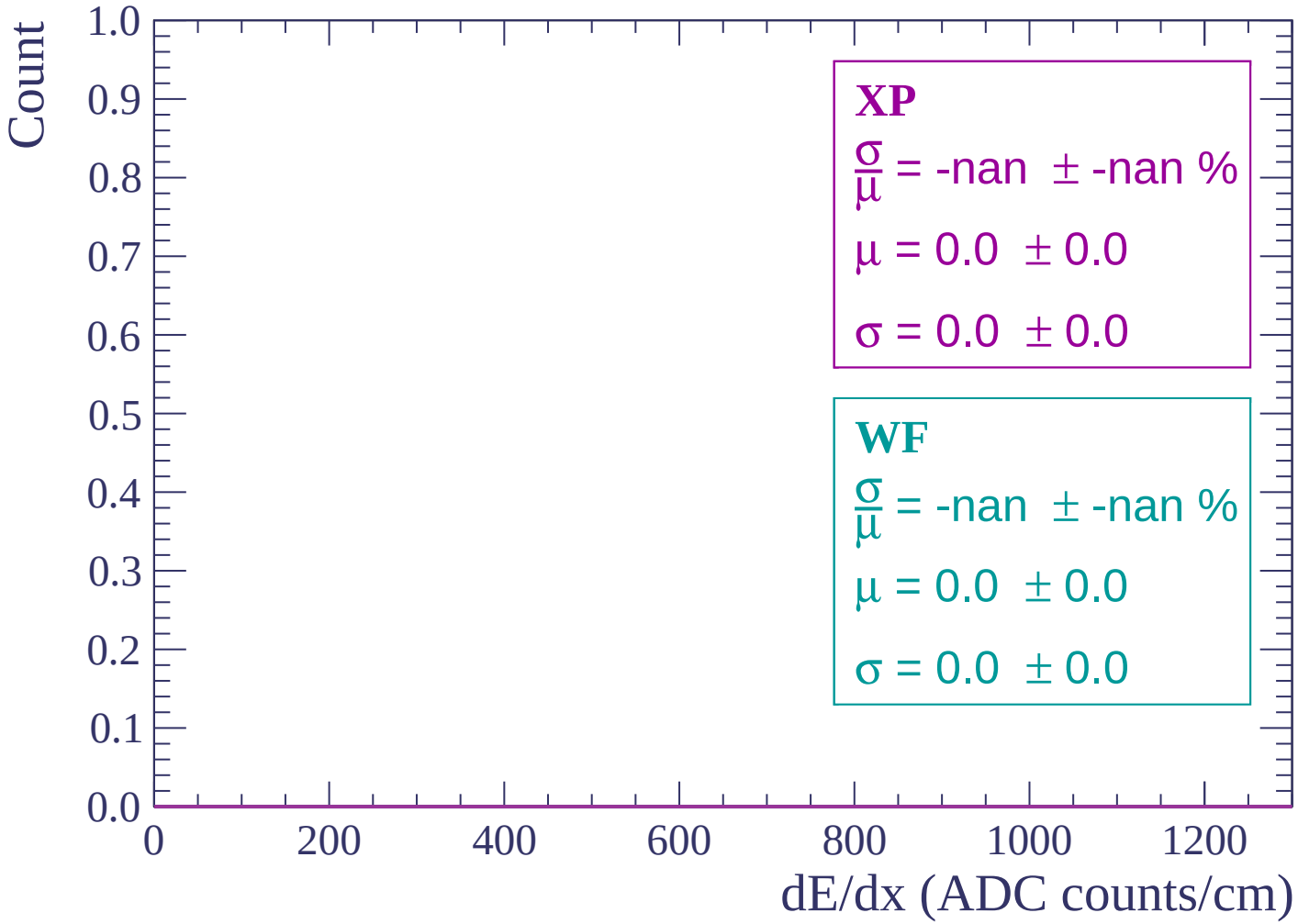
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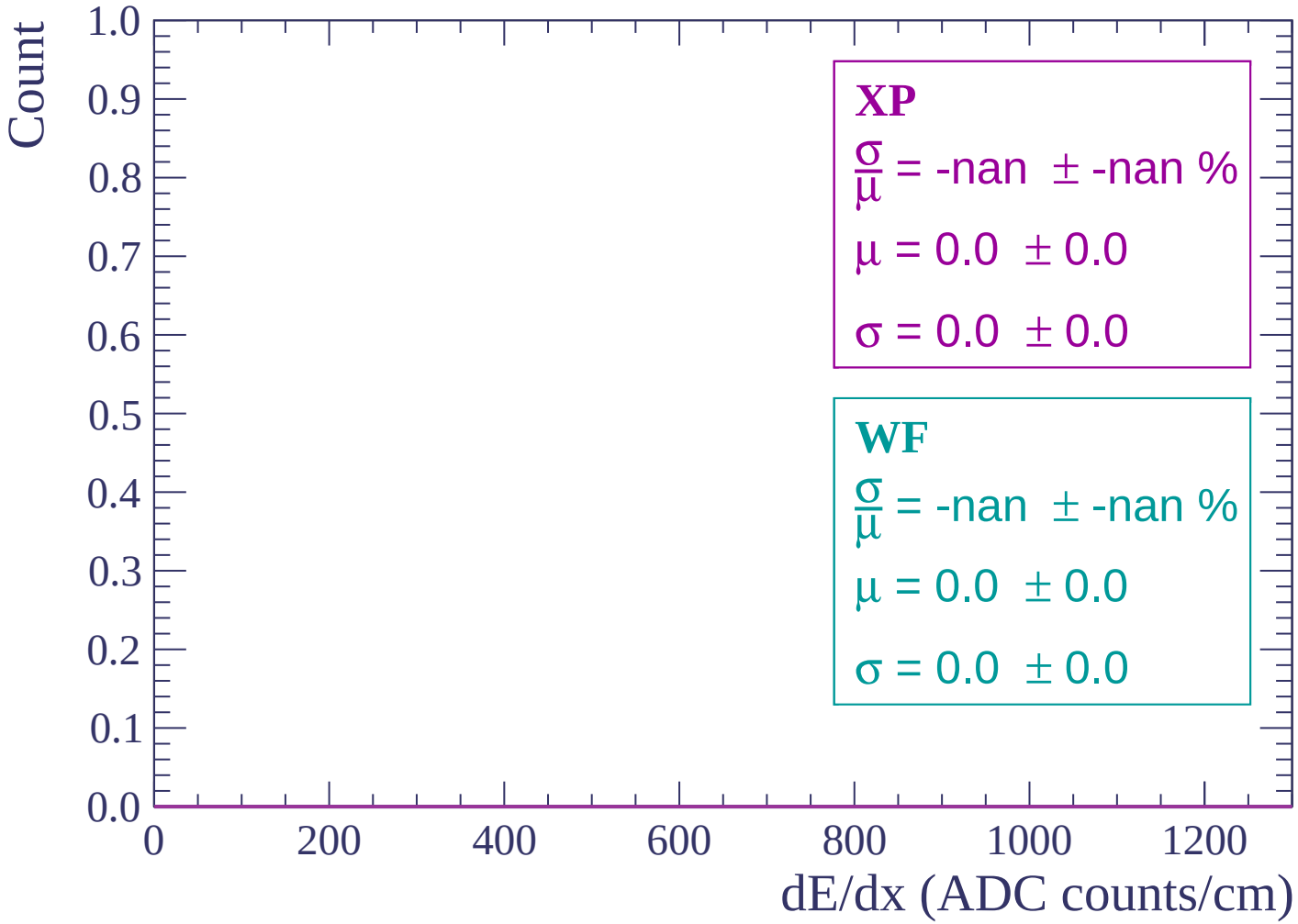
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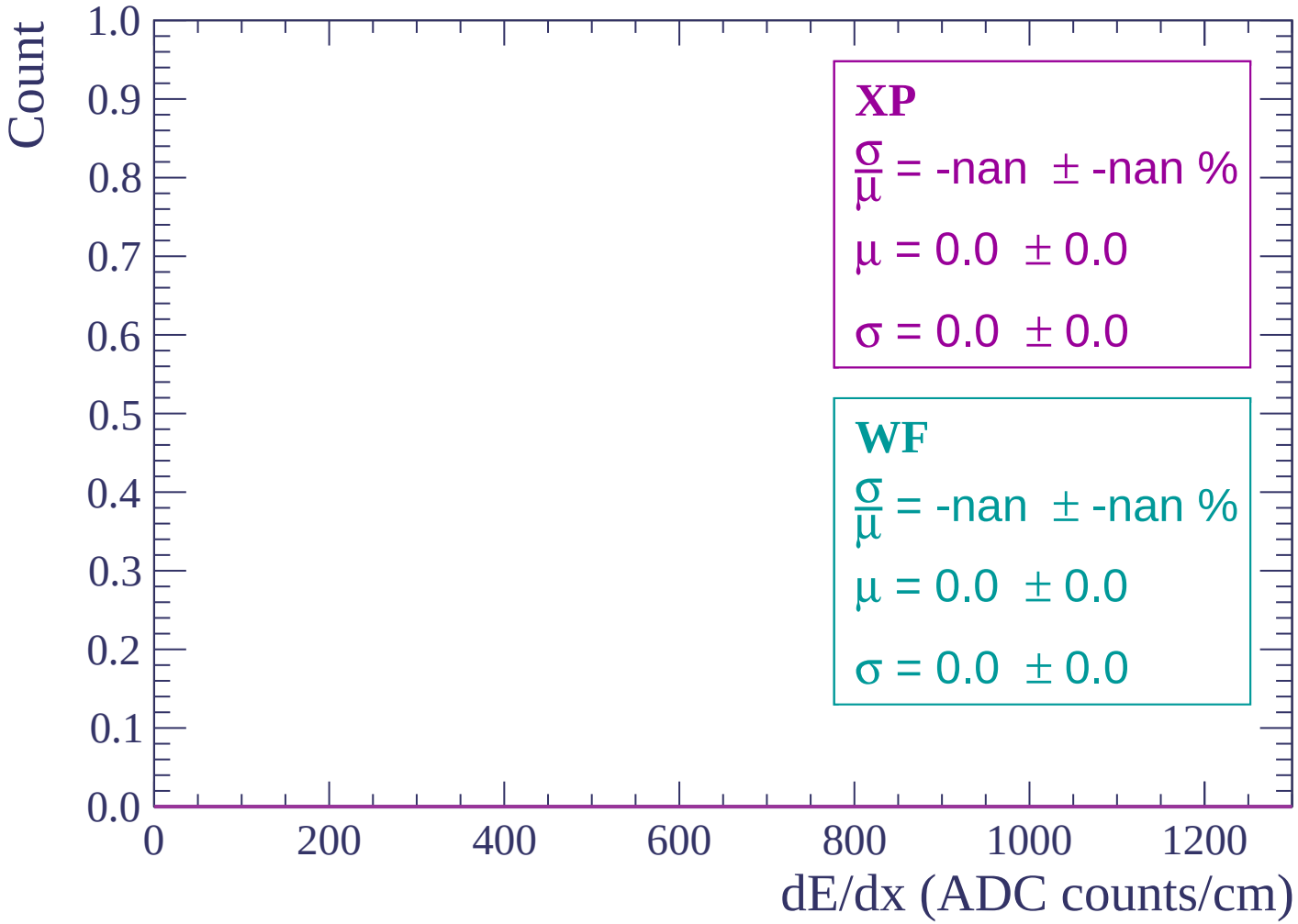
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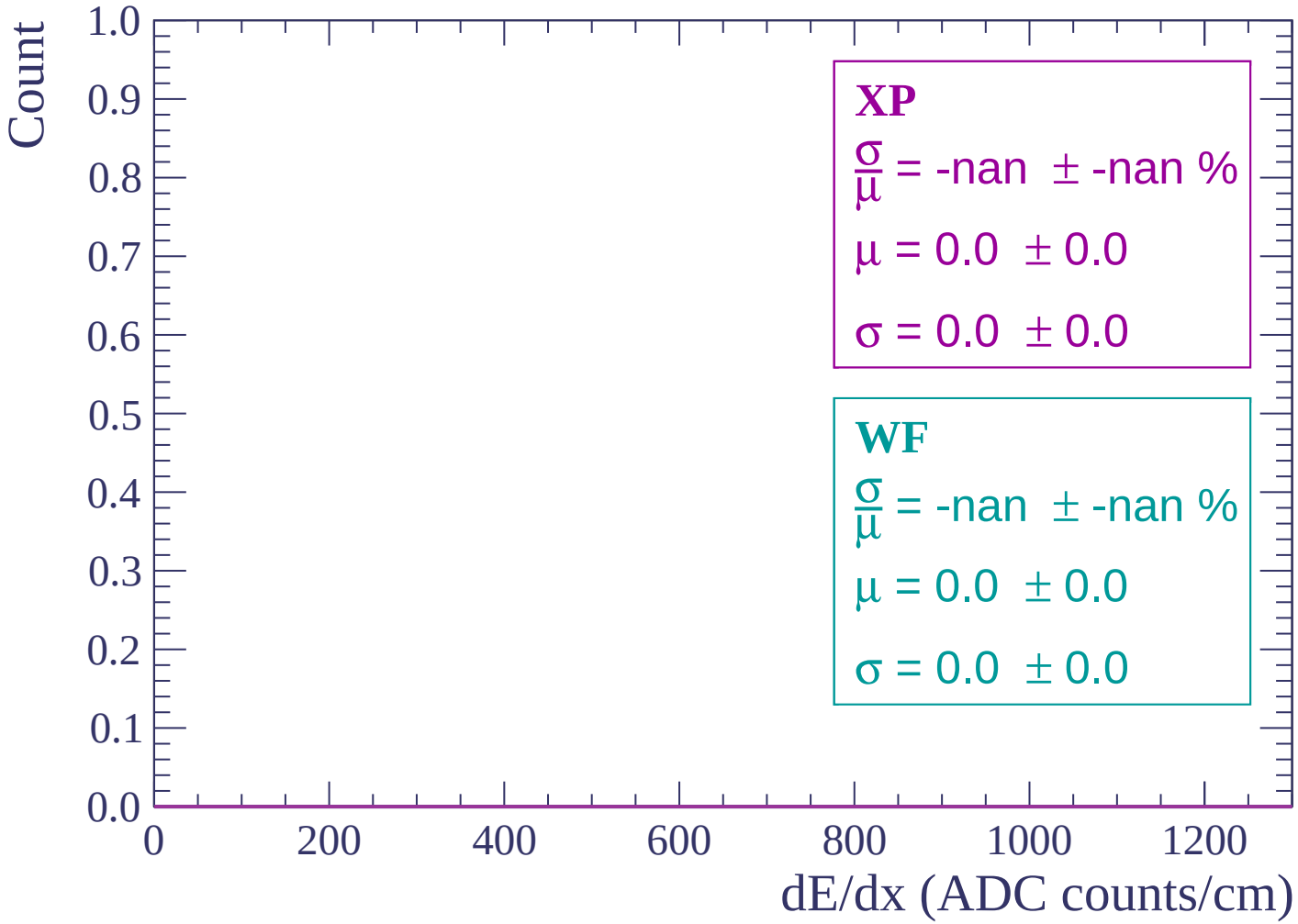
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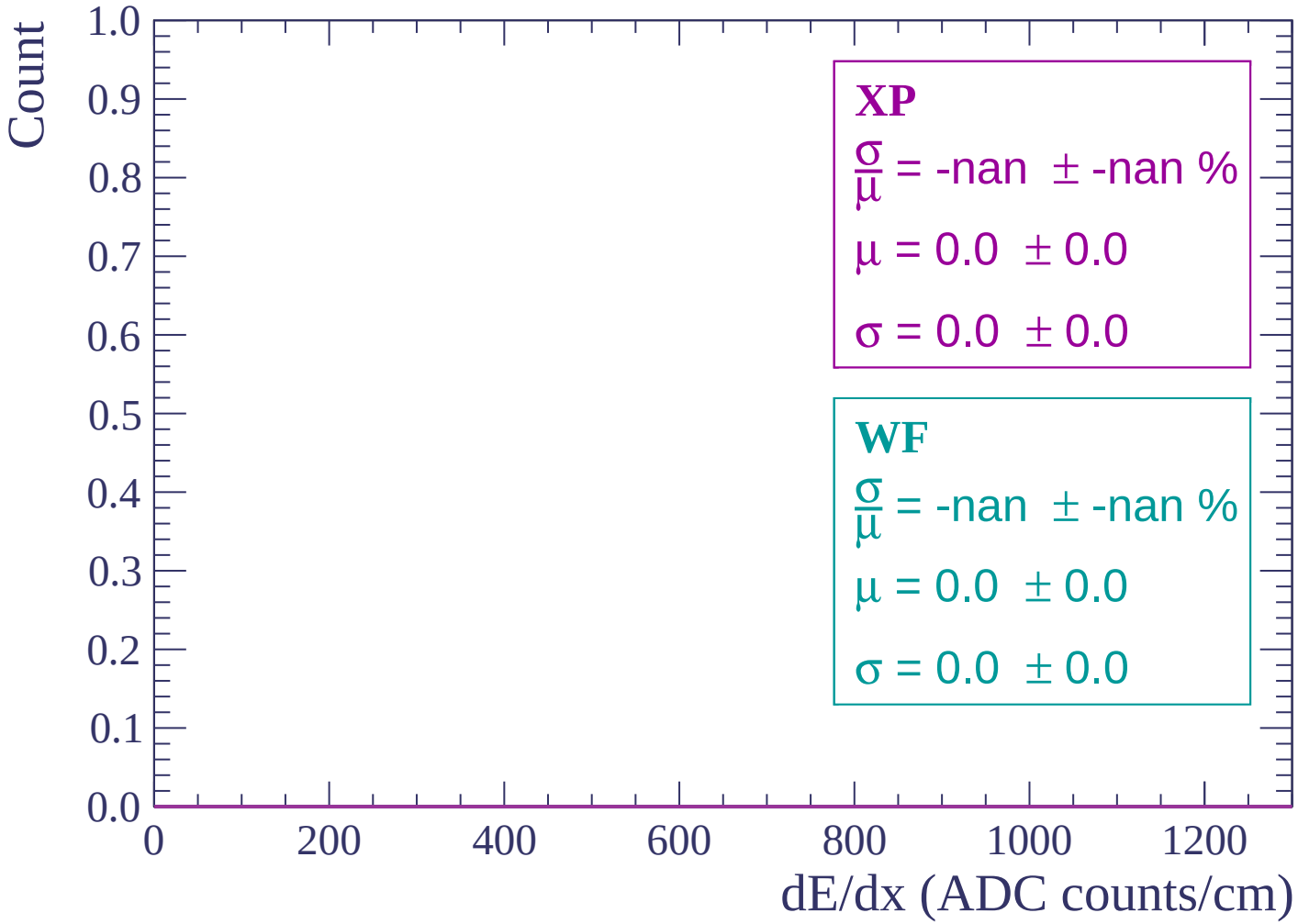
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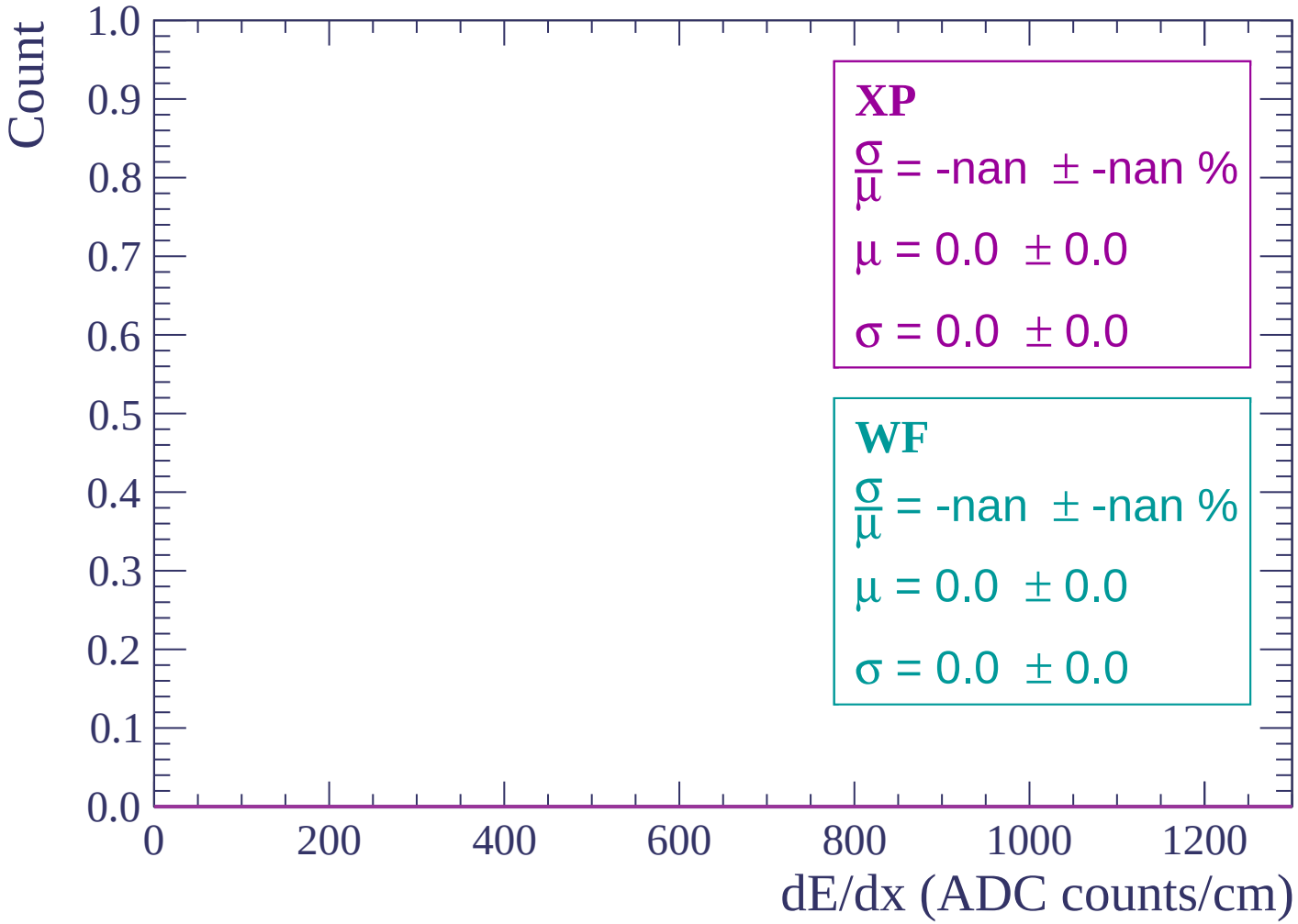
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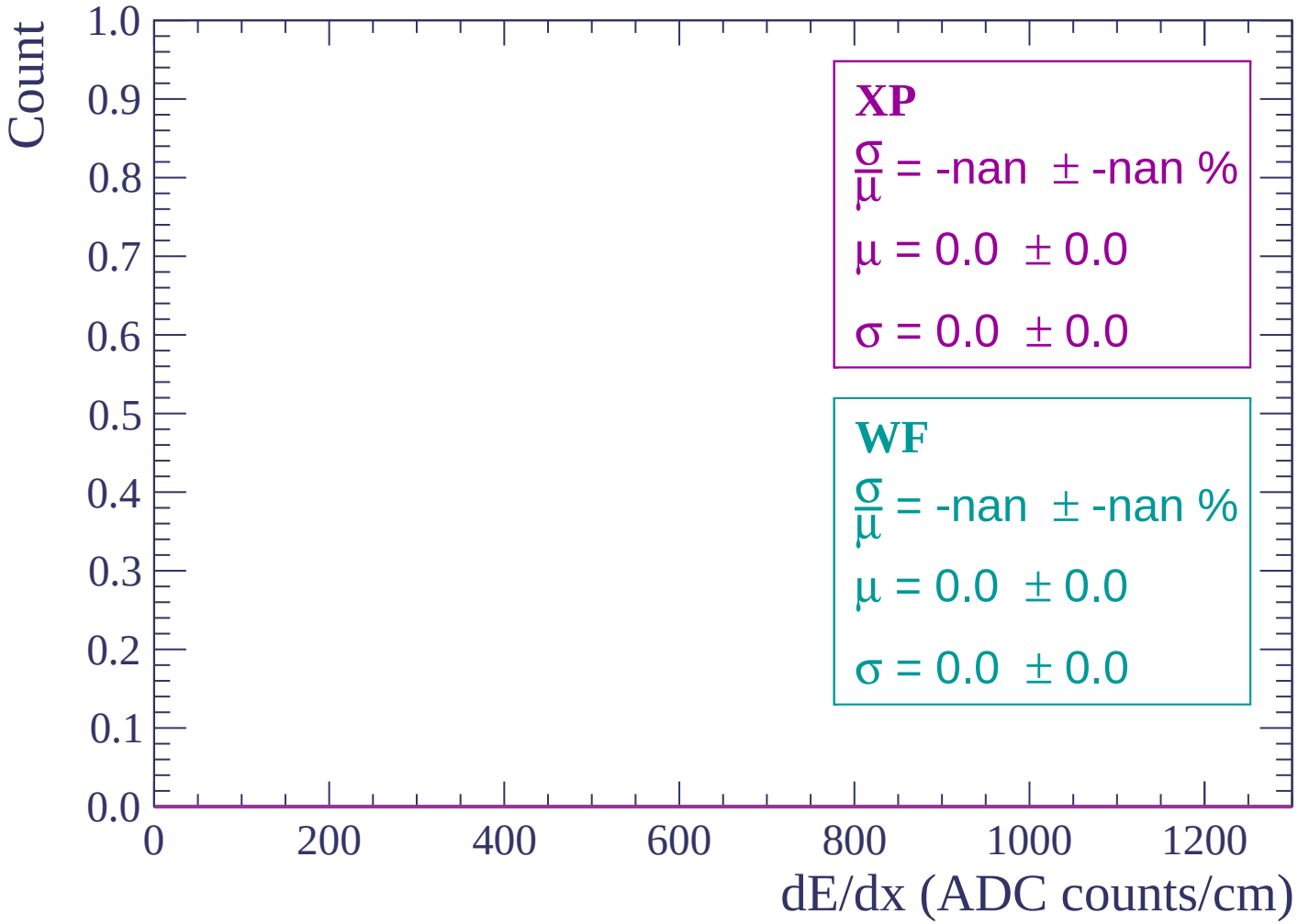
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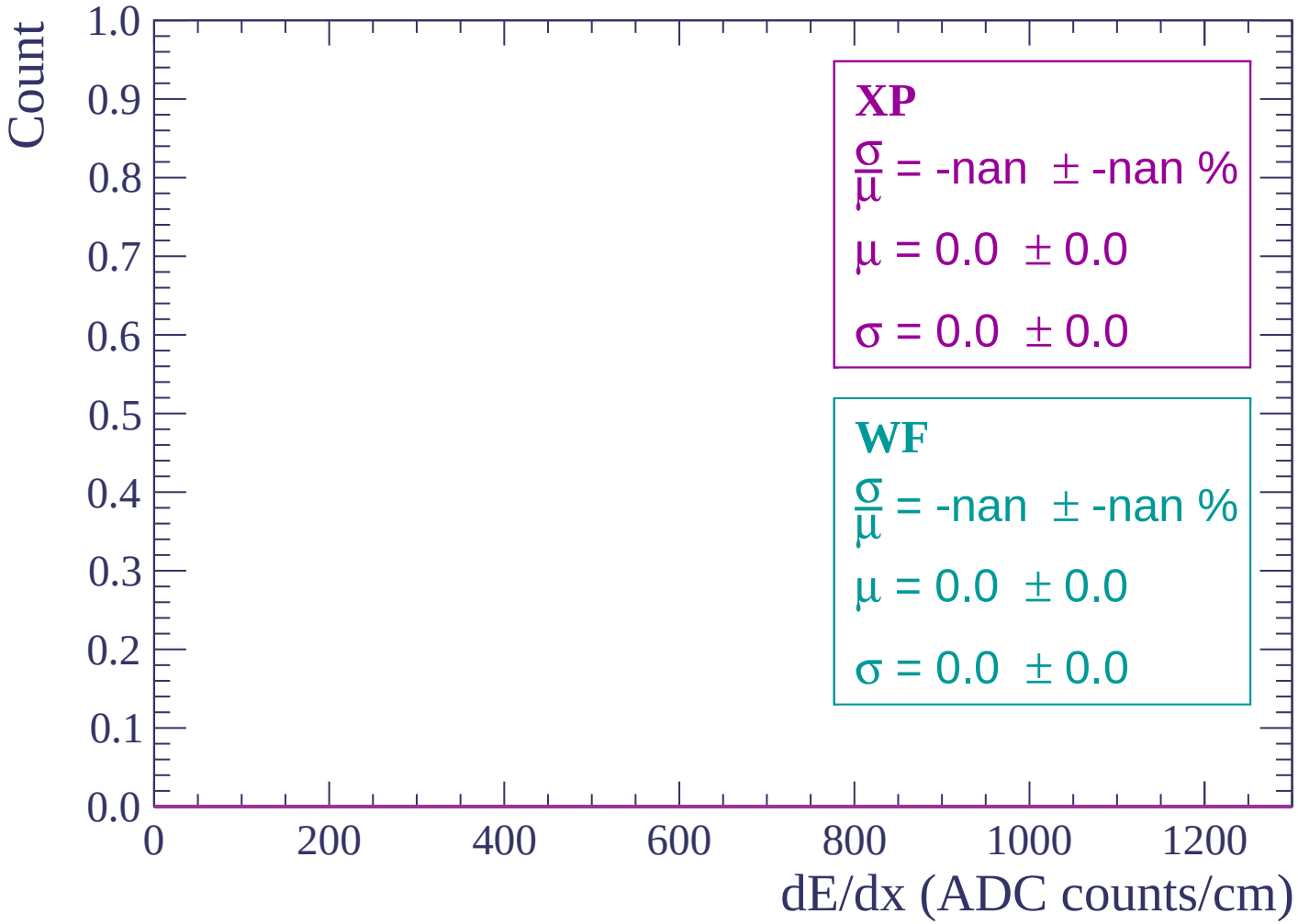
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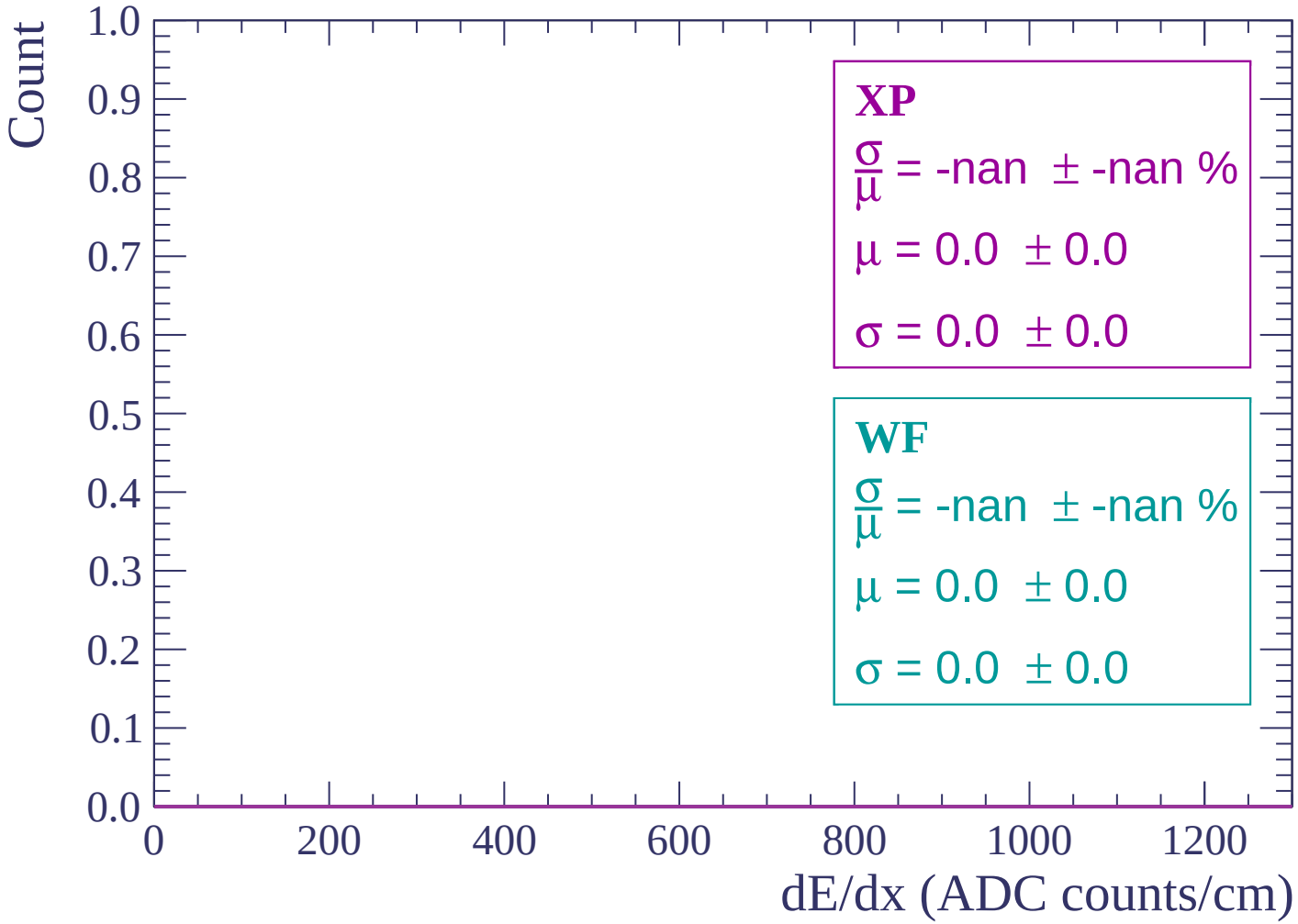
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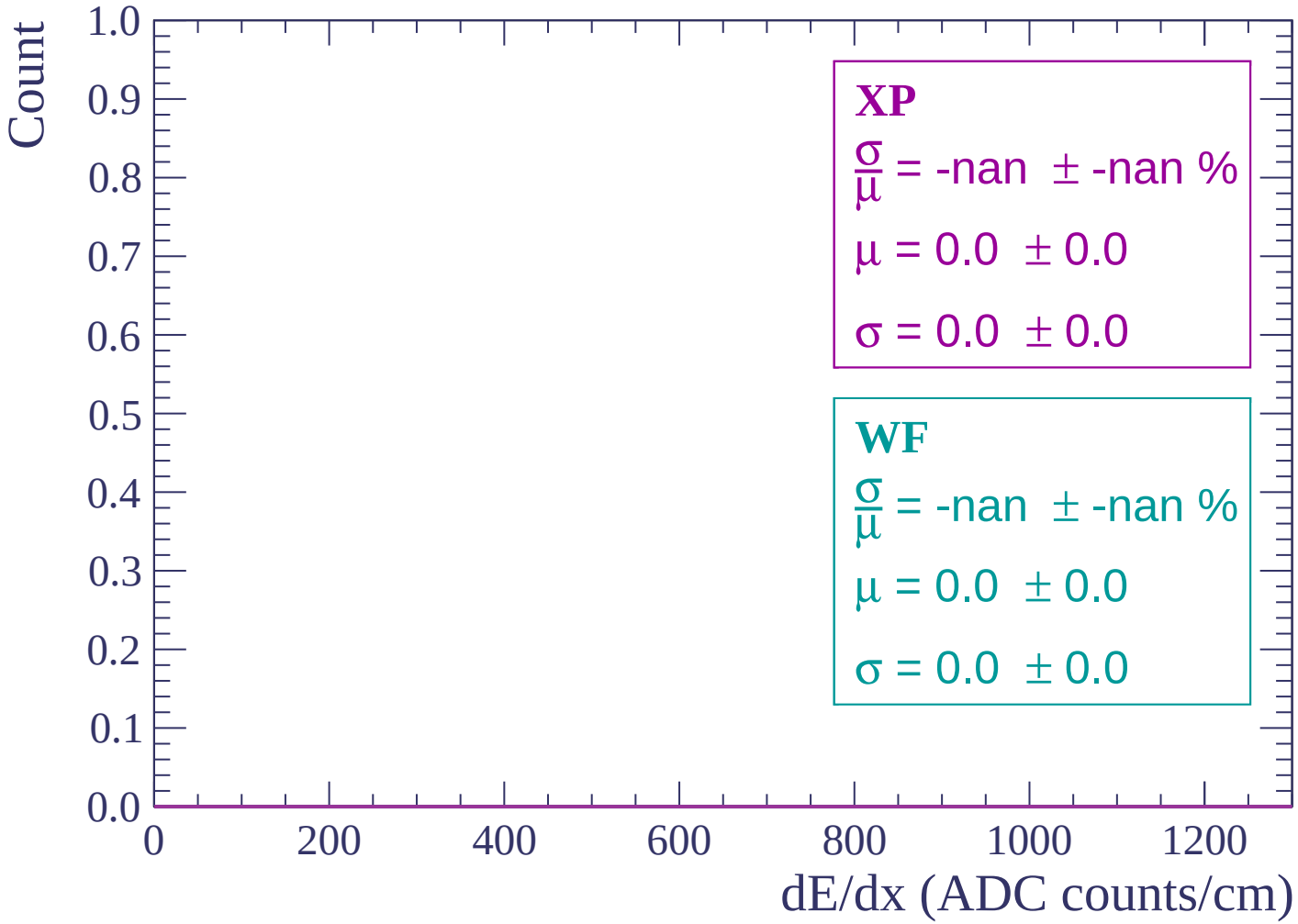
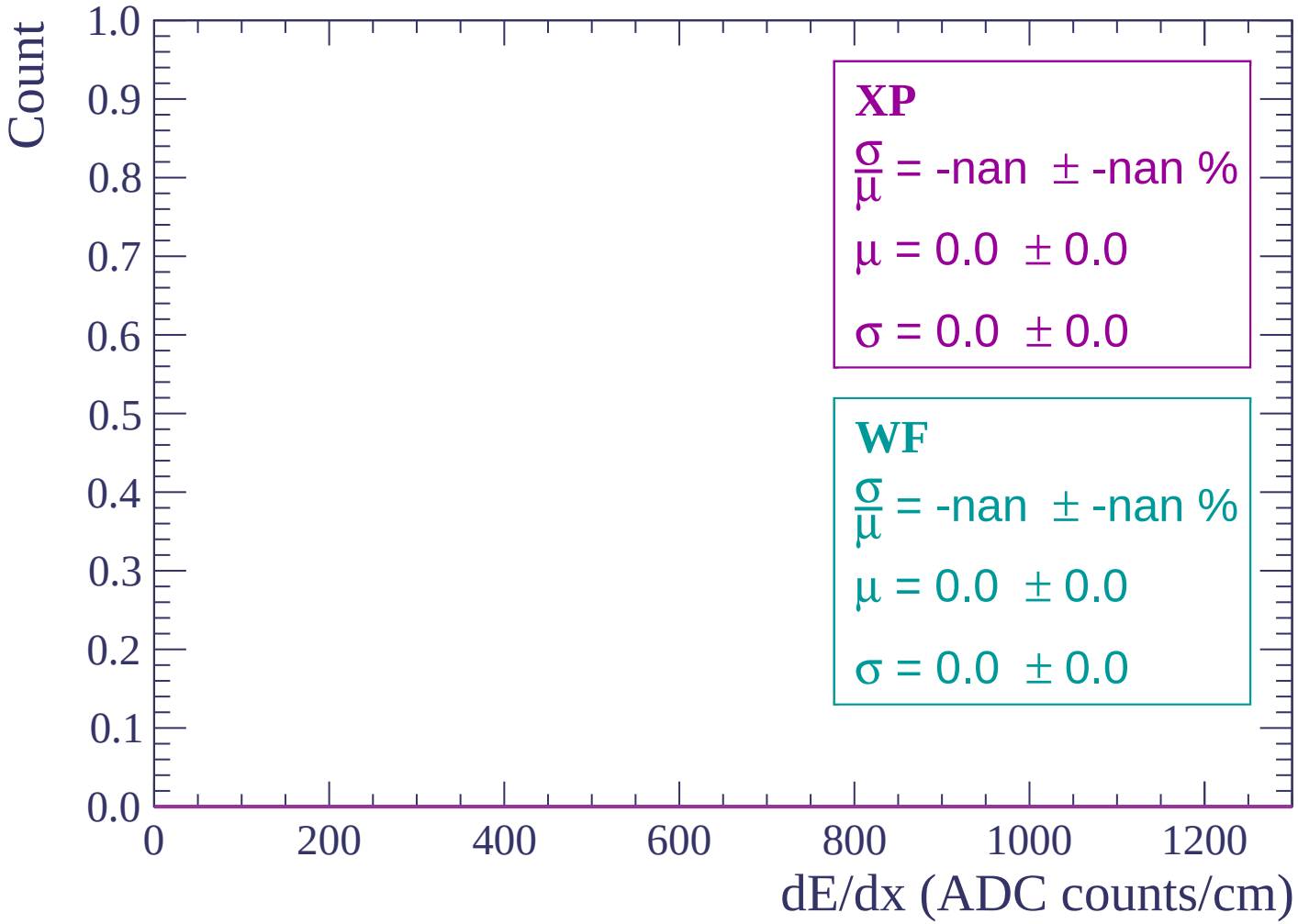
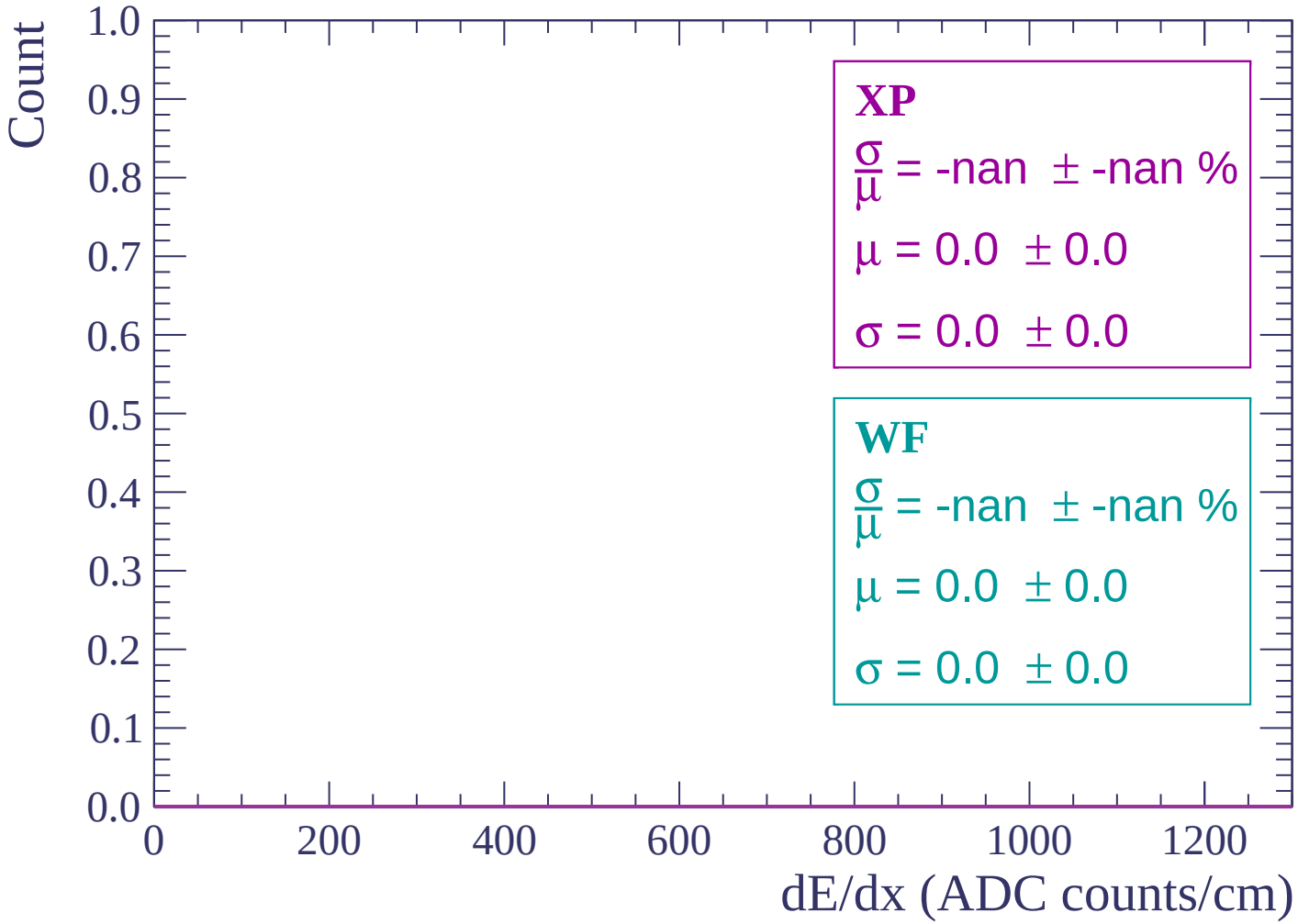
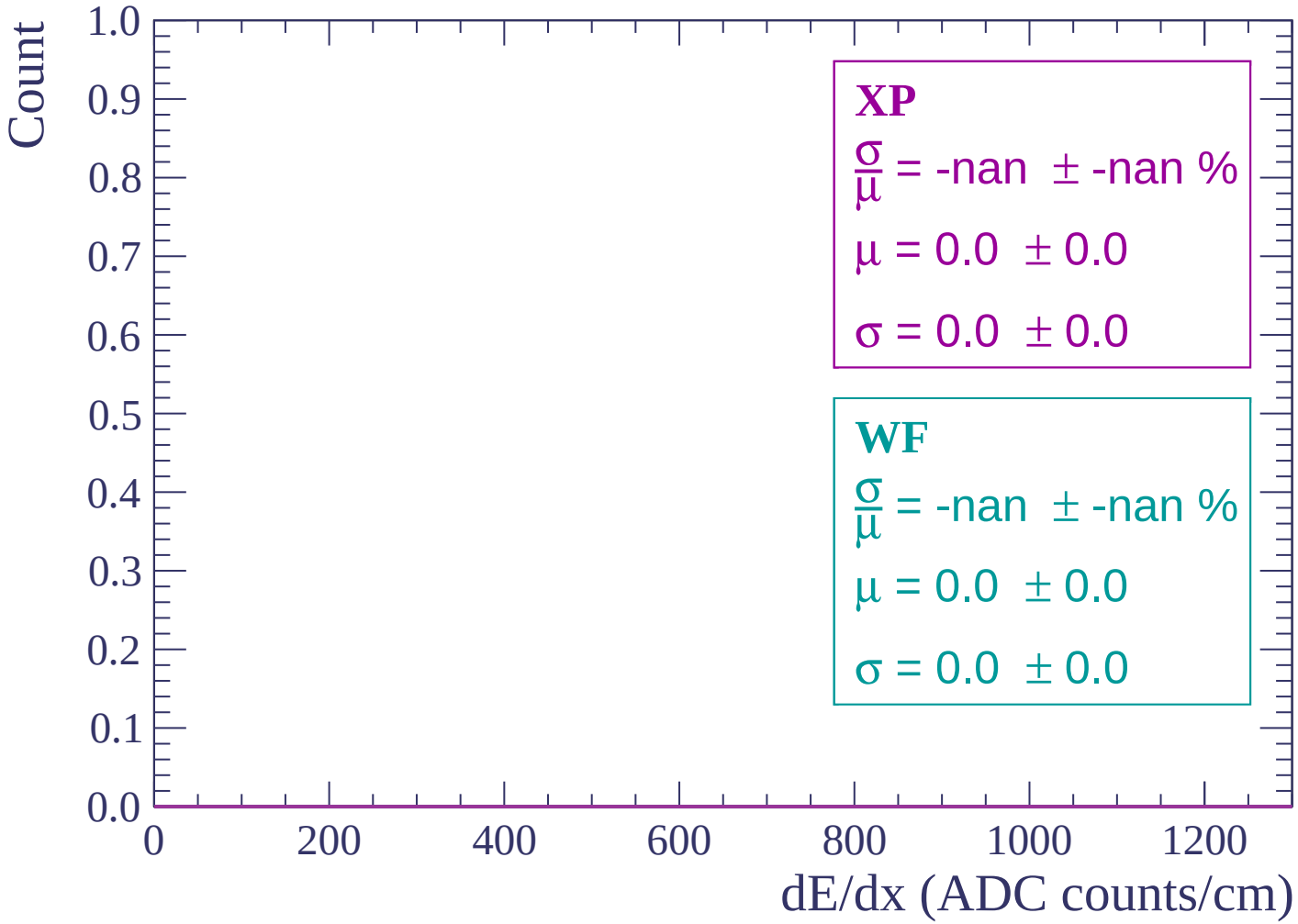
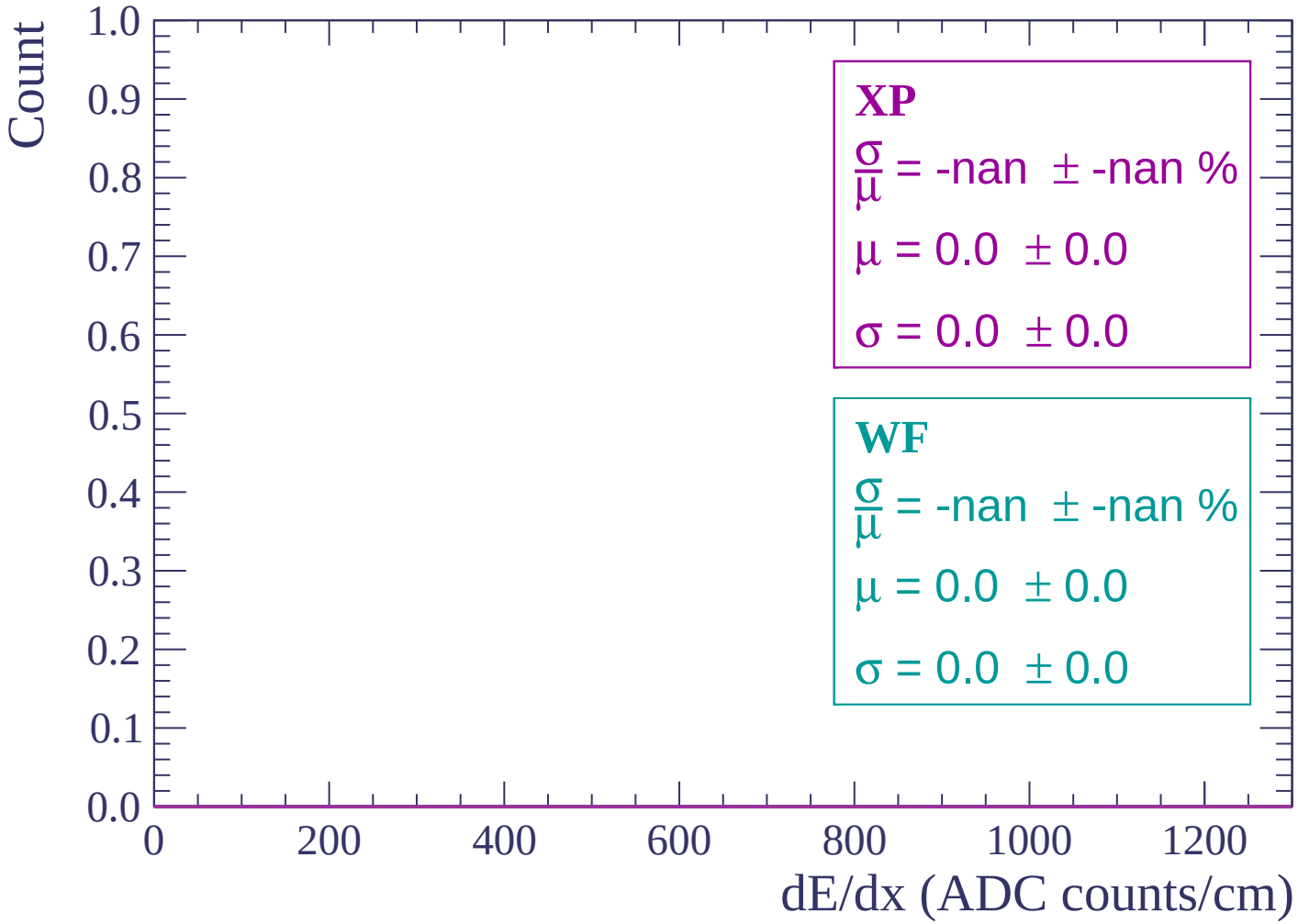
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

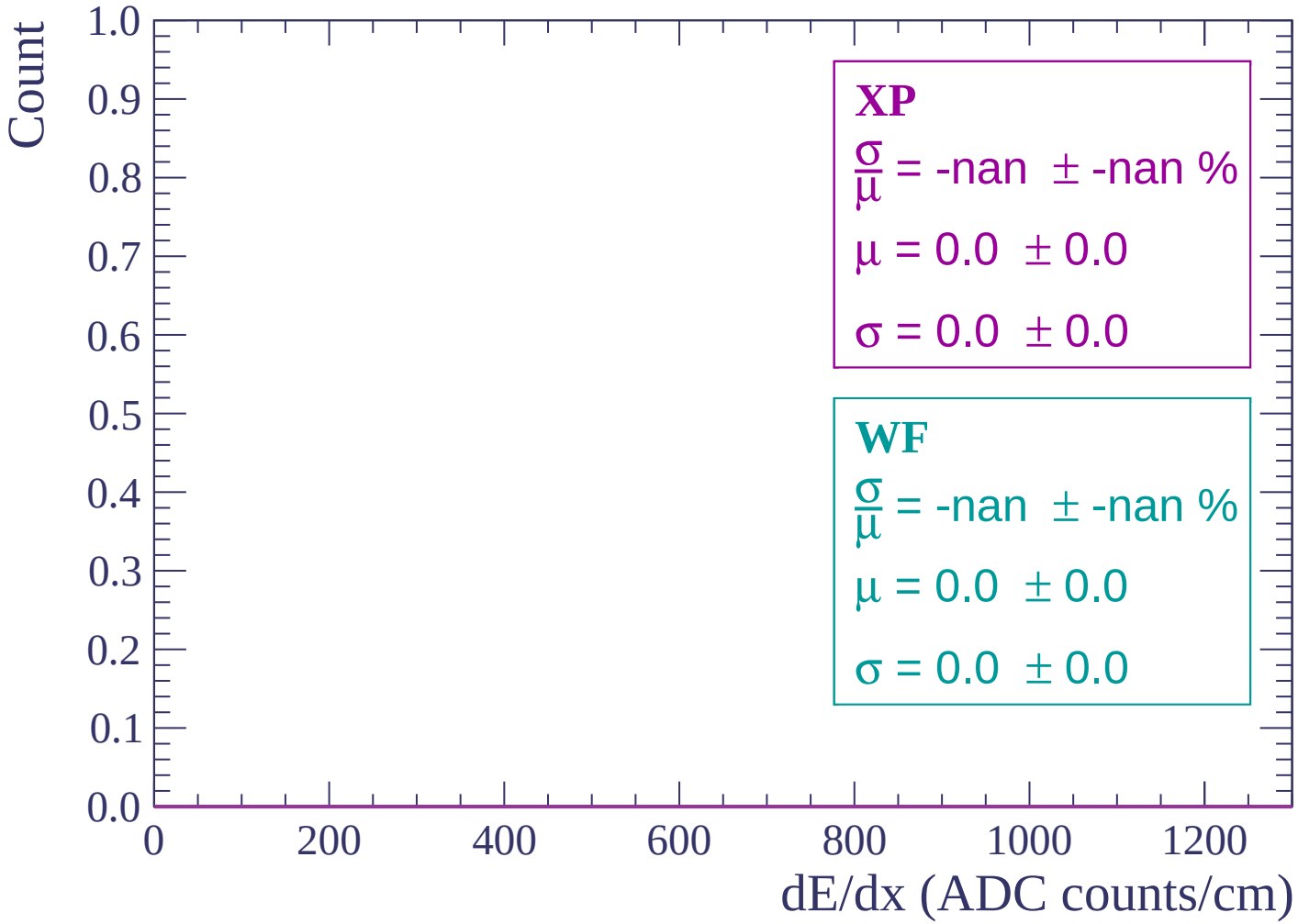

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

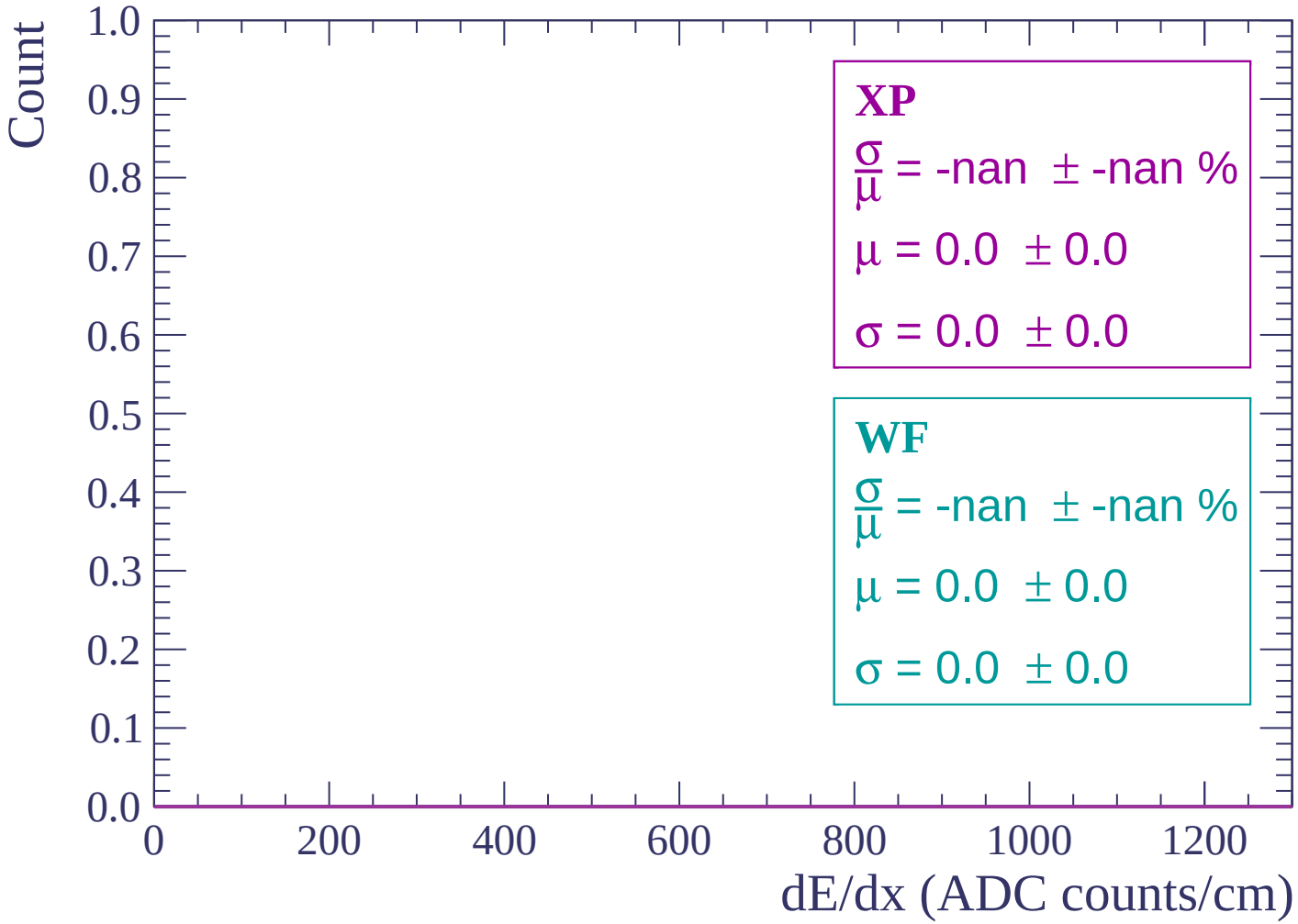

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

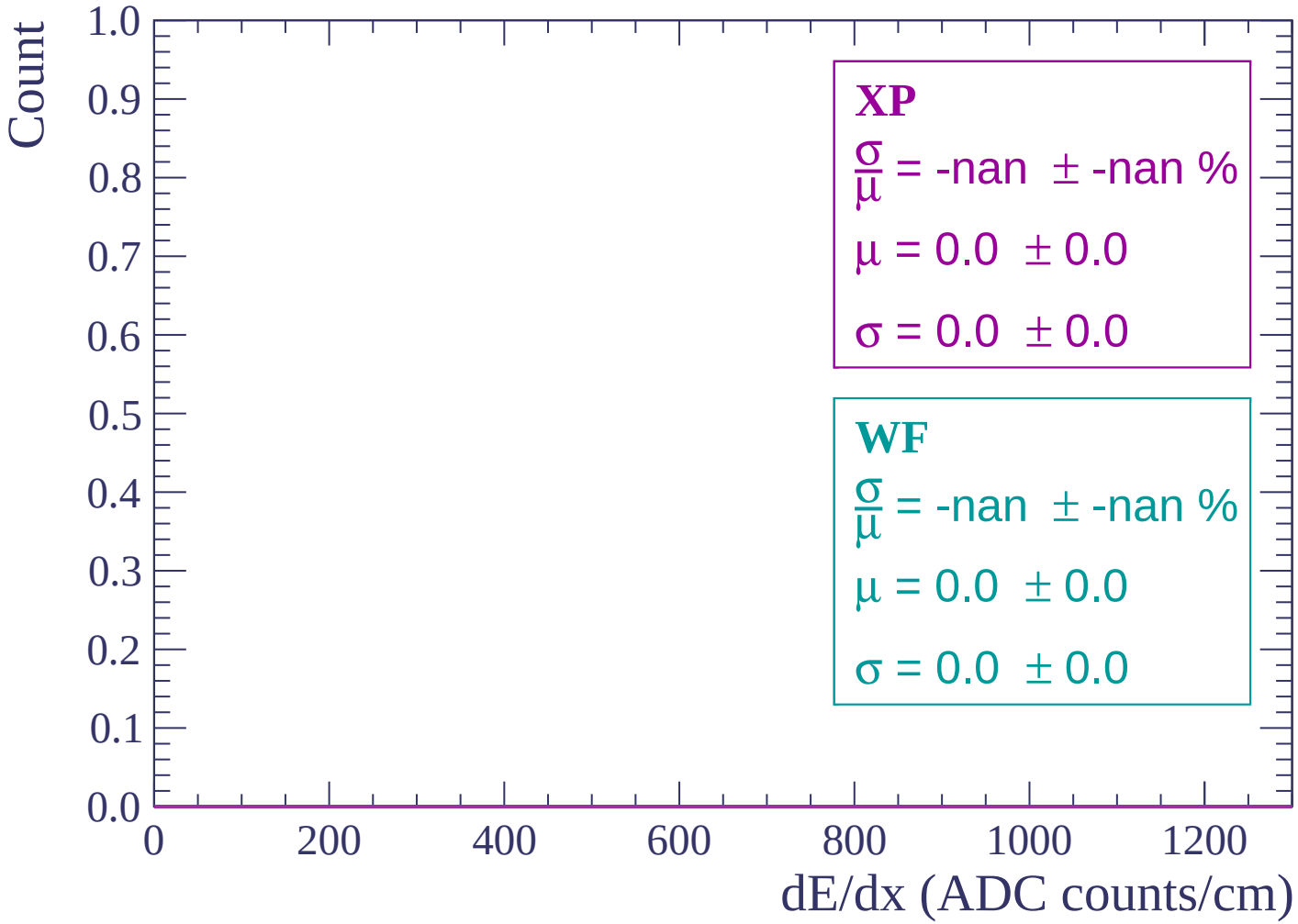

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

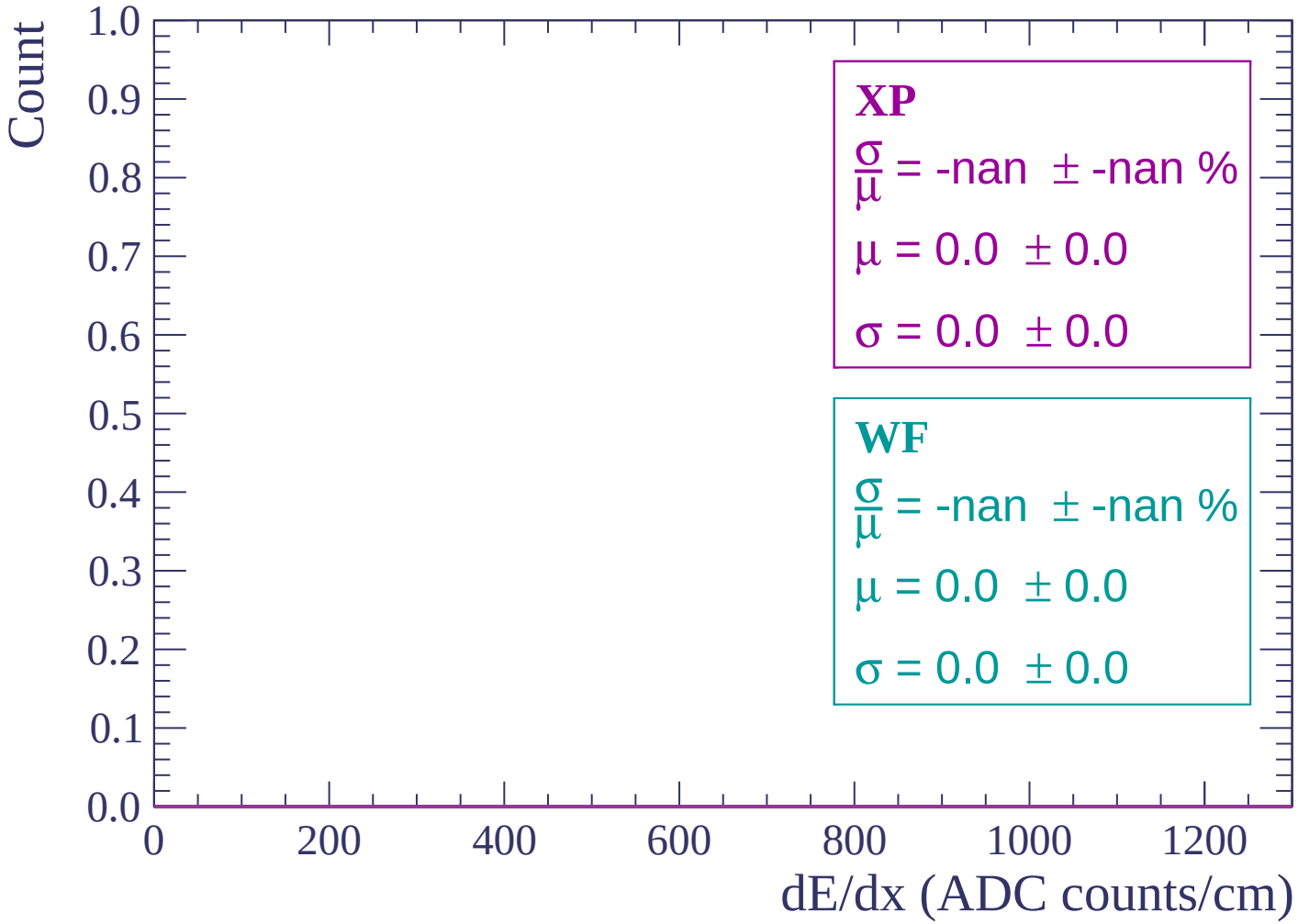

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

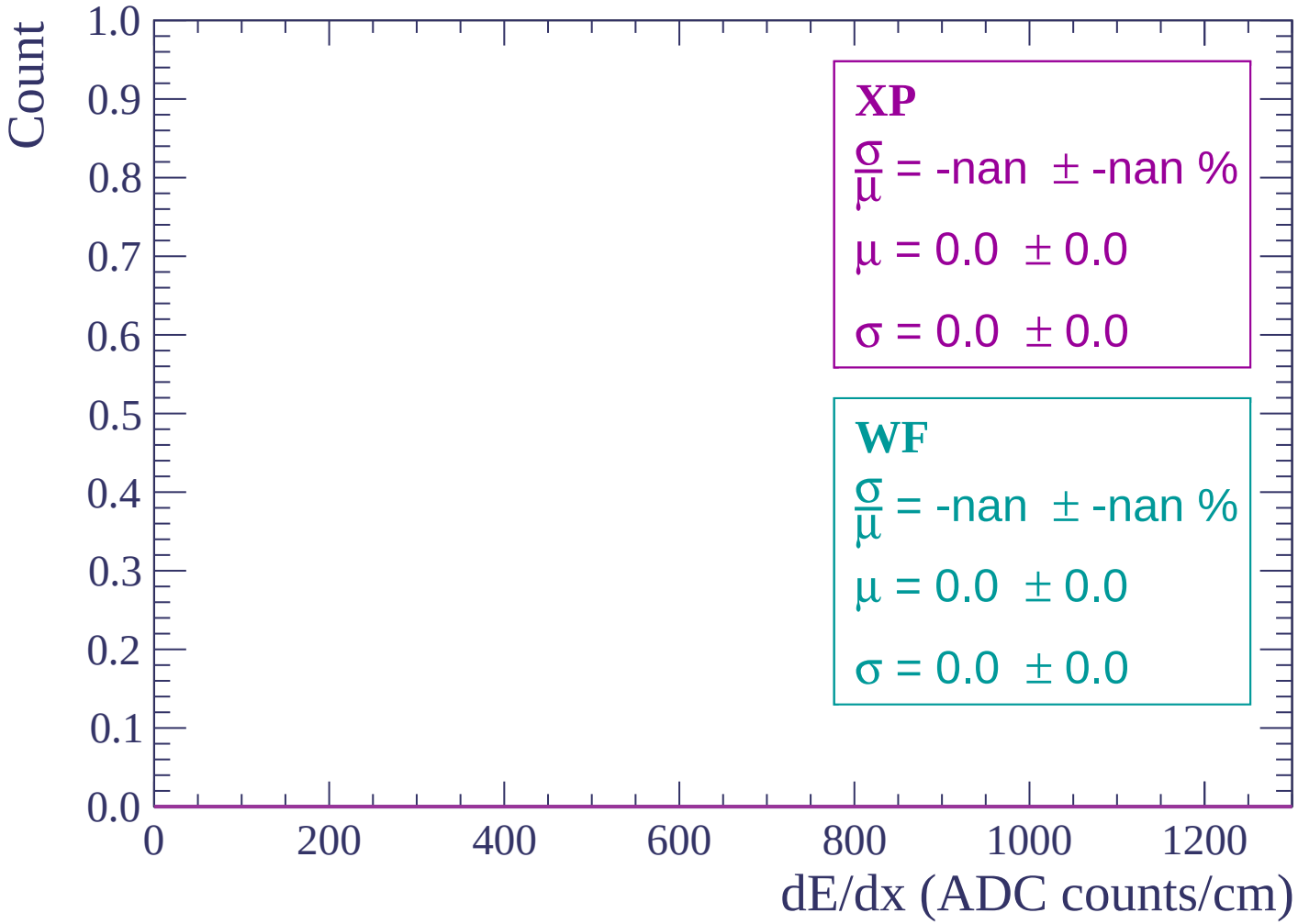

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

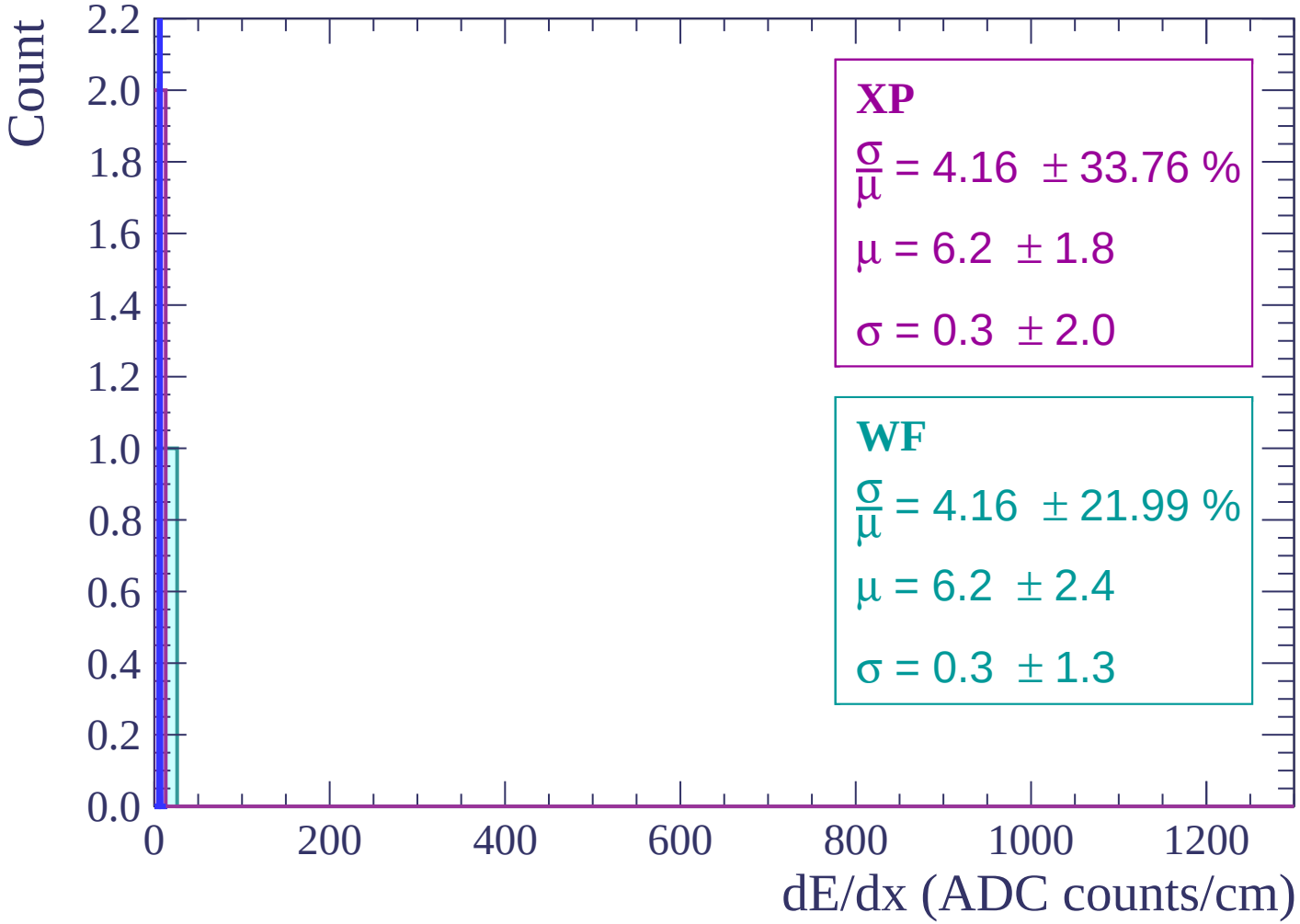

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

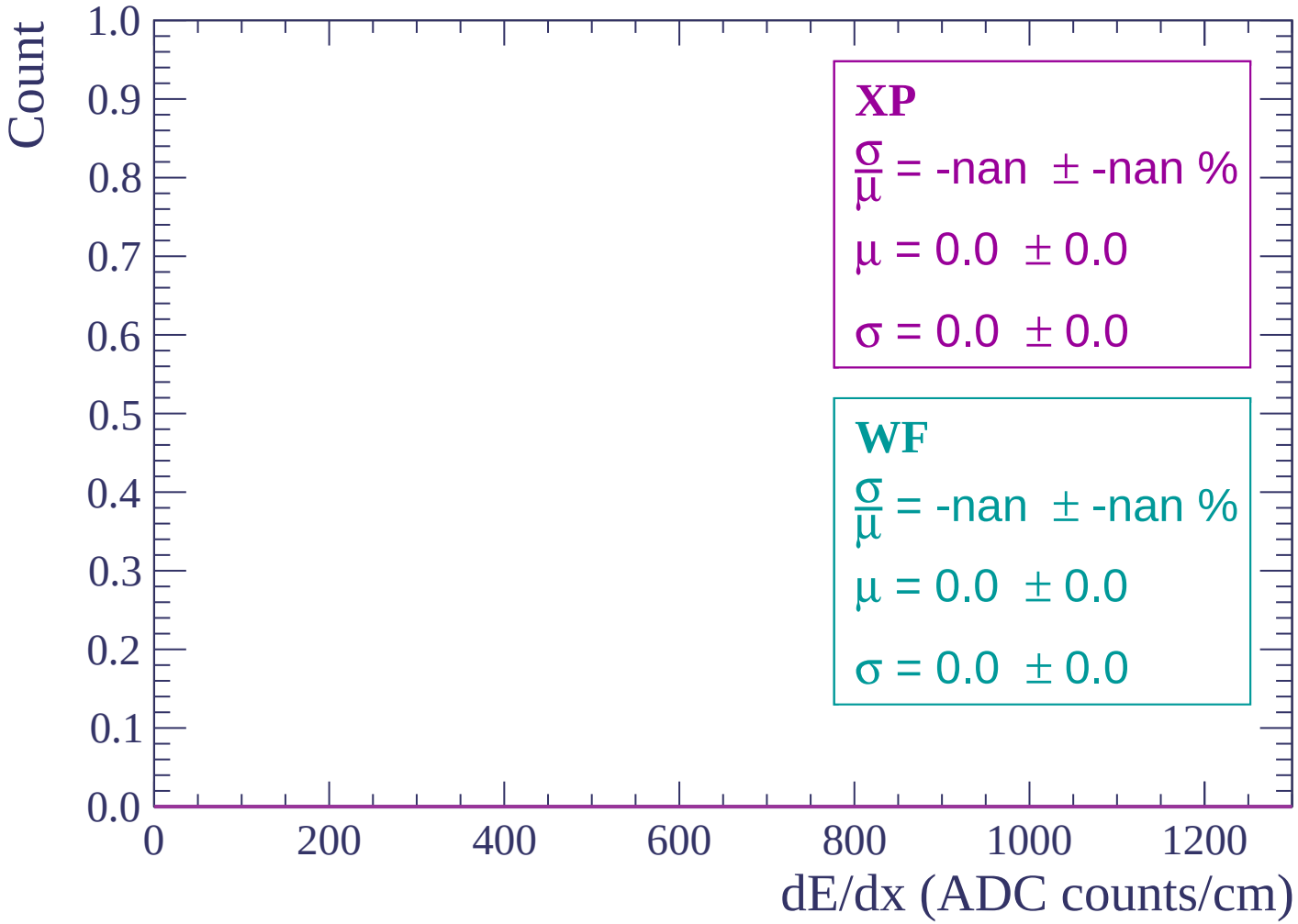

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

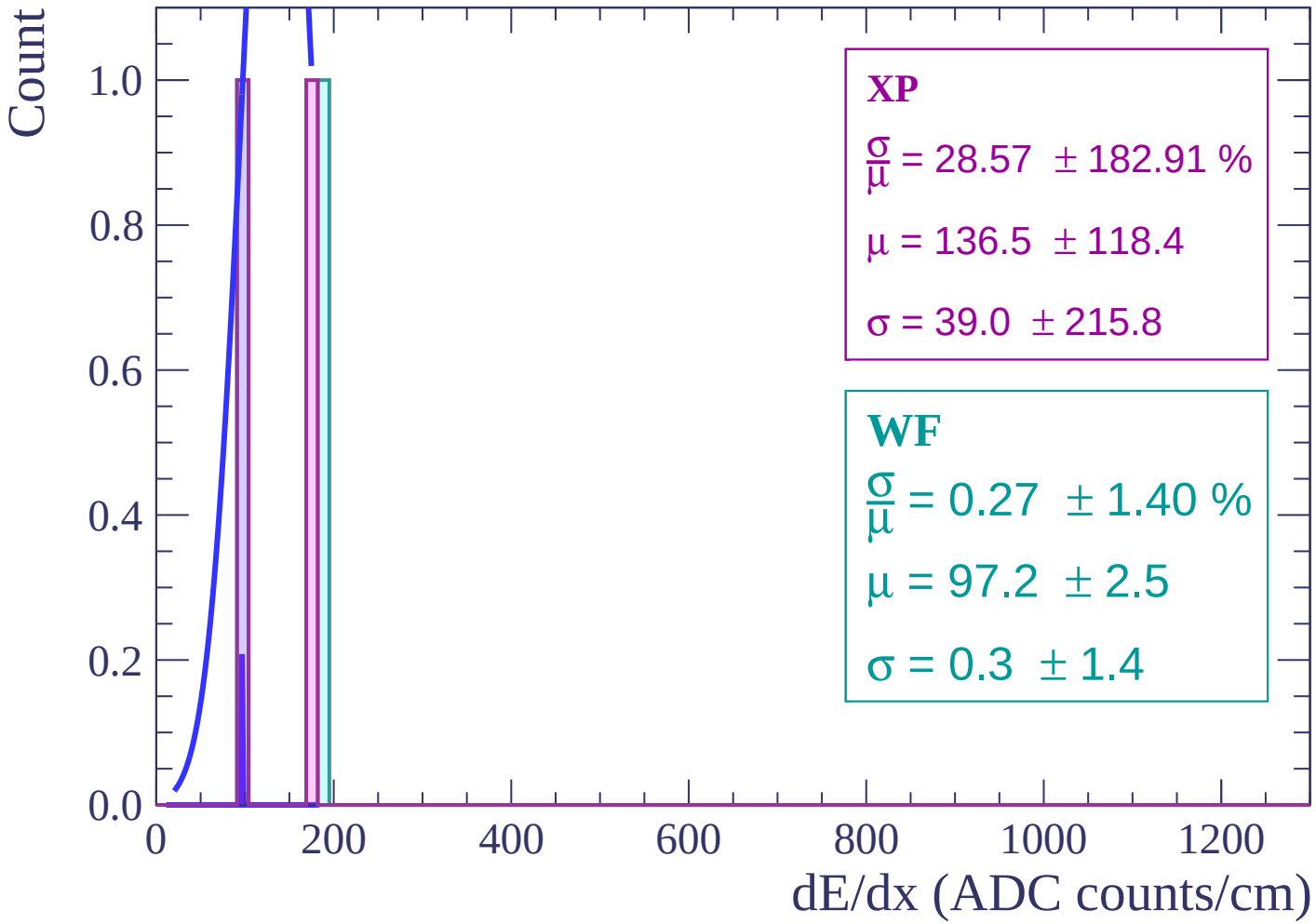

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

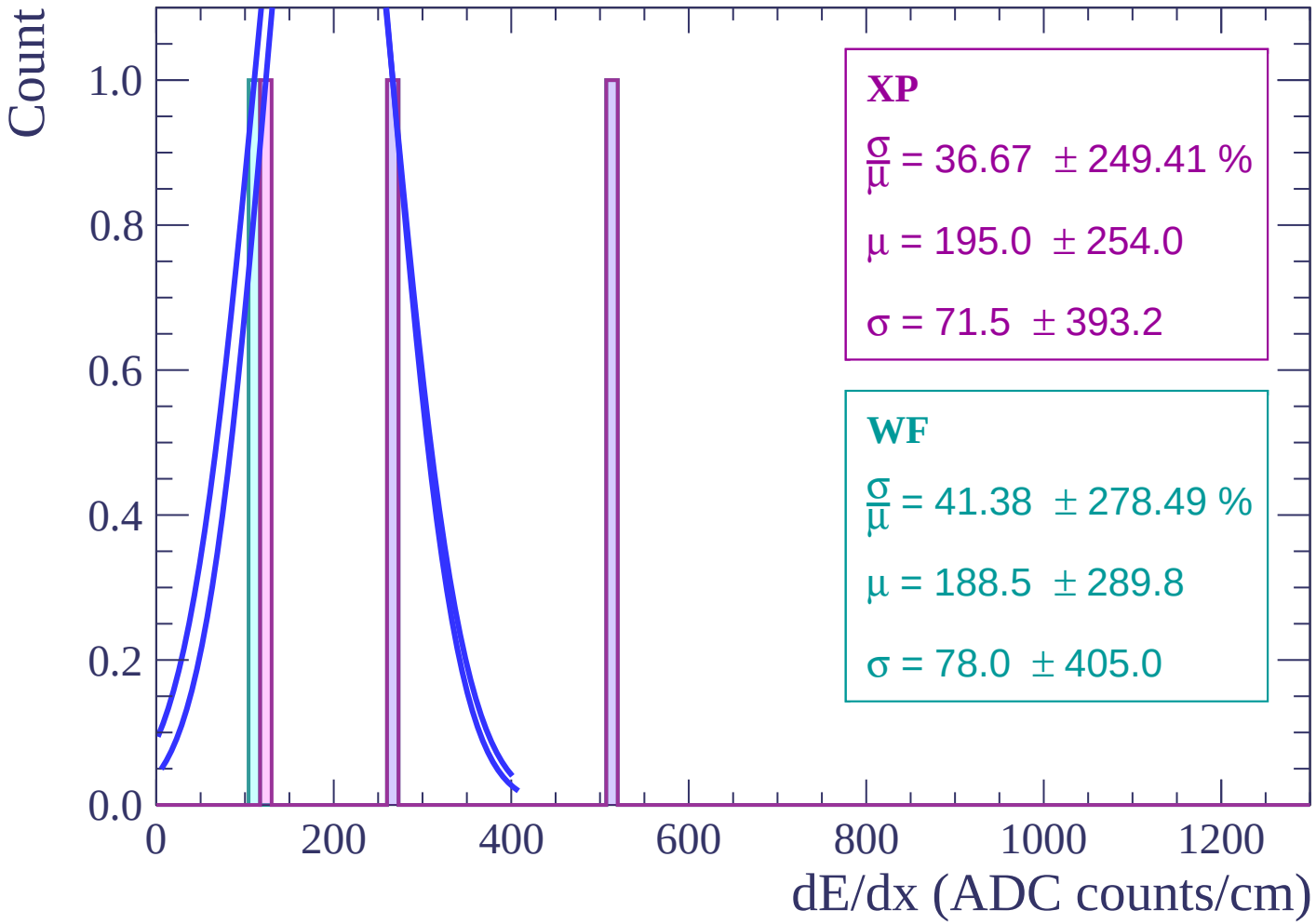

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

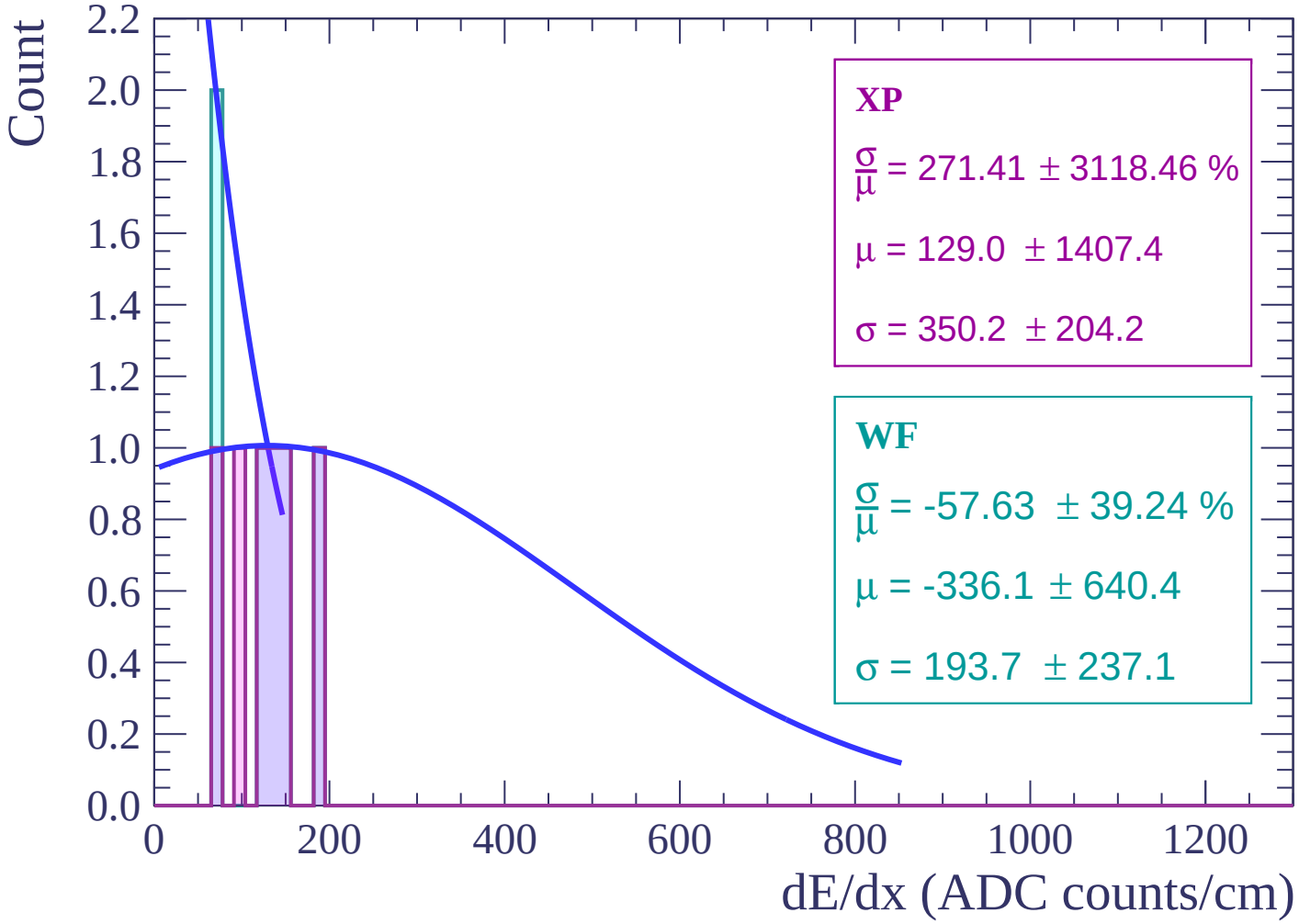

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

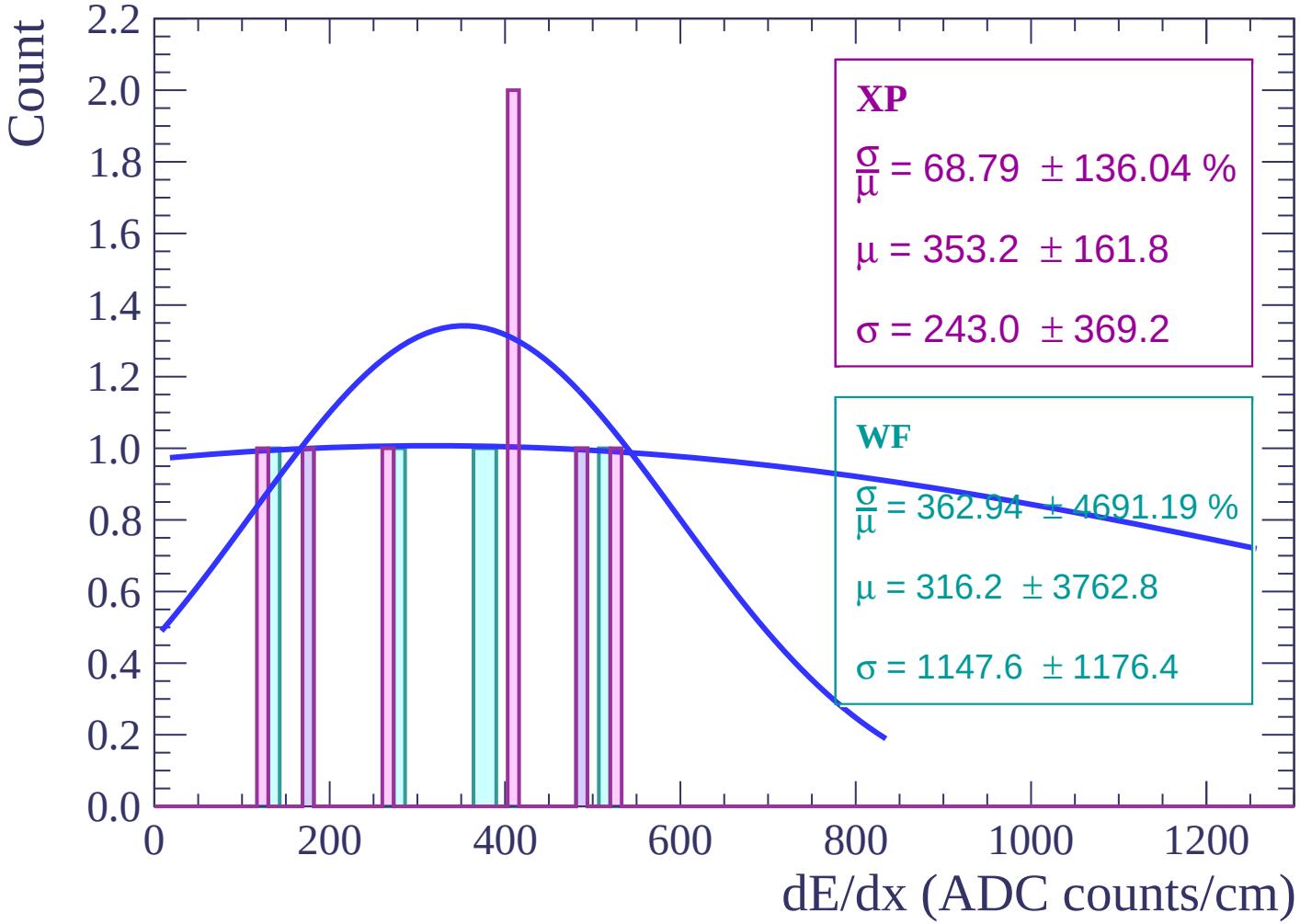

Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 4.16 ± 33.76 %
μ = 6.2 ± 1.8
σ = 0.3 ± 2.0
WF
σ/μ = 4.16 ± 21.99 %
μ = 6.2 ± 2.4
σ = 0.3 ± 1.3

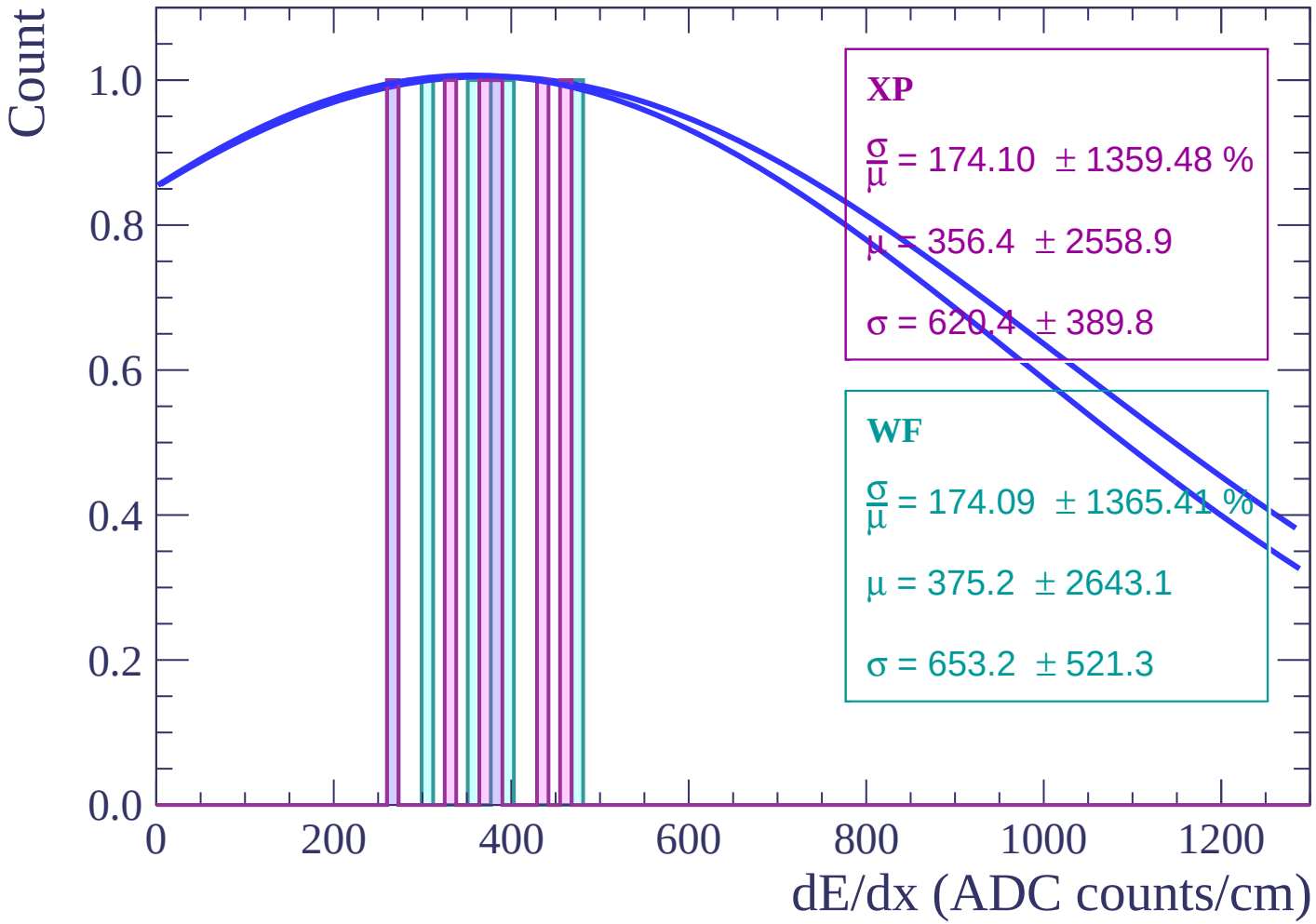

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

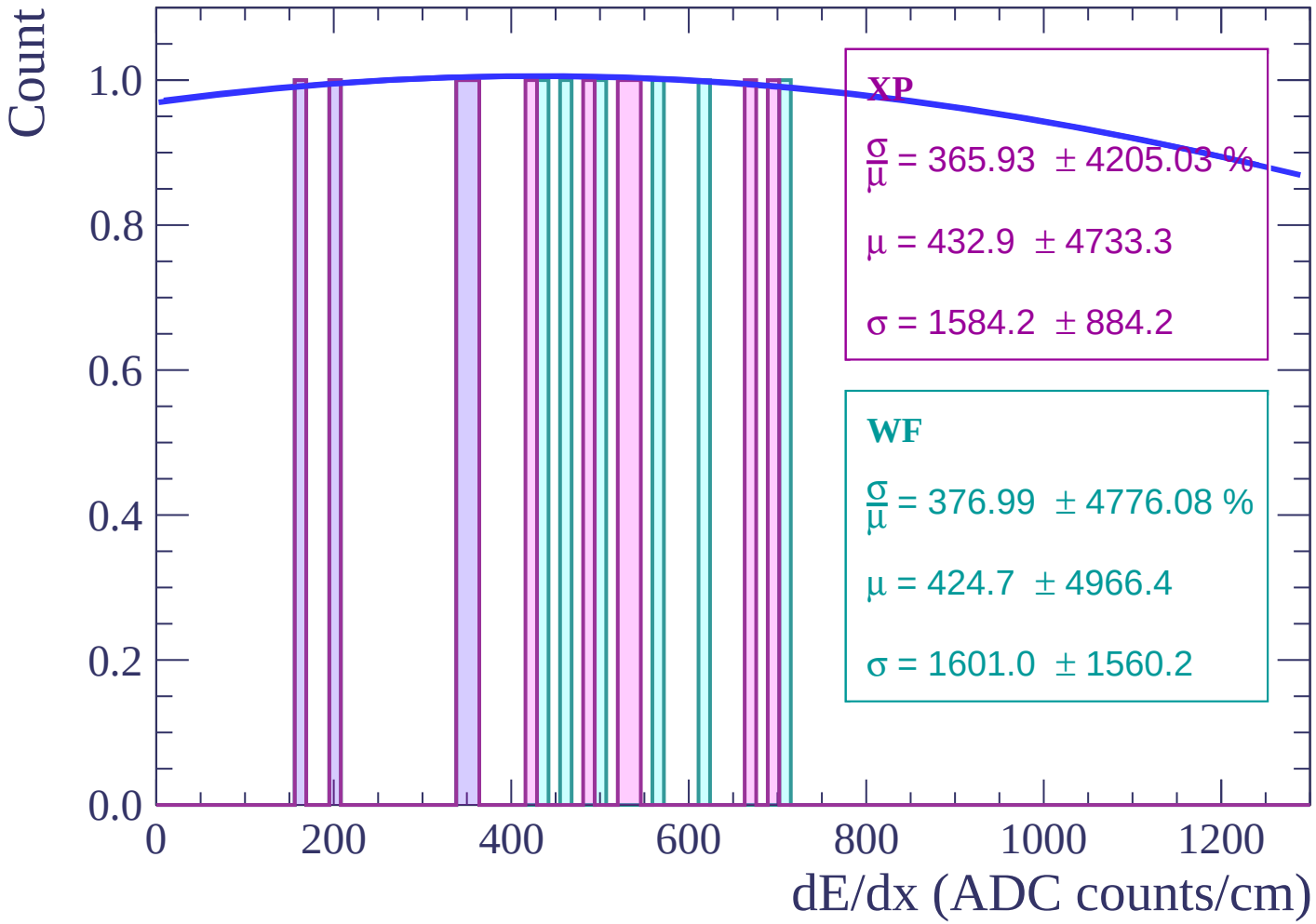

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 28.57 ± 182.91 %
μ = 136.5 ± 118.4
σ = 39.0 ± 215.8
WF
σ/μ = 0.27 ± 1.40 %
μ = 97.2 ± 2.5
σ = 0.3 ± 1.4

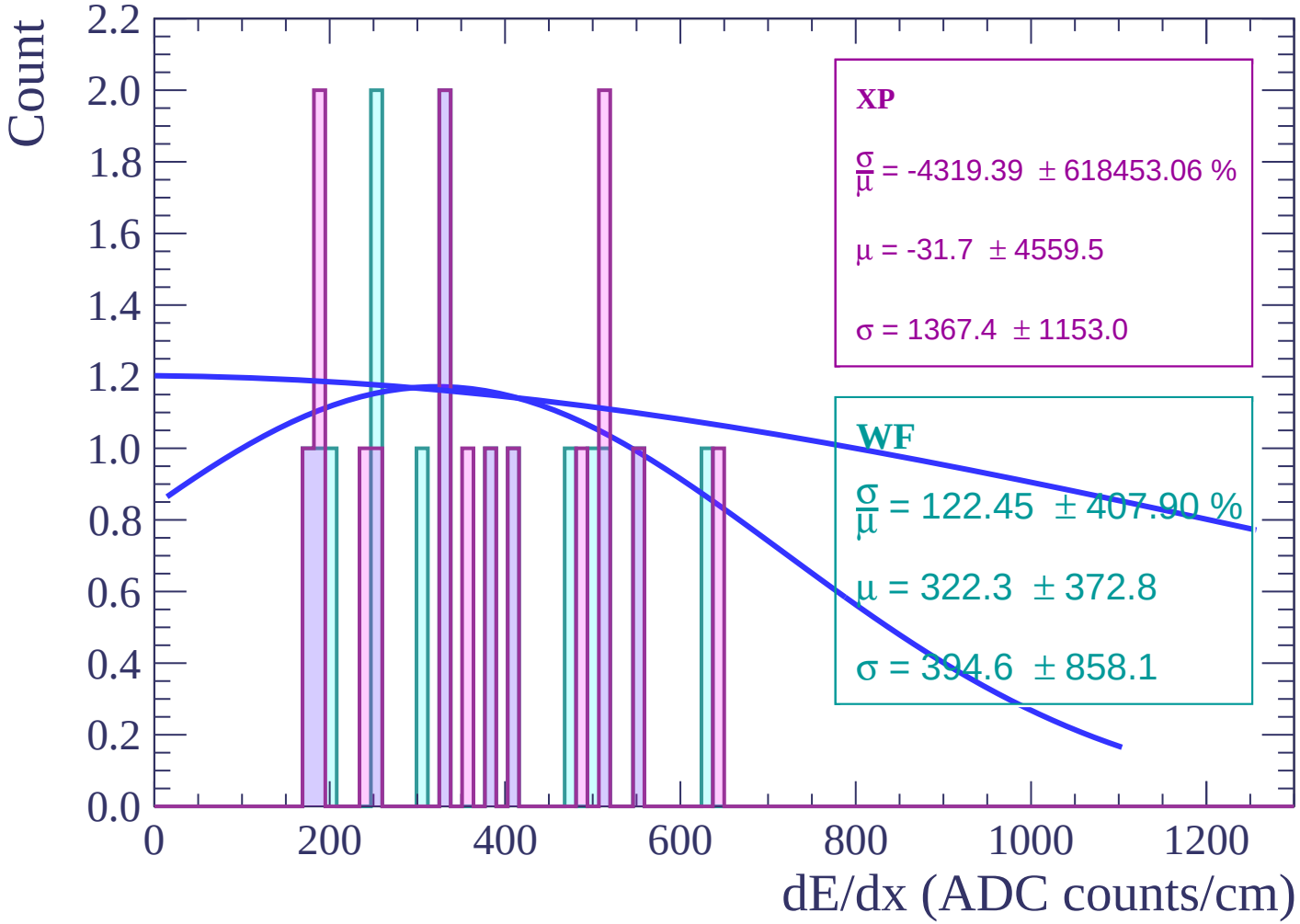

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 36.67 ± 249.41 %
μ = 195.0 ± 254.0
σ = 71.5 ± 393.2
WF
σ/μ = 41.38 ± 278.49 %
μ = 188.5 ± 289.8
σ = 78.0 ± 405.0

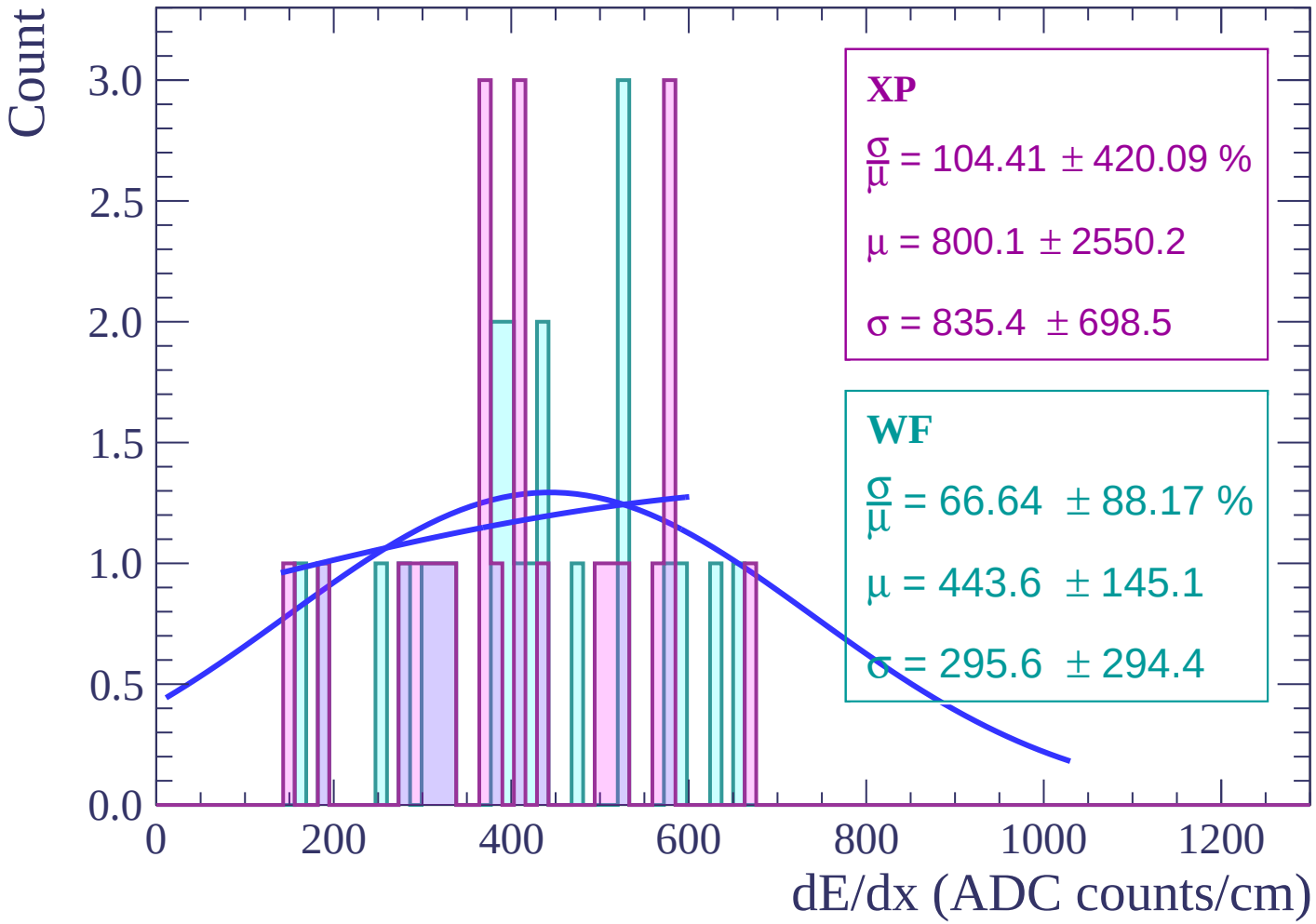

Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 271.41 ± 3118.46 %
μ = 129.0 ± 1407.4
σ = 350.2 ± 204.2
WF
σ/μ = -57.63 ± 39.24 %
μ = -336.1 ± 640.4
σ = 193.7 ± 237.1

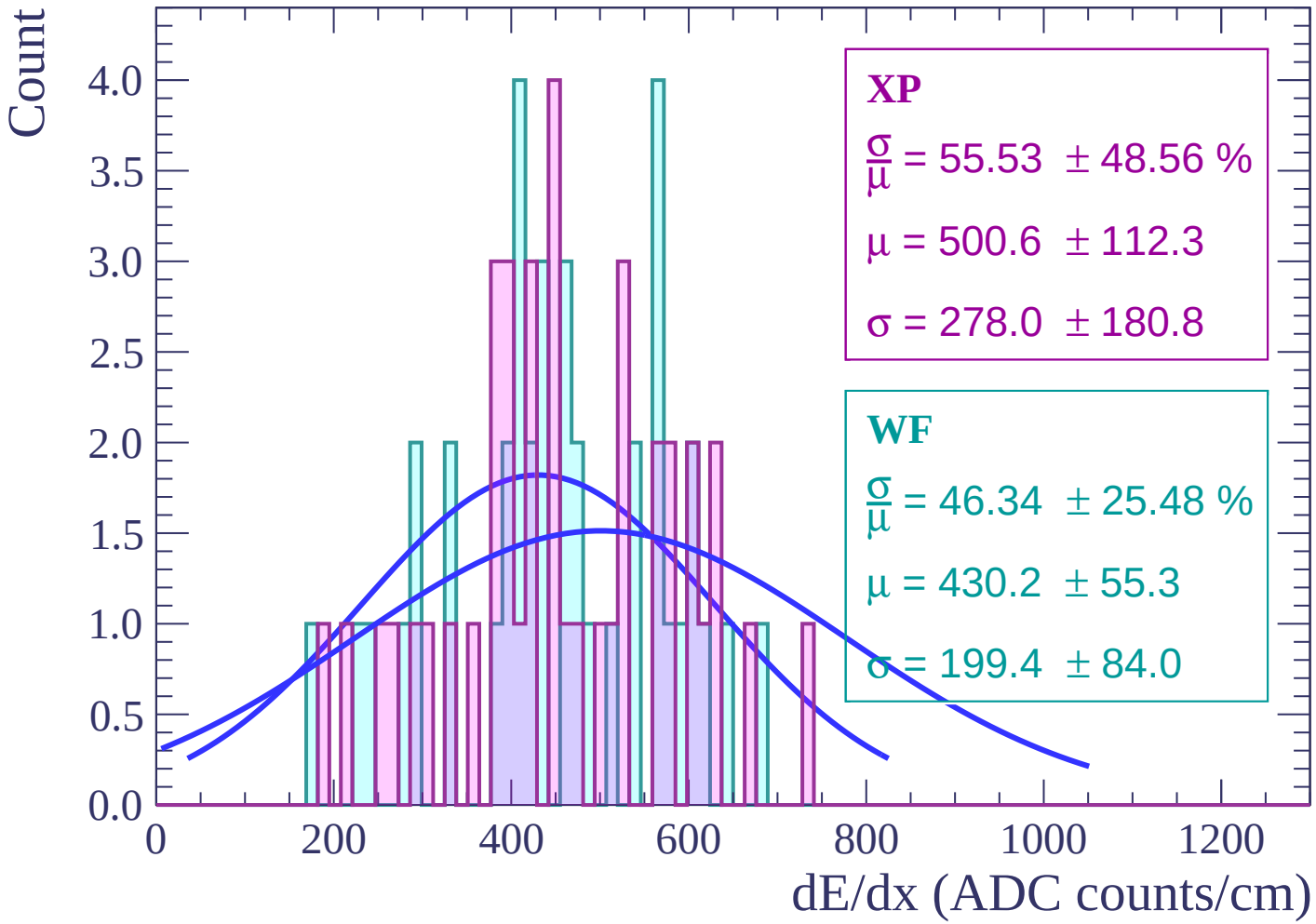

Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 68.79 ± 136.04 %
μ = 353.2 ± 161.8
σ = 243.0 ± 369.2
WF
σ/μ = 362.94 ± 4691.19 %
μ = 316.2 ± 3762.8
σ = 1147.6 ± 1176.4

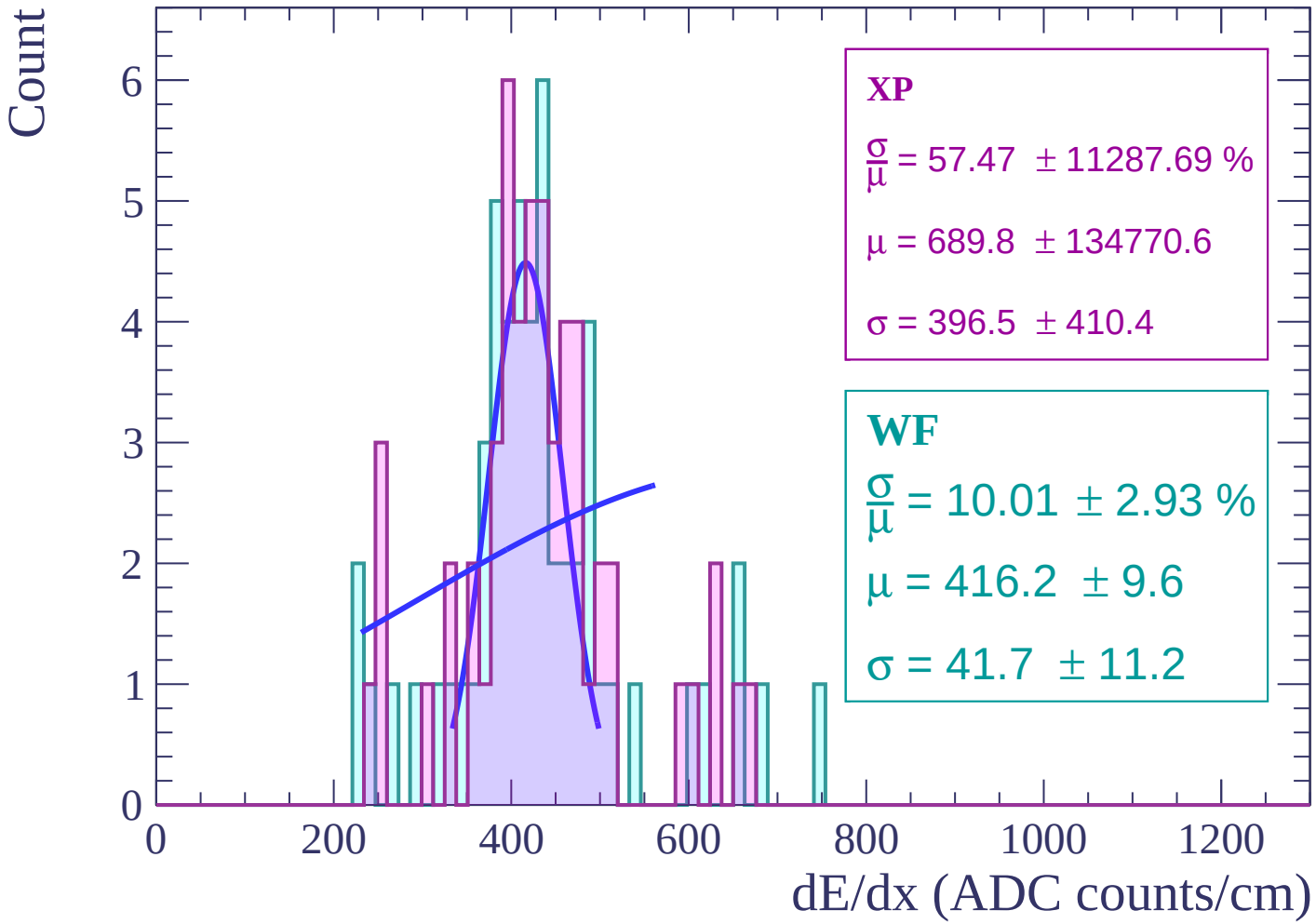

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 174.10 ± 1359.48 %
μ = 356.4 ± 2558.9
σ = 620.4 ± 389.8
WF
σ/μ = 174.09 ± 1365.41 %
μ = 375.2 ± 2643.1
σ = 653.2 ± 521.3

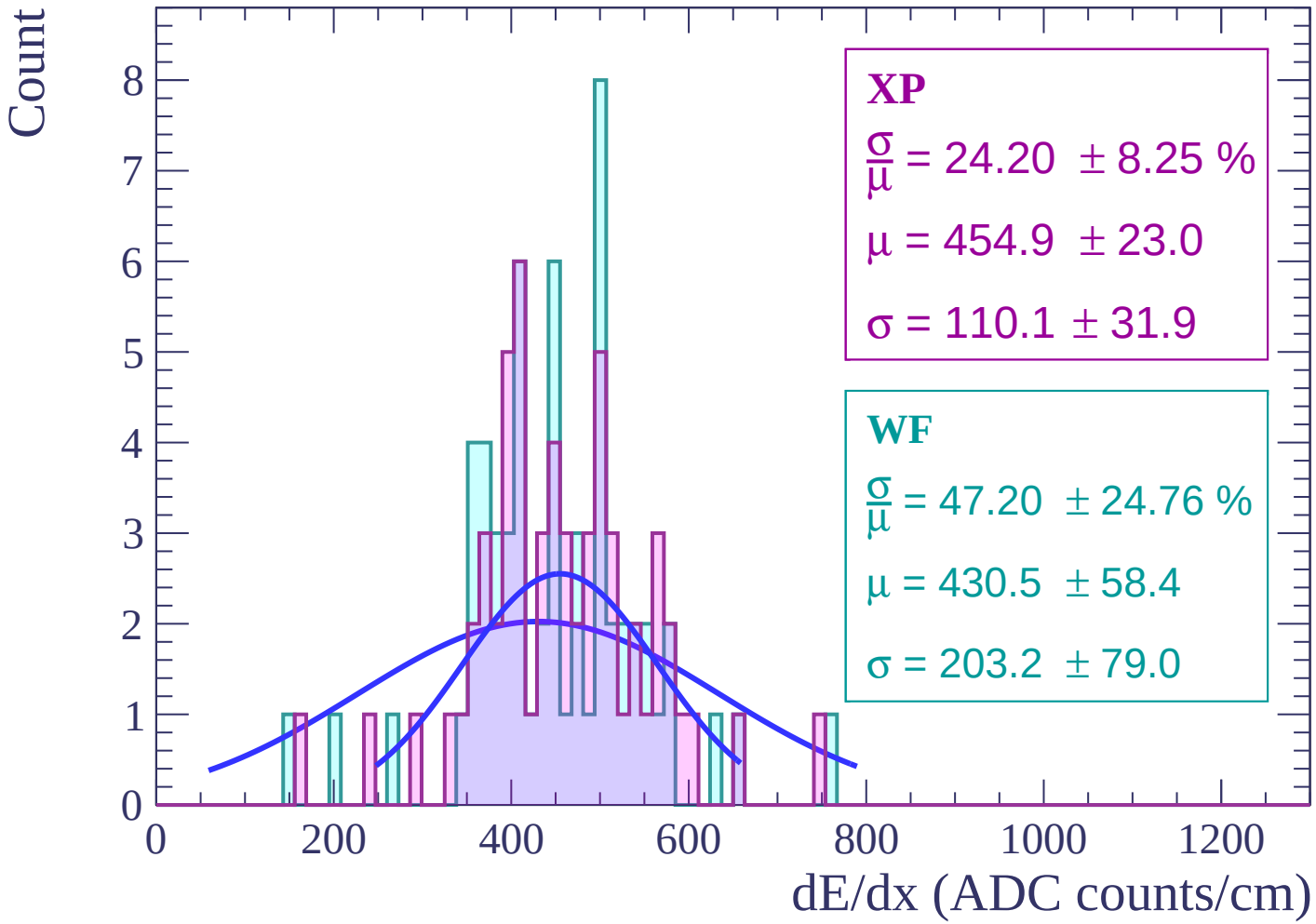

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 365.93 ± 4205.03 %
μ = 432.9 ± 4733.3
σ = 1584.2 ± 884.2
WF
σ/μ = 376.99 ± 4776.08 %
μ = 424.7 ± 4966.4
σ = 1601.0 ± 1560.2

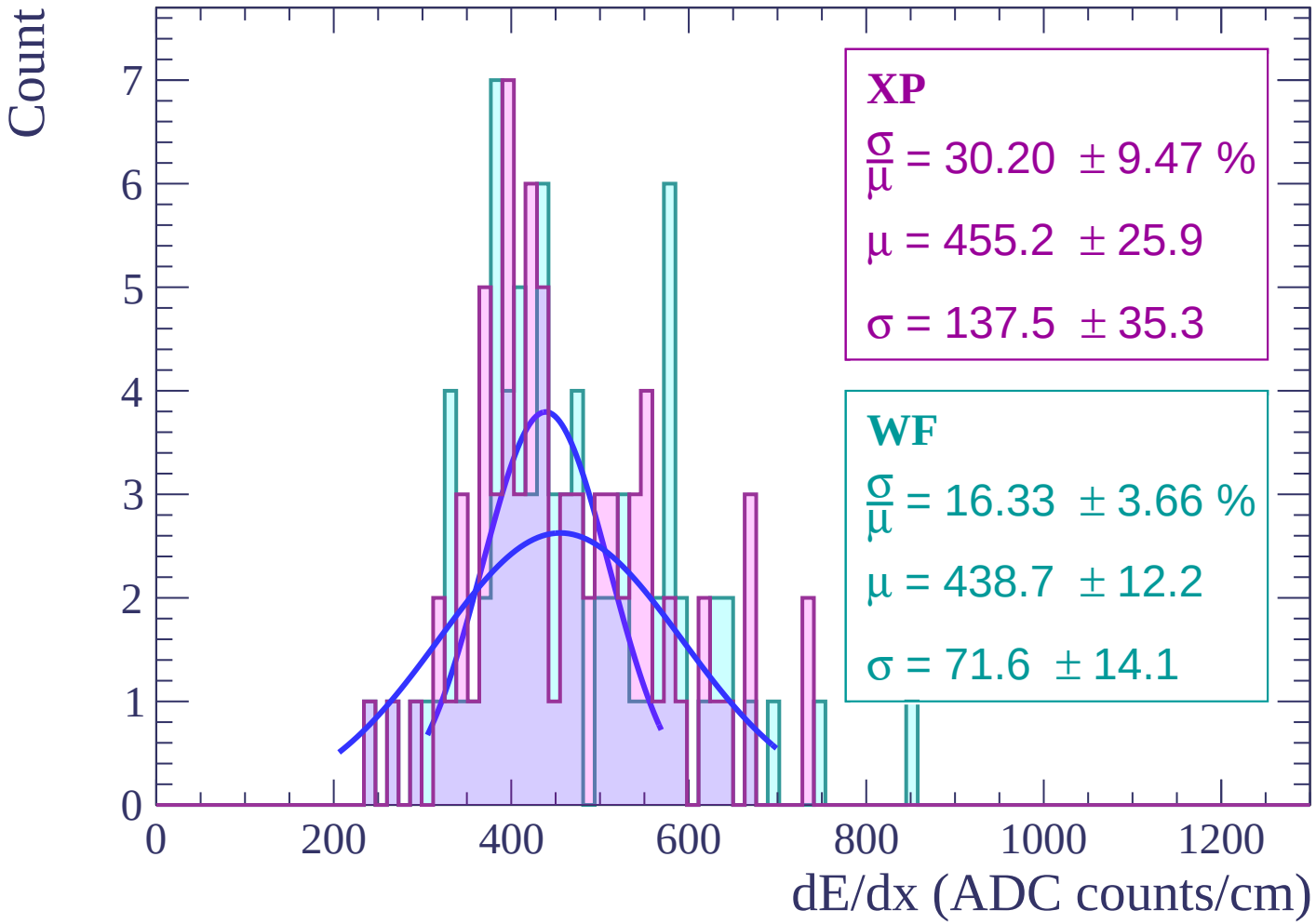

Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -4319.39 ± 618453.06 %
μ = -31.7 ± 4559.5
σ = 1367.4 ± 1153.0
WF
σ/μ = 122.45 ± 407.90 %
μ = 322.3 ± 372.8
σ = 394.6 ± 858.1

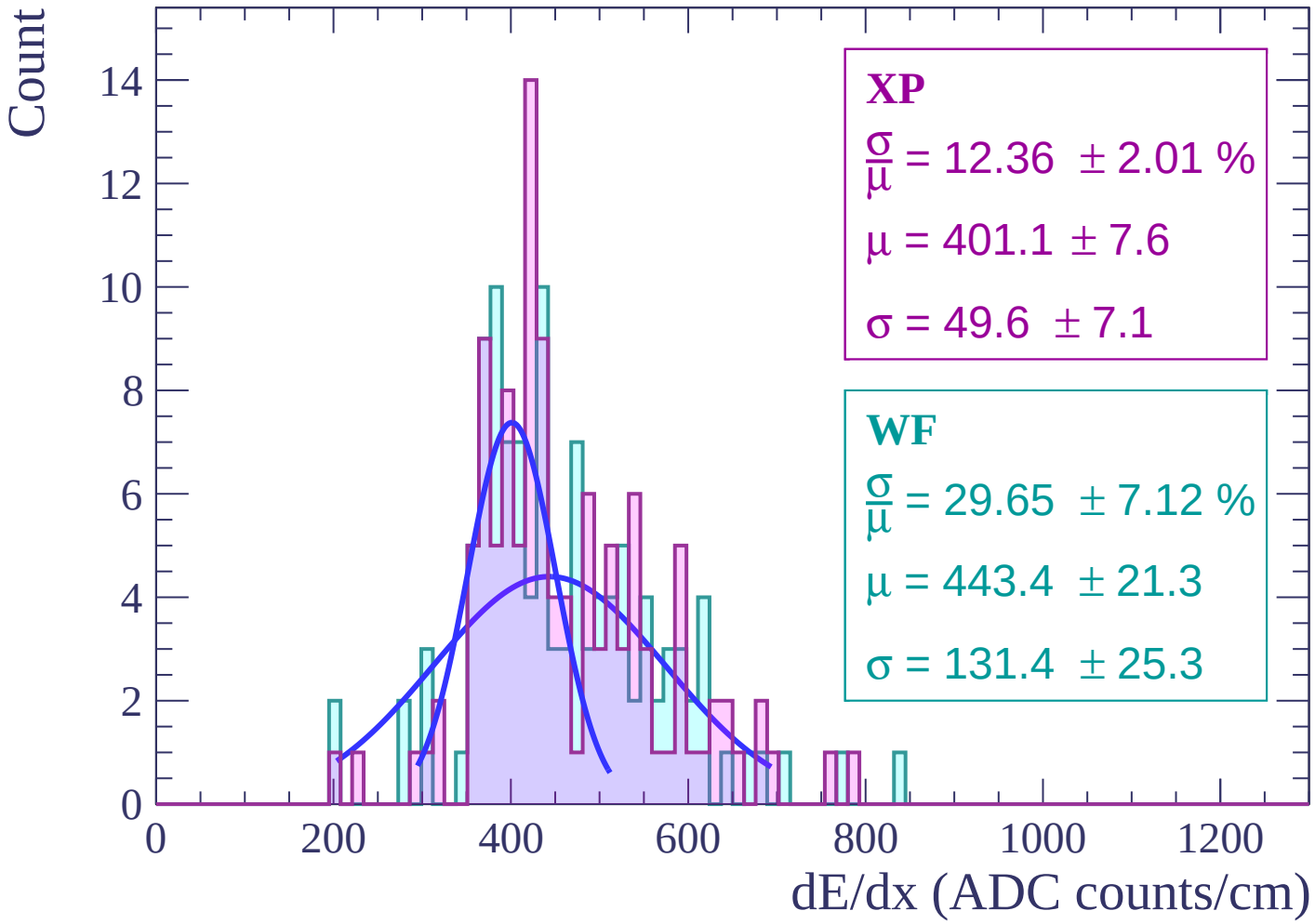

Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 104.41 ± 420.09 %
μ = 800.1 ± 2550.2
σ = 835.4 ± 698.5
WF
σ/μ = 66.64 ± 88.17 %
μ = 443.6 ± 145.1
σ = 295.6 ± 294.4

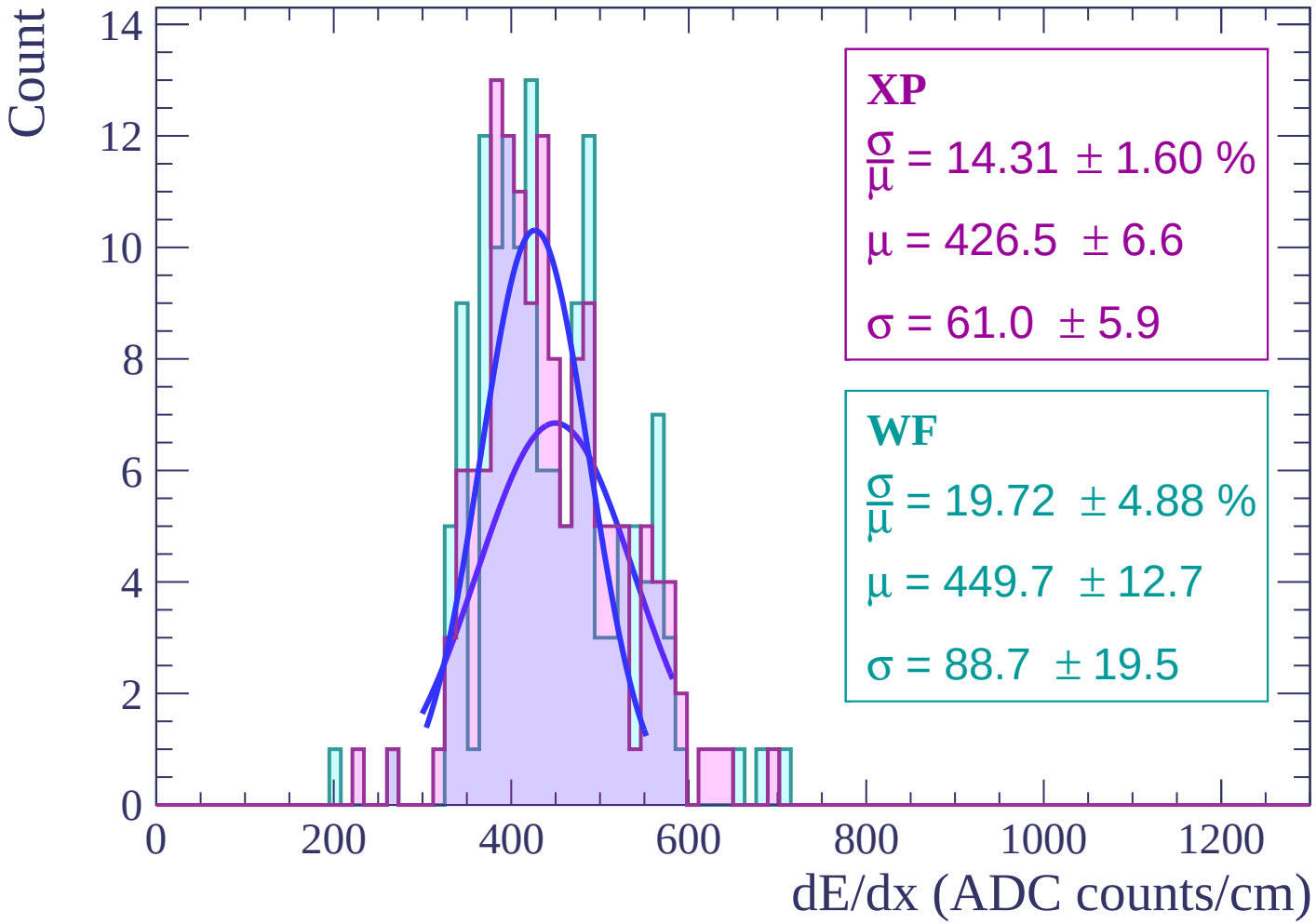

Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 55.53 ± 48.56 %
μ = 500.6 ± 112.3
σ = 278.0 ± 180.8
WF
σ/μ = 46.34 ± 25.48 %
μ = 430.2 ± 55.3
σ = 199.4 ± 84.0

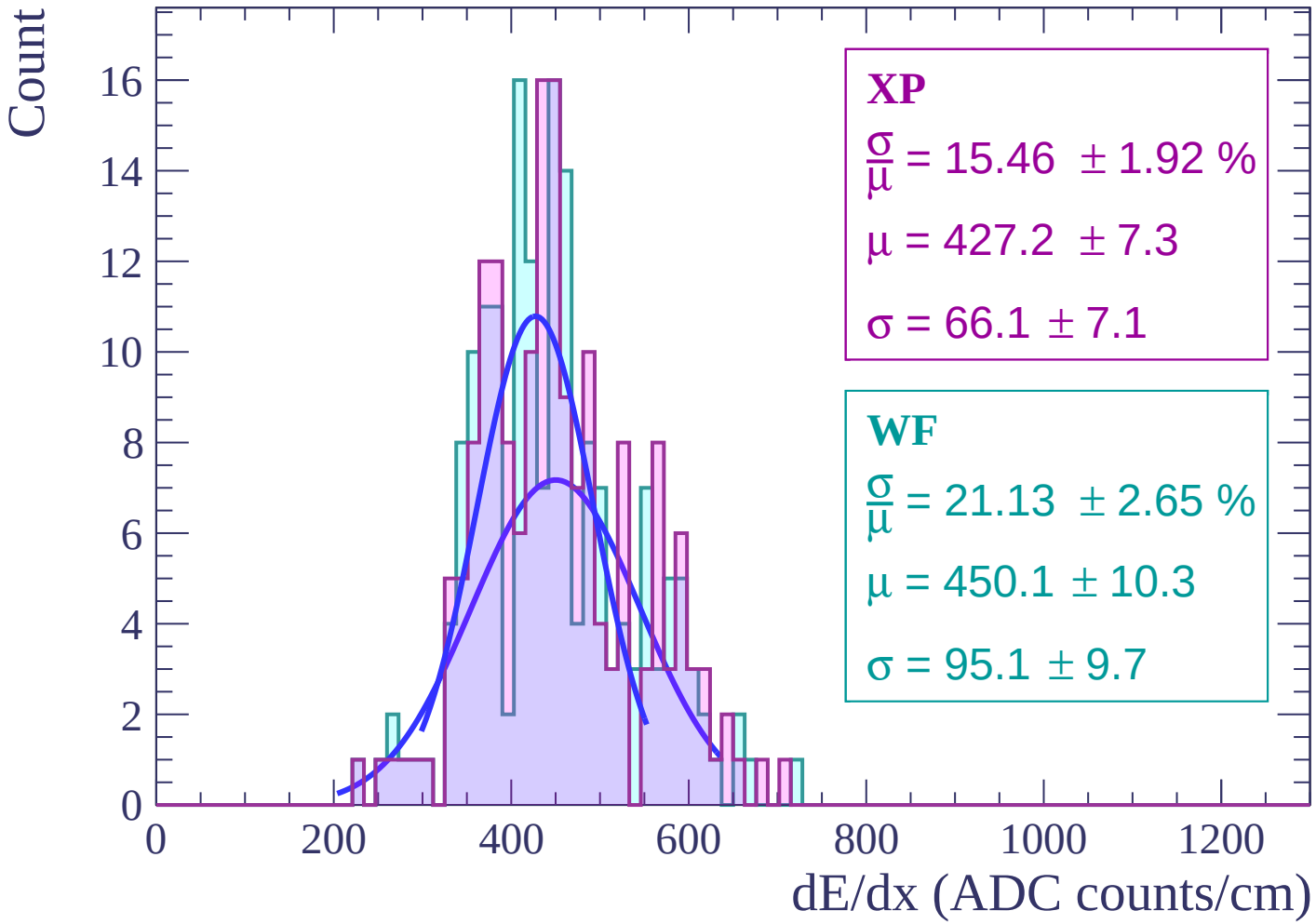

Count
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 57.47 ± 11287.69 %
μ = 689.8 ± 134770.6
σ = 396.5 ± 410.4
WF
σ/μ = 10.01 ± 2.93 %
μ = 416.2 ± 9.6
σ = 41.7 ± 11.2

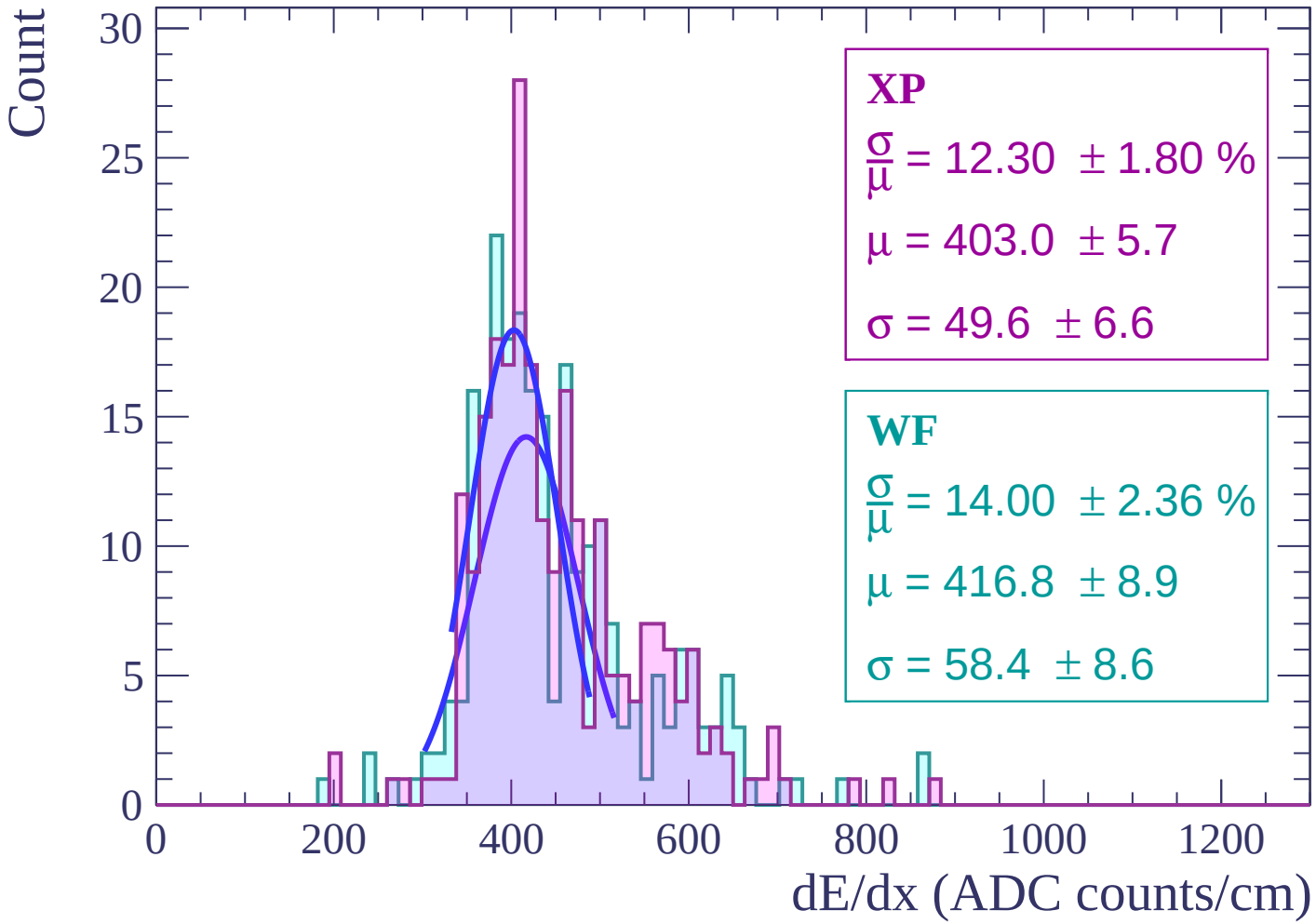

Count
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 24.20 ± 8.25 %
μ = 454.9 ± 23.0
σ = 110.1 ± 31.9
WF
σ/μ = 47.20 ± 24.76 %
μ = 430.5 ± 58.4
σ = 203.2 ± 79.0

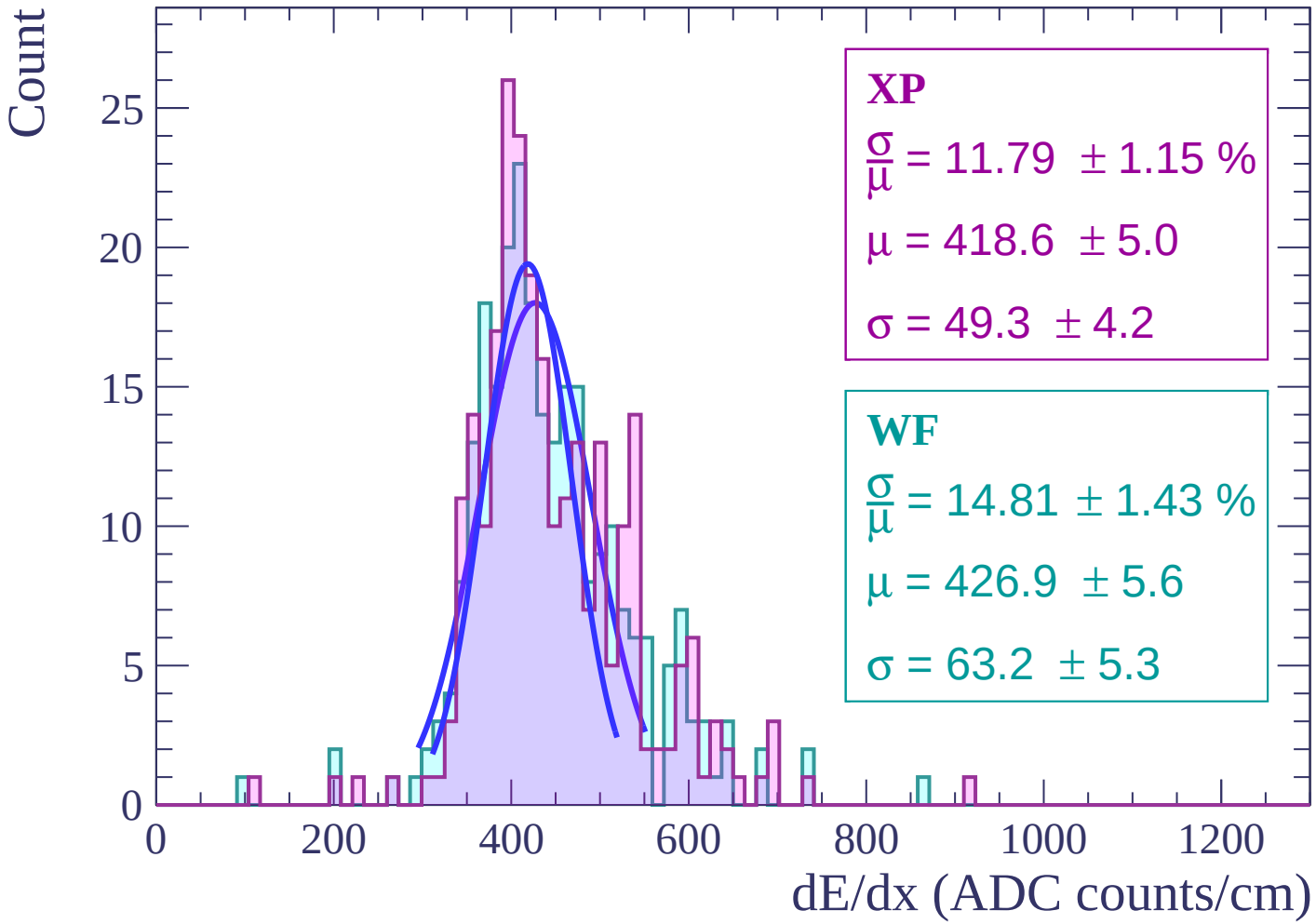

Count
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 30.20 ± 9.47 %
μ = 455.2 ± 25.9
σ = 137.5 ± 35.3
WF
σ/μ = 16.33 ± 3.66 %
μ = 438.7 ± 12.2
σ = 71.6 ± 14.1

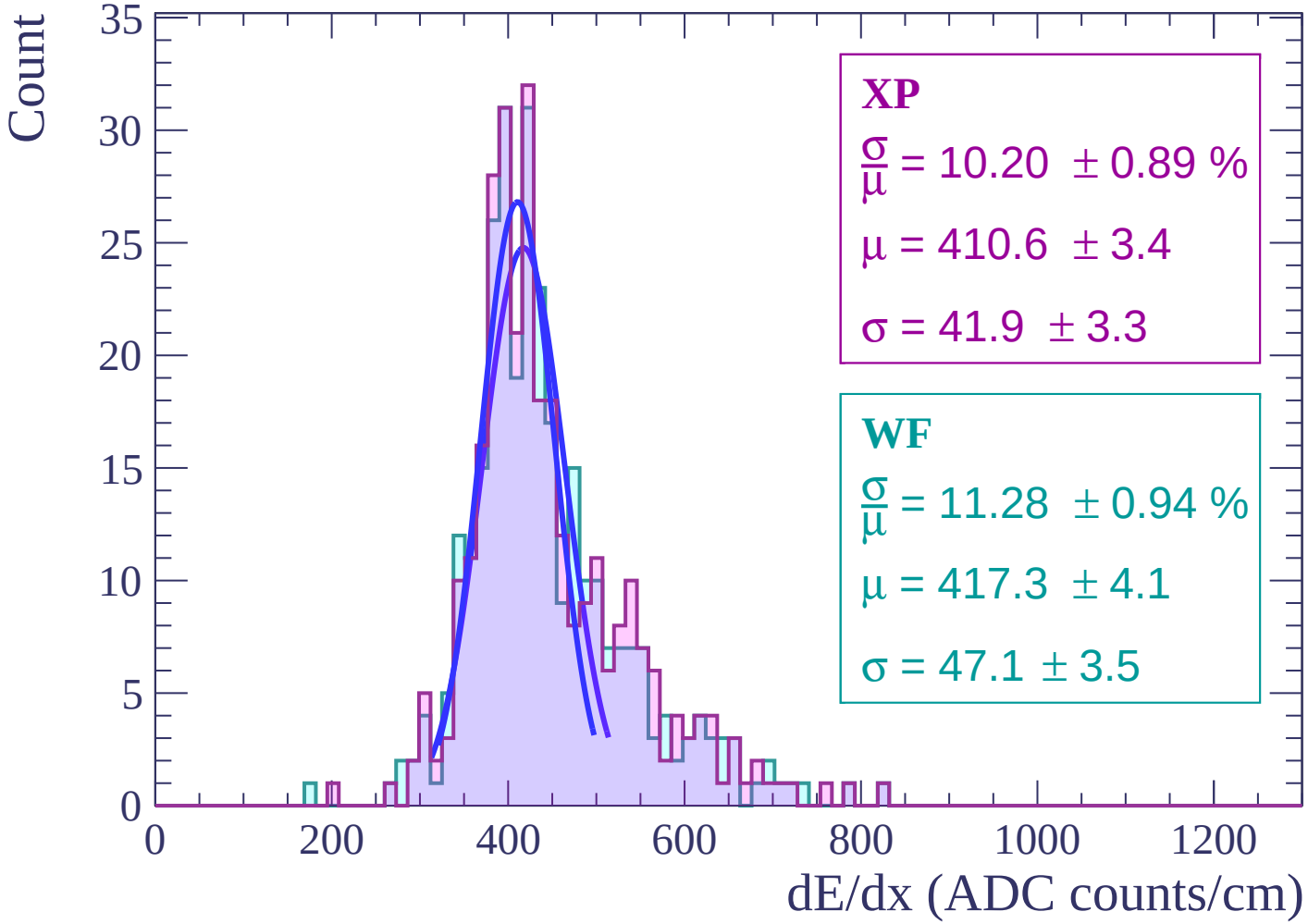

Count
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.36 ± 2.01 %
μ = 401.1 ± 7.6
σ = 49.6 ± 7.1
WF
σ/μ = 29.65 ± 7.12 %
μ = 443.4 ± 21.3
σ = 131.4 ± 25.3

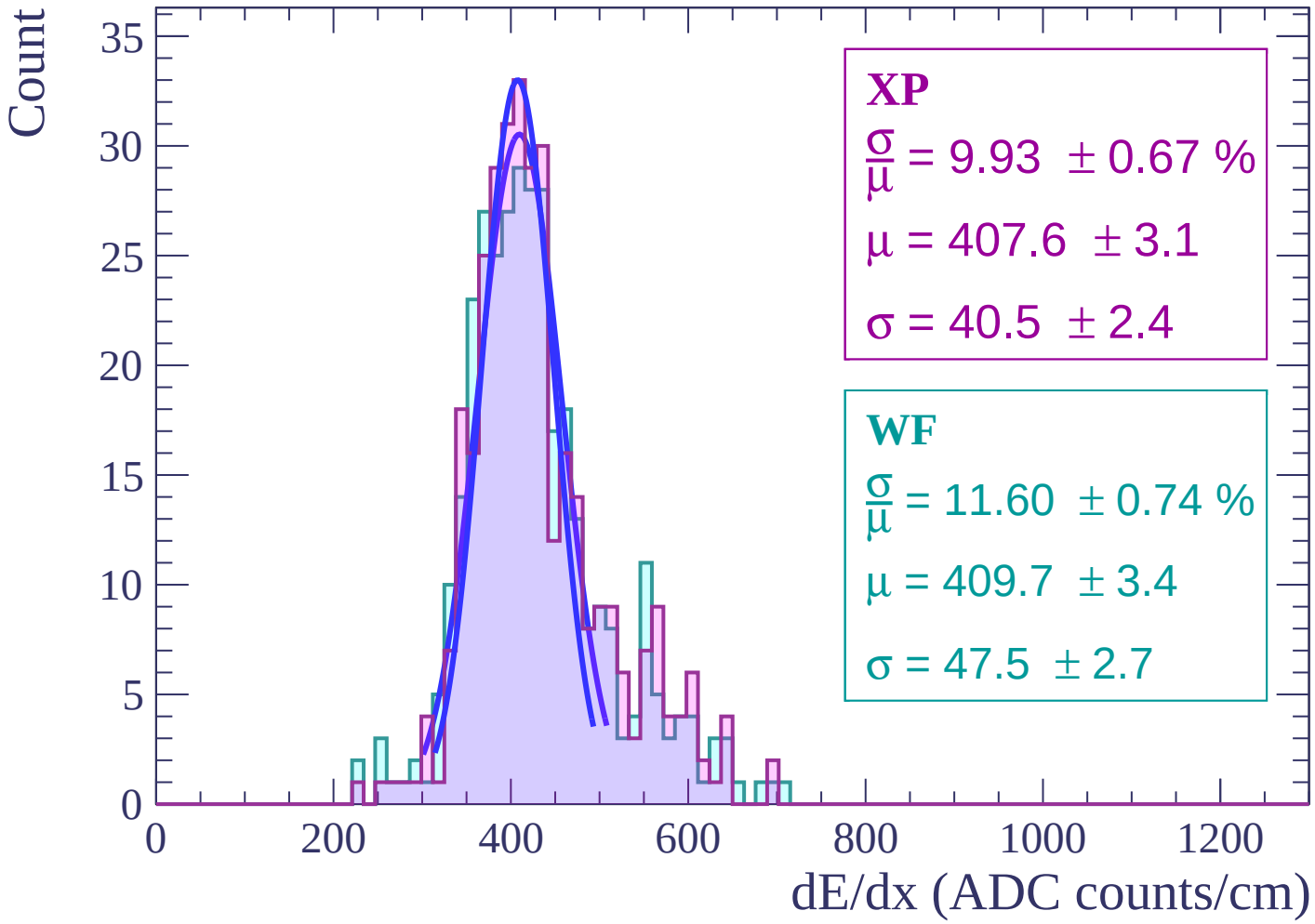

Count
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.31 ± 1.60 %
μ = 426.5 ± 6.6
σ = 61.0 ± 5.9
WF
σ/μ = 19.72 ± 4.88 %
μ = 449.7 ± 12.7
σ = 88.7 ± 19.5

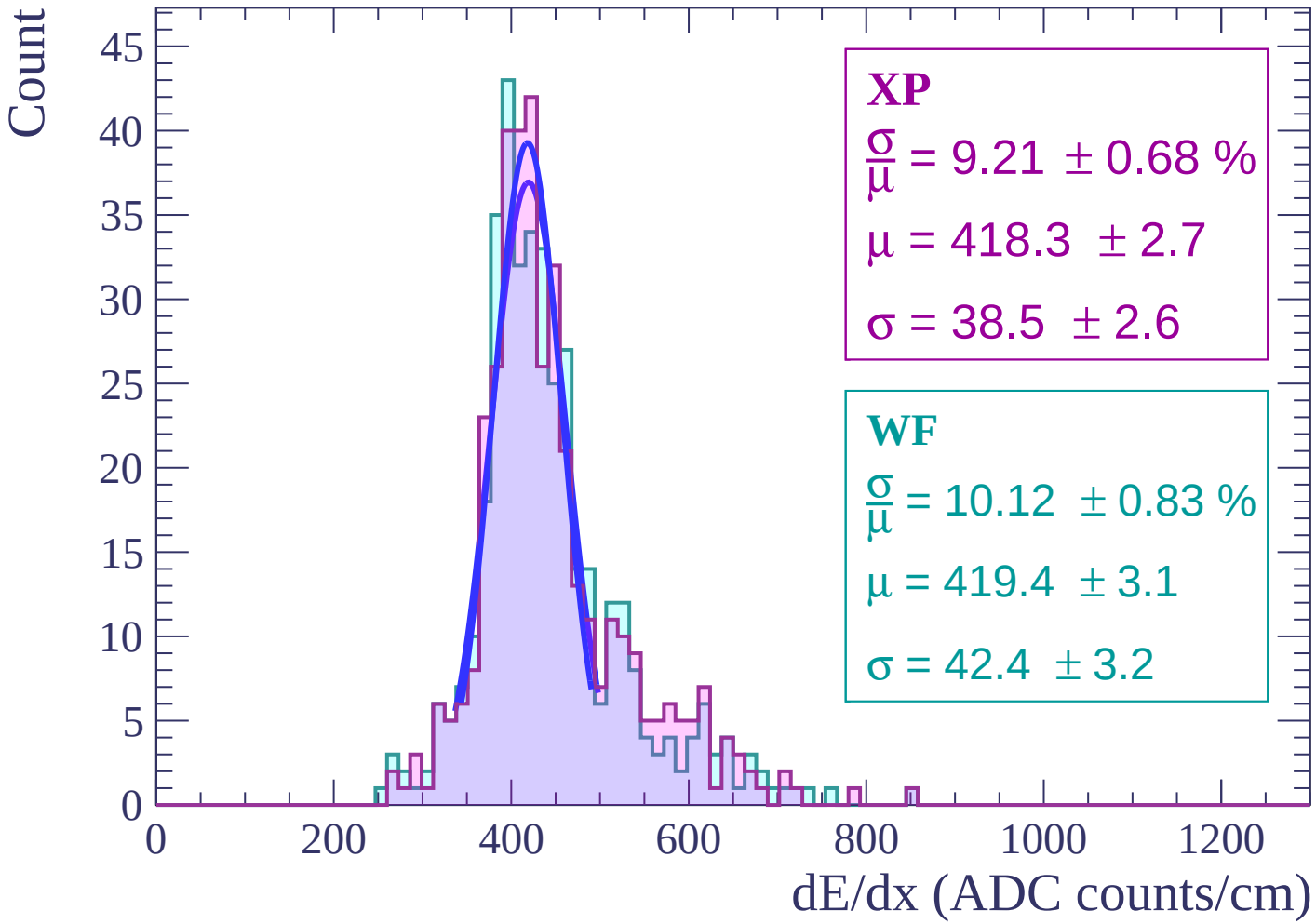

Count
16
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.46 ± 1.92 %
μ = 427.2 ± 7.3
σ = 66.1 ± 7.1
WF
σ/μ = 21.13 ± 2.65 %
μ = 450.1 ± 10.3
σ = 95.1 ± 9.7

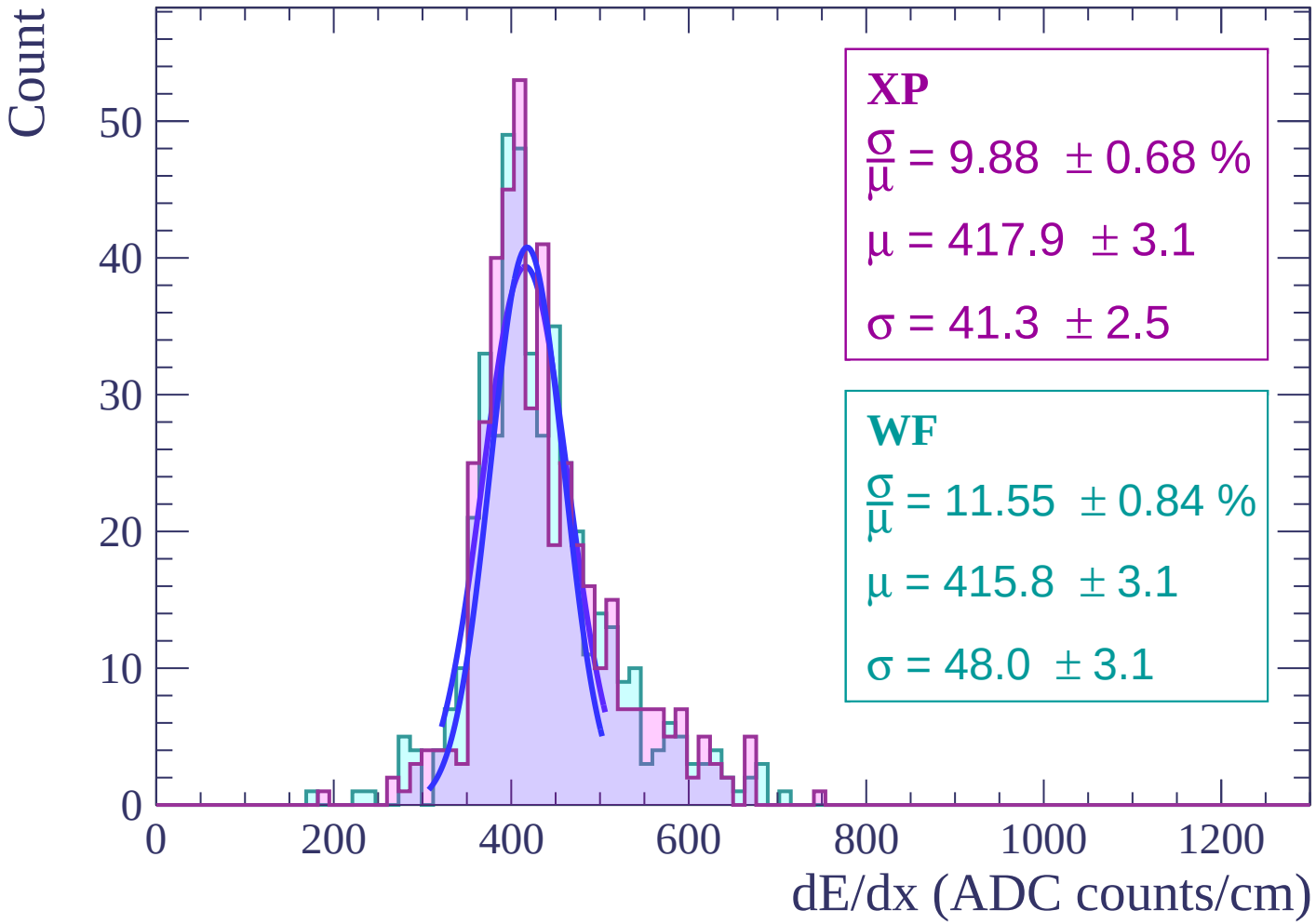

Count
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.30 ± 1.80 %
μ = 403.0 ± 5.7
σ = 49.6 ± 6.6
WF
σ/μ = 14.00 ± 2.36 %
μ = 416.8 ± 8.9
σ = 58.4 ± 8.6

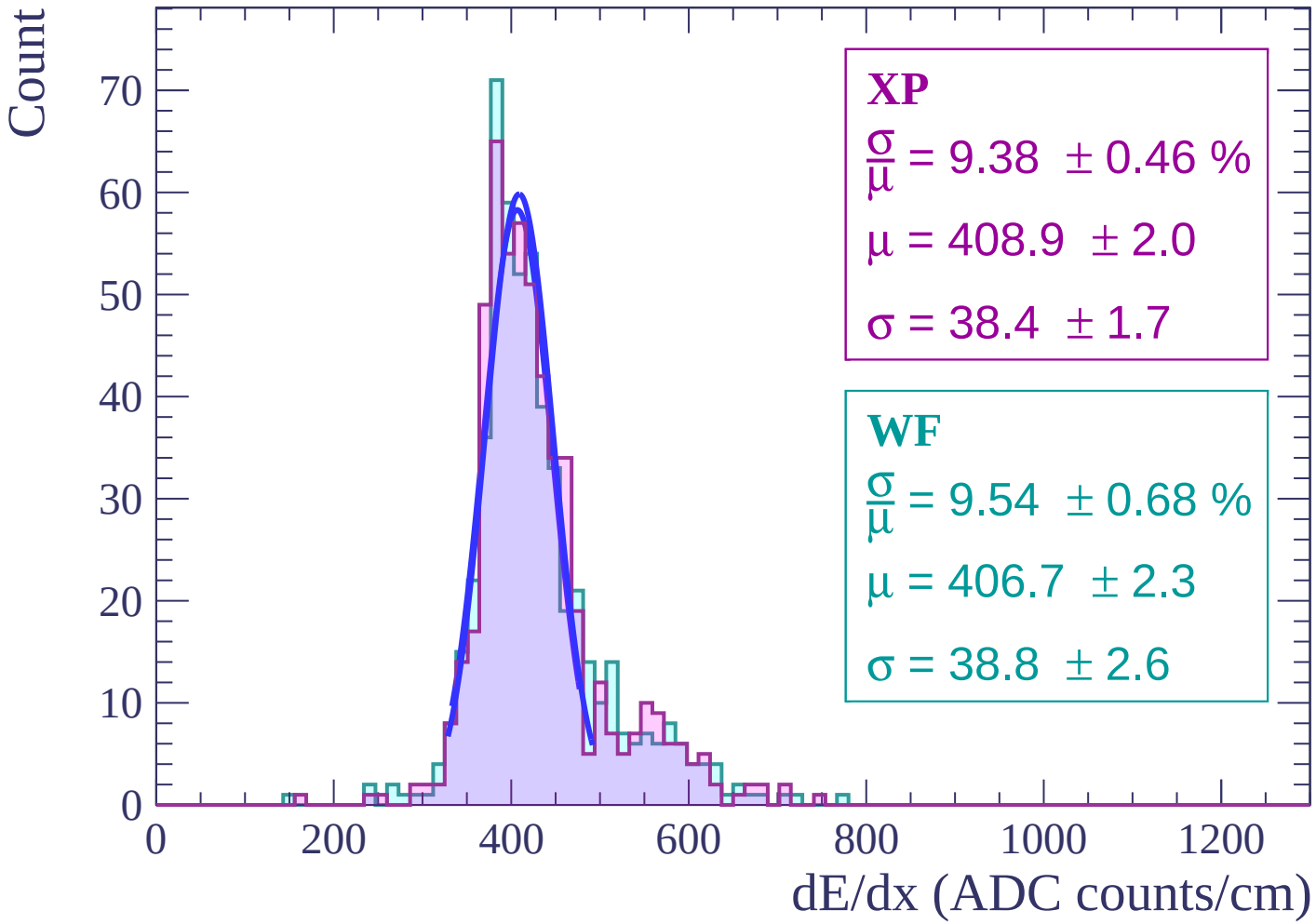

Count
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.79 ± 1.15 %
μ = 418.6 ± 5.0
σ = 49.3 ± 4.2
WF
σ/μ = 14.81 ± 1.43 %
μ = 426.9 ± 5.6
σ = 63.2 ± 5.3

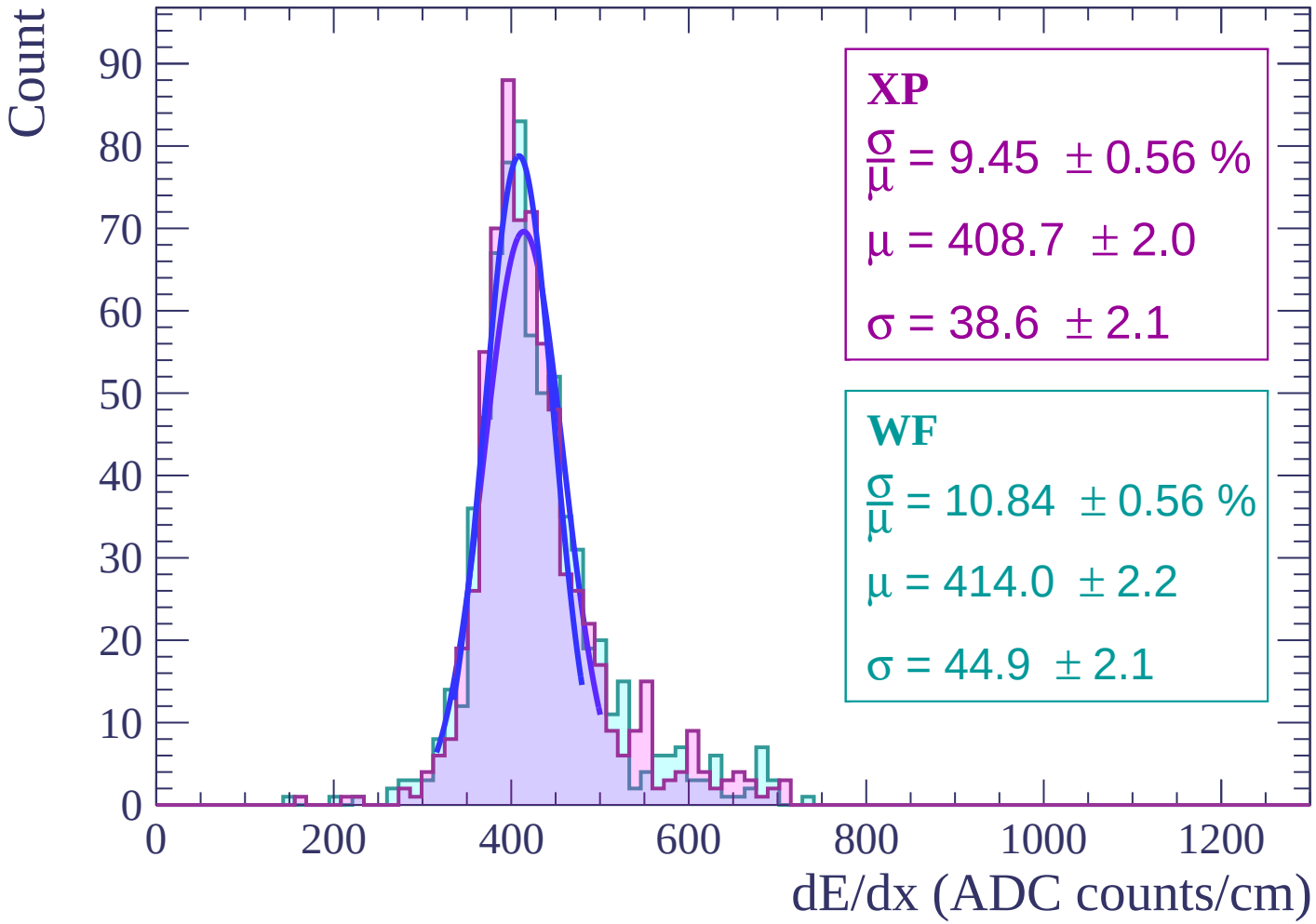

Count
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.20 ± 0.89 %
μ = 410.6 ± 3.4
σ = 41.9 ± 3.3
WF
σ/μ = 11.28 ± 0.94 %
μ = 417.3 ± 4.1
σ = 47.1 ± 3.5

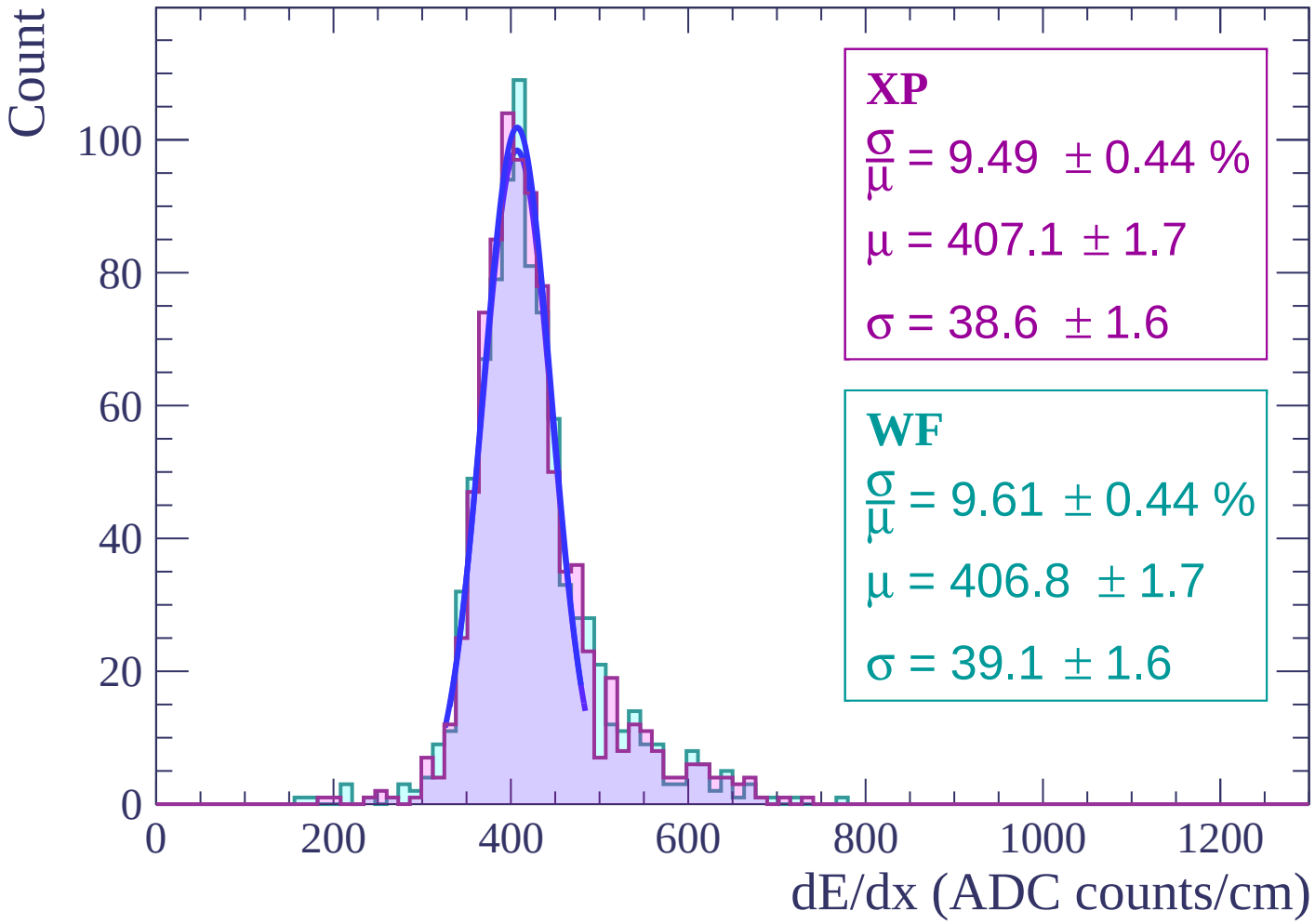

Count
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.93 ± 0.67 %
μ = 407.6 ± 3.1
σ = 40.5 ± 2.4
WF
σ/μ = 11.60 ± 0.74 %
μ = 409.7 ± 3.4
σ = 47.5 ± 2.7

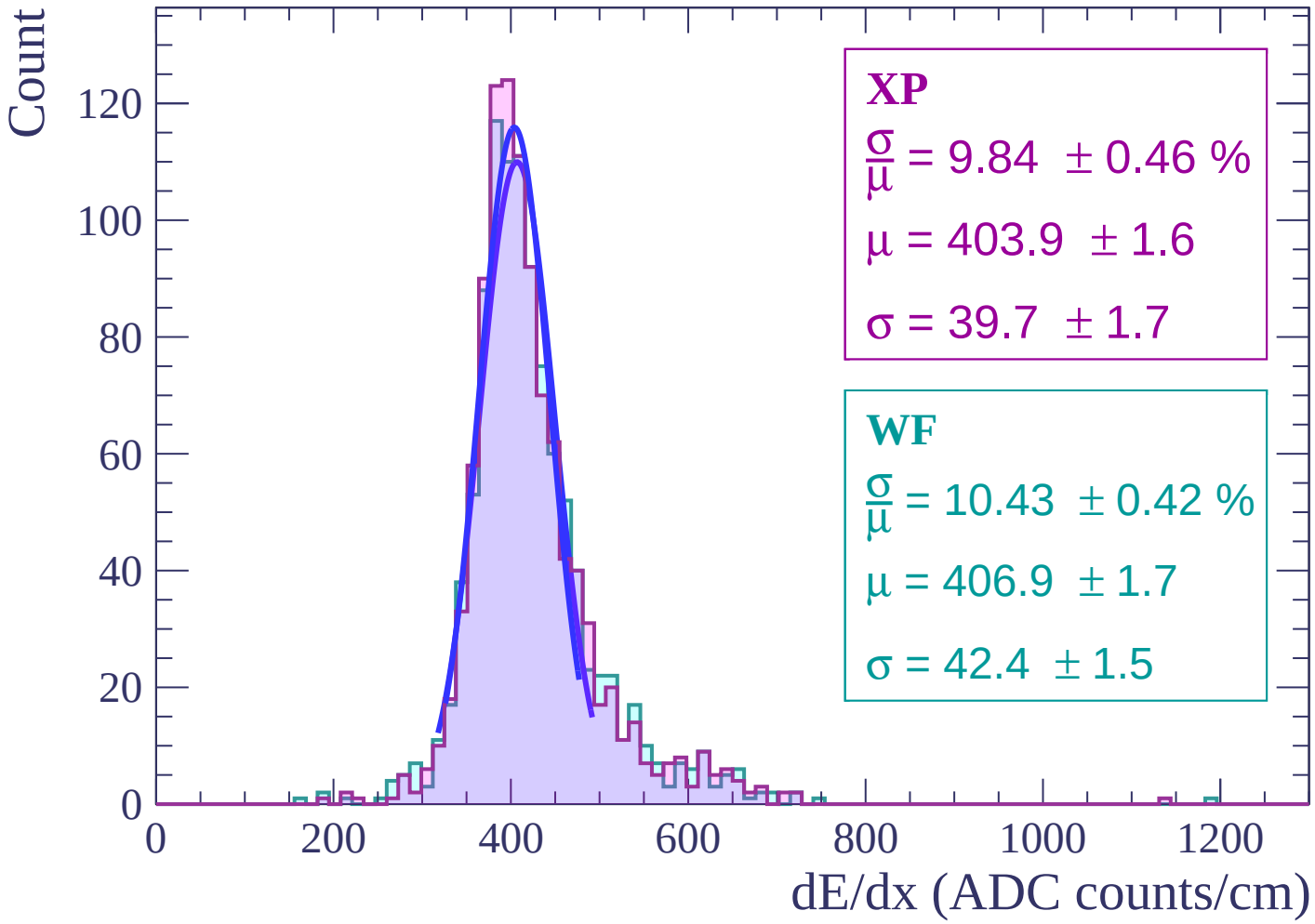

Count
45
40
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.21 ± 0.68 %
μ = 418.3 ± 2.7
σ = 38.5 ± 2.6
WF
σ/μ = 10.12 ± 0.83 %
μ = 419.4 ± 3.1
σ = 42.4 ± 3.2

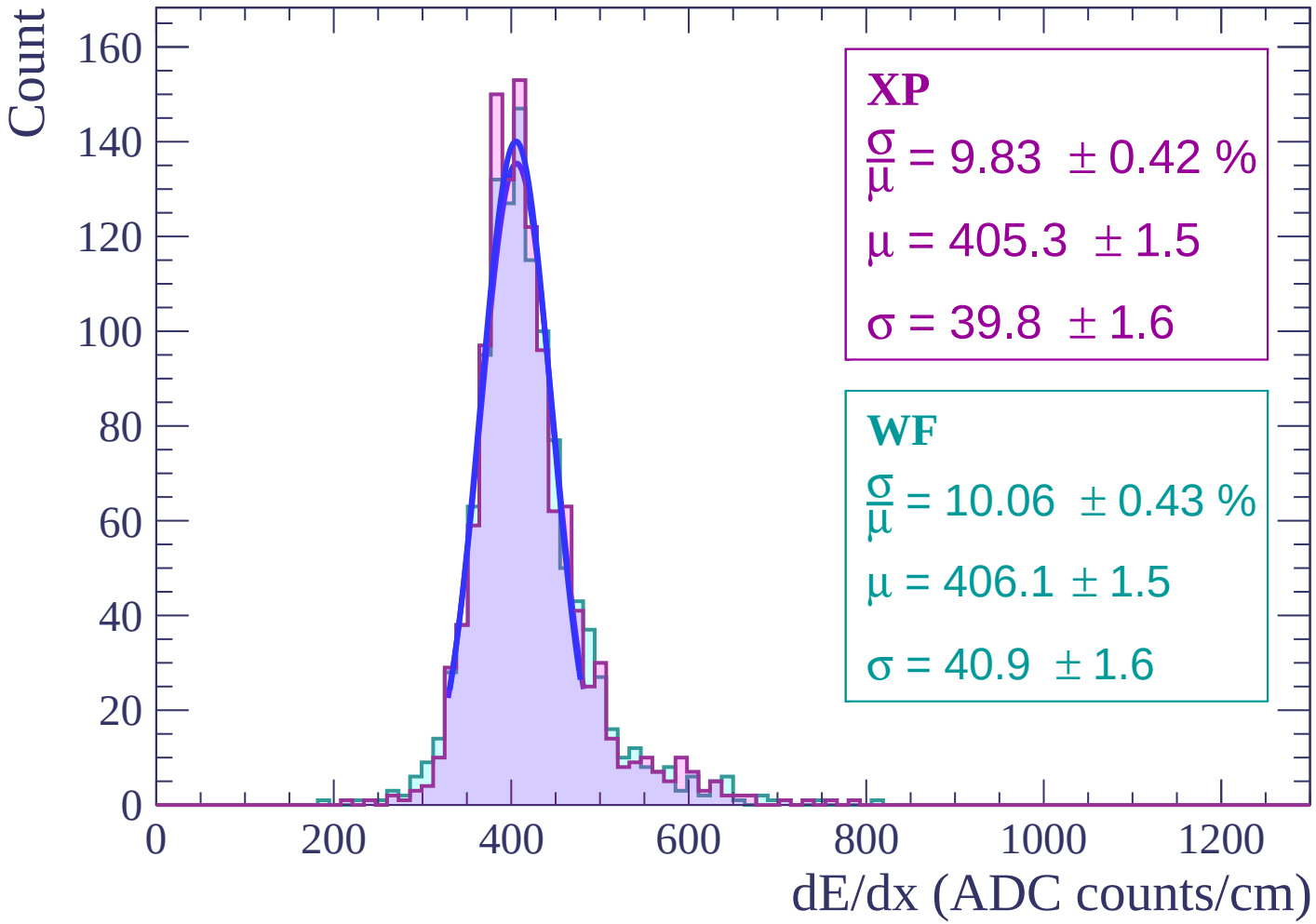

Count
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.88 ± 0.68 %
μ = 417.9 ± 3.1
σ = 41.3 ± 2.5
WF
σ/μ = 11.55 ± 0.84 %
μ = 415.8 ± 3.1
σ = 48.0 ± 3.1

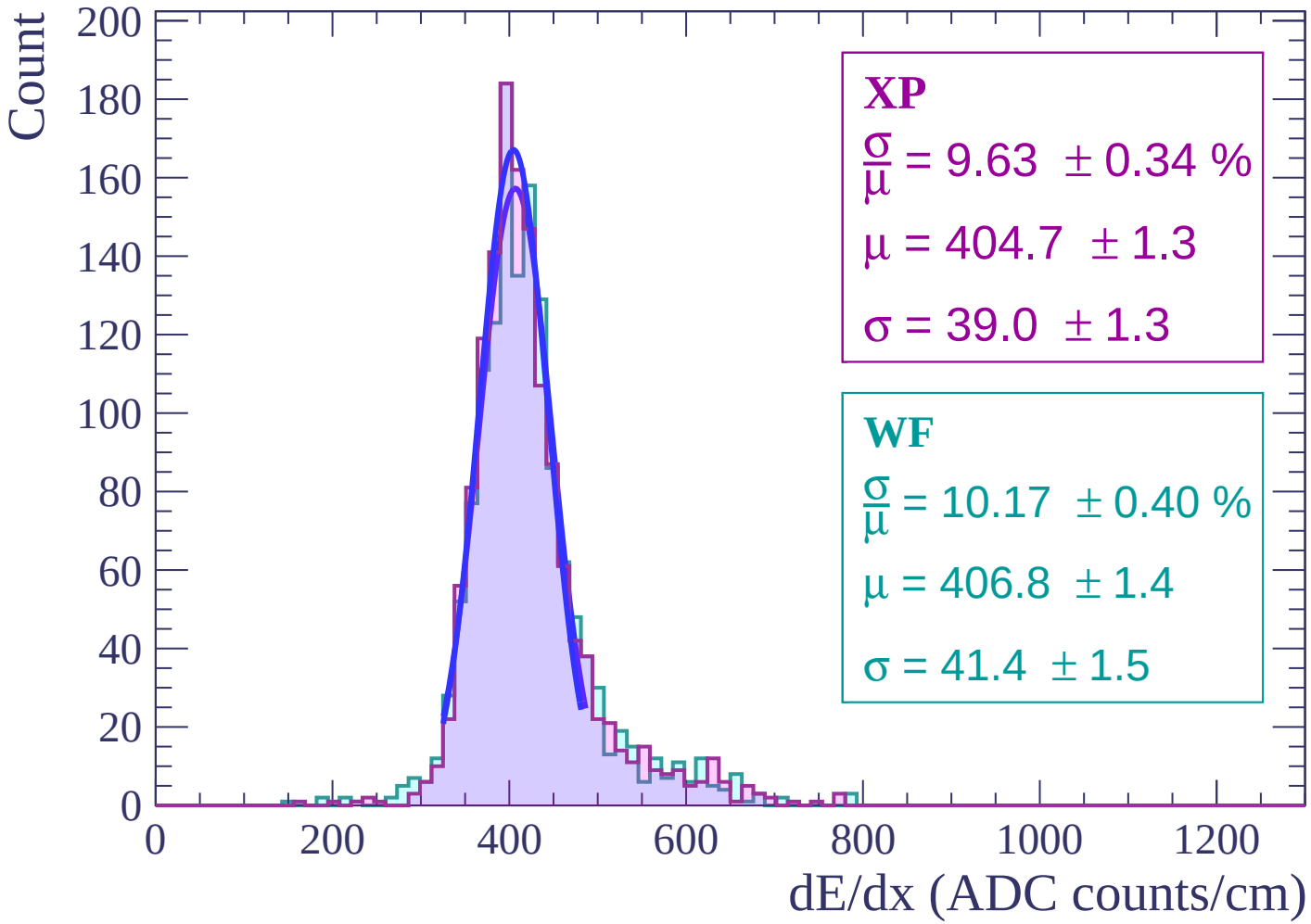

Count
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.38 ± 0.46 %
μ = 408.9 ± 2.0
σ = 38.4 ± 1.7
WF
σ/μ = 9.54 ± 0.68 %
μ = 406.7 ± 2.3
σ = 38.8 ± 2.6

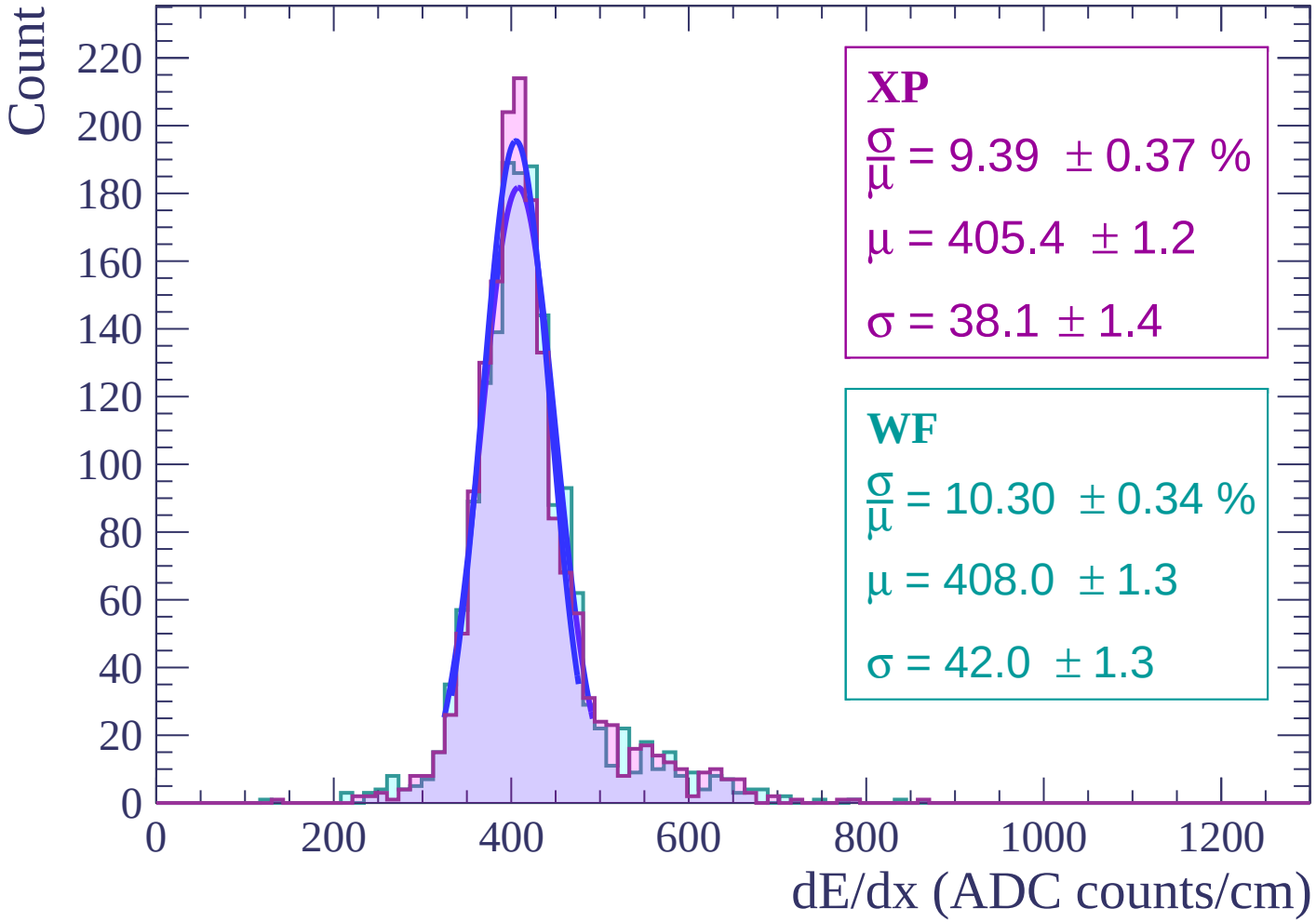

Count
90
80
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.45 ± 0.56 %
μ = 408.7 ± 2.0
σ = 38.6 ± 2.1
WF
σ/μ = 10.84 ± 0.56 %
μ = 414.0 ± 2.2
σ = 44.9 ± 2.1

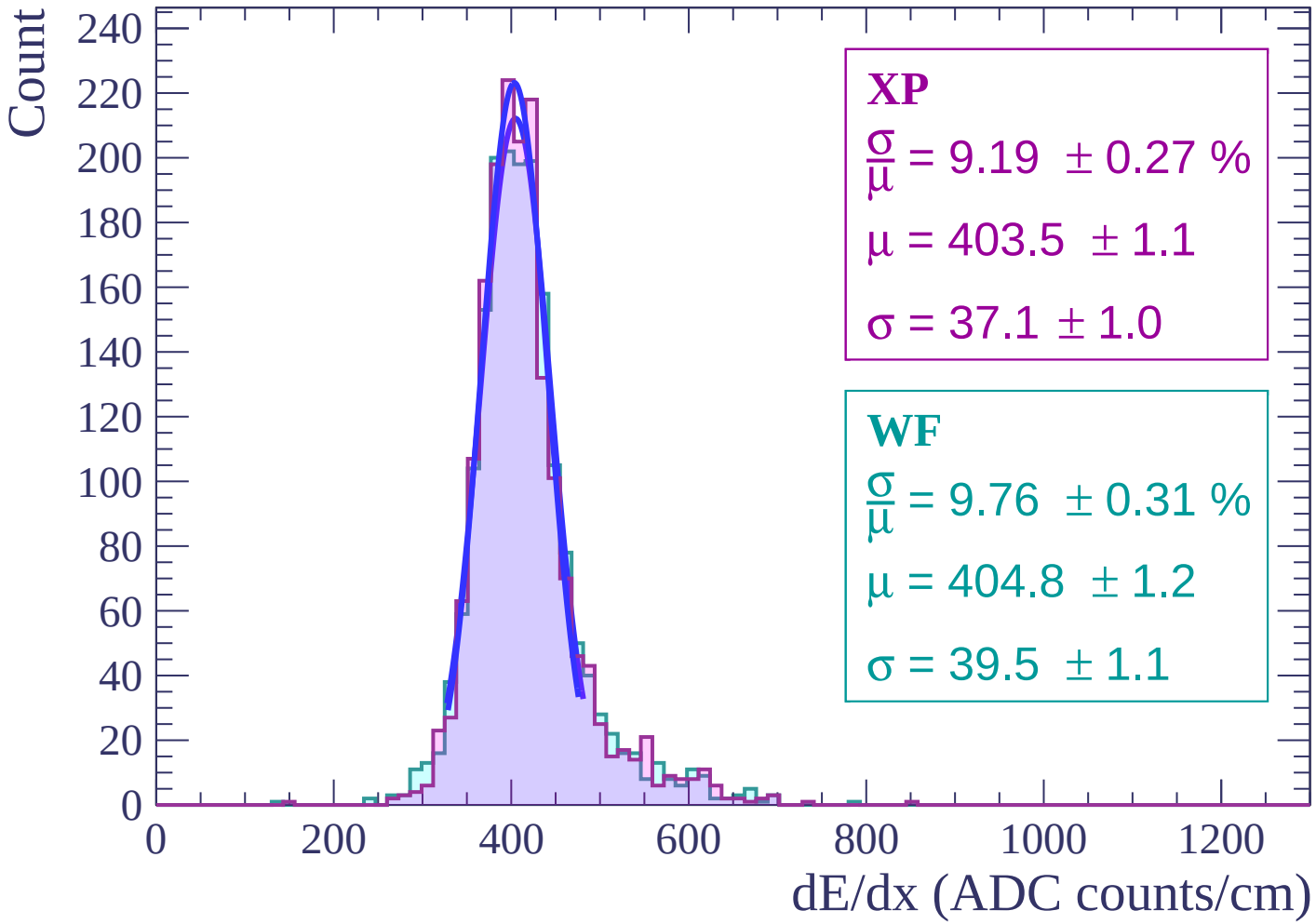

Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.49 ± 0.44 %
μ = 407.1 ± 1.7
σ = 38.6 ± 1.6
WF
σ/μ = 9.61 ± 0.44 %
μ = 406.8 ± 1.7
σ = 39.1 ± 1.6

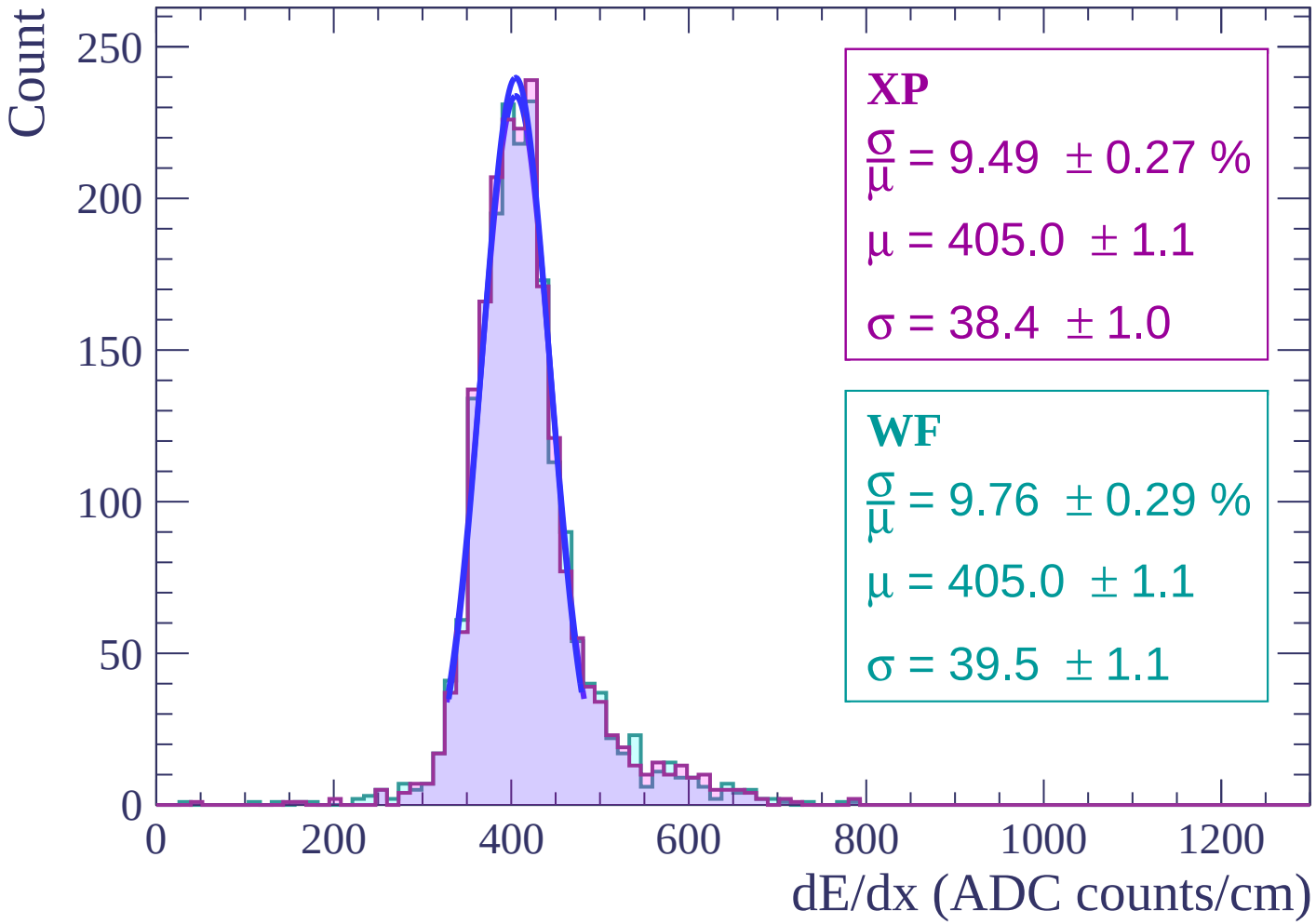

Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.84 ± 0.46 %
μ = 403.9 ± 1.6
σ = 39.7 ± 1.7
WF
σ/μ = 10.43 ± 0.42 %
μ = 406.9 ± 1.7
σ = 42.4 ± 1.5

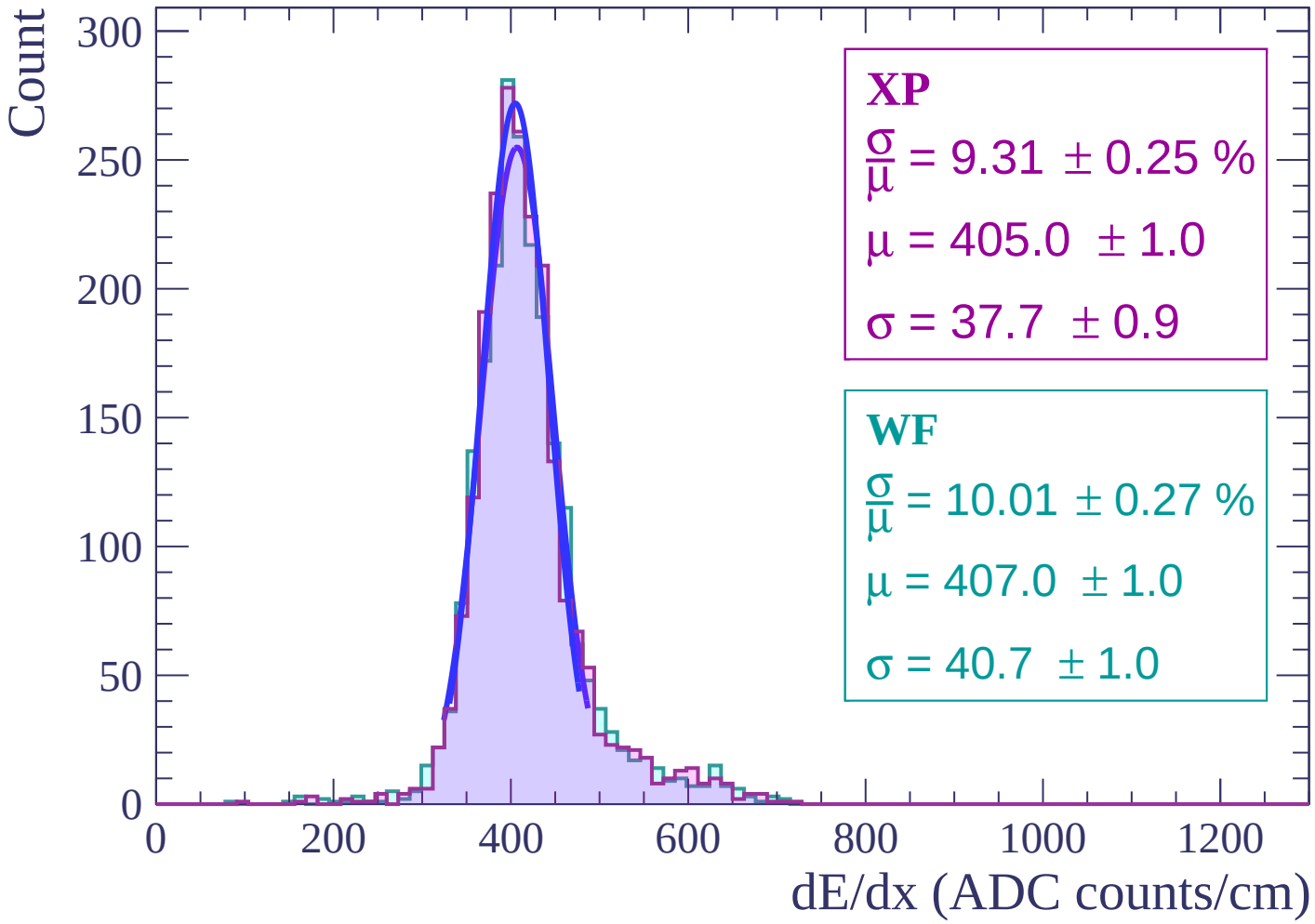

Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.83 ± 0.42 %
μ = 405.3 ± 1.5
σ = 39.8 ± 1.6
WF
σ/μ = 10.06 ± 0.43 %
μ = 406.1 ± 1.5
σ = 40.9 ± 1.6

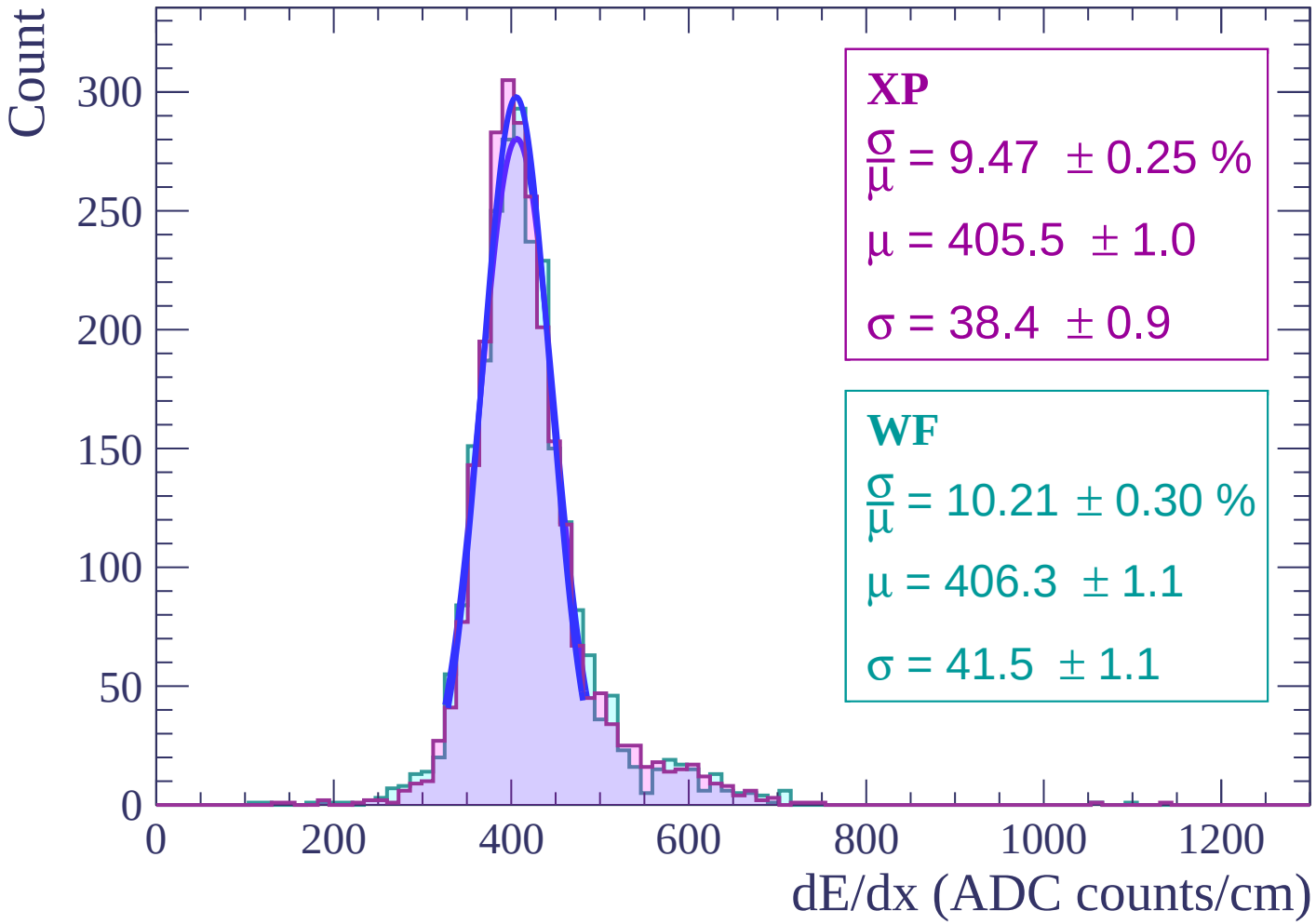

Count
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.63 ± 0.34 %
μ = 404.7 ± 1.3
σ = 39.0 ± 1.3
WF
σ/μ = 10.17 ± 0.40 %
μ = 406.8 ± 1.4
σ = 41.4 ± 1.5

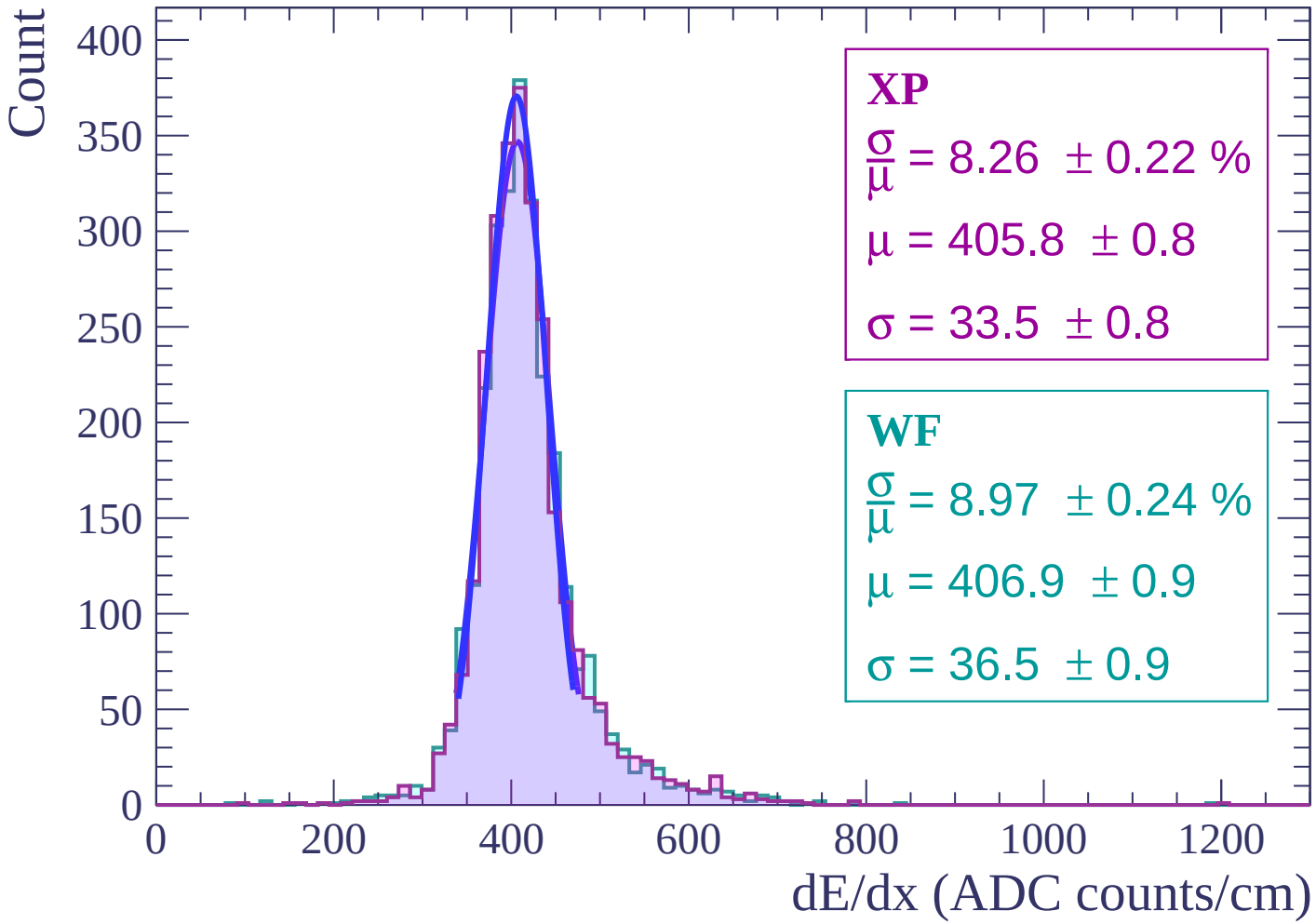

Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.39 ± 0.37 %
μ = 405.4 ± 1.2
σ = 38.1 ± 1.4
WF
σ/μ = 10.30 ± 0.34 %
μ = 408.0 ± 1.3
σ = 42.0 ± 1.3

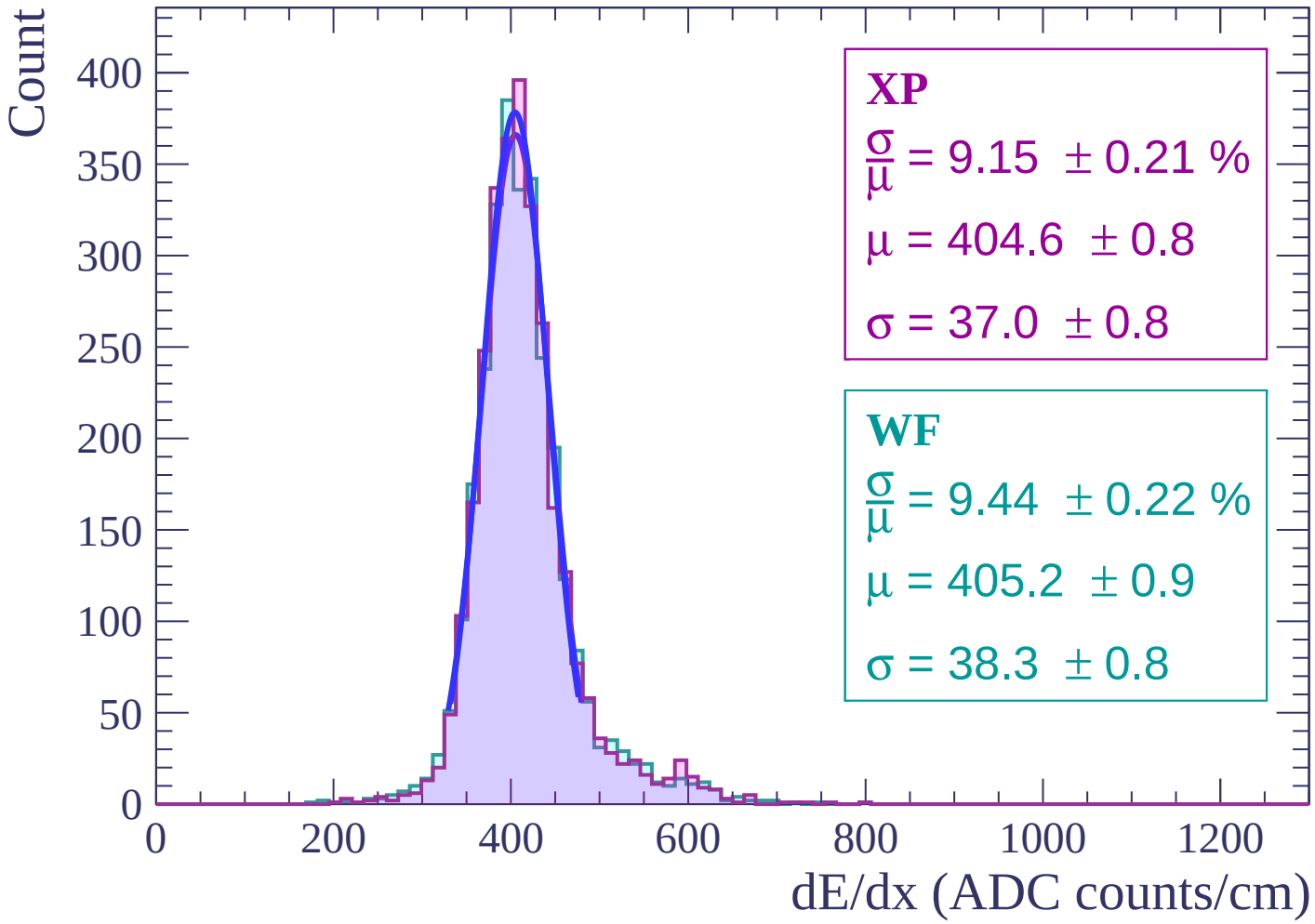

Count
240
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.19 ± 0.27 %
μ = 403.5 ± 1.1
σ = 37.1 ± 1.0
WF
σ/μ = 9.76 ± 0.31 %
μ = 404.8 ± 1.2
σ = 39.5 ± 1.1

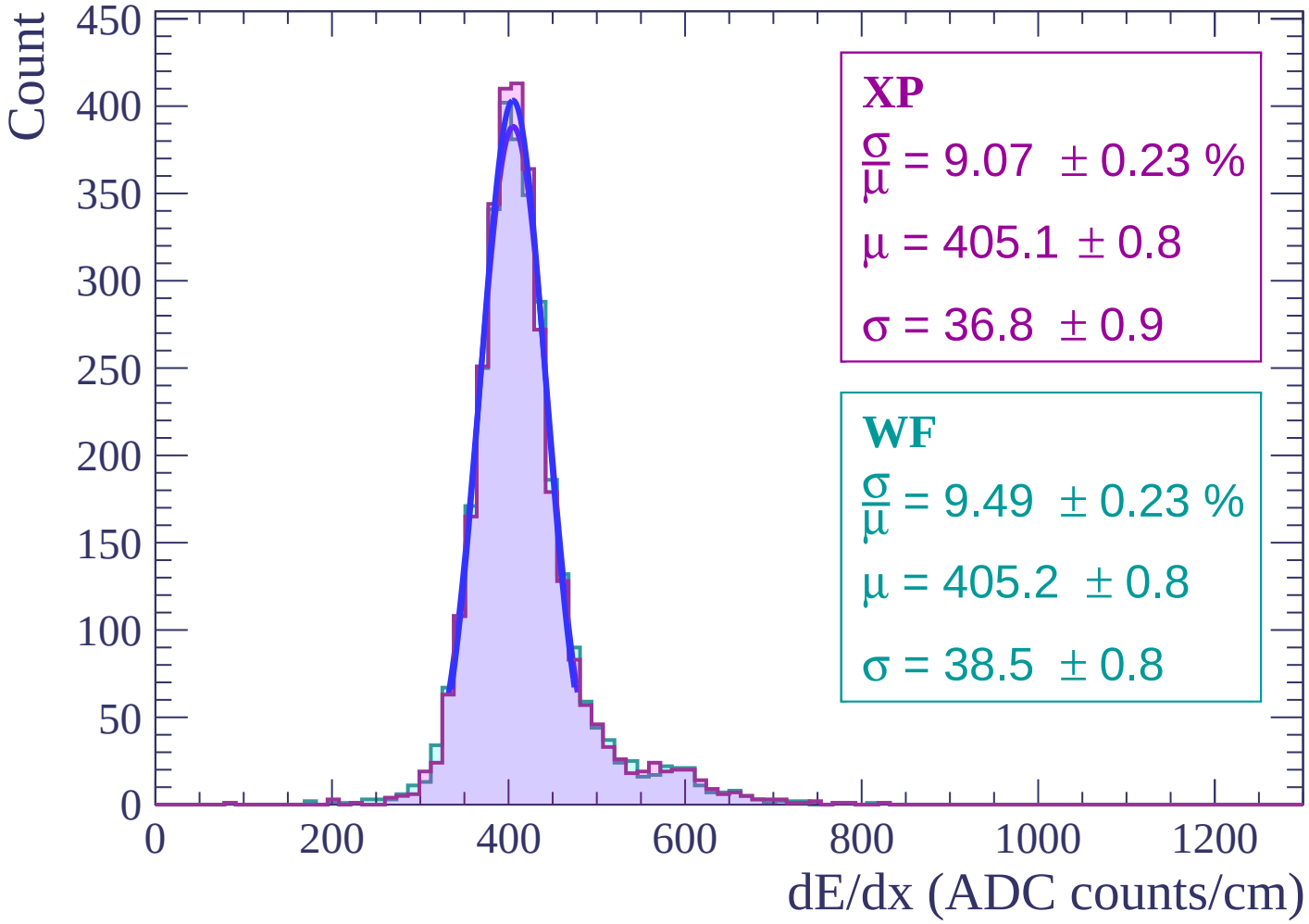

Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.49 ± 0.27 %
μ = 405.0 ± 1.1
σ = 38.4 ± 1.0
WF
σ/μ = 9.76 ± 0.29 %
μ = 405.0 ± 1.1
σ = 39.5 ± 1.1

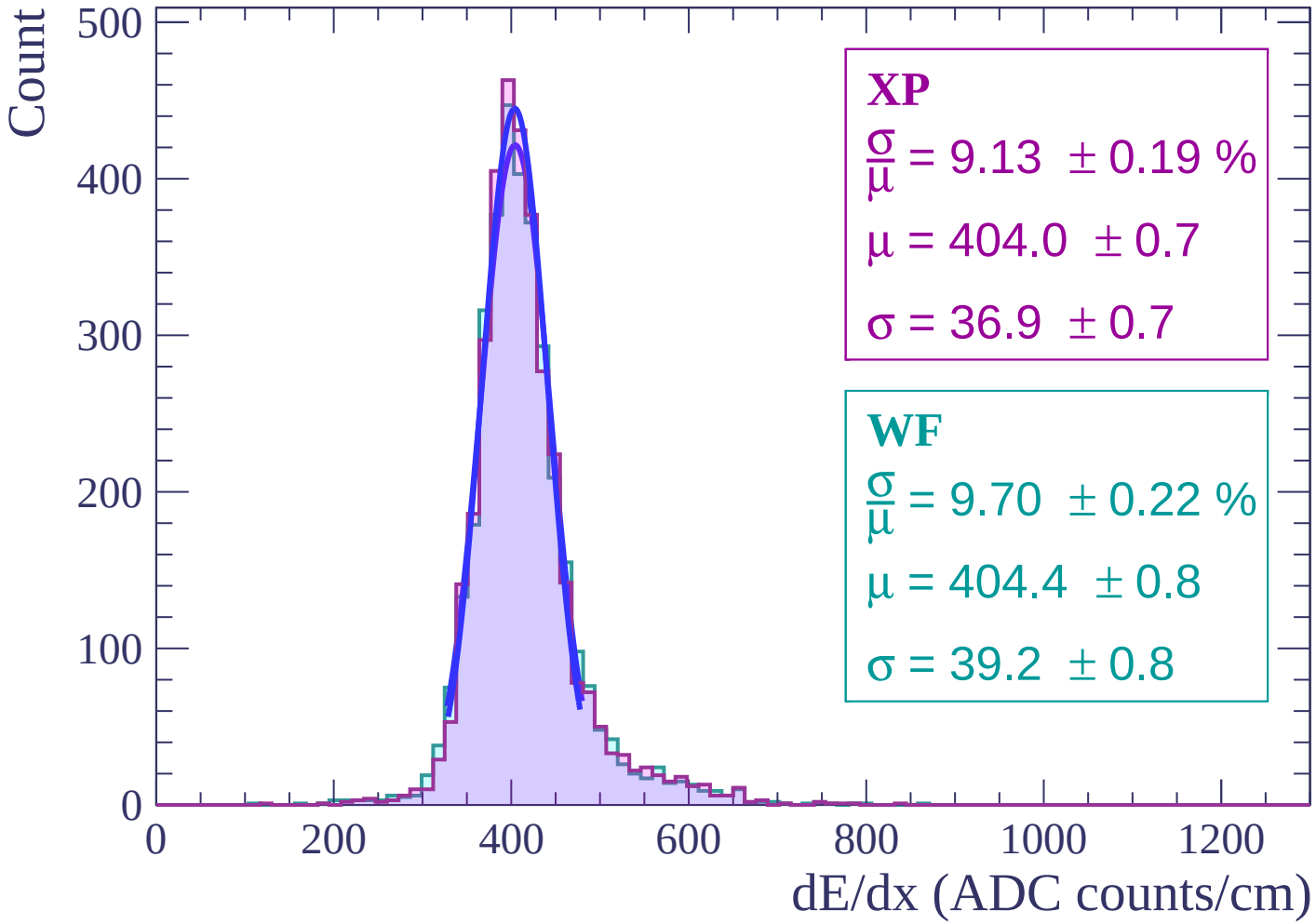

Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.31 ± 0.25 %
μ = 405.0 ± 1.0
σ = 37.7 ± 0.9
WF
σ/μ = 10.01 ± 0.27 %
μ = 407.0 ± 1.0
σ = 40.7 ± 1.0

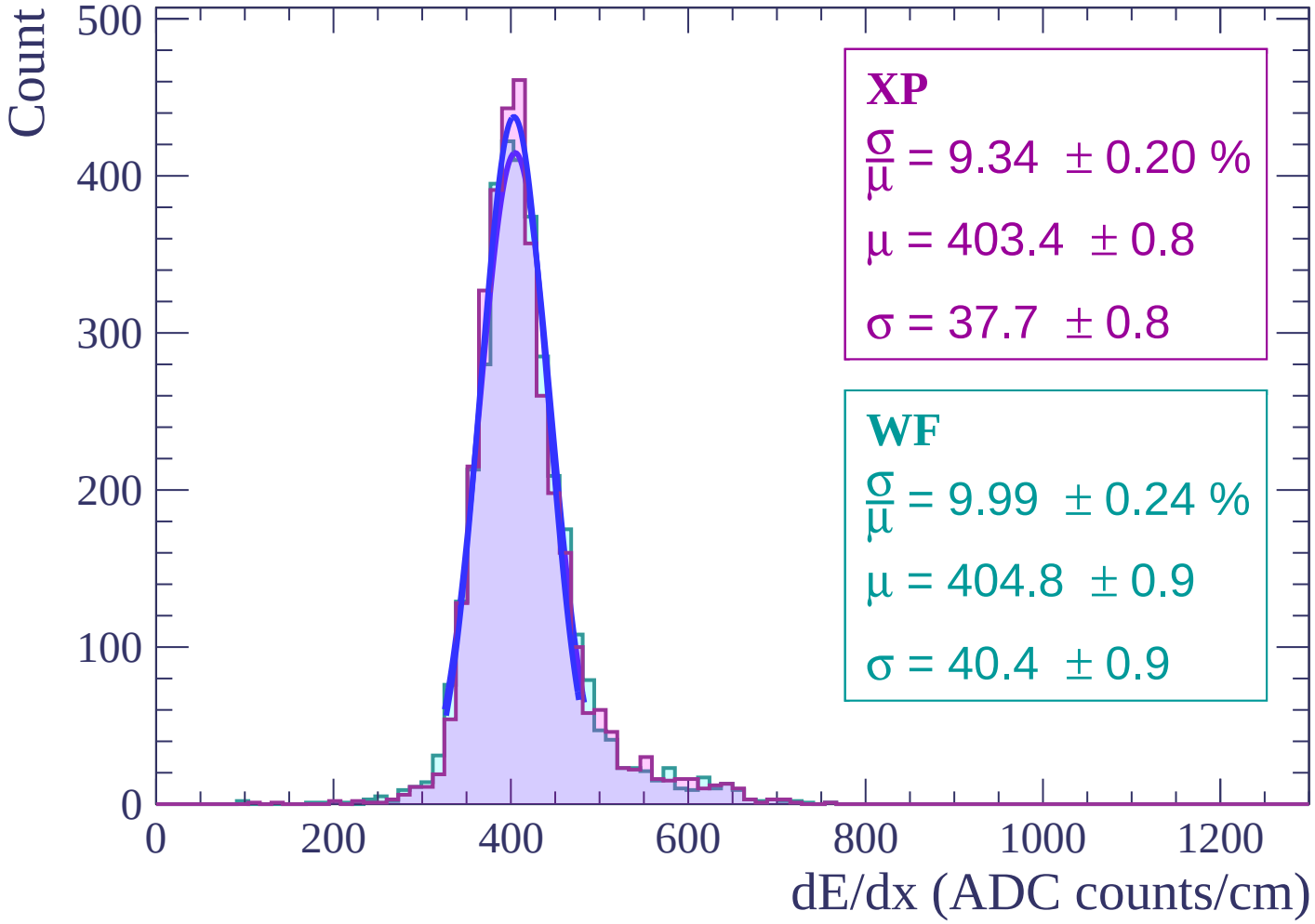

Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.47 ± 0.25 %
μ = 405.5 ± 1.0
σ = 38.4 ± 0.9
WF
σ/μ = 10.21 ± 0.30 %
μ = 406.3 ± 1.1
σ = 41.5 ± 1.1

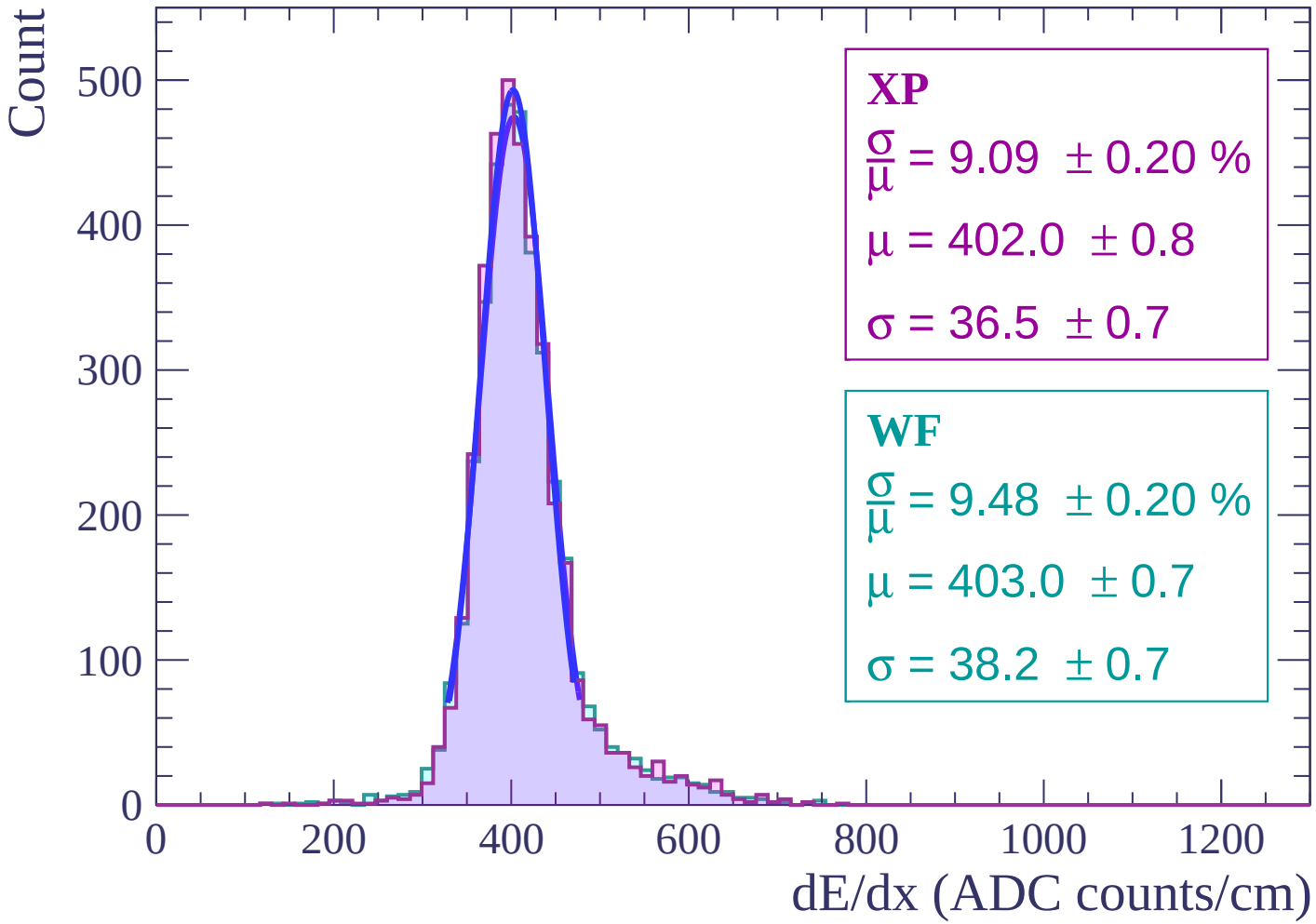

Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.26 ± 0.22 %
μ = 405.8 ± 0.8
σ = 33.5 ± 0.8
WF
σ/μ = 8.97 ± 0.24 %
μ = 406.9 ± 0.9
σ = 36.5 ± 0.9

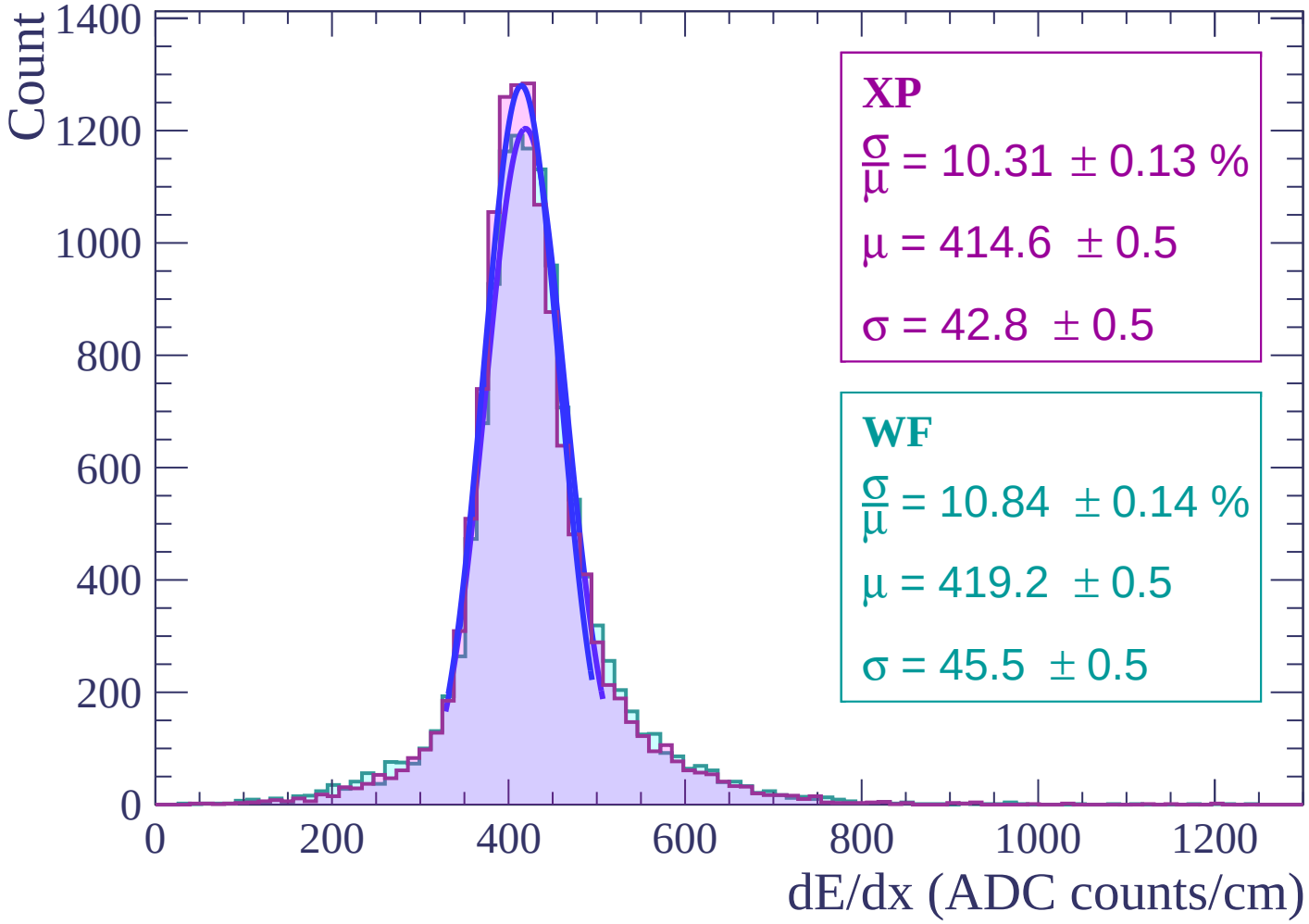

Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.15 ± 0.21 %
μ = 404.6 ± 0.8
σ = 37.0 ± 0.8
WF
σ/μ = 9.44 ± 0.22 %
μ = 405.2 ± 0.9
σ = 38.3 ± 0.8

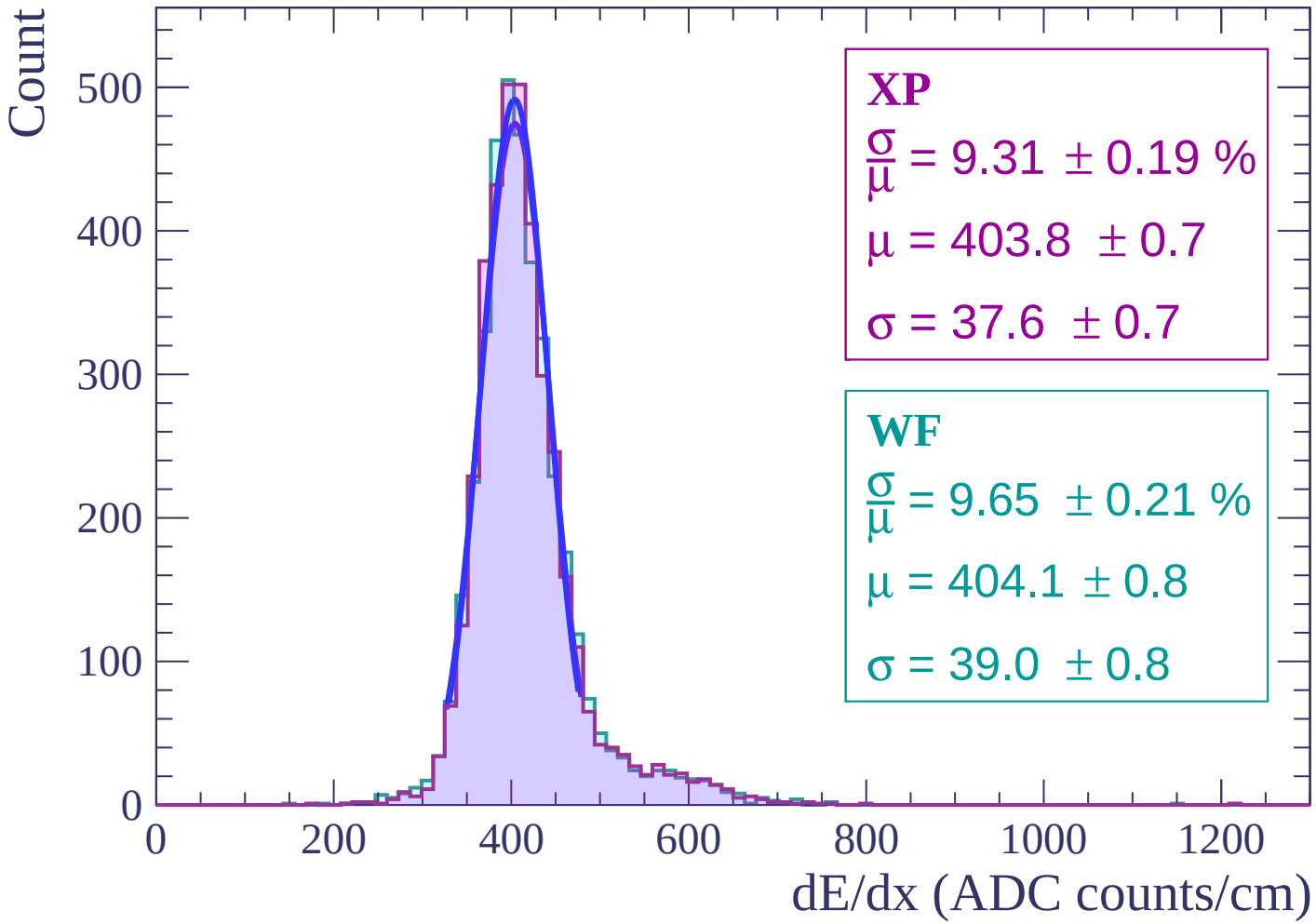

Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.07 ± 0.23 %
μ = 405.1 ± 0.8
σ = 36.8 ± 0.9
WF
σ/μ = 9.49 ± 0.23 %
μ = 405.2 ± 0.8
σ = 38.5 ± 0.8

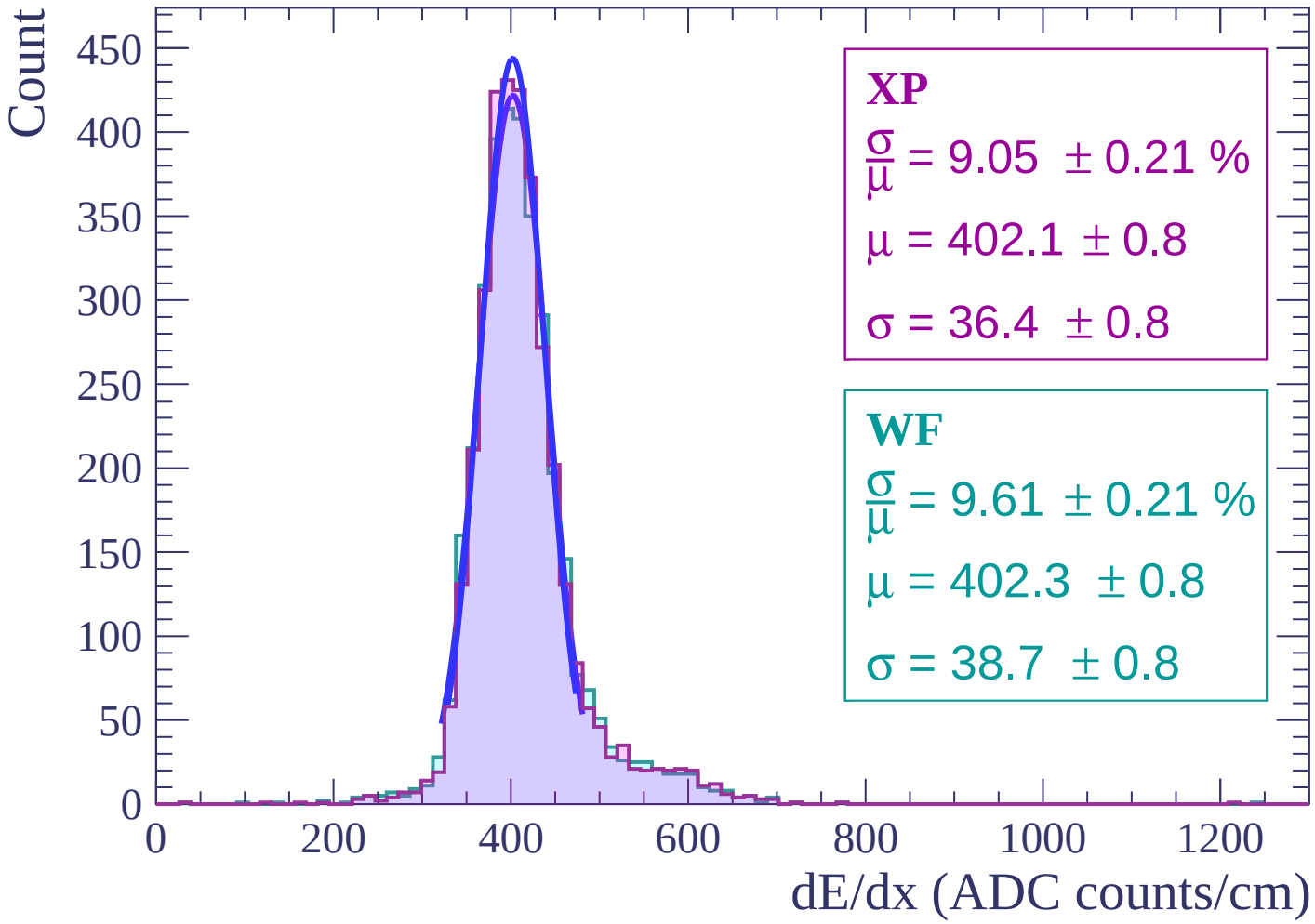

Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.13 ± 0.19 %
μ = 404.0 ± 0.7
σ = 36.9 ± 0.7
WF
σ/μ = 9.70 ± 0.22 %
μ = 404.4 ± 0.8
σ = 39.2 ± 0.8

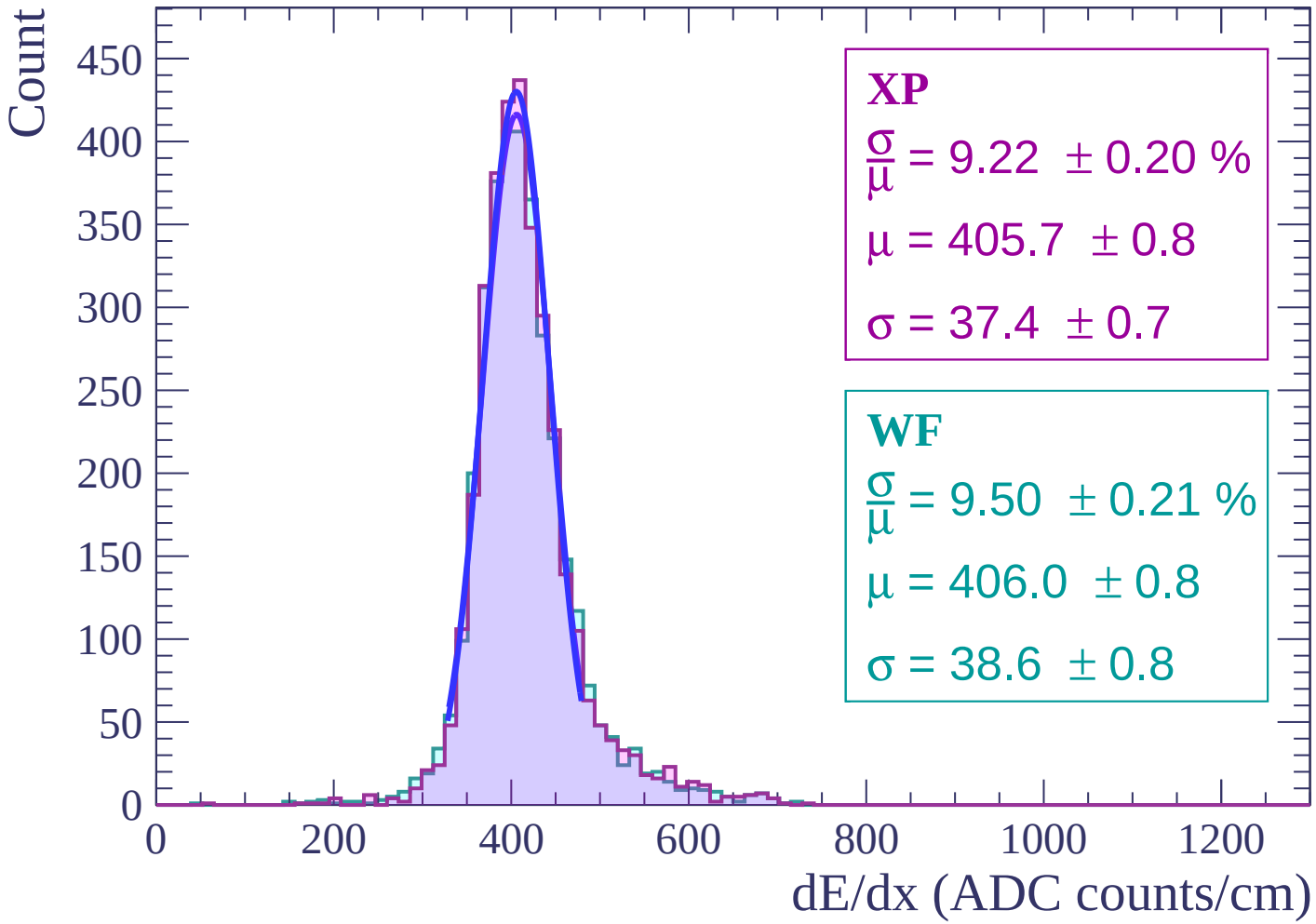

Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.34 ± 0.20 %
μ = 403.4 ± 0.8
σ = 37.7 ± 0.8
WF
σ/μ = 9.99 ± 0.24 %
μ = 404.8 ± 0.9
σ = 40.4 ± 0.9

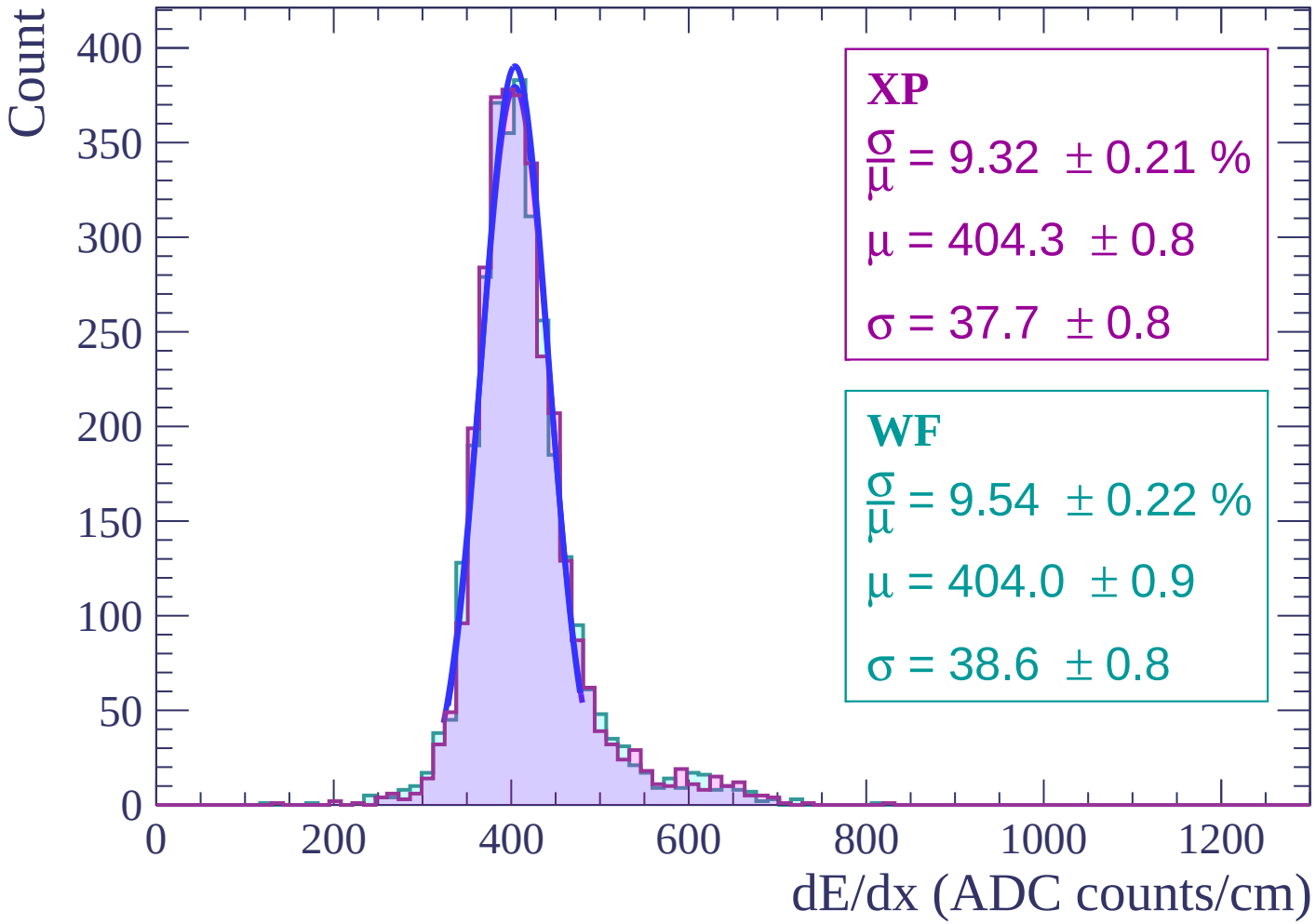

Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.09 ± 0.20 %
μ = 402.0 ± 0.8
σ = 36.5 ± 0.7
WF
σ/μ = 9.48 ± 0.20 %
μ = 403.0 ± 0.7
σ = 38.2 ± 0.7

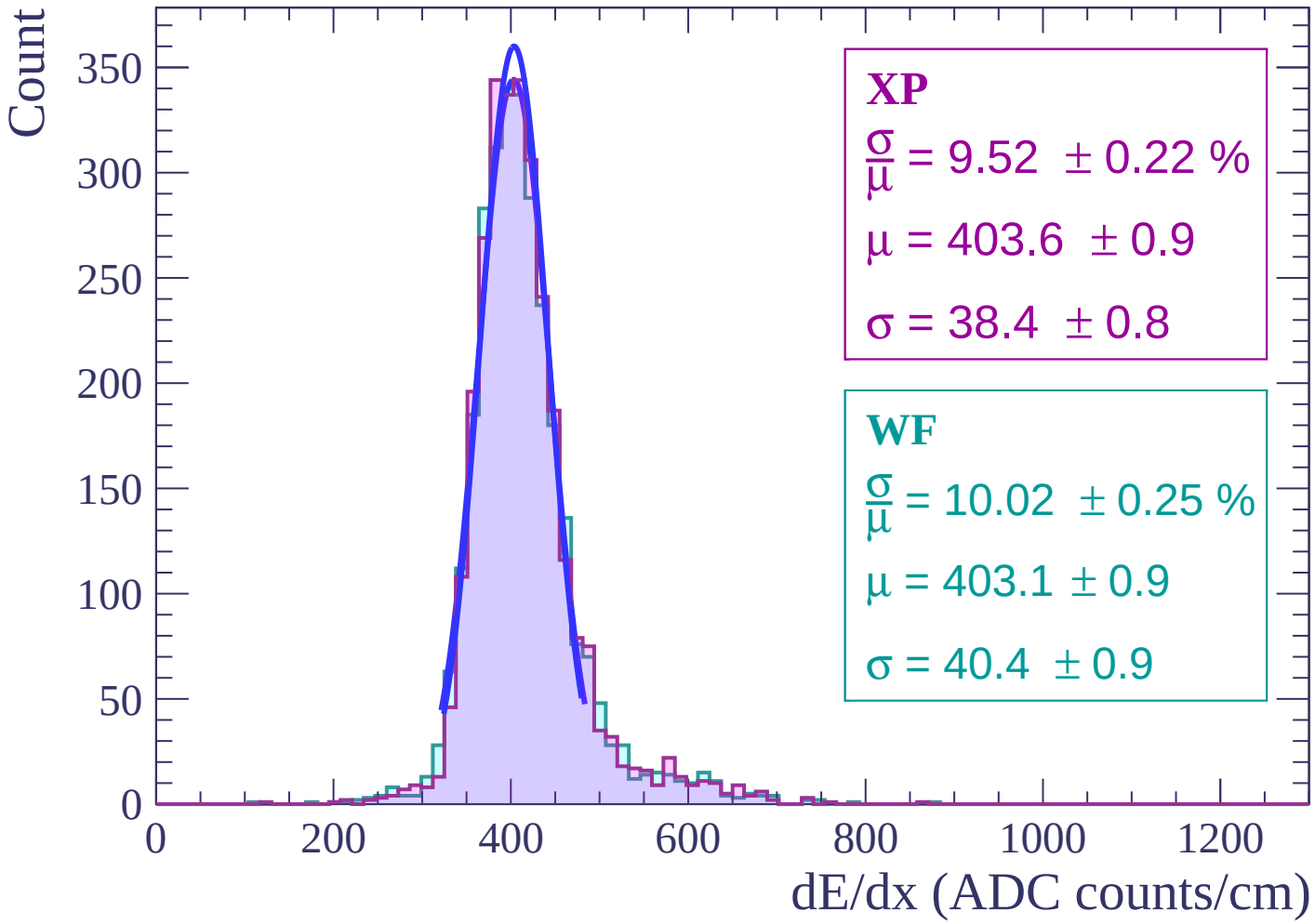

Count
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.31 ± 0.13 %
μ = 414.6 ± 0.5
σ = 42.8 ± 0.5
WF
σ/μ = 10.84 ± 0.14 %
μ = 419.2 ± 0.5
σ = 45.5 ± 0.5

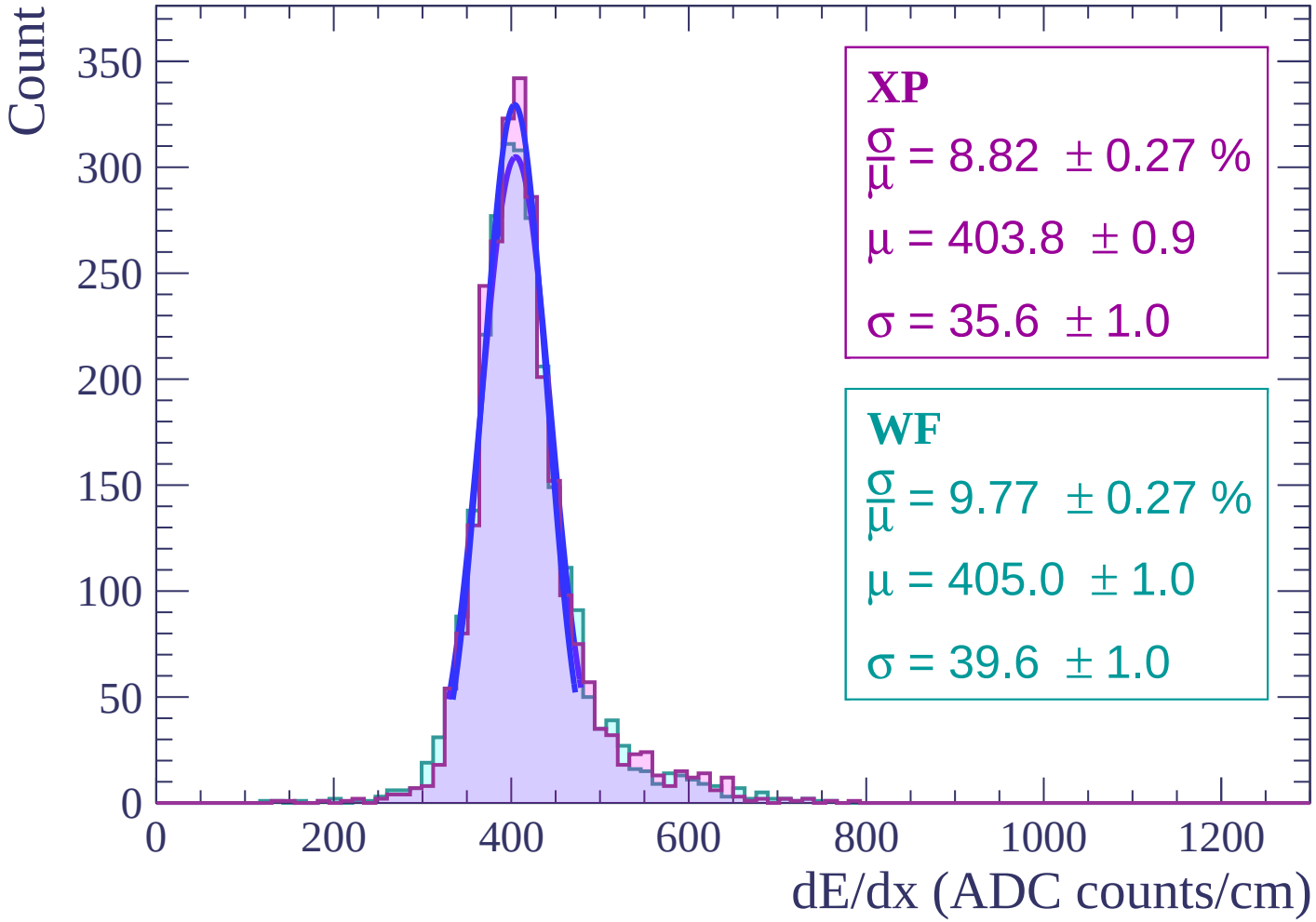

Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.31 ± 0.19 %
μ = 403.8 ± 0.7
σ = 37.6 ± 0.7
WF
σ/μ = 9.65 ± 0.21 %
μ = 404.1 ± 0.8
σ = 39.0 ± 0.8

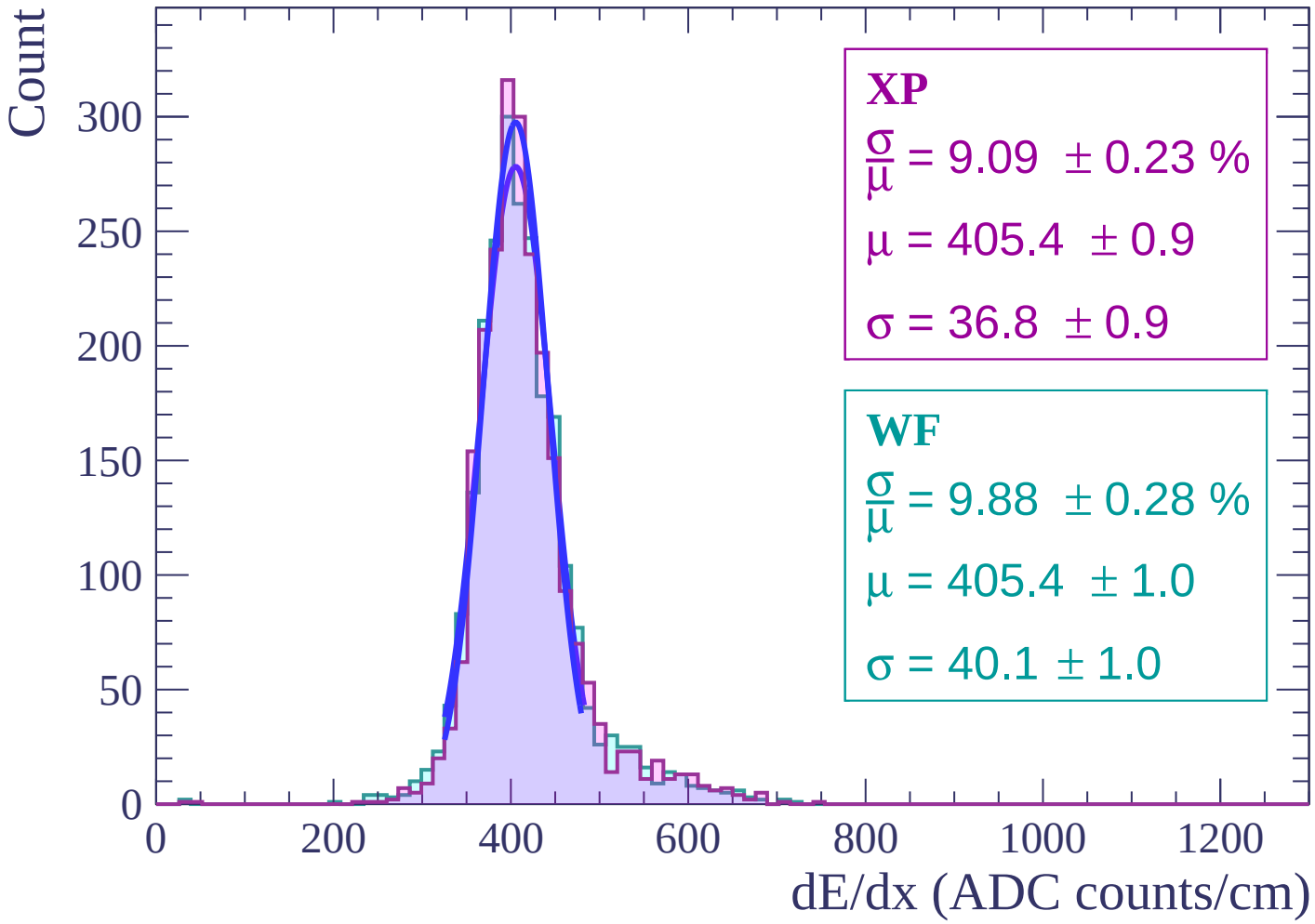

Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.05 ± 0.21 %
μ = 402.1 ± 0.8
σ = 36.4 ± 0.8
WF
σ/μ = 9.61 ± 0.21 %
μ = 402.3 ± 0.8
σ = 38.7 ± 0.8

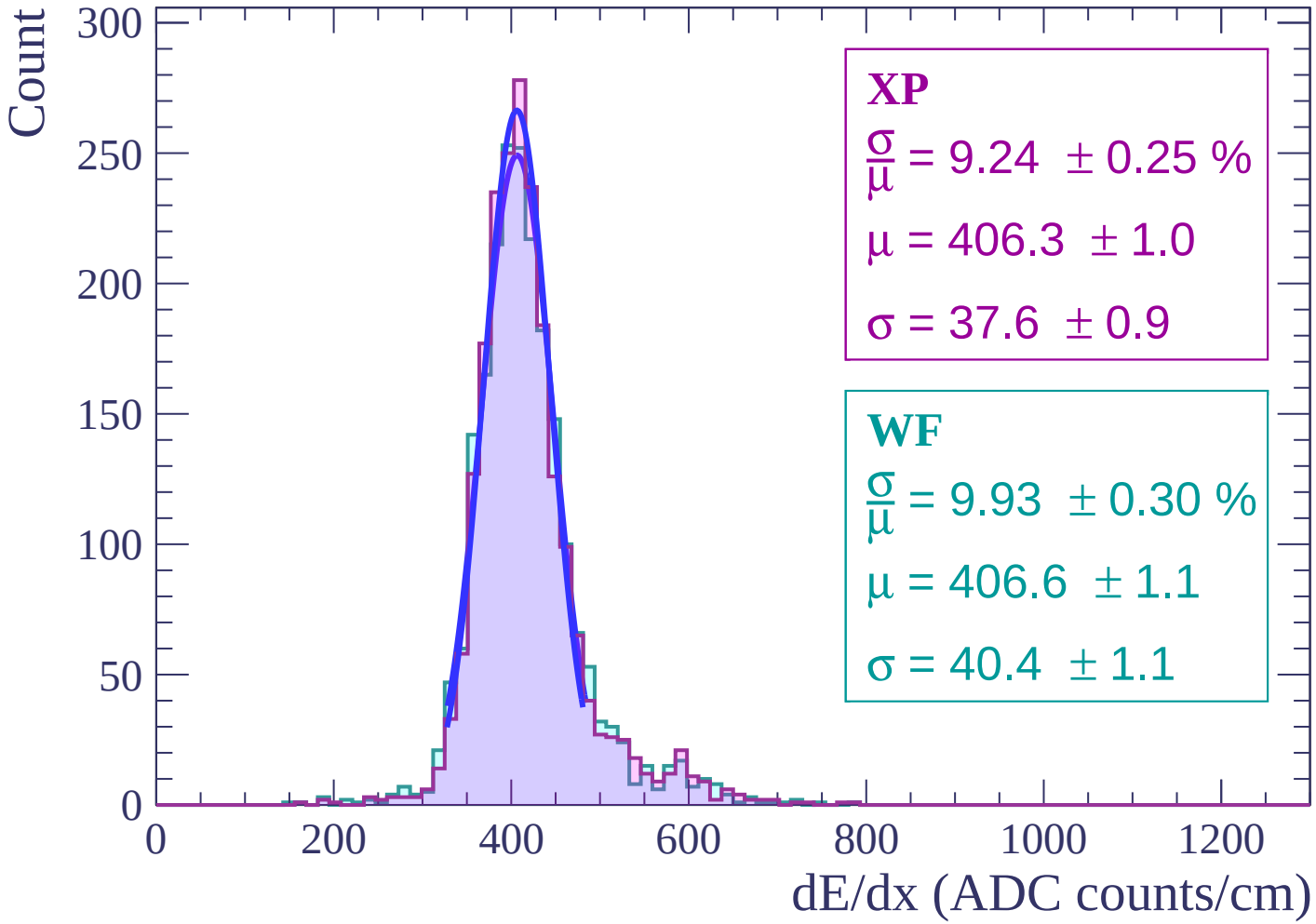

Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.22 ± 0.20 %
μ = 405.7 ± 0.8
σ = 37.4 ± 0.7
WF
σ/μ = 9.50 ± 0.21 %
μ = 406.0 ± 0.8
σ = 38.6 ± 0.8

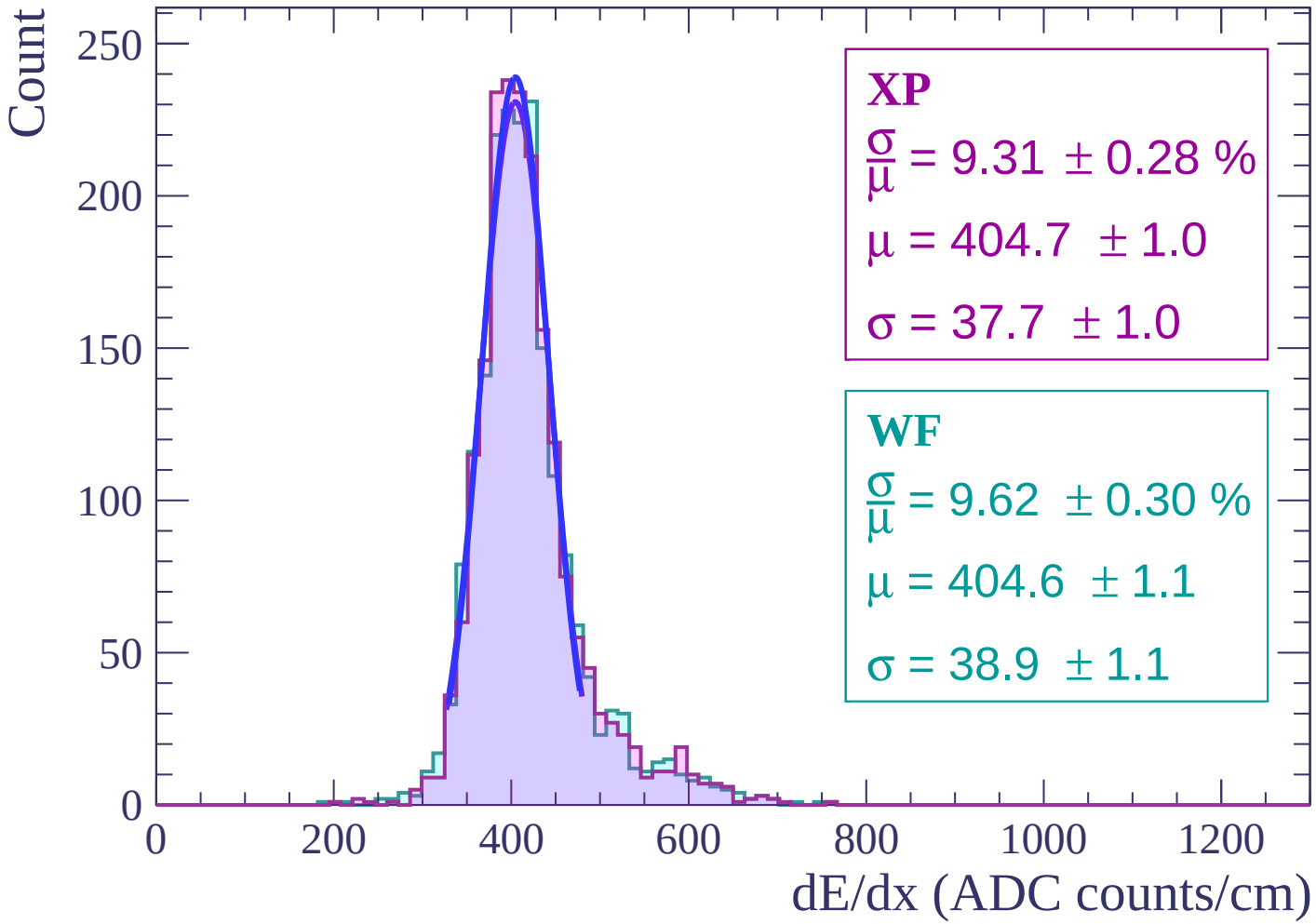

Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.32 ± 0.21 %
μ = 404.3 ± 0.8
σ = 37.7 ± 0.8
WF
σ/μ = 9.54 ± 0.22 %
μ = 404.0 ± 0.9
σ = 38.6 ± 0.8

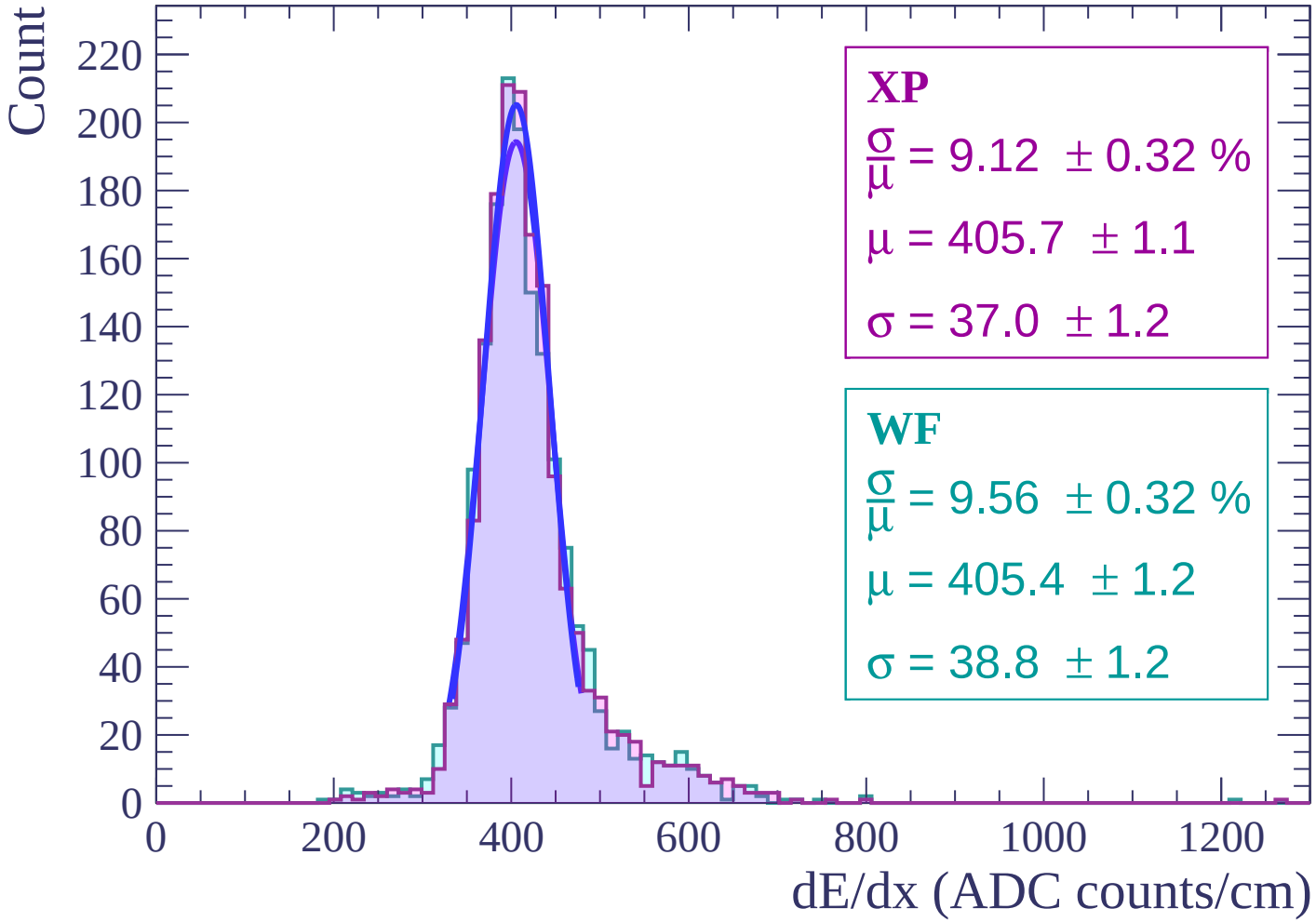

Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.52 ± 0.22 %
μ = 403.6 ± 0.9
σ = 38.4 ± 0.8
WF
σ/μ = 10.02 ± 0.25 %
μ = 403.1 ± 0.9
σ = 40.4 ± 0.9

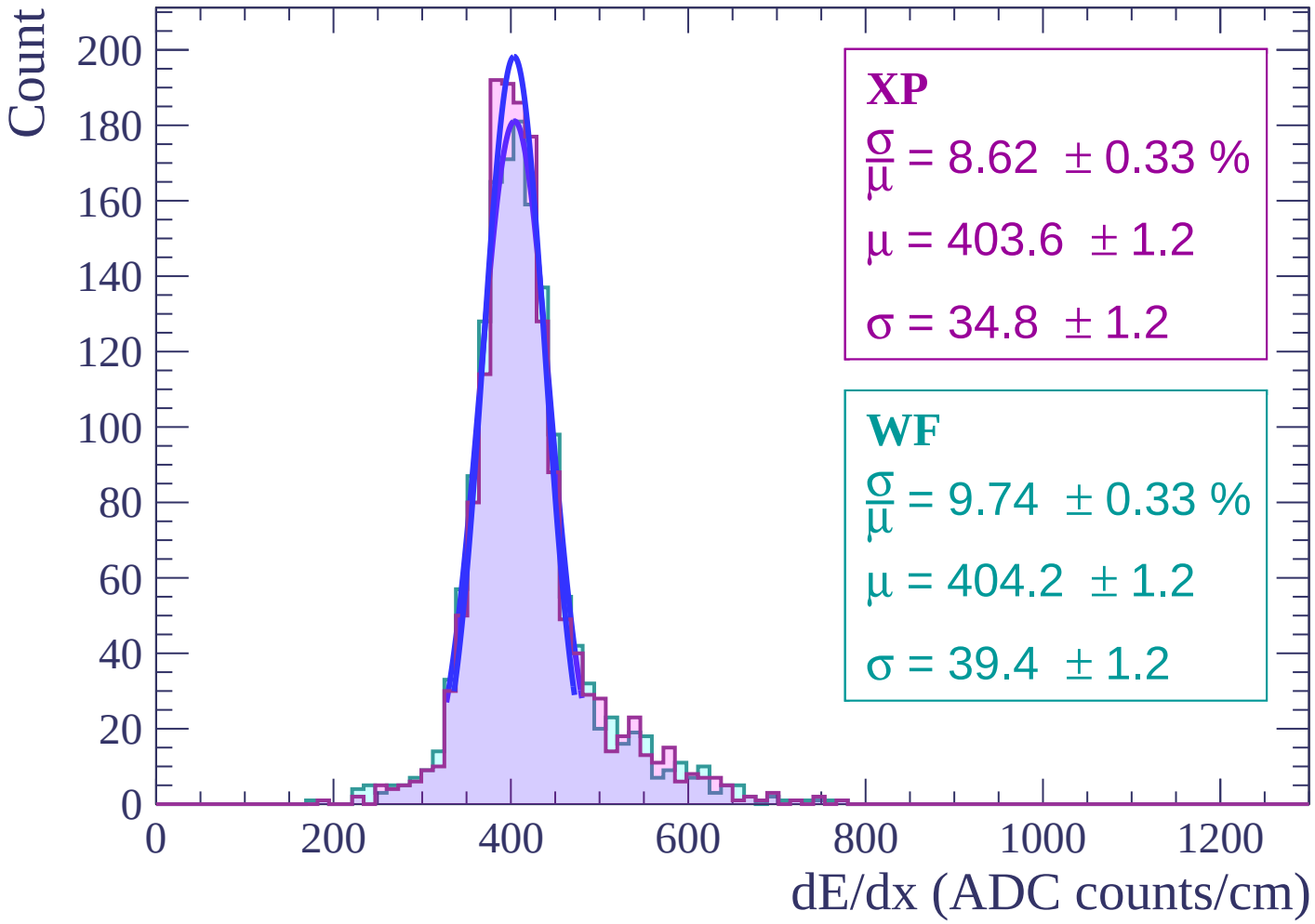

Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.82 ± 0.27 %
μ = 403.8 ± 0.9
σ = 35.6 ± 1.0
WF
σ/μ = 9.77 ± 0.27 %
μ = 405.0 ± 1.0
σ = 39.6 ± 1.0

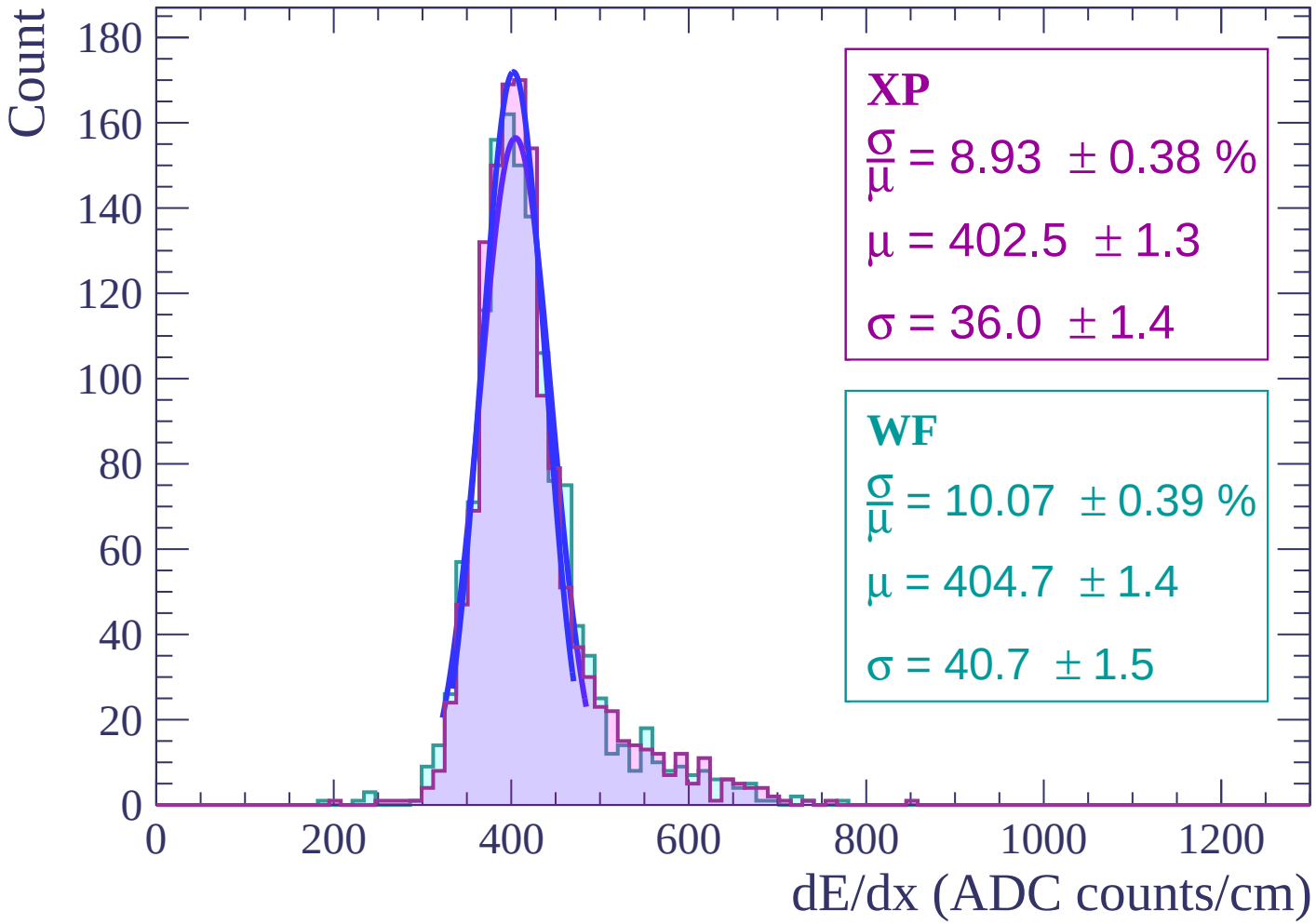

Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.09 ± 0.23 %
μ = 405.4 ± 0.9
σ = 36.8 ± 0.9
WF
σ/μ = 9.88 ± 0.28 %
μ = 405.4 ± 1.0
σ = 40.1 ± 1.0

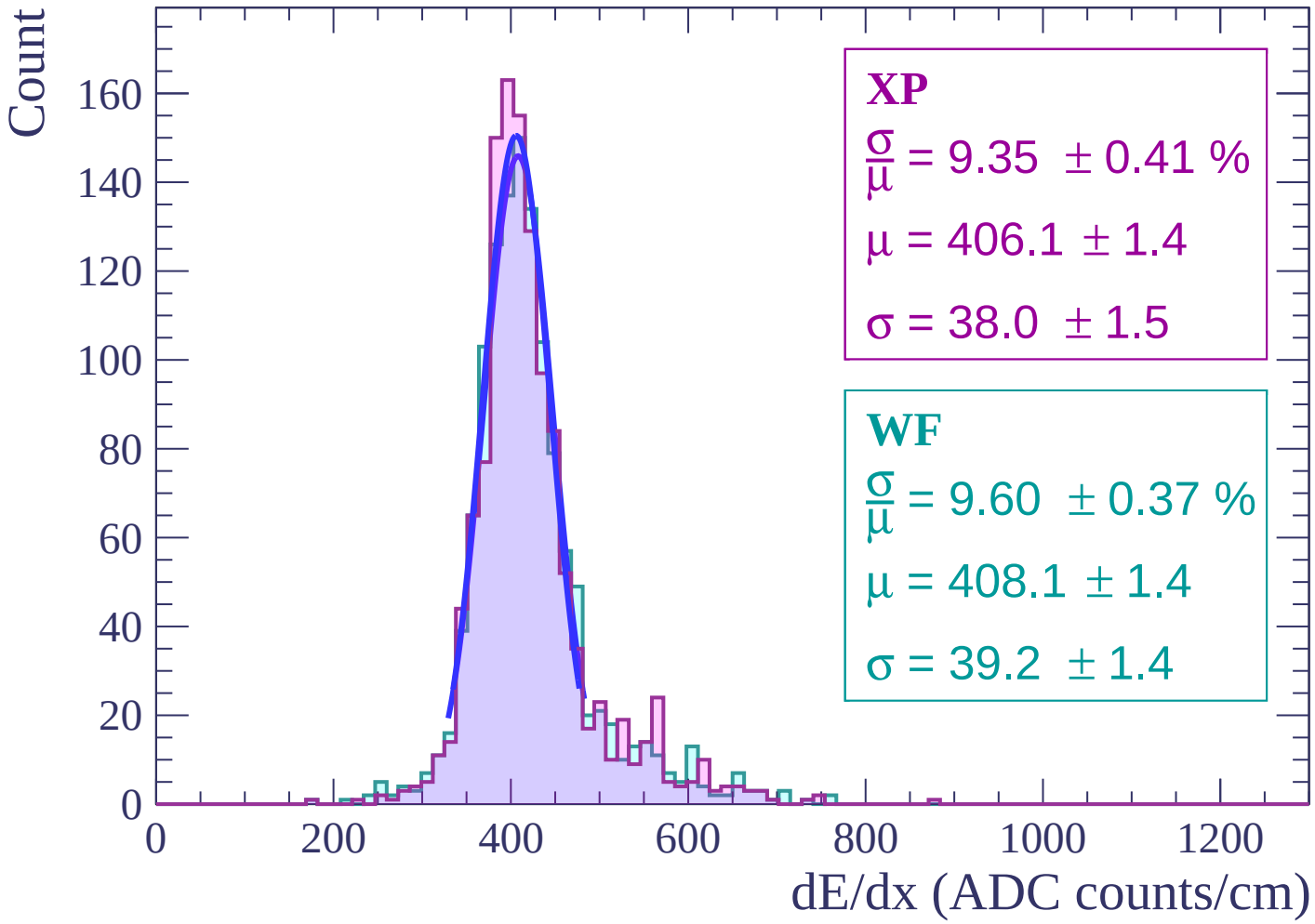

Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.24 ± 0.25 %
μ = 406.3 ± 1.0
σ = 37.6 ± 0.9
WF
σ/μ = 9.93 ± 0.30 %
μ = 406.6 ± 1.1
σ = 40.4 ± 1.1

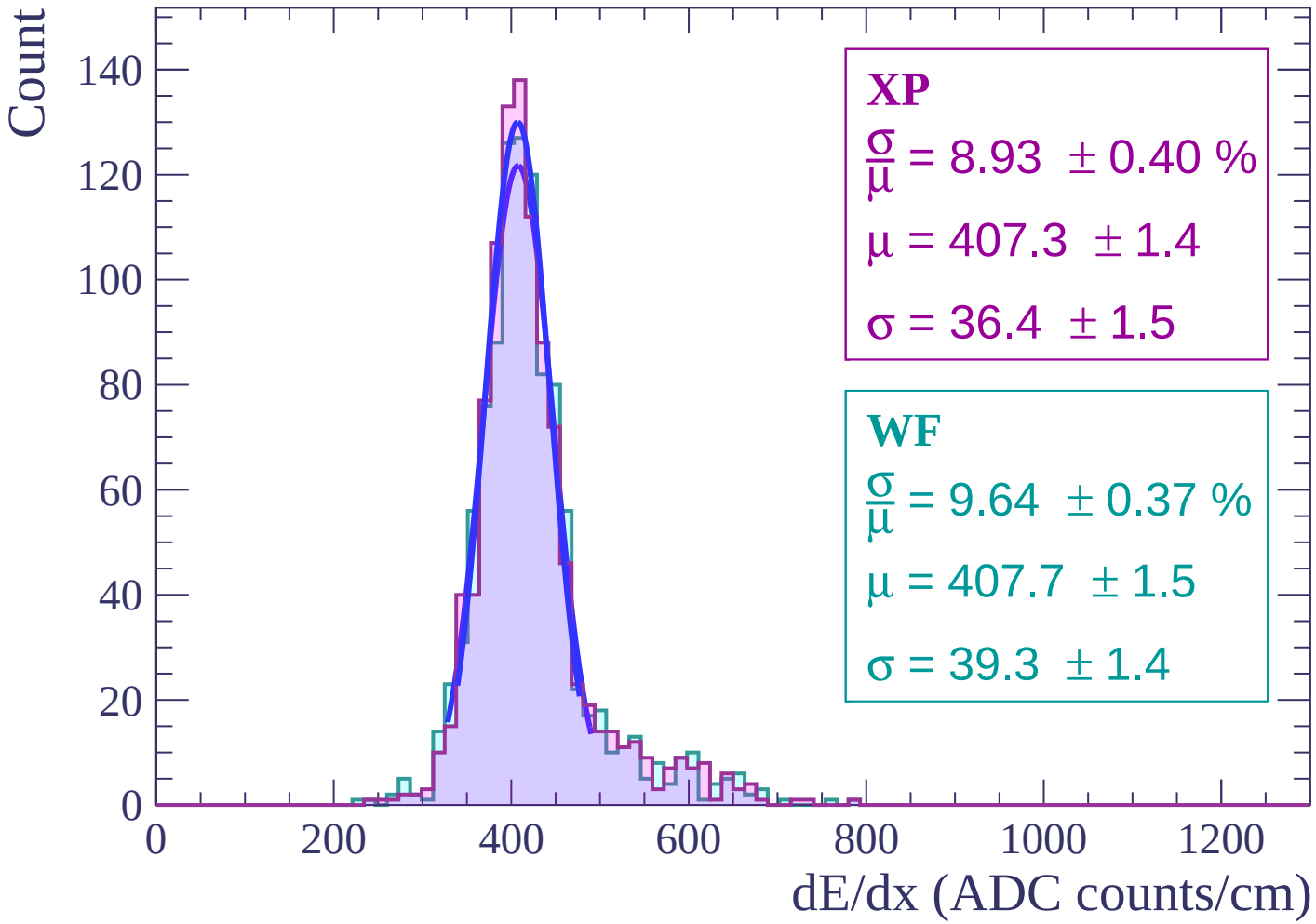

Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.31 ± 0.28 %
μ = 404.7 ± 1.0
σ = 37.7 ± 1.0
WF
σ/μ = 9.62 ± 0.30 %
μ = 404.6 ± 1.1
σ = 38.9 ± 1.1

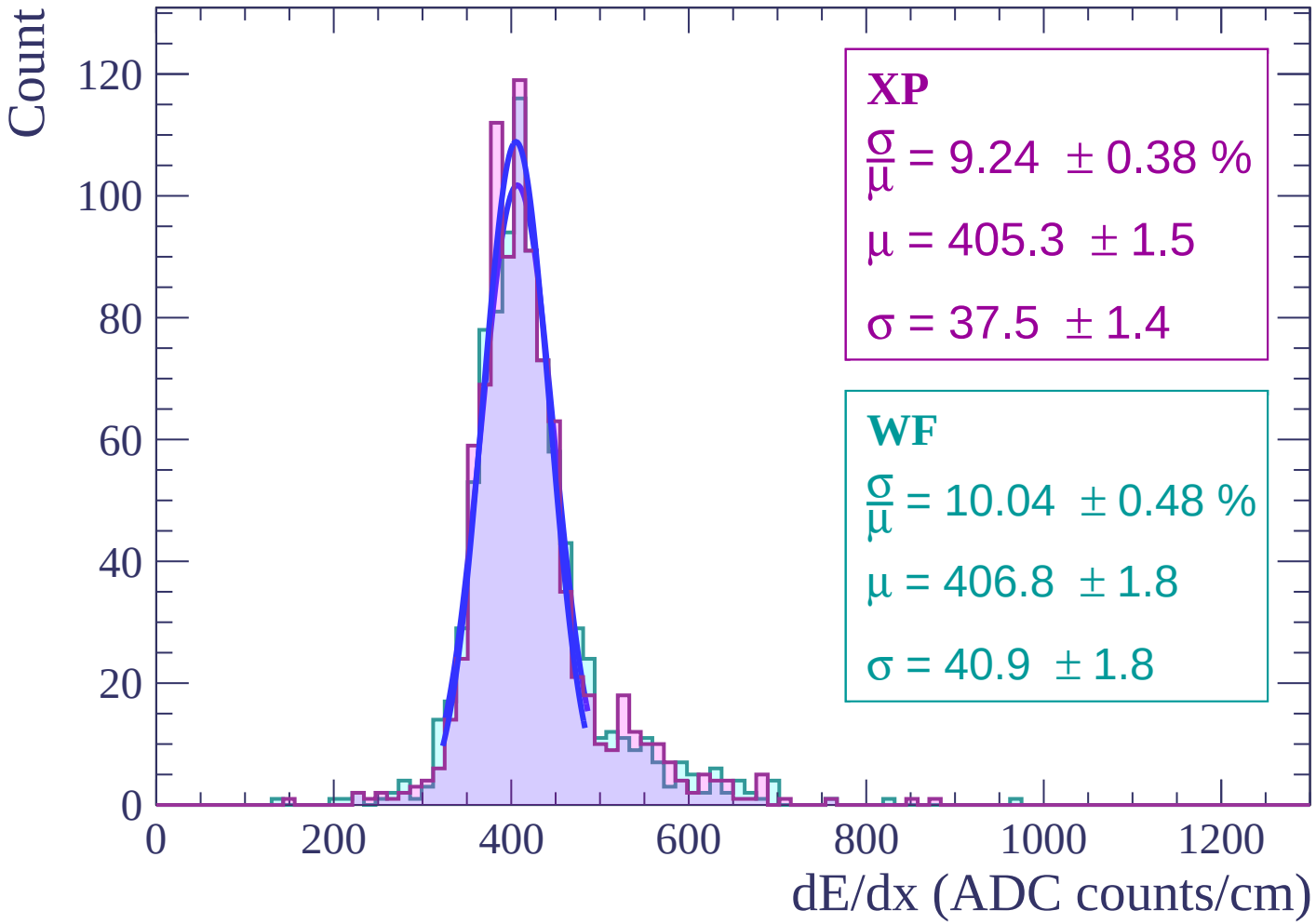

Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.12 ± 0.32 %
μ = 405.7 ± 1.1
σ = 37.0 ± 1.2
WF
σ/μ = 9.56 ± 0.32 %
μ = 405.4 ± 1.2
σ = 38.8 ± 1.2

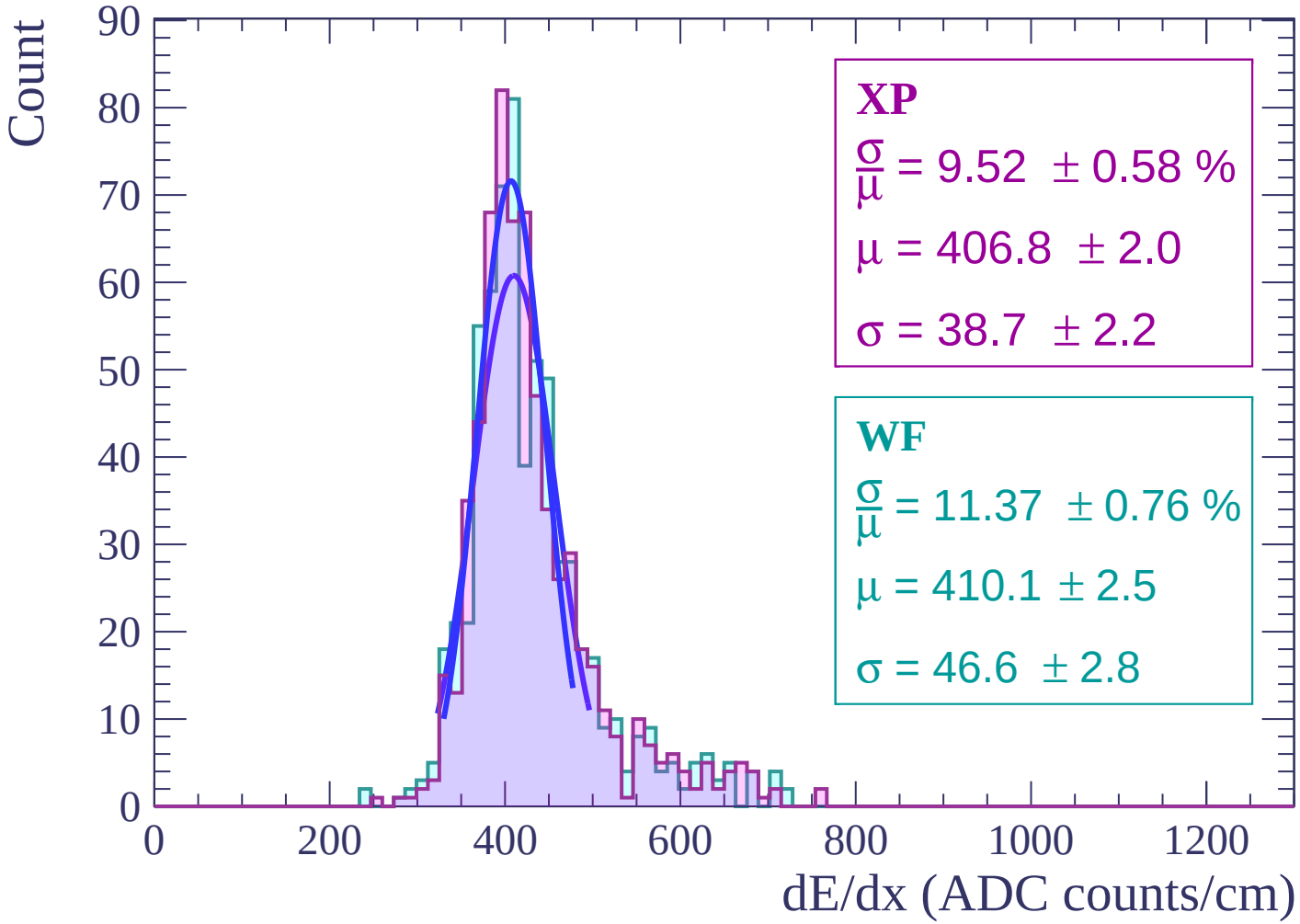

Count
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.62 ± 0.33 %
μ = 403.6 ± 1.2
σ = 34.8 ± 1.2
WF
σ/μ = 9.74 ± 0.33 %
μ = 404.2 ± 1.2
σ = 39.4 ± 1.2

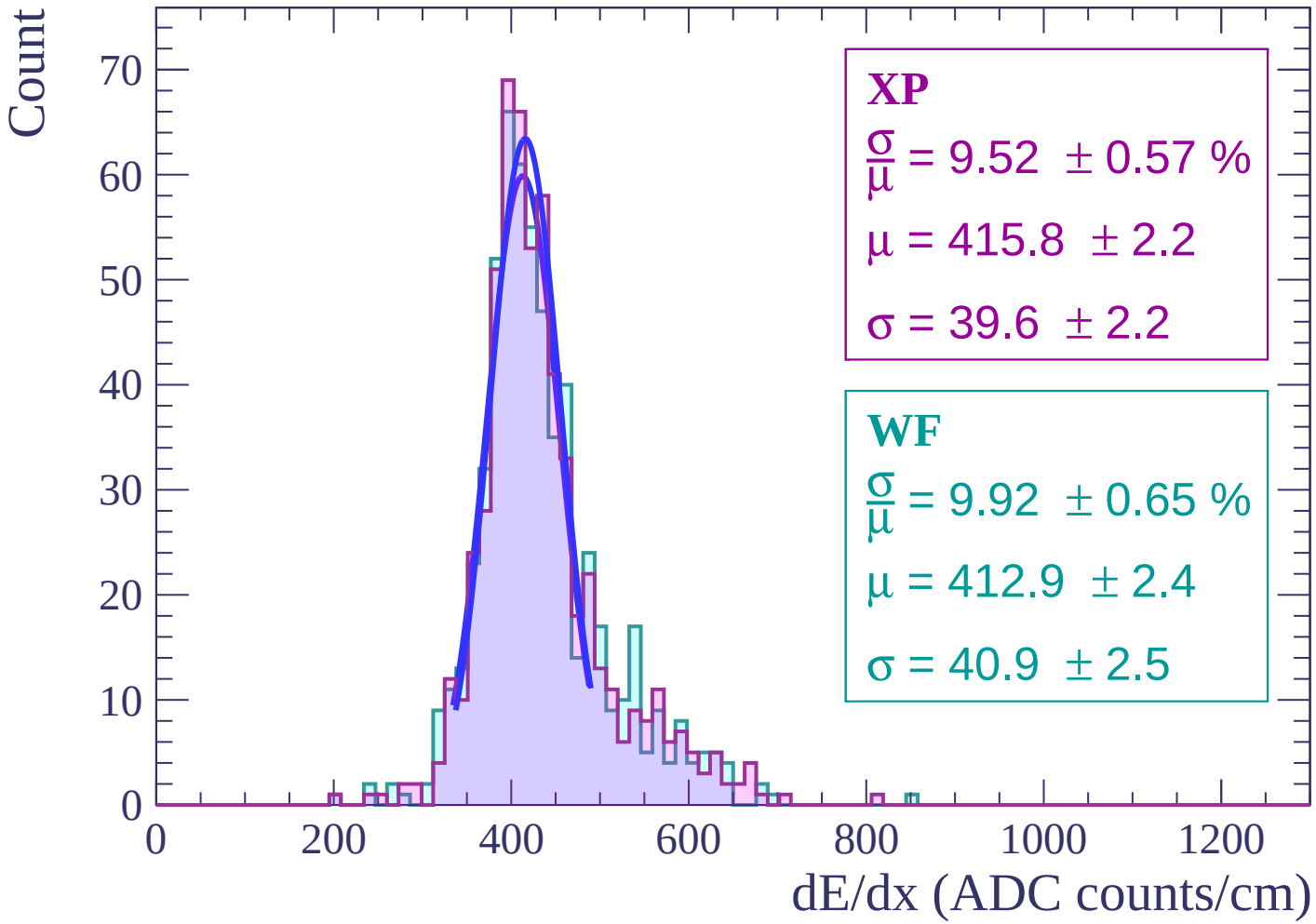

Count
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.93 ± 0.38 %
μ = 402.5 ± 1.3
σ = 36.0 ± 1.4
WF
σ/μ = 10.07 ± 0.39 %
μ = 404.7 ± 1.4
σ = 40.7 ± 1.5

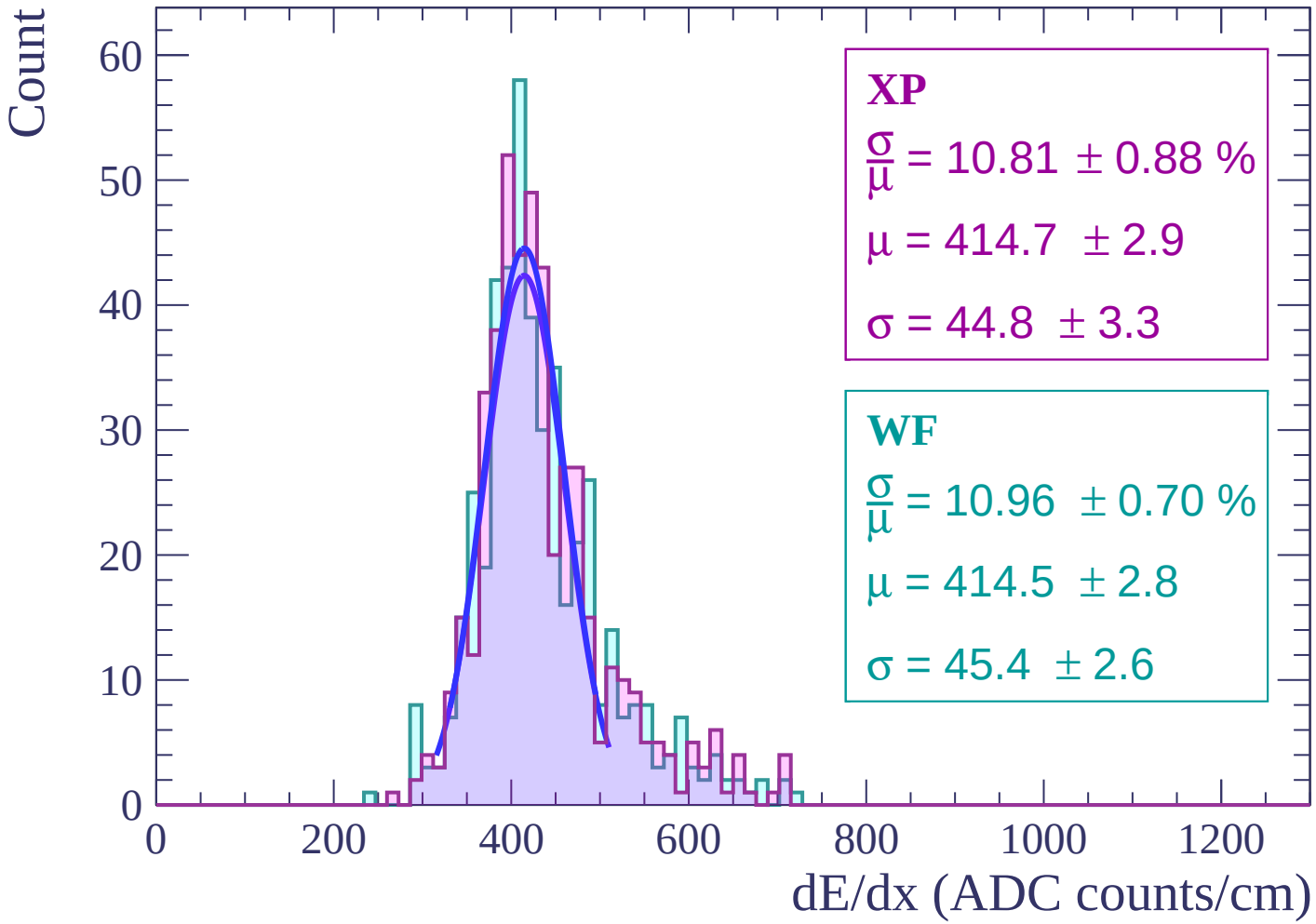

Count
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.35 ± 0.41 %
μ = 406.1 ± 1.4
σ = 38.0 ± 1.5
WF
σ/μ = 9.60 ± 0.37 %
μ = 408.1 ± 1.4
σ = 39.2 ± 1.4

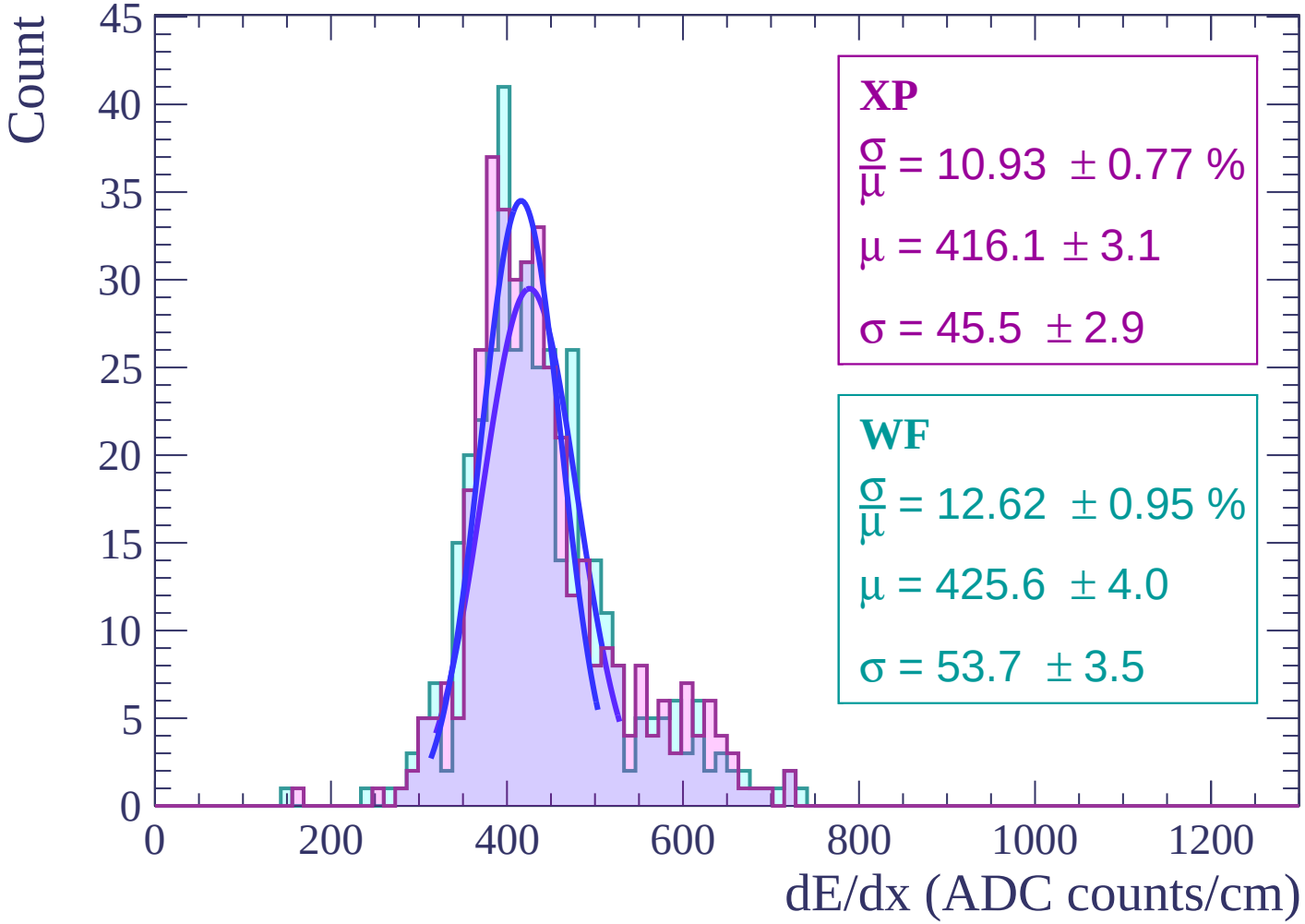

Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.93 ± 0.40 %
μ = 407.3 ± 1.4
σ = 36.4 ± 1.5
WF
σ/μ = 9.64 ± 0.37 %
μ = 407.7 ± 1.5
σ = 39.3 ± 1.4

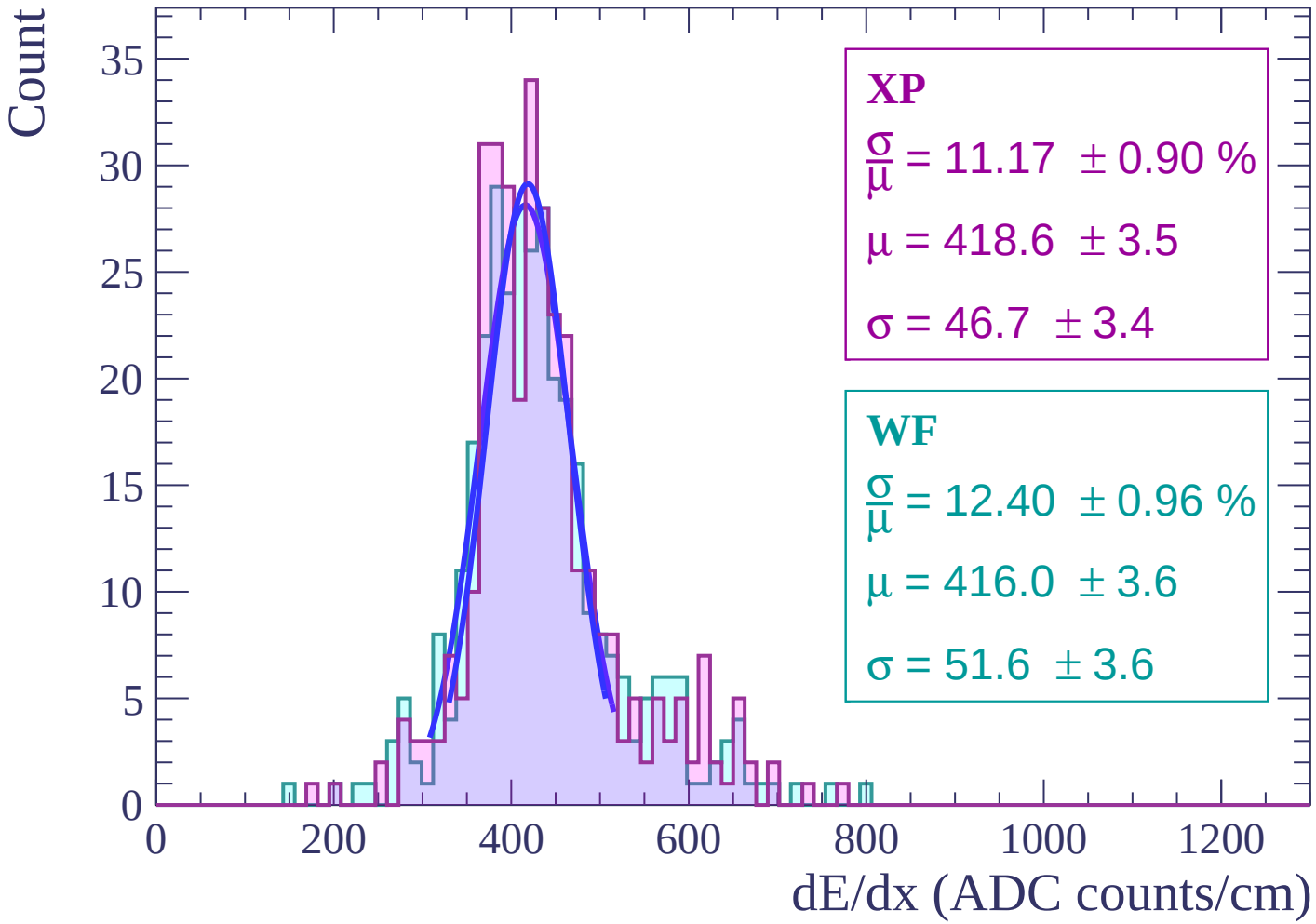

Count
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.24 ± 0.38 %
μ = 405.3 ± 1.5
σ = 37.5 ± 1.4
WF
σ/μ = 10.04 ± 0.48 %
μ = 406.8 ± 1.8
σ = 40.9 ± 1.8

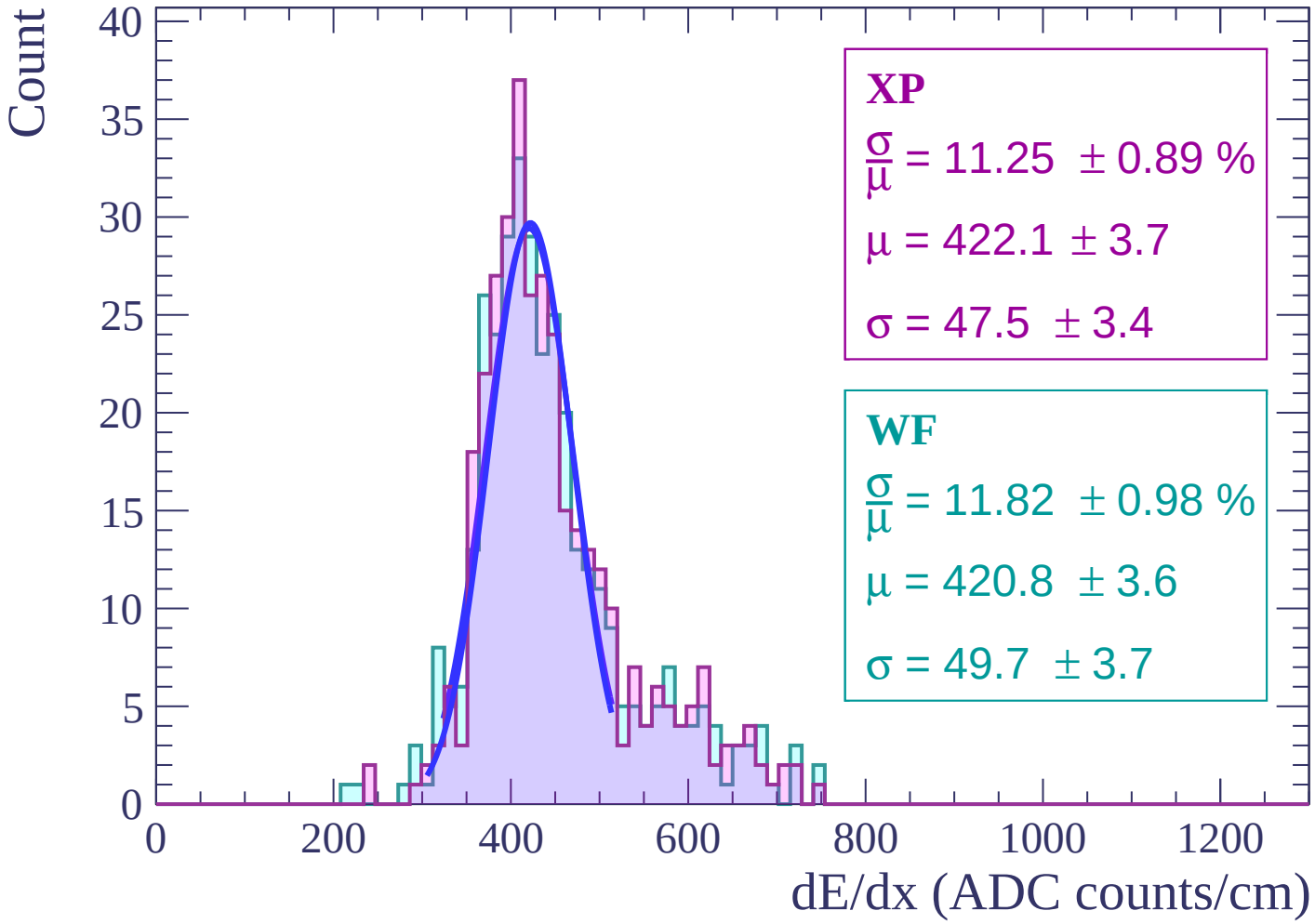

Count
90
80
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.52 ± 0.58 %
μ = 406.8 ± 2.0
σ = 38.7 ± 2.2
WF
σ/μ = 11.37 ± 0.76 %
μ = 410.1 ± 2.5
σ = 46.6 ± 2.8

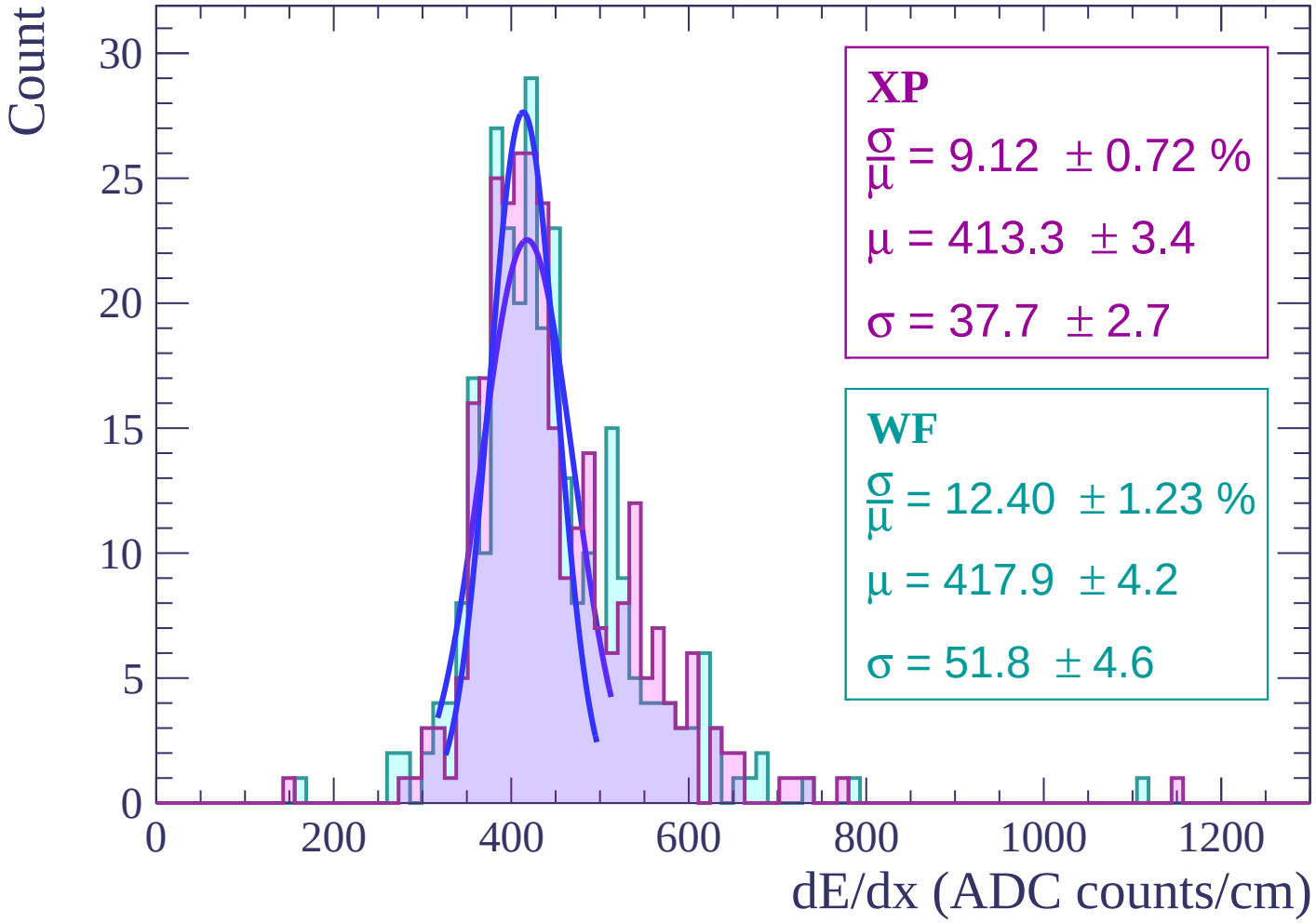

Count
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.52 ± 0.57 %
μ = 415.8 ± 2.2
σ = 39.6 ± 2.2
WF
σ/μ = 9.92 ± 0.65 %
μ = 412.9 ± 2.4
σ = 40.9 ± 2.5

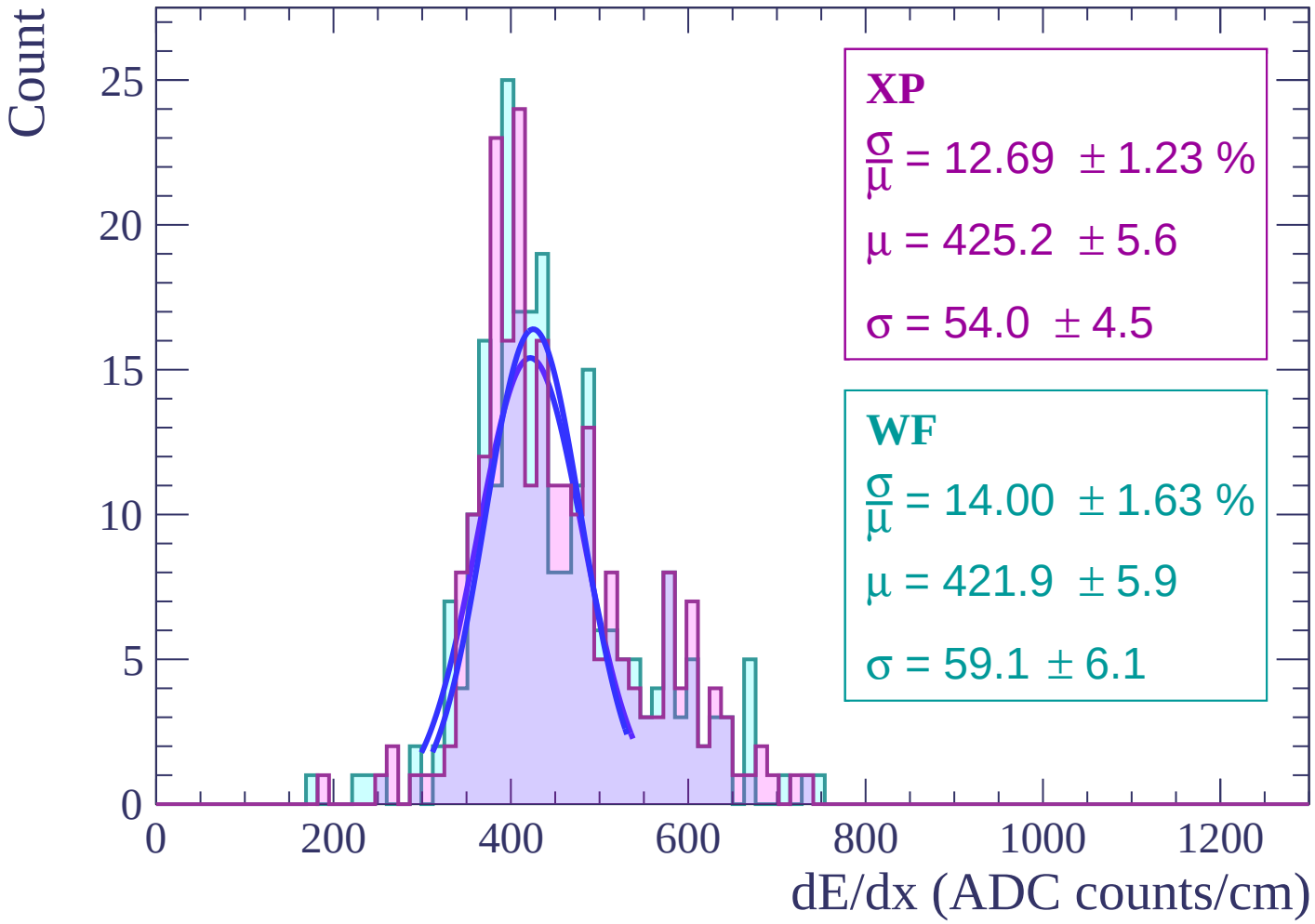

Count
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.81 ± 0.88 %
μ = 414.7 ± 2.9
σ = 44.8 ± 3.3
WF
σ/μ = 10.96 ± 0.70 %
μ = 414.5 ± 2.8
σ = 45.4 ± 2.6

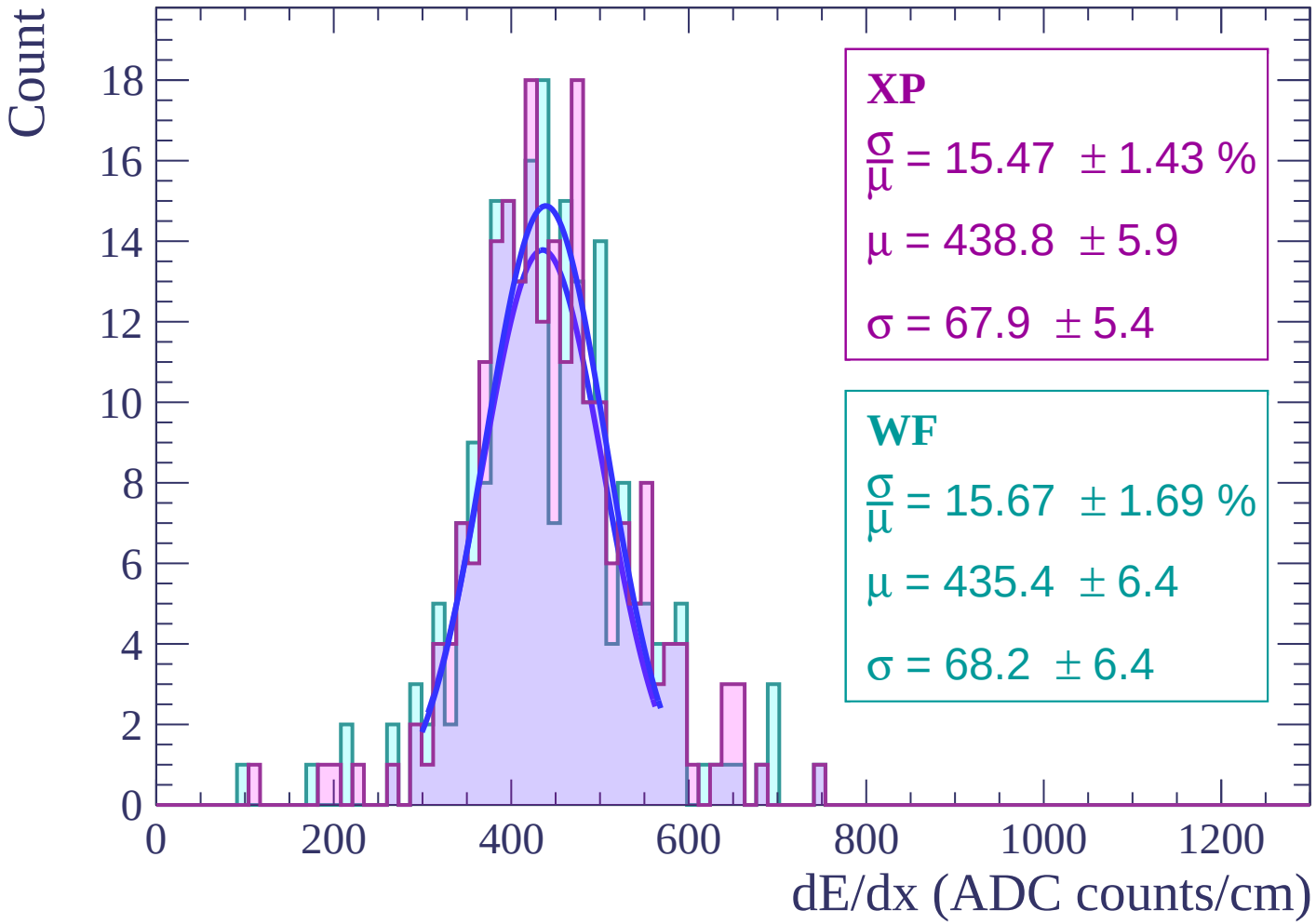

Count
45
40
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.93 ± 0.77 %
μ = 416.1 ± 3.1
σ = 45.5 ± 2.9
WF
σ/μ = 12.62 ± 0.95 %
μ = 425.6 ± 4.0
σ = 53.7 ± 3.5

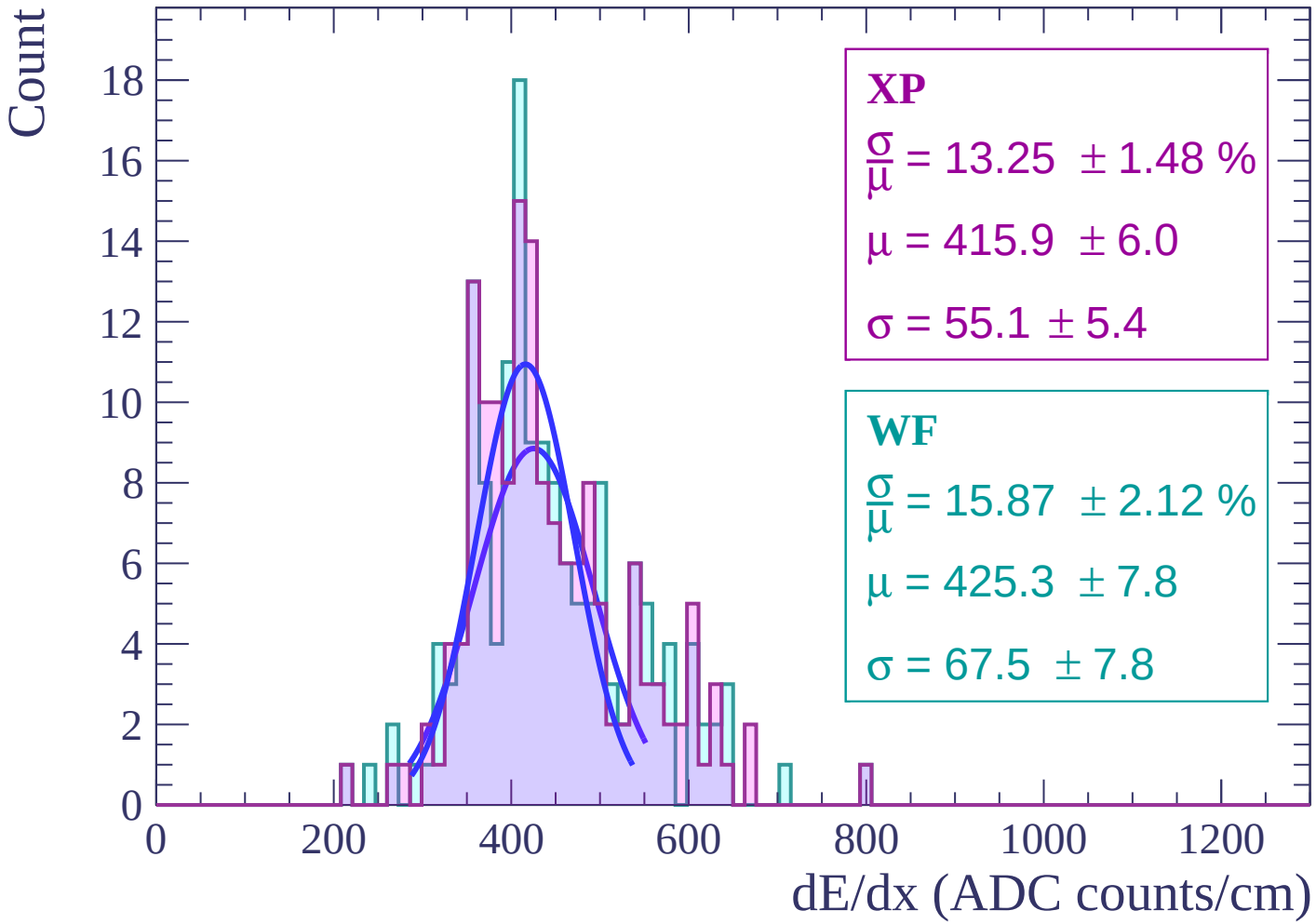

Count
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.17 ± 0.90 %
μ = 418.6 ± 3.5
σ = 46.7 ± 3.4
WF
σ/μ = 12.40 ± 0.96 %
μ = 416.0 ± 3.6
σ = 51.6 ± 3.6

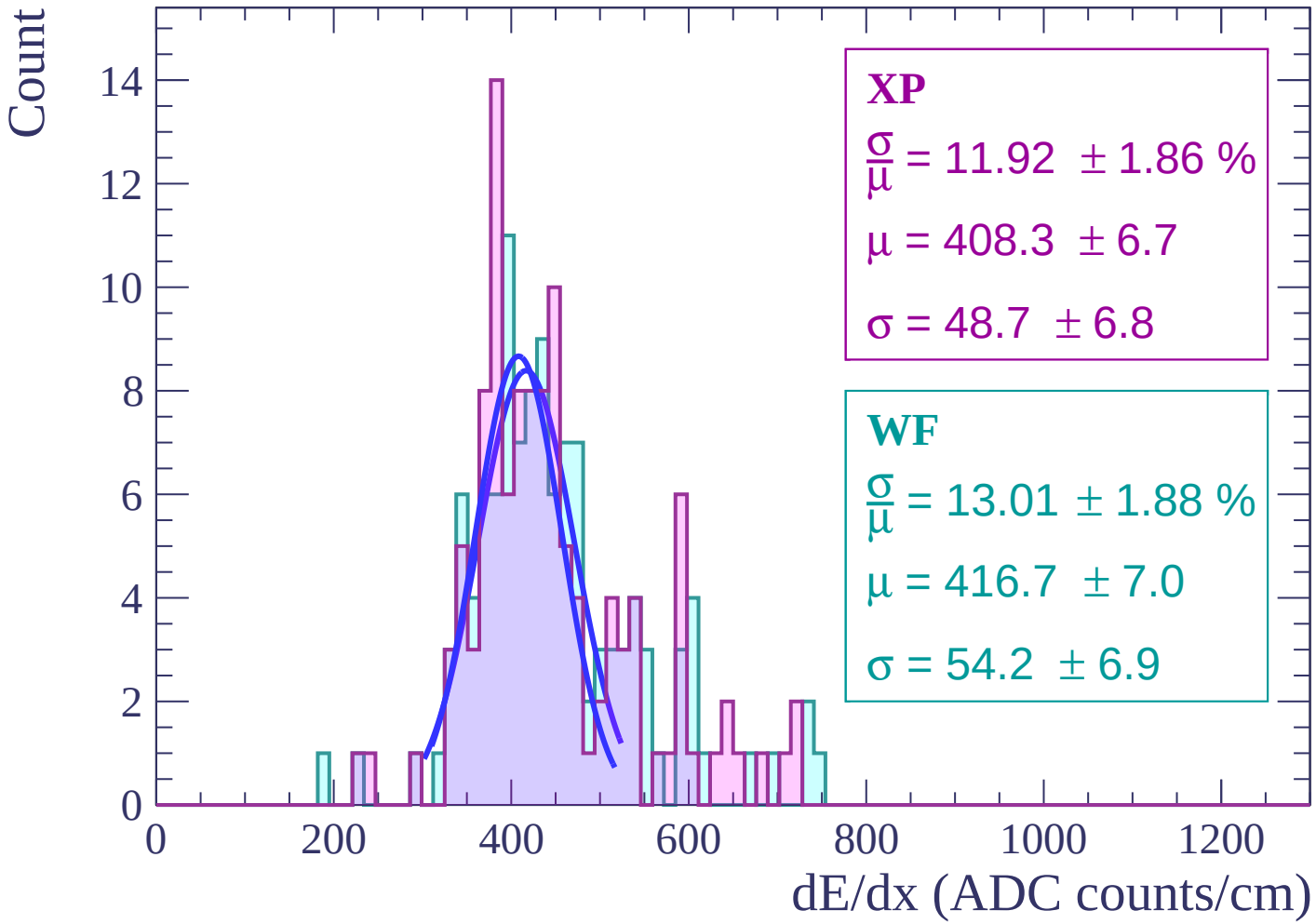

Count
40
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.25 ± 0.89 %
μ = 422.1 ± 3.7
σ = 47.5 ± 3.4
WF
σ/μ = 11.82 ± 0.98 %
μ = 420.8 ± 3.6
σ = 49.7 ± 3.7

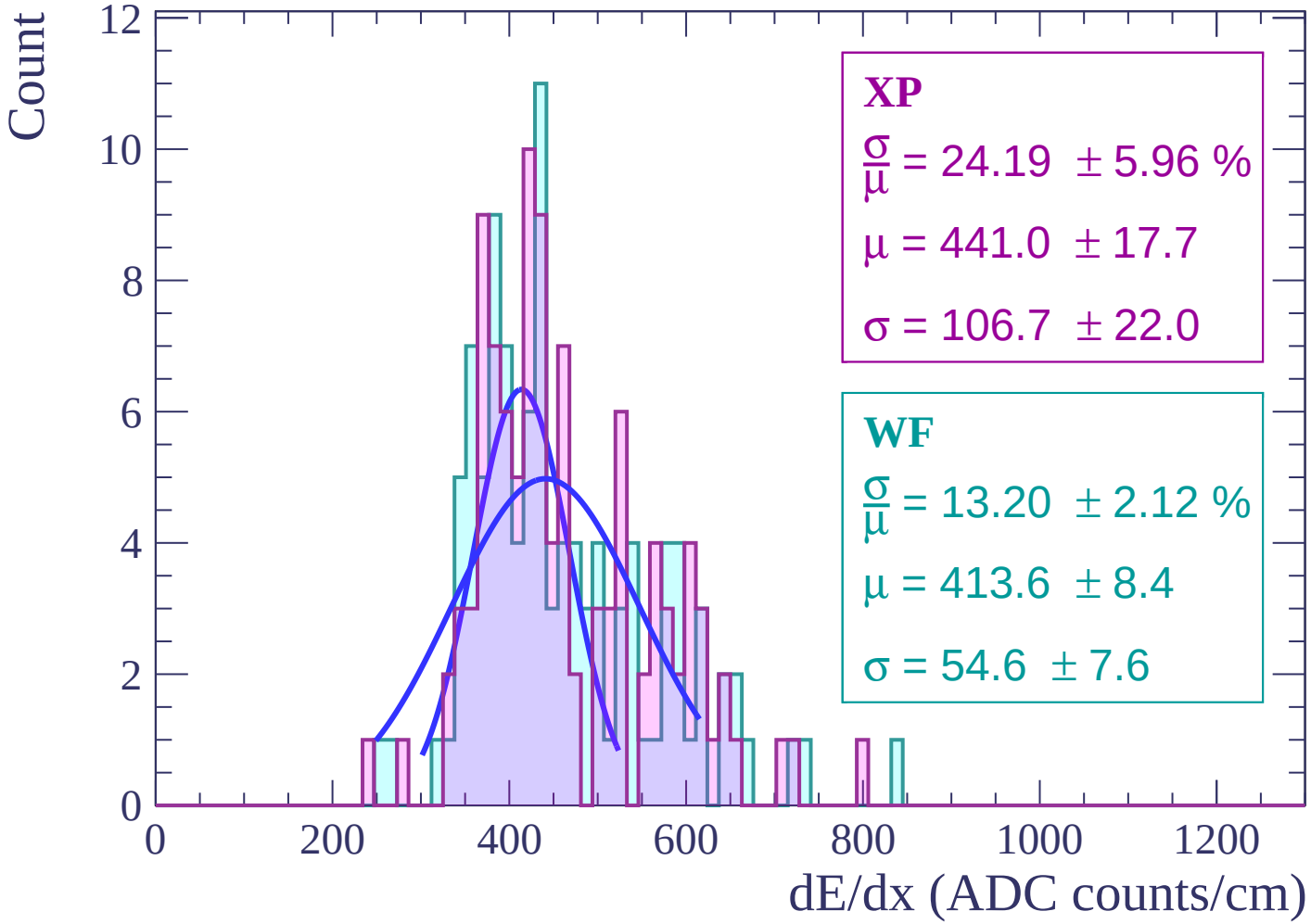

Count
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.12 ± 0.72 %
μ = 413.3 ± 3.4
σ = 37.7 ± 2.7
WF
σ/μ = 12.40 ± 1.23 %
μ = 417.9 ± 4.2
σ = 51.8 ± 4.6

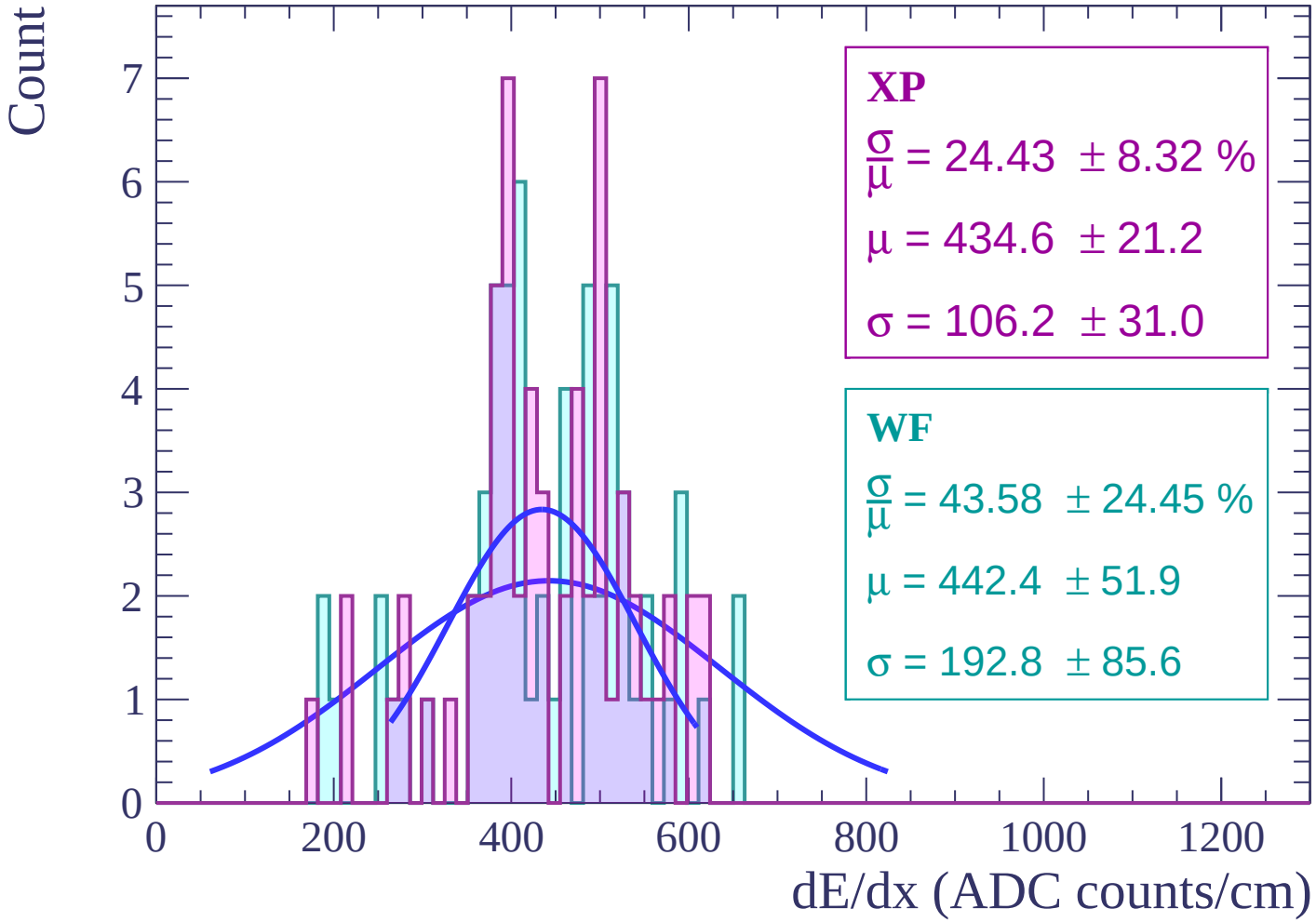

Count
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 12.69 ± 1.23 %
μ = 425.2 ± 5.6
σ = 54.0 ± 4.5
WF
σ/μ = 14.00 ± 1.63 %
μ = 421.9 ± 5.9
σ = 59.1 ± 6.1

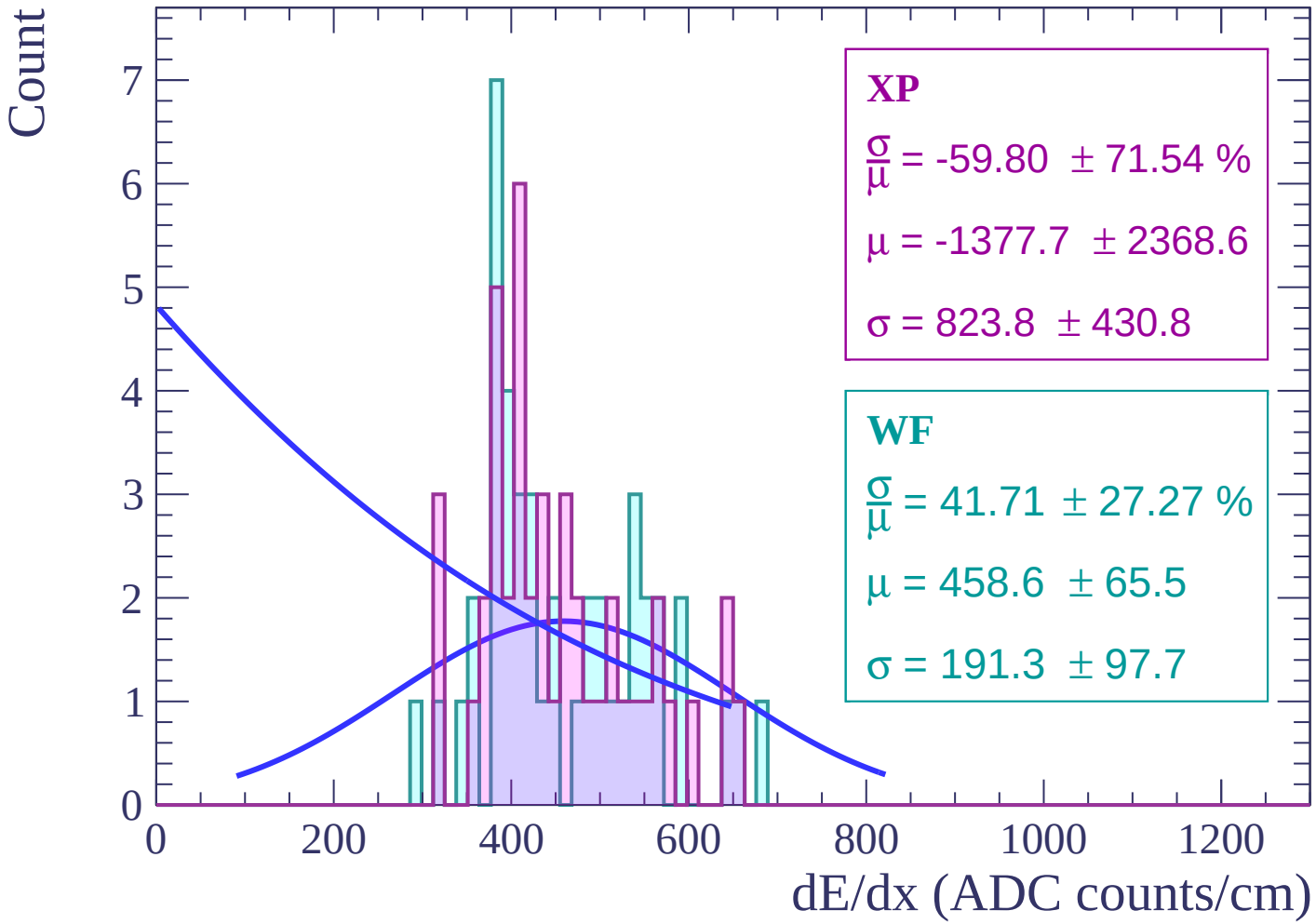

Count
18
16
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.47 ± 1.43 %
μ = 438.8 ± 5.9
σ = 67.9 ± 5.4
WF
σ/μ = 15.67 ± 1.69 %
μ = 435.4 ± 6.4
σ = 68.2 ± 6.4

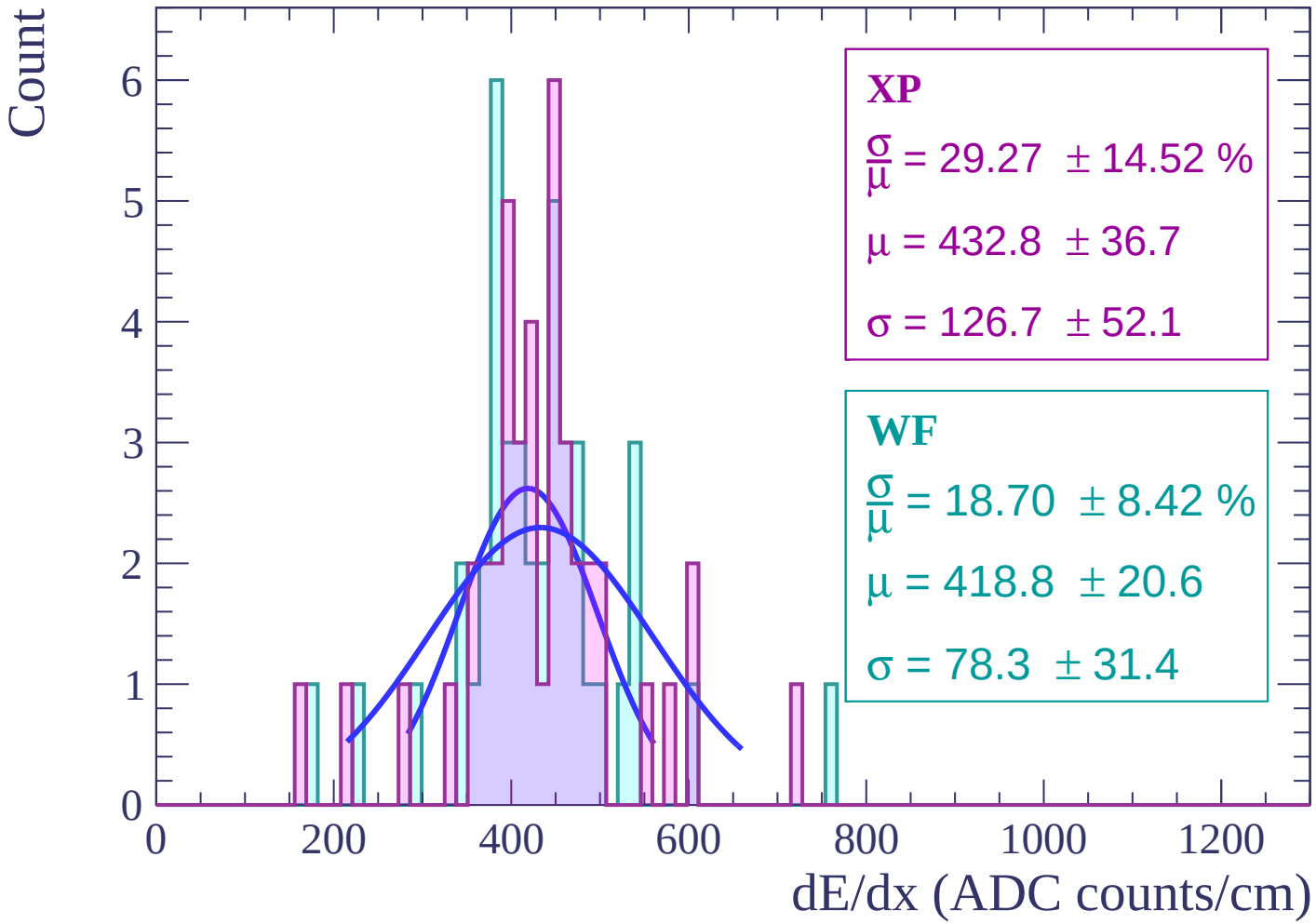

Count
18
16
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.25 ± 1.48 %
μ = 415.9 ± 6.0
σ = 55.1 ± 5.4
WF
σ/μ = 15.87 ± 2.12 %
μ = 425.3 ± 7.8
σ = 67.5 ± 7.8

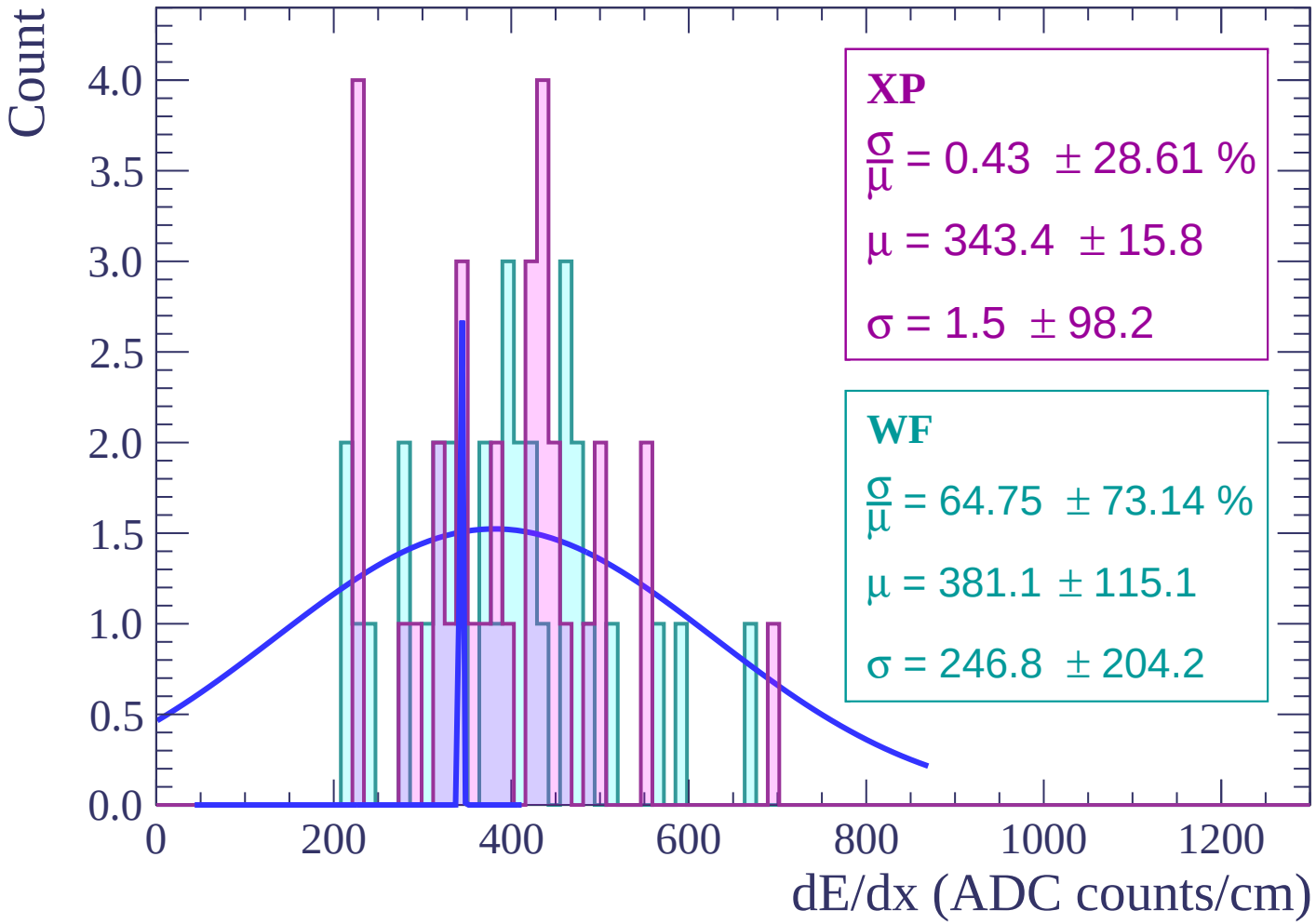

Count
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 11.92 ± 1.86 %
μ = 408.3 ± 6.7
σ = 48.7 ± 6.8
WF
σ/μ = 13.01 ± 1.88 %
μ = 416.7 ± 7.0
σ = 54.2 ± 6.9

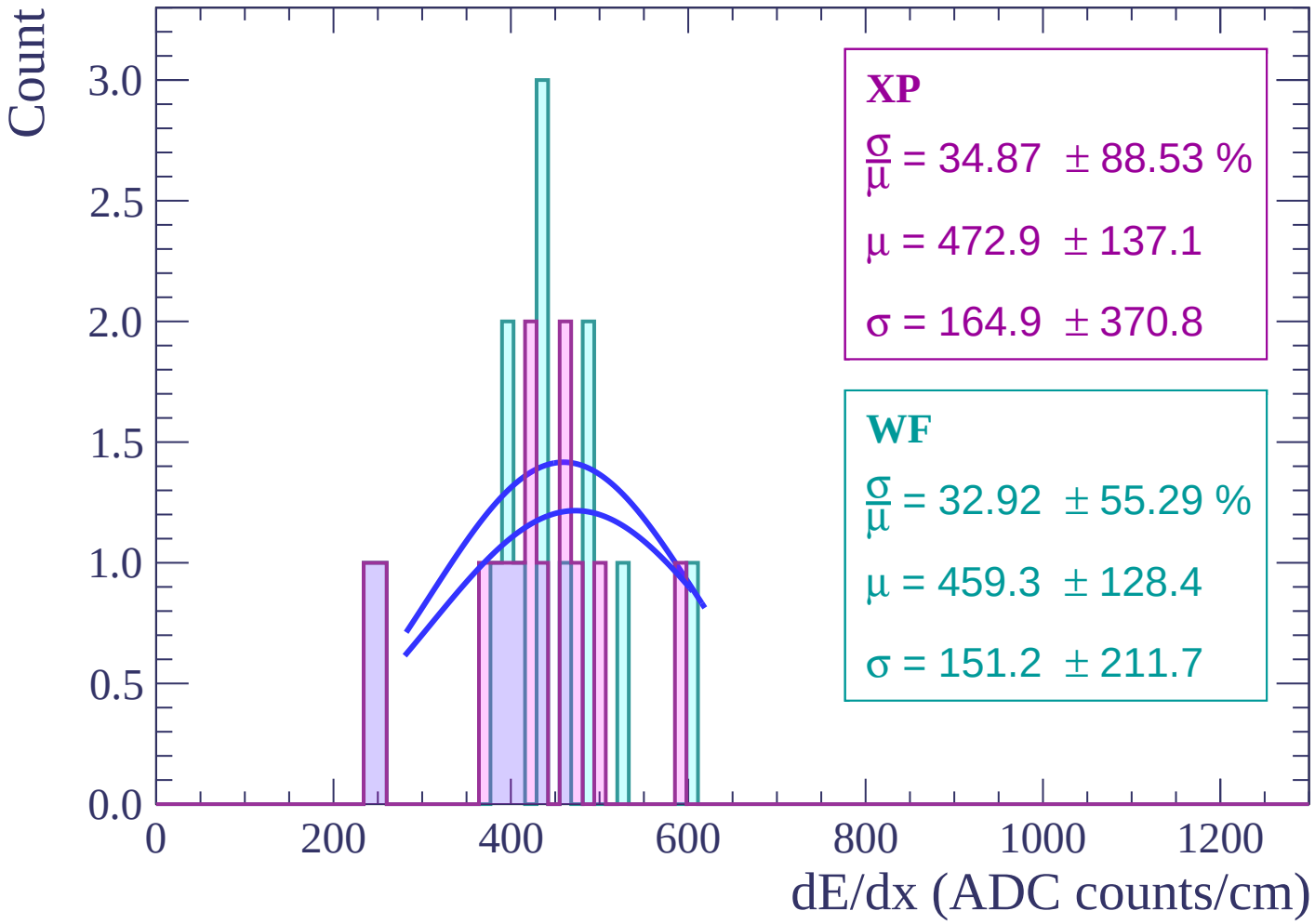

Count
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 24.19 ± 5.96 %
μ = 441.0 ± 17.7
σ = 106.7 ± 22.0
WF
σ/μ = 13.20 ± 2.12 %
μ = 413.6 ± 8.4
σ = 54.6 ± 7.6

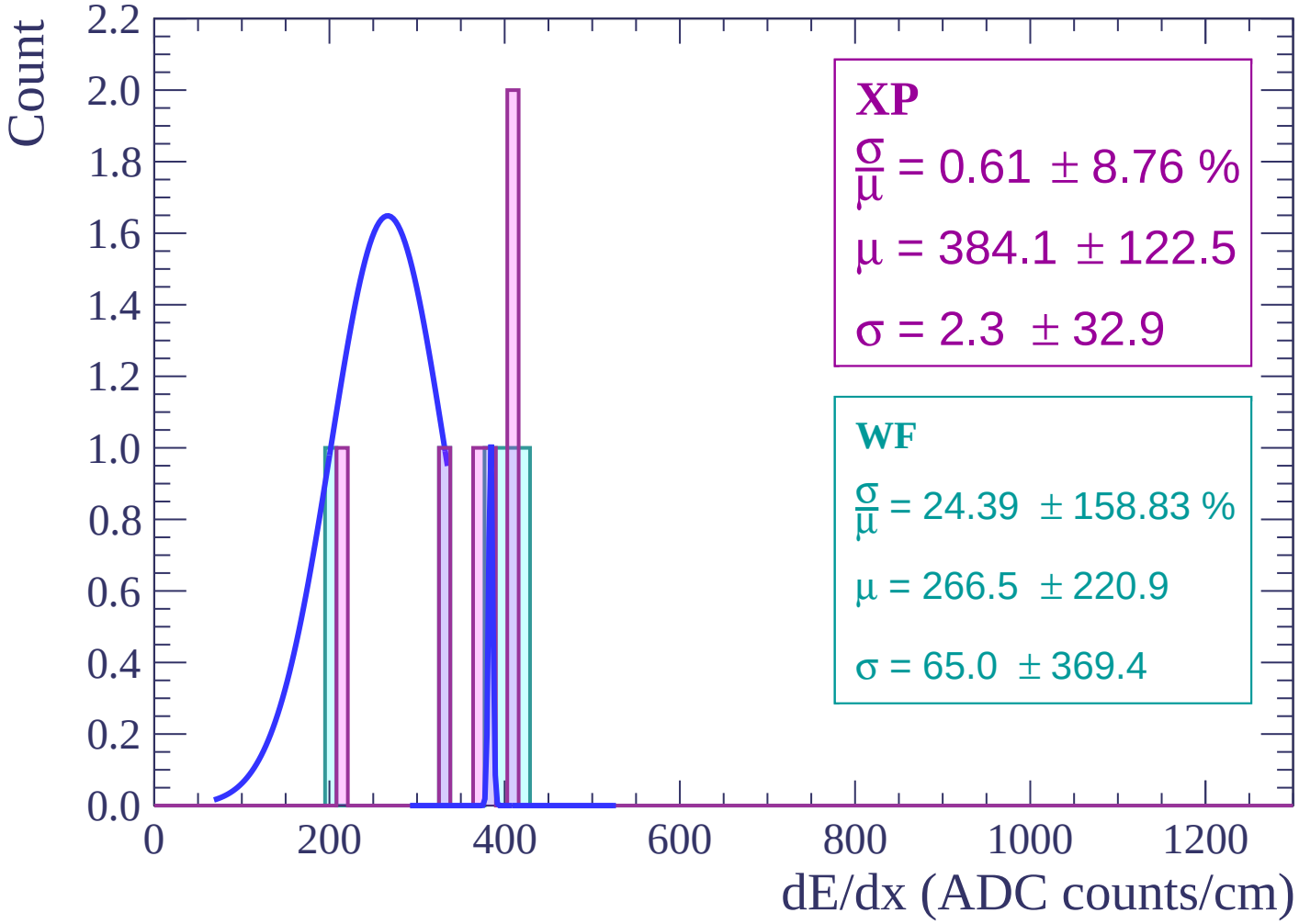

Count
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 24.43 ± 8.32 %
μ = 434.6 ± 21.2
σ = 106.2 ± 31.0
WF
σ/μ = 43.58 ± 24.45 %
μ = 442.4 ± 51.9
σ = 192.8 ± 85.6

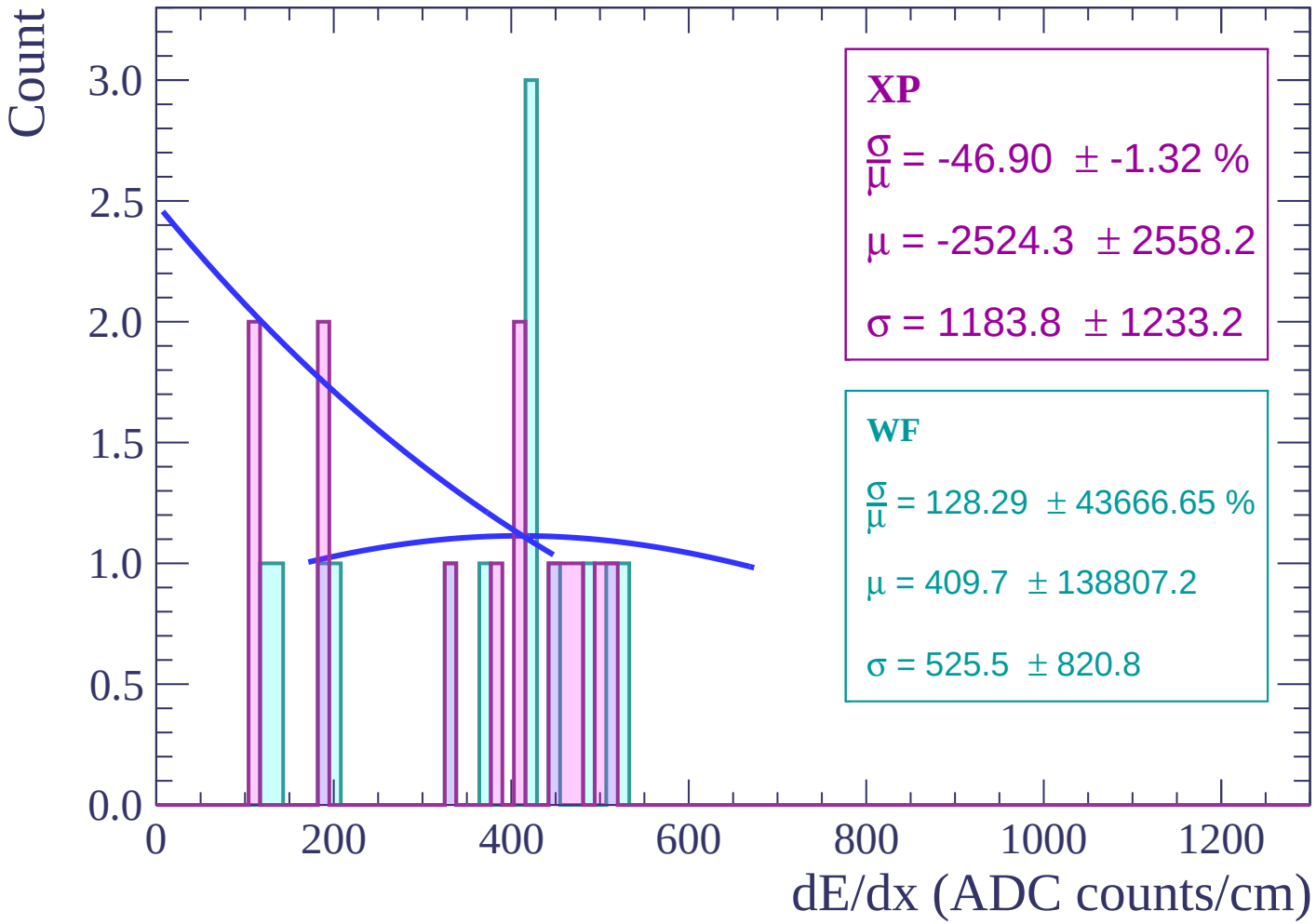

Count
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -59.80 ± 71.54 %
μ = -1377.7 ± 2368.6
σ = 823.8 ± 430.8
WF
σ/μ = 41.71 ± 27.27 %
μ = 458.6 ± 65.5
σ = 191.3 ± 97.7

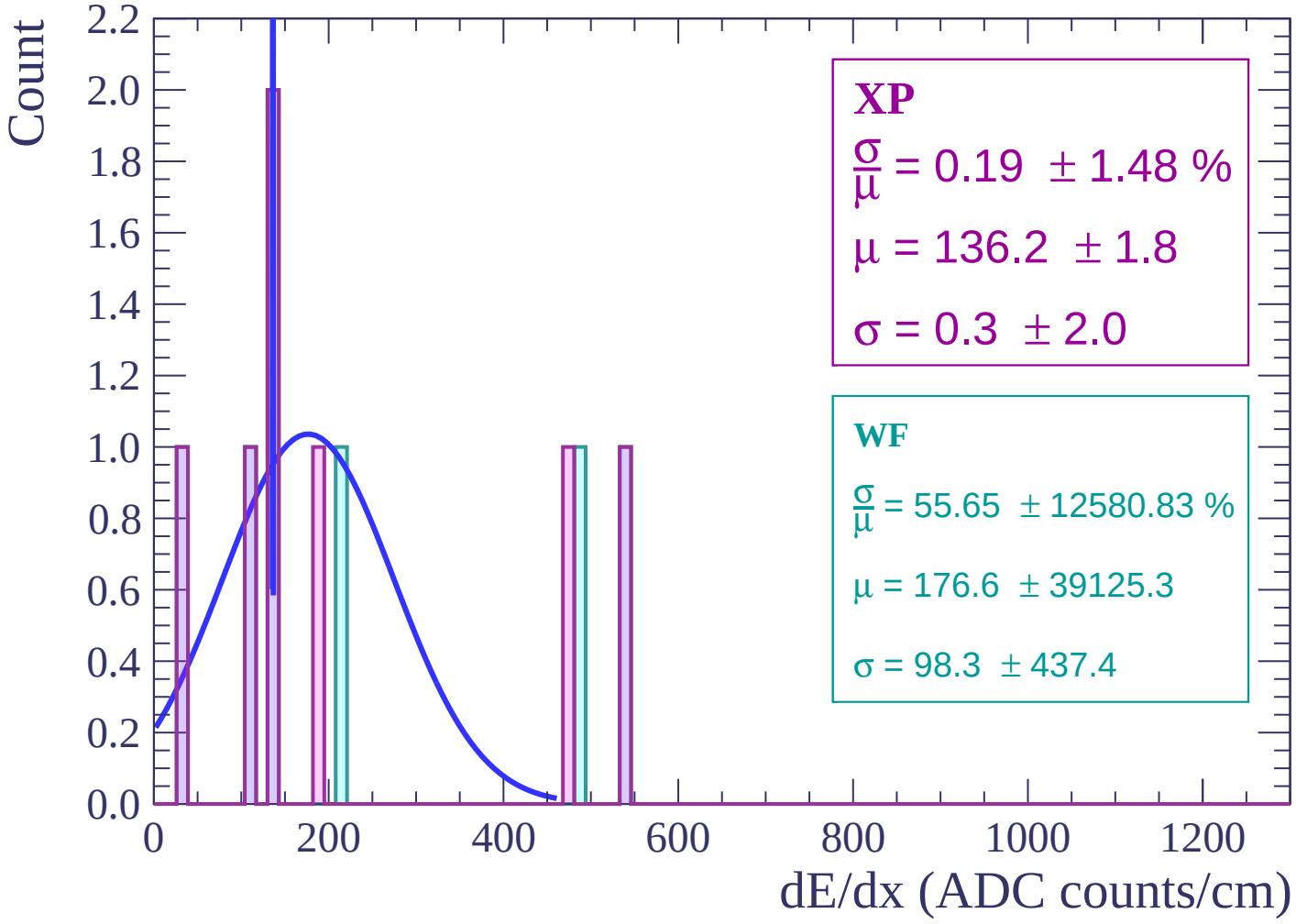

Count
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 29.27 ± 14.52 %
μ = 432.8 ± 36.7
σ = 126.7 ± 52.1
WF
σ/μ = 18.70 ± 8.42 %
μ = 418.8 ± 20.6
σ = 78.3 ± 31.4

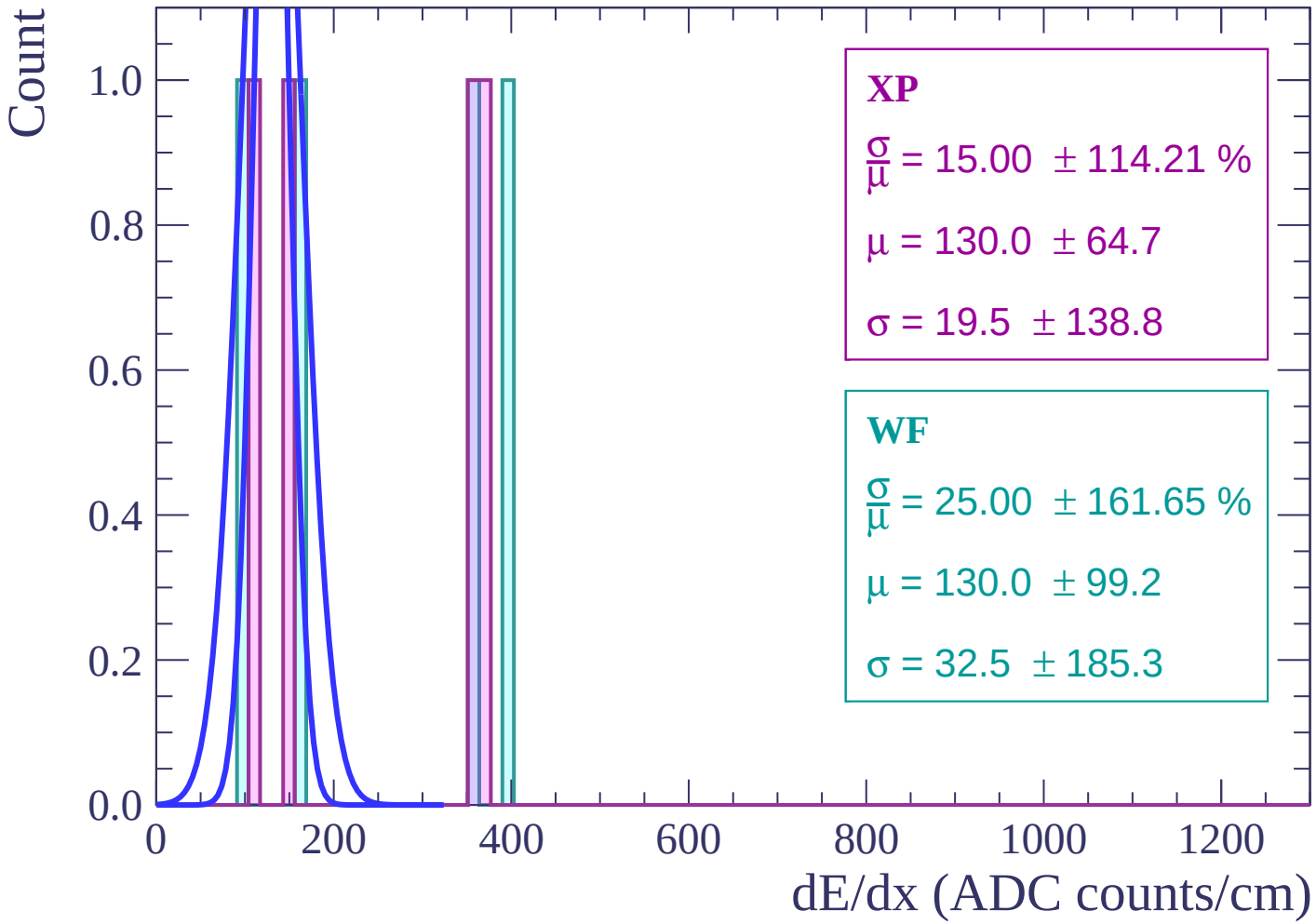

Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.43 ± 28.61 %
μ = 343.4 ± 15.8
σ = 1.5 ± 98.2
WF
σ/μ = 64.75 ± 73.14 %
μ = 381.1 ± 115.1
σ = 246.8 ± 204.2

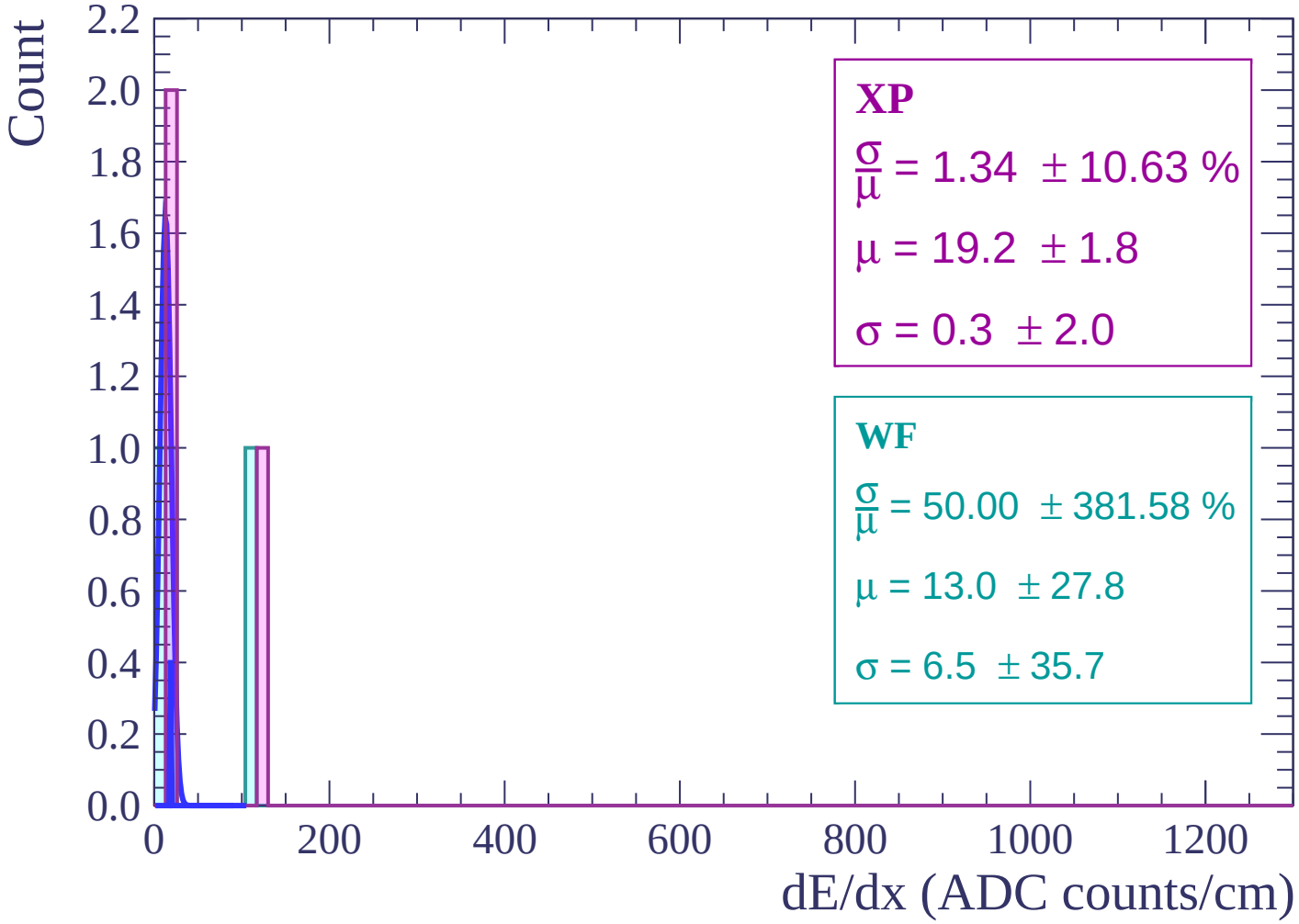

Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 34.87 ± 88.53 %
μ = 472.9 ± 137.1
σ = 164.9 ± 370.8
WF
σ/μ = 32.92 ± 55.29 %
μ = 459.3 ± 128.4
σ = 151.2 ± 211.7

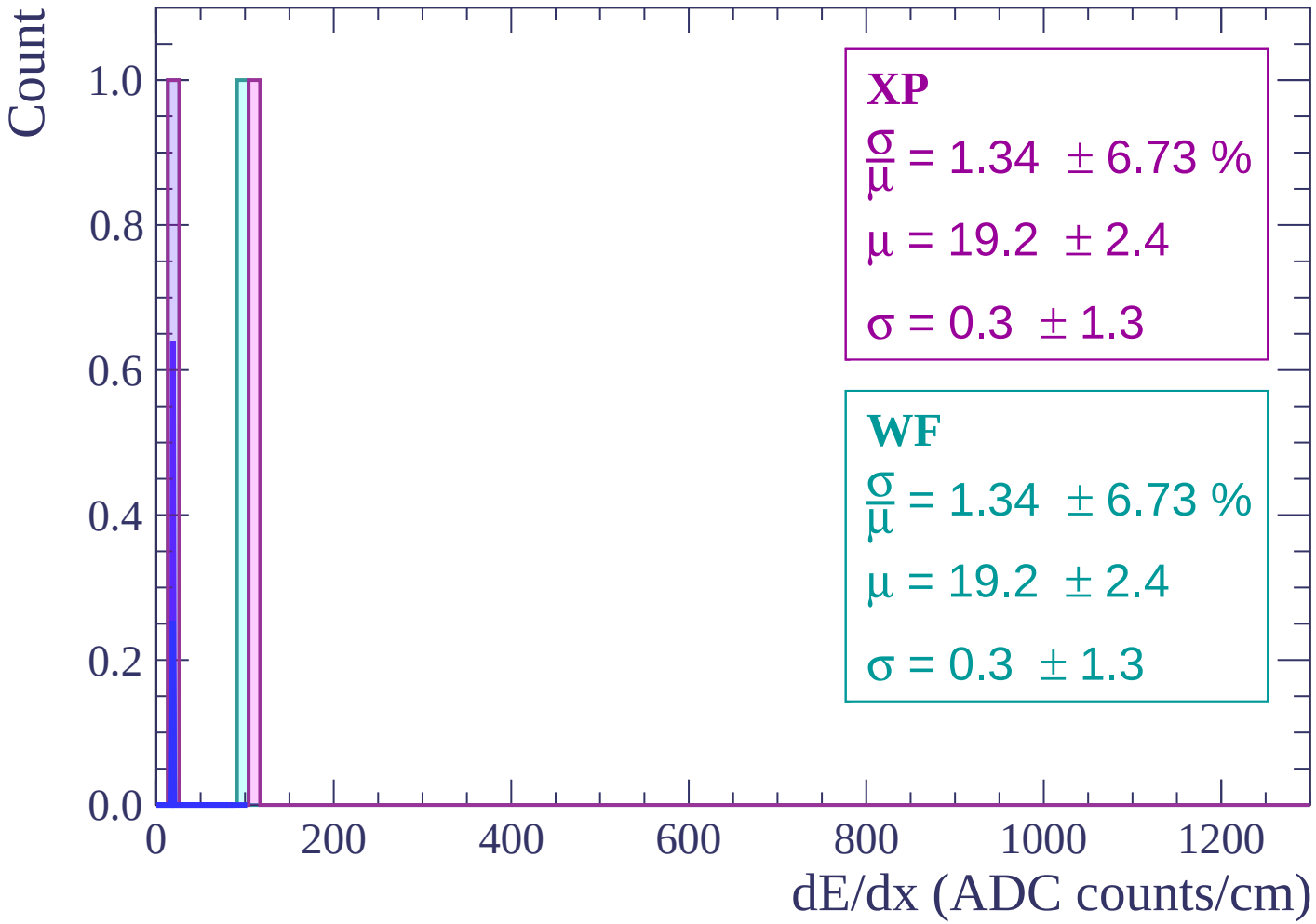

Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.61 ± 8.76 %
μ = 384.1 ± 122.5
σ = 2.3 ± 32.9
WF
σ/μ = 24.39 ± 158.83 %
μ = 266.5 ± 220.9
σ = 65.0 ± 369.4

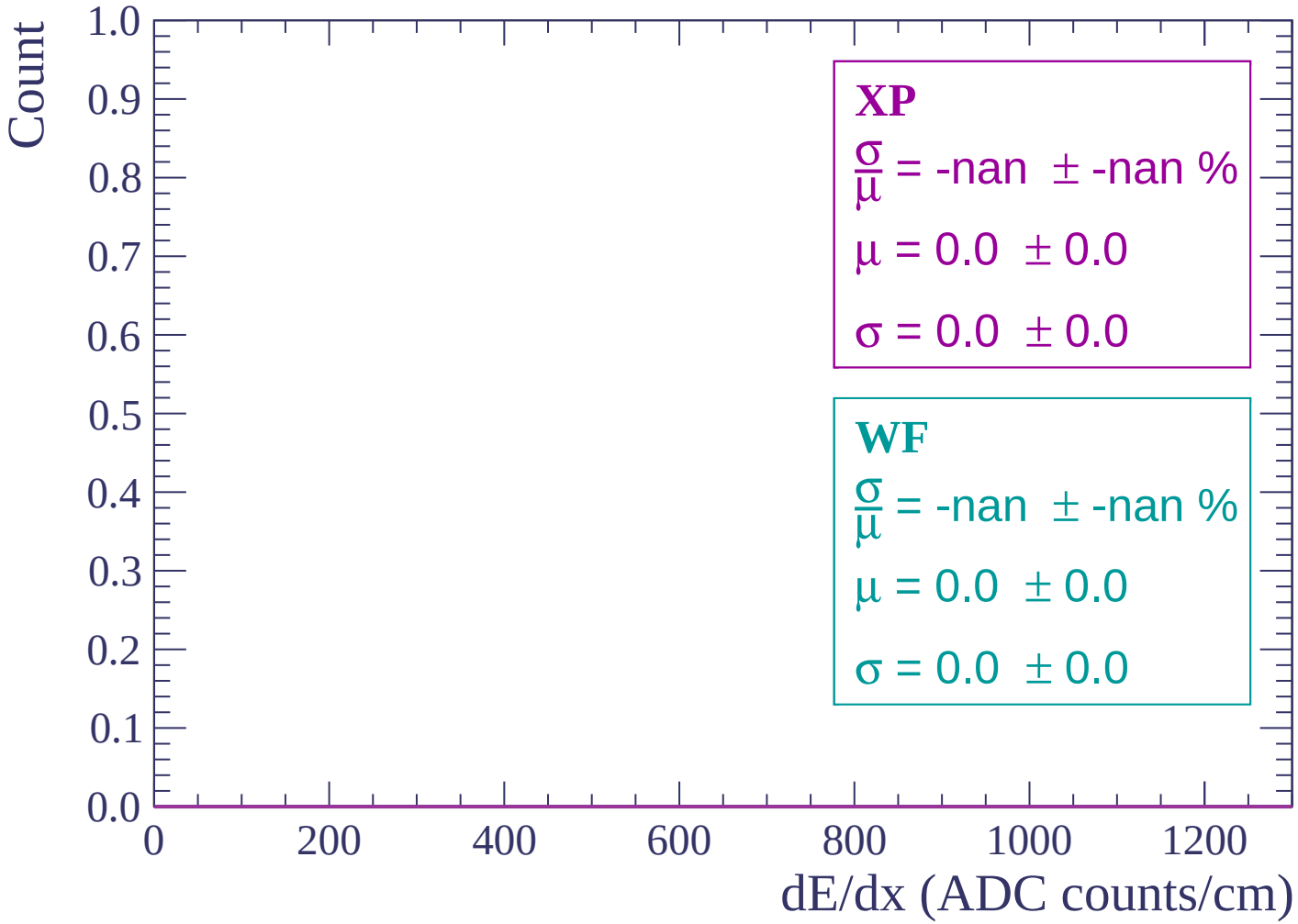

Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -46.90 ± -1.32 %
μ = -2524.3 ± 2558.2
σ = 1183.8 ± 1233.2
WF
σ/μ = 128.29 ± 43666.65 %
μ = 409.7 ± 138807.2
σ = 525.5 ± 820.8

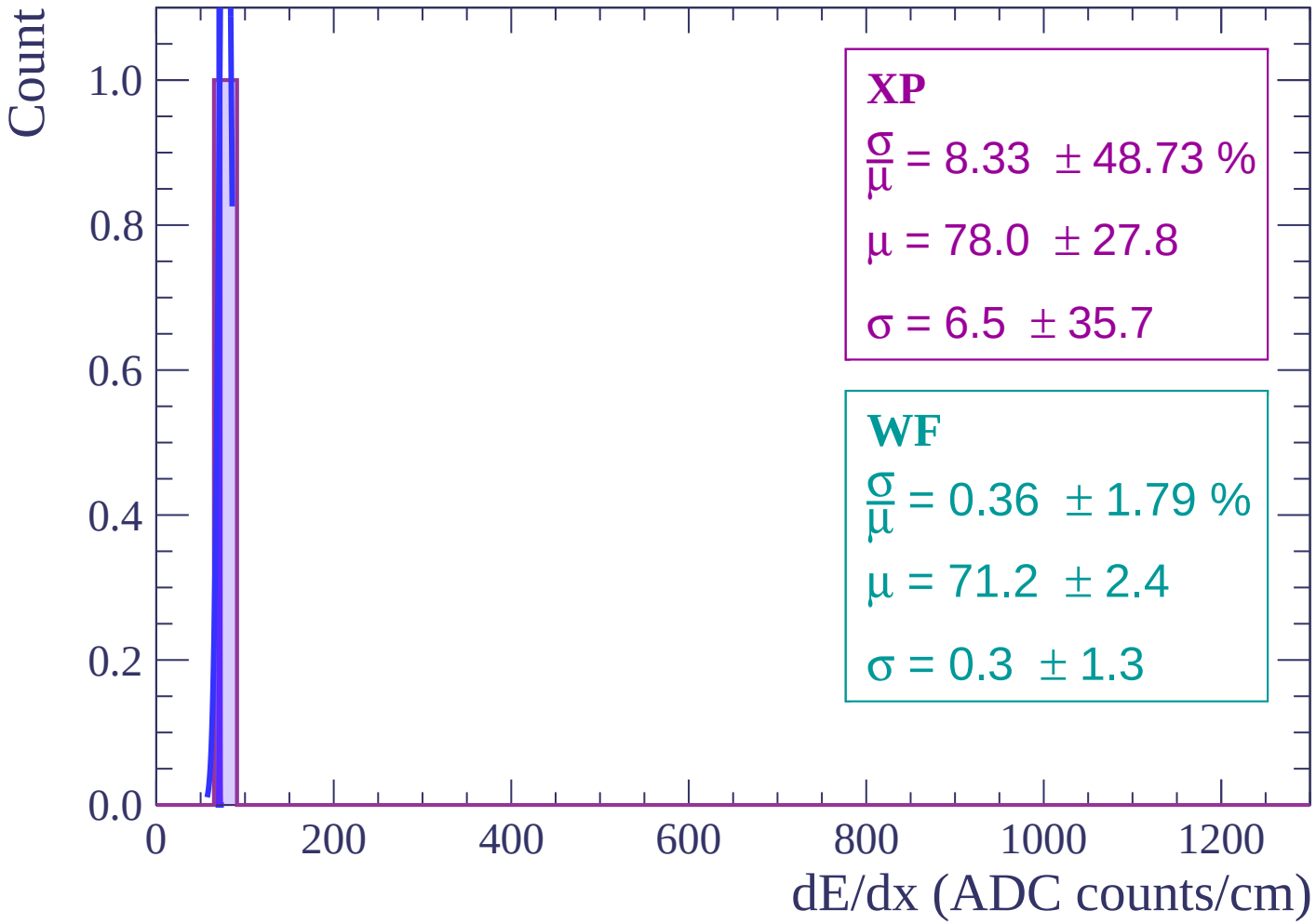

Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.19 ± 1.48 %
μ = 136.2 ± 1.8
σ = 0.3 ± 2.0
WF
σ/μ = 55.65 ± 12580.83 %
μ = 176.6 ± 39125.3
σ = 98.3 ± 437.4

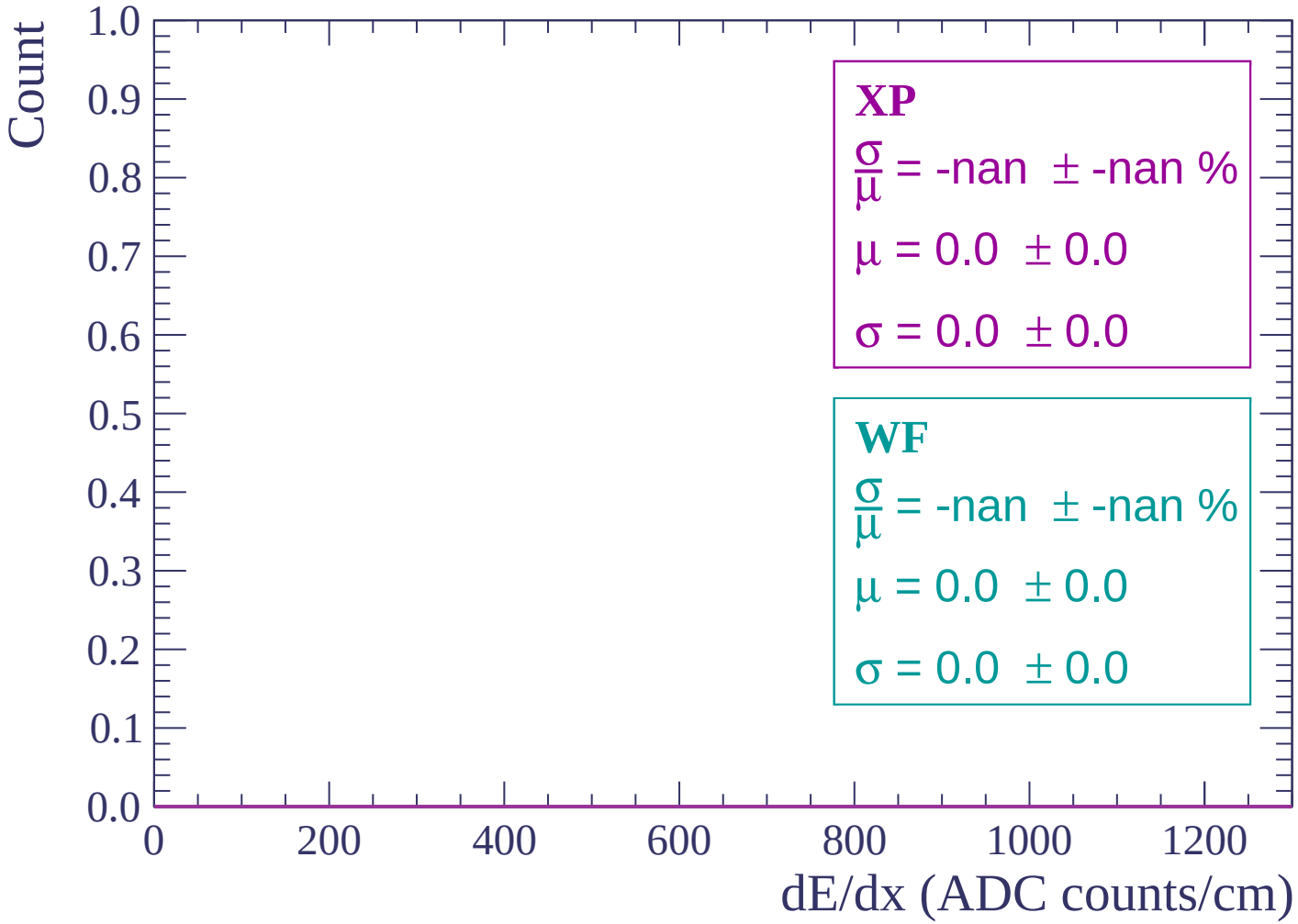

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.00 ± 114.21 %
μ = 130.0 ± 64.7
σ = 19.5 ± 138.8
WF
σ/μ = 25.00 ± 161.65 %
μ = 130.0 ± 99.2
σ = 32.5 ± 185.3

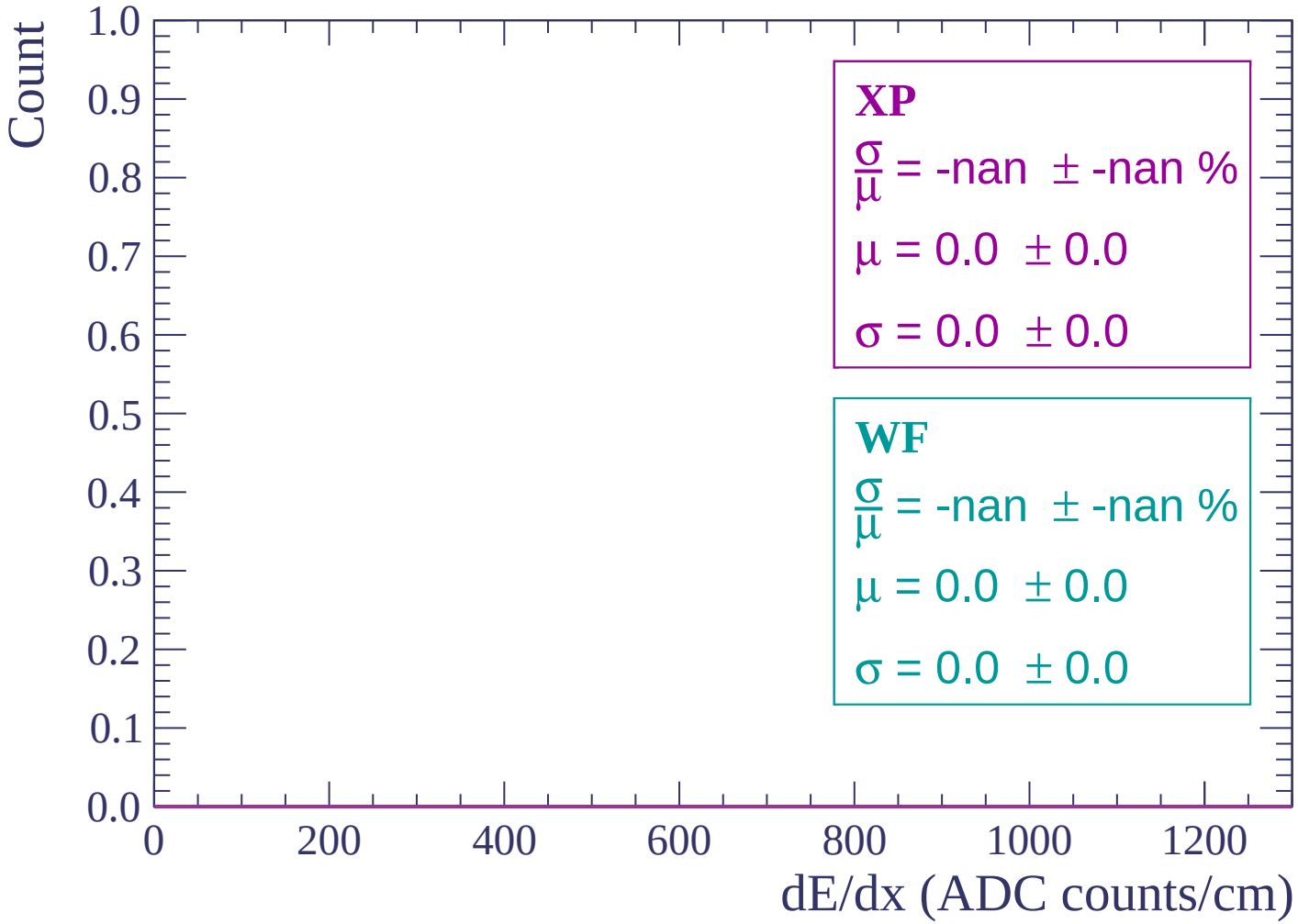

Count
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 1.34 ± 10.63 %
μ = 19.2 ± 1.8
σ = 0.3 ± 2.0
WF
σ/μ = 50.00 ± 381.58 %
μ = 13.0 ± 27.8
σ = 6.5 ± 35.7

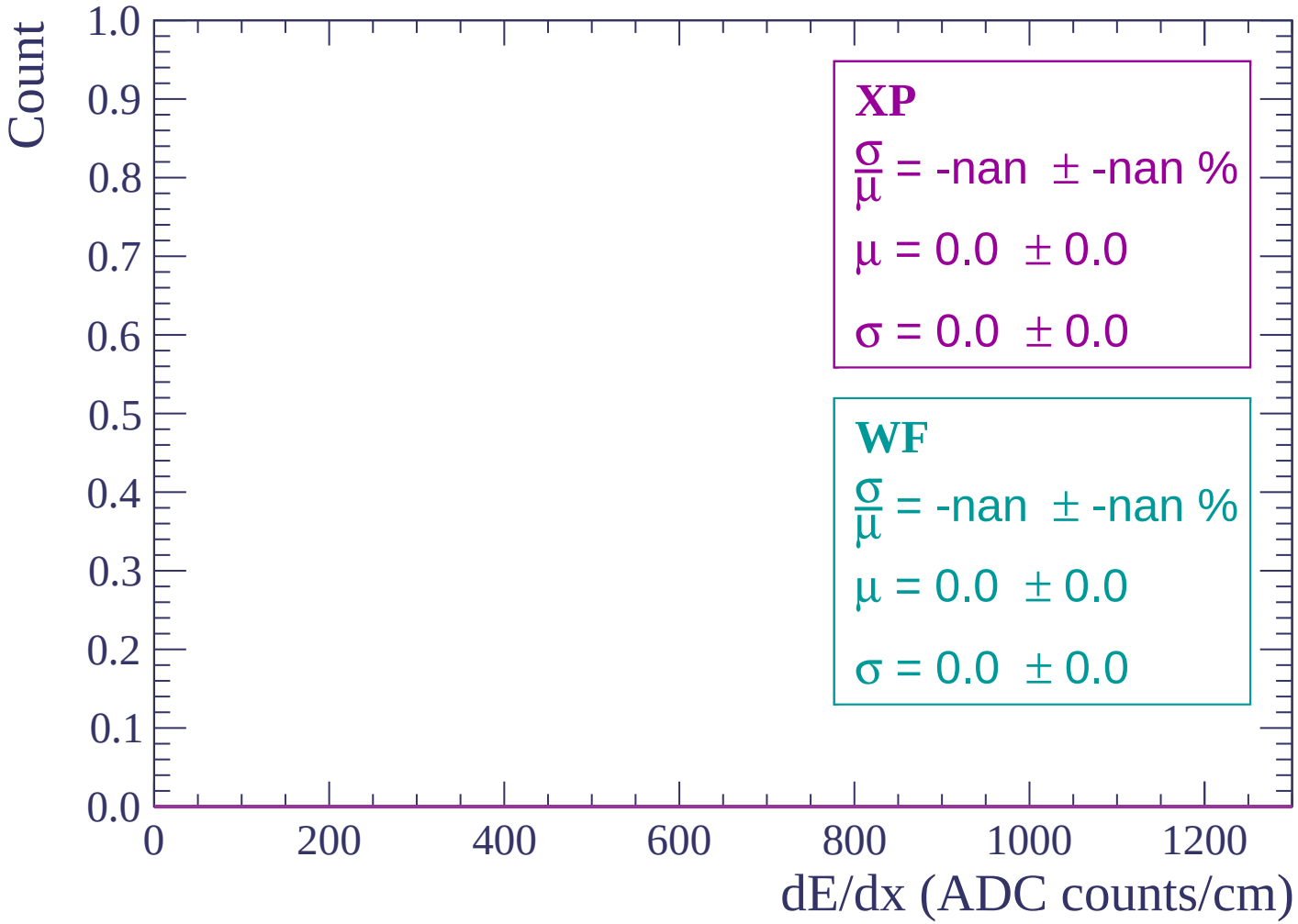

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 1.34 ± 6.73 %
μ = 19.2 ± 2.4
σ = 0.3 ± 1.3
WF
σ/μ = 1.34 ± 6.73 %
μ = 19.2 ± 2.4
σ = 0.3 ± 1.3

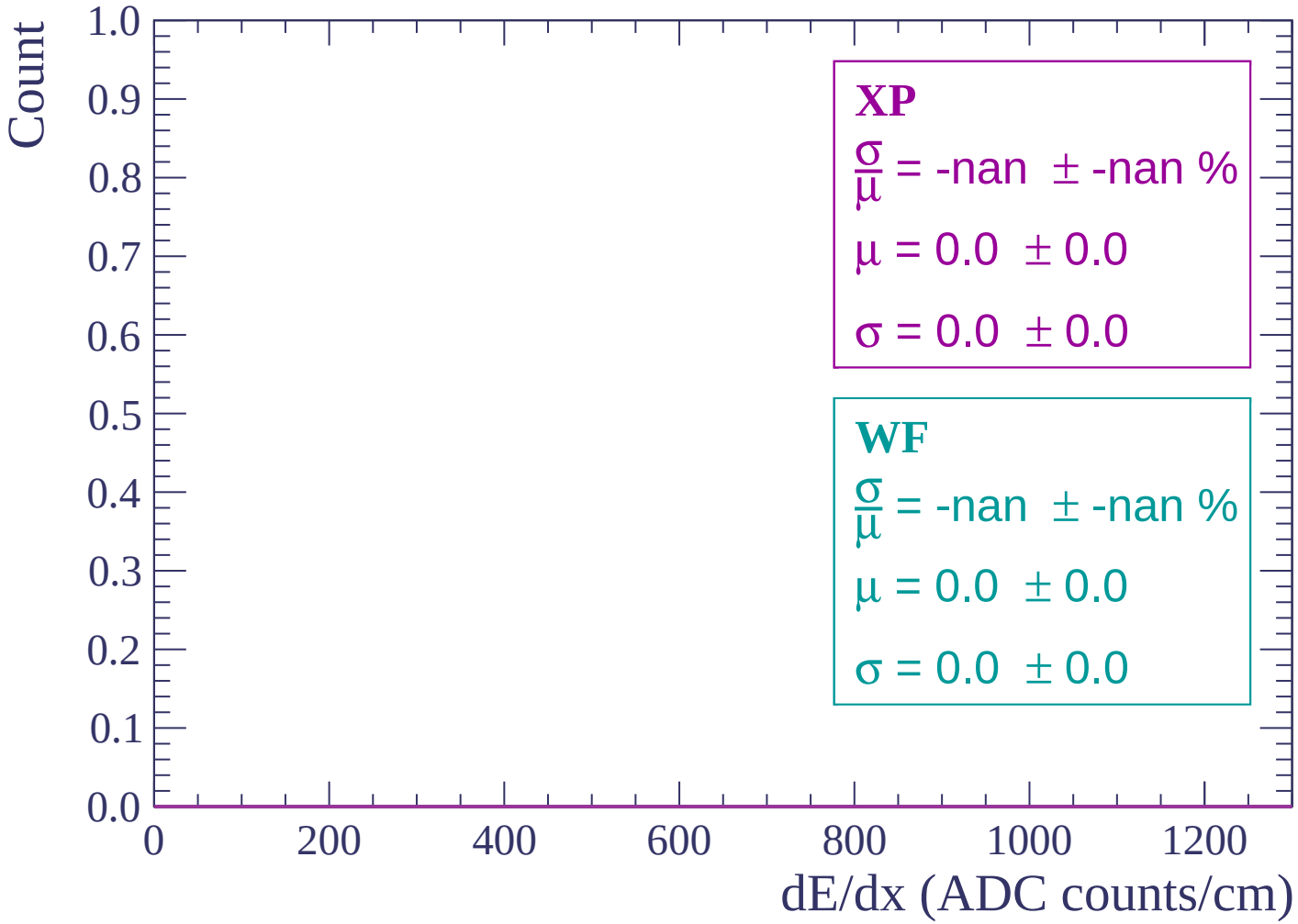

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

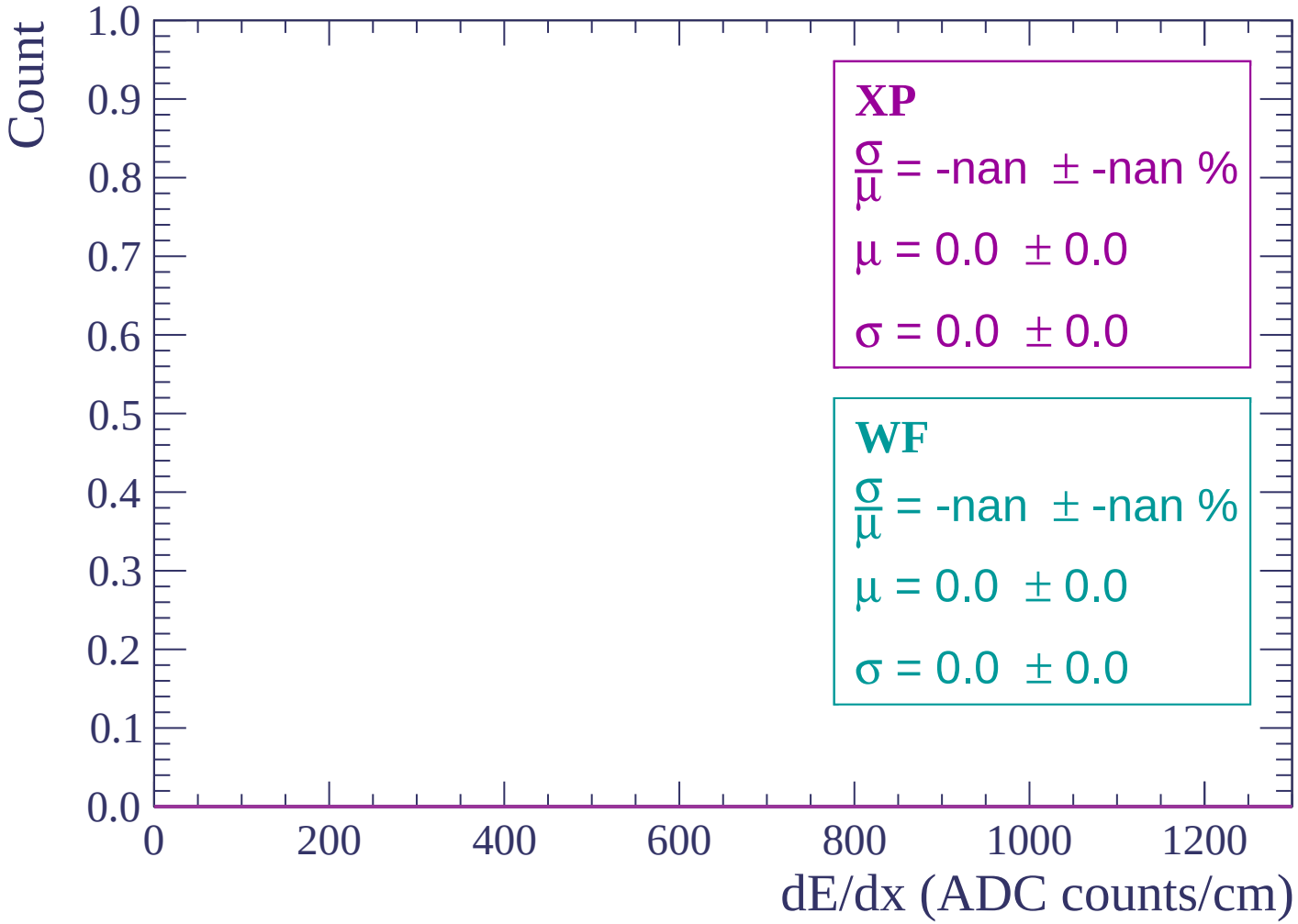

Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.33 ± 48.73 %
μ = 78.0 ± 27.8
σ = 6.5 ± 35.7
WF
σ/μ = 0.36 ± 1.79 %
μ = 71.2 ± 2.4
σ = 0.3 ± 1.3

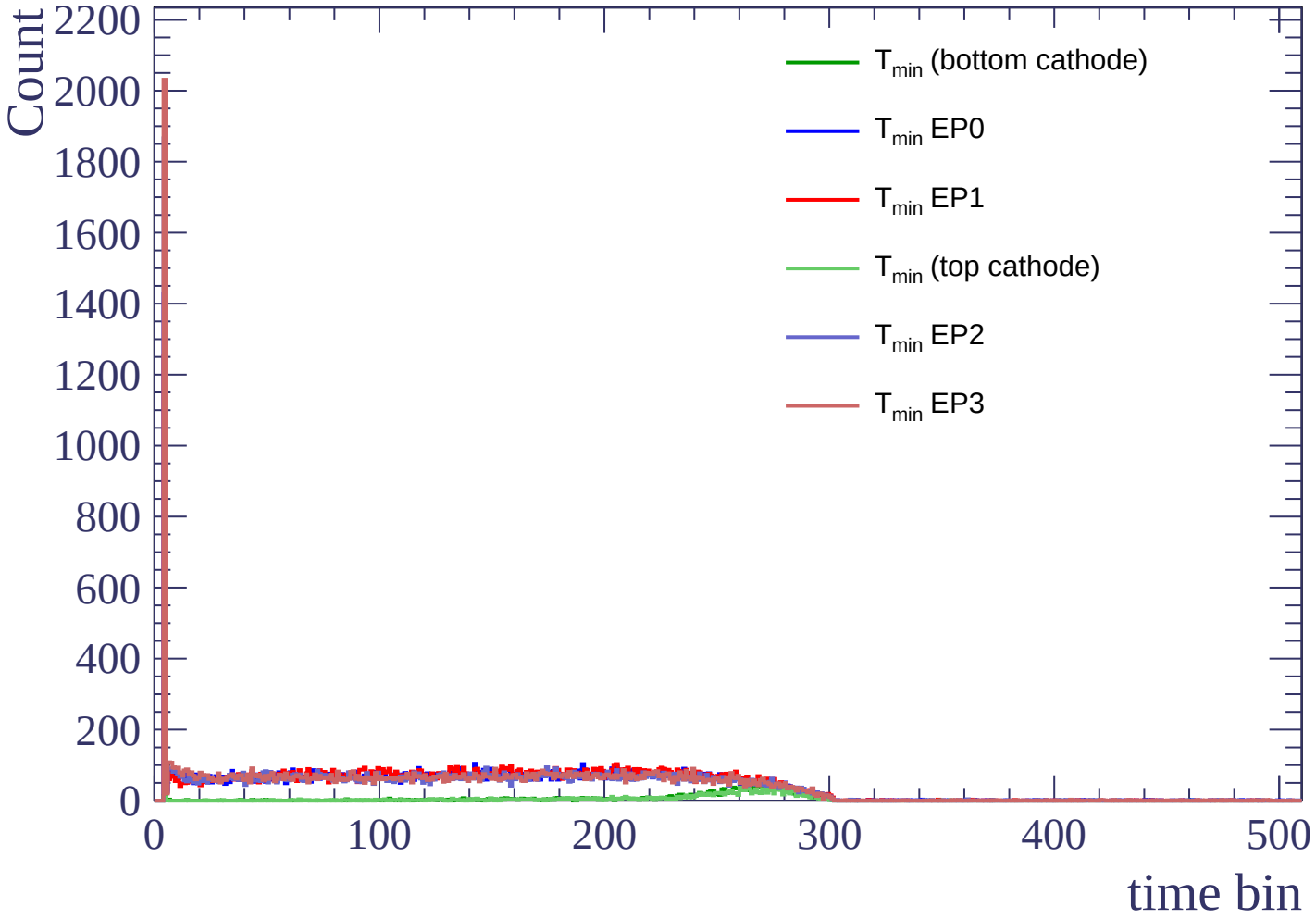

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0

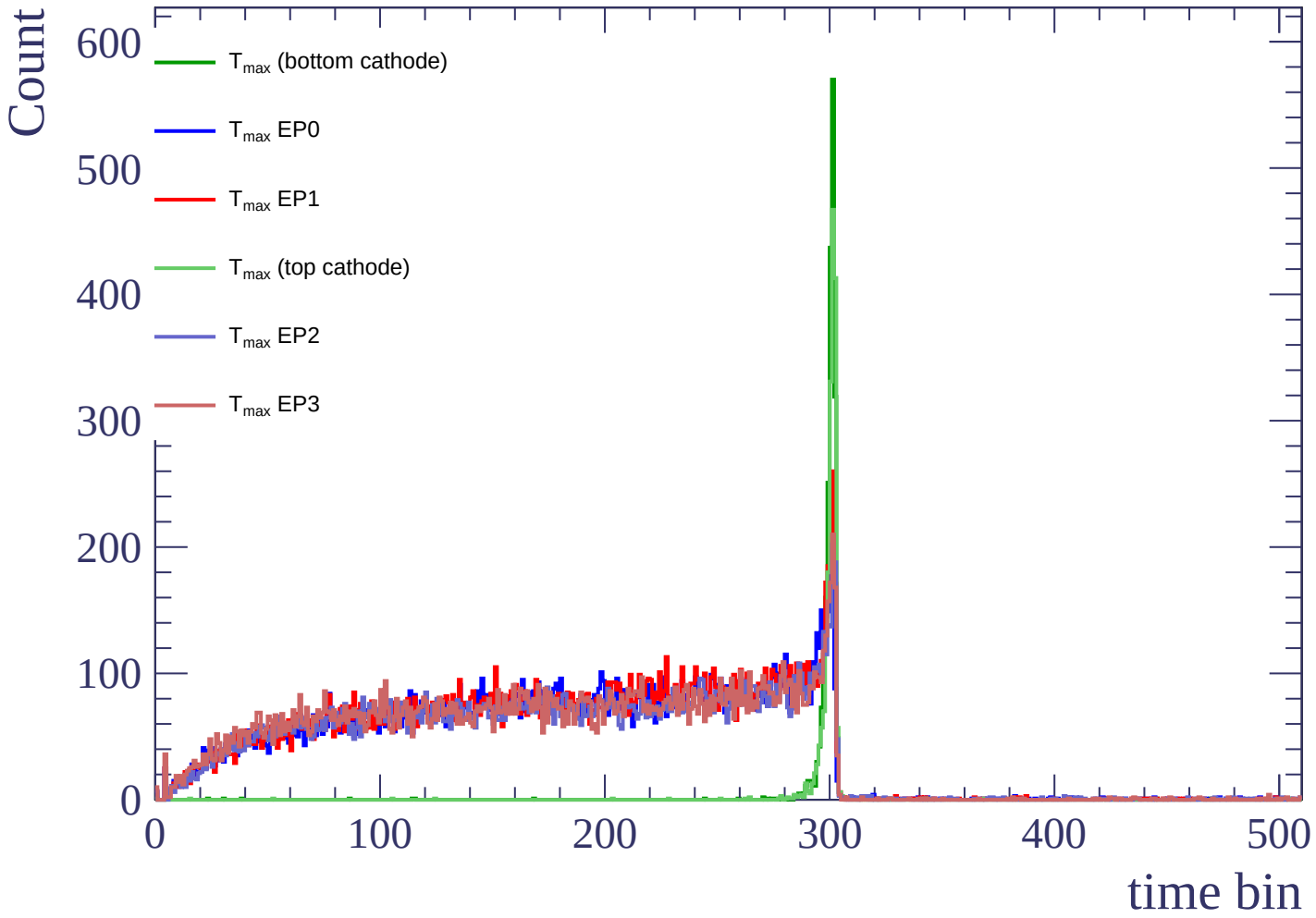

Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0
WF
σ/μ = -nan ± -nan %
μ = 0.0 ± 0.0
σ = 0.0 ± 0.0


Count
2200
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3


Count
600
500
400
300
200
100
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3